
SPEED SECRETS
SPEED SECRETS 365
A year's worth of daily tips to help you drive faster, smarter, and more confidently.
1 Speed Secret at a time.
EBOOK

Important Stuff

You know that motorsport of any kind is dangerous. Therefore, I provide my advice for you to use in the way you choose, and I can't be held responsible for anything that might happen as a result. Hey, you're a grown-up, right?! You're responsible for yourself, and by reading on, you are accepting that responsibility.

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Having said that, I love helping drivers perform better – I enjoy sharing what I've been fortunate to learn through experience, study, and observation. I hope you learn from this eBook, and become an even better driver because of it.

Keep learning and having fun!

Ross Bentley

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Introduction

What's the best way to eat an elephant? One bite at a time!

What's the best way to improve your driving? One bite at a time. Or, one tip a day.

That's the purpose of *Speed Secrets 365*: to provide you with one driving tip each day of the year.

All of these tips have been taken from what I wrote over nearly ten years in *Speed Secrets Weekly*, my weekly e-newsletter. It's a bit of a "best of" album; I've gone back through the 500-plus articles and chosen what I think are my best pieces of advice. As with driving, though, there's always more… there are always ways to improve. So, I've revisited all of these tips and given them a tune-up.

I recommend reading only one tip per day, using this just like you would an old-fashioned printed calendar. If you're like me, whenever I got one of those *Far Side* calendars (you know, the ones that were two inches thick), you sat there and flipped through most of the year's worth of comics. Then, each day you'd rip one off, laughing as though you'd never seen it before.

You can take the same approach with *Speed Secrets 365*, flipping through and reading whichever day's tip you want (you won't be laughing like you would with a *Far Side* calendar, though). But again, I recommend that you then take them one at a time, one day at time. The most important thing is to spend a little time with each one, thinking about the tip and how you can use it. If all you do is read it without internalizing it, reflecting on it, you won't be getting the most value from it.

I've purposely presented the tips in random order, instead of clumping them together by topic or theme. Why? Experience tells me that if you read too many things with the same theme, you tend to gloss over them and not think about the key take-aways. By mixing them up, I hope to almost surprise you with a different topic.

Keep in mind that some of what you'll read here was "of a time." In other words, I may reference a particular driver or event that happened a while ago, and yet, I think the message still applies to today's driving.

Okay, let's go. Green, green, green!

Keep learning and having fun!

Day #1: The more references you have, the better you'll drive.

Track references have become my life! And they should be yours, too.

The more sensory information you take in visually, auditorily (hearing) and kinesthetically (touch, feeling g-forces, balance), the more reference points you're taking in. And the more reference points you take in, the sooner you'll recognize when you're doing the right or wrong things in the right or wrong places.

Let's look at a simple example. When you first began driving on track, you probably had just three references for each corner: the turn-in point, apex, and exit or track-out point. If you made a mistake and turned in early for a corner, it was probably around the apex that you knew you were off-line. At that point, the correction you needed to make was a big one!

As you gained experience, you gained references - a seam or crack in the pavement, a surface change, the angle your car's at in relation to the curbing, a particular sound your tires make, the feeling of hitting a bump or compression in the track, and so on. With each of these references, the sooner you're able to identify whether you're in the right place on the track, what you should be doing with the controls, and where you're headed, the more consistently fast you'll be. If you make a mistake, you recognize it early, and therefore the size of your correction is small.

One of the differences between the best drivers I spend time with and every other driver is the amount of references they have - and the detail.

And just to reinforce what I've said before, references are not only things you see. They're also things you feel and things you hear.

The other point about references is that they're what Peter Krause refers to as places to "index" what you're doing. Just because a crack in the pavement, the sound of your engine echoing off a wall, or the feeling of your car compressing into a hill section of the track is not exactly where you want to do something, that doesn't mean you're not indexing off of it. For example, your engine's sound echoes off the pit lane wall; you pass the end of the wall, and your internal clock subconsciously counts, "one, two, three, now" and you begin to brake for Turn 1. Or you start braking four car lengths past the beginning of the curbing on the right. That's indexing off a reference point.

Preparing to drive a track is not just understanding that the first corner is a right-hander, and the second corner is a left-hander. It's not just knowing the radius of the corners. It's not just knowing when there's an elevation change. Preparing to drive a track is studying the circuit, and collecting as many references as you can. It's driving the track and soaking up more references. It's committing those references to memory, to your mental programming, so you no longer have to think about them — you simply take them in and use them.

As a challenge to you, take time to study the next track you're going to — watch videos, read articles about it, look at photos of different sections of it — and write down as many references as you can... before you even go there. And then when you get on track, soak up even more, and write them down on a track map or in your notebook.

Make soaking up references a way of life!

Day #2: Balance positives & negatives in your mind.

Positive self-talk is a commonly-recommended self-improvement technique. Just look at the number of self-help books touting its effectiveness. Based on this advice, you should simply stand in front of the mirror each morning and tell yourself over and over again, “I am fast. I’m the fastest driver in the world.” Then go to the track and drive faster than everyone else.

There’s an old Saturday Night Live skit, Daily Affirmations with Stuart Smalley that makes fun of positive self-talk, and with good reason. However….

Whether your belief in using this kind of self-talk lies on the spectrum from “It works” to “It’s total BS” is up to you, but there’s no doubt that it is on the spectrum. That is, there is some value to it, and that’s been proven through numerous scientific studies. However, not even I believe that simply standing front of a mirror and telling yourself that you’re faster than everyone else is going to make you World Champion. But I also know that self-talk does impact your performance, to some extent.

Like driving a car at the limit, it’s all about balance. Self-talk without any mental and physical practice will not work. I can tell myself all day long that I’m the best basketball player in the world, but unless I grow a foot taller, go back fifty years or so, and practice shooting hoops every day, it’s not going to work.

Similarly, if I tell myself over and over again that I’ll never be a fast driver, it will impact my belief system – what I truly believe about myself, deep down inside. And very rarely do we out-perform our most deep-rooted beliefs about ourselves.

So, there’s a balance to all of this self-talk.

Ultimately, our self-talk does become part of our mental programming. And there’s a balance with mental imagery (most often referred to as visualization), too.

Mental imagery is very powerful. I doubt there's a high-performer in the world who does not use it in some form, and the best — the elite — use it in a very deliberate way. They are clear on what they’re going to program before they ever begin. They close their eyes, slow their breathing, relax their mind so it becomes more receptive to the learning. Then they imagine whatever they’re wanting to mentally program in great detail, using more than just their visual sense; they feel it and hear it, too. They make it as real as possible, for your mind doesn’t know the difference between a real and an imagined experience — if you imagine it with enough detail.

Still, mental imagery alone will not do it all. You also need to be aware of where you are now in relation to what you’re trying to achieve. Mental imagery alone is a bit like standing in front of the mirror and telling yourself you’re the fastest driver in the world. It may lack the necessary level of reality, and that’s where awareness comes in.

You may already know where I'm heading here… to my "learning formula," MI + A = G.

"MI" stands for Mental Image — the very clear, detailed picture you have in your mind of what you're trying to achieve. "A" represents Awareness — where you are now in relation to this imagined goal. When you have both, your mind will do an amazing thing, and that is bring the two together, achieving the Goal (represented by the "G" in the equation).

To me, the awareness provides the reality check, the balance between expecting "magic to happen," and being bogged down in "this will never happen."

The winner of more Olympic gold medals than anyone in history, Michael Phelps, was taught early in his career (by his coach Bob Bowman) how to use mental imagery to develop skills and prepare for events. Phelps was taught to see in vivid detail exactly how he was going to win his races. And he devoted massive amounts of time to deliberately pre-playing his developing skills, mindset, and how he would perform in upcoming events.

During the 200m butterfly event at the 2008 Beijing Olympics, Phelps' goggles filled with water the moment he dove in to start the race. He swam blind for the entire race, but still came up with a gold medal-winning performance. When he was asked what it was like to race blind, his response was, "Just like I imagined it would be."

See, Phelps had gone beyond his coach's advice and done mental imagery of things going wrong, and then adapting. No, he didn't imagine his goggles filling with water, and then struggling to deal with it. Instead, he saw himself managing the problem, and overcoming it with a near-perfect performance.

Prior to some of my clients racing in endurance events, I've had them write a short piece describing how they would deal with fading brakes, a gearbox stuck in one gear, an engine that has dropped a cylinder, and even being so far in the lead that they could reduce their pace to put less stress on the car. After writing it out, I'd have them do mental imagery of the various scenarios.

Some will tell you that all of your mental imagery should be positive, but I disagree. Some of it should be you dealing with problems. You might say that is being positive about a negative, and "that's okay" (to quote Stuart Smalley!). To me, it's a balanced approach to using mental imagery and self-talk. You're preparing yourself for whatever happens. And even though the exact problems that you've imagined may never occur, the fact that you've "pre-"practiced dealing with those issues in your mind will help you feel more confident in your ability to deal with them in reality. Your mindset and your belief system will be ready to deal with the problems.

Day #3: Get real.

Let's look at four different scenarios, and then what we can learn from them.

Scenario #1:

Me: "Where are you beginning to brake for Turn 6?"

Driver #1: "At the 200 board."

Driver #2: "Just before the 100."

Driver #3: "Wow, I can't seem to brake any later than about the 300 marker."

Me: "Interesting. When I'm watching from the side of the track, I see your brake lights coming on at the same place for all three of you. So, I just checked the data, and I was right – you all begin braking at the exact same place."

In this case, perception is not reality. How is that possible?

Driver #1 began braking just as the nose of his car passed by the 200 marker. Driver #2 begins braking when she sees the 100 marker (while passing the 200). Driver #3? He identifies the 300 as he's passing it, and then begins braking (at speed, he's now at the 200 marker when his foot actually applies the brakes).

Let me guess. You've had an experience similar to this, in which you and a couple of fellow drivers are talking and what they're saying just doesn't seem to match what you're experiencing. And the worst part is that you can't help but question your abilities when you hear what other drivers are saying.

Scenario #2: A driver tells you that he never trail brakes, and yet you know that you are. So, you can't help but wonder whether you should stop, thinking that maybe you'll be faster if you didn't trail brake.

When your data is compared to the other driver's, you see that you both trail brake the same amount. The difference, in this case, is that the other driver used to drive a car that rewarded lots of trail braking – almost to the apex. Now that he's driving a car that doesn't require as much trail braking, he thinks that he's not doing it at all.

Scenario #3: Next, you're talking to a fellow driver who tells you she aims her car for the tallest tree on the horizon when cresting the hill leading into Turn 5. You think about it for a while, and can't help but question the way you're driving that turn because you didn't even know there was a tree on the horizon! Are you on the wrong line?

Scenario #4: You're racing wheel-to-wheel with another driver. You have the line into the braking zone of the left-hander Turn 2, on the inside. Just as you turn in, you feel a tap in your right-rear, causing you to wash a little wide as you head towards the exit of the corner. And that's when you get tapped again, but this time it's in the left rear, and causes you to spin.

Back in the paddock, the driver who hit you twice comes up with his in-car video, demanding an apology. Confused, and less than thrilled with his verbal attack, you tell him that it was all his fault. Of course, that leads to a lengthy and heated argument, only to be “broken up” by another driver pitted next to you. He sits down and watches both in-car videos – the other driver’s, and yours – and comes away with the conclusion that it was “a racing incident.” Oh, did I mention that this driver was following both of you and saw it all in real-time? Yeah, despite what was obvious to you, the perception of what happened and who was at fault was very different for three different people.

Take-Away: It all depends on your perspective. And that’s why it’s important to get real information before making a decision on what you should or shouldn’t be doing (or have done). I’ve witnessed far too many drivers hear just a bit of conversation between two experienced drivers, and take that piece of communication as gospel, changing the way they approach a corner, a driving technique, and/or a racecraft situation.

It reminds me of how some drivers/engineers/teams will hear that the car that’s on pole is using 900 lb/in rear springs, so they figure that they should change theirs to match. And that’s just plain stupid if you’re not also running the same front springs, the exact same shocks and settings, the same anti-roll bars, same ride heights, same camber, same caster, same toe, same aero settings, and so on. Oh, and that assumes that the drivers in both cars drive in pretty much the same way.

To take a small piece of information and make big conclusions from it is a mistake waiting to happen. It’s absolutely okay to listen and learn from others, but get real information before making changes – or even doubting your abilities and what you’re currently doing.

Day #4: The advanced trick stuff is just doing the basics really well.

It's time to go back to some basics. I'm seeing many experienced drivers making mistakes - and they're mostly because of bad habits.

What I've seen:

- Steering with one hand, leaning or bracing the other arm against the door, or resting one hand on the shifter.
- Steering with hands in the "praying mantis" position (at 10 & 2, or even 11 & 1), rather than 9 & 3 (resting the thumbs where the steering wheel designer meant them to be).
- Turning the steering wheel way too much, in and out of a corner.
- Snapping the foot off the brakes, rather than smoothly releasing the pedal (probably the number one bad habit I see).
- Looking from cone to cone, rather than through the corner.
- Focusing too much on the Begin-of-Braking point, and not on the End-of-Braking point.
- Hovering the left foot above the clutch pedal, rather than bracing against the dead pedal (to the left of the pedals).
- Leaning the whole body into turns, rather than keeping the head and body upright, so your eyes provide your brain accurate information.

It's easy to say to yourself as you read this, "Okay, but I don't do any of those things." Guess what? When I pointed these things out to the drivers I saw doing them, their first reaction was, "I don't do that." So maybe you do.

If you use an in-car camera, watch for subtle (and not-so-subtle) habits slipping into your driving.

If you work on improving these basics, you'll probably find that the advanced stuff doesn't matter. In fact, the advanced stuff is really just doing these basics very well!

I can't count the number of times that a driver has asked me for the super trick advanced technique that's going to make them the half second faster that they're looking for. But I can tell you the number of times I've told someone what that super trick advanced technique is: zero. That's because there isn't some secret that only a handful of "gifted" drivers know. Okay, that's not entirely true. The "gifted" drivers know that the secret is just doing the basics better than anyone else.

Day #5: Make the car show its weakness.

Something I see way too often is a driver wanting to make set-up changes to their car before they're driving anywhere near the limits of the car.

Okay...there is the very rare instance where a car is handling in such an evil way that something needs to be done before the driver can get anywhere near the limit. But that's the exception to the rule.

Drive the car with the mindset that you're trying to find its weakness — understeer or oversteer on entry, mid-corner, or at exit; instability while braking; inability to put the power down; "lazy" turn-in; etc.

If you get out of your car after a session and you honestly don't know what the car needs to make it faster, then you know that you need to focus on your driving, first. Keep pushing, keep working at driving faster (in a strategic way) until your car tells you it has a weakness somewhere in its handling. Then, and only then, can you start making adjustments to the car's setup.

Day #6: Performance driving is 3-dimensional.

Performance driving is a very complex activity, where one thing interacts and affects something else, and that thing impacts something else, and that relates to something else, and... well, you get the idea. It's definitely not a situation where this affects the next thing, and that affects the next, and that affects the next thing in an orderly, logical fashion. No, everything is interrelated and has an impact on something else, and in a non-linear way.

What?

See, I've been thinking lately about driving being a 3-dimensional undertaking:

- The first plane is the cornering line.
- The second plane is how you use the controls (steering wheel, throttle, brakes, gearbox).
- The third plane is how you combine these two.

Okay, if you're racing, there's actually a fourth plane, and that's traffic and racecraft. But, for now, let's leave the fourth dimension to Max Verstappen, Scott Dixon, and Stephen Hawking!

See, the line is not static. It changes —slightly, but definitely — based on how you use the controls (and conditions).

Using the controls is not static, either. It changes, too, based on the line you want to drive (which is not always the line you actually drive).

Hey, my 3-dimensional analogy may not be perfect, but you get the idea. My main point is that the line you drive is somewhat dependent on how you use the controls, and vice versa. Change your line, you have to change how you use the controls; change how you use the controls and you have to change your line.

For example, if you release the brakes rather abruptly when entering a corner, you're more likely to need a later turn-in and apex to get through the corner quickly and onto the next straight. If you release the brakes more slowly — trail brake more — you can usually turn in slightly earlier because you're rotating the car due to how you have the car balanced. One way may not be any better than the other — it's just different.

If you've ever wondered why one driver can make an earlier or later turn-in and apex work for them, and you can't, it's because of a difference in how you both use the controls.

I hope there are a couple of take-aways for you from what I've written here:

- If you only focus on the cornering line, you're missing a big opportunity to improve your driving — further developing and fine-tuning how you use the controls to match the line you're driving can lead to big gains.
- There is more than one way to drive fast.

Day #7: Many tiny gains lead to a big gain.

Count to a tenth of a second. You can't do it, right? It's such a tiny amount of time.

What do you need to do to gain just one-tenth of a second in one corner? Not much, right?

- If you were to enter the corner with half-a-MPH more speed, you would gain a tenth of a second.
- If you used an additional few inches of track at the turn-in, apex, and exit of the corner, you'd gain a tenth of a second.
- If you were to get to full throttle a car's length earlier exiting a corner, you'd gain a tenth of a second.
- If you compressed your brake zone approaching a corner by a couple of car's lengths, you'd gain a tenth of a second.
- If you began releasing the brakes a car's length earlier (but not necessarily finishing braking and taking your foot completely off the pedal any earlier — in fact, maybe you'd even keep a little bit of pressure on the brakes longer), you would carry a little more speed into the corner and still be able to get the car to turn in and follow the line you want — and you'd gain a tenth of a second.

Let's look at this the other way around.

- If you turn in to a corner from a foot away from the edge of the track, barely get to the apex, and are not right at the edge exiting the corner, you will lose a tenth of a second.
- If you release the brakes too quickly at the end of the brake zone and cause the car to become less balanced, you will lose a tenth of a second.
- If you hang on the brake pressure too long — trail brake too long — you will drag off too much speed, and you will lose a tenth of a second.
- If you begin applying the throttle too soon, resulting in having to ease back up off of it before going to full throttle, you will lose a tenth of a second.
- If you don't unwind the steering enough exiting the corner — "pinching the exit" by holding the car in the corner too long – you will lose a tenth of a second.
- If you "crab" in from the edge of the track at the entry to a corner, you will lose a tenth of a second.
- If you jerk the steering into the corner too abruptly, you will lose a tenth of a second.
- If you ease up on the throttle in that fast corner when you could take it at full throttle, you will lose a tenth of a second.

Now, imagine you're driving a track with ten corners. And you gain a tenth of a second in each corner. Just a tenth of a second! You've now improved your lap time by a second. What would you give to gain a full second in lap time?!

Many drivers, when looking to improve their lap times, go looking for too much. They believe they're a full second off what the car is capable of, so they try to find that entire second in one place. Inexperienced drivers may be able to find a second in one or two places, but as you know, that low hanging fruit gets picked fairly quickly. And then it's a matter of finding little gains in time.

As you improve your driving, the gains become harder and harder to find, and they also become smaller and smaller.

How do you eat an elephant? One bite a time. If you're a writer, you must read the book *Bird by Bird*, by Ann Lamott (even if you're not a writer, you should read it!). In it, she relates the story of her father (a well-known writer) giving her brother advice about writing a school report on birds when he couldn't get started. He said, "Bird by bird, buddy. Just take it bird by bird." Driving is no different. Approach improvement one bite of the elephant at a time, bird by bird, looking for a tenth of a second here and there.

Day #8: You do what you do because you're programmed to do so; you don't do what you want sometimes because you either don't have the right programming, or you accessed the wrong program at the wrong time.

The way you're sitting (or standing, or laying down) right now is because it's a habit. And a habit is a mental program.

Where did this mental program come from? Mostly from doing something over and over again, either physically or mentally. Sure, you can argue that some of your mental programs are part of your DNA, and you were born with them. And you'd be right — at least for your core activities, like walking. But the bulk of what you do is because you've built habits from doing them, over and over. Oh, and you also develop mental programs by observing and imitating others, often without even knowing you're doing that. If you've ever noticed yourself do something that you've seen your parents do, you know what I'm talking about.

How you react to what people say or do to you is no different. And often, the reaction triggers an emotion - or an emotion triggers a reaction.

Have you noticed that some drivers react in very emotional ways in the car, and others do not? In NASCAR, how often did you see Jimmie Johnson have an emotional response while racing? How about Kurt Busch? In Indy Car, compare Scott Dixon to Will Power. In F1, how about Lewis Hamilton and Sebastian Vettel?

Were "calmer drivers" born with the ability to control their emotions better than others, or is that something they've learned — mentally programmed — over many years? I'm not going to argue over how much is nature versus nurture here, but I will state as fact that some amount — even a large amount — of that ability has been, and can be, developed.

There was a point in my racing career where I needed to improve my ability to adapt my behavioral traits in certain situations. For example, there were times I needed to crank up my assertiveness, times I needed to become more patient, and times I needed to be more outgoing. Of course, there were times I needed to do the opposite, and I also needed to manage emotions like anger, frustration, jubilation, empathy, and even fear.

What I knew then was that hope was not a very effective strategy! So, I didn't just hope that I'd get better at managing my emotions and behavioral traits. Instead, I laid out a program of mental imagery, and visualized myself in a variety of situations, all requiring a different behavior. For at least twenty minutes every morning, and every night, I saw, felt, and heard myself — in my mind — adapting to these different situations. I mentally rehearsed my reaction to a driver blocking me, to having a diminishing lead with a couple of laps to go in a race, to a crew member saying something that pissed me off, to dealing with a journalist asking about our team's poor results, to making a mistake and losing a position on the track, and so on. In each scenario, I saw, felt, and heard myself staying calm, relaxed but energized, and focused on what I needed to do — and how I

would react to whatever happened. I built a program around this mindset: "I'm ready for anything — nobody can upset me. I'm cool and calm. Focused."

Critically, I tied in a "trigger" to each behavior, to each situation. The trigger was a phrase that I'd say to myself that clicked on my mental program's icon, and launched that application. For me, I used "Car dancing" as a trigger to focus on the act of driving, and "Watch this!" for cranking up my assertiveness, for triggering my go-for-it mindset.

How do you manage your emotions on the race track? By deliberately mentally rehearsing various scenarios, mentally managing your emotions, using a trigger word or phrase, over and over again, until they become your default mental program - your default reaction, your habit. And then, when on track, triggering the right mental program for each situation by using your trigger word or phrase.

Whether they realize it or not, drivers like [pick your favorite driver] have achieved success, in part, because they've programmed themselves to manage their emotions, their behavior.

You have a choice:

1. You can let your emotions manage you, and race poorly.
2. You can hope that you get better at managing your emotions.
3. You can deliberately program, and then trigger, the way you want to react when emotions happen.

Day #9: Use quick hands to control a slide.

I've had my share of opportunities to watch in-car video of drivers spinning and crashing. No, not from a gory entertainment perspective, but from one of helping drivers learn from their experiences. I help them understand what they could have done differently to control their car better.

Rarely is a crash the result of an isolated mistake. Most often, it's the end of a series of mistakes, with the early one(s) not corrected soon enough. In fact, experience tells me that if a driver corrects the early mistake soon enough – and appropriately – the following mistakes don't happen, and a crash never happens. I'd go so far as to say that one of the biggest differences between a very experienced driver and a less-experienced one is the more accomplished driver will catch the early mistake immediately; there is nothing to deal with after that. The newer driver often doesn't begin fixing the error until they're dealing with the second, third, or fourth mistake in the series – and that's too late!

The key is sensing and identifying the earliest signs of a mistake, and beginning to correct them immediately.

Think about your own experience. How often have you made a mistake, caught it, and carried on? And how often have you, for example, gone wide exiting one corner, causing you to lift out of the throttle, resulting in the rear of the car stepping out; then you didn't get enough steering correction in to stop it, so around you go in a spin, sliding off the track, and maybe even hitting something? If you'd corrected the very first mistake quickly and effectively, the other mistakes would not have happened.

I want to focus today on correction with the steering.

When watching literally hundreds of drivers' in-car videos, nine times out of ten, the real cause of the spins or crashes were not catching the rear of the car with enough steering input quickly enough, or not putting in enough steering correction. The rear of the car steps out in oversteer, and the driver doesn't turn the steering enough with opposite lock, or they do it too slowly. If they turned the steering wheel quickly, and more, the slide would have been caught, and the spin/crash would not have happened.

I see it waaaaaay too often! I think part of the problem is that drivers are told to be smooth with their steering, so they're reluctant to turn the wheel quickly enough to correct. Or, they just have a habit/program to turn the wheel slowly. Or, they don't sense the need for a correction soon enough. Or, they're afraid that turning the wheel quickly will make the car do something really bad. Or, they're not looking in the right direction. Or, they think they can control it with a bit of steering, only to find out they can't. That's a lot of parts!

Here are four free or relatively inexpensive ways to improve your chances of catching a mistake early and appropriately:

1. **Skid pad training.** There's no doubt this will help, and it's way underutilized. But skid pad training does require good coaching, too, otherwise the driver will just practice more of the same (not turning the steering enough or quickly enough; and not using their vision appropriately). The key is to practice moving the wheel quickly and enough to correct. That usually requires someone to observe and provide immediate feedback to really make this most effective.
2. **Sensory Input Sessions**. This is something I've written about extensively, so I won't go into great detail here, but the basics are going on track and focusing solely on one of the three senses we rely on when driving (visual, kinesthetic, auditory). This means an on-track session simply focusing on soaking up more auditory input; then a session focused on kinesthetic input; and then a session focused on visual input. By focusing solely on taking in more sensory info at a conscious level, you're training your brain to do this at a subconscious level. And when your brain has more sensory information to work with, it performs better. Yes, a huge part of reacting properly to your car sliding is sensing it earlier, and doing these Sensory Input Sessions will fine-tune your ability to this. In my experience, there is no more effective way to become sensitive to what your car is doing than by using Sensory Input Sessions.
3. **Simulator**. For those of you with a simulator, change the car's setup so that it oversteers dramatically, and then practice "quick hands" to correct any slides. Oh, and this can be fun!
4. **Mental imagery**. You know, close your eyes, relax, breathe, and visualize yourself catching mistakes, slides, and near-spins with quick hands. Practice in your mind. It's super-important to physically move your arms and hands while imagining catching a slide (if you have a steering wheel laying around, hold and turn it while doing mental imagery). And then a key is to build the trigger, "Quick hands." Make that phrase part of your mental programming, so that whenever the car steps out, you use "quick hands." (*Mental Imagery Guide for Drivers* eBook available at eBooks)

I could have included driving autocross as a way to improve, and for sure it will help. But the one downside of this strategy is that most corners are of relatively short duration. Compare a typical autocross corner to Turn 1 at Mid-Ohio, for example. The advantage of a skid pad is that a driver can spend a long time in a slide, and therefore learn how to manage the slide (with quick hands). This is one reason I've had road racers whom I've coached go to DirtFish Rally School — so they can slide a car in gravel for a long duration. Rather than a "what just happened?" moment, it becomes a "okay, that's oversteer, and I need to catch it" longer moment.

Having said that, if you want to improve your car control skills, autocrossing will definitely help, so add that to the list if you have the opportunity. The message that you need to program in your mind is this: "Initiate slowly, react quickly." Your goal should be to turn into most corners with slow hands, but catch any slide with quick hands.

Day #10: Use deliberate practice to fine tune your ability to sense the limits of your car.

Let me share some specific, deliberate exercises that you can do to improve your ability to sense your car's limits.

Sensory Input Sessions – I've discussed this exercise many times in *Speed Secrets Weekly*, my books, videos, and everywhere else I go. Why? Because it works. It's one of, if not the, most effective method of improving one's driving that I know of. And since I've covered it a lot in the past, I'm not going to go into great detail here (please go to those other sources if you need more info - start with this YouTube video I put together a while ago).

Take a session on track to focus solely on soaking up more auditory information, then another session to focus on kinesthetic input, and finally on your visual sense. Again, these are separate sessions, as what you're doing is focusing on just the one sense at a time. By doing this, you get better at providing your brain with better information that it can work with. And by doing this at the conscious level for a session, subconsciously you'll be doing this better in the future.

By the way, this (like all of the exercises I'm suggesting here) is not something you do once and never have to do again. I recommend you make these a regular thing. No, not every single session, but at the beginning of the season, and at least a few times during the year. They're particularly effective when learning a new track, after you've gotten down the basic direction and rhythm of the track, as they'll help you learn the track faster than just about any other way.

Traction Sensing Sessions – Devote a set amount of laps or time to simply focusing on how much traction/grip your car/tires have in each corner. Drive the track while asking yourself, "On a scale of 1 to 10, how much of the tires' traction am I using?" If the tires are at their limit, that's a "10." If you feel you've gone slightly beyond the limit and the tires are slipping too much, that's a "10.5" or "11" (hopefully, you don't go beyond that!); if you feel you're not using all of the grip available, that's something less than a "10." For reference, I would hope that you rarely go beyond a "5" while driving on the street; the average driver probably drives closer to a "2" or "3."

Now, here's where this exercise really pays off: After you spend ten to twenty minutes on track, solely focused on how close your tires are to their traction limit, sit down with a track map back in the paddock and make notes of those ratings in each corner. Better yet, break the corners down into entry, mid-corner, and exit phases, and rate how close the tires are to their limit in each part of the corner. You may find that, for example, you're using the tires' traction limits from mid-corner out to exit (a rating of "9.5" or "10"), but not during the entry phase (a "7" or "8"). By simply writing your ratings down on the track map, it'll make you more aware of where you're leaving time on the table. Often, drivers go out in the next session and immediately begin driving closer to the limit, just because they're more aware of where they can push a little harder.

Focus on Steering Effort & Feedback – Much like the Sensory Input and Traction Sensing Sessions, devote a session (at least a significant part of a session) on focusing on the feedback you get through the steering wheel. Keep a light grip on the wheel and notice the vibrations you feel, while also being sensitive to how the steering effort changes as you go through the corners. Now, imagine taking the same map that you made notes about Sensory Input and Traction ratings on, and writing down the various things you felt through the steering wheel. Better than imagining it, do it. Yes, the map is getting full of notes. Fantastic! Think about how valuable this map is becoming! It's a map of how you sense the limits of your car. And because you wrote these notes down, instead of simply relying on your thoughts and memory, it'll stick with you.

Focus on Weight Transfer – Okay, now you're devoting an on-track session to becoming more aware of the weight transfer that's occurring while you drive. In this case, actually call out the tire that has the most load on it while you're driving. Yes, talk to yourself while driving. Don't worry, no one will know (unless you record video while driving, which is actually a brilliant thing to do because you can go back and review it later).

For example, as you approach, turn in, and accelerate out of a left-hand corner, you might say, "Fronts" (under braking), "Right front" (as you trail off the brakes while turning into the corner), "Rights" (while in steady state cornering), "Right rear" (as you accelerate out of the corner). You might be thinking, "Why do I need to actually say it, as I know where the weight is going to be – just as you've written it out here"? The act of verbalizing it makes it stick in your mind more; you'll be more focused on it while talking it through; it'll keep you on task better (rather than thinking about other things, such as chasing that car in front of you); and you may be surprised at exactly where around the track the load changes.

Yaw Angle Sensing – Deliberately make your car rotate on corner entry, even if this makes you slower. Pick safe places to practice this, typically in slower, tighter corners. These corners can sometimes reward an exaggerated rotation, anyway, so deliberately moving your brake zone well into the corner will make the car's yaw angles extreme. Move your Begin-of-Braking (BoB) and End-of-Braking (EoB) further ahead and into the corner, to the point where your EoB is at least two-thirds of the way to Apex, well past the Turn-in.

Test Grip Levels – While driving through a corner, turn the steering wheel more. You may need to do this rather quickly. If you slowly turn the wheel, and you're not quite or just at the limit, the car will respond and tighten up the radius. But if you turn the wheel quickly, you're likely to induce some amount of understeer, if even just for a moment. And that tells you something. It tells you where the grip level is. Relate the feel back through the steering wheel with the response to the "change of direction request" you've put into the steering.

As you exit slower corners, exaggerate when and how hard you go back to full throttle, in the attempt to get wheelspin. Learn how much throttle it takes to make the driving wheels spin, and even induce some power oversteer (or understeer in a FWD car).

Again, pick your test corners, so you manage the amount of risk you're taking. I like doing this in slower, tighter corners where I can make a quick adjustment and get the car back on line easily, or in long, sweeping corners where I have enough time to correct my line. Ideally, do these in corners where the consequences of making a mistake are relatively low.

Skid Pad Training – Whether you have access to a proper skid pad, have to MacGyver one in a parking lot, or you go to a rally school where you can slide a car around in the dirt, any kind of "low grip" driving will help. Skid pads and loose-surface driving allow you to experience slip for a longer period of time, giving your mind time to process, calibrate, catalog, and react to changes in yaw, steering angle, visual pictures, weight transfer, and slip angles much better.

While coaching elite-level racers, I've taken them to skid pads and rally schools to fine tune their car control skills, so this is not something relegated only to inexperienced drivers.

Street Driving – You knew this was coming, right? You know I believe the main reason we drive on the street is to practice our track driving! Practically everything I've suggested above can, and should, be practiced (legally) while driving on the street. The great thing is because the speeds are lower, the loads, forces, angles, and feedback are all lower. So, if you're able to sense them on the street, just think how much more sensitive you'll be to them when you get on track.

You'll notice that there is a definite theme throughout all of my recommendations above: It's all about deliberate practice. If football teams practiced the way most race and performance drivers do, they'd show up for a practice and play a game. But they don't do that, do they? Instead, they break the game down into individual skills, such as blocking, kicking, running, and passing drills to fine tune their technique. That's exactly what I'm suggesting here. By focusing on each of these exercises, or drills, you will fine tune your ability to sense the limits of your car.

Or, you could leave it to wishing you were better. :)

Day #11: Drive everything you can get your hands on... as long as it's safe.

I'm a great believer in cross training, and that applies to driving as much as it does to anything else. Drive everything you can get your hands on: from go-karts to rally cars; from open-wheel cars to off-road trucks; and everything in between.

The only rule I have is to make sure what you're going to drive is safe. And this is a tricky one, because in the excitement of driving something new, we often make decisions that are not well-thought-out.

"Yeah, that looks like a fun car! Okay, the roll cage is a bit iffy, but I don't plan to use it."

"The car is poorly-prepared, and looks a bit ragged, but they told me they'd go through it one more time before I get in it."

"Wow. The first time I sat in the car in the pits, the brake pedal went to the floor. But they're going to bleed the brakes before I drive it, so it'll be okay."

These are all things that I've heard said by a driver. How do I know? Because they were said in MY head. I was so excited to drive these cars that I overlooked serious safety issues. Fortunately, I'm still here to talk about it, and pass on my regrets.

Having said that, the more different types of cars you drive, the better a driver you'll be. I recently had a driver ask me what I thought he should do in the off-season: take his car south and drive at some new-to-him tracks, or rent some cars to drive elsewhere. Obviously, the best answer to this question depends on the driver's experience and skill levels (and budget), but all things being equal, the greater the variety, the better. In this case, I suggested time at a rally school (developing car control skills) and being coached in different types of cars than he usually drives as the best options.

I'm often asked by HPDE or track day drivers "What's the best way to shave half a second off my best lap time?" My usual answer? Rent a drive in a car that races in one of the low-cost endurance series, such as WRL or Chump Car. A driver will learn more about different lines, different techniques to deal with different lines, and how to adapt in one race than they will in a season or two of track days. In the process, the driver will go back to their car during track day events and be lapping at least half a second faster than ever before. The experience will have helped the driver get outside of their comfort zone and usual ways of driving.

My advice here doesn't just apply to racers. If you're participating in track day events, do what you can to drive different types of cars. Go to a driving school. Drive different tracks. Variety makes you use different "muscles."

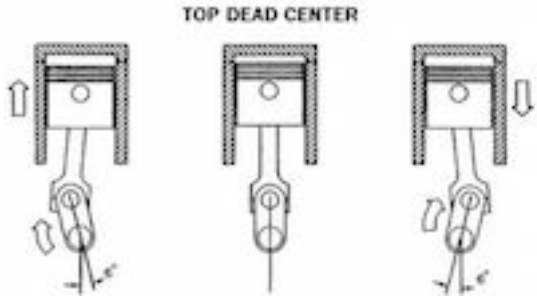
So, go find something new to do, but never forget to make sure what you're driving is safe. If you're not sure, then don't drive it (or ask someone who does know, for guidance).

P.S. - Take a look at the variety of what Colin Braun has raced: quarter midgets, go-karts (every type and level), open-wheel formula cars, production GT and sports cars (RWD, FWD, rear engine, mid-engine, front engine), prototype race cars, rallycross, NASCAR trucks and cars — and some things he can't talk about publicly. Variety is one of the reasons he's as good as he is.

Day #12: To drive at the limit, you need to go beyond Top Dead Center.

For those of you who have built a piston engine, you know what Top Dead Center (TDC) is. For those who have not, let me explain.

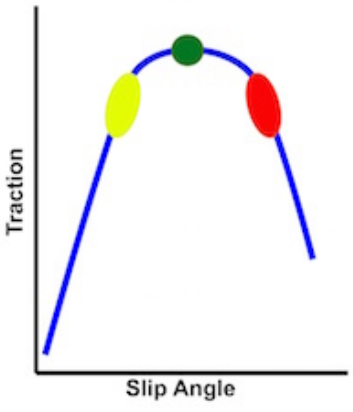
As I'm sure you're well aware, the engine in your car has some number of cylinders, and each one has a piston that moves up and down in it (unless you have a Mazda rotary engine!). Without going into great detail, the camshaft (which essentially opens and closes the valves) must be in sync or time with the movement of the pistons. To do that, during assembly of the engine the builder must start with one piston at the very top of the stroke (the up and down movement of the piston). As I suspect you also know, the piston is connected to a crankshaft by a connecting rod (I wonder how they ever came up with that name!), and when the crankshaft is turned, the piston moves up and down in the cylinder. As the engine builder turns the crankshaft the piston moves up in the cylinder, it gets to TDC, and then begins moving back down again. By turning the crankshaft back and forth in incrementally smaller movements, the engine builder can hone in on the exact point where the piston is no longer going up, but has not quite started moving back down yet. That's TDC, when the piston is at its peak.



Driving the limit is just like finding Top Dead Center.

You work your way up towards the limit, then go over just slightly, then back up, over the other side… making each of the "over the top" excursions smaller and smaller until you're spending all your time at Top Dead Center. Or, in this case, at the limit.

One thing that is critical to notice when you look at a tire's slip angle versus grip/traction curve (below - the limit is the green area; under the limit is yellow; over the limit is red) is this: You have just as much grip just over the limit as you do under the limit.



For many drivers, that's a comforting bit of knowledge. I'm sure every driver knows this at a gut level, but until they become consciously and logically aware of it, there is this fear of the unknown over the top, over the limit. The fear is that once our tires reach their limit, they give up completely and the car will immediately spin or slide off the track. Obviously, when we think

logically, we know this isn’t the case, but we often don’t think logically when driving at high speeds around a race track!

This is why I think it’s important that you have a good, strong mental model for what the limit is. Thinking about a piston moving up and down in the cylinder, reaching TDC, going past, back to the other side… and honing in on that exact TDC is important. It's no different from driving with your tires at their limit.

Day #13 Use a process when searching for – and driving – the limit.

Something that many drivers wonder about is how to get closer to the limit without taking steps that are so big that they result in a crash. They wonder how best to "inch up on the limit."

Let me provide one (there are many) step-by-step approach to honing in on driving the limit. To do that, let's look at braking later, since that's an area that many drivers consider first when trying to go faster.

Step 1: Figure out where you're braking now. That might sound simple and easy – and it is – but it's not something that many drivers do. By this, I mean identifying where you're currently beginning to brake for the one corner you're wanting to brake later in. Note a brake marker, a crack in the pavement, your position compared to some curbing, or whatever.

Step 2: Decide how much later you want to start braking. I recommend thinking in "car lengths," as that's easier to picture in your mind than some measurement, while traveling at high speed approaching a corner. Even if you use data acquisition analysis to determine that you could brake twenty feet later, turn that into car lengths. In that case, you'd say to yourself that you're going to brake two car lengths later. Unless you know absolutely for sure that you need to brake much, much deeper, I recommend moving your brake point in units of two car lengths for starters.

Step 3: This step is critical, so don't skip it. While standing in the paddock, look and identify what two car lengths is. Get a strong image in your mind of what that distance looks like.

Step 4: Take the "where do I brake, now?" image and merge it with the "two car lengths later" image. Imagine what that would look like on the track. Play it through in your mind a few times — the more, the better.

Step 5: Go on track, and on your out lap (at lower speed), drive two car lengths past the place where you used to start braking (where you imagined in your mind), and start braking. Even if it takes two or three laps at a slightly lower speed, ensure that you brake two car lengths past where you used to brake.

Step 6: Increase your speed until you're driving at your normal fast pace, ensuring that you continue to brake two car lengths later.

That's it. Notice that I didn't suggest moving your brake point ten car lengths later. You said you wanted to inch up on the limit, and that's what we're doing. Once you're comfortable with two car lengths from your original braking point, then step it up another two, then another two, until you feel you're reaching the ultimate limit, or your late braking is negatively affecting your cornering. You may want to even fine-tune with one car length.

Also notice that you didn't go on track at your normal speed, and then begin thinking about braking two car lengths later. You thought it out, you made a plan, you built a mental image/model of what you're going to do, and then you started out doing exactly what you wanted to do at a lower speed. This last part is critical. You're building a new mental program — one of braking two car lengths later — so getting into a habit of applying the brakes at the place you want while driving slightly slower is critical to making this work.

Of course, the entire time you're going through this process you should be focusing your eyes in to your End-of-Braking (EoB) point, and being aware of the timing and rate of release of the brakes as you turn in. Combine all this with the mental model of honing in on the limit, just as though you're finding TDC with a piston, and you'll definitely move closer to braking at the limit.

Of course, this step-by-step approach can be applied to inching up your cornering speed, applying the throttle sooner, or anything else. Use it, and enjoy consistently driving a little closer to the limit.

Day #14: What you believe is what you get.

More than once I've written and talked about a driver's "belief system," and its importance in being fast. What might not be clear is what I mean by belief system, so let me explain.

A person's belief system is everything that they believe. It's their core beliefs, and it's more than just their level of confidence. Sure, confidence is part of one's belief system – it's where your confidence lives – but there is more to it than just confidence.

You have beliefs about how you should act in certain situations. You have beliefs about certain people and what they say and how they act. You even have beliefs about certain products and services that certain companies produce. And your beliefs are very likely to be different from other people's beliefs, especially people who grew up in other parts of the world.

Where do your beliefs come from? Primarily from past experiences, your environment, chance happenings, and what you've been told and taught. That's why your beliefs are different from mine, which are different from someone else's.

What does this have to do with driving? You have certain beliefs about your ability to do things, including driving a car. And that will have a major impact on how you perform when behind the wheel.

As Henry Ford once said, "If you believe you can, or believe you can't, you're probably right."

All sorts of research and studies about the power of one's beliefs point to how that impacts performance. My favorite is based on a sleep study, where researchers took people into a lab and wired them up to an EEG machine to measure brainwave activity while they slept. With an EEG, researchers can determine how deep a sleep someone has, and that was what they did. But before they did, they split the group of people into two, and had one sub-group drink warm milk; the other drank coffee.

In the morning, they interviewed each test subject, asking them how well they slept. Across the board, the individuals who drank warm milk said they slept well, and the ones who drank coffee said they had a restless sleep. When researchers checked the EEG data, it correlated perfectly with what the individuals reported – the ones who said they slept well showed lots of signs of Delta wave activity, a sign of deep sleep; the ones who drank coffee and reported a restless sleep showed less Delta wave activity.

But here's the real interesting thing: the researchers had put caffeine in the milk, and the coffee was de-caffeinated. We know that caffeine is a stimulant, and yet these people's beliefs about the effect of what they were drinking would have on their sleep over-powered the stimulant.

That's more than just confidence. That's a deep-down-inside, strongly-held belief that impacted their sleep patterns.

And that's what I'm talking about when I say that a driver's belief system is everything. If a driver truly believes deep down inside that they're capable of doing something with their car, it's more likely to happen. On the other hand, if a driver doesn't believe they're capable of doing something, such as taking a corner at full throttle, then they won't. Yes, this is confidence we're talking about, but it's the real deep confidence – or lack thereof.

But a driver's belief system also covers their beliefs about how they react or behave in certain situations, and the types of decisions they make.

The challenge with your belief system is that I (or anyone else) can't just come along and tell you to believe in yourself, or that you can take Turn 1 flat in top gear. It might help if I tell you that you can do it, but it's going to take time to program that belief. Of course, that means that, with deliberate practice, you can change your beliefs through the use of mental imagery.

* * *

I follow an interesting blog by a fellow named James Clear (JamesClear.com), and it's mostly about performance: how we as humans perform, and specifically how we can perform better. In a recent post he wrote about the "placebo effect."

Clear mentions a study conducted in 1972 with weight lifters, where a group was told they were receiving steroids, and were compared to a control group who were given nothing. Over a 4-week period of believing they were taking steroids (when they were really just taking a placebo), the weight lifters increased the amount of weight they lifted by seven times! Obviously, that's a huge gain, and it's attributable to these athletes believing they were taking steroids, enabling them to make these gains.

As Clear writes, "The evidence was clear. Every athlete got stronger simply because they believed they were on steroids. They expected to improve and so they did."

I find it interesting that when I talk to drivers about their beliefs, many seem to categorize this mental part of driving as "not real." It's like we're talking about "that woo-woo stuff" – if it's just in your mind, it's not real.

But as Clear points out in his post, "Placebo improvements are still real improvements. Those weren't real steroids the athletes were given, but those were real gains that they made in the gym. Whether they lifted bigger weights because they actually took steroids or because they merely believed they took steroids, either way they lifted bigger weights."

So, your beliefs – what you believe deep down inside – is real.

Where this gets really interesting in the performance/race driving world is where one's beliefs and physics collide, or diverge. Or, to put this in real-world perspective, if you believe strongly enough that you can take that hairpin corner flat in fifth gear, can you? If physics says you can't, but your belief system says you can, what happens? I suspect most of you reading this will side with physics, saying that ultimately it controls just how far (or how fast) we can go with something.

In other words, we can't go beyond the laws of physics. I won't argue with that. However, do we know all of the laws of physics? Or are we still learning about them, and most important, how our minds interact with them?

There was a time when scientists believed that if a human ran a mile in less than four minutes, their body would fail. Then Roger Bannister ran the first sub-four-minute mile, he didn't "blow up," and our understanding of human performance, physics, biology, and who-knows-what-else changed.

Here's what I believe – what my belief system says: We're capable of much more than we think, as long as we truly, deep down inside, strongly believe that we're capable of much more than we think.

Day #15: Know when it's okay to coast.

In my first *Speed Secrets* book, which was published way back in 1998, I wrote, “Your right foot should either be on the brakes, squeezing the throttle down, or flat to the floor.” Since that time, I’ve had drivers tell me that I wrote “Never coast,” which I’m not sure I ever did (if someone can tell me where I did, I’d appreciate knowing — and be happy to admit to it!). Either way, I can see how my original advice would be taken as saying “never coast.” I apologize for this.

Before I get into an explanation, understand that back in the mid-nineties (when I actually wrote the first book), left-foot-braking was not a popular technique, which is why I’d written it with the focus on the right foot. It’s also why, in 2003’s *Speed Secrets 2*, I devoted a whole section on left-foot-braking. In *Speed Secrets 2*, I also wrote, “In theory, you should never be coasting in a race car — you should either be braking or applying the throttle. In reality, coasting is sometimes (although rarely) necessary.”

Finally, in *Ultimate Speed Secrets* (published in 2011), I was able to combine and fine-tune a lot of what I had learned between the time both of the earlier books were published – and add new content.

My point here is not to give you a history lesson on my *Speed Secrets* books, but to clarify a few things.

First, what I call "Speed Secrets" are not rules. Rules, to me, work 100% of the time. Very few pieces of advice about performance driving work 100% of time, and therefore my Speed Secrets are guidelines. They do work more often than they don’t, but rarely can anything be taken as a rule.

Second, if you interpreted my guideline about what your right foot should be doing as a rule that says, “never coast,” it was not meant that way.

In some cars, in some corners, it’s best to coast. And since I still don’t like the word “coast” being used in performance and race driving, I’m going to use the word “hesitate.” In this case, it refers to the act of hesitating after releasing the brakes, and before beginning to apply the throttle. Why would you ever want to hesitate, and not be either on the brakes or on the throttle? To help the car finish rotating — so it’s pointing in the right direction — so you can get to full throttle sooner.

Have you ever begun to apply the throttle, only to have to either pause and hold it where it is, or even ease off it a bit to ensure you don’t run off the outside of the corner at the exit? I bet you have! So has every driver in the world. In these cases, it’s because you didn’t allow the car to change direction enough (rotate) so that it’s pointing on a path that allows you to quickly squeeze back to full throttle.

When I say, “quickly squeeze back to full throttle,” that’s a relative term, as well. Even though I say quickly, it could be relatively slow. Or it could be a very quick, almost

slamming it down to the floor. In most cases – but again, not all, as this is a guideline – what matters most is when you get to full throttle, and not when you simply begin applying throttle. I've seen cases where one driver begins to apply the throttle later than another driver, but gets to full throttle sooner. Guess who's the fastest? The driver who gets to full throttle the soonest. Again, there are exceptions, but this is a good guideline.

So, yeah, there is a time and a place for coasting…er, hesitating... before applying the throttle.

Day 16: Turn things around.

Ask drivers how to shave moments off their lap times, and you'll find they've most often been told to "Brake later." Let's say that you currently brake at the "2" marker approaching a 90-degree, large radius right-hander and you take it in 4th gear. Following the "Brake later" advice, your thought and goal is to go back on track and begin braking inside the "2" marker, perhaps at "1.5."

On track, as you approach the brake zone, what's your mindset? How are you feeling about braking later? Many drivers in this situation tense up a little, feeling a bit anxious about braking that much later. So, first, the tensing of the body leads to less sensory information being soaked up, and, therefore, less feel for the limits of the tires and car. And secondly, nine times out of ten, a driver will apply more brake pressure as they go past the usual brake point. Braking harder is very common, and a somewhat natural response when braking later.

By braking later, the driver has increased the length of the straightaway, which will shave time off the lap, assuming that it doesn't negatively impact minimum corner and exit speed. But unless the driver is braking way early, the improvement in lap time from braking later is typically not very much – perhaps a tenth of a second or two. Still, every tenth is worth it. But is it the best way to approach this corner?

As I said, when drivers brake later, they typically brake harder, too. They compress the brake zone, and the BoB (Begin-of-Braking) is later, but the EoB (End-of-Braking) is in the same place. This usually does one of two things: results in the corner entry speed being the same as in the past (and sometimes even slower) and the driver locks up the brakes (assuming no ABS), or results in the driver simply over-shooting the corner from too much brake pressure while trying to turn into the corner.

This is where a coach who considers the counter-intuitive approach comes in handy. Oh, and that coach could be you — yourself — self-coaching.

What if, instead of braking later, you braked lighter? I hear it all the time when drivers are given this advice: "What? I thought driving fast was all about braking late and hard?!" Not always.

What is the advantage to braking at the same starting point – the same BoB – but lighter? Higher corner entry speed, right? Now, before you bring up the "In slow, out fast" advice, understand that I'm not talking about entering the corner ten MPH faster. No, maybe just one or two MPH faster. Use the following as a way to think about the amount of brake pressure you're using: a "1" is barely touching the brake pedal, and a "10" is threshold braking or getting into the ABS. So, if you were using a "9" pedal to begin braking at the "2" reference point, use an "8." It's not a lot lighter; it's just a bit. Enough to result in turning into the corner a couple of MPH more, at the most.

But what if you already felt that you were turning in with the most speed the car could take? It's possible that you were, but only because of how you had your car balanced – or in this case, not balanced. Remember that a car that is perfectly balanced, with load equally distributed over all four tires, has more overall traction than a car that's in a load transfer situation, such as more load on the front tires than the rears. Or more load on the outside tires than on the inside tires. Or, a combination of both of those, as when turning into a corner with some amount of braking still going on (trail braking), and most of the load is on the outside-front tire, and the least on the inside-rear. Phew! Get all that? The car, in that situation, will have less overall traction than one that's more perfectly balanced. (For a bit more insight into this, watch the video I created, Why Smooth is Fast).

By braking lighter, the car will be better balanced – closer to the chassis being flat to the track surface – and will therefore have more grip. That grip could be used to turn into the corner with a couple of MPH more speed.

In this example, the driver could begin braking at the same place as in the past (at the "2" marker), but a little lighter, carrying a couple of MPH more into the corner. And because the car has more grip from being better balanced, that speed is carried all the way through the corner, and the driver is able to begin accelerating from this slightly higher speed. Which would you rather begin accelerating from: 66 or 68 MPH?

Yes, braking lighter is not normal, but what is normal in our sport?!!

The next time you think about braking later for a corner, consider what would happen if you braked lighter, instead. And don't be afraid of turning things around and looking at what you're doing in a less-than-normal way. Rarely have breakthroughs been made in any area of life without having a different perspective, and this applies equally to your driving.

Day #17: Use knowledge to build confidence.

A short and sweet message today.

There's no doubt that confidence has a huge impact on your driving performance. Some would say that it's hard — even impossible — to control your confidence, that it just comes and goes. I disagree. Confidence is programmable through the use of mental imagery. And you can start to build it by improving your knowledge.

I was reading *Think Like a Rocket Scientist: Simple Strategies You Can Use to Make Giant Leaps in Work and Life* by Ozan Varol. In it, he writes about and quotes Canadian astronaut, Chris Hadfield (who, incidentally, once sent me an email from the space station, with a photo of Vancouver from about 200 miles above the earth). Here's what Varol wrote:

“Astronauts maintain their calm not because they have superhuman nerves. It’s because they have mastered the art of using knowledge to reduce uncertainty. As astronaut Chris Hadfield explains, 'In order to stay calm in a high-stress, high-stakes situation, all you really need is knowledge.… Being forced to confront the prospect of failure head-on—to study it, dissect it, tease apart all its components and consequences—really works.'”

“'Fear comes from not knowing what to expect and not feeling you have any control over what’s about to happen,” writes Hadfield. “When you feel helpless, you’re far more afraid than you would be if you knew the facts. If you’re not sure what to be alarmed about, everything is alarming.'”

Think about that. Then improve your knowledge.

Day #18: Follow the basics when tuning your car's handling.

I fully admit that engineering a car's setup is not my main area of expertise, but I've had to learn how to do this, either through necessity of not having anyone else to help me, or something I've done while coaching drivers. Fortunately, I've had some great "teachers" along the way to help me with both the detailed technical understanding and the overall strategy or approach to tuning a car's handling.

Probably the most important lesson I've learned from these engineers, and from my own experience, is that one size does not fit all. In other words, what works on one car may not work on another. And sometimes what would seem to be the logical thing to do does not work, either. With that in mind, understand that I – or anyone else for that matter – cannot tell you, for example, to raise your tires' pressures two psi to increase grip. If someone gives you a blanket statement like that, run in the other direction. That might be good advice sometimes, but other times it won't be, so if you follow it blindly, you'll end up fooling yourself.

The following are general guidelines - my Top Ten to consider when tuning the handling of your car:

1. *Make big enough changes that you're sure to feel*. I often see drivers making a one psi tire pressure change, and most drivers – even some pros – cannot feel that small of an adjustment. And the worst thing you can do then is think you should notice it, and have your brain make something up. If you don't feel it, admit it. Then make a bigger change. It's okay to raise or lower tire pressures by 5 psi. The same goes for shock settings, spring changes, aero settings, and so on. Make big enough changes, and understand that what you're mostly looking for is a direction. If the change helped, you can always reduce the amount you changed it – but at least you know which direction seems to be helping.
2. *Work with what you have*. This was a simple but "Duh, why didn't I think of that" lesson I learned from Jeff Braun. He says (and I'm paraphrasing), "If the only tool you have is a tire pressure gauge, then tune your tire pressures; if you have the ability to adjust shocks, ride heights, anti-roll bars, springs, and aero settings, then be prepared to adjust them."
3. *Do A-B-A tests*. It's important to go back to your baseline setup every now and then to check that the changes you've made have actually helped. Sometimes you go faster throughout the day as you're making setup changes, and while you think it's due to the tuning you're doing, it may just be your driving that's improved (or track conditions have changed). That's not a bad thing from a driving perspective, but it can fool you into believing the car changes helped. Go back to the original setup and check. You may find that it's actually faster.
4. *Take notes*. Every single change, no matter how small, must be written down. Now. Right away. Not later, when you get to it. Do not rely on your memory. If necessary, bring a friend or family member along to the track with the sole responsibility of writing down those changes. I guarantee you will regret not writing things down.

5. *Work on the end of the car that needs improvement - then the opposite end if you can't fix it*. If your car is understeering, do everything you can to improve front grip. If, after you've tried everything you can to help the front grip, then do something to the rear to reduce grip. I know the idea of reducing grip seems counter-productive, but see my next point below.
6. *Balance is more important than overall grip* (another Jeff Braun maxim). You will almost always drive a well-balanced car faster than one with more grip, if that grip is biased too much to either the front or rear. That is why reducing rear grip in a car that understeers will sometimes be faster. Yes, you have less grip, but driving a car that is better balanced will be easier and give you more confidence. A confident driver will always be faster than the one struggling to make the car do what they want.
7. *Learning what doesn't work is as important as learning what does*. I often hear drivers complain that they made a bunch of changes, none of which helped, and so they're disappointed. But guess what? They learned a lot, so they should be happy about that. If you try something and it doesn't help the car go faster, at least you now know what not to do. You're smarter than you were before. And you can eliminate that change from the menu of things to try in the future (at least, in the same car, same conditions, same track – that particular change may work on another day, car, track, etc.).
8. *Avoid pre-conceived ideas*. Just because something didn't work in the past, don't write it off. It just may not have worked with that car, those tires, that track, those conditions. And there are many exceptions to the rules. Just because something should or shouldn't work in theory, that doesn't mean it will or won't. Keep an open mind, make logical changes, and observe the results. And maybe, just maybe, every now and then make an illogical change.
9. *Understand why something works*. Making a change, having it make the car perform better, but not understanding why it helped is a waste of an opportunity. Take the time – even if that's after the day or event, sitting down with your notes – to think through why each change did what it did. If you can't figure out why, ask someone about it. They may not know, but in the process of asking, I bet you gain more of an understanding. Simply talking about it will help.
10. *Make one change at a time*. If you make two changes, then go out and turn faster laps, which change made the difference? The first one? The second one? Both? If you make one change at a time, you'll know what does what. There are exceptions to this guideline, but usually that involves having a really good, experienced engineer who can separate the results from two or more changes. Most of us don't have that engineer. If you did, you may not be reading this!

The above guidelines are certainly not everything you need to know to tune your car's handling, but it's a good start. And again, it's not the specific adjustments or settings that matter – it's the process that will result in you finding the right adjustments and settings.

Day #19: Practice doesn't make perfect. Practice with sleep makes perfect.

Okay, that's a new twist on what I've said in the past ("Only perfect practice makes perfect"). But I can't tell you the number of times I've told a driver I was coaching that they just needed to sleep on what they'd learned that day to take their driving to the next level – to shave half a second or more off their lap time. It works.

In the November 13, 2018 episode of the Hidden Brain: Eyes Wide Open: Part 2 podcast, neuroscientist Matthew Walker (who's written the book, *Why We Sleep*) talked about the importance of sleep – not just for rest, but to allow the brain to work through what it had worked on during the day. He explained, "It's practice with a night of sleep that makes perfect."

Listen to this podcast - it's fascinating. You'll hear stories about Keith Richards of the Rolling Stones essentially creating the guitar riff for *Satisfaction* in his sleep; the creation of the periodic table; and more.

There are two types of sleep: REM (Rapid Eye Movement) sleep, which is when we dream, and non-REM sleep. Non-REM sleep cements memories, as Walker refers to when he says, "We imprint information during the day... deep non-REM sleep almost hits the save button on those recently acquired informational pieces."

The next time you're struggling to find another half a second of lap time, go to sleep. Ideally, for the night. And even better, just before going to sleep, do some mental imagery of what you want to achieve the next day. Simply thinking about being half a second quicker is not what I'm talking about. No, you need to imagine doing what will lead to being half a second quicker. You need to mentally rehearse making the changes to your driving that will result in you being faster.

For example, imagine adjusting your braking: compressing a brake zone by applying the brakes later and harder, applying less pressure to roll more speed into the turn, trail braking further to help rotate the car, or releasing the brakes earlier. Imagine how that would look, feel, and sound. And imagine it in specific corners.

Imagine committing to full throttle just a fraction of a second earlier, spending a tiny bit more time at full throttle in between corners, or breathing the throttle less in that fast sweeper corner. How would that look, feel, and sound?

Mentally adjust the line you're driving through specific corners, seeing, feeling, and hearing how that allows you to get through that section of track quicker and faster down the next straightaway.

Imagine that. Then relax and go to sleep. Your mind will continue to program the specific changes you imagined, and when you wake up, you'll be faster.

Day #20: Experiment. It's the best way of learning to drive faster.

Jeff Braun, one of the best race car engineers in the world, told me something so brilliant that I had to share it with you.

Jeff said, *"If I make a change to the car and it does what I expect, I haven't learned anything - I didn't get any smarter. If I change something and it does something different from what I expected, then I think about why that is, and I become smarter."*

I shared that with a group of drivers, most of whom had more than ten years of driving on the track. What I had noticed among them and others with that level of experience and ability is that they don't experiment enough with their driving. It's almost as if they are afraid to do something that wouldn't work. If you buy into what Jeff says, as I do, they're not getting any smarter. Perhaps, they getting better at what they've always done before, but they're not going to make significant improvements. And what do you get when you do the same thing you've always done in the past?

Interestingly, after having this group of drivers try different things, use different techniques, and learn what does and doesn't work, the most common comment I heard at the end of the day was, "Wow, that was amazing. I knocked over two seconds off the best lap time I've ever driven — and it's easier now."

Most of these drivers tried things they've never tried before. Some things worked, and some didn't, but they all got smarter.

A difference between you and Max Verstappen, for example, is that you have a relatively small number (compared to the World Champion) of techniques and methods for taking a corner. Verstappen may have ten times as many as you do. Obviously, every single corner in the world is different, and therefore each one requires a slightly different approach. If you have one way of driving all corners, what are the odds that one way is the right way for the next corner you drive?

When drivers talk about needing more seat time, one thing they're really talking about is building a larger database of techniques and methods of driving different corners. It's about having more tools in the toolbox. Most drivers develop this larger database over many years, whereas other drivers deliberately set out to do it. Guess which drivers build their database faster?

I'm sure you've been told to drive consistently. Good advice. Up to a point. And then, when you're able to be consistent, practice being inconsistent. That is, experiment. Deliberately do different things. Brake earlier, later, lighter, harder. Release the brakes sooner, later, faster, slower. Turn the steering into a corner quickly, slowly, progressively. Hesitate when transitioning from the brakes to the throttle, then immediately go from brake to throttle. Squeeze the throttle down, and then apply it very quickly. Turn in late, early; unwind the steering early and slow, late and fast.

Learn what works, what doesn’t, and most important, what the result is of what you’ve done. What doesn’t work in one place might in another. See, it's just as important to find out what doesn't work as it is to learn what does work.

Experiment. Add tools to your toolbox.

Day #21: Develop your ability to focus.

It's possible to develop your ability to focus, and specifically focus on what's important. And you should. How? Here are four "exercises" that'll make you a better driver:

1. **Auditory Focus**: After reading this article, sit back and relax. Close your eyes. Listen for every sound you can possibly pick up. Do you hear traffic outside? The sound of the heating and cooling system in the room? Other people talking or moving around? Movement of something mechanical? What else? Your breathing? Your heart beating?

2. **Kinesthetic Focus**: Next, become aware of what you can feel. Notice the pressure points between your body and the chair you're sitting on? Feel the weight of your feet, your legs, your hands, your arms, and your entire body. Do you feel the air moving? Be aware of, and focus entirely on what you can feel.

3. **Visual Focus**: Gradually open your eyes and pick one object to focus your vision on. Laser focus on it. Zoom in and see only that object, and nothing else. After about twenty seconds or so, zoom out while still looking directly at the object. But be aware of everything else you can see. Use your peripheral vision, and even imagine what's immediately behind you and out of view. Spend twenty or more seconds in this expanded focus view. Now, switch back to the hyper-focused state, taking in just that one object and all of its detail for twenty seconds. Then back to the expanded focus. Alternate back and forth, spending about twenty seconds on each, and do this at least five times.

* * *

When you can focus to the point where you can hear your own breathing and heart beating, you've tuned up your ability to listen (a good thing!), as well as your focus. If your mind becomes distracted before you get to the point of hearing yourself, then practice more. You're practicing focus.

Reading what your car is telling you is critical to driving fast, and practicing being sensitive to what you feel is the key to doing that. Practice feeling what you feel.

Performance driving requires you to switch from using a narrow visual focus to a broad one and back again; the quicker and more effortlessly you can make that switch, the better you'll be. Practice changing visual focus.

4. **Motion Focus**: One last exercise to improve your ability to sense the limits of your car: Sit in a moving object — a car, on a rocking chair, riding a bike, in a plane, on a boat — and pay hyper-attention to the motion. Notice it rock forward and backward, and side-to-side. Note whether it moves up and down. The better you get at feeling the motion of any object, the better you'll be at sensing the limits of your car when driving at or near the limit.

Obviously, all of these simple “exercises” are done without ever being on a track, and they’re free! That means there’s no “I can’t afford it” excuses….

Day #22: Balance loading the car on corner entry with a disciplined turn-in.

There's a fine line between adding a small amount of load to the car before turning in and "crabbing" in to the corner with a slightly early turn-in. Yes, the smallest micro-movement of the steering wheel can shift the initial load to the outside tires and may help. But it may also hurt.

There has been talk of some F1 drivers doing a bit of the "load the steering" thing. Some have been coached to do this. What I'm talking about is just before they initiate the turn into a corner, the driver turns the steering just a tiny bit. The idea is that it puts a little load on the outside tires and it's almost as if those outside tires are not being "shocked" with a big load when turning in. And it can help the car take a set earlier, so you know what you have to work with.

For drivers at the F1 level, this can work because they have the discipline to ensure they're not crabbing into the corner. I define "crabbing" as a driver angling the car away from the edge of the track just before turning into the corner - it's not a good thing. Why do drivers do this? For two reasons: First, when the driver gets near the turn-in point, he looks to the apex. As we know, our hands follow our eyes, so looking to the apex can get the hands initiating a turn before we really want it. And second, sometimes a bit of anxiety (or even fear) sets in due to the speed the driver is carrying, and it feels safer to be further away from the edge of the track.

The big problem with crabbing into the corner is that it reduces the radius of the corner, meaning you can't carry as much speed. Even being a foot or so from the edge of the track at the turn-in point makes a big difference in the amount of speed you can flow through the turn – it can make the difference of three or four MPH and that can lose a few tenths of a second in just that one corner.

Back to loading the car at turn-in. When word spreads that some F1 drivers turn the steering wheel a tiny bit to load the car, many other drivers think they should do the same. And that could be like a novice skydiver trying to "fly" like an experienced pro.

I've seen some casual track day drivers who drive just a few times a year try to copy this approach of loading the car on entry, and they end up doing more harm than good by turning in early. So it can be a good thing, but only if you have the discipline to make sure you don't turn in early. I highly recommend checking whether you're doing that with data and/or video, as most drivers who crab in or turn in early don't even realize they're doing it.

That advanced stuff is usually just doing the basics better. In this case, it's making sure you're doing the basics right before ever doing something advanced like loading the car with a tiny bit of steering before turning into the corner.

Day #23: Understand the difference between weight and load.

As drivers, we all know that when we brake, the front of the car dives down; more load is put on the front tires, and therefore they gain traction. The opposite is true: under braking, the rear tires become less loaded, and therefore their traction is reduced.

When accelerating, the rear tires gain traction due to the additional load on them as the car squats, and the traction on the front tires is reduced. When going around a corner, the tires on the outside of the turn have more load on them, and therefore gain traction; the tires on the opposite side of the car (on the inside of the turn) have less traction due to the reduced load on them.

The obvious conclusion from understanding this concept is that the more load on a tire, the more traction it will have; the less load on the tire, the less traction it will have. Okay, we're all good so far.

The next obvious conclusion, then, is that the heavier the car, the more traction it will have (since all that weight is pushing the tires down onto the track surface), and ultimately it will be able to go around corners faster.

But, no. Colin Chapman would roll over in his grave if this was true. (If you don't know who Colin Chapman is, he founded Lotus, and was famous for saying this about car design/development, "Simplify, then add lightness.")

Wait, what? Why not?

Hang on, and I'll explain, but first let me clarify a few things.

Most performance/race driving instructors, as well as the drivers who have been taught by them, use the phrase "weight transfer" when describing when the car nose-dives under braking, squats when accelerating, and rolls or leans when going around a corner. And yes, I use "weight transfer" when I shouldn't, and even used it in my books. All the engineers and physics experts who have heard me talk or read my books want to string me up for this! More accurately, we should be using "load transfer."

If your car weighs 3,000 pounds, and you stand on the brakes as hard as possible, your car still weighs 3,000 pounds. Sure, your front tires do have more traction, but that's because there is more load on them.

As Randy Beikmann wrote in his excellent book, *Physics for Gearheads* (if you know anyone interested in STEM, get them this book!), "Because we rely on it for traction, we should really think of vehicle weight as gravitational downforce." And, "Note that although load transfer is commonly called weight transfer, it is not: the weight of the car is still acting at the CG, straight downward."

Okay, with that out of the way, let's get back to explaining the impact of load transfer on traction.

As load increases on a tire, the amount of traction it has goes up. But it's not a linear relationship. In fact, the tire doesn't gain as much traction as the additional load would suggest. For example, if you double the amount of load on a tire, the amount of traction it has does not double. The traction will increase, but not to the same extent. To once again quote from *Physics for Gearheads*, "The lateral force capability lost by the inside tires is greater than what is gained by the outside tires."

For many, this is more than a bit counterintuitive, and the often-asked question is, "Where does that traction go? Does it just disappear?"

Think of it this way. Depending on your fitness level, going for a 5-mile run might be easy, or a huge amount of work. No matter how fit you are, though, if I asked you to carry an extra 100 pounds on your back, the 5 miles would be even more work. It's similar with tires. The more load on them, the more work they have to do. And that's why (mostly - and certainly enough of a reason for our purposes here) the load versus traction relationship is not linear; more load, more traction, but also more work needing to be done, so at some point the gain in traction is not increasing at the same rate as the additional load.

Many years ago, while teaching a high-performance driving classroom session, the idea of the Traction Unit Number popped into my head as a way of explaining what really matters to us drivers. No, it's not a super-scientific, physics-driven explanation… but that's why so many drivers have found it easy to understand – and use. I wrote about it in my Speed Secrets books, have used it over and over again in classrooms, and created a video called Why Smooth is Fast (you can see it here).

The big take-away from all of this is that the less load transfer, the more overall traction your car is going to have. Of course, you can't drive a car at any speed without causing some load transfer, and the faster you go, typically the more there will be. In fact, the easiest way of causing as little load transfer as possible is to drive very slowly – but that defeats the purpose of what we love to do! However, if you drive smoothly, you will cause less load transfer, even if you're going fast.

While I'm clarifying some things… when your car leans or rolls (and no, I'm not talking about rolling completely over on its roof, but just rolling to the side slightly), that is not necessarily causing load transfer. Yes, the load transfer is causing the chassis to lean, but a mistake some people make is to think that a car that is stiffly sprung, that doesn't roll or lean much, is not transferring as much load as a car that leans a lot. That's not true. In fact, it's the opposite. I think the easiest way to understand this is to think that a softer suspension soaks up the load transfer, resulting in the tires on the inside not losing as much as they would with a stiffly sprung car; the overall effect is that the car has more overall traction.

I don’t expect the entire driving community to change their language because of this article and begin using “load transfer” instead of “weight transfer.” I don’t expect you to. In fact, I’m sure that I’ll use “weight transfer” every now and then when I’m not thinking about it, too. But that’s the point of this article – to get you to think in terms of load, rather than weight. And maybe you and I will use “load transfer” more often. More importantly, it’s how this change in thinking affects your driving.

As one website I visited said, "Load is a term frequently used in engineering to mean the force exerted on a surface or body." And it’s this force you exert on the tires each time you brake, turn the steering wheel, or apply the throttle that makes the difference between being fast, or very fast.

Day #24: Leave a gift for the fish… er, driving god.

Ancient Hawaiians would leave a gift for the fish god, as they believed that would result in better fishing. Silly superstition, right? Or maybe not. See, it affected their motivation. If you truly believed that leaving a gift for the fish god would result in more fish, you would be more likely to stick it out and wait, and fish longer. Fish longer, and you'll likely end up with more fish. After all, you knew the fish god would look after you; it was only a matter of time. If you didn't believe in the fish god, you would likely give up when fishing was not so productive – and would come home with fewer fish. A self-fulfilling prophecy.

Do you have a superstition that helps with your motivation?

Let's get something straight right here and now: superstitions are not silly. They work. Why? Because they're simply a part of your belief system. If you truly believe that wearing your lucky socks will help you drive better, you will be in a better state of mind, and drive with more confidence. There's nothing wrong with that. To deny that is to throw away a valuable mental game tool.

The only time a superstition can backfire is when you can't control it. For example, believing that you only drive well on cloudy days means you're going to have days when you don't perform at your best. Even your lucky gloves can be lost, or will wear out and need to be replaced.

What's most important is not thinking that the simple routine you go through before driving is silly and unimportant. If it works, go with it. Heck, if you knew that the placebo effect would help you survive a disease, would you write it off and not use it?

Embrace your superstitions. Accept them and use them, whether you're fishing or driving.

Day #25: Time your downshifts.

Imagine that you're at top speed on a long straight approaching a tight hairpin corner. You're in 5th gear, and the corner ahead should be taken in 2nd gear.

How and when do you downshift?

First, the timing of the downshifts is pretty much the same whether you're driving an H-pattern manual or a paddle-shift, whether you use the clutch when downshifting or not. I say "pretty much the same," because there are slight differences between practically every car you'll ever drive. In fact, there may be as much difference between two H-pattern cars as there are between an H-pattern and paddle-shift car. Obviously, the big difference is going to be the non-use of the clutch in the paddle-shift car.

Let's start with some basic guidelines:

- Brake… then downshift. In other words, let the car slow a little before starting to downshift. Why? If you begin braking and immediately start downshifting, you're likely to over-rev the engine because the car (and therefore, the engine RPM) has not slowed enough.
- Have your final downshift completed and your foot completely off the clutch and back on the dead pedal (assuming you're using a clutch) before you start turning the steering wheel into the corner. If you're using a paddle shift, think of the same thing — you must be finished the final downshift before turning in, and not downshift in the corner. Why? Because if the driven wheels get a "blast" of engine compression (as you let out the clutch, or the downshift is completed in a paddle-shift car) when you've turned into the corner, those tires will have a momentary over-slow, causing them to partially skid (actually, just turn slower than the speed of the car). This often results in a spin or near-spin; at least a slight reduction in grip, and therefore cornering speed).
- Use the brakes to slow down, not the engine. Why? Because the brakes work on all four wheels, whereas the engine braking effect only works on two (unless you're driving an all-wheel-drive car). If you're driving a vintage car that has a stronger engine than it does brakes, then — and only then — can you use the engine to assist in slowing the car.
- Skip gears only when you're ready. If you're going from 5th to 2nd, you don't have to go through all the gears, if you don't want to... and if you're very, very good with the timing. Why? Going from 5th to 4th, 4th to 3rd, and 3rd to 2nd takes up time, and mental capacity. Every time you do something, you're using some little amount of brain power, and that brain power could be used for more important things. Plus, the more you do, the more chance of making a mistake; the less you do, the less your chances are of making a mistake. However, it's easy to make a mistake when downshifting directly from 5th to 2nd - you time it wrong (most likely, too early) and over-rev the engine. That's why you have to be

proficient at going through all the gears before you ever begin to think about skipping gears on the downshift.

- Blip the throttle on downshifts. Unless your car is equipped with some form of "auto-blipper," which revs the engine on its own as you downshift, you must blip the throttle to match revs. If you're driving a car with paddle-shifts, you don't have to worry about this, as the car will do it for you. But if you're driving an H-pattern car, you need to heel & toe downshift. Why? Because you need to match the revs with the engine while braking. You will never be as good a driver as you can be without using heel & toe. I know there are some drivers who say they can downshift without matching revs, but it definitely is not as good – it puts more wear and tear on your car, upsets its balance, and causes some near-lock-up of the driving tires (do any of those sound good to you?). I'm not going to go into the details of how to heel & toe here (there are plenty of resources out there explaining how to do it, as well as how to do it better than you do already – my books, and this video I made may be good places to start).

Okay, so you're approaching the corner. Where are you looking? Glance ahead into the corner to see that everything is good – no cars spinning in front of you, track surface looks okay, etc. As you approach the brake zone, a quick focus on the approximate place you will begin braking (BoB, Begin-of-Baking) then ahead to identify the turn-in point, and most importantly, the End-of-Braking (EoB) point. You'll pick up the BoB point in your peripheral vision, and then comes the physical tasks of braking and downshifting (remember the timing). Part way along the brake zone, you'll glance further through the corner to check that all is well, and then your main focus will move from the turn-in point to the EoB point, to the apex. As you pass the turn-in point, your peripheral vision will check how close you are to where you want to be – if you notice that you turned in before or after the turn-in point, you've identified an error right then. You can make small adjustments to what you're doing in the moment, instead of after the apex where the adjustment will need to be a big one. This "glance, focus, and check with your peripheral vision" is an ongoing process.

But… it's very difficult to focus your mental and visual attention on what you should be doing if you're having to think too much about your braking and downshifting. These techniques need to be "programmed" in your mind so deeply that you barely have to think about them – you just do them.

The more you practice your brake application, downshifting (timing and technique), the better you'll be. Of course, it's critical that you practice the right things, and that's why I've tried to explain the process here.

Day #26: Mental imagery is a skill like any other - it takes practice.

When I was fifteen, I played tennis pretty seriously, and was committed to improving — no matter what it took. At one point, I had a small piece of making tape stuck to the ceiling in my bedroom, and I would practice tossing a tennis ball up to that point as I practiced my serve movement. Over and over again, I'd toss the ball hundreds of times a day, making sure it hit that exact spot. I was building muscle memory of that one little part of the serve movement, knowing that if my ball toss was consistently accurate, the rest of my serve would mostly take care of itself.

Around that time, I walked into my Phys-Ed. teacher's office at school and noticed on his desk a magazine with a cover story about the emerging new training approach that Olympic athletes were using: *visualization*. I stuffed the magazine into my gym bag and walked out, went home and read that article over and over again. Yup, I admit to stealing that magazine (I don't remember, but I might have taken it back a few weeks later - but honestly, I can't remember!).

In addition to tossing a ball to that spot on my bedroom ceiling, hitting hundreds of forehands and backhands against a wall every day, and playing as many games as possible each week, I played thousands of shots in my mind every night. And thus began my life of formally using what I would later learn was really mental imagery (since I naturally started moving my body, in addition to simply seeing a visual picture in my mind).

When I really think back, though, I had begun using mental imagery much earlier than that. I had "dreamt" of being a race driver since I was five years old. Rarely had I gone to sleep without imagining driving (often replaying an event my Dad had taken me to the previous weekend at one of the local short oval tracks I grew up at), with me replacing an actual driver in the race.

By the time I was eighteen, I was back to mentally racing cars every single night, using my imagination that I'd kept fresh. From that time until now, there has rarely been a day or night that I haven't used mental imagery to prepare for something. I've practiced it a LOT! I think that's why I was confused when drivers told me they had a difficult time staying focused long enough to imagine driving one single lap, or that they couldn't see the details, or that they couldn't move their bodies along with the imagery, or that they couldn't put themselves into the same state of mind when they drove at their best in the past, or that they couldn't recall what a certain part of a track looked and felt like in their minds, or....

I just couldn't get it because, for me, it was so "natural." But I'm convinced that I was not born with a superior "imagination" gene, that my DNA meant I could imagine any better than anyone else.

A few years back, I watched the TED Talk that Sir Ken Robinson did, where he spoke about how we often suck creativity and imagination out of children. Please, please watch his talk by clicking here.

In his talk, Robinson tells a story about a young girl in school who has been asked to draw something. As the teacher walks around the room, she stops to look and talk to the young girl, and asks, "What are you drawing?" The girl replies, "I'm drawing a picture of God," to which the teacher says, "But nobody knows what God looks like." The little girl's response is perfect: "They will in a minute."

Ask a young child to draw something, and they will. Ask an adult to draw something and most will say something like, "Oh, I can't draw!" Where did the imagination go that the adult had as a child? It's still in there. Too often, it's been restricted by fear of failure, by fear of embarrassment, by fear of judgment.

Guess what? No one is judging your ability to do mental imagery. You can't screw up mental imagery. There is no reason to feel embarrassed to do mental imagery. Just do it. Practice it. Make it a part of your daily routine. Make it a way of life.

Day #27: Use all the track…except when you shouldn't.

I've been thinking lately about two conflicting concepts that we, as drivers, make decisions on with every single corner we drive. And we make these decisions while driving at speed, in a split second. So, deepening our understanding of the pros and cons of each decision is critical.

Concept 1
We all know that the larger the radius we can drive through a corner, the faster we can drive. That's why we drive our cars from the outside edge of the track, to the inside, and back out to the outside, following a path as straight as possible. It is fact.

Of course, we often modify that perfectly large radius to enable us to begin accelerating as early as possible by turning in and apexing later. But let's strictly stay with the "largest radius" concept for today.

This fact is why a common piece of advice — and a Speed Secret in one of my books — is, "Use all the track since you paid for it." Many drivers approach a corner and ease their car some distance away from the edge of the track just before turning in (I call this "crabbing in"). Or, they don't use all the track when tracking out at the exit of a corner. Either way, this reduces the radius of the corner, meaning the absolute limit speed is reduced. But how much?

Back in the mid-90s when I was writing my first *Speed Secrets* book, I did the calculations to figure out exactly how much of a difference using just a little less than maximum track width does to a car's speed through a corner. As I've been thinking about this subject lately, I tried to find my original calculations…and couldn't. I'm sure they're on a storage disk somewhere in my office closet, but I need a storage disk to keep track of all the storage disks I have in there! So, I had to start fresh, which is not a bad thing, especially since I knew I could count on Samir Abid to help me do the math! He did.

I started with the 180-degree hairpin corner I have mapped out on the whiteboard in my office, and used that as a model. The radius of the corner, using a geometrically perfect arc, is 45.3 meters (148.622 feet). I often see drivers turn in about a meter (a touch over 3 feet) from the edge of the track, so when I subtract that amount of track and calculate the radius, it's now 44.7 meters (146.655 feet).

Taking that reduction in radius into account (and a car capable of pulling 1.0 g of lateral acceleration), the resulting maximum speed that can be carried through that corner has been reduced from 89.8 KPH (55.8 MPH) to 89.2 KPH (55.4 MPH). Okay, a reduction of 0.6 KPH (0.4 MPH) doesn't sound like a lot, but depending on how long the straightaway is after that corner, it could easily cost a tenth of a second or more. In one corner! Simply by not using all the track's width on the entry. A tenth of a second doesn't seem like anything to worry about, but experience tells me that if a driver crabs

in on one corner, they are likely to do it in others, as well. And it doesn't take long before you've lost a full second. Again, just by not using all the track available to you.

If using all the track approaching a corner was difficult to do, then I'd buy the argument that it's not worth focusing on for fractions of a second. If it took time and money to develop a new skill to drive your car to edge of the track at the turn-in point, yeah, maybe it's not worth it. But it's not difficult, and you don't need to develop anything new in the skills department. It's not like learning to left-foot brake! And it doesn't raise the risk/danger level at all. In fact, driving a tighter radius by not using all the track increases the opportunity to spin and damage your car.

Concept 2
The flip side of the argument for using all the track is that a shorter distance is a faster way of getting from point A to point B (like from the Start line to the Finish line).

Of course, there are many examples of corners where you don't use all the track at the exit because you need to get back to the other side of the track to set up for the next (more important) corner. I'm not really focusing on these ones, because they're more about sacrificing speed in one place to gain in another. I'm talking about corners where a shorter distance outweighs the benefits of a larger radius.

Some examples of corners where not using all the track is faster that immediately come to mind are:

- Entry to Turn 7 at Portland International Raceway
- Entry to the Carousel at Road America
- Entry to the Carousel at Watkins Glen
- Entry to Turn 2 at Thunderhill East

In these corners, driving out to the edge of the track increases the distance you're driving, and the benefits of the larger radius don't outweigh the time it takes to travel further. Can you think of other corners where the shorter distance is more important than the larger radius?

Keep in mind that the speed at which you approach a corner is also a major factor in determining how much track to use; you're trying to get the ideal compromise between distance and radius.

Think about this. In fact, I encourage you to think beyond the obvious (consider the shortest distance), but not give away the simple, easy stuff (use all the track). Pull out your track maps for the circuits you'll be driving this year (you can download most of them at TrackMaps) and consider whether a shorter line could be a benefit, and where you might not currently be using all the track that you should.

Day #28: Study the track to gain subtle advantages.

Let's talk track camber, not suspension camber.

As I've shared already, there is a trade-off and balance needed between driving the ideal line and the shortest distance. The ideal line uses all the width of the track, from the outside entering a corner, to the inside at the apex, and back out to the outside when exiting the corner. It's the largest radius, and we all know that the larger the radius, the faster we can drive. However, the fastest way to get from one point to another is by driving the shortest distance. And this contradicts the "drive the largest radius" strategy. Sometimes the shortest distance is the fastest, and sometimes the largest radius is.

But there are more factors to consider when determining the fastest line around a track, such as:

- Camber/banking
- Surface
- Elevation changes

I started my road course racing career at Westwood Motorsport Park, just outside of Vancouver, Canada. Before being turned into a housing development in 1991, it was the scene of some of the best racing ever in Formula Atlantic, Trans-Am, club racing, and even the NASCAR regional series. My memories of watching Gilles Villeneuve destroying Keke Rosberg, Bobby Rahal, Danny Sullivan, Kevin Cogan, Price Cobb and others in the rain in the 1976 Formula Atlantic race will never leave me. But back to my point…

If you looked at a map of Westwood, you'd assume that Turn 1 was maybe a 3rd gear corner – but you could also imagine taking it in 2nd gear. Being about 160-degrees, it almost looked like a large hairpin. But driving it was a different story: I'd often take it in 4th gear in my Formula Ford. And that was simply because of the 15-degree banking, which was as steep as any road racing circuit's.

Built in 1959 as Canada's first permanent road racing circuit, the pavement was repaired more than once, and that played another role in the grip level around the track, but especially in Turn 1. There was a strip of pavement that was best to avoid in the dry, but you absolutely had to have your left-side tires on in the rain (and it did rain more than once at Wetwood… I mean Westwood). Combined with the banking, you wanted to turn in much earlier than a classic line would have ever suggested. The goal was to throw the car into the banking, and let the surface and track camber grab hold of you. It took a huge amount of trust to believe that it would do that, too, and that's why you could also tell when someone was new to Westwood – they just didn't believe that it would grip that hard!

Oh, and on a flying qualifying lap in my Formula Ford, I'd dip my inside tires inside the track surface, and hook the inner sidewalls on the edge of the asphalt. It was a bit hard on the car, but worth it for a lap or two.

With the marketing slogan, "Mountain High Racing," and literally built on the top of a mountain, there was no shortage of elevation changes. To be really fast there, you used the changes in elevation to your advantage, and minimized their disadvantages. The entry to Turn 1 was uphill after a very long (but twisty) straightaway, so the braking zone was about the length of your car. But braking into Turn 2 and 4 were downhill, so you had to manage that. Turn 3a/3b was a double-apex downhill sweeper that led onto the back straight. Whether it was actually off-camber or not, it absolutely felt that way – both to the driver and the car (I was hired to drive two of the very first Acura NSXs for promotional purposes, serial numbers 002 and 003. Not long after I drifted 002 through Turn 3 a few times, I heard a knocking sound from the engine. I was told by the overseeing Honda engineers that Ayrton Senna had tested the cars at the Nürburgring, so not to worry about oil starvation. Five laps later, a connecting rod exited through the side of the engine block…).

I was so very fortunate to have learned to race at a track like Westwood because it taught me so much about how to use camber, different surfaces, and elevation changes to my advantage, while minimizing its negative factors. I learned to often put these factors ahead of the largest radius and shortest distance.

Besides having fun sharing some stories about Westwood, my goal here is to challenge you to think about ways to use or minimize the effects of track camber, surface changes, and elevation. When you really study a track, you'll find all sorts of subtleties that can have a huge impact on the speed you're able to carry.

Day #29: Train your mind like you train your body.

If you were to ask one hundred performance and race drivers of all levels, from novices participating in their first few HPDEs to elite-level pro racers what their single biggest mental challenge is, guess what most would answer?

Oh look, a squirrel! No, not squirrels – it's just that I saw one out the window of my office as I wrote this. Back to the question.

But before I get to answering the question about… ahhh, what was the question? Oh yeah, something about the biggest mental challenge. Which makes me think about the time I was conducting an Inner Speed Secrets seminar/workshop in Estonia. Wow, the city of Tallinn is beautiful, even in the middle of February.

Back to my question. Which was… ahhh, the biggest mental challenge most drivers face. And it is…

Yes, focus. Maintaining focus, that is.

Why don't more drivers focus better? Because they think losing focus is a bad thing, and they don't have a strategy for regaining it.

Let's take this example to the track, and make it personal. You're heading into Turn 1 of your favorite track. You brake, downshift, and while looking up and through the corner, you turn in while easing off the brake pedal – trailing in to help rotate the car, just a little bit. The car feels good, you've got grip, so it's time to get back to power and get a monster exit out of the turn. But as you get on the gas, the car loses some front grip and begins to understeer. You instantly realize that if you stay in the throttle, you'll run out of track, so you ease up. In fact, you have to lift completely out of the throttle for a fraction of second to get the front tires heading where you want them to go. And in doing so, you know you messed up your exit speed, and you know that's going to hurt all the way down to Turn 2. You made a MISTAKE.

"Argh, messed that up. Should have been more patient with the throttle. Next lap. Stupid me. I needed this to be a good lap, and I screwed it up in the first bloody corner. Okay, time to get my act together. Focus. But what a stupid thing to do back there. All I needed to do was be a little more patient with the throttle and… whoa, missed that apex in Turn 2. Okay, come on, focus. Gotta get Turn 3-4 right. Just don't do what I did in Turn 1, cuz that was stupid. Not going to do that next lap. No, next lap, in Turn 1 I'm going be more patient with the throttle. That's what I'm going to do… oh crap! I just messed up Turn 4, now! Man, I gotta get it together here. I always seem to lose my focus when I'm trying to get my best lap. I suck at this…."

If you've had even some tiny little part of that conversation with yourself, then you know what I'm talking about. Once you make a mistake, it's easier to focus on the mistake you made "back there" than it is to focus on what's going to happen up ahead. Maybe

it’s because it’s something real that has happened, and what’s ahead of you is not real yet. But looking back – and thinking back – is often easier than focusing forward.

Of course, at the speed we travel on a race track, we can’t go back and fix something right then and there. It’s not like the typing I’m doing right now: if I mistype something, I can immediately backspace and re-type it, and it doesn’t really make that much difference. Or does it? If you write at all, you know that sometimes when you go back to fix a typo or a grammatical error, you forget where you were going with that sentence, paragraph, or full thought/idea. That’s why the most common advice writers give is, “Just get it down. Don’t worry about the grammar and typos. You can fix them later.” That advice applies to driving on a track, too. You can fix your driving grammar on the following lap, but if you’re thinking about fixing that "comma" right now, you’re not thinking about the next corner. And that’s all that matters in the moment.

Having said all that, focus can be trained. As I always say, it’s not your ability to maintain focus that is most important; it’s your ability to quickly regain it. The faster you are at regaining your focus, the more consistently fast you drive.

Developing and using a Pre-Planned Thought (**PPT**) is critical. It’s the key to triggering re-focus. The more you use it, the better you’ll be at regaining your focus.

Most drivers would agree that the mental game is the number one most important factor in their performance. So why don’t drivers do more about it? Why do drivers spend an unimaginable amount of money on car parts, and yet little to none on what’s in their heads? Why will many drivers spend hours each week in the gym, working on their physical fitness, but not spend even a few minutes on mental training?

Could it be because they don’t know what to do, how to train mentally?

The “downtime” when you can’t drive on track is the perfect opportunity to hit the “mental gym” and train your mind.

Day #30: Experiment with four approaches to fast corners.

I was coaching at a car club event and a discussion about how to best approach a fast corner came up. Four drivers and I talked about four options when faced with a corner that you don't really need to slow down very much for:

1. *Brake lightly, but longer* - the advantage being that the car stays better balanced, therefore having more grip.
2. *Brake a little harder, but for a shorter period of time* - the advantage being that you stay at top speed longer.
3. *Don't brake at all, but simply lift off the throttle* - the advantage being that the car may stay more balanced.
4. *Stay at full throttle, but squeeze the brakes with your left foot* – the advantage being that the car doesn't transfer weight forward as much, keeping it better balanced.

My experience shows that all of these options work, and it comes down to the specific corner and the car you're driving as to which is the best option. The only way to know for sure is to experiment with them. Try one approach for at least half a dozen times to ensure you're doing it as well as you can, then switch to the other option, and then the other. And don't rely simply on lap times, because what you do the rest of the lap may counter what you do in that fast corner. Sure, use lap times as one metric, but comparing data and segment times are the only way to know for sure (as I wrote about in this column a few weeks ago, one of the simplest ways of measuring segments times is with an in-car camera – hold one finger up in from the of the camera and try option #1 for 5 or 6 laps, then two fingers in front of the camera and try option #2 for a few laps, and so on; then use a stopwatch while replying the video, measuring a segment of track from just before the start of the fast corner section to well down the straight afterwards).

While the pros and cons were being discussed, I mentioned another point about approaching fast corners: when to make your speed adjustment. In most cases (more than 90% of the time), when approaching a fast turn that requires a small speed adjustment, make it early - lift or brake early and get back to throttle at turn-in.

Our natural response to trying to go faster is to rush into the corners, leaving any speed adjustment (brake or throttle lift) as late as possible - it makes us feel like we're going faster.

Day #31: Adapt your throttle application style.

How you apply the throttle exiting a corner may be just as important as when you apply it – maybe even more important.

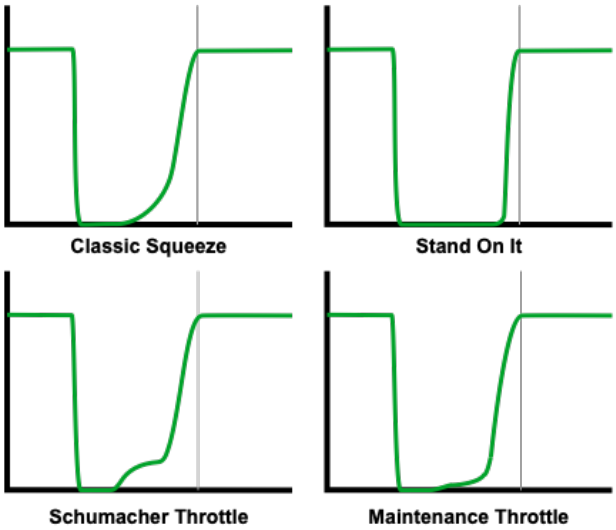
While the details of how you can apply the throttle are infinitely variable, I think there are three main styles:

1. The Classic Squeeze
2. Stand On It
3. The Schumacher Throttle

The Classic Squeeze is just as it sounds - a progressive application from zero to full throttle. This is what we're taught when we first learn to drive on track, and it's the perfect place to start. Smooth is fast, right? And just because it's the perfect place to start, that doesn't mean that it's not the right approach at all levels. It can be. But it's not the only way to apply the throttle.

With the Stand On It approach, you wait until it's time to apply the throttle, and then whack it wide open. There's less finesse involved with this approach, and while that may sound like the wrong way to drive, it can be the right way. Smooth is not necessarily fast, right? If you're driving a 700-HP car on narrow tires, it may never work. But if you're driving a car with more cornering grip than power (we usually pick on Miatas at this point in the conversation…), there are times when you just need to slap the throttle to floor and get on with getting out of a corner.

I used the term "Schumacher Throttle" in past articles and books after digging into some writings that described (and showed) data traces from Michael Schumacher, as well as thinking through some observations of my own while watching the seven-time World Champion in Montreal at the Canadian GP. What Schumacher did – often, but I don't believe all the time – was this: As he released the brakes, he immediately tipped in 10-30% throttle and held it there for a "moment" (fractions of a second to perhaps a full second), then squeezed or stood on it to full throttle. It was as if there was a short plateau before going to full throttle. Now, don't confuse this short plateau with maintenance throttle. Maintenance throttle simply maintains the car's speed so it's not decelerating; Schumacher's plateau was generating a very small amount of acceleration. This slight acceleration meant an increase in corner minimum speed, which meant that he was beginning to accelerate from a slightly higher speed.



Look at the graphs where I illustrate what the data traces look like for each of these throttle styles (including maintenance throttle as a reference). Note that in each, the release of throttle approaching the corner, and full throttle are reached at the same points. And yet, how the driver got to 100% is very different. Also, keep in mind that the rate of application can change for any of these, but the basic trends of each style stay the same.

Okay, I know what you're thinking: Which one is best? And I bet you can predict my response: It depends. But depends on what?

- Acceleration versus cornering ability.
- Radius of the corner.
- Whether you need to change the car's balance to affect the handling.
- Elevation, camber, and surface changes in the track.
- How your car reacts to weight transfer.
- Whether your car is impacted by aerodynamic load shifts caused by a change in the rake of the car.
- What gear you're in.
- Your car's throttle responsiveness.
- How your car puts the power down exiting corners.
- Track and ambient conditions.

The one approach to exiting a corner that rarely, if ever, works is beginning to apply the throttle, and then lifting back out of it before going to full throttle (this could be called "The Oops, Wait" style!). In most cases, that means that you either began applying the throttle too early, or too quickly. Either way, you probably would have been better off waiting for a fraction of a second before beginning to apply the throttle, or being a little more gradual with the way you squeezed it on. If you did, you would likely have gotten to full throttle sooner without having to breathe back out of it. Breathing back out of the throttle when exiting a corner causes what may seem like subtle shifts in the load – weight transfer – and that lessens the overall grip level of the car.

I've been making the point to learn to be more adaptable. Mostly, I've focused various articles on braking – the initial application, and timing and rate of release – and challenged you to be adaptable, learning that different cars, corners, and track conditions require different styles or approaches. This is the same – there is more than one way to apply the throttle. And being deliberate and strategic in the way you develop your ability to adapt is the key to being a consistently faster driver.

You do what you do because you're programmed to do so. You have habits. And especially in the higher stress levels of driving fast on a race track (the kind of stress we love!), you tend to rely on those programmed habits. If the only style you have for

applying the throttle is The Classic Squeeze, or Stand On It, then that's the only way you'll drive – even when a different approach would be better.

I'm challenging you to spend some practice time – mentally and physically – learning (which is programming) different throttle application styles. First, look at the images below and imagine drawing them with your right foot. Spend some focused mental imagery time rehearsing each of them to the point where you can do any of them without thinking. Close your eyes, imagine driving the same corner, but use each of the three different styles. And move your right foot while imagining them, building the muscle memory. Then go to the track and practice using them in a deliberate way. Spend a session or two just using one style, then the next style, and then the next.

The advantage is that while on track at speed, you can easily adapt to the ideal throttle application style – the one that best suits your car, the corner, and the conditions. That's where a good portion of that last little bit of consistent speed comes from.

I've seen too many drivers use the exact same throttle application style, no matter what they're driving, what corner, what conditions. That means they'll be quick in some cars, some corners, some conditions... but not in others. If you've ever wondered why some drivers are consistently faster than others, this is a big part of the reason.

Ultimately, your goal is to accelerate as fast as possible out of every corner. To do that, you need to adapt your throttle application style.

Day #32: Performance driving vision is not static - it's a process.

Anyone who tells you that you only need to look farther ahead when driving on track or autocross course is doing you a big disservice. And worse, if they specifically tell you to look X distance ahead (i.e., two reference points, or seven or twelve seconds ahead), run away. Why? Because how you use your vision is not a static thing!

I've heard instructors tell students that they should look about two references ahead. They're recommending that you should be looking at the apex before you approach the turn-in point, for example. That would be nice if performance driving were that simple, but it's not. Now, I'm not saying that it's super complex, either, but to look "two references ahead" is missing the point.

Your vision must not be static.

Now, before I go any further, I have to confess that I've told thousands of drivers to "Look further ahead." As a general objective, there's nothing wrong with it, but I've learned a lot since I first started suggesting this. It's a very incomplete bit of advice, just like telling someone who wants to be healthier or stronger to exercise. Well, duh. But how should that person exercise? What type of exercise?

Approaching a corner, I could look for the turn-in point before the Begin-of-Braking (BoB) point, and then the apex before the turn-in point, and so on, and that would be looking two references ahead. But that would also mean my vision was static — that it was focused a specific point or some set distance ahead.

There is a process to how the best drivers use their vision, and it's far from static.

The second through fourth steps in the process are:

2. Glance
3. Focus
4. Peripheral

All of these are preceded by what I call ***mental vision*** — that's the #1 on our list. In a way, this is your memory of what's coming up ahead, what you've learned from previous multiple laps. Once you've driven a track or course once (or more often), there is a mental program for the direction the circuit goes, the shape of the corners, the elevation and camber of the surface, and so on. Sure, after just one lap or run, the program is pretty hazy, but with each repetition — whether physically driving it or mentally imagining driving it — it becomes a deeper part of your memory or programming. So, when you approach a section of track or course, you absolutely should be imagining what's up ahead before you're actually able to physically see it.

Let's look at the Glance–Focus steps of the process (no pun intended), using a ninety-degree corner at the end of a straightaway as an example. Heading towards the brake

zone, you glance ahead (hey, two references ahead is not a bad place to start, but it will depend on the shape of the track, elevation, etc.), checking to see that the track is clear and everything is where you remember it. You then bring your focus back to the next reference point, which is likely to be the BoB; then glance ahead towards the turn-in and End-of-Braking (EoB) — as that's how you judge where your BoB should be — back to focus on the next reference; and so on. And here's the thing: your glance is very quick in comparison to the time you spend focusing on the next reference. As a very general guideline, you should be spending about twice as long focusing as you do glancing. So, if you glance ahead for half a second, then come back to what you're focusing on for about a second. Then it's back to glancing ahead, and then focusing again, and so on.

Do you notice what's happening as you do this? Right - you'll be scanning from what you're glancing ahead to see, back to what you're focusing on, glancing ahead, and back again. When you're vaguely told to scan with your vision, this is what is meant.

Now, what about your peripheral vision? It's used to determine "how am I doing?" Think about it. As you pass the apex of a corner, you don't turn and look directly at (focus on) it, do you? No, you use your peripheral vision to notice how close your car is to the apex, and the angle it's at while passing it. That information is used to help you decide how hard you can accelerate. At least, it should be.

Okay, how to practice this? First, on the road. As you take a route you've driven before (heading home, for example), use your mental vision to picture in your mind the path you want your car to follow. Glance to where you'll turn in and end your braking (EoB) for the intersection up ahead, then quickly back to the car in front of you and where you're going to begin braking. Glance ahead, flick your vision back to where you need to focus, glance ahead, focus… and so on, moving the scanning zone further down the road as you drive. At the same time, deliberately use your peripheral vision to check on your progress, as well as your mirrors. In a conscious, deliberate way, practice this process. Sure, it will feel odd at first to be thinking so much about this, but over time, it will become part of your subconscious mental programming, and you'll be using this process as if on auto pilot, without thinking about it.

How about on a simulator? Yes, but here's the thing. I don't have enough evidence to prove my theory, but it's possible that sim driving is the cause of not turning the head enough! See, for many sim setups, due to the positioning of the screen(s), you don't have to turn your head to look through a corner as much as you do on the real track. VR glasses can help this, as long as the simulated track is accurate enough. But, if you deliberately practice the vision process while even exaggerating your head movement, you'll build the right mental habits.

"Eyes up — look ahead." You can still use this as a trigger phrase to set your vision process mental software in action. Think of it as a catchphrase that captures and triggers all phases of the vision process.

Day #33: Be proud of your basics.

You've heard all the sayings about a house needing a solid foundation… blah, blah, blah. The reason you've heard them so often is that they're true — as is the case for needing a solid foundation of skills for any activity, including driving.

Smooth is fast. In slow — out fast. Late apex. Drive "the line" before you try a different line. All great pieces of advice. Up to a point... until they're not. But they are the foundation, and if you can't do them, you'll never do the stuff that'll make you really fast. You can't skip the basics and go directly to the "fast stuff." Well, you can, but you're likely to cause some harm (personally, financially, mechanically) if you do.

Some drivers seem to want to move past the basics and dig into the really juicy, fun, advanced, trick stuff too soon. I was (and still am) on a mission to help drivers focus on the basics before chasing the dream of acquiring that one super-secret technique that will take their driving to some mystical, Senna-like level.

A while ago, I received an email from a reader who shared the following reasoning for drivers wanting to skip the basics:

> "The somewhat less generous possibility is that it can be all too easy to see the basics as "boring." You've been watching videos of drivers flinging their cars around the Nürburgring, or you've been eating up a season of Formula One or Indy Car, and then you get to the track and… you're just going to work on braking a little later for turn three all afternoon? Sometimes it doesn't seem like much. (This dovetails with your observation about people wanting to focus on "cool tricks.")
>
> "Perhaps the solution lies in not only, as you point out, recognizing that advanced stuff is usually just a finessed form of basic stuff, but also seeing sound basics as something to be proud of in and of themselves. Rock-solid basics may not be flashy, but they'll put you in a position to do well — and incidentally, other experienced drivers will recognize and appreciate your accomplishments with the so-called "easy stuff" even if some spectators may not."

I love it!

Be proud of your basics! Embrace them, in fact. If you want, show them off. Let the world notice just how good you are at the "foundation."

Imagine being able to say:
"When it comes to simply applying the brakes, there's no one better than I am."

"I can turn a steering wheel sweeter than anyone."

"The way I apply and release the gas pedal is a work of art."

Recently a friend made a comment about the way I shifted gears in the manual transmission rental car we shared in Europe. It was not a big deal (and I only share the story here to illustrate a point, not to brag about it), but the comment stuck with me. I'm proud of the way I can shift gears smoothly. It's basic stuff, but it's rewarding to be really good at it.

So, go for it. Tune up your basics, and be proud of them. Show your basics to the world.

Day #34: To consistently drive the limit, you have to step over it every now and then.

I've heard people give this advice to drivers: the only way to know where the limit is is to crash the car every now and then.

I don't buy that. But I do buy the fact that you have to go over the limit at times.

You can go over the limit without crashing the car, though. That's the important message — and your goal.

My point is that before you can consistently drive the limit, you have to be able to drive beyond the limit. Once you've driven beyond the limit, and kept the car on the track, it is much easier to dial it back a little, back to "at the limit."

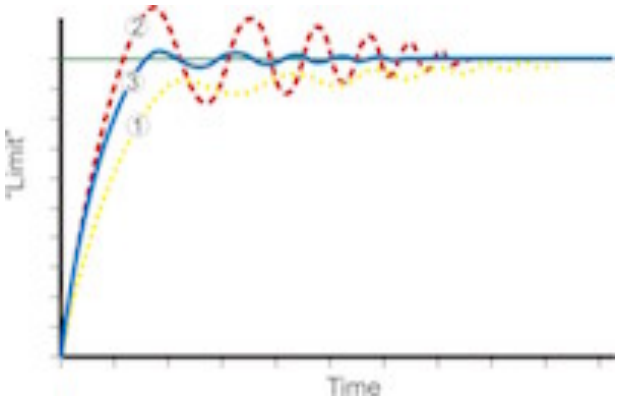
Many successful autocrossers use this approach to the three runs they get: the first one is a little conservative, just to learn the course; the second run is driven beyond the limit; and the third and final run is dialed back just enough so that the car is as close as possible to its limit. If the driver never over-drove the car and course on the second run, it's likely that they wouldn't get to the limit.

If you can't overcompensate, or over-drive, you'll never be able to home in on the limit. If you can't over-drive the car, you'll never be able to drive the limit consistently.

Some drivers can do this for one corner or one lap. But to do this consistently, lap after lap, is the goal, right?

For many drivers, it's keeping the car consistently at the limit.

In the graph below, there are three different drivers' approaches to driving the limit.



Number 1, the yellow line, creeps up on it slowly, taking small steps. Number 2, the red line, takes big steps, often going way beyond the limit. Number 3, the blue line, takes medium steps, going slightly beyond the limit and bringing it back, homing in on the limit.

If your goal is to drive the limit, then the approach shown by the yellow line is not going to cut it. Well, it might take a long time to reach the limit, at least. That's not all bad, especially if you're not getting paid millions of dollars to drive within the last hundredths of a second of the limit. But if you want to consistently drive at the limit, then you're going to need to be a little more aggressive in your approach to reaching it.

The problem with the approach shown by the red line is some of the peaks are way beyond the limit. In fact, they may be so far beyond the limit that a crash is imminent.

Sure, the blue line approach requires a well-developed ability to sense the limit, but if you use the suggestions I made in my main article, you'll get there. What's most important is to practice sensing the limit in a very deliberate way. I also highly recommend using the Sensory Input Sessions that I've written about in the past.

Day #35: Smile when you drive.

“Attitudes affect moods — and moods affect what you think — and what you think affects the 'how' of what you do! Feelings of excitement, happiness, and satisfaction improve the belief in one’s own capabilities to successfully make a change.” – Dr. Cecilia Brantley, Adjunct Professor at Argosy University

That’s an interesting quote, isn’t it? I think we all know that our mood, or our state of mind, affects the way we perform at anything. If you’re in a bad mood, how well do you perform your job? If you’re in a good mood, what happens?

So... be in a good mood and you’ll drive better. How’s that for good advice? It’s accurate, but me telling you to be in a good mood is no different than me telling someone who is about to jump out of a plane to skydive for the first time, “Don’t be afraid.” It’s one thing to tell someone how to feel, but a completely different thing for that person to actually feel that way. Hey, if it were that simple, we could rid the world of anti-depressant drugs by just telling people to be happy!

But can we control our mood, and what would happen if we did?

One of the many very interesting studies reported in the fascinating book, Endure: Mind, Body, and the Curiously Elastic Limits of Human Performance by Alex Hutchinson, showed that when marathon runners smile, their running efficiency improved by two percent. Just by smiling. Of course, smiling near the end of running 26.2 miles is a challenge for even the most elite of marathoners. Eliud Kipchoge of Kenya, the 2016 Olympic marathon champion, has begun to deliberately smile when running. It’s something he learned through the research project he’s involved in with Nike, called Breaking2 — the goal is to run the marathon under two hours.

You might be thinking, “That’s easy! I’m smiling all the time when I’m on track. How could I not be, as it’s the most fun thing I could ever do.” You might be right, you might be smiling. But maybe you’re not. And if your mind and body performed even two percent better because you deliberately smiled while driving, wouldn’t that be a good thing to do?

Here’s something else to consider: I’ve written a lot about the negative impact that holding your breath has on your performance. What if you deliberately practiced “breathe with a smile”? What if you mentally practiced that before ever going to the track? What if you practiced this every day when you’re driving on the street? What if you took part of a few sessions on track to deliberately “breathe with a smile”? What impact would that have on your driving?

A technique to control your state of mind (mood, emotions) that I’ve shared here in the past is to replay a past success. That success doesn’t need to be while driving a car, either. If, before you go on track, you recalled something in your past that made you feel great — some time when you performed at your best, something that happened in your

life that made you feel successful, an incredibly positive experience — you will be in a better mood. The more details you recall when replaying this experience in your mind, the more you'll find yourself smiling. If you're able to put yourself back in that place and time, and experience the feelings you had then, you will have controlled your state of mind.

Replay a past success right now. Take a few minutes to recall something in your life that made you feel fantastic, and imagine all the thoughts and feelings that you had at the time. Breathe. Smile. Enjoy.

Now, head out onto the track. Breathe with a smile.

Day #36: Use the throttle to tell you whether you're driving the best line or not.

Every now and then, I get a driving question that really makes me think deeply. Okay, I'm not the smartest person in the room! But still, this question got me thinking to the point where I had to write about it to figure it out (there's a tip for you right there: to figure something out, try writing out the solution).

Paraphrasing, the question is: *"I share a car in autocross with another driver who is a tiny bit faster than me, and it's mostly because of the line he drives and how that relates to how much time he spends on throttle. What is the best way I can learn more about car position and how to know if I am positioning the car in the most efficient way, since the distinction is subtle enough that I cannot usually tell much of a difference in our line from the data?"*

In this case, the driver had someone else's data to compare to. Essentially, he's asking how he can determine what line to place the car on to maximize his speed, knowing that he wasn't spending as much time at full throttle because of his line.

For sure, data doesn't do a perfect job of pointing out exactly where your car is positioned in relationship to another. GPS data is accurate, but not super-accurate when compared to the subtle differences two drivers/cars can have through a course.

But data does hint at the answer. In fact, with enough experience, and enough analysis, data can tell you a lot. However, when I thought more about this question, I wanted to look at it from a different angle.

Rather than looking to the data to "draw" the ideal line, look to the throttle, speed, and longitudinal g-force traces to identify where there is a lack of acceleration compared to another driver, or even a previous on-track session.

Too many drivers drive the line they do because they're following what someone else either told them, the data told them, or it's what they saw someone else use. To use data, or even another driver, to tell you where to position the car might be short-term thinking. Turn things around and focus on spending more time at full throttle — and what it takes to do that. I've used this approach when coaching drivers and they find a better line by focusing on doing what it takes to be at full throttle more often.

Think about how a driver finds the ultimate line — the one that leads to the fastest lap or run in an autocross course. The best drivers don't depend on another driver telling him where to position the car. They don't rely on the data to tell them. They don't simply copy the line of a driver they watch in a YouTube video. So what do they use to determine the ideal line?

The overall process looks something like:

- Start driving what feels like the right line, based on the basic geometry of the track/course

- Fine-tune the line based on observation (what other drivers are doing, what the data and/or video suggests)
- Experiment with small adjustments to the line, and notice what works and what doesn't (using lap/course times and data as a metric, and compared with what feels faster), focusing on what allows more full throttle

That seems simple, right? Well, there's a lot going on in each of those steps, but that's really it. It's the final step that makes the ultimate difference between the fastest and less-than-fastest. And, ultimately, it's finding that if the car is positioned "here," you can get to full throttle sooner. So, by experimenting with what it takes to spend more time at full throttle, the positioning/line comes to you.

What sounded like a very difficult question to answer comes down to a very simple answer: Keep asking yourself what line the car needs to be on to spend more time at full throttle. This approach may not work one hundred percent of the time, but in my experience, it will nine times out of ten.

Take time to think about this. The line is defined by what results in the fastest run or lap time. Most times that relates pretty closely to the lap/run that you spend with the throttle flat to the floor the most. Again, there are always exceptions to the rule, but focusing on doing what it takes to spend more time at full throttle is rarely a bad thing. When you're trying to decide which line to drive — or even whether a very subtle difference in angle of the car relative to an apex, for example — deliberately apply more throttle, and see what happens. If you run out of track or run wide, can you adjust your line so you can apply that much throttle and end up where you want?

It's this process of focusing on what it takes to spend more time at full throttle that the best drivers use to find the ideal line. You can rely on data, video, or others to tell you where to drive, or you can use your own car to tell you.

Day #37: Think (and practice) strategically.

For many drivers, seat time is the key. It's what separates them from being the best at a track day, or winning the F1 World Championship. I can see you right now, nodding your head, thinking, "Yeah, that's right. If I could practice more, I'd take Max Verstappen's seat — and I'd even do if for less than he makes, maybe only $25 million a year!" But, as you've heard me say many times here, quality will almost always win over quantity. That's why it's important to think strategically about how you develop your abilities as a driver.

I'm going to give you a few specific — and possibly surprising — approaches to practice that have proven to be very effective when I've used them to coach other drivers.

Use taller gears. Here's a simple one. Drive the entire track for a session in a higher gear than you usually do. If you typically take Turn 1 in 3rd gear, now use 4th. If you drive Turn 12 in 4th gear, now use 5th. Turn 7 in 2nd gear, now use 3rd. Your goal is to drive as quick a lap as you can, as close to your usual lap times, but by rolling more momentum through the corners so your engine won't bog down so much. Two things will come out of this approach. First, you'll learn that you can carry more minimum speed through some corners (but not all). And second, you may find out that using a taller gear in a few corners makes you even faster (despite it feeling slower). For the next on-track session, go back to using the gears that you usually use, and see what happens. You may be surprised what you learn.

Brake lighter. This is especially effective if there's a corner that your gut tells you that you could brake later for, but you just can't seem to muster up the you-know-what to do that. That's okay, just begin braking where you have in the past, but do so a tiny bit lighter — let's say 5% lighter. Or think in terms of a scale of 1 to 10 (with 10 being maximum), threshold (or just starting to activate the ABS). Ask yourself how hard you're braking for that corner now, then knock one point off that and brake that hard. For example, if you're using a 9 pedal to brake for Turn 1 right now, tell yourself to brake at an 8. By doing so, you'll arrive at the corner going just a little bit faster, and learn how the car reacts. After some number of laps like this, learning that you can make the corner just fine with the lighter braking, it'll seem relatively easy to begin braking harder and later again (with the 9 pedal).

Pop your foot off the brakes. Yup, I said it (and it was hard for me to actually type those words!). I learned this approach from reading a golf instruction book years ago (and no, I don't golf – but let me on that golf cart!). The advice was for golfers who slice the ball when hitting off the tee. It suggested that the golfer go to a driving range and purposely figure out how to slice the ball, so that they would learn what causes it, and then they can fix it. If you drive a session on track popping your foot off the brake when entering corners, you'll learn what happens — and I bet a lot of money that after doing that for a little while, you'll stop, and you'll know why you don't want to do that. Oh, and you'll also learn when you might just want to pop your foot off the brake pedal — or, at least, release the pedal more quickly. It's an amazing learning experience.

Make your car handle poorly. This exercise does two things. First, it helps you learn to adapt to different handling characteristics, and second, it makes you more sensitive to what the car is doing. Depending on what adjustments you can make to your car, do whatever it takes to make your car understeer (disconnect the rear anti-roll bar, raise the front ride height, decrease the negative camber on the front wheels, reduce front tire pressures by 6 psi, or whatever). And ideally, understeer more than just a little bit, as in, make it handle like a pig (okay, I have no idea how a pig handles, but I can imagine…). Then spend a session learning how to minimize the effects of the understeer (turn in more gently, trail brake more or less, etc.). Experiment with your driving technique until you find a way to minimize the understeer's impact on your speed. Then do the same, but with oversteer. Doing this exercise, you'll become much more adaptable with your driving, and in most cases it leads to being an overall faster and more consistent driver, too.

Overdrive the car. Okay, this one depends on where you are with your development as a driver, and specifically how good your car control skills are. However, most drivers can do this – but they don't because they're afraid of what might happen or what someone might say about them. The idea here is to spend a session where you select certain parts of the track where you're going to deliberately overdrive the car – brake too late and run wide entering the corner; ease off the brakes too early so that you carry too much speed into a corner; turn the steering wheel too much in the middle of the corner; or get on the throttle too early. It's hard to know where the limit is, and stay at or below it, without going over it sometime. The objective here is to go over it in a safe place, in a safe way.

Focus on just one thing. Doesn't matter what, just pick one. This is not such a radical idea, but what is radical is just how narrowly focused you should be with this exercise. Instead of planning to drive smoother, focus on just one part of your driving that you're going to make smoother, such as throttle application. And then, focus on a smoother throttle application in just one corner. And then, determine just how much smoother you're going to be by sitting in your car in the paddock and practicing it. And then, think through at what point in the corner you're going to begin applying the throttle, and by what point will you be at full throttle. And then, practice that sitting in the paddock. And then, practice all of this in your mind, visualizing what that would look, feel, and sound like. And then… finally, go on track and focus solely on this one task. You might be surprised at what it does for your driving.

Okay, that's enough for now. I can hear a few drivers saying, "I don't have time to do things like this. I only get four sessions throughout an entire event, so to use up one or two sessions to do these exercises… well, that's a waste of my limited time." Is it? I know that you're nodding your head and agreeing with me that there may not be any better, more effective way of using your limited track time! Well, that is if you're interested in learning and improving.

Understand that there are times when you'll need to go backwards to go forward. Two steps backward for three steps forward. And, what is right, right now, might not be right, later. In other words, think strategically in terms of the process of learning.

Day #38: Use segment times and data to really know what's going on.

Often, a driver will make a change to their car's setup, or to their driving, and decide whether it was a good change or not simply by looking at the lap times. That's like telling a cop that you were driving the speed limit because you were only in fourth gear. The gear might give you an approximation of what speed you're driving, but it's not accurate enough.

Your lap time is not accurate enough to determine whether a change in one corner helped or hurt because there are too many variables throughout the rest of the lap.

Instead, use segment times and data.

I'm not going to go into a lengthy discussion about the use of data to compare one technique or line in a corner. That's for another day, but what I will say right now is by looking closely at the lap time comparison tool that practically all data acquisition software has, you're getting a much more accurate measurement of how you did in that one specific corner.

Segment times might seem old-fashioned, but that's not necessarily true. The old tried-and-true method is to have someone stand in a position where they can see your car enter the brake zone for the corner in question, and some distance down the following straightaway after the corner, then they simply time each lap from the start to the end of that segment. Because the segment is short, the difference between laps may be a tenth of a second or less. So that means that the person doing the segment timing needs to be consistently accurate with the use of the stopwatch.

A "newfangled" method is to combine video with segment timing. Simply sit down with your in-car video after a session where you tried different techniques and/or lines, find a spot on the track that you can consistently see on the video each lap as you approach the corner, and a spot well after the corner. Ideally, this will be a brake reference point, a change in the pavement, or some other marker that is easy to see on your in-car video. Then, just use a stopwatch while viewing the videos, and take segment times from the first marker to the end marker.

To make life easier for yourself, prior to going on track, make note of the two different options you're going to use (a technique or line, for example). When you're driving and have established a baseline of a couple of laps, hold up your hand in front of your in-car camera and show one finger — this is option number one. After you feel that you've gotten the most from this approach, hold up two fingers in front of the camera, and do enough laps to know that you've gotten a solid representation of that method. Afterwards, sit down with your video and take segment times, comparing the two options.

Day #39: Don't use data.

Oh, this is going to piss off a few of my data guru friends, big time!

Recent research has shown that the use of GPS and smartphone apps telling us how to get from point A to point B is having an effect on how our minds work. It's making our brains lazy, and our spatial awareness and positional mapping abilities are changing (not for the better).

If you're old enough to remember a time when we used paper maps to figure out directions to where we wanted to go, you know how you got a mental model of where things were. You knew that the race track was northeast of the airport, and the hotel was south of the track. Or, at least, the track was "up and to the right" of the airport, and the hotel "was down over there" from the track.

How many times in the past few years have you plunked the address of the race track into your smartphone map app, pressed "Go," and followed the "Take the second exit at the roundabout," "In one mile, use exit 189 onto Main Street," and yet not really known in which direction you've driven? I bet you didn't really have a mental map of where you were going, or where you were when you got there! You followed the directions, but no longer had an image of where you were positioned, relative to other things.

Using GPS makes our minds lazy. So, too, can data acquisition/telemetry systems.

Without data, we have to figure out how to drive faster. With data, we can follow the directions, like we're following what our smartphone tells us.

As an example, I get in your car and drive some laps, and I'm faster (hey, it could happen!). Afterward, we look at my speed trace on the data analysis software, and notice where I'm faster (as well as where I'm slower). You look at that and think, "Okay, if he can go that much faster right there, so can I." You head out and you try to go faster "right there." And maybe you do. Or maybe you don't. Even if you do go faster, do you know why you went faster? If you do know why you went faster, can you apply what you learned to other areas of that track, or another track? And most important, if you went faster, was it because you sensed that the car could be driven faster, or were you just trusting that since I went faster, you could, too?

One of the wonderful things about performance and race driving is the mental exercise we get. There's no doubt that technology is mostly making our lives easier and that's a good thing. But in doing so, in some ways, our minds are getting out of shape.

So, about the advice to not use data… I was kidding. Kinda. But not completely. There's a time and place for data, just as there is a time for a seat-of-the-pants feeling for what the car is telling you. The next time you put a destination into your map app, take the time to expand the map so you get a global view of where it is in relation to where you're starting. Then, as you follow the directions, imagine how you're getting to your

destination.

Same thing with your data. Instead of just following the directions, expand your view to a global one and understand why you need to do what you need to do, and why you're not doing that already. Are you lacking that mental map of what you need to do, that seat-of-the-pants feel for what's needed? Use the data to point you in the right direction, and maybe even use it as a safety check, so you don't get lost. But think.

Day #40: Be a rain Meister.

In 1979, at the U.S. Grand Prix held at Watkins Glen, Gilles Villeneuve put his stamp on the very-wet sessions, driving his Ferrari. He produced lap times that were unbelievable to his competitors, to the point where some of them got out of their cars and walked down to Turn 1 to watch. Can you imagine Max Verstappen or Lewis Hamilton being eleven seconds faster than everyone else? Not a chance, right? But that's what Villeneuve did that day forty-three years ago — he was eleven seconds faster than his teammate, Jody Scheckter, who won the World Championship that year. And before you think it was because his car was superior, or the spread across the field was much bigger than it is today, the rest of his competitors were all within a couple of seconds of each other, and they were all more than eleven seconds behind Gilles. In the wet-drying-wet-drying, changing conditions of the race, Villeneuve won by nearly a lap.

Did Villeneuve know a special line around the Glen in the rain? Did he know of a different technique? Did he possess more skill or talent? Or was there something else going on in his mind?

I wasn't there that day, but there are things I've learned through my experience driving and coaching in the rain that'll make you more Gilles-like.

The Wet Line
The typical advice for driving in the rain is to drive the "wet line," off the usual "dry line." The thinking is that the track surface on the dry line is highly-polished and embedded with rubber, and therefore has less grip when it's wet. True — usually, until it's not. You have to consider the age of the track surface, and even the type of asphalt used; if the surface is relatively new, it hasn't had time to be polished and smoothed out, and therefore the wet line may still be on the dry line. You have to test the surface to see where the grip is.

Sometimes, the grippy wet line is a "rim shot" around the very outside of the corners. Yes, there is less margin for error when you're just a few feet or meters from the edge of the track, but often there is so much additional grip out there that it will feel as though there's a berm that your tires hit and run along – it may feel as though you could never slide beyond them and off the track.

With most tires, you lose more lateral grip than you do longitudinal grip on a wet track. That means that your outright cornering grip will be lower than your braking and acceleration grip, and that's why you want to "make" as many straights as possible.

Approaching corners, brake off-line to the inside of the usual dry braking line — where there will be more grip. Then, cross the cornering line so you can run at least a car's width wide of the apex, and finally, turn the car so you can drive as straight as possible exiting the corner. Essentially, drive a road course like it's a series of drag strips: accelerate as fast as you can in as straight a line as possible down the straightaways,

brake, spend as little time as you can turning the car, and accelerate again as straight as you can.

Curbs may be your friend or enemy in the rain, and the only way to know is to test them. But do so gently and smoothly. Some painted surfaces, like curbs, are very slippery; other curbs have a material like sand embedded in the paint, and it has the grip of sandpaper. Your job is to find out.

Ultimately, the wet line is all about driving where there is more grip, and avoiding where there isn't grip. For example, Turn 4 at the end of the backstraight at Mid-Ohio is one of the most challenging bits of track you'll ever face in the rain (like most of Mid-Ohio when it's wet!). You'll probably find that it's best to brake to the inside (to the right) of the dry line, but as you begin to turn in, you have to cross a section of track that has the grip level of ice. But that's okay because on the other side of that section is a different grip level — it's like finding the berm to lean a dirtbike into around a tight corner — the car just grips up and you can turn. Of course, you have to trust that it's there, and be okay with your car sliding across that section of ultra-slippery track before you get to the grippy stuff. That's what I mean about using the track where it has grip, and minimizing where there isn't grip.

Rain Driving Technique

The number one technique to put into play in the rain is to make the car "do something" — ideally, understeer from the very instant you turn into a corner. That way, you know what it's going to do. What many/most drivers do is turn into corners expecting the car to begin to slide, but waiting until it does to react to it. The best drivers proactively make the car do something the instant they turn in, and that way they're not surprised by it. In fact, because they made the car do it, they're ready to use what they did and what the car does.

This requires turning the steering wheel with purpose, and being deliberate with the timing and rate of release of the brakes to make the car do what you want. For most drivers, they want the car to understeer a little, so they release the brakes early enough to begin unloading the front tires, while turning the steering wheel quickly enough to induce some understeer. Then, they can work with that understeer though the remainder of the corner.

Getting and keeping heat in the tires is a challenge in the rain, especially if the ambient temperature is a little (or a lot) on the cold side. But the harder you "work" the tires, the more heat you'll generate, the more grip you'll have, the more you can "work" the tires… and so on. The more tentative you are when braking, turning the steering wheel, and accelerating — and carrying speed through the corners — the less heat you'll have to work with.

The initial application of the brake pedal should usually be a little more smooth and gentle than what you do in the dry, but not as much as you might think. And remember that much of the heat you build in the tires comes from the brakes — the harder and

more you use the brakes, the more heat that is generated which transfers up through the upright, into the wheel, and to the air inside the tire. Heating the tire from the inside — and keeping heat in the tire — is the best thing you can do.

Visibility, or the lack thereof, is the most intimidating part of driving in the rain, and it's one of the main reasons some drivers don't enjoy it. I've noticed two different types of drivers: those who really struggle with the lack of visibility, and those who don't. The main difference between them is that the first almost solely relies on references in front of them (in the dry or wet), and the latter uses many references off to the sides of the track. What does that tell you? That you should be picking up and adding side-of-the-track references to your mental database. But you need to start doing that while driving on a dry track, and ideally comparing and contrasting these "side references" with the ones in front of you. The key is finding these side references before you need them.

Changing Conditions

Any talk of wet weather driving needs some attention on what most drivers feel is the most challenging condition: mixed, when it's changing from wet to dry, dry to wet, and the in-between stuff.

When it begins raining, keep in mind that until water builds up on the surface, you still have a lot of grip, especially if your tires are hot. Many drivers see a few raindrops on the windshield or helmet visor, and immediately slow down. Of course, then the tires cool down, the grip goes away because there's less temp in the tires, and the grip level spirals down – a case of confirming the feeling of having less grip.

It can be surprising how much grip a hot tire — even a racing slick with no tread — will have on a wet track that has no build-up of water.

If you're on "rain tires" — proper wet weather racing tires, especially, but even street tires with deep grooves — and it begins to dry, you first start to feel the tires squirm, then you begin to feel that you're losing grip (when you're not). It's a bit like driving on a dry roadway with snow tires; you feel the treadblocks twisting and squirming, and that makes you feel like you're sliding. You need to accept that the car will feel like it's sliding, and live with it moving around.

(Of course, too much of the squirming on the treadblocks, the tires will overheat, and could even start tearing the tread off the tire carcass).

I like to think that on a drying track, if you're not going faster, you're going slower. Every lap should be a little faster – and sometimes a lot faster if the track is drying quickly. And if it's going from dry to wet, the goal is to minimize the downward trend of slower laps. Much of this is tied to your mindset.

Rain Driving Mindset

First, if you're the type of driver who refuses to go on track when it's raining, I'm asking you to reconsider. Driving in the rain seems like it's riskier, but I'd argue that it isn't.

Having been at thousands of tracks when it's raining (and many, many multiples of those many times when it's not raining), with drivers of every level of experience and ability, the number of crashes on wet days is no more than when it's dry — and I'm pretty sure it's even less. No, I don't have statistics on this, but I'm confident with my observations.

Plus, when there is an incident in the rain, it's not as severe as others in the dry since the speeds are lower.

The learning value of driving on a wet track cannot be beat. Period.

If you think about it, what you learn from driving in the rain (great car control skills) actually makes it less likely that you'll crash in the dry, so the overall risk of having a crash over the course of a season, or all the years you drive/race on track has been lowered.

In the rain, most drivers focus on the lack of grip. But the best focus on the grip they have, rather than the grip they don't have. This is a key mindset shift (if you're not already there). Look for the grip, and use it, rather than struggling with the areas of the track where you don't have the grip you're used to (in the dry).

It's this "be happy with what you have" attitude that makes the difference between being an okay or a great wet weather driver.

Ultimately, your attitude or mindset — and your confidence — makes the biggest difference. Going back to Gilles Villeneuve. Was it the line he drove, the technique he used, and the talent he had that made him eleven seconds faster than everyone else that day at Watkins Glen? Sure, those things mattered, but ultimately it was his belief in his abilities to use the line, the technique, and his talent that made him so fast.

Gilles had grown up racing snowmobiles, so he was used to sliding a vehicle. But he intentionally made whatever vehicle he was driving slide (some say that he even did that in his helicopter!).

Not only was Gilles confident in his abilities in the rain, but he enjoyed it. For me, that's the most important thing; more important than the line, technique or anything else. You have to love driving in the wet, even if you have to fake loving it for a while.

Again, was it his line, his technique, his talent, his mindset/confidence? Yes. Oh, and all the practice he had in slippery conditions. Don't waste your opportunity to learn and develop your wet weather abilities by avoiding driving in the rain. Be like Gilles.

Day #41: What you practice on the street results in habits you rely on when driving on the track.

Powering out of Turn 3 and onto the back straight at Westwood Motorsport Park (yes, this was a long time ago, before they turned one of the greatest road racing circuits into a housing development), me in the right seat, my student behind the wheel of his 5-Liter Mustang, I noticed an irregular surging of acceleration and deceleration. It was if the driver was flooring the throttle, lifting off, flooring it again, lifting off, and so on. “What engine problem could cause that?” I wondered. I then looked towards his feet, and it was not “as if” he was doing this. He was doing this.

“Keep your foot to the floor on the throttle,” I calmly said. I love it when I can make a difference in how someone drives with a simple observation and suggestion. Low hanging fruit is tasty.

“I can’t. I can’t,” my student shouts back at me, looking down at his leg as if it’s not attached to him, his face with a “surely, it belongs to someone else” look.

I can’t say that the rest of his driving was much better. In fact, he certainly got my FULL attention.

After many, many laps, and repeated reminders, my student finally got so he could drive on the straightaway holding his foot in one position. He had his leg under control. And at that point, I realize just how powerful the habits one develops while driving on the road can be. It has had a huge impact on my own driving, as well as my approach to training drivers ever since. So, thank you, Mr. My-leg-is-not-attached-to-me.

What “bad” driving habits do you have, that you formed, and possibly are reinforcing with your everyday driving? Take time to be aware of anything that you do when driving on the road that might be negatively affecting what you do on the track. Also, flip that around and think about the habits you want to build for tracking driving, that you could practice and build on the street.

Day #42: Every driver has doubts.

Have you ever doubted your driving abilities? If your answer isn't "yes," then you're either lying or you're fooling yourself.

Every driver has doubts about their abilities, at some point.

Do you think World Champions doubt themselves?

Read what Jean Todt (former Ferrari F1 team manager) had to say about Michael Schumacher (as reported in the December 5, 2017 edition of Autosport magazine):

"Another thing to show what Michael is like: when he was finishing the 2000 season as champion, we were starting the 2001 season. He asked me, quite shy, because he's a shy guy. He looks arrogant but he's shy. He asked me, 'would you allow me to do some testing in Fiorano to make sure I am still able to drive?' He was always full of doubt, whether he was a good driver or not. He made his testing, and he was not too bad...."

Not too bad?! But even Michael Schumacher doubted himself.

What does this have to do with you?

First, just being aware that even the best drivers in the world doubt themselves makes it a little easier to accept your own feelings.

But second — and most important — it's what you do about these feelings that count.

- Ask yourself if any of the talent and skills you have stored away in the programming in your brain has dropped out of your head overnight as you slept. Okay, a bit sarcastic, but think about it: you don't lose talent and skill. You might not be accessing all of it right now, but it's still there — you haven't forgotten how to drive.
- Even if you're currently not driving at your best, or you don't seem to be improving, take time to think about all of the things you're doing well. Make a list of them. Come on, I know there's more than that. Write down every single little thing that you're doing well, right now, as a driver.
- Imagine yourself without any doubt. What would that feel like? How would you think? How would you stand and walk? See yourself like that. Feel those emotions. Spend at least ten minutes just imagining what it would be like if you didn't have any doubts.
- Go drive. And don't be judgmental. Don't pick apart what you're doing, just do; be aware of how the car feels and sounds. Be hyper-sensitive to that. Notice how you're using the steering wheel and pedals. There's nothing good or bad about the way you're using them. No judgment. Just be aware of them.

- After you've driven, and before you look at any lap times, data or video - and before talking to anyone - ask yourself how you felt. How did you drive? If you felt good, great. If you didn't feel good, that's okay, too. Just make note of how you felt after driving that session.
- I'd say that there's a pretty good chance that you're feeling okay with the way you're driving. You may or may not be faster than you have been in the past. Take time to consider what's different from when you've been faster (if that's the case). Are track conditions different? What about your car? Your tires? Air temperature? The mental state you're in? How much sleep you've had? How you've been eating? My bet is that you're comparing apples to oranges — your lap times now compared to your past best times. And that could be for the better or the worse. But remember, you still have all the talent and skills you've always had — and likely even more since you've practiced more.
- One more time, imagine yourself feeling confident in your abilities. It's a deep down inside kind of confidence — a strong belief in yourself. Imagine that. Really imagine it. In fact, imagine it so much that it's now real.

Just like Michael Schumacher, you believe in yourself. Sure, you doubt yourself every now and then, but that's just being a human — and knowing that that little niggling thought of "Am I still okay?" is driving you on to being even better.

Sure, sometimes we don't access all of our talent when we want, and one sure way of making that happen is losing confidence and trying harder. The harder we try, the worse we perform; and that causes more doubt, and hurts our confidence even more.

If you drive on a track, there will come a day and time when you have doubts about your driving abilities. If you feel as though you've taken a step backwards, I suggest taking a page out of the Zen meditation book and just "let go." It's obvious that trying harder will not help, so why not get all "woo-woo" and relax? It can't hurt, right?

As you let go, keep in mind that you never lose talent. It's still there. When you let go, it's more likely that your talent will be easier to access.

When you compare your results with others, that often leads to a loss of confidence. But as we all know, when a car's performance is part of the package, it's quite likely that our own performance is not the entire problem. I know it's easier said than done, but focus on your own performance, rather than on the results. Know that the results are very much driven by your car, and there are times when your car is not a match for someone else's.

And at all times, keep in mind that World Champions have doubts and lose confidence every now and then. So, it's okay....

Day #43: You have to be close to take advantage of luck.

How many times have you seen a race come down to the last lap or two, and for it to be decided by one driver's misfortune?

Never give up, no matter how far behind you are, no matter how unlikely it seems you will catch the competitor in front of you. Keep pushing until the checkered flag falls. You never know if the competition is having problems that might be terminal if they have to drive hard to fend you off. How many times have you seen the leader of a race have a mechanical problem with only a few laps to go? You will never be able to take advantage of their problems if you are not close.

I once won a race, taking the lead in the last corner leading onto the front straight, and winning by just a couple of feet. Ten laps earlier, I had been nearly fifteen seconds behind — I could barely see the leader, and it seemed a bit hopeless trying to catch him since that gap had not changed for a while. The thought went through my mind, "Just hang on and protect second place — that's not bad." Then another thought hit me — the Speed Secret that I'd shared with many other drivers: "You have to be close to take advantage of luck." So, I pushed hard. I figured that no matter how unlikely it was that I'd catch the leader, I was going to give it a shot, and make it as close as possible. And, maybe, just maybe....

Two laps later, I noticed I was making up ground on the leader, and yet it was still a stretch to catch him. He got held up passing backmarkers, and I got through cleanly — a big gain. Then I noticed that he was struggling through a particularly fast corner — maybe his tires were going off. And another pack of slower cars held him up more than they did me — his misfortune. I pushed hard, and on the last lap, I was just a few car lengths back. In the fast sweeper, I held the throttle flat (I'd never done that before), and found myself right on his rear bumper going into the last corner. More important, he noticed I was on his rear bumper, and overcooked the turn just the tiniest bit. And that gave me the opportunity....

If I hadn't pushed all the way to the checkered flag, I wouldn't have been close enough to take advantage of the slower cars holding him up a little. Lesson learned. Lesson proved.

Day #44: Focus on the possibilities rather than expectations.

The author of the book, *Atomic Habits*, James Clear wrote:

"Anticipate, but don't expect.

“Anticipation: You're excited for what the future holds, but you don't try to control it.

“Expectation: You try to predict the future and restrict your happiness to one outcome.

“Always be excited about the possibilities. Never be entitled to them."

When I read this, it reminded me of the way I coach drivers: I work with them to focus on what's possible more than what they expect. It's related to focusing on one's performance, as opposed to the result.

Have you ever put a target lap time in your mind, matched it, only to find that everyone else on track went faster? See, it's rare that we will exceed what we put in our minds as a goal or target.

If so, you're not alone. Even F1 drivers do this at times. I shared the following in my Ultimate Speed Secrets book:

Karl Wendlinger, who was severely hurt (he spent weeks in a coma) driving a Sauber F1 car at the 1994 Monaco Grand Prix, provides a perfect example of the power of mental imagery and expectations in Christopher Hilton's book, *Inside The Mind Of The Grand Prix Driver*. In this case, he is referring to when Wendlinger went back to testing for the Sauber team in 1995, having been away from the cockpit for almost a year.

In the words of Wendlinger, “Because I'd only done a little driving I had time to prepare my body, to do training. I did a lot of concentration exercises. Then I arrived at Mugello. It was a two and a half day test and, the evening before, I thought ‘OK, one minute 30.4 would be a good time.’ I concentrated, I closed my eyes and as I crossed the line into the lap I started my stopwatch. I did a whole lap in my brain and looked. The stopwatch said 1:30.4. The next day on the track I did 1:30.4.

“Then I said to myself, ‘it was too easy, tomorrow you have to do 1:29.3.’ The best lap that Heinz-Harald (Frentzen, Wendlinger's team-mate) did in Mugello all year was 1:29.0 and because I had done so little driving I thought 1:29.3 was competitive. I sat in the hotel again, closed my eyes and started the watch. I ‘drove’ the lap and looked. 1:29.3. Next day I did 1:29.3. You know the best thing was — and this fascinates me about what you can do with your brain — I only did the 1:29.3 because I made a mistake and lost three-tenths. If I hadn't made that mistake I would have done 1:29.0 — but the evening before I had fixed in my brain 1:29.3 and not zero, and that's what happened. If I had fixed 1:29.0 maybe I wouldn't have made the mistake and I'd done the time easily.”

Expectations are limitations, and you rarely exceed your expectations. Wendlinger proved that, along with the power of mental imagery. Expectations program results into your mind, and your mind is very efficient at running those programs — sometimes too efficient! In Wendlinger's case, it sounds as if he had expected to turn a 1:29.0, he would not have made the error that ultimately cost him three-tenths of a second. That's the (negative) power of expectations at work.

Certainly, this applies to your on-track driving performance, but perhaps in these times of uncertainty, keeping the possibilities in mind are more important than ever. And remember... you can choose what you focus on. You decide what is possible.

P.S. - After spending weeks in a coma following his crash at Monaco, Wendlinger did make a comeback. He drove in F1 in 1995, then went on to race and win in sports cars. I wonder what he was focusing on throughout his recovery...?

Day #45: Soften your car's setup in the rain.

As I've said many times, there are no hard and fast rules that apply to every adjustment to either your driving or your car's setup. But a pretty good guideline for when you're going to drive in the rain is to soften the suspension setup.

What does that mean? Soften what?

That depends on what you can soften. For many cars, the first thing would be to soften your anti-roll bars, especially the rear one. And maybe even disconnect the rear bar altogether.

If you knew for sure you were going to spend the entire time on a wet track, and you can do this, install softer springs.

If you have adjustable shocks, soften the settings on them.

Again, these are general rules that apply more than half the time (but not necessarily all the time).

What about tire pressures? That's an even less sure thing. Often, reducing the pressures will help, especially if the track is wet, but not puddling. Lowering the pressures is almost like softening the suspension.

But, if there's lots of water build-up, and you're experiencing hydroplaning, then consider increasing tire pressures. And I mean increase them a lot. The idea is it create a crown across the tread of the tires, effectively reducing the width of the tires, improving its ability to cut through the deep puddles of water on track.

Okay, the big question is, "Why?" Why do you usually soften the suspension setup? The simple answer has to do with load? transfer — and the reduced amount of it in the rain.

When cornering on a dry track, the forces are higher than in the rain, and therefore there's more load? transfer. In the dry, you need a stiffer car to make the transition from one direction to another responsive. In the rain, because grip levels are reduced, you can't transition (or corner) as fast, and therefore the responsiveness is not as important. What's most important is overall tire grip, and a softer suspension allows the tires to follow the track surface better, providing better grip.

As a general rule, that is....

Day #46: To be fast, you need to be a weight manager.

No, this is not an ad for Jenny Craig - it's not about weight loss. It's about getting the most out of your car.

Driving a car at the limit is all about managing its weight, its balance.

Smooth is fast, right? Why? Because it manages the car's weight better, resulting in more overall grip.

Want your car to turn into the corners better? Manage the car's weight, keeping the right amount (some, but not too much) transferred onto the front tires, by adjusting the timing, and rate of release of the brakes while entering the corner.

Need to increase or decrease the car's radius through a fast, sweeping turn? Squeeze on or ease off the throttle to transfer weight to the rear or front tires.

Too much understeer or oversteer exiting a turn? Change the timing and how hard you apply the throttle.

Want to carry more speed through a turn, but you feel as though the tires are at their limit of adhesion? Keep the car "flatter" — more balanced, with the least amount of weight transfer, by being smoother with the steering wheel and pedals.

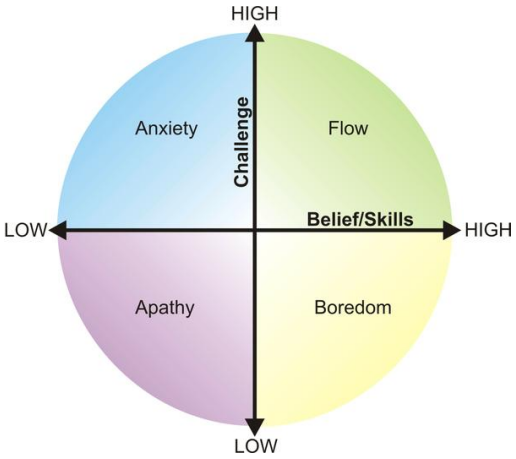
Managing your car's weight is like dancing with it. It's about finesse, coordinating your hands and feet with the timing of Helio Castroneves.

Day #47: Balance the challenge of driving with your belief, and you're more likely to get in the zone.

For sure, racing is challenging. And that can be a trigger for a state of mind that we, as humans, naturally strive for.

In his groundbreaking book, *Flow*, Mihaly Csikszentmihalyi, talks about what causes us to get into a state of flow, or “the zone.” Considered the grandfather of the topic, Csikszentmihalyi knows what he's talking about, as he's spent most of his adult life researching what causes a person to get into this state.

Being in the flow is something that most everyone has experienced at some point in their lives, whether it’s in sport, playing music, driving, business, parenting, public speaking, or pursuing a hobby. For a few - the superstars of an activity — it's almost a way of life. For others, it's a fleeting experience, there and gone in a flash. But experienced once, it's something we crave to experience again and again. Track driving gives us an opportunity to flirt with being in the flow — it's accessible, unlike many other activities we spend time doing. Where else can we test ourselves to the extent that a race car at speed on a track provides?



One factor that Csikszentmihalyi claims will often trigger a flow state is a balanced sense of challenge and belief. If you're faced with a task that is not challenging, and you don't believe you can handle it, you're likely to feel ambivalent towards engaging in it. You probably don't even want to do it, and it's unlikely that you'll get into the flow doing it. If you feel you're faced with a big challenge, and yet don't have the belief in your ability to handle it, you're likely to feel anxious and not perform at your best. If you're facing a task that seems to offer no challenge to you, and you're super confident in your ability to handle it, it'll seem almost boring — that will not trigger a flow performance, either.

However, when you feel challenged, and yet have a deep-down-inside sense of belief in your ability to handle it, that's when you're most likely to perform in the flow. A sense of challenge balanced with confidence leads to performing in the zone.

What does performance or race driving do? It challenges you. And most successful drivers have a pretty strong sense of belief in themselves (or they're working their way towards that feeling, especially when they've prepared well for the driving task). That triggers a flow state, often. And we're attracted to this state. In fact, we're more than just attracted to it. It's a basic human need. We want it, and we want more of it!

Performance and race driving do that. They provide an opportunity to experience a flow state, triggering all sorts of "reward chemicals" in our brains. Like dopamine. Flow triggers the release of dopamine in the brain, as do certain drugs. So driving in the flow is more than like a drug, it is a drug! While it's questionable whether it's less expensive than a drug addiction, it's definitely healthier, and legal!

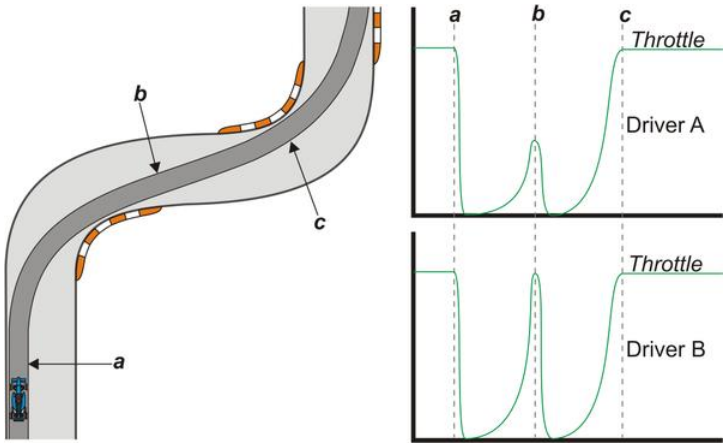
The workout that comes from performance driving, mentally and physically, is an attraction. Like a runner's high, a driver's high is triggered by the release of chemicals in our brains.

Day #48: Hustle the car where it matters.

Good enough is not good enough. At least, not if you want to be fast.

Many tracks have what I call hustle zones: areas of the track, usually in between corners, where there are opportunities to get to full throttle for a fraction of a second or more. But the thing is, many drivers don't give the throttle the squirt that it needs. Instead, at a subconscious level, their mind thinks, "There's no point in going to full throttle when I'll just have to lift off of it again in a second. Giving it a little throttle is good enough."

In the illustration below, I show the throttle traces from two fictitious drivers. Driver A is like the example I just mentioned - giving the throttle a little application, but certainly not going to full throttle. Driver B, instead, gives the throttle a good squirt, getting to full throttle for a brief amount of time.



You might wonder how much of a difference this bit of hustle has on a driver's lap time. Usually, a significant amount. At least if anywhere from a quarter to half a second is significant to you.

Not all tracks have these hustle zones, but good examples of them are: between Turns 3 and 5 at Sebring; Turns 6 and 7 at Road Atlanta; 3A and 3B at Pacific Raceway; 5A and 5B at Pacific Raceway; 3 and 3A at Sonoma; and 7 and 8 at Grattan. Okay, not all cars can get to full throttle in these areas, but many can. It's up to you to find out.

The next time you're at the track, think about whether there are any hustle zones on the track — places where you can get back to full throttle for even a fraction of a second. It's worth it.

Day #49: Tell the car what you're going to do next.

My guest on episode #148 of my Speed Secrets Podcast (SpeedSecrets.com/Podcasts, or on your favorite podcast platform) was Rob Wilson. Rob is the guy who has spent a lot of time training Formula One drivers such as Valtteri Bottas, Kimi Raikkonen, Nico Rosberg, Juan Pablo Montoya, and others — using a box-stock Vauxhall Astra.

As is usual, towards the end of our recorded conversation, I asked my guest for his Speed Secret — the one piece of advice he'd give drivers to help them perform better. Rob said, "Always tell the car what's coming next."

During our podcast conversation, we'd talked about the technique of turning the steering just slightly before actually initiating the turn-in for a corner. The idea is that by doing so, you're not "shocking" the tires; you're telling them, "Hey, we're about to change direction, so get ready."

In my first Speed Secrets book, I shared the message, "Build up the tire's cornering force slowly." I then went on to use the analogy of breaking a thin piece of string. If you held each end of the string in opposite hands and quickly snapped the ends apart, the string would break. But if you very slowly pulled each end of the string, it would take more effort to break it. It's the sudden snapping of the string that makes it break more easily.

The same is true with tires and the grip they have with the track surface. If you come up to a corner, and very quickly jerk the steering wheel into the turn, the front tires will likely break traction and begin to slide. But if you slowly, and progressively turn the steering wheel, the front tires will hang on longer before breaking away.

Watch in-car camera views of various top race drivers and you'll likely see hints of this type of steering action. When I say "hints" of it, I say that only because it's a very subtle movement, one that is not easy to notice. But most of the very best drivers use this technique, and my experience tells me that many don't even realize they're doing it. They just do it because it works, and they may only understand what they're doing after picking up on it later by noticing the small differences in the squiggly lines of the steering trace on a data analysis.

As Rob and I continued our conversation about this technique (and others — it's always fun to spend time talking with a fellow coach who is as committed to learning and helping drivers as I am), I shared with him my one main concern with teaching this technique to some drivers: crabbing.

I've written about "crabbing into a corner" in the past, but just to be clear, here's what I mean by this phrase. Imagine approaching a right-hand hairpin corner. About a car's length or so before the turn-in point — that point on the track where you initiate the turn of the steering wheel — rather than keeping the wheel pointing straight, you rotate it just a little bit to the right. This pulls the car away from the left edge of the track, as if it's

moved slightly sideways (like a walking crab). And this reduces the radius of the upcoming corner because you're not using all of the track width.

Crabbing is a habit that I see many drivers commit, and it does cost lap time.

But, there's a difference between crabbing (a no-no) and what Rob and I talked about, which is giving the tires a slight warning before fully turning the steering wheel (a go-go).

There are three different ways you can turn the steering:

1. *Linear* – The rate of rotation, or speed at which you turn the steering wheel, stays the same from beginning to full lock.
2. *Digressive* – You quickly initiate your turn of the wheel, but then slow down the rate of rotation.
3. *Progressive* – You start off relatively slow with the rotation of the steering, but then speed it up.

What Rob talked about when he said to "tell the car what's coming up," and what I wrote about back in 1997 in my first *Speed Secrets* book, is essentially progressive turning of the steering wheel.

If you head to the track this coming weekend, determined to "tell the car what's coming up" by "Build(ing) up the tire's cornering force slowly," be aware of whether you crab into the corner. Turning the steering wheel progressively is great, but not if you're turning in from a car's width (or even less) from the edge of the track.

Day #50: It's all about the pressure.

Apply the throttle. Release it.

Apply the brakes. Release them.

Turn the steering wheel. Straighten it.

Each of these seem like a position-sensitive movement of the controls. Push down, lift up, turn left, turn right.

I'm pretty sure that I wasn't the first one to say that the gas pedal is not an on-off switch. It's more of a rheostat. The same goes for the brake pedal. And the steering wheel.

It's easy to think of each of these controls as something you move the position of. Turn the steering X number of degrees and your car turns a specific amount. Push the gas pedal half-way and you're at half throttle. Push the brake pedal and then let it off.

But I'd like you to start thinking in terms of pressure, as opposed to position. There's a certain amount of pressure you apply to and release from the gas and brake pedals, and it's the amount of rotational pressure you apply to the steering wheel that makes the difference.

What's the difference? Smoothness, for one.

Precision, for another.

For a moment, think about the feel you get through the bottom of your foot when you apply the brake pedal. There's a certain amount of pressure you feel as you quickly squeeze it down, and then as you gradually release it. It's that pressure that matters, that tells you how you're controlling the brakes. Most critical is the pressure you slowly release as you approach and enter corners. Think about managing brake pressure to help the car do what you want it to do by managing the pressure your foot is applying to the pedal.

As you turn the steering wheel, think of applying pressure to it in either direction, rather than turning it. You pressurize the steering!

For those of you who were told to either push up with one hand to turn the steering wheel, or pull down with the opposite hand, toss that instruction out the window. It's silly stuff. You have two hands, so use them equally. You'll be smoother that way. Apply pressure up with one hand while you apply pressure down with the other — increasing and decreasing pressure on the wheel, resulting in a smooth, controlled movement of the wheel.

The throttle is also pressure-sensitive. The release of pressure at the end of a straightaway may be very quick, and even the increase in pressure as you feed in the throttle exiting a corner may be fairly quick. But still, you manage the pressure of the throttle pedal.

You will have trouble managing pressure on the controls if your body is tense. The more relaxed your hands and arms are, the better your feel is. If you squeeze the steering wheel hard, you'll get less feedback through it. You want to hold the steering with just enough strength as to not lose grip of it. That's a fairly light hold. The steering is not just an output device, where you turn it to make something happen. It's also an input device — it feeds information back to your hands and arms, and ultimately, your mind where it contributes to sensing whether the tires are near, at, or beyond their limit of traction. And that's a pressure thing. The steering wheel gets heavier or lighter as the tires reach their limit.

Think about some of the other activities you do, whether it's skiing, tennis, playing a musical instrument or even flying small aircraft. There's what you do, and then there's the pressure. On skis, you can carve a turn without managing the pressure, but when you do your skiing goes to the next level. Your tennis racquet grip, and how hard you hit the ball (pressure) is as important — or more important — than where you hit the ball. I don't play music, but I do know enough to say that the pressure one places on the strings of a guitar is as important as where they are. For sure, when controlling an aircraft (plane, helicopter), the pressure you apply to the yoke or cyclic is critical. You move the controls with pressure, rather than positioning.

There's the "what to do" (i.e., apply the brake pedal), and then there's the "how much" (i.e., the pressure). The use of the controls is all about the pressure. Pay attention to how much you're applying to them, and how much they feed back to you.

Day #51: Focus on the grip you have, not the grip you don't have.

I know that sounds a little too philosophical to be good advice for when you're driving in the rain, but it's pretty practical advice if you really think about it.

I often hear drivers of all types and levels "complaining" about a track's lack of grip in the rain. They're focusing on the lack of grip. While I understand why drivers do this, it doesn't help. I also think to myself, "Yeah, isn't that great?" See, a car sliding on a wet track is a blast! But only if you focus on the grip you do have, and use it.

Whether you enjoy driving on a wet track or not, it's what you do with the grip that's available that is critical — not what you can't do with the grip that is not available.

If you find yourself complaining about the track's lack of grip, switch your focus to where the track has grip. Look for, search out, and be aware of where there is grip. Then use it.

Day #52: Self-coaching is all about what you think about.

After not driving a purpose-built race car for about fifteen years, I drove a LMP3. Over the previous decade-and-a-half, I'd driven production-based and GT cars. And for various reasons, I had a period of about two years where I didn't drive a single lap on track. So, when the opportunity to drive the LMP3 car came along, you could guess my first thoughts... absolutely, let me at it! No hesitation at all. Then a little while after that, I did begin to wonder if I was up to the task, physically, and skill-wise.

I was pleasantly surprised at how it went. Sure, with it being an endurance race (Thunderhill 25-Hour), I never pushed for the last few tenths, but I did okay. So, I've been thinking about why that was, and how that info might help you.

Here are some of the things I did:

- I made sure I stayed full throttle until lifting for braking.
- I focused on a smooth brake release.
- I focused on getting the car rotated early so I could commit back to full throttle.
- I was patient enough with the throttle so that when I went to throttle, I went all the way to full.
- I focused on committing to throttle, knowing that once I was above 60 percent or so of throttle, I might as well go to 100. The amount of engine power delivered from 60 plus percent throttle to 100 is not enough to use up too much more of the tires' grip — but every little bit helps with corner exit speed.
- I focused on spending as much of the lap as possible at full throttle, looking for every opportunity to give the throttle even a short squirt of full — I "hustled" the car.
- I focused on looking as far ahead as I could, and thinking, "Get there!"
- I used different timing and rates of release of the brakes (along with different rates of turning the steering wheel into different corners), rather than trying to drive every corner the same.
- In a few corners, I deliberately released the brakes early to roll momentum through the turn.
- I focused on managing the platform, keeping the car as flat and balanced as possible, even if this meant I didn't brake quite as late as I could have.
- I made deliberate decisions about where to push and where not to, weighing up the risk versus benefit.
- Finally, I used a lot of mental imagery to pre-program and reinforce what I needed to do.

All of the above points sound simple, don't they? And obvious. But, as simple and obvious as they are, it's easy to not focus enough on them. They're the basics. And it's the basics that count. By coaching myself on the basics, I wasn't slow in the LMP3 car.

Driving fast is often knowing what to not do. Approaching driving from a coach's perspective — knowing what to focus on and what not to — made a world of difference.

Oh, and I had a blast! In fact, I focused on doing what felt like the most fun. And that most often results in the best speed.

Day #53: Don't be a crab.

A while ago, I raced in a World Racing League (WRL) race at Road America. It was really fun to meet in person many drivers with whom I've traded messages in the past (funny how so many of them don't look like what I imagined them to look like from their emails...), and get so much positive feedback from so many people. Thank you to everyone I talked to at Road America.

Of course, it was also fun to drive and mix it up with so many drivers.

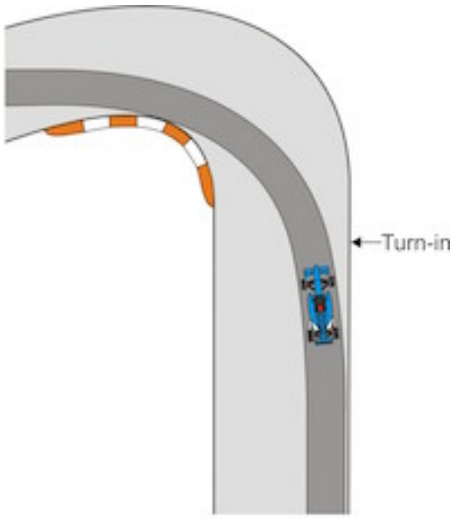
Being on track with some sixty or so cars, driven by a variety of drivers (with a variety of driving experience and skills), I learned a few things. One thing that stood out was a common error that I witnessed so many times that I lost count: crabbing.

I use the term "crabbing" to describe what I illustrate below. It's when a driver steers the car away from the edge of the track just before the turn-in point, and it's extremely common. It makes you slower.

Why do drivers do this (I've asked myself many times)? There are a few reasons:

- It feels safer. When approaching a corner at speed, it feels safer to be closer to the inside because there is less to hit there!
- Before the turn-in point, the driver has turned their head and is looking toward the apex. As you know, your hands follow your eyes, so when you look to the apex you steer toward it. Looking to the apex before the turn-in is a great thing — as long as you simultaneously have the discipline to hold your line right down the edge of the track to the turn-in point.
- The driver began braking early, got near the turn-in point and felt they were going slow enough, so "why not begin turning in?" It's almost funny how this works (it would be funny if it wasn't slowing the driver down so much).

You know why crabbing into a corner is bad, right? It reduces the radius of the corner, meaning you can't travel through it as quickly (the smaller the radius, the slower you need to drive; the larger the radius of a corner, the faster you can drive).



Watching drivers in the WRL race, there was rarely a lap that went by that I didn't notice some driver do this. And yes, I bet I did, too. But some drivers do it as a habit, over and over again; some do it drastically, costing them big chunks of time.

The next time you're on track, really be aware of whether you're consistently turning in from right at

the edge of the track or not. Just being aware of what you're doing is the first step in improving.

If you're crabbing, make sure it's not due to braking so early that you feel you've slowed down enough that you might as well start turning in. Make sure you're disciplined with your turn-in.

Don't be a crab!

Day #54: A diamond can be a driver's best friend.

In long corners, such as 180-degree turns, consider the diamond line over the classic line. Some examples of these types of corners:

- Turn 2 at Laguna Seca
- Turn 11 at Sonoma
- Turn 2 at Thunderhill East
- Carousel at Road America
- Carousel at Mid-Ohio
- Carousel at Grattan
- And some others I've forgotten!

In the classic line, you brake mostly in a straight line approaching the corners, trail brake in a little (while turning in late), and head towards a late apex. This is the classic "In slow, out fast" line.

The diamond line means beginning your braking a little later, but angling it in so that you're braking well into the corner; you apex almost a third of the way into this long corner, let the car move out from the inside in the middle, and then apex a second time around two-thirds of the way through the corner (the same apex point as the classic line uses). I like to think of the diamond line as, "Drive it in hard, collect it in the middle, and drive it out hard," or "In fast, out faster."

Let's look at the pros and cons of each approach:

Classic line

- It fits with what most drivers were taught from the very beginning, so it makes sense — and it's a habitual line to drive.
- From the moment you turn in to the point where you're able to begin accelerating again, it's likely longer than with the diamond line — and that means more time just maintaining speed, rather than either decelerating and accelerating.
- As it's the "in slow, out fast" line, you spend more time going slow than you do with the diamond line.

Diamond line

- This might be a bit more challenging, as it requires you to use the brakes to turn the car (rotate), rather than just the steering wheel. You also need to have a little better feel for the compromise between braking and cornering — a more sensitive feel for the brakes.
- While maybe (but only maybe) more difficult, it actually provides a bit more margin for error as you're never quite as close to the edge of the track.

- It’s a shorter distance, as you don’t go out as wide in the beginning of the corner. A shorter distance takes less time to travel.
- You usually begin braking later, meaning you’re able to maintain top speed on the straight approaching it for longer.
- You’re almost always either decelerating or accelerating — there is very little time spent simply maintaining speed.
- Most times, the diamond line is faster overall.

Guess which line I like to use? Not always, but, for sure, I'm going to try it.

Day #55: Balance and confidence usually win.

As a driver, feel and balance is most often more important than overall grip. Confidence is everything.

There are exceptions to this, but if the car gives you confidence that you can sense what it's doing, you'll drive faster than when you're driving a car that has more grip but scares you because it has a lot of understeer or oversteer!

There are two main things you need to consider when thinking about feel:

1. Is the car communicating what it's doing, and can you sense it?
2. How responsive is the car to your inputs?

A softer car will generate more overall grip, as it doesn't transfer as much weight from the inside to the outside tires (keeping the load on each tire more equal). With that in mind, you would think that a softly-sprung car would always be better. In some cases, that's true — especially on a low grip surface, such as slippery pavement, in the rain — or to the extreme — when racing on ice. This is why the general guideline for tuning your car for driving in the rain is to soften the springs, anti-roll bars, and shocks.

If you've ever driven a very softly-sprung car (think of an old Buick!), you know its handling is not responsive. When you turn the steering into a corner, it takes a while to respond. And if you have a section of track where you quickly go from a right turn to a left turn and back to the right again (tight Esses, for example), you won't like it. To make a car more responsive, the general guideline is to stiffen the springs, anti-roll bars, and shocks. Yes, the opposite from what might help generate grip.

Ultimately, your goal is to find the ideal compromise between overall grip and responsiveness. There is no magic formula telling you how much grip you should sacrifice for responsiveness, and vice versa. This is where driver preference really comes into play, and why two drivers may prefer a very different setup on the same car.

Whether you're adjusting the anti-roll bars, springs, shocks, tire pressures, or suspension alignment, think of it as a funnel. Begin with big changes, and work your way down until you're fine-tuning with small changes. If your car has a drastic handling problem, there's no point in making a one psi (pounds per square inch) tire pressure change, or going one click stiffer on the damper compression settings. No, make a big change, and that's more likely to be with the anti-roll bars or springs. Again, look to get a direction from a large change, and then begin making smaller and smaller changes as you home in on the handling you're looking for.

Tuning is an ongoing process, and part of the reason is that your driving will change over time. As you make the car's handling better, you'll drive with more confidence, meaning you'll drive faster, which will expose the next weakness in your car's handling. It's a never-ending loop from car to driver to car to driver.

Day #56: More effort doesn't necessarily lead to a different result.

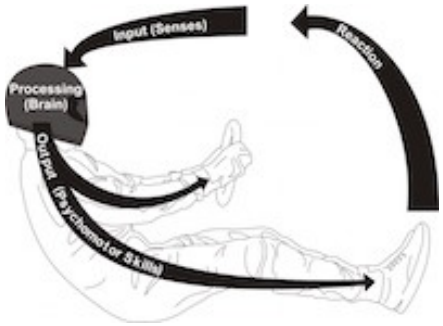
How many different areas of your driving can you work on to improve? Let's make a list...

- *The line* — turning in earlier or later; turning in more quickly or more gently; holding the apex longer or releasing the car immediately; unwinding the wheel quickly or slowly; straight-lining a section of track or following the path of the track; driving the ideal line or using a different line that takes advantage of camber in the track.
- *Braking* — your initial application (ramping up slowly or a quick almost-spike); whether you hold pedal pressure for a while or immediately start releasing it; how you manage the last 5% of the brake release (quickly release it or trail off the pedal even longer); trail brake or don't trail brake.
- *Steering* — crisply turn-in or use a more progressive turn of the wheel; unwind the wheel quickly or slowly.
- *Throttle* — hesitate slightly after releasing the brakes or immediately begin squeezing it on; use some maintenance throttle to balance the car or not; tromp on the throttle or gently feed it in.
- *Managing weight* — using your brake release to rotate the car entering a corner; breathing the throttle in the middle of a corner to make the car do what you want; adjusting how you apply the throttle to maintain front grip while accelerating.
- *Managing fast corners* — when faced with a corner that requires a very small speed adjustment, do you just ease off the throttle, lightly brake for a fraction of second, or brake against throttle?
- *Vision* — practice moving your visual focus, from as far ahead as you can see, to the reference point just in front of you, back to the distance, to somewhere in the middle, scanning to the sides and behind; looking farther ahead than in the past; being aware of any little movement in your mirrors while looking directly ahead, using your peripheral vision.
- *Adapting* — using different techniques, or variations on the techniques to best manage different corners, track conditions, and your car's handling characteristics.
- *Shifting* — using the engine to help slow the car or not; skipping gears when downshifting; short-shifting in between corners to avoid having to shift in the middle of a turn; ensuring your shifting at the best RPMs for maximum acceleration.
- *Sensing the limits* — take time to simply focus on sensing the limits of the tires/car, deliberately focusing on listening to them, feeling them, and seeing what the limit looks like.

That may not be a complete list of everything you can do, but as you prepare your test plan focused on driver development, consider these. Compare them to how you would

answer this question: If I could improve just one technique in my driving, what would it be?

Take time to answer that question, then think through what you can do to improve it. Remember, doing the same thing with more effort is not going to give you a different result.



This process requires self-awareness, and honesty. The benefits of going through this process are huge, though.

Of course, this list only covers physical techniques, and ignores the whole mental game of driving. It also ignores how you can use your mind in very deliberate ways to help you learn new techniques faster, as well as fine-tune existing ones. Ignoring the mental game will slow your driver development progress. I'll come back to that topic some more in the future.

Day #57: Build a trigger for, and practice, regaining your focus.

The best drivers in the world lose their focus while driving just as much as you or anyone else does. The difference? They regain it more quickly. What they're really good at is quickly and almost imperceptibly regaining it.

What's important is not losing your focus. It's regaining your focus. But how? That's the $64 million question, right?

With a trigger.

I'm sure you've seen someone wearing an elastic band around their wrist. Perhaps you've done that. It's used to trigger a mental program, to take the person's mind off of nail biting, smoking, worrying, or whatever. With repetition, the snapping of the elastic band triggers an alternative behavior. It programs the person's mind to behave a different way.

Losing focus is like nail biting, smoking, or worrying — it's a behavior, one programmed through repetition. The more you lose focus, the "better" you get at losing focus. The good news is that the more you practice regaining focus, the better you get at that, too.

I recommend using a trigger phrase, such as "Look ahead," or "Slow hands." Each and every time you recognize yourself losing focus, say one of these phrases and re-focus by focusing on looking farther ahead or slowing your hands down with the steering wheel. Focus on a specific aspect of the act of driving. Do this in your daily life, at work, while driving down the highway, when reading a book or watching a movie, or out for a walk or run. The more you do this, the quicker you'll get at regaining your focus.

Day #58: You don't need to crash a car to know whether you're driving the limit.

I recently watched the video that McLaren produced to celebrate the 40th anniversary of the team giving Gilles Villeneuve his F1 debut at the British Grand Prix. It's very interesting, with interviews with Leo Wybrott (Villeneuve's race manager at McLaren), and Gilles' son, Jacques. You can watch it by clicking here.

Having seen Gilles race Formula Atlantic cars, I was a fan. I was excited when I heard that he was getting his F1 opportunity, one that would ultimately lead to being hired by Enzo Ferrari.

It came as no surprise to those who knew Gilles that he would purposely spin the McLaren F1 car, multiple times, to learn the limits of the car. He had little to no fear of anything.

That got me thinking about the comment one often hears around tracks. It goes something like this: "The only way to really know whether you're driving the limit is to go over it." While there is some truth to this statement, it can easily be taken too literally.

True, that approach was the one that Gilles took — with everything in his life. And it's part of the reason he's still admired as almost God-like by so many. But look at drivers like Jim Clark, Jackie Stewart, Aryton Senna, Michael Schumacher, Lewis Hamilton, Sebastian Vettel, or Max Verstappen. Did they purposely spin the car to learn the limits? No. They approached the limit in more subtle ways!

I was thinking about Jim Clark and Ayrton Senna, in particular, and their deaths. There was never any thought that either one of them had made a mistake; everyone "knew" immediately it was a mechanical failure that caused them to crash. It had to be — they were too perfect to have made a mistake and crash. In Gilles' case... most people kinda "knew" that the big one was going to happen someday. It was only a matter of time.

Am I suggesting that Clark, Stewart, Senna, and Schumacher never drove their cars beyond the limit? Not at all. Did they make mistakes? Absolutely. But did they purposely throw the car off the track, put it in a spin, and take huge chances to find the limit? No. Instead, they worked their way up to the limit, bit by bit. Their ability to sense small changes in speed and g-forces were finely honed, through a lot of deliberate practice. So, when they wanted to go a little faster, they knew exactly what "a little faster" meant. And that's what they did.

Gilles may have just been a little less patient than those other drivers! He didn't want to take time to gradually build up to the limit — he wanted to be there RIGHT NOW!

There's a great story in Gerald Donaldson's biography of Villeneuve about qualifying at the U.S. Grand Prix at Watkins Glen. Gilles was driving for Ferrari, then, and the weather was not cooperating... unless you were Villeneuve. It was pouring rain. And guess how much faster Gilles was than all the other F1 drivers in that session? If you're

guessing a couple of seconds, that would be understandable. But he was 11 seconds faster than every other driver. Apparently, other F1 drivers got out of their cars in pit lane and walked down to Turn 1 just to watch Gilles in the rain. They couldn't believe their eyes.

My point? Unless you're Gilles Villeneuve, don't fall for the "purposely driving the car over the limit is the only way to drive the limit" advice. Develop and fine-tune your ability to sense speed and g-forces, and gradually increase them. (By the way, consider how much time you spend focused on driving the perfect line around a track, and then how much time you practice sensing speed and g-forces, and then think about which will result in consistently-faster driving)

Having said that, I believe that a driver must have the ability to over-drive their car before they can consistently drive at the limit. You need to have the ability to drive over the limit — by just a little bit — and then dial it back slightly to the limit.

That is, if you want to drive the limit. And guess what? Even if you don't want to drive right at the limit, the same approach applies. If you want to drive consistently at 8/10ths, then you're going to need to be able to drive at 9/10ths every now and then, and then dial it back to 8/10ths.

It was, and still is, huge fun to watch Gilles Villeneuve (while you're on YouTube watching the video from McLaren, poke around and watch some other clips of Gilles' brilliance). There may never be a driver as committed to being the fastest driver than he was. And maybe no one more willing to risk it all.

Day #59: Do your homework.

Driving on the track is both expensive and hard to get. Well, duh! Like I'm telling something you don't know.

Despite all the talk/writing about how anyone can become an expert at something with 10,000 hours of practice (just Google “The 10,000 hour rule”), it’s not really accurate. In fact, it’s been blown way out of proportion, and twisted from some research that Dr. Anders Ericsson did. And adapting this message to our sport, it's practically impossible to ever reach the 10,000 hour range of practicing on the track. Okay, so let's just give up, right?

Nope. Not me.

See, while track time is difficult to get (and expensive), we can practice as much as we want in our minds. Of course, I'm talking mental imagery. If Lewis Hamilton doesn't have 10,000 hours of physical practice on a track, I bet he does if you add in the time he's driving on track in his mind. Easily that many hours.

Simulators — once purchased and set up — are next to free to drive, too, so they add dramatically to the amount of practice time one can get.

And the reason we drive on the road is to practice our track driving, right? Yes, while it's next to impossible for any driver to ever get as much practice as many other athletes — what do most of us do almost every day? Drive on the road.

The key is making the most of every hour — no, every second — we spend on track. Anything you can do to soak up information from other sources before you get to the track — and then turn that into "mental practice" — makes a huge difference.

Why do some drivers improve faster than others? I'm not going to say that natural ability is a non-factor, but experience and research proves that it's what a driver does with their natural abilities that makes the biggest difference. If I have two drivers with equal natural talent, but one does more "homework" — reviewing and analyzing data and video, reading, sim driving, and especially, mental practicing — who is going to improve the most?

Do you know that tennis coaches in England have noticed a trend: for a few weeks following the Wimbledon Championships, the players they've been coaching for months improve their game performance. And that's with no additional practice time on the court; it's simply due to the amount of time these players have been watching — and mentally modeling — elite tennis players.

One disadvantage of our sport is that the actions, movements, techniques, and skills that elite-level drivers make and have are not as visible as a tennis player's. In fact, most are not visible at all. So we have to use our imaginations even more, noticing the

point where a driver begins braking, how quickly the car decelerates and changes direction entering a corner, the cornering line itself, the attitude of the car (do you notice changes in pitch and yaw?), when you hear the driver applying throttle, and so on. When we notice those things, imagine what the driver is doing to make them happen. It's the driver who is causing every action of the car. Mentally put yourself inside the car, and drive it.

Make the most of every second you have on track by doing your homework. Fortunately, it's not the kind of boring homework that we had in school!

Day #60: Take balance over responsiveness & grip most often.

A balanced car will be easier to drive than one with more grip that is not balanced well.

Many drivers relate handling to responsiveness, and that's why they think stiffer springs and/or anti-roll bars and shocks result in better handling. But it's important to separate your thinking - initially - between responsiveness, balance, and overall grip.

- As a general rule, stiffer springs will result in a more responsive car. You turn the steering wheel, and the car will respond. It feels crisp on turn-in.
- As a general rule, softer springs will result in more grip. Yes, the car will roll (lean) more in corners, but it will be generating more overall grip.

Yes, these contradict each other.

To complicate things even more, you could have a car with stiff springs up front, and soft on the rear, relative to each other. This might make the car feel responsive when you turn in to the corners, but it will likely be unbalanced. In this case, it's likely to understeer shortly after turning into the corner — so, you've traded balance for responsiveness.

Whenever you're tuning the handling of your car, think in terms of balance, responsiveness, and overall grip.

Having driven thousands of cars on many tracks, I'll choose a balanced car first, over one that is responsive, or has more overall grip. Of course, I'd rather have it all — more grip and responsiveness in a balanced car — but if I had to choose one over the other, balance most often wins. Why? Because I can make the car do what I want, increasing my chances of making up for the lack of grip.

But if I had to choose just between balance and responsiveness? That depends on the extent of either one, but I'd still go with balance first (for the same reason — it's easier for me to work with a balanced car). Having said that, if the car is very unresponsive, that makes it difficult to make the car do what I want (which, most often, is change direction quickly).

When you finish an on-track session, take time to debrief with yourself to determine which is most needed: balance, responsiveness, or overall grip.

Day #61: Minimize steering movement.

“Race car drivers are minimalist. They put the least amount of input in.” - Johnny O'Connell

That quote says so much — in a minimum number of words (fittingly).

Right now, hold your hands up as if you're holding your steering wheel. Pretend you're approaching Turn 1 of your favorite track. At the turn-in point, turn the steering as you would when driving through the corner, and along the line past the apex and out to the exit point.

Oops, you rotated the wheel one degree more than you needed to. Did you notice that? Yup, just one degree too much, and then you quickly straightened it back up that little bit.

That one degree cost you speed.

Here's a true story worth repeating: I was once coaching two drivers in the same team, both driving Formula Mazda cars. After a practice session where they had turned lap times that were within hundredths of a second of one another, I was reviewing the data and in-car video. I noticed that one driver used a LOT of steering angle. I was even surprised at how much more steering angle he put into his car than the other driver did.

I asked the team engineer to write a math channel for the data system that would add up the total number of degrees of steering angle that was put into the car on each lap (writing that math channel was way beyond my capabilities!). Even though they had turned almost identical lap times, one had used more than a hundred more degrees of steering angle over the course of his fastest lap.

For the next practice session, I asked him to focus on minimizing the steering input, and I gave him a numerical target: "You can only use 740 degrees (which is what the other driver was using) of steering angle per lap, instead of the 855 or so you're currently using."

While he didn't get all the way to 740, he did reduce it to about 760 degrees. And his lap time? About 0.6 seconds faster than his previous lap, and nearly 0.7 faster than his teammate in that session.

Over half a second of improvement by doing less.

P.S. — Yes, the other driver was not happy, and wanted me to find a similar technique to make him faster. That was the next session's priority.

Day #62: MI + A = G.

If you have a very clear and strong Mental Image (MI) of what you want to do, and an Awareness (A) of how close you are to doing that, your brain will make you achieve that Goal (G).

For example, you want to take Turn 1 at full throttle, and you have a clear/strong MI of doing it. As you drive through Turn 1, you're very Aware of what your foot is doing with the throttle. With both the MI and A, your brain does its magic and literally makes you achieve your Goal — taking Turn 1 at full throttle.

However, if you have a clear/strong MI, but are not Aware of how close you are to your Goal (i.e., you don't know how close your foot is to flat on the throttle), you won't accomplish what you want.

If you're Aware of how you're doing, but don't have a clear/strong MI of what you want to do (i.e., flat Turn 1 - being able to imagine every detail of how that looks, feels and sounds), you won't accomplish what you want.

Got it? Make sense?

You need a clear/strong Mental Image of what you want to do. You need to be Aware of how you're doing, compared to that MI. When you have those two, your brain will do what it takes to make them match — and you'll achieve what you want.

MI + A = G. That's your mental math homework. Use it. Get a clear/strong MI, and then become super Aware of how you're doing. Your brain will take care of the rest.

Day #63: Drive the car, not the track – revisited.

“Drive the car, not the track.” What, exactly, do I mean by that?

I talk about driving the car more than the track, and here’s the origin. I see too many drivers overly-focused on driving the perfect line, and they do it to the point of driving the car under the limits of what the car is capable of. They’re so caught up with clipping the ideal apex, for example, that they either drive slow to get there, or scrub off speed getting there. By “driving the car,” I’m talking about driving it at its limits — and worrying less about the line. Yes, the line is important, but driving the car at the limit is more important.

If you drive the car at the limit and are off the line a ways, most times you’ll still be faster than if you drive the perfect line, but with the car/tires below their limit. So, the idea is to focus on driving the car at its limits first, and then fine-tune the line from there.

And here’s the important thing: when you drive the car at its limits, it tells you where the ideal line is; when you try to drive a line that someone told you was the ideal line, most times you won't be driving at the limit — and you will be slower.

Get a group of drivers together for more than fifteen minutes, and it’s almost guaranteed that a discussion about the ideal line will come up. And, at some point, one driver will say something like, “Okay, that might work in your car, but I drive a ____, and the line is completely different for it.”

I’ve had Colin Braun on my podcast (Podcast.SpeedSecrets.com) multiple times in the past where we talked about this topic (episodes #7, 66, and 94). Particularly in episode #66 (https://speedsecrets.com/speed-secrets-podcast/066-colin-braun-learning-tracks-the-secrets-to-his-speed/), we talked about how to learn a track and get up to speed quickly. He’s shared more than once that in the cars he’s driven at Road Atlanta (that’s the track he used as an example), he drives pretty much the same line in all of them. And by “all of them,” he means prototype sports cars, front- and rear-engine GT cars, front-wheel-drive production cars, Spec Miatas, Formula Mazdas, and even NASCAR Cup cars.

Colin’s point — and my point — is that if you drive the car at the limit, the line doesn’t change that much, and where it does differ comes from what the car is telling you. If you’re not driving the car at the limit, moving your line a foot in either direction is not going to make much, if any, difference in your lap time. Move your brake zone in by a car’s length, carry one MPH more through a corner, or get back to full throttle a car’s length sooner…now that’s going to make a big difference. Sure, the line can help or hurt your ability to do any of those things, but it starts with what the car tells you.

Perhaps, when drivers get together for fifteen minutes and talk, they should be focused on how to drive their cars closer to or at the limit, rather than where the ideal line is.

I realize that not every driver reading this has the same incentive to drive at the very limit. Yes, you may not make your living racing cars at the very limit, like Colin does. And while I'm talking about how the car will tell you where the line is when you drive it at the limit, you don't have to be right at the very limit to have your car tell you where the line is. But you do have to pay attention to how close the tires are to their limit. You have to sense weight transfer, balance, g-loads, and rotation. Put as much focus and attention into that as you do into driving the ideal line and you'll see faster lap times.

Day #64: See the line.

In the trailer for the movie *Ford v Ferrari*, Christian Bale (playing Ken Miles) says to his son, “Look out there. Out there is the perfect lap. You see it?” His son replies, “I think so,” to which Bale/Miles responds, “Most people can’t.”

This is Hollywood’s way of suggesting that there is some mystical way that the gifted few in the world can see an imaginary line through corners, and that’s the secret to being fast.

Okay, there’s some truth to that. The line is critically important. But is it everything? No. Still… it’s a cool concept. And it got me thinking.

See the line.

It’s difficult to write the following without it sounding like bragging, so excuse me. I’m pretty good at figuring out how to get around a track in a fairly short amount of time. Why? I seem to be able to look at a corner or series of corners and see the line. Factoring in the shape of the corner (radius, length), what comes after it, and any elevation or camber, it becomes obvious to me the line my car should be on. Okay, there's always some fine-tuning over the first few laps, and then some possible changes as the grip level changes, but the line is something that comes easy to me.

Why? Spatial awareness. Geometry. Thinking like a car. Direction memory. Mentally envisioning it.

Recently, I was speaking with an architect and the conversation got around to the differences between his first experience drawing and the now-ubiquitous use of computer applications. I shared how I had spent a few years in school taking drafting classes (paper, pencil, erasers, rulers!), and had the ability to draw a 2D image, but see a 3D version of it in my mind. I also shared that I had spent many years drawing imaginary driving lines on track maps from various circuits around the world (I would spend hours doing this instead of the homework I “should” have been doing). And when I traveled — before GPS — I would print out a map that included the airport, hotel, and track that I’d be going to.

Some of the drivers I coach today have a difficult time finding their way from the hotel to the track without Siri giving directions — after three days of driving the route! When I ask them to draw a track map, I’m surprised at how inaccurate it is.

I don’t believe that I was born with anything different from these drivers. I do believe that the endless hours of thinking about the cornering line has contributed to my learning tracks quickly. And that means that any driver can develop and improve this ability, no matter what level they’re at now:

- Look at the geometry of a corner, or series of corners. What’s the radius? Is it constant, decreasing, or increasing?

- What comes after the corner? A straightaway, another corner? If there's a straight section, how long is it? Enough to make corner exit speed the focus, or should the focus be on corner entry speed?
- Can the corner or series of corners be straightened out? While the track may bend, meander, curve back and forth, or whatever, can you straighten that section out?
- Are there elevation changes? Can you turn the car more in the areas of track where there's a compression, giving the car more grip? When cresting a hill, where do you have to position the car to have the steering as straight as possible?
- What about banking or camber in the track? Can you take advantage of it by turning more in the banking? Can you steer as straight as possible when there is no camber in the track?
- Can you shorten the distance you drive? What's the shortest way to get from one point to another? A straight line. Are there places on the track where you can drive a straighter line?

Think like your car. Where do you want to go? Picture yourself as the car, and think, "Where do I want to be on the track?" You crave grip, and want to change direction as little as possible where there isn't grip. You want to drive straight to maximize speed. You know where to compromise one corner for another, and where not to. See the line. Drive the line.

Day #65: Smooth is fast.

How many times have you heard that advice: "smooth is fast?" More than once, I'll bet. But why?

You know that tires gain grip when more load is put on them, right? When you brake, the front tires gain grip because weight is transferred forward and onto them. When you accelerate, the rear tires have more load put on them from the weight transfer, and therefore gain grip. When you go around a corner, the tires on the outside of the curve gain grip because weight is transferred onto them.

But here's the big deal: The tires that become less-loaded lose more grip than the loaded tires gain. And that means that whenever there is any weight transfer, the overall grip level for the total of the four tires on your car is reduced.

The more smoothly you drive, the less weight transfer you'll cause, and therefore the more overall grip your tires will have. That's why smooth is fast. Simple as that.

But how do you drive smoothly? Look way ahead, and make your inputs (turning the steering wheel, applying and releasing the brake and gas pedals) as smooth as possible.

Of course, one way to be smooth would be to drive very slowly. But that wouldn't be any fun, would it?!

Day #66: Smooth is fast - but not always.

There are at least three questions should come to mind when you hear the advice, “Smooth is fast.”:

1. Why?
2. How do I get smoother?
3. Does smooth always pay off?

I'm not going to get into the Why? question because I created a video that's posted on YouTube that covers that. Click here to watch it.

How do you become a smoother driver? I suggest two things:

1. *Awareness*: How smooth are you now? That's kind of a tough question to answer, isn't it? So rate yourself on a scale of 1 to 10 (with 10 being what you picture as perfection, and 1 means you suck!). After every session on track, make a note of this number. And I mean it when I say make a note — actually write it down. Keep a notebook with you, write down the date, track, session, car, weather and track conditions, and then your smoothness rating. Write other notes about the session, too, and then how and what you're going to do the next session to be even smoother. Then make notes after that session. This only takes a few minutes, but will make a huge difference.
2. *Deliberate practice*: Break your driving technique down into small pieces and refine them. Let's say you want to improve the smoothness of your brake release (which is probably a good thing). Focus on just that part of your driving for a period of time. Practice it driving on the street, and on the track. Since you spend more time driving on the street, and the consequences of not being smooth are harder to sense, make focusing on smoothness there a way of life. I'm a pretty smooth driver, and I know I wasn't born that way! But as a teen, I would go out and drive for hours in the evening, practicing being smoother with each and every technique, one at a time. Forty-something years later, I still practice driving smoothly on the street. To simply say, "I'm going to work on being smoother," doesn't really work. It's not defined clearly enough. I mean, how are you going to drive smoother? Pick just one technique (turning the steering wheel, applying the throttle, releasing the brakes, etc.), and focus on just that — and nothing else. That's how you become smoother.

So, is there a time when you can be too smooth? Yes. I wrote about this in the past, and will again, but for now think about where you might be too smooth — so smooth that you're slow. But as I’ve said before, one way of being smooth is to drive extremely slow! And that would be no fun at all. There comes a time when you need to push a little harder, and that means being less smooth. And then, when you're fast, work at smoothing out what it took to become fast.

There's always more....

Day #67: Integrate your brain to get into the zone.

Outside the performance and race driving world (yes, it exists!), a lot has been written about the type of information each side (or hemisphere) of our brain processes. It's generally known that your left hemisphere is where you process factual, logical, and detailed information; it's where your language is processed. Meanwhile, your right hemisphere is your creative, intuitive, and big picture processor.

What's rarely discussed is the time orientation each hemisphere has. Your left hemisphere is mostly concerned with the past and the future, while your right seems to be focused on the present — the here and now.

To be successful at anything, one must be able to learn from the past, look and think ahead to the future, all while performing in the moment. Many people get bogged down in the past and future, as they operate almost entirely with their left hemisphere. Others — although far fewer from my observation — allow their right brains to drive their life. When I say "allow," I really mean that. One cannot tell the right hemisphere to do anything. In fact, the best way to allow your right hemisphere to perform is to get out of its way. To do that, you need to "turn the volume down" on your left hemisphere.

In a battle of hemispheres, it seems the left wins most often. It dominates most peoples' thinking. True, we live in a world that rewards left-brain thinking. To get ahead in most disciplines, you need to be logical, factual, use language (verbal, mathematical), and pay attention to the details. Look at our education system: few courses are provided that teach people how to be more intuitive and to think big picture. Even creativity is rarely taught, and when it is, it's a side-effect of a course in the arts. Typically, creative and intuitive thinkers are seen as having a gift, but one that has not been trained.

So, the left brain tends to over-power the right brain. But I believe that we all have that intuitive and creative ability — we were all born with it. The difference with those few people who seem to have the intuitive/creative gift is they have learned how to not restrict that type of thinking. They have learned how not to inhibit their intuition and creativity — by slowing down their left brain thinking and being in the moment.

I'm not saying that left brain thinking isn't good. It is, when appropriate. And I'm not saying that your left hemisphere cannot help in intuitive and creative processes. In fact, the best intuitive/creative thinkers can use their left hemisphere to make more sense out of it. Intuition and creativity alone — without logical thinking — would be incomplete.

As with any peak performance, whether it's a physical activity or a mental task, the best comes out when you use both sides of your brain. Accessing the right hemisphere, though, often requires you to stop your left brain from getting in the way.

Some people are too smart to perform well. Huh? Peak performers in any activity know when to "be smart," relying on their left brain to think logically and factually while paying attention to the details; they also know when to "dumb down" their left brains and let

their right brain operate. To perform at one's peak, there comes a time when one must almost turn off the thinking brain and just trust the "doing brain" (right hemisphere) to perform in the moment.

Peak performers strive to get "into the zone." One definition of being in the zone is that magic moment when your left and right hemispheres work together in perfect harmony, in perfect balance. The left brain is not leading or dominating the right, or vice versa. It's when you're aware of the past, looking and thinking about the future, while performing in the moment. The best drivers have the ability to be logical when they need to be. Then they can shut off their conscious thinking and just drive.

Of course, telling you this, and you doing it are two entirely different things, right? But by giving you a target — a balance between driving with your left brain and your right brain — my hope is that you'll get into the zone more often. And most often, that means letting your right brain in on the action.

Power to the Right Brain!

Day #68: Listen for & feel your shift points.

One day, I picked up a friend in my road car (at the time), a Mini Cooper S. After a few miles he asked, “Why do you shift where you do?” He was referring to the RPM at which I changed gears.

The question seemed a bit like someone asking me, “Why do you breathe?” Shifting is just something I do, and when I shift is not something I’ve thought a lot about for many years. At least not while driving on the road. Driving on the track is another matter. But let’s go back to street driving for a minute.

When I first learned to drive, and for a few years afterward, shifting was a major focus of mine. My goal was to always shift as smoothly as possible, whether up- or downshifting, to perfect the timing. What do I mean by "perfect the timing?"

In the perfect world, we’d all have the horsepower and torque curves for our engines, along with the gear ratios, and from that, we could calculate the exact ideal RPM to shift at. I think I’ve seen dyno sheets for two of my engines over the past fifty years, so unless your relationship with your engine builder is better than mine, I suspect you haven’t many more than I have. If any.

If you did have a dyno sheet, you could figure out that you’d want to upshift so that you kept your engine in the RPM range between maximum torque and horsepower. The idea is to keep your engine in the rev range that produces maximum acceleration, so yes, you’d also want to know how many RPMs your engine drops when you upshift from one gear to another (and that varies depending on gear ratios). With a little bit of math and probably a spreadsheet, it’s not difficult to calculate the exact RPM that you should upshift at in each gear.

But since most drivers don’t have access to engine dyno sheets, what do we do? Pay attention to the longitudinal g-loads, or acceleration forces.

If you drive your road car on the track, this is actually easier than if you drive a purpose-built race car, as you can figure most of this out on a long stretch of highway (one on which it’s is safe to accelerate fully). Ideally, you’d do multiple full-acceleration runs, using various RPM points to upshift at, and with some practice you’d sense when you feel maximum acceleration. Often, shifting at redline is not the ideal shift point for maximum acceleration, and that’s why I’d suggest you vary shift points by 500 RPM at first, and then when you begin to find what feels best, you narrow it down by varying them by 250 RPM, and then 100 RPM. In my experience, most drivers usually cannot sense the difference in acceleration in RPMs of less than one hundred.

Again, keep in mind that your shift points can be different in each gear. For example, shifting from first to second gear at 5800 RPM might be ideal, then 5900 for second to third, 5900 for third to fourth, and 6000 RPM for fourth to fifth. It could even be a bigger spread between shift points, but if you really pay attention, you’ll notice the difference.

Of course, you could take this process one step further towards objectives measures, and use a stopwatch or data system. Using a stopwatch can be difficult if you're doing it on your own (clicking the start button, watching RPMs and the speedo, and clicking it again at the exact right time). If you have a data system in your car, you can do multiple acceleration runs, then compare the longitudinal g and speed channels to see which shift points resulted in maximum acceleration.

Far better than a deserted stretch of road is a drag or air strip. Hey, if you plan things out properly, you could go to your local drag strip on their public night, pay your twenty or fifty bucks, and make as many runs as you can, using your various pre-planned shift points. Who knows, you might even win a few drag races!

If you're driving a car that isn't road-legal, then ideally you'd use a data system. In this case, you'd spend a few track sessions trying different shift points (always keeping track by having a pre-run plan and making notes), and then taking the time after each to compare RPMs (hopefully your data system is logging them) to speed and longitudinal g loads. With a little planning, experimentation, time, and thought, you'll quickly identify what RPMs you should shift at for each gear. Then, I suggest you spend a little more time focusing on what the engine and wind noise sounds like at each shift point, and file that away in your mind for when you can't look at the tach.

Ultimately, though, you want to get so you that you can sense the ideal shift points. If you really pay attention, you'll be able notice subtle differences in acceleration forces, relating them to the sound of your engine and wind noise, knowing that you can always shift at or close to the ideal RPMs without even looking at the tach.

I answered my friend's question by saying, "When it feels and sounds just right." Oh, and since I wasn't using full throttle on the street, the ideal shift point was different (and lower than if I was trying for a max acceleration run). I've developed a feel for different shift points depending on how much throttle I'm using, and even the elevation changes in the road. It all comes down to listening for and feeling for what your engine is asking for.

Day #69: Measure your improvement.

“What gets measured, gets managed.” And improved. When it comes to measuring improvement for us track drivers, it’s easy, right? Lap times! What else is there?

The problem with lap times is that things change. Track conditions, car performance, and weather all impact lap times, so you could be a second per lap faster than ever before (or the other way around), but not know it because the track had lost grip. So, lap times are just one metric and not always the most reliable.

There’s also how you achieve those lap times, and that may be just as important as how fast you went. If you don’t know what led to an improvement in lap time, it’s unlikely you’ll know how to continue to improve. And that would suck, because who doesn’t want to continue to improve?

Usually, the first step in learning what led to the improvement is knowing where the improvement occurred. For example, did your lap times improve because you compressed your brake zone, or moved the entire zone further into the corner? Did you release the brakes slower or faster? Did you trail brake more or less? Was your corner entry speed faster or slower? Did you get back to full throttle sooner or later? Did you change your line, and if so, in what way? This questioning process can go on for a long time, and it will also lead to some important insights.

After asking yourself these types of questions, that’s the time for data and/or video — and not before. See, asking yourself these kinds of questions gives you more awareness — more than you’ll get from data/video alone — and awareness is the first step in making any improvement. Plus, data and video can distort your gut feel, and that feel is just as important as the raw facts coming out of the data/video. It’s one thing to have the data tell you that you lifted the throttle for a corner, and it’s a completely different thing to dig into what you were feeling as you did that (and whether you felt that you had lifted).

I won’t go into the details of how to do it here, but you can use segment/sector times from your data system as a tool to determine where you went faster or slower. It’s the most efficient way of learning the “where.”

Another super-effective tool to help determine whether you’ve improved or not is self-ratings. Asking yourself to rate your overall performance, your smoothness, and your level of calmness on a scale of 1 to 10 brings awareness to each of those factors. And when you become more aware, you can improve. You can even rate the amount of tire noise on a scale of 1 to 10. Not enough noise — not close enough to a 10 — and you’re not driving fast enough; too much noise — an 11 on your 1-10 scale — and you’re over-driving the car (sometimes called “Spinal Tapping” the tires, which some of you will understand…).

Lastly, what if your best lap time ever was a 1:40.0, but you were feeling like you’re on

the ragged edge to do that time? Then, on another day, you turn a 1:40.0, but it's almost effortless. Which would you prefer? Being able to easily turn your best lap times, consistently, without stressing out over it would be a good thing, right? But other than an objective self-rating, what else can you do?

One thing I've done is used a heart rate monitor. Generally, the more stressed (and the harder you're trying), the higher your heart rate. Keeping track of peak and average heart rate over the course of a session, and comparing that to your best and average lap time, is a very useful way of measuring performance.

Again, if you could consistently turn your fastest lap times, but with a lower heart rate, that's a good measure of improvement. It also provides another way of targeting your performance improvement plan. Instead of focusing on driving faster lap times for a session, simply work on turning the same lap time, but with a more relaxed approach — a lower heart rate will show whether you did it or not. And hey, nine times out of ten, you'll also drive those laps more smoothly, putting less wear and tear on your car. Then, it often happens that lap times begin to fall again, without trying. Your heart rate might go up for a while, and then you work at turning your new best times with a lower heart rate, keeping the cycle going.

When you measure your improvement, use more than the usual lap time.

Day #70: If it seems you're not improving, you are about to.

If there's one thing that is common to all drivers, it's this: They improve with time.

Okay, I know you might be thinking that at some point, at some age, this isn't true anymore. I'm not going to argue that point here, because when, where, how, and why that might happen is beyond the scope of this piece.

As humans, we do not follow what most people refer to as the learning curve. Instead, we learn in steps. We usually start off reasonably strong, learning on a pretty steep angle. Then, we taper off, and sometimes seem to be on a plateau for a while. This could be a short period of time, or what feels like forever. But while we're on this plateau, our brains are making all that we know part of a mental program. It's solidifying what we've learned.

I have never seen a person stay on a plateau forever, though. After each plateau, there is another burst of learning. It might be a long, steep burst, or a shallow and short one. But it's heading in the right direction! That is, if you don't get frustrated. If you stay relaxed, expect that you will have another burst of learning, and stick to the basics of what you're working on improving, you will learn.

After every plateau of learning, there's a new burst of learning. Relax and it will happen.

Day #71: Focus on doing things right and your speed will follow.

“By concentrating on precision, one arrives at technique, but by concentrating on technique one does not arrive at precision.” – Bruno Walter

Okay, Bruno Walter never drove a race car (that I'm aware of, anyway), but I love his quote above.

What does this have to do with driving? Think about what he said: "By concentrating on precision, one arrives at technique, but by concentrating on technique one does not arrive at precision." If you follow Walter's thinking, could it be that we spend too much time focused on the specifics of the various techniques we use when driving around a track, and not enough time focused on what we're really trying to achieve: spending as little time as possible to go from the start line to the finish line, and doing that in a very precise way?

But is it also possible to be overly precise? Certainly, I think so. I believe that some drivers focus too much on hitting the perfect line. In doing so — in trying to get the car to hit the perfect apex, for example — the driver actually has to slow the car, even just a fraction of a MPH (and that makes a difference).

Sometimes you have to just let the car find its own path (within reason!), allowing it to carry speed and momentum, and not be so concerned with the precise line.

Am I contradicting what Bruno Walter said? Not really. See, I think if we focus on being precise with sensing what the car is telling us, the techniques will work out just right. The techniques we use will be just right to ensure that we achieve our end goal, but it all starts with being precise with what the car is telling us. And not, necessarily, with hitting the precise apex point, for example.

In fact, I'd argue that focusing on hitting the precise apex is the technique. What we should focus on is the precision of sensing what the car is telling us.

Day #72: Mentally program being lucky and you will be.

I once coached a driver who seemed to be unlucky. If there was going to be a crash on the track, he'd find a way to be in it. If another driver was going to crash, he somehow managed to include "Bob." The crew actually looked to see if there were any large magnets on the car that attracted other cars!

Eventually, "Bob" began to believe he was unlucky, and the problem got worse. The more he thought he was unlucky, the unluckier he got. It became a self-fulfilling prophecy. The more he told himself he was luckless, the more he got collected in others' incidents.

I'm not one of those people who suggest that you need to look in the mirror and recite positive affirmations. But I do know that our minds have this amazing ability to make sure we achieve what we put in it. If we internalize and focus on something happening, our brains work to make that occur.
If we think we're lucky, our brains will do whatever it can to ensure "lucky" things happen to us.

Over time, I had "Bob" list everything that went his way, no matter how small or seemingly insignificant. At first this was difficult for him - he had a hard time seeing anything as being lucky, or positive in any way. But eventually he began noticing all the small things that did go his way, and ultimately he started to believe he was lucky. Guess what happened? "Bob" rarely ever got caught up in others' incidents any more. And when unfortunate incidents did happen to him — as they do in the real world — he was able to see them for what they were (typically, a logical series of events leading to a negative situation). Most importantly, he saw the reason behind what happened — and it wasn't that he was unlucky.

Ultimately, "Bob" had changed his beliefs. The funny thing is that he began to see himself as a lucky driver, and that contributed to some very good results.

Day #73: Ask more of your tires.

For drivers wanting to drive at the limit, a common “leaving speed on the table” error I see many make is simply not asking enough of the tires.

What do I mean by “asking enough of the tires”? Mostly, turn the steering wheel more. But it could be braking harder, and applying the throttle earlier and more aggressively.

Who hasn’t been told that smooth is fast? It’s doubtful that anyone reading this hasn’t heard that at least a hundred times (because I’ve probably said it at least ninety times). And that can sometimes lead to a driver not using the tires, not asking enough of them.

Every now and then a driver will ask me to look at their in-car video, and to give them feedback on how well they’re driving. More often than not, I’ll watch the video and report back, “Your line looks great — no suggestions there. But it looks to me like you’re just not pushing the car hard enough — you’re not asking enough of the tires.”

What tips me off and leads me to this response? Usually, it’s steering movement that is too smooth. Yup, I’m the guy who wrote in a book that “smooth is fast.” But if you want to be fast, you can be too smooth. If I notice that a driver turns into a corner, and the steering wheel barely moves from the initial rotation until it begins to straighten (and even then, it’s a perfectly smooth unwinding of the wheel), I know that the tires have more to give.

I know that not all drivers want to drive at the very limit of their tires and car. Some want to give themselves a margin for error, a cushion. That might be a conscious decision, due to a lack of experience, some level of self-preservation, and/or a misinterpretation of what driving smoothly and at the limit means.

While driving through a corner, have you ever deliberately turned the steering wheel more? You know, just given the wheel a solid but smooth crank of rotation, and then brought it back to where you had it? If your car responds immediately and changes direction when you do this, you haven’t been asking enough of the tires, and you could drive faster. If, when you do this, there is little to no change in direction of the car, then you can be fairly certain that you’re close to or at the limit.

Now I’m not suggesting that in every corner that you just give the steering wheel a snap of more input. I am saying that you should test the limits every now and then. Pick the best places to do this, and I’d suggest starting this process in longer corners where you have more time or in corners with more runoff room, in case you need it to correct your direction.

In most cases, if you test your tire grip level in one corner, you’ll learn what they have to give, and be able to use that in every other corner, too. Obviously, that doesn’t necessarily apply if you test it in a corner that has a large concrete patch in it, as that surface will likely have a different level of grip from the rest of the track.

Even if you want to give yourself a cushion, and not drive at the very limit, this is a worthwhile exercise. By testing the grip level in a safe place, by asking more of the tires, you'll have a better understanding of where the limit is, helping you stay below it. If you don't know where the limit is, how can you ever consistently know you're driving under it?

I'm also not suggesting that you just wiggle the steering wheel back and forth. I see drivers do that, saying that they're "feeling the tires." I don't buy that. It's simply wiggling the car around, and actually reducing the grip level. If you're going to test the tires' limit, you need to turn the wheel more than joggling it a bit.

Give it a try. Smoothly, deliberately, and using discretion, ask more of your tires' grip level.

Day #74: Improve your pattern-matching skills.

Some drivers seem to be better at making good on-track/behind-the-wheel decisions, while others are not so great. Some drivers are more consistent than others. Some drivers are just better drivers than others. The great thing about performance and race driving is there is always something more to learn, something to continually improve.

If you knew what made some drivers better than others, you could adopt what they do to improve your driving. In a conversation with fellow coach, Mike Zimicki, he wondered aloud: “If Scott Dixon is the best current Indy car driver, why don’t more drivers copy what he does, both on and off the track?”

So, what is it? What makes the best so good? One of the things that separate the best from the rest is the ability to notice patterns.

When you approach an intersection on the street and you think “I just knew that red car was going to pull in front of me,” that’s pattern-matching. It’s what most new teen drivers are not very good at, because they haven't yet seen as many scenarios. But two drivers who have been driving for twenty years are not necessarily equally good at pattern-matching, either. Some drivers are either more naturally-gifted with this all-important skill, or they’ve developed the ability through practice.

Approaching a corner on the track, many drivers look for singular reference points to determine where to brake, turn-in, apex, and exit the turn. But the best drivers use the pattern made up of multiple references to give them a more detailed cue as to when to begin braking, turn in, ease off the brakes, and arc in toward the apex. Think of it as watching a low-resolution (single reference) versus high-resolution (the pattern made up of multiple references) video.

This is especially true when dealing with other cars in a passing situation. How many variations are there in the positioning of just two competitors going into one corner? An infinite number of scenarios, right? So how do the best consistently make the right decisions? Pattern matching. While no two patterns in a racecraft scenario will ever be the same, there are similarities (as you’ve proven when you’ve identified a car that was going to pull out in front of you on the street).

The good news is that your pattern-matching abilities can be developed. How? First, be aware of the pattern-matching that you’re currently doing. I suspect you’re doing it more than you think you are. So just become aware of how many patterns in the mental snapshots you have as you drive around a track. The more you’re aware of it, the more it’ll sink in and become part of your deep mental programming.

Again, we think that we use references for where we begin to brake, turn into a corner, apex, and track out of a corner. But it’s not that simple – at least, it shouldn’t be. Imagine stopping on a track just before the brake zone, and look ahead and into the corner. What do you see, besides your usual references? I bet when you really look at

it, you'll see various angles crossing, merging, and aligning. Again, it’s the relationship of one reference to another, then to another, and another, and so on. It’s that image made up of many references that lead to accurate, consistent, fast laps.

The next time you’re watching your in-car video, stop the video just prior to the Begin-of-Braking (BoB) point, and look for the pattern. Take a snapshot in your mind of what it looks like.

- If you were to extend the edges of the track past the turn-in point, where would they intersect with the horizon?
- How far to the right is that tree on the horizon from the turn-in point compared to the edge of the track on the right?
- If you extended a line perpendicular to the BoB reference, what would it line up with on the other side of the track?
- Look for the angles from the front edges of your car to the horizon — where do they cross over the apex curb?
- Do you notice subtle changes in direction of the fog lines on the edge of the track?

Forward the video to just before the turn-in point, and do the same. Then just before the apex, at the apex, as you begin to unwind the steering wheel, and just before the track-out curb.

I know this is going to take time, but do this for multiple laps, until you notice the patterns. The pattern might be when you’ve gotten the car lined up just right, or you might notice that the pattern changes from lap to lap, telling you where you’re consistent or not.

The interesting thing is the more you do this, the better you get at pattern-matching — at speed. You’re improving your ability to pattern-match while driving at speed.

As you take mental snapshots from the various points around the track to improve your pattern-matching, also pull back and take a big, overview snapshot. It’s like using your peripheral vision to give yourself a bigger view. Seeing the details of the patterns are critical, but so is seeing the big picture. Learn to trust your peripheral vision to pick up on the subtle clues off to the sides of the track, as well as off in the distance.

Day #75: The steering wheel is a suggestion box.

When I was involved with the founding and operation of DirtFish Rally School, some of the instructors with lots of rally experience shared the following phrase with me: “The steering wheel is merely a suggestion.”

That struck a chord with me. The idea is to simply use the initial turn of the steering wheel to start the change in direction, but to then use the weight transfer — managed with the feet working the brake and gas pedals — to control when, where, and how much the car turns.

This advice connected with me because I’ve often said (kind of in a joking way, but not completely joking) that my ultimate goal in life is to drive a single lap of a road racing circuit without ever turning the steering wheel and only changing the direction of the car with my feet. In other words, managing the weight transfer of the car in a way that will lead me to steering the car with my feet.

Hey, I wrote an entire chapter in *Ultimate Speed Secrets* about this topic! That's how strong this fantasy/goal is of mine.

Somehow, I suspect that you’ve experienced the ability to change the direction of your car by modulating the brake and gas pedals. For example:

- Entering a corner, you trail brake so much that the rear of the car begins to slide — managed oversteer, or rotation — because most of the weight of the car is on the front tires and less on the rears. Just after turning into the corner, you realize that you’ve straightened the steering wheel — it’s pointing straight ahead — and yet the car is turning on the path towards the apex.
- In the middle of a long, sweeping, fast corner, you squeeze on the throttle, only to find that the car begins to understeer and you’re now on a larger arc than you want — you’re not heading where you want. So, you ease up just slightly on the throttle — you breathe it — and the nose of the car tucks back in on the line you want. Now, the car is heading too much in that direction, so you squeeze back on the throttle and the car is now more neutral — following the line you want. All this time that you’ve been changing the direction of the car by smoothly modulating the throttle, you haven’t turned the steering wheel one single bit.
- Exiting a corner on power, you realize that unless you turn the steering wheel more, you’re going to run out of track. But you don’t want to turn the steering more, since you’re trying to unwind it to maximize your exit speed off the corner. Besides, when on power like this, it feels as though more steering won’t make a difference — the car would just understeer. So, you “pause” the squeezing back to full throttle. It’s not a lift of the throttle — you just don’t apply more. There’s a plateau in the way you add throttle. And a fraction of a second later, without changing the steering wheel angle, the car turns just enough to avoid driving off the outside of the track.

You'll notice a common theme among all of the above examples; the steering wheel never moves. Okay, there's a tiny bit of movement, but it's just nudging the car to change direction. That fits my fantasy of driving a lap without ever turning the steering wheel! Yes, I might need to use it to "suggest" that the car change direction, but from then on, it'll be up to my feet to steer.

Beyond the fun of this challenge, there's a benefit, too. The less you turn the steering wheel, the faster you'll be (another of my original Speed Secrets from my first book).

Who's with me? Do you want to join in my "Drive a road course without turning the steering wheel" challenge?

P.S. — This is not all fantasy-talk here. Learning to manage the weight transfer to the point where you could almost drive a track without turning the steering wheel is hugely valuable. If you actually did it to the extent I'm talking about here, you might be sliding the car too much, and ultimately you won't be as fast as you could be. But in the process of learning how to do this, you're slightly over-driving the car, and learning the subtlety of using the pedals to steer the car. And that would, absolutely, make you an even faster driver.

Day #76: Different lines for different cars? Not necessarily.

"Yeah, that line might work for you, but I drive a Porsche and that means I need a completely different line." "But I drive a front-wheel-drive car, so that means I need a later apex than you."

Those are just a couple of the comments I've heard among drivers, comparing the line they drive with another driver. The message is: Every car requires a different line.

I disagree.

I talked with Colin Braun on one of my Speed Secrets Podcasts. After I asked him about the difference between the front-wheel-drive Kia he raced in World Challenge and some others cars, he said something very interesting: "I've been fortunate enough at a place like Road Atlanta to drive a NASCAR Sprint Cup car, I've driven Prototype Challenge cars, Daytona Prototypes, GTLM cars, GTD cars, Porsche Cup cars, Ferrari Challenge cars... you name it [he didn't mention his Chump car Miata, but he could have included it, too]. And I don't think I've ever felt like, oh the line in this specific car is different than that specific car. Again, it's more about doing the inputs and technique differently for each car, but the line just takes care of itself."

I agree.

Too many drivers spend too much time worrying about driving the line, and not enough time focused on driving the car. As Colin says, the line will take care of itself if you focus on driving the car, and on the techniques it takes to do that.

Having said that, there is one thing that changes a lot, depending on the car you're driving: the speed at which you're traveling. And that can change the line you drive — a little.

If you're driving a car with relatively low top speed, but with lots of aerodynamic grip, the way you approach a corner (and yes, maybe even the line) might be different from a car with a high top speed and less overall cornering grip. Turn 1 and the approach to Turn 2 at Sonoma Raceway is a perfect example. But unless we're comparing a small-bore open-wheel car to a production/street car, this is a non-issue.

The main take-away and key message to keep in mind here is this: focus on driving your car, on your inputs, on your technique, and the line will be obvious. And, if you're driving your car close to, or at, the limit, you'll find the line doesn't change much — if at all — from car to car. If you focus on driving a different line than someone else because you're in a different car, you'll likely not be driving at or near the limit - and you'll be slow.

Day #77: Although your skill level may have physical limits, your mind's potential is limitless.

Okay, if your DNA has determined that you're 7 feet tall, then driving a race car may not be in your future. That's a "physical limitation" on what you can do in racing.

But your mind does not have limits. In the book, *The Brain That Changes Itself*, Norman Doidge, M.D., shares research and true stories about people who have what many would consider limits — learning challenges (ADHD, Aspergers Syndrome, etc.), strokes, and even visual restrictions — who have adapted their brains. If these people can do what they've done, performance or race drivers can do so much more with their minds than they're currently doing.

Most would agree that performance/race driving is more mental than physical. In fact, one's body (physical) does not do anything unless the brain tells it to do so — that makes this sport 100% mental.

My point is that you can always improve your mental abilities — and don't buy into the age argument. Sure, it takes specific strategies, but take time to work on them yourself: prepare by using mental imagery, consider computer mental training applications, practice getting into your perfect performance state of mind, and so on.

As with your lap times... there's always more.

Day #78: Manage the timing of your downshifts.

I've often encouraged drivers to get all of their downshifts completed prior to turning into corners, and that's still accurate — usually.

Let's take a step back here for a minute. The reason for downshifting is to simply get the car in the best gear for accelerating out of the upcoming corner, and not for slowing the car down. Usually. The thinking behind this advice is that your brakes are for slowing the car — and operate on all four wheels — and the engine is for accelerating the car — and usually operates on only two wheels. So, use the brakes for slowing down, and the engine for accelerating.

In most cases, your goal should be to time your downshifts throughout the brake zone, from BoB (Begin-of-Braking) to the turn-in point. Yes, you want to have your final downshift completed by the time you turn into the corner. You don't want to be finishing up a downshift, and letting out the clutch, as you're turning in. If so, you can upset the balance of the car so badly that it causes a spin. And even if it doesn't cause a spin, it can unbalance the car enough to result in less overall tire grip, meaning you have to drive slower. That's why I strongly say complete your final downshift in time so that you have your foot completely off the clutch pedal before you turn the steering wheel to head into the corner. If you follow that guideline, you'll never really go wrong.

Of course, there are exceptions to guidelines like this, so let's take a look a few.

First, if you're a driving an automatic transmission car, it's almost a non-issue because the transmission is not going to make a rough downshift in the middle of a corner. Okay, some automatics are pretty clunky with their downshifts, and are bad where they do them. But even the worst of those are rarely going to cause enough of an unbalancing moment that they'll cause you to spin. But could it cause an unbalancing moment? Maybe....

But if you're driving a paddle-shift semi-automatic transmission car, one that doesn't require a clutch to downshift, then you have a bit more leeway. If you downshift just after turning into the corner, it's unlikely going to generate a big upset of the car, and won't cause a spin. The benefit of doing it this late is that the downshift, and resulting engine braking, usually won't cause you to over-slow. As you know, sometimes that final downshift to the lowest gear can lead to you slowing more than you actually want, so by delaying it as long as possible, it has less of this tendency.

You have another option with a paddle-shift car: you downshift a little earlier, using the engine braking effect to help slow the car just before turning into the corner. When I refer to this technique, I'm assuming that your car has an auto-blip feature, meaning that its computer will rev the engine enough to ensure that you get a smooth downshift, while also making sure the engine is not over-revved.

Again, this is an exception to the guideline of not using the engine to slow the car. Remember that in a rear-wheel-drive car, that engine braking effect is only being applied to the rear wheels. If you're already braking hard enough to have them right at their very limit, and then you add the extra retardation from the engine braking, they will try to lock up. If your car has ABS, the rear wheels won't lock up, but they can still upset the balance the car, resulting in less overall grip from the tries. So, I'd recommend using this technique in rare situations (same with the idea of making your final downshift after turning into the corner — it's an exception to the rule).

One final note regarding the timing of downshifts. The first one you make as you approach a corner should usually be as late in the brake zone as possible. If, for example, you're approaching a corner with your engine at or near maximum RPMs, and you suddenly downshift immediately as you begin braking, you'll over-rev and damage your engine. That's why I give the advice of "Brake, brake… wait... downshift…". And even then there are exceptions, but so rare that it's only through experience that you'll learn when they're an exception. Oh, and with some paddle-shift cars, if you downshift too early, the computer will deny it because it would over-rev the engine if the downshift actually happened.

The timing of when you downshift can make a big difference to how controlled and fast you're able to drive. Think about it.

Day #79: Communication is a two-way street. Ask questions and listen.

As an instructor or coach, this Speed Secret is pretty straightforward. At least its meaning is. As a driver, what does this mean to you?

If you have an engineer or crew chief with whom you can debrief about your car's handling characteristics after driving on the track, that's pretty self-explanatory. It should be a 2-way conversation, and not just you telling your engineer about all the problems you're having with the car's handling. Let your engineer ask questions — encourage them. *You* ask questions. The more questions you both ask, the more you'll be digging to the core of the problem, and therefore finding the real solutions.

If your engineer is someone you see every morning when you look in the mirror, then you're going to need to have that two-way conversation with yourself (sure, you might not want to do that, loudly, while pushing a shopping cart at the grocery store all by yourself!!!). But, once again, ask questions — of yourself. Listen for your own answers, and then ask more. The more questions you ask, the more you'll be digging to the core of the problem.

Could this two-way conversation be between you and your car? Yes! In fact, that's perhaps the most important conversation you'll ever have. You know that your car is talking to you, through its sounds (engine, brakes, tires, wind passing the car), feel (vibrations, chassis pitch and roll, g-forces, feedback through the steering wheel and pedals), and what you can see (the rotation of the car, change of perspective when the chassis leans or pitches forward/rearward). If you ask yourself, "What do I feel? What do I hear? What do I see?", your car will answer… if you're paying attention — if you're "listening"—and communicating.

Day #80: Balance your focus on the result with process.

For most drivers, lap times are the most obvious, and certainly the easiest metric of success — or lack of it. A faster lap time means a better result, therefore you drove better, right? Lap times are a clear, objective metric. Nothing subjective or judgmental about them. They don't lie. They are what they are.

But there are many factors that impact lap times, to the point where they become more subjective than we think. Track conditions are a huge variable, as is the condition of your tires, the overall performance of your car (how well is your car set up and handling? how's your engine performing?). I've experienced lap times changing by two or more seconds from the morning to the afternoon due to ambient temperature changes. While some of these variables are controllable — handling setup, for example — others are not. Well, unless you have the ability to control the weather! And even your car's setup isn't something you can do a lot about while you're driving.

So, lap times are results that are difficult to control. But what leads to these results? The process, your technique, your performance. The process is the only thing you can control, and yet, even that is difficult.

With focused practice, your technique, your skills, should improve. But they don't improve in a linear fashion. As you know, you have days when you make big jumps in your technique improvement, and others when you don't.

Even if your techniques improve in a perfectly linear fashion, does that mean the results — the improvement in lap times — would improve in a linear, consistent fashion? Not unless you can control those uncontrollable factors (which you can't).

Hey, wouldn't it be great if you could control everything, and your improvement was a yes or no thing? You would simply look at the results – lap times – and know how much you improved. That would make things so much easier! But it's unrealistic. And maybe that would take away some of the fun, some of what makes our sport so challenging.

I think it's the uncontrollable, the variables, the non-linear fashion of improvement that makes our sport worth doing. It's what brings us back for more and more. And I think it's this attitude that leads to improvement. If you try to force results through sheer willpower… well, that works up to a certain point, but beyond that it often leads to frustration, anger, disappointment, and, all too often, crashes.

Don't get me wrong, willpower and desire are super-important to success (if you define success as improving one's lap times, for example). When I'm asked about why a certain race driver is not performing as expected, I often ask, "How hungry are they?" By that, I'm questioning their willpower, because without a strong desire and willingness to do what's needed, they won't improve or perform up to their potential.

But again, willpower, desire, hunger, whatever you call it… well, that alone will not overcome focusing on the wrong things. It's like the saying, "Doing the wrong thing with more effort will not lead to the right result."

Driving requires an amazing amount of discipline. There are places on a track where you need to slow, and be patient, and that can feel counter-productive to trying to turn a faster lap time. But the best drivers know when and where to be patient in order to go faster.

The best drivers also have the discipline to focus on what will lead to the desired result, and that's the process. The process – the techniques and skills - is what leads to the desired result. The result is the goal, and it's critical to have that in mind as the target, because that's what motivates you to focus on the process.

After decades of driving and coaching, I've noticed something that is more than a simple trend: Better process most often leads to better results.

Okay, process, results, techniques, skills, lap times, desire, willpower, controllable and uncontrollable factors… there's a lot to think about, isn't there? That was my goal: to get you thinking more deeply about your approach to your improvement.

Day #81: Build up your toolbox of driving techniques.

Does every corner reward the same technique? Should you trail brake, for example, into every corner? Should you brake as late and as hard as possible on the approach to every corner? Does every corner reward focusing on exit speed?

When I look in detail at the driving techniques used by many (maybe even most) drivers — other than high level pros — I see a common trait. Drivers use the same technique in every corner! They drive slow corners like they do fast corners; they drive fast corners like they do slow corners. They aim to threshold brake every time they slow their car. They turn the steering wheel the same way in each corner.

Colin Braun made a comment to me shortly after his first race in the Global RallyCross series about the idea of a driver's toolbox. He related that on road and oval tracks, he has a full toolbox of techniques he can use to get around a corner quickly, but driving a rallycross car on dirt he felt he only had a hammer and screwdriver (that changed very quickly as he developed more techniques that made him one of the quickest in that form of motorsport, too).

Let's look at just a few different techniques you should have in your driver's toolbox:

- In fast corners, make your speed adjustment early and be on throttle as close to turn-in as possible. In slow corners leave your braking late and trail brake into the turn to rotate the car. This means you need an early, light, and long brake application for fast corners that require less change in direction of the car, and a hard initial application then trail-off of the brakes in tight corners.
- Some corners reward a progressive turn of the steering wheel, while others a linear, or even digressive turn. In other words, start to turn the steering wheel at one speed and then either speed that rate up, keep it constant, or slow it down.
- Sometimes, you want to use a point-and-shoot style for a corner, and other times you want to carry momentum by not slowing as much.
- Approaching a fast corner that requires a small adjustment in speed, you can use a light brake or simply breathe the throttle.

It's usually at this point when a driver asks me for the silver bullet, wanting a works-all-the-time rule for when to use one tool, and not another. Sorry, but it isn't that simple. Believe me, after all the thought, experimentation, data and video review, talking to drivers and coaches, working with engineers... and the codifying of that knowledge that I've done over the past four decades, if it was possible to write some rules for this, I would have.

Well, actually, there is a rule, and here it is: You need to try different techniques and approaches to figure out what works for the specific track, corner, car, and conditions you're driving.

I'm all for working on being consistent with your driving, but not to the point where you're simply doing the same thing in every corner because "it's just the way I drive." Yes, you need to be able to drive consistently in one way before you start trying different techniques. But again, you also shouldn't just keep doing things the way you've always done them. You need to try different:

- Amounts of pressure on the initial application of the brakes.
- Timings and rates of release of the brakes.
- Timing of when you brake in relation to the turn-in point.
- Rates of speed of turning the steering wheel.
- Amounts of time between when you finish braking and when you begin applying the throttle.
- Rates at which you unwind and straighten the steering wheel exiting corners.

One-trick ponies are fast in a few corners, but not in all corners. Add more tools to your toolbox.

Day #82: Practice in your mind before practicing on the track.

How many laps will you do on track this season? How many laps will you do mentally, in your head, this season? Or, perhaps I should ask, how many should you do?

I don't need to remind you of the cost of every single lap on track. But what about the cost of every lap you do in your mind? It's free! And free is not something we see a lot of in our sport!

Here's a challenge for you: Get out a stopwatch, and time your mental imagery of a lap of your favorite track. Close your eyes, relax, and then as you mentally approach the start-finish line, start the timer. Mentally drive a lap, and then click the timer again as you pass the start-finish line. Check your lap time.

How did you do? With practice, you should be able to get within a second of your real lap time. In fact, this is a good form of feedback about how well you do mental imagery. If you're many seconds off of your actual lap time, then you definitely need to practice in your mind more. And even if you're within half a second, keep doing it. The more you practice in your mind, the better you'll be on the track.

Day #83: Get to know a track's personality.

A few general observations about driving race tracks.

- *Some corners need to be attacked, while others require a gentle approach; some need you to flow speed into them, while others reward being patient and slower on the entry*. Not all corners are created equal, so you have to adapt your driving to their different types. I see too many drivers drive every single corner the same way. For example, they might brake as late and hard as possible, trail brake into the corner with a relatively quick and crisp turn of the steering wheel, aim for a late apex, then immediately begin feeding in some throttle the nano-second they're off the brake pedal, driving the car out the edge of the exit curb. That works for many corners, but not all. Some like a light brake on the approach; some reward a little hesitation between coming off the brakes and beginning to apply throttle; in some you don't trail brake at all. Adapt your driving technique to the corner.
- *The shortest distance is often the fastest way to get somewhere, including the end of a lap*. The classic advice is, "Use all the track." That typically means driving from the outside edge of the track on the entry, the inside at the apex, and back out to the outside as you track out. Good advice — except for when it's not. The problem with this is it means you're driving the longest distance (even though it's the largest radius, usually a good thing). But sometimes there is just too much track available, so short-cutting the distance is an advantage. Experiment with that.
- *Grip is usually more important than the theoretical best line*. Rarely is this not true (but there are always exceptions to the rule). This is another example of where it might not be worth driving the ideal outside-inside-outside line, but instead go where the most traction is. For example, some corners have more banking or camber in them down closer to the inside. Go there, and use that banking. Stay in it for as long as you can. The same applies to surface changes. Tracks tell you where to drive, but you have to pay attention. Drive the grip.
- *Tracks have a rhythm, so learn to feel it, and move with it*. But understand: the best tracks mix it up. You can go from the smooth flow of classical music to the start-and-stop of punk rock — all in one lap. Turns 1 through 6 at Road Atlanta have a very different rhythm than 7 does (hint: that's why Turn 7 is challenging, despite it being a fairly easy near-90-degree right-hander).
- *Every track has thousands of references*. These are not just points, but areas, things, cracks, bumps, and even sounds that you should use to trigger your actions. The more of these that you have filed away in the driving database in your mind, the more consistently fast you'll be. Spend time on track searching for them, allow your brain to soak them up, and then make note of them (ideally written on a track map).
- *Lap times will vary*. Period. Accept it. You can have the exact same car, driven exactly the same way, and from morning to afternoon the lap times can change by two seconds. Same for the time of year. Track and ambient temperatures

make a huge difference to grip levels, and therefore, the line (not to mention your car's engine power). I've seen too many drivers bang their heads against the wall, chasing a lap time they did in the morning, or even two months ago. I've also seen many of these drivers end up banging their cars against a wall in a failed attempt to get back to that lap time. Drive your car to its (and the track's) limits, but no more.

When people ask me what my favorite race track is, I usually respond, "The next one." I love them all! And it's the different personalities that make it so. It's why the Milwaukee Mile oval is as much fun as the fast and flowing Mosport (Okay, Canadian Tire Motorsport Park); the point-and-shoot of Long Beach's street course is as challenging as the variety in Australia's Phillip Island; the bumps (but no elevation change) of Sebring are as cool as Eau Rouge at Spa. Oh, and throw in the parklike atmosphere of places like Mid-Ohio, Road America, and Monza. They're all different, but equally challenging and enjoyable.

Be honest. Have you ever met a race track you didn't like?!

Day #84: Eliminate time pressure to trigger flow.

When I ask drivers to describe what being in the zone, or flow, is like, one of the main factors they mention is how time changes. It slows down for most drivers. It's as if one has all the time in the world to do what needs to be done. On the flip side, when a driver is not performing well, they feel rushed, as if there is not enough time available.

Is this change in how we perceive time — what flow researchers refer to as "dilation of time" — a result of getting into the zone, or a trigger for it?

Steven Kotler, who wrote the book *The Rise of Superman* (a great read, all about how extreme athletes get into flow), suggests that time may be as much a trigger as it is an outcome. I agree.

Recall a time in your life when you were faced with a task, let's say a work project, or working on your race car. You dug into it without a specific timeline in mind, but thought you'd be done in an hour or so. Instead, four hours later, your growling stomach told you that you'd been totally engaged in the process of working on the project — you'd gotten lost in it, forgetting all about time.

If, just before you dove into that project you were told that you had just thirty minutes to complete it, and by the way, "Get it right!" would you have gotten lost in the task? Would you have gotten into flow? Or would you have been checking the time while thinking about whether the result — "getting it right" — was doable?

I suspect you're thinking along the same lines I am: Without the time constraints, and without the pressure to deliver the "right" result, it's actually more likely that you would perform at your best, and get into the zone.

So why, then, do we focus so much of our attention on lap times when we want to perform at our best when driving? I know, I know, driving fast around a track is all about competition, even if it's just competing with ourselves to turn a faster lap than ever before — our personal best. And that's where the great balancing act of performance driving comes in. We think that balancing the car is the biggest challenge when track driving, but no. It's balancing our desire to beat the competition (other drivers, the stopwatch, ourselves) with doing what's best to perform so as to do that. Whew! If this was easy, everyone would do it.

If time can negatively affect our ability to get into the zone, rather than it just being an outcome, should we remove it from our thinking? Hey, maybe we should tell ourselves that we have all day to complete one lap! Maybe that would trigger a zone performance, and, in the process, result in our driving in such a way that we turn a faster lap than ever before.

As Kotler says, "Neurobiologically, I'll bet this causes a decrease in cortisol and norepinephrine. It lowers stress levels, which allows us to increase the size of the

challenge we're willing to tackle. In short, time availability helps us tune the challenge-skills balance."

Lap times are a result, something to be aware of after the fact. They're not the objective.

If this lack of focus on time triggers better results through better performance, how do I explain a driver who seems to do their best when chasing another car? Isn't that focusing on the result?

I'd argue that a well-performing driver chasing another car is not focused on the competition or the result. I'd argue the driver is focused on what's going on right then and there, in the moment. The driver is very focused on where they're going, and not so much on the result. There is actually less focus on the mechanical/analytical thought of "do this now, then that then, and turn here, and on the gas here..." and more on "just do." The driver is in the moment.

Of course, some drivers, when chasing another car do get overly caught up in thinking about what the other driver is doing, and that is as bad for performance as thinking too much.

I wish there was an easily-identified procedure for balancing these competing factors, but there isn't. And to make matters more challenging, the balance tips slightly one way or the other, depending on who you are and what works best for you.

How do you balance the focus on lap times or results with what it takes to get into the zone. Do you think about lap times or your position? Are you aware of them? Do you put them out of your mind completely until after the event/session? Is time a trigger for your zone performances, a result of it, or neither?

Day #85: Focus on learning, and you'll never stop improving.

Given a choice, which would you rather have:

- A race win where you didn't learn anything?
- A great learning experience, where you gained knowledge that you'll use the rest of your career, but it came with a 10th place finish?

Okay, that's a bit of a trick question. But think about it.

No matter how talented you are, or how successful you are, the more you improve, the better your chances of winning in the future.

Unfortunately, I've met and observed far too many drivers who felt they had all the natural talent in the world, but were eventually beaten by drivers they felt were less skilled. Why? Because those "less-skilled" drivers focused on learning.

Here's a challenge for you: No matter what type or level of motorsport you're involved in, make learning your number one objective. Focus on learning. Do everything you can to learn.

If you take this approach, and stay disciplined and focused, I bet you'll be performing better and better as the season goes along — and you'll get better results because of it.

Day #86: Performance driving is a series of compromises.

Let’s build on the idea adding to your toolbox of driving techniques. I'm talking about compromises, such as:

- Trade off corner entry speed for exit speed.
- Less grip for trail braking to rotate the car.
- Lighter and longer braking for more overall grip.
- Downforce and lower top speed for increased cornering speed.
- Softer spring setup that improves grip for a stiffer one that improves transitions.
- And so on....

The compromise I want to talk about here is line for grip. There are many corners where the ideal line has, for example, a big bump in it. Therefore, you're better to drive around the bump, compromising the perfect line for improved grip. Or a part of a corner has more banking/camber, so you drive through it rather than on the ideal line. Or a certain part of the track surface has more grip than another because of a newly-paved section.

The big question is how to know when to make the compromise? When is it better to give up the ideal line for a different line where the track provides more grip?

Experiment. Sound familiar?

While I appreciate people who drive the line through a corner because they're following the advice I gave in my books, there comes a time when even I say toss the book out the window and just drive. Focus on where your car grips the track more than if you drove on a specific line that I or someone else told you to drive. Stop driving by the textbook and start driving with your senses.

How do you sense whether a certain part of the track has more grip than another? Pay attention. Be aware. Use your senses.

You have something called the proprioceptive system that can sense when g-forces build up and then begin to taper off. As long as you feel those g-forces building, then you know grip is increasing; when you feel them falling away, you're losing grip. (I know... well, duh!)

If you put as much focus into being aware of the g-forces increasing and reducing as most drivers do into driving the perfect line they've been told to drive, you would drive on the line that is fastest for your car, in that corner, on that day, in those conditions.

Oh, and different grip levels result in different sounds, too. So, if you focus more on what you hear, you'll be more sensitive to where the grip is.

Compromise focusing on the ideal line you were told to drive for being aware of and sensing grip levels. That's often the best trade-off you can make.

Day #87: Drive in your mind before driving on the track.

I want you take a few minutes to use your imagination. But first, I want you to relax your body. Take a few slow, deep breaths. Feel your feet relax, your legs relax. Breathe. Feel your hands relax, your arms relax. Breathe slowly and deeply. Relax your torso, and feel your body sink into whatever it is you're sitting on.
Breathe.

Now, imagine that a lemon is sitting on a table in front of you. It's a bright, shiny, yellow lemon. Imagine picking it up and holding it in your hands. Feel the shape of the lemon, and the texture of its skin. Place the lemon back on the table in front of you, where there is a knife laying alongside it. Pick up the knife in one hand, while holding the lemon steady with the other hand, and cut the lemon in half. As you cut it, you can hear the knife slicing the skin of the lemon, and you see the juice on the shiny blade of the knife. It drips onto the table.

Once you've cut all the way through the lemon, pick up one half of it, and give it a squeeze. See the juice dripping onto your hand, and feel it dripping down your fingers. Bring this half of lemon up to your nose and inhale the smell of the lemon juice. Give the lemon half another squeeze, then bring it to your mouth and lick the face of the lemon, the lemon juice.

What just happened? Perhaps, your mouth puckered, and it filled up with saliva. Why? Because your brain can't tell the difference between a real and an imagined event, if it's imagined with enough detail — and enough senses. If I had only asked you to imagine the look of a lemon, you probably would not have had the same reaction. But because you used all your senses: visual (you saw the lemon, you saw the juices), feel (you felt the shape, the texture, the dripping juice), hearing (you heard the blade of the knife cutting through the skin), smell (you smelled the juice), and taste, (licking the lemon juice), it made it "real" for your brain.

You've just completed a great mental imagery session. That's how you should be using what is usually referred to as visualization to program various techniques and mental states of driving.

Day #88: Brakes make you faster.

The following are random observations about braking.

1. *Why do we brake?* The most obvious answer is to slow down for a corner. The next most obvious answer is to manage load transfer. But a less obvious — but critical — answer is to help us get back to full throttle sooner. I like to step back and think about the why behind what we do. Think of the brakes as a tool to help you get to full throttle sooner. When you think about the end goal — and you keep that in mind — it helps you do what you really want to do. Someone/something — a coach, driver or data — can tell you to do so-and-so here and there (left-foot-brake, right-foot-brake, use the ABS or don't use the ABS, squeeze the pedal or hammer it, etc.), but if you really understand what the end goal is, you tend to do the right stuff more naturally. Successful companies keep their BHAG (Big Hairy Audacious Goal — from the classic *Good to Great* book) in mind at all times; successful drivers do the same. What is your BHAG for braking for Turn 1? Is it the same for Turn 5? Keep your BHAG in mind and the brakes becomes even more than a tool to get to full throttle sooner – they become a weapon!

2. The advice about quick, hard (but not too hard) application of the brakes is important, as is threshold braking. There's a time and place for that, just as there is for early, light braking. Huh? Yeah, I know, that doesn't seem very race-y, does it? I was coaching two highly-paid pro drivers at a test in a GTLM car. One driver consistently out-braked the other — later and harder braking in seven out of the eight corners on the track that required braking; the other driver braked earlier and lighter. The second driver — the one who braked earlier and lighter — was consistently half a second faster than the other. Interesting, right? Sure, if a driver brakes very early and lightly, they're going to be slow. You have to get close to braking at the limit. But once you do, sometimes keeping the car's platform more balanced by braking lighter helps. Now, if these two drivers were in a different car, the results could have changed, as the hard/late braking might have been a benefit. But don't assume that you have to brake late and hard to be fast. My point is to reinforce what some drivers know: you have to read and adapt to the car and track — and even the specific corner you're driving (one corner might reward late/hard braking; another might reward earlier/lighter).

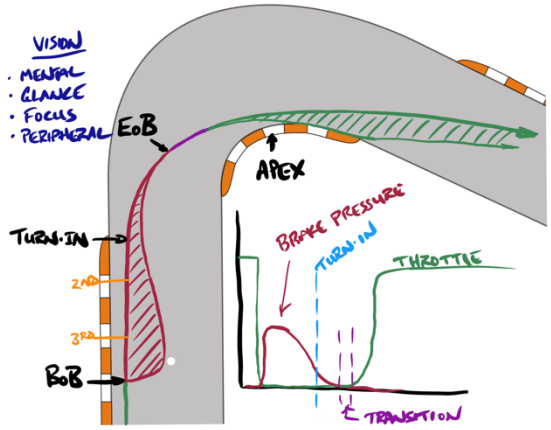
3. *Let's jump to the other end of the brake zone: the release*. For almost all drivers (and this applies as much to novices as it does to experienced drivers), if they put as much focus on the timing and rate of release of the brakes as they do on when and how hard to apply them, they'd be faster. Period. Think about that.

4. *We tend to think of the brake pedal only as an input device*. We input pressure on it, and it does something. The best drivers use the brakes as an output device, also, to tell them what the car is doing — it's a form of feedback. Focus on what the brakes are telling you, and you'll be consistently faster.

5. *Remember, we do what we do because we're mentally programmed to do so; we sometimes don't do what we want because we either don't have the right programming*

yet, or we access the wrong mental program (we make a mistake). Practice is programming. What we do when driving on the street programs what we do on the track. We can mentally practice using mental imagery (using more than the visual sense; actually moving our feet), simulators, sitting in our car in the garage, or driving on the road.

6. *Finally, while my favorite data channel/trace is brake pressure, I've learned as much or more by watching a driver's foot on the brake pedal. I love using data, but remember it's showing us the result*. But what happened to cause that squiggly line? And why? I once mounted a GoPro in the footbox of a car and learned more in a couple of laps from the video than all the data in the world. Why was the data showing inconsistent brake traces? Heck, I'd told the driver to be consistent many times! But still, he wasn't. A video can show whether a driver's foot is resting and pivoting on the floor (which provides much more control), whether a heel stop is needed, the timing and movement compared to throttle release, whether the foot is bouncing around and in a different position each time, and can even show a driver's muscle control. Lift weights for thirty minutes and then try drawing something in fine detail with a pencil and paper. I bet your lines will be a bit shaky because your muscles are tired. Same thing happens if a driver's foot, ankle, and calf muscles are not fit enough. I once coached a driver to work on his foot/ankle/calf strength, and his braking improved.



I hope you take a moment to think about these observations before your next time on track — and then while on track. To help with that, I'll leave you with this illustration.

Day #89: Driving harder is lazy. Drive smart, not harder.

Tim Ferris, author of *The 4-Hour Workweek*, said, “Hard work is lazy thinking.” Which got me thinking (notice how I did that?). It’s no different with driving. Doing the same thing with more effort is not going to lead to the result you want. Often, what you need to do is put less effort into it — as long as you did enough preparation up front, and/or dug into what it is you really need to do.

In a repeat of something that happened a few years ago, an instructor at a recent track event told me that a particular driver was having a difficult time with the line around the circuit. I hopped in the right seat, prepared to instruct for a couple of laps. On track, I immediately noticed a bigger problem than the cornering line. In fact, it was one that contributed to the driver’s line being inconsistent. The problem? The driver popped his foot off the brake pedal faster than he was driving. In doing so, the car felt unbalanced, not giving the driver the confidence to do what he really wanted to do — which was drive the line.

No matter how hard the driver worked at driving the ideal line, it wasn’t going to work until he got smart about his driving. And no matter how hard the instructor worked at getting the driver to drive the line, it wasn't going to happen until the driver used the proper driving techniques. This goes back to what I’ve said in the past about the trick advanced techniques are simply doing the basics better. Releasing the brake pedal at the end of the brake zone, then entering a corner, is a basic skill. And until it’s done correctly, nothing else much matters.

If you find yourself not improving the way you think you should — perhaps feeling as though you’re on a plateau — it’s time to drive smarter, not harder. Spend more time debriefing with a coach/instructor, a fellow driver, your spouse (it works), or by yourself. Make more notes on a track map. Ask yourself questions about what’s working and not working, and each time you answer a question, go to the next level deeper and ask another. Dig down until you’ve found what the real cause of a problem is, or what’s holding you back from improving.

Then do your homework. Read, watch video, analyze data, study… and think. Mentally prepare, by visualizing what it is you want to do. That’s driving smarter. Then do it,

The following are a few prompts to get you thinking about areas of your driving where you might be working too hard, and not driving smart:

- Trying to make a certain line or reference point work, despite it not feeling right for you (but you were told it was what you needed to do).
- Struggling to make the car go where you want, when the bigger problem is related to how you're holding and turning the steering wheel.
- Trying to be smooth, but not breathing and relaxing enough.

- Over-slowing for corners and trying to brake later to make up for it, when the real problem is you're not looking into the corner and at the End-of-Braking point.
- Trying to carry more speed through a corner (because the data shows that you're slower than another driver) — but you're "crabbing" on the entry (not using all the track, reducing the radius).
- Over-slowing for corners (again), trying to make up for it by braking later or lighter, when the real problem is you're downshifting to too low a gear.

Drive smarter, not harder.

Day #90: Do something different.

After two full days of watching a driver make the same mistakes over and over again (to the point of being really, really good at doing the wrong thing), I had to put some words to paper (or screen). I was working with a coaching client at the time, but I couldn't help but see this other driver doing the wrong things repeatedly. If I could have, I would've gone over and said, "Stop it!"

Okay, end of rant. But it's the beginning of my message. If what you're currently doing isn't doing what you want, and nothing is changing, do something different.

The driver I observed over those two days of testing was consistently downshifting to far too low a gear, which was causing him to over-slow for most corners. To be fair, my assumption is that the driver wanted to improve. Sure, it's possible that he didn't want to go any faster, but that's not the way I think! And as I had coached someone in the same type of car a year earlier, I was curious how this driver's lap times compared to what my driver had turned. He was consistently much slower than I'd seen the previous year. I was moving around to different corners to observe my driver, and I couldn't help but hear the mistakes — over and over and over and over again.

While I didn't go anywhere near the above driver over the course of the two-day test, I could imagine him "working on the line" in an attempt to go faster. Or maybe tuning the handling of the car with tire pressure and suspension adjustments. And based on the lap times I randomly took over the two days, nothing helped. Of course, the reason was that he hadn't identified the cause (downshifting to too low a gear, which resulted in over-slowing of the car) of the problem (not improving lap times). It looked as though he was putting a cast on his arm, as if dealing with a broken arm was going to help when he had kidney stones!

I must admit, the driver sure sounded fast! Over-slowing by downshifting too much seems like it works, but it rarely does. The downshifts, the high-revving engine, the quick upshifts coming out of corners… all sounded wonderful. And I'll bet the driver felt he was going fast because he did accelerate quickly out of the corners. The problem was that he was beginning to accelerate from at least 10 MPH slower than he should have been, and no matter how early he got on the throttle, no matter how much acceleration the car had, he was not going to make up for the slow minimum speed.

If you ever find yourself in the situation where you feel you're not improving, do something different. In this case, if I could have told him, "Go out and do one fifteen-to-twenty minute session where you run one gear taller in every corner of the track," I know for a fact that he'd have made a big improvement in his lap times.

Why do I feel so sure of this? Because I've said that exact thing to drivers I was coaching. Making improvements in lap times in the range of two-to-five seconds is not surprising in a situation like that.

By the way, some of my coaching clients who use the taller gears report over the radio — and then again when they come into the pits — that they're slower. And yet, the lap times prove otherwise. Sometimes what feels and seems slower is actually faster.

Day #91: Know whether you're driving at 10/10ths or not.

The whole concept of 10/10ths being the limit — and anything below that as being less than or beyond that being over the limit — is a pretty standard way of thinking in motorsport. There are few drivers who don't understand what's meant by driving at, let's say, 8/10ths. At least, conceptually.

But how does one know when they're actually driving at 7/10ths versus 10/10ths versus 11/10ths? I don't think this is an easy thing to put into words (otherwise I probably would have already done so a long time ago), but I'm going to try here.

First, 10/10ths is when the tires are consistently at their limit, at least in the areas where they should be. There are times when we're driving at the limit but we don't actually want the tires to be at their limit. Huh? Imagine driving through a bend in a straightaway. The more you straighten the bend out, the less load you're putting on the tires — the further they are away from their limit — and the faster you'll be. Your goal in that part of the track is to actually have the tires as close to 0/10ths as you can!

If you've made the decision, even if at the subconscious level, to drive at 8/10ths because that's the level of risk you're willing to accept when driving on track, how do you know you're there and not at 7/10ths or 9/10ths? That's the big question, right?

Consistency is a factor. Many drivers are not able to drive an entire session or race at 10/10ths, or even 9/10ths. Sure, top-level pros can because they practice driving there all the time. That's why they get paid the big bucks. But for drivers who get on track a dozen or so times per year, driving consistently at 10/10ths is a lot to ask. And unlikely to happen.

So, if you can drive at a certain speed, a certain lap time for example, every lap for an entire session, it's likely that is not 10/10ths — unless you're a well-above-average driver. If you can consistently turn a lap time within a couple of tenths of a second, there's a good chance that you're not driving at the limit — at 10/10ths. That's not always the case, but most often it is. It's more likely to be 9/10ths or less. That's just one way of knowing whether you're driving at 10/10ths or not.

If you find that you're never experiencing any understeer or oversteer — or you're not able to sense any amount of slip in the tires — I would say that you're driving at less than 7/10ths. That's my rating, so I'm not saying that's a law! But it's a good guideline. No slip equals less than 7/10ths.

Okay, so we've defined 7/10ths and 9/10ths. Now all we need to do is define 8/10ths in a way that is more specific than simply saying it's between 7 and 9/10ths.

The single most important tool you have to determine how close you are to driving the limit is your sensory awareness. If you notice that the tires are making THAT sound — you know, the one where you're feeling a little slip in the car, but the g-forces are high

as well — then you're close to the limit. If you feel the g-forces building up, but never begin to taper off, then you're not quite at 10/10ths yet. In other words, if you pay enough attention to what the car and tires are saying to you, you'll know if and when you're near or at the limit.

If you have a data system, you can use it to help calibrate your ability to sense the limits. For example, if there's a section of track where you're braking at maximum — you're on the verge of locking up or activating the ABS — make note of the longitudinal g-forces on the data at that point. That's a good measure of the tires' maximum grip level. If you notice that in other brake zones that are relatively equal (same surface, no big bumps or elevation changes, etc.) you're at 80% of that maximum g-force, then you're at 8/10ths in that section of the track (again, that might be the fastest way to get around the track — as I said earlier, there are places where using the tires at less than maximum is what you want). If you notice that your lateral g-loads are at 70% of the maximum, then that's a good approximation that you're cornering at 7/10ths.

Between really paying attention with your eyes, ears, and body — as well as checking your data system — over time, you'll become more sensitive to knowing when you're driving at 7/10ths, 8/10ths, 9/10ths, 10/10ths or over the limit. And the more consistent you get at that, the more consistent you'll be able to drive at whatever limit you choose.

But remember, it's one thing to drive one section of the track — let's say from the braking zone through the exit of one corner — at 9/10ths, and it's a whole other thing to have the car at 9/10ths all the way around the track, lap after lap.

Day #92: Program your breathing. Mentally practice breathing.

"Wow, I really sucked it up, held my breath, and went for it on that lap!" That's something we hear race drivers say every now and then after a great qualifying run. But I can practically guarantee you that no race drivers got the best out of themselves while holding their breath.

Do you think your muscles are more or less relaxed when you hold your breath? They actually tense up when you don't breathe, right? And do you think that restricts the amount of feedback you get about the car - sensing the limits of the tires? You bet it does.

So, why do we sometimes hold our breath when driving? Because we're facing some things that, perhaps, our brains weren't originally designed for, things like very late braking at the end of a long straight, entering a super-fast corner, and passing another car in a tight gap. In fact, I often notice that drivers hold their breath in all these situations.

And that's why we need to practice breathing. Much like in meditation, we need to be aware of our breathing. But we also need to program ourselves to breathe in those "scary" situations. You can do this with mental imagery (visualization), and you can practice it on the street. For example, every time you begin braking or enter a fast freeway off-ramp, deliberately exhale. Every time you do a lap of the track in your mind, exhale as you hit those difficult areas.

Practice your breathing until it becomes a habit, or program, in the most challenging areas of the track.

Day #93: Data never lies. But sometimes it doesn't tell the whole truth.

Many data experts have said, “Data never lies.” However, it doesn't always tell the whole story. Or perhaps more accurately, it's difficult to dig deep enough into the data to reveal the whole story.

Let me share a simple example.

In my career, one of my favorite drivers to race against was James Weaver. Generally accepted as one of the fastest and toughest sports car drivers ever, he was always fair, a true gentleman, and had a wicked sense of humor. When you beat him, you knew two things: First, you'd done your job that day, because beating James was a real feat. And second, he'd be the first to congratulate you on doing so.

When I wrote my first *Speed Secrets* book, I asked a few people to contribute some comments, and James was one of them. I gave him my draft manuscript and just said, "Go for it. Make or add any comments you wish." And one area that he commented on was in my brief discussion about data acquisition.

James said, *"One of my favourite* (I spelled "favourite" here the English way, as he would appreciate) *pieces of information the data acquisition system can give is a full-throttle histogram — telling me what percentage of the lap I'm at full throttle. If the percentage increases after changes to the car or driving technique, I'm bound to be faster."*

Later that year, during the 1997 IMSA season, he and I shared our throttle histograms when we were racing against each other at Road Atlanta. We were both driving Riley & Scott Mk. III World Sports Cars, his powered by a Ford, and mine by a Chevy. We had turned lap times that were within a tenth of a second of each other, and yet there was quite a difference in what the throttle histograms were saying.

James was at full throttle for something like 78 percent of the lap, and I was at full throttle for around 69 percent of the lap. How could that be? How could there be so much difference, and yet our lap times be so close?

Looking closer at the histogram, we noticed that while James was at full throttle 78 percent of the lap, he was at zero-throttle (fully off the throttle) almost 20 percent; I was at full throttle for 69 percent of the lap, and yet was at zero-throttle less than 10 percent of the lap.

In other words, James was either at full throttle, or no throttle — he used the throttle like an on-off switch. And while I was at full throttle less than he was, I was at part throttle more, squeezing the throttle down in places where he was off the throttle.

What this demonstrated so clearly was our very different driving styles. As anyone who watched James drive knows, he was in attack mode at all times, King of the Late

Brakers, and absolutely stood on the throttle. Me, I used a smoother style, carrying momentum, flowing speed. And in the end, both were fast.

What this also demonstrated was that if you only took the data on face value, one could be fooled. Without digging a little deeper, one could have determined that there was only one solution. In the end, James went away thinking about how to increase the time he used part throttle, and I was focused on spending more time at full throttle. It helped us both.

Of course, all things being equal and relative — like comparing to yourself, as James referred to it — an increase in the time spent at full throttle is almost always faster.

That's why I say that data systems never lie, but sometimes they don't tell the whole truth. It's only when you dig deep to fully understand what's really going on do you gain the full benefit from it.

Day #94: Check your conclusions.

I want to challenge your thinking today.

You've likely heard the term "spurious relationships," and know what it means. To quote Wikipedia (the source of all real knowledge, right?): "In statistics, a spurious relationship or spurious correlation is a mathematical relationship in which two or more events or variables are not causally related to each other, yet it may be wrongly inferred that they are, due to either coincidence or the presence of a certain third, unseen factor."

Here are a couple of examples of spurious relationships taken from the excellent book, *Never Stop Learning* (which just happens to be one of my life mottos!): "From 2000 to 2009, per capita cheese consumption in the U.S. showed a 94.7% correlation with the number of people who died by becoming entangled in their bedsheets. From 1999 to 2010, the marriage rate in Kentucky showed a 95.2% correlation with the number of people who drowned by falling out of a fishing boat."

Correlation doesn't imply causation.

- Variable #1: Your car understeers early in a tight, slow (hairpin) corner.
- Variable #2: You've been told to trail brake more to keep more load on the front tires, to increase front tire traction.

Okay, you need your front tires to grip more to reduce the understeer, so more load on them makes sense. You think, "I'll trail brake more — that'll help." That's worked every other time — or at least somewhere between 94.7 and 95.2% of the time. But it doesn't this time. In fact, the more trail braking you use, the worse the understeer gets.

Check your conclusions. Ask "Why?" Understand what's really going on.

I'm not saying the amount of trail braking you use and the amount of understeer your car has are not related — that they have a spurious relationship — but it's easy to come to the wrong conclusions by looking at a problem at a surface level.

If more load on a tire does not result in more traction — if more trail braking does not reduce understeer — why not? Could it be that the front tires are being over-worked? Could it be that less trail braking would mean the front tires have more traction to work with? That giving some of the traction that you're using for braking back to the tires for turning, that you'll experience less understeer? It's very possible.

Short of making adjustments to your car's setup, what can you do as a driver to reduce understeer? More or less trail braking; more or less load on the front tires. Remember, driving fast on a track is not easy! There are rarely any black-and-white, always-do-this rules. And that's why sometimes more trail braking will help reduce understeer by keeping more load on the front tires, and why sometimes less trail braking will help by asking less of the front tires.

I was recently talking to what I'll call a "typical track day driver." He spends between twenty and twenty-five days each year on track, driving his own car, a well-prepped BMW 3-series. We were discussing shifting, and specifically downshifting. I asked if he used the heel and toe technique. His response: "No. Every time I do, my lap times are slower."

Is heel and toe downshifting the cause of the slower lap times, a symptom, or something unrelated? Relating heel-and-toe with lap times might be smart. Or it might be like correlating the marriage rate in Kentucky with the number of people who drown by falling out of a fishing boat!

With some coaching, the "typical track day driver" and I figured out that because his heel and toe technique was so poorly refined, he braked significantly earlier when he attempted to do it. So, my first reaction — my conclusion — was to help him refine his technique, to help him improve his heel and toe through some specific, deliberate practice. But rather than stop there, I spent a bit more time digging into his mindset to understand the problem more deeply. What we/I discovered was that the biggest problem was not the actual physical skill, but really his fear of having his foot slip off the brake pedal, causing a crash.

Absolutely, with more physical practice the driver would have refined his heel and toe skill to the point where he would begin to trust it more. But without addressing the fear factor, it would have been a longer, more difficult challenge. By challenging my own initial conclusion, and making a plan to address the real cause, we sped up the process of making improvements. In fact, by the end of that very day he was almost a second faster than he'd ever lapped before, while using heel and toe. And in fact, the heel and toe was one of the causes/reasons for the improvement.

Very often, the outcome is not the cause. Check your conclusions. Look for the real cause.

Day #95: Pick the low-hanging fruit.

Every now and then it's important to take a step back from your driving and check to see if you're looking after the basics.

Having spent time this past summer picking fruit from the trees in our yard, I really get the whole idea of going for the low-hanging fruit. I spent more than a few hours perched on the top rungs of a ladder, stretching, reaching for that last apple or those perfect-looking cherries way out there on a branch. It was risky stuff. It wasn't easy. And to be honest, I'm not sure it was worth it. Sure, they were tasty, but no more so than the ones that were easy to reach — the low-hanging fruit.

Same goes with our driving. The stretches, the reaches, the risky, difficult stuff often is no more helpful in shaving time off a lap time than going after the easy stuff. So, as a reminder, I'm going to share what I believe is often the low-hanging fruit for many drivers. These are things I've seen more than one driver do more than once. And the interesting thing is most of those drivers were not even aware that they were doing these things. So, in no particular order…

- Easing off the throttle before braking for a corner.
- Not bracing the left foot on the dead pedal.
- Not keeping the heel of their right foot on the floor when pivoting from throttle to brake.
- Being in too much of a hurry to switch to left-foot braking.
- Not spending enough time and effort to make sure their seating position is correct.
- Not spending the time and money on a proper seat.
- Spending money on making their car faster before spending time on making their driving better.
- Cutting corners on safety equipment (especially head & neck restraints).
- Releasing the clutch for a downshift after turning into a corner.
- Being overly-focused on looking far ahead, to the point where you lose sight (no pun intended) of what's in your peripheral vision.
- Driving without two hands on the steering wheel.
- Not using all the track surface (turning in with the car a distance from the edge of the track, not clipping the very inside of the track at the apex, not using all the track surface at the exit/track-out)
- Crabbing into a corner (easing the car away from the edge of the track before turning into the corner).
- Releasing the brakes too abruptly.
- Shifting gears too much.

- Only ever driving the line that they were taught, and not trying alternatives.
- Being too focused on driving the line and not enough on feeling the car and driving it at or near its limit.
- Not spending enough time doing mental imagery.
- Not focusing on mental preparation before getting in the car.
- Not doing any form of physical warm-up to get the body ready before getting in the car.

Okay, that's a pretty good list of the most common low-hanging fruit opportunities to improve one's driving. But as I said earlier, many drivers are unaware that they're doing one or more of these things. Mounting a video camera so you can see your footwork, watch your hands on the steering wheel, see how much track is available at turn-in and exit, using data, having someone ride with you, or having a coach work with you — all of these will help you pick those last few apples and cherries without too much risk and effort. Use the tools to help you turn high-hanging fruit into low-hanging fruit

Day #96: Balance radius with grip.

Why do drivers drive over the inside apex curbing, and the exit curbing? No, it's not just to look cool, bouncing the car over the curbs like they do with the DTM, British Touring Cars, and Aussie V8 Supercars! It's to increase the radius of the corner.

The straighter the line you drive through a corner, the faster you can drive, right? So, driving from the very outside edge of the track at the turn-in point, over the most inside edge of the track at the apex (including driving over the curb, or even inside the curb), and then out and over the exit curbing (and sometimes more!) leads to the straightest line — the largest radius possible.

If all curbs were "friendly," then you would always want to drive over them. But some are not friendly. In other words, some curbs have significantly less grip than the track does, or they upset the handling of the car due to the height and shape.

There's no magic formula that tells you when you should use a curb, and when you shouldn't. You have to test them to see just how friendly they are. In most cases, there's a compromise to be made: trade grip for larger radius; trade radius for grip.

One thing is for sure: if you don't test the curbs and compare them to the amount of grip your tires have, you'll never know for sure which ones to use, and which to avoid.

Day #97: Use mental imagery to program your entire mental mindset.

I've been thinking a lot about the use of mental imagery — what's typically called visualization. And two separate email exchanges with two different people contributed to my thinking.

One email lead me to read a research paper, *Positive Fantasies About Idealized Futures Sap Energy*, by Heather Barry Kappes and Gabriele Oettingen in the Journal of Experimental Social Psychology (fun reading!). The basic conclusion is that imagining a positive outcome saps energy because it doesn't motivate you to put the effort in that it takes to make it happen. Or flipped around, imagining a less-than-positive outcome — a more realistic outcome — energizes you to put in the effort that it takes to make it happen.

Now, I didn't read the research paper as saying that you should imagine everything going wrong, and resulting in a disaster, such as crashing your car! But instead of simply imagining yourself driving around the track, then coming into the pits at the end of a race and standing on the podium, imagine what it would take to get to the podium. Be realistic. If you typically run in 20th place, imagining yourself winning is just a dream, and will actually cause you to put less effort and energy into doing what it takes to move up the finishing order. But imagining battling for 15th or 12th — a realistic improvement — and all that it would take to get there is much more than a dream. It's a goal if you imagine what it'll take to get there. Once you're running in 15th or 12th for a while, then move that image up to 8th or 5th, making realistic steps.

My take-away from this paper, and others that I've read, framed it more as simply being realistic with your mental imagery. If you see the negatives, the challenges, then you'll begin to see what you need to do to overcome them.

This relates to what Mihaly Csikszentmihalyi said in his legendary book, *Flow*. As the leading researcher in the field of studying what causes people to get into the flow or zone, his research showed that when you're faced with a challenge, but have a belief in your ability to handle the challenge, you'll perform at your best — in the flow or zone. But if you feel confident, with not much of a challenge, you're unlikely to put in the effort — or energy, as Kappes and Oettingen refer to it — that it takes.

It also relates to my advice of focusing on the act of driving, or your performance, rather than the results. That way you're focusing on the challenge, the hurdles you need to overcome, and you're more likely to perform at your best.

If I sit and imagine a million dollars dropping out of the sky and into my lap, how likely is it to happen? Not very, right? That's dreaming. That's being overly-focused on the positive. Instead, if I imagined a million dollars, and then all the challenges it will take to get it, I will become more energized. I will be more likely to perform at my best because I'm focused on the act of getting it — what it takes — rather than the result.

Finally, I think it relates to awareness, too. If you have a strong mental image of what it is you want to achieve, and an awareness of where you are right now in comparison, your mind will do what it takes to bring them together. Simply having a mental image of where you want to be without an awareness of where you are now will get you lost! It's like having the goal of going to Shanghai (since I'm writing this while in Shanghai), but not knowing where you are right now — where you're starting from. That would make it difficult to purchase a plane ticket to fly to Shanghai if you didn't know where you were, where you were starting from, wouldn't it?!

And that's what I've been thinking about: How mental imagery, focusing on your performance instead of the result, balancing the challenge with your belief in being able to handle it, and a realistic awareness of where you are right now in relationship to where you want to be are all related and interrelated. Get the mix of them just right, and magic happens. Actually, it's not magic at all. It's using mental imagery to program your entire mental mindset instead of simply dreaming about a result.

I hope this makes you think about how you use mental imagery.

Day #98: There are those who can drive fast, and then there are those who can race.

I don't know if anyone else said, "There are those who can drive fast, and then there are those who can race" before me, but I couldn't find any other reference to it. And that thought just hit me one day... a long time ago, shortly after I began racing.

See, there was a driver who was fast, and would qualify well for every race. But I could beat him almost every time in a straight-up, wheel-to-wheel battle. He was fast, but not a great racer.

Don't get me wrong, to win races you need to be fast. But driving fast is just the beginning. And it is the beginning. So if you're going racing, start by learning to drive as fast as you can. Then... once you start racing, you'll learn to drive even faster; once you spend time (closely) around other cars, and you're chasing them, you'll find more ways to go faster. That's why, when a track day driver asks me the easiest way to knock a second off their best lap time, I tell them to go racing!

(It almost goes without saying, but I'll say it again... make sure you have the very best safety equipment in your car and on your body that you can possibly get. Don't cheap out here.)

When you do go racing, pick a class and type of racing where there will be lots of opportunity for you to learn. And that means racing in a class that has lots of cars. In a class like Spec Miata, Spec E30 (and now E46), Spec Racer Ford (notice the theme to the class names?), most junior formula car classes, and the low-budget endurance racing series, there will always be someone to race, no matter how fast (or not) you are.

Besides any class of car with "Spec" in its name, I'm a big fan of "low-cost endurance racing" as a way to get started. I don't know of any other way of getting as much experience for the dollar.

I love the philosophy that Jeff Braun suggests: Race where you have the biggest budget. If you have $5,000 to race for a year, find somewhere where most drivers only have $4,000. Be the best-funded driver in the series. I'm not saying you should "buy your wins." What I'm saying is there is nothing more frustrating and limiting than racing with your wallet — being afraid of crashing and hurting your car. If I could go back and change something in my racing career, this is what I'd change.

Find a mentor. No matter what level you're going to race at, there is someone who would be flattered to give you some guidance — and many who are well worth listening to. Ask questions of them, follow their advice, watch what they do, learn from them. And this applies to not only what happens on the track, but off the track, too. Especially off the track with things like whether to rent or buy a car, how to prep your car (or have someone prep it for you), what hotels to stay in near the tracks you're going to race at

(more importantly, which hotels to avoid!), transporting to and from the track, how to apply for your racing license and enter a race, and so on. In fact, there is more to learn about what happens off the track than on.

Finally, focus on learning and having fun. Don't be in a rush to win (when you're not in a rush, you'll actually get there sooner). Be in a rush to learn, though. Enjoy the relationships you build with other racers. And never forget, no matter what happens on track between you and another driver, show respect for the other drivers and the officials. Your behavior off-track will have a big influence on your learning and level of fun.

Day #99: Focus on one thing that will help in other areas of the track.

How often have you come off the track and asked yourself what you should do to improve in your next session? More than once, I suspect! To prioritize, you might ask yourself

- What will make the single biggest improvement?
- What's the easiest thing to improve?
- What can I do to get a quick win?

Here's a question I'd suggest you ask yourself: "What will help in more than one place?"

Let's take a look at that approach.

Often, if you improve one thing on the track, you improve many more. Why? Because you sense what the car can do and can apply it everywhere.

For example, let's look at braking. Remember, there are at least four things you can do with the brakes:

1. Compress your brake zone by beginning to brake later, but ending at the same place.
2. Move your whole brake zone in, by beginning and ending later.
3. Release/end your braking earlier to carry more corner entry speed.
4. Brake lighter to increase corner entry speed.

It's easy to look at braking and think the only option we have is to begin braking later — in other words, compress the zone. But that can be a mistake. Instead, moving the zone in, releasing sooner, or braking lighter are better options. So don't go for the obvious — braking later. Instead, think about those other options, and look for the one that will help the most in all areas of the circuit.

Looking at your track map (you do have a printed map to make notes on, right?), count the brake zones. How many of them would benefit from compressing your brake zone, how many from moving the zone in, how many from releasing the brakes sooner and carrying entry speed, and finally, how many would benefit from braking lighter? If you were to do just one thing differently, how many places would it benefit your lap times? Find the one thing that will help you in the majority of places, and that's your priority.

Even if this one thing seems to only help in two places on the track, my bet is that it will actually help more. Why? Because you will learn what your car is capable of, and apply that limit to other areas of the track — and you can do that without trying.

Day #100: Improve your adaptability through knowledge and practice.

Handling problems: Let me count the ways...

- Corner entry understeer
- Entry oversteer
- Mid-corner understeer
- Mid-corner oversteer
- Exit understeer
- Exit oversteer

Throw in some instability under braking and that's a pretty good list of handling issues you can face with your car.

Each one of them can be improved by making adjustments to your car or your driving.

The best drivers are able to adapt their driving to minimize the negative effects of these handling problems, and maximize their speed, despite them.

But how? I'm not going to even attempt to cover all the various adjustments you can make to your car to improve it (that's not my area of expertise), so I'll focus on your driving.

Ultimately, there are three things you can adjust to adapt to handling problems:

1. *Your braking* — how hard and the timing of when you apply the brakes; the timing and rate of their release as you enter a corner.
2. *Your steering* — how crisply or quickly you turn in, the line you scribe through the turn, and how you unwind the wheel.
3. *Your throttle* — when and at what rate you apply the throttle, and whether you modulate it to change the balance of the car.

If you look at these three areas, you're really just adjusting the line through the turn, and the balance of the car.

Then, if you think about what it is you want your car to do, and how you can help it with the brakes, steering wheel and throttle — adjusting the line and balance of the car — the best response is relatively easy to identify.

Of course, identifying it and doing it are two different things — especially at speed! But the more you think through these various handling issues, and how you can help your car do what you want it to do, the more likely you'll do them while on track.

I also recommend going onto the track to practice adapting. If you only ever drive with one technique, you won't be very good at adapting. Learn, through practice, to change:

- When and how you apply the brakes.
- When and how you release the brakes.
- When and how quickly you turn into the corners and then begin unwinding the wheel.
- When and at what rate you begin accelerating.
- Whether you can change your car's line by modulating the throttle.

I challenge you to really think through each of the handling problems I've identified above, then decide how you would adapt your driving to them. Better yet, after you've thought through each one in a logical way, close your eyes, relax, and then do a mental imagery session where you adapt to the new approach. If you do that, I'll guarantee you'll be better at adapting in the future — and that will make you a better all-round driver.

Day #101: Follow a process to move beyond a plateau.

What are the practical, hands-on "things" you should do to get out of a slump or move beyond a plateau?

1. *Break your driving down into small, digestible bites.* You can't improve everything about your driving at one time. Dig deep in asking yourself questions (or ask for input from another driver or a coach): what one single skill, technique, or mental aspect of your driving could you isolate and work on that would make the biggest difference to your driving? Focus on just that, and nothing else.

2. *Do something different — experiment*. As Albert Einstein once said, "A sure sign of insanity is doing the same thing over and over again and expecting something to change." If you're on a plateau, almost by definition you want to improve (otherwise you wouldn't call it a plateau). And to improve, you're going to have to do something different than you're doing now. Brake earlier or lighter. Enter corners fast and don't worry about exit speed; focus on your exit speed and don't worry about momentum. Turn the steering wheel quickly and abruptly. Drive un-smooth! Do the opposite of what you're doing now. That may not ultimately be the fastest way to drive, but you will learn something in the process, and that's what you need to move beyond the current plateau.

3. *Give yourself a break*. In other words, relax. Understand that every driver gets on plateaus or in slumps. It's part of being human. The more you resist them, the harder you try to force improvement, the worse it gets. Be patient. Let the improvement happen. And understand that no human ever learns in a steadily-progressive manner — we all learn in steps, and if we're patient with ourselves and don't try to force progress to happen, it will.

4. *Get coaching*. Finding the right coach (not everyone is ideal for you — I answered the question, "How do I find & select a driver coach?" on my *Ask Ross* page here) is critical here, but getting someone else's feedback and input may make the biggest difference. And understand that you don't have to have someone who is a better driver than you to help with this. In fact, sometimes you just need someone to ask you questions and get you talking, to the point where you figure it out yourself.

5. *Be okay with being uncomfortable*. Ask yourself how many times you've made a mistake when driving on the track. How many? Now, how many of those mistakes have turned into something ugly, like hitting something? If you're like most drivers, the percentage of mistakes that led to a crash is very small. Think logically about this. You're probably pretty good at making a mistake, and the worst thing that happens is you run wide of the apex, get a little sideways, or drop a wheel off the edge of the track. When you think about it this way, doesn't it make you feel a little better about pushing the limits a little? Go on track and drive just a little outside your comfort zone. Not over the top crazy, just a little over your normal comfort level.

6. *Between time on the track, go back to the basics*. Read about driving techniques, watch videos, learn. This helps your mind make sense of everything you've been practicing. Often, the main factor stopping your improvement is not letting your mind turn the struggle of driving into mental programming. Study, then sleep on it. And as I've said many times, the super-trick advanced stuff that elite drivers do is just the basics done very well. There's no need to look for anything trick. Focus on the basics.

Now... is it possible that if you follow my advice, you'll improve? Of course it is, right? Take a little time and let that sink in. You now have a plan, and you're preparing to improve. And feeling prepared also contributes to improvement. When you've prepared, you feel more confident. Feel more confident, and you're more likely to improve. That is why taking even fifteen minutes each day to read about driving, view training videos, listen to drivers and coaches, and study will help you get to a new level of driving.

And finally, have fun! Focus on how much pure enjoyment you get from driving on the track. I know for a fact that you will improve with this process.

Day #102: Your brain doesn't like a vacuum. Fill it with productive thoughts.

Have you ever noticed how easy it is for your mind to fill up with useless thoughts?

Picture this: It's between on-track sessions, you pull out a chair, sit down to relax, and drink some water. Your mind begins to replay the previous session's laps. You see yourself behind another car, and in each of the braking zones, it seems to pull a half a car's length advantage on you. Then, coming off each corner, it out-accelerates you by another car length or two. You think, "They must have better brakes and more horsepower."

As you sip your water, the thought of your car having less-than-ideal brakes and its engine being down on power rumbles around in your head. "Yeah, I really need to try those PFC pads next time because I know what I have now aren't as good as everyone else's. And my engine... well, I know it's down on power. There's no way that I'm being out-driven off the corners, so it's got to be the engine."

By the time you've finished your bottle of water, you've convinced yourself your car is at least a second slower than everyone else who's running the same car as you.

You get up and wander through the paddock, talking to friends. After a couple of conversations about where you and those friends get to full throttle coming out of turns 3 and 7, you begin to wonder about the line you're driving through the corners. For sure, your engine is down on power, but maybe the line you're driving isn't helping — maybe it's causing you to get to full throttle later than some other drivers. "I'm an okay driver, but I've never been that great at exiting corners. Maybe that's it — maybe I'm not that good getting to full throttle. If I'm honest with myself, I kinda suck at that. I wish I was better at corner exits."

HEY, HEY! WAKE UP!!

There's something about our mind... it doesn't like to be left idling. It doesn't like to be left empty. It'll fill in that empty pretty quickly with something, anything. And it's usually easiest to fill it with negative thoughts. Unproductive thoughts.

There's nothing wrong with identifying a weakness, an area where you can improve. That's a good thing. But what's most important is what you do with that information. Once you've identified it, you need to fill your mind with what you're going to do about it. Because if you don't, your mind will fill itself up with more of the same thoughts — more and more and more thoughts about what's wrong.

Identify what's wrong, and then fill your mind with what you're going to do about it.

This is one of the many reasons I strongly recommend making notes, and specifically, what your plan is for the next on-track session. The act of writing down what you're

going to do fills your mind with positive, productive thoughts. And when your mind is full of these, it doesn't have room for useless, negative thoughts.

Remember, your mind will fill itself up with negative thoughts more readily than it will positive, productive thoughts. Left alone for a few minutes and it'll start occupying the vacuum in your head with things that do you no good. Get in the habit of filling your head (but never over-filling) with productive thoughts.

As you sip your water and take a break between sessions — and before you go talk to friends in the paddock — take out a track map and pen, and make notes of what you did in the previous session. By this, I mean write down where you begin braking; in what gear you take each corner; what reference points you use; what you do with the release of the brakes (trail brake, no trail brake, quick of slow release); where you're at full throttle exiting each corner; where you're looking; what parts of the track have the most grip; what curbs you can use and which ones you shouldn't; and so on. And then — and this is the most important part — make a list of what you did well during the previous session. I know, it's easier to make note of what you want to improve, but that's my point. Make note of what you did well. Give your mind something it can focus on that it can use, because if you do more of what you did well, nine times out of ten you'll fix the problems.

Again, I'm not saying you shouldn't work on improving areas of your driving that are causing you problems. You should. But there's no point obsessing about them. In the process of writing down what you've done well, I'll guarantee that your mind will identify and place the "things we need to improve" in your head — and you'll access them when on track, where you can do something about them.

As a coach, this is one of the most critical things I do: make sure my driver has the right stuff, and the right amount of the right stuff in their head. Too little and I know it'll begin to fill up with useless crap; too much and it'll overflow. I'm constantly aware of — and testing — what's going on inside my driver's head. It's a quantity and quality thing, though. It's all about managing what the mind is focusing on.

Even without a coach, it's a choice you have. You can let your idling vacuum of a mind fill up with useless stuff. Or you can choose to fill it with what you're going to do, instead. Your choice.

Day #103: Manage aggressive behavior. Be assertive, not aggressive.

I need your help in addressing an issue in most forms of motorsport, from track days to racing. An email from a subscriber triggered my thoughts below. Here's what he said:

> *"There are lots of new drivers that don't understand there are consequences to their actions. As far as I can see, no one has explained this to them. One area that I feel should be addressed is the unfortunate training in aggressive driving we have found taught at some schools. There is nothing worse than going out with some novice driver that has been taught that he or she should intimidate other drivers or perform dangerous moves just to get ahead. It has been said that driving wheel-to-wheel and not touching in any car takes greater skill and judgment than banging doors. It seems that the bang and crash style of driving is getting worse with the younger drivers."*

I like to race another competitor as hard as anyone, but I believe in doing it in a fair and respectful way, so I agree with his comment. And I think this is a big issue, as it's throughout motorsport worldwide. Some would say that the cars and tracks have become so safe that too many drivers take too many risky chances. That sets an example, and others follow. Then it becomes the norm and the culture… and spreads... and now it seems that more and more drivers think it's okay to run another driver off the track, bang wheels, or just outright push someone off track.

I hope to do my small bit to help the situation by writing about it and talking about it. But I'd like your help, too. If you race, set a good example. Be assertive, but not aggressive.

To me, there's a distinct difference between being assertive, and being aggressive. You could think of assertiveness as "controlled aggression." Aggressive behavior lacks a certain amount of control, whereas being assertive means taking control, but in a deliberate, well-thought-out, smart, decisive kind of way. Assertive drivers are tough, but fair. In Indy car racing, Scott Dixon is assertive; he wins a lot. In NASCAR, Jimmie Johnson was assertive; Kyle Busch was mostly aggressive. In F1, Max Verstappen was a little on the aggressive side at times, but has learned how to be assertive.

While preparing for a webinar on racecraft, I asked for input from many drivers; I was surprised at how aggressive some of the responses were. And this is not just from young, up-and-coming-future-pro-driver-wannabes (or, as some suggest, "sim racers who don't understand there isn't a Reset button in real life"). Even some of the responses from HPDE and track day drivers has made me think that this is a cultural issue. Sure, the aggressive behaviors may have started in the most cutthroat of racing series, but the example set has trickled down to the way some drivers expect to pass on a fun track day!

Drivers are not the only ones to blame, as race officials and track day organizers allow a culture to develop within their events.

There’s a driver who’ll remain nameless here whose aggression got him, as you can guess, banned. Some would say that ban should have lasted longer, and the series should have done more to make a statement. I think they should have, as there is no excuse for what this guy did. If I were in charge, I would have given him a lifetime ban from racing, for two reasons. First, his behavior was inappropriate, out of control, and dangerous (more than once). And second, by letting him back into the sport the officials have, in effect, said that it’s kind of okay to do what he did.

That’s why I need your help. It’s one thing to talk about this, but it’s action that makes a difference. I need you to set an example. You might be thinking that some aggressive pro drivers, and all the young drivers following in their footsteps are not paying attention to you. Okay, you’re right. But…

Setting an example can happen from the grassroots level. It doesn’t need to start with F1. Based on what we’ve seen from drivers like Hamilton and Verstappen, we can’t count on them. We need to send a message to other drivers that it’s not okay to be aggressive, and put other drivers at risk.

If you play any type of driver training role, I’m asking you to hammer home a clear message of respecting other drivers. If you’re an official, don’t let drivers get away with aggressive behavior. If you're an event organizer, establish a clear line, and don't let drivers cross it. As a driver, support tough measures to penalize aggressive drivers. And finally, set an example on the track.

In the risk management world, there's a concept of "near misses." Basically, it says that statistically speaking, some number of near misses will result in an incident; some number of incidents will result in a serious one, and eventually a death. How many near misses have you seen over the past year? I've seen dozens of them, at HPDE events, club races, and every level of pro racing. Some have turned into major incidents, and people have been hurt. Some have turned into minor incidents, and people have escaped without injury. And one thing for sure, I haven't seen them all. So, for every one that I have seen, there must be many more that I haven't.

Even beyond serious incidents where people get hurt (and what could be worse than that?), there's something especially meaningful in what our subscriber wrote to me: "It has been said that driving wheel-to-wheel and not touching in any car takes greater skill and judgment than banging doors."

Take pride in being assertive, but not aggressive, because it demonstrates a higher level of skill. If respecting others is not enough, then do it for yourself — take pride in being the best you can be, and keeping yourself safe.

Day #104: Driving the limit means driving the limit everywhere.

Many drivers drive the limit. In fact, most drivers do. But here's the thing: they don't do it enough. They don't drive the limit everywhere.

One major difference between elite level pro drivers and every other driver is they drive the limit everywhere, whereas many/most drivers only drive the limit in certain parts of the track.

For example, typical places where drivers are below the limit are:

- From turn-in to the apex (despite being at the limit from the apex out)
- In the short sections between corners where the driver's mind is saying to itself, "Why bother going all the way to full throttle for a fraction of a second, when we're just going to lift out of it again?"
- Early in brake zones by not braking hard enough initially

Now, here's the interesting thing: Even though my last example above was not braking hard enough to use up all of the tires' grip to be on the limit, sometimes the best thing to do is brake lighter — not at the limit — so the car is better balanced, and therefore has a higher limit for corner entry. Ahh, that elusive perfect compromise that performance driving is all about!

When you sit down with your physical track map after each session, break the track up into sections:

- Brake zone from BoB to Turn-in point.
- Trail brake zone (if there is one), from Turn-in point to EoB.
- First third of the corner, from Turn-in point until about 1/3rd of the way into the corner.
- Second third of the corner.
- Final third of the corner.

Actually, make small tick marks on the edge of the track map with a pen or pencil, and then ask yourself how close the tires are to their limit in each section. Try to find a part of the track where you're not at the limit. Dig for it.

Don't get me wrong. As I've written about in the past, I understand that you may not want to drive right at the absolute limit, as you could be in your daily driver car and want to make sure you get home at the end of the day. That's okay. Then, instead of evaluating whether you're at 10/10ths in each of those sections, ask yourself if you're as close to the limit as you want to be. For example, if you want to drive at 7/10ths, then dig to see where you're at 7/10ths, and where you're not (or even over 7/10ths).

Awareness is an amazing tool. Once you become aware of something, you very often change your behavior. If you become aware of driving below the (or your self-defined) limit, your mind will typically do what it takes to get it to where you want it to be. But the key is taking the time to evaluate how close you currently are to the limit. Oh, and it's okay to be honest with yourself!

Day #105: Every inch of track you're not using is costing you speed. You paid for it, so use it all.

If you decrease the radius of a turn by not using all the track, your maximum speed will be significantly reduced. For example, by entering a corner even one foot away from the edge of the track, the radius of the turn may be reduced by as much as one percent. What's one percent worth? As much as half a second on some road racing circuits. And what's that half second worth to you?

There is no reason not to use all the track surface on the entry to a turn… or at the apex… or at the exit.

Okay, there are turns where using all the track actually costs you time (by increasing the distance you have to drive), or there's a bump you want to avoid. But these are exceptions to the rule, and choosing not to use all the track should be a deliberate action — not one that happens because you're unaware of the track's edges.

You "paid" for it — use it.

Day #106: Brakes slow you down, and that's not always a good thing.

Let's take a look at threshold braking:

- When MBI Publishing (who published all of my *Speed Secrets* books, and has now become Quarto) asked me to write another book — *The Lost Art of High-Performance Driving* — much of the focus of the book was on the technology in modern cars to aid the driver (the "nannies"). As I've thought about my experiences (and those of others I asked) using the brakes when driving on a track, the obvious differences between ABS-equipped and non-ABS cars come into play. And threshold braking is a big part of driving a non-ABS car.
- While conducting some classroom and break-out debrief sessions with drivers at a few different track events lately, I found that the number of drivers who either have the wrong perception of what threshold braking is, or when to use or not use it, was surprisingly high.
- Some of the one-on-one coaching I've done recently has focused on drivers moving away from using threshold braking to go faster.

To review, threshold braking is where you brake so hard that the tires of your car are at their limits of adhesion, just before either locking up and skidding (in a non-ABS car) or activating the ABS. It's when the tires are on the threshold of locking up or going into ABS mode, thus the name.

Driving fast, to many drivers, means threshold braking every time they use the brakes. And that couldn't be more wrong. These drivers are surprised when they learn how often the very best drivers brake lighter than they do, even if that means braking earlier for a corner.

The natural thinking is that you need to brake as late as possible, as hard as possible, to be fast around a race track. To many people, that's what high performance or race driving is all about. But that's not accurate.

In fact, high performance driving is all about the correct timing, application of pressure, and release of the brakes — and how to adapt that to suit the corner, car and conditions. In other words, it's knowing when to use threshold braking, and when not to.

One of the most challenging times to control your braking is when you're looking for those last few tenths of a second of lap time. When I ask drivers of all levels and types of driving what they could do to improve their lap times by half a second or so, most reply with something along the lines of, "Brake later for…."

In reality, braking lighter — not later — is often the best approach to going faster. After all, braking harder (which is what most drivers do when they brake later) slows you down!

Braking lighter has two advantages:

- First, it usually results in faster corner entry speed. If you're able to carry even one MPH more through a corner — and still get back to full throttle as early as before — you will have made a big improvement in your lap times.
- Second, when you brake lighter, your car stays better balanced — flatter, with less weight transfer. The less weight transfer there is, the more overall traction your car has, which means you can carry that extra amount of speed into the corner, without it having any negative effect.

A few years ago, I was coaching high-level pro drivers in a GT3 car. Two drivers were testing at Sebring, and one was about half a second quicker than the other. When we looked at where the faster driver had an advantage, it was in the brake zones. Practically everywhere they were exactly the same (line, when they got to full throttle, etc.), except when braking. And the faster driver applied the brakes earlier and lighter. Rarely did he threshold brake.

Often, when I first suggest to a driver that he brakes lighter and carry one or two MPH more into the corner, I get pushback that the car is already at the limit, so any more speed won't work. So I then have to remind him that with lighter braking the car will be better balanced, and therefore have more grip. With more grip, the car will turn into the corner, hit the apex, and he'll be able to get to full throttle as soon/early as before — but he'll be beginning to accelerate from a slightly higher speed.

Of course, there's a limit to this. If you brake so lightly that you carry too much speed into the corner, you'll be late to getting back to full throttle exiting it. That's most often a poor compromise.

I also need to point out that this advice is aimed at drivers who are braking near — or at — the limit. If you're braking way early — let's say, one hundred feet or more — then yes, focus on braking later. But don't feel that you have to be the driver who brakes the very latest. Get it close, then start experimenting with braking lighter. Ultimately, you might then brake a little later and harder — but only after you've got your corner entry speed just right.

Think about this the next time you're on track: Rather than braking later and harder, try braking exactly where you are now, but lighter.

Day #107: Racing in the rain, and being fast in it, is more about your mindset than any technique or skill.

While there are techniques specific to being fast when driving in the rain, your speed will come more from your mindset — and your belief system — than any of these skills or techniques.

Henry Ford once said. "If you think you can, or think you can't, you're probably right." I wonder if he was talking about being fast when driving in the rain?

Drivers who walk around saying that they hate driving in the rain, and then wonder why they are slow on a wet track… well, they just don't get it.

So, get it. Get that your attitude and belief about driving in the rain will impact your performance more than anything. If you don't like driving on a wet track, change your programming. Start liking it, even if you have to "fake it until you make it." Tell yourself how much you love the feeling of the car slipping and sliding around, and the lack of visibility. Use mental imagery to change your mental programming, your belief about your love for driving in the rain. The more you work on your mental programming, the more you take the attitude that you love this, the faster you'll be. And of course, that will lead to you loving racing in the rain even more.

I can't think of anything more fun that driving a car at the limit in the rain.

Day #108: Practice how you plan to race, and then you'll race as you practiced.

How can you expect to drive at the car's limit if you never practice driving at the car's limit?

I see drivers do this all the time: They go out for a practice session and drive at 90 or 95%, thinking they're reducing the chances of risking a crash. Well, they might be during that practice session, but they might also be increasing their chances of a crash in the race. How? By not having practiced the way they plan to race. If, in the race, they take their driving up to close to 100%, they've spent little time at that level. And that might increase their chances of a crash in the race.

So, the next time you think you'll take it easy in a practice session, think again. Practice like you plan to race.

Day #109: Learn through trackside observation.

An often-overlooked opportunity for learning to improve your driving is the act of observing other cars and drivers on track from the spectator areas — or even from areas that spectators can't normally get to, but are accessible by drivers (with the right credentials).

Fun story: In 2000, I was coaching a young driver competing in the North American Formula Atlantic Championship series, and we were racing as a support event at the Canadian Grand Prix in Montreal. I was dressed in the team's gear, a light blue polo shirt and black pants. I also had my "hard card" credential around my neck, a radio clipped to my belt, and a headset. Prior to our second practice session, I walked out to turns 6-7, a left-right section, and arrived while the F1 cars were still on track. As I walked up to the fence designed to keep spectators back from the track, I noticed a small, gated opening, so I headed towards it. As I passed a security guard, I realized that the shirt I was wearing was identical to the one he was wearing. So close in appearance, in fact, that the security person give me a nod and wave, as if I was "one of them." Thinking this might be an opportunity to get a little closer to the track and be in position to observe my driver more closely, I walked with authority through the gate and stood right next to the turnworker station. A flagger gave me a nod, and I stood there with my headset on and watched closely as the F1 cars zipped by, less than 30 feet away. What I saw left a lasting impression.

Now, I've spent many weekends of every year of my life since I was five years old at a race track. If you watch enough cars closely enough for enough years, you get pretty good at noticing very subtle differences in line, a change of direction, in attitude of the chassis, and in speed. I immediately recognized a difference in speed among the cars of Mika Hakkinen, David Coulthard, Jensen Button, Jacques Villeneuve, and Rubens Barrichello... and one other driver. The speed they approached the brake zone, the rate of deceleration, corner entry speed, mid-corner speed, and the speed and rate of acceleration out of the corner were all things I picked up. And one car stood out from all the rest in the very short section between turns 6 and 7 — the mid-corner. It was the Ferrari driven by Michael Schumacher. Every other driver was within a MPH of each other at that point on the track, but Schumacher was significantly faster, by at least 3-4 MPH (I've learned to be able to tell the difference in speed of cars within a MPH).

From my privileged viewpoint, I began my study of the different drivers and cars. What allowed Schumacher to carry more speed there than anyone else, even his teammate (Barrichello) in the same car? Was it his line? No. I thought deeply about it as I watched the attitude of the chassis — how the front compressed under braking and then raised up slightly as the driver released the brakes while turning in — and there was definitely something different going on in this one Ferrari. But, what? The more I watched, the more I was convinced that Schumacher did something different with the way he released the brakes while turning in. But, what? Again, I watched closely, and came to the conclusion that he managed his brake release just a tiny bit better than everyone else — he balanced the car on the edge between too much brake and not enough

brake, the right amount of load on the front tires, combined with just the right blend of steering input. I remember imagining what was going on in the cockpit of his Ferrari, picturing the way his left foot came off the brake pedal while his hands turned the steering wheel.

Fast forward a few years and I'm doing some presentations for a car club at Watkins Glen. At one point during the day, I found myself with a thirty-minute break. Not one to waste time at a track, I told three drivers standing nearby that I was going to go out and watch some cars in "the Boot" — turns 6, 7 and 8. We hopped in my rental car and away we went.

As we stood on the hill overlooking turn 7, I asked my fellow observers what they noticed. At first there was very little response, so I asked again what they noticed. Hmmm… not much — maybe some slightly different lines. I asked what they noticed about the different speeds, could they see the rate of change of direction when the drivers initially turned into the corner; did they notice the sound of the engine as they got back on the throttle; how much track were the drivers using at the exit (and whether they noticed the drivers deliberately unwinding the steering wheel or did it seem that the acceleration was driving the car out to the edge of the track); what they learned from observing the gap between the top of the tire and the fender (relating to the change in ride height, which was impacted by bumps in the track surface, as well as how the driver was managing the balance of the car).

In less than ten minutes, the three drivers standing there with me started to pick up very subtle differences, some seeming to be good, and others not-so-good. As they noticed the differences, I asked them to think about what that meant, and more importantly, how they could use that information. I also asked them to imagine what it was like inside the car (as I did when imagining what was going on inside the cockpit of Schumacher's Ferrari). I could barely take in all the responses, they were now coming at me so fast!

Understand one very important thing: The obvious — the differences in lines being driven through the corner — were not the thing that got the most attention or the most Aha! revelations. It was the chassis attitude, the rate of change of direction and speed, and the sounds they heard that had the biggest impact.

For sure, this was not the first time any of these three drivers has stood and watched from the side of a track. But I'm pretty sure it was the first time they had really been "forced" to pay deep attention and to really think about what was going on.

A couple of hours later, shortly after my new friends had finished their next on-track session, they each came to me and said they had shaved a significant amount of lap time off their very best time. For one driver, he had improved over two seconds. And that from observing intensely, asking questions, and thinking about what it all meant — and not actually driving. Interestingly, I had not told them one single thing to do to improve their driving. It had all happened because of what they learned from observing at the side of the track.

Day #110: Have a plan. Use a plan.

Let's pretend that a goal you have for this season is to improve your car's handling. How are you going to do that?

1. Hand your car over to an experienced tuner/shop/engineer and tell them, "Make it better."
2. Go to the track and drive your car, and when you notice a problem, figure out how to make it better.
3. Put a plan and schedule in place for designated test days, ones focused specifically on tuning your car's performance.

There's nothing wrong with any of these options, and there are others, too.

If you're going to hand your car over to someone else to develop it, keep the following in mind:

- How they set up your car may not be ideal for you. What works for them may not be good for you. It could be a matter of experience (they're tuning the car for a pro-level driver, and you're not there yet; they're tuning the car to make it comfortable for a relatively inexperienced driver, and that's not you, either). It could be a driving-style thing — perhaps you prefer a car that oversteers or understeers more than how they set it up.
- Understand each and every change they make to your car. Learn why they're doing what they're doing, what the effects should be and why, and whether there are fine-tuning adjustments available to what they're doing. Once you get the car to the track, unless your tuner is there with you, you need to understand what you can and can't do, and why.
- Who have you hired/contracted to do this work, and is this the right person/shop for you? They may have a great reputation, but they may not be the right fit for you. Some shops are great at setting up a car, but will not honestly share with you what they've done (they want to protect their "secrets"). While that's a quick fix, it won't help you long-term. If you don't understand what's going on, and why, you're going to have to keep going back to this shop/person (Hmmm... now you know their "secret"). Find someone who is willing to help you understand what they're doing to your car.
- Don't have someone tune your car if they're not open to different approaches. If their approach is "With that car, we always put this set-up on it; we've got a proven set-up that we always run on those cars," I'd recommend looking around for someone else. I'm not saying that their proven set-up is not right, but they should be open to trying something new. It's okay to start with the proven set-up, but being open to options is good.
- Reputation is super-important. But just because someone has the reputation of being the "set-up god" for your car, that doesn't mean that's the only way to go.

Some shops with great reputations have passed their sell-by date. Having said that, if someone has a great reputation, there's probably good reason for it.

- Check with others who have used this tuner. What was their experience like? Did it improve their car? How was service? Was what they promised delivered on time (this is a big one!)? How were they in terms of helping the driver understand what they did and why?

If your approach is go to the track, drive the car, and respond to it, great. But have more of a plan than simply reacting to it. Take time before your day at the track and think about all the possible handling issues you might experience — make a list of them. Then, jot down one, two or three things that you could do to help improve each one. Remember, the best place to start when tuning your car is to use what tools you have. If the only tool you own is a tire pressure gauge, then the option you have for every handling issue is "adjust tire pressures." If your car has adjustable anti-roll bars, you can alter ride height front and rear, you can adjust camber and toe at the track (without messing things up), you can change spring rates, you can alter aero settings front and rear... well, the list of options is a long one. Take time to pre-play these options before going to the track, and prioritize them. It'll make the limited amount of time you have at the track more productive.

Speaking of limited time, anything you can do to make changes more efficient, the better. Having someone, or a few people, who have been briefed and trained at the track and ready to make the changes for you is better than you having to hop out and quickly roll under the car to make a change.

If you've decided that increasing the camber on the front wheels would be a good adjustment, for example, practice that change before you ever go to the track. With many changes, you practice them at home, measure it, and then when at the track you know "Turn the adjuster one full turn and that will result in a toe change of a 1/16th of an inch." If you want to add 1/16th of an inch of toe-out and you haven't done that until you're at the track, it's going to take a lot longer to make an adjustment, measure it, adjust, check it, adjust, measure, and so on. And it's unlikely to be very accurate. Practice set-up changes before you ever get to the track.

Obviously, what I've done is begun to merge the third option I presented above into the second. Develop a plan for a designated test day, with a well-thought-out test plan. The plan may be as simple as:

- Write out the various handling issues you are likely to experience with your car.
- Write out what options you have for adjustments to your car (i.e., tire pressures, ride height/rake, toe, camber, springs, anti-roll bars, shocks, aero).
- Write down what resources you have (tools, people).
- Write out one to three options that make sense to you that you can do to the car to address each of these handling problems.

- Write out the schedule for the day — how many sessions, their scheduled times, and how long each session is.
- Write down what areas of your car's handling you'd like to focus on during each session (which corners, what part of the corners).
- Prepare a notebook to keep track of every single bit of feedback you have about the car's handling after each session (and if a previous change made it better or worse), the track/weather conditions, time of day, condition of tires, and what changes you made to the car.

Now you have a plan! Use it. Keep in mind that a change that didn't make the car handle better is almost as valuable as one that did help — if you make note of it, think about why it didn't work, and learn from it.

Finally, if you're in car development mode, your driving needs to be consistent and focused on car set-up — not on your lap times. Sure, lap times should be recorded and used as a metric to help you understand whether a change helped or hurt. But learning that a certain set-up is slightly slower but easier to drive may be as valuable as finding something that makes the car a few tenths of a second faster. So, if you're constantly adapting your driving to what the car is doing, you're going to have a tough time tuning your car's handling.

Day #111: Do everything possible to maximize your vision when racing in the rain.

I once raced in a World Racing League event at COTA — in the rain. What a blast! I spent two hours and forty-five minutes playing in the wet, all of it pure enjoyment. (I posted a video of the first few laps of the race, with my “inner voice” narrating over top of it — watch it here)

Well, there were a few moments when the windshield completely fogged up that were slightly less than pure enjoyment....

With that strong memory of driving blind, let's talk about what you can do to maximize visibility. In this case, I'm talking visibility through the windshield of production-based race cars, many of which have the factory-installed heater/AC unit removed.

The big challenge our two team cars had was that the insides of both windshields fogged up. Judging by the number of other cars that stopped in pit lane, we weren't alone.

At one point, I lined the car up next to the Armco on the inside of the pit lane entry and looked directly out the side to keep the car the same distance from it, as I assumed it would guide me around the corner and into pit lane (and, as I hoped, there were no other cars stopped in front of me with the same problem). Once in pit lane, I loosened my shoulder belts enough to reach forward and give the windshield a wipe with my glove, giving me a small window so I could find my crew.

Had I put together a list of things to do to reduce the chances of the inside of the windshield fogging up, it would have looked like this:

- Seal the engine compartment so that hot air cannot get through to the inside of the windshield. Especially when water hits hot headers, the steam that comes up through the dash and onto the inside of the windshield is unstoppable.
- Add an electric defogger on the windshield. Many teams install one of the electric defoggers that are used for rear windows. Our car had it, but it couldn't keep up.
- Clean and dry the inside of the windshield before the start of the race. This is crucial, no matter what else you do.
- Once clean and dry, coat the windshield with an anti-fog treatment. There are a variety of products out there that do a good job, and a few that don't. Also, wiping on and then polishing in dish soap (I hear Dawn is particularly good) works well. I'll leave it to you to do the research and testing to find out what works best.
- Airflow. This is critical. If you're able to get some amount of airflow to the inside of the windshield, it will make a big difference. Scooping air through a side window opening, or using a fan will help a lot.
- Rag on a stick! Yup, if all else fails there's nothing like having a squeegee with a rag wrapped around it that you can reach up and wipe the windshield with while driving. You hope you never need it, and finding a place you can securely store it

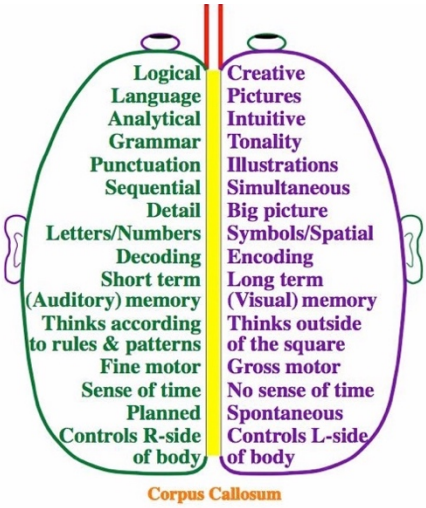
while driving is a hassle, but there were more than a few drivers this past weekend who wished they'd had one.

There’s nothing better than racing in the rain! Especially when you have visibility.

Day #112: Switch on your brain to perform at your best.

Your brain is made up of two hemispheres. Your left hemisphere tends to process the factual, logical, detailed information; your right brain is responsible for the intuitive, artistic, big picture information. Just like using the steering wheel is a two-handed activity, driving in general is a two-brained (or, at least, a two-hemisphere) activity. To perform at your best, you want to use both sides of your brain. There are times to be detailed and times to see the big picture. There are times to be logical and factual, and times to be intuitive and artistic.

Between the hemispheres there is a bundle of nerves called the corpus collosum. It acts like a cable between a computer and a printer, transferring information back and forth, and is the only thing connecting both sides of the brain. In fact, it is the bio-electrical communication that occurs between the hemispheres of the brain that dictates how well we think, how well we process information, and how quickly we do this. There are times in our lives when it's as if there is a rheostat, dialing up and down the amount of bio-electrical communication between the hemispheres. When there is some restriction, we tend to think more left-brained, or right-brained, but not whole-brained — not integrated. When there is no restriction, our brain is integrated and we perform at our best.



What restricts communication between our two hemispheres? Our state of mind, for one: Fear, anxiety, lack of confidence, sadness, anger, frustration to name a few states. Do any of these creep into our minds while on the track? If you're human, yes. But also, certain words and actions represent past memories, triggering reactions - our habits, our programming — and they can either turn up or down the bio-electrical communication between the hemispheres of the brain, impacting your performance. For example, holding your breath triggers a fight or flight response, and in the moment there will be some restriction of communication between your right and left brains — and you will not perform as well.

So how do we manage this? What can we do about it? The simple Cross Crawl activity forces communication between the hemispheres of the brain, because the left side of the brain controls the right side of the body, and vice versa. A Cross Crawl is done by standing in place, lifting your left leg like you're walking in place, touching the left knee with your right hand, then doing that with the opposite side — raise the right knee and touch it with your left hand. Continue doing that, alternating like you're walking in place, touching the knee that's raised with the opposing hand. Do that for 10 to 30 seconds and your brain will be more integrated; your corpus collosum will be transmitting information back and forth from and to each hemisphere, and you will perform better.

Prior to getting in the car, do some Cross Crawls. You can even do them while seated in your car, on the grid awaiting to go on track. Yes, it's a bit more difficult to do while seated and strapped into your car, but even a slight movement of the leg, and touching it with the opposite hand will trigger brain integration.

This is a long topic to cover here, but I cover it in detail in my *Ultimate Speed Secrets* book if you want more information. But for now, just do Cross Crawls every morning, prior to doing anything that requires you to be at your best (at work or school, making a presentation, playing a sport, etc.), and just prior to getting in your car (like Formula One, NASCAR, Indy car, and many amateur race drivers do). That is, if you want your brain to perform at its best.

Day #113: Understand the difference between education and training.

To illustrate the difference between education and training, a friend of mine likes to use the question, "Would you like your teenager to attend a sex education or a sex training class?" He makes a good point, doesn't he? There's a difference. While "sex training" may not be a good thing, driver training is!

Information without practical tactics to change and/or build behaviors or mental programming is not very useful. But training tactics... that's what it takes to make a difference.

Books contain a lot of information. If you know what to do with that information, you can make improvements. But if you don't, and the book doesn't provide you with specific, deliberate, practice tactics, you're not going to improve.

Wouldn't it be great if every single piece of information you read also came with practical tactics for implementing it? Well, in a way, it does. Okay, it takes some effort from you, but it's almost always there.

Where?

In your head. The next time you read some information, or watch something that's meant to transfer information from the screen to your mind, take time to reflect on it. Take time to imagine actually doing it. Take time to actualize it. Take time to mentally practice it.

I seem to get two different kinds of feedback about my *Speed Secrets* books:

- "I've gotten a lot out of reading them."
- "I don't learn much from reading books."

I completely understand that people learn in different ways, and reading is not as effective for some as for others. But given the cost of every lap driven on a track versus what can be learned from reading a book before ever getting there, I think it's worth thinking about why some drivers "don't learn much from reading books."

That's the question I asked myself. And no, this is not all about books, as the question could be aimed at watching videos, driving simulators, listening to a coach/instructor/driver, or attending a course/seminar/webinar.

When I thought deeply about the question, one conclusion I came to was a lack of deep thought. It's one thing to read the words on a page, and another to turn them into an image in your mind. It's one thing to watch a video, and another to imagine oneself in that video. It's one thing to hear the words coming out of the mouth of a coach, instructor or driver, and another to "actualize" them – to live them, as if you were doing what was being said.

The difference involves imagination.

I've had numerous drivers (and non-drivers too) tell me they don't have a great imagination, and that's why they don't use mental imagery, or visualization. They claim that they've never been very good at imagining things, and therefore have a difficult time closing their eyes and seeing themselves driving on a track.

I don't believe them. The use of imagination is a practiced skill. "Use it or lose it" applies to your ability to imagine things. I'm sure that as a toddler and small child your imagination was alive and well. Perhaps you were encouraged to use it. Or, perhaps you lived in an unfortunate environment where it was not encouraged — and that meant your imagination skill withered. You were told to "grow up," or "get real." Aren't you glad that Sir Isaac Newton, Albert Einstein, Bill Gates, Steve Jobs, Michael Jordan, and Ayrton Senna didn't stop using their imagination?!

The next time you read a book (or even this eBook), listen to a fellow driver or instructor, watch a video or drive a simulator, take it beyond surface level. Think deeply about it. Use your imagination and put yourself in the picture.

Use your imagination. Turn education into training.

Day #114: The more you learn, the better you get; the better you get, the more you win. Focus on learning, and you'll win more often.

What if you looked at every session you spend on a race track as simply a learning opportunity... what if you looked at your objective as being to learn, instead of "how fast can I go?" or "winning a race"? What if you focused entirely on learning?

Perhaps you would learn how to go faster, and that would make you happy. Perhaps by going faster, you would place better in the races you compete in.

So often, when we focus on our speed, our lap time, our race result, we don't learn as much. Our focus is on the result, and not on the act that leads to the results — not on improving.

When you improve, it's more likely you'll get the result you want.

Think about that. Then focus on what you can learn.

Day #115: Improve the quantity of quality information you take in with your senses.

I received the following in an email a while ago:

> *"I've learned from you that sight, hearing, and seat of your pants are the three senses we need most to drive quickly. I have had the most difficulty developing the third one until a 24-hour race at High Plains Raceway. In last run of the day I was driving our car with a tranny oil leak hitting the exhaust and causing some smoke in cockpit. Breathing smoke led to watery eyes, cough, sore throat, etc. — including my ears completely plugged up so I couldn't hear anything. Was driving in silence. For first time was feeling full effect of Gs in a turn, vibrations in the steering wheel never recognized before — especially during braking and turns and could literally feel tires slipping in turns through my seat. This followed my first effort at night trying to be quick in the car. Couldn't really sense speed through the turns by seeing how fast the car was moving, but I knew and I think it was what I felt in the seat. For what it's worth… I don't know if this can be reconstructed to help training but it was certainly an eye opener. I'll leave it to the pros to determine if this experience can be leveraged to help others."*

I won't go into the dangers of this situation, and what the driver could have done. Instead, let's take a look at what we can learn from this.

It's fascinating that when some of our senses are restricted, the others will pick up the slack — and that ability stays with us. During practice sessions, I've purposely restricted feel for a driver by having them use very thick gloves and extra padding in the seat; I've had them use extra ear protection to restrict what they could hear. And I gave them back their full senses (normal gloves, ear protection, seat), they immediately went faster. I've been tempted to restrict a driver's vision for a few laps, but that seems just a little too dangerous! But that's what the driver who emailed me did, which is a great learning experience.

Of course, this ties into the Sensory Input Sessions that I've written and talked about extensively. Anything you can do to improve the quantity of quality sensory information your brain soaks up, the better you'll perform.

Many people ask me why one driver is better than another, and the simple answer is the best driver soaks up better quality information through their senses.

I'm not suggesting that you purposely loosen the drain plug on your transmission so oil smoke gets in your eyes so you can focus on feeling the car more! Instead of restricting one sense so the others make up for it, focus on using just one sense for a while on track, and notice how you become more sensitive to that area of your driving.

Or, look at this another way. Sit in your stationary car (you can do this in the shop/garage), with all of your safety equipment on. Then, take ten minutes to simply think about:

- What might be negatively impacting the amount of quality visual information you're able to take in.
- What might be negatively impacting what you feel.
- What might be negatively impacting what you hear.

The inability to move your head, a gap between your hips and the seat, an uncomfortable ear plug, mirrors that can't be adjusted exactly the way you want, a dead pedal that is not angled to fit you, and so on are all restrictions. Make note of everything that might have even the slightest negative impact on what you see, feel, and hear.

Then change some things. Now, in the off-season (if there is one), is the right time to be thinking about — and doing something about — these things.

The time to prepare is NOW. Of course, that could be said about any time of the year, but let's go with RIGHT NOW.

Day #116: Develop and use a Pre-Planned Thought (PPT).

How many times have you found yourself driving through Turn 4, while thinking about a mistake you made in Turn 1? If you're human, I'll bet it's happened!

The problem is that we cannot not think about something. Telling yourself not to think about your Turn 1 mistake doesn't help. In fact, all it does is reinforce it — it'll make you think about it even more. As proof, right now do not think about a pink elephant.

And while you're driving through Turn 4, telling yourself not to think about your mistake and to get focused, how far have you traveled? At speed... a long way, right?

Rather than taking up valuable time thinking about stuff you shouldn't be, develop what I call a Pre-Planned Thought, or **PPT**. A PPT is a word or phrase that brings you back into focus — now. You could use "Look ahead," "Be smooth," "Slow hands," or anything that brings you back into the moment and the act of driving.

But the key to making this work is this: you can't wait until that moment on track to start thinking about what you're going to say to get your focus back. No, you need to pre-plan it. You need a Pre-Planned Thought. The more you use it in your everyday life, the more you practice using it, the more quickly and naturally you'll regain your focus on Turn 4 — and not on that mistake you made back in Turn 1.

Day #117: Build & work with your team.

Whether you're a professional driver, club racer, or tracking your car for pure fun, you are part of a team. Sometimes that team consists of your spouse, one friend, or a few people you pit next to in the paddock; sometimes they're professional crew. Either way, they're your team.

Obviously, there are many aspects to being a good team player, working well with your team, and bringing out the best in your teammates and yourself. Here, I'm going to list a few:

- *Work on strengths*. And I'm talking about your own, and those of your teammates. Put people in positions where their strengths will be most valuable to the team. Use, and reward an individual's strengths, instead of focusing on a person's weaknesses. If an individual uses their strengths enough, their weaknesses become non-issues.
- *Understand team energy*. I'm sure you've experienced a situation where someone came into a group (a meeting, for example), and just sucked the energy out of the room. You've probably experienced the opposite, too, where one person energized a group. Be that energizer, not the de-energizer. Encourage others to share their good energy.
- *Be honest*. There's nothing like honesty (or dishonesty) to build (or destroy) trust. Interestingly, the time when you can most easily build trust, and a stronger team, is when you've made an error — but only if you step up and admit to it. You will gain more respect and trust from your teammates by doing that than you will by trying to seem super-human and never making a mistake.
- *Communicate*. A lot. Don't assume that a teammate knows what you're thinking, or even the details of what's expected of them. Tell them. Be clear — especially when it seems so clear to you. It's when it seems obvious to you that most misunderstandings occur. Communicate. A lot. Did I mention that you need to communicate a lot?
- *Lead through example*. No matter your position and role within a team, be a leader. And there's no better way of doing that than to stand up and set an example. If a few team members are speaking poorly about another teammate, don't join in. Instead, lead by example, and do what you'd like others to do if they were talking behind your back.

As I said, there's more that could be added to this list, but it's a good start.

Of course, a big challenge is dealing with a teammate who is very negative, and destructive to the team. What do you do?

First, read what I said above, and communicate with that person. Be honest. Don't expect that they know what you and others are thinking. Lead through the example of addressing the issue.

And remember, you and your teammates cannot not think about the negative person on the team. But what you can do is focus on what you want. Develop what I call a Pre-Planned Thought (**PPT**) that you can use to focus on what you want, not what you don't want (in this case, the negative person).

Day #118: Use your awareness to notice what you're doing well as much as what you need to improve.

You may be thinking that you've gotten worse as a driver over the past little while.

As drivers, as instructors, the more we improve, the more we tend to be tougher on ourselves. That has to do with being more aware, which is a good thing. However, if all your newfound awareness does is make you more critical of yourself, it may not be so good.

But if your awareness includes a healthy dose of what you're doing right — in balance with what you're doing wrong — then that's not so bad.

The next time you feel you're not improving, or even feel as though you're going backwards, I suggest pulling out a piece of paper and start writing down all the things you're doing well. If you're honest with yourself, and give yourself the credit you're due, you might find that you're doing a lot right. And when you focus on what you're doing right, you'll likely do more of that right stuff. And if you do more of the right stuff, it's likely you won't have time to do the wrong stuff.

Be aware, but also be aware that you may be too hard on yourself. If you recognize that, start being aware of what you're doing right.

Day #119: Manage your apexes, apex angles, min speeds, corner priorities.

Let's start with a few definitions. Admittedly, these are my words, my definitions, but they're based on many years of discussion with many smart drivers, instructors and coaches, so they should stand up fairly well. As with any definition, though, there are subtleties that one person may add, delete or change — rightly or wrongly.

Apex: The Apex can be thought of as that point where you're no longer driving into the corner, but you're beginning to drive out of it. It's the place where the tires on the inside of the curve you're driving through clip past the inner edge of the track, the most inside point or area of the corner. Using the basic "the line through a corner goes from the outside edge of the track, to the inside, and back to the outside" advice, the Apex is that "inside" point or area of the cornering line. As I said, an Apex can be a point, or an area of some small distance. An Apex can be right in the middle of the corner (geometric Apex), earlier than that (early Apex), or after that (late Apex). The Apex is also sometimes called the "clipping point."

Apex Angle: This is the angle of the car as it passes the apex point, in relation to where you want to go, towards the Exit point of the corner. Often, a driver will clip past the ideal apex point, but the car will still be pointing in a less-than-ideal direction because it's not at the right apex angle.

Clipping Point: As I said, the Apex is also sometimes referred to as the "clipping point," in reality, they may be slightly different, but for our purposes here, let's use Apex and Clipping Point interchangeably. Personally, I don't think the differences are great enough to worry about. Having said that, I recommend you watch Nico Rosberg talking about driving COTA, and specifically Turn 1 in this video: https://youtu.be/CamWIp5i9_E. Go to about the 3-minute mark to notice how he separates the Apex from the Clipping Point.

Min (Minimum) Speed: There is the point in every corner where you're at your minimum velocity — the min speed. Typically — but not always — it's just about the EoB (End-of-Braking) point. That is, it's most often after the turn-in, around where you finish braking, and just before you begin to apply the throttle.

WHAT TO DO

Let me be very clear about one thing: your min speed is not necessarily at the apex or clipping point of a corner. Sometimes your min speed will be before the Apex (usually in slower corners — the "in slow, out fast" type), sometimes at the Apex, and sometimes after the Apex (usually corners that don't have much of a straightaway afterward — corners where exit speed is less important).

Speaking of types of corners, let's review what Alan Johnson wrote in his *Driving in Competition* book way back in 1971 (still one of the classic and most important driving books). He identified three types of corners:

- Type 1: Corners that lead onto a straightaway

- Type 2: Corners at the end of a straightaway
- Type 3: Corners in between other corners

Type 1 corners are all about exit speed, and are usually the most important of all the corners on a track. The priority for Type 2 corners is entry speed because there isn't a straightaway afterward of enough length to matter. And your focus for Type 3 corners — which is usually a series of corners, such as Esses or a chicane — is to set up for the final part (which is a Type 1 corner).

This identification also led to the over-simplification of which corner is the most important on a track, saying that the corner that leads onto the longest straightaway is it. Like I said, this is a bit of an over- simplification, because that's not always true. It works a lot of the time, and is a great place to start, but sometimes other corners are most important. In fact, one way of prioritizing the importance of corners is to simply look at which corner you're losing the most time in.

Often, the corner leading onto the longest straight is a relatively slow one — and that's the one that everyone tells you to focus on.

But think about this. Which corners do you (and other drivers) usually have the most difficulty with? The slow corners or the super-fast ones? I bet an honest answer is the fast corners. Why? Because they're fast! They're scary! And if you answered "the slow corners" I suspect you're missing something — like big speed in the fast turns. So much so that you don't even realize it.

If you want to be really fast, focus on the fast corners first. You know, the scary fast corners.

See, the difference between the speed of the quick and the super-quick drivers in slow corners is usually not much — a few MPH at most. But in the really fast corners it can be 10 MPH or more. How do I know? Because I've seen the data of many hundreds of drivers in this situation over the past couple of decades.

So, the next time someone tells you to focus on that slow corner leading onto a long straight, nod your head in agreement. Then head out on the track and focus on maximizing your speed through the fastest corner on the track, whether it leads onto the longest straight or not (by the very nature of fast turns, it will almost certainly lead onto some type of straightaway). When you have an advantage of a couple of MPH over every other driver, then you can spend a little more focus on that slow corner.

When looking at data, your min speed point (where your min speed is in the corner) is an important piece of information to consider. What's most important is not just how fast it is, and where it is, but also how long you spend at it. In some corners, a lower min speed helps with your exit speed ("in slow, out fast"); in other corners rolling momentum through it is more important, and therefore a higher min speed is what you're looking for; some corners reward very little time at min speed; and with others, spending a little

more time at min speed is okay (this typically goes along with a higher min speed, or rolling more momentum).

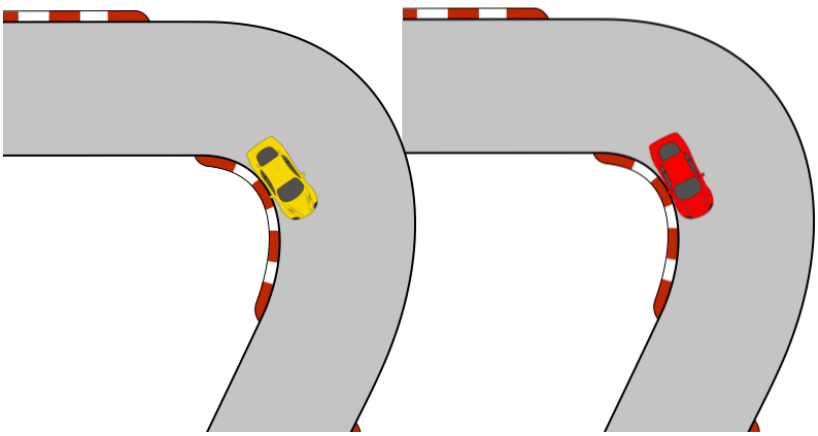
Note: When looking at a data trace of speed, notice the difference between corners where the speed trace looks like a “V” (little time spent at min speed), and when it is more of a “U” (more time spent at min speed). Typically, slow corners of less than approximately 60 MPH reward a “V” shape, and faster corners reward a “U” shape. Thank about what this is telling you as a driver: slow corners reward getting the car slowed, turned, and accelerated out of them; faster corners are more about rolling momentum, slowing the car as little as possible.

Since we’re talking general guidelines (but not rules… unless you plan to break them every once in a while), here are a few more:

- The min speed in Type 1 corners is usually prior to, or at, the Apex.
- The min speed in Type 2 corners is usually at or past the Apex.
- The min speed in Type 3 corners is… well, it depends on what’s coming up next, but usually in the middle of this “connecting corner,” because you want to be accelerating toward, through, and out of the Type 1 corner that it’s connecting to.

Let’s go back to the angle your car is at when clipping past the Apex. It's not just where you apex that matters, but also the angle your car passes by the apex.

One thing that often comes up when I walk a track with a driver is the exact location of the apex. Drivers will point out where they're apexing, but the most important discussion I like to have is not the exact location of the apex, but the angle, or direction, that the car is aimed in at the apex. That's more important, and something many drivers don't think enough about.



For example, compare the angle the two cars in the illustration. You could argue that the yellow car is apexing earlier, but a better way to think about it is that the yellow car is on a different angle than the red car as it’s apexing (the Apex being the second red panel of the inside curbing). In this case, I'm not going to say that one is necessarily any better than the other — but they are different.

It's important for you to recognize the angle that your car is on while passing the apex. To change the angle, you can turn in earlier or later, you can turn in quicker (more "crisply") or slower, you can use your brake release to change the balance of the car to help it rotate quicker or slower... or some combination of these.

Just because you're "hitting the apex," that doesn't mean you're on the right line, as the angle you pass by it is even more important.

IT'S ALL IN YOUR HEAD

As much as there is a technical aspect to this topic that these guidelines and techniques apply to, to me the mindset is more important. For Type 1 corners, the mindset should be on exit speed; Type 2 corners the mindset is on entry speed; Type 3 corners is all about discipline in the middle to maximize either the exit or entry (usually the exit). The technical aspect is around how to do that, but without the right mindset, the "how to" doesn't matter much. In fact, I'd say that with the right mindset, most drivers can apply the right techniques to make the right things happen. But that will rarely happen if you don't understand why you're doing what you're doing.

In the past, I've written and talked about the Why behind the line you drive through any corner. Many drivers drive the line they do because that's the line they were told to drive! And who's telling them to drive this line is a fellow driver, instructor, coach, or some other human… and not their car. In the end, it's your car who should be telling you what line to drive, not a person who is basing it off what they do, or what they were told by someone who was told by someone else, who was told by someone else, who was told by someone else….

I'm not saying the line will be massively different from car to car. In fact, it's usually not very different. But your apex angle, where your min speed is, how long you spend at min speed, and how you prioritize corners does depend on your car and what it needs. Yes, "it depends!" So, you have to listen to what your car is telling you, and what it's asking for. And again, that'll never happen if you don't have the right mindset to each and every corner, because each and every corner wants something a little different.

When I say you have to listen to your car, that also means you need to listen to what your overall lap times, your segment times, and the data says. There are times when something feels faster, but the data says otherwise, and there are times when a lap time is faster, but the segment time through the specific corner you're working on is slower. That's why you need to look beyond the obvious, and truly listen and read into the information you have to see what's working, and what's not working.

Apexes, apex angles, clipping points, min speeds, corner types, and mindsets. Who knew there were so many things, all going on at the exact same time, that you need to manage just right. Of course, that's what makes it so much fun, right?!

Day #120: Don't be so quick to turn off those driver aids.

When I'm asked to drive a car around a race track, some drivers are surprised that I don't immediately turn off the stability and traction control. There are two big reasons why I rarely turn them off.

First, it's out of respect for someone else's car. And remember, this is coming from a person whose answer to the question, "What's your favorite car to drive on a race track?" is "Someone else's."

It's the second reason that I think is most important — and the one I want to talk about here.

As a performance driver, I can use the stability control to maximize my performance. I don't think about it as something that intrudes — it's a limit. My job is to drive within the car's limits, and if the stability control cuts in where I could possibly drive it faster with it turned off, well... that is the car's limit. Again, my job is to drive the car to its limits, no more, no less.

And just like I can drive all day long on a track without activating the ABS, there's no need for me to turn it off. In fact, it allows me to use it... if only just the slightest amount. That's what I like to do with stability control — lean up against it.

If you've braked hard and gotten the ABS to activate just the slightest amount — perhaps feeling just a pulse or two of the pedal - that's what I mean by "leaning up against it." It's not pounding the pedal and letting the ABS do all the work, and it's not so light that the ABS doesn't ever activate.

When I can do the same with stability control, that means I really know the limits of the car I'm driving. And I can use it to my advantage in places. Imagine coming out of a corner just below the limit, knowing that there is a place on the track that has less grip. As I approach it, I could limit my acceleration to anticipate the lack of grip coming up. Or, I could anticipate it, and know that the stability control is going to help me through it.

That's knowing the limits of my car, and using the driver aids. That's leaning up against them.

If a driver is unable to do what I'm talking about here — knowing the limits of their car with the driver aids on — there is no way they should turn them off.

As an instructor, I don't want a driver I'm coaching to ever turn off any of the driver aids unless I'm 100% convinced that they know exactly when things like stability control are helping them. If the driver can demonstrate to me that they're doing what I talked about — leaning up against it and using it — and knows exactly when it's activated, then I'm okay with them turning it off.

If the driver is able to verbally tell me exactly when the stability control is activating — at speed — and it matches with what I feel and see (when the light comes on on the dash), then that's a driver who is capable of using the system. And I'm okay with them turning it off.

But I'm still going to ask "Why?" If a purpose for track driving is to understand the limits of one's car, shouldn't the driver drive it in the same way that they do on the street?

One final thing to consider: Many cars are designed to handle well with the driver aids engaged, and when you turn off the electronic systems, they're not particularly well-balanced.

For example, if you turn the ABS off on many modern cars, the brake bias (the balance of braking force, front-to-rear) would not be right. The same is true with stability control.

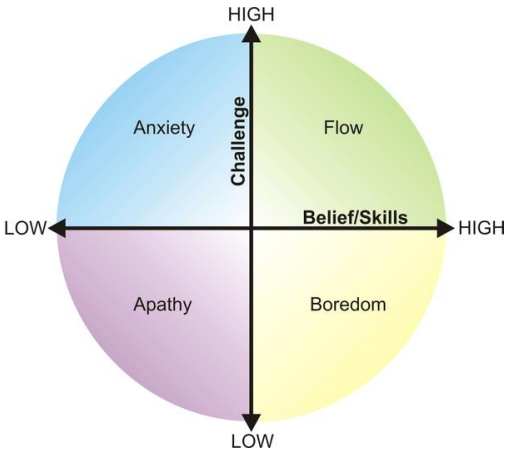
In fact, there are some very good handling cars that have suspension geometry that is a big compromise to suit packaging (larger storage areas, room for fuel tank, etc.), and yet they handle well because the electronics look after things. In other words, the designers sacrifice ideal suspension design for other factors, knowing that they can "fix" it with the electronics. That's why a very good handling car can turn into a beast when the electronic nannies are turned off. Now imagine what might happens if you're not fully prepared for that change....

Day #121: Challenge + Belief = Zone.

Think of the times that you've performed at your very best, when you've felt as though you were in the zone.

I'll bet that you went into that activity, at that time, feeling challenged. Not overwhelmed-challenged, but challenged just enough.

I also bet you felt you could handle the challenge. You believed you could take the challenge on and come out on top.



Research has shown that people in all walks of life perform at their best when they feel challenged, and yet have a belief in their abilities to handle the challenge.

If you look at the illustration here, you'll see an X-Y graph, with the Y-axis representing a person's sense of challenge. At the bottom, there is little challenge; towards the top, the challenge increases. The X-axis represents a person's belief in their ability to handle the challenge, with little belief at the left, and a strong belief to the right.

Note the four quadrants:

1. The bottom-left, when a person does not feel challenged, and doesn't have much belief in their abilities, they feel apathy towards the activity. They don't even want to do it.
2. In the bottom-right, when challenge is still low, but the person believes strongly in their abilities, there's boredom. It's a "Oh, this is easy. I don't even need to put any effort into this" kind of situation.
3. At the upper-left, there is lots of challenge, but a lack in one's beliefs. This is a situation where the person isn't confident in their ability to tackle the challenge, and that leads to anxiety.
4. Finally, in the upper-right, where there's both a sense of high challenge and high belief, the person performs in the zone, or "flow."

"So what?" you say. What this tells you is that if you can frame an activity, like performance or race driving, as highly challenging, while you strongly believe you can handle it, you're more likely to perform at your best. Sure, there's no guarantee that this will trigger a "zone performance," but I can guarantee you that if you lack either a sense of being challenged, or the belief that you can handle it, you will not get into the zone.

The great thing is that you can control this. If you're faced with a track session that you don't think will be difficult (not very common, but possible), take time to consider all the techniques and skills you need to put into play at just the right time to be successful. Think about what makes it challenging. Make it challenging.

Then, take time to consider how you have the talent, knowledge, and skills to handle the challenge. Logically think through why you should believe in your ability to handle the challenge. Don't minimize the challenge by telling yourself that it will be easy, but put it into perspective so you can see why you should feel confident.

When you balance the challenge with your belief in your abilities, and hit the track with this mindset, you're more likely to perform better.

And yes, this applies to more than just driving….

Day #122: Manage your driving personality.

Every race driver has a reputation, which is based on the perception of past performance and behavior.

The same thing applies to non-competitive track day and HPDE drivers. It applies to you. You have a reputation as a driver. Your "driving personality" is known to other drivers, and it has an effect on how they drive or race around you.

Have you ever used one of those personality traits instruments (they're not really a "test"), such as Meyers-Briggs, DiSC, Birkmann, Big 5, Enneagram, etc.? The idea behind all of these is to discover your true basic behavioral traits. In other words, how you will behave in certain situations, often when under stress or pressure. And as you know, driving on a track can cause some stress and pressure (fortunately, it's usually the good kind of stress and pressure).

You probably know someone who is a very laid back, calm, relaxed, and patient type of person off the track, but once they put their helmet on, they turn into someone very different. Or someone who seems very confident and assertive, but on the track is a little tentative. Hey, one of those drivers might be you!

With most of these personality instruments, traits such as introvert/extrovert, dominance/non-dominant, analytical/intuitive, patience/impatience, conformist/non-conformist are part of the mix, with each one being on a spectrum. As you know, how you behave in one environment may not be the way you behave in another, and yet we do seem to have a base from which we adapt to different situations.

From working in this area with many, many drivers, and lots of observation and even testing of various levels of them, the ones who are most successful — whether in competitive or non-competitive driving —are the ones who can adapt. Champions have the ability to be more or less patient, more or less dominant/assertive, more or less analytical, and so on. For every driver, there comes a time when you need to dial up your level of assertiveness, and a time you need to dial it down; times you need to be patient, and times you need to be impatient and get on with the job; times you need to focus on the details, and other times when you need to let go and trust your gut (your mental programming) to just drive.

And here's the thing: if you always behave in one way on the track, you're going to get a reputation for being that way. That may be a good thing, or maybe a bad thing. What's most important is to know that you can learn to be more adaptable in how you behave. You program your mind to adapt to what will best suit the situation.

Race drivers who have a reputation of being smart, tough racers typically have learned to adapt — they know when to give up a spot, and know when to go for it. Aggressive drivers tend to always do the same thing: be aggressive, and that often leads to crashes or other drivers racing them in a certain way because that's their reputation.

Think about the top five drivers in any race series (F1, Indy car, IMSA, WEC, NASCAR, WRC, etc.) and ask yourself what the reputation is of each driver. How does the way they behave on track impact the number of races they win? What would happen if any one of them was able to dial up or down one or more of their behavioral traits?

What's your reputation when you're on the track? If I was to ask other drivers you shared the track with to describe the way you behave when driving, what would they say? Are you okay with your driving reputation? If you could change anything, what would it be?

If you want to change the way you behave, you can use mental imagery to do so. Close your eyes, relax, and put yourself in different scenarios where your behavior will impact the outcome. Then, imagine yourself adapting your behavioral traits to suit the situation, and how, over time, your reputation changes to what you truly want it to be.

Day #123: Avoid the Change-in-Speed crime.

I recently witnessed a crime. What was it? Over-slowing a car, then assuming the limit was THE LIMIT. It wasn't. In the three separate corners where I observed the offending driver, the car was being driven approximately 4-5 MPH under what the car was capable of carrying through these turns. And yet, the car reacted like one being driven at the limit, and the driver swore up and down that he was a "driving God" who had it on the ragged edge.

What was going on?

Imagine a 90-degree right-hand corner with no banking, consistent pavement throughout, and which leads onto a straightaway that's long enough for most cars to get into top gear. Most drivers use 3rd gear, and the minimum speed (vMin) is about two-thirds of the way between the turn-in point and apex — just slightly after the EoB (End-of-Braking, the point where the driver's foot has completely released the brakes — the point where the brake lights would go out). With an EoB about half to two-thirds of the way to the apex, that means there's a fair amount of trail braking going on to help turn the car (rotate the car).

In this imaginary corner (not unlike one in which I observed the above-mentioned driver), the ideal vMin is 60 MPH. By ideal, I mean that if the best driver in the world spent a day practicing and testing all sorts of setups, that 60 MPH would be as high a vMin as he could carry.

Before moving on, let's think about what the ideal vMin really means. I define it this way: It's the minimum speed somewhere between the turn-in and exit/track-out points that is the ideal compromise between being slow enough to allow the driver to get back to full throttle as early in the corner as possible, but not so slow as to make it so that the early full throttle acceleration cannot make up for the slow vMin.

We've all been told that "In slow, out fast" is the way to drive, and that whoever begins accelerating first is going to turn the fastest lap. But one way of beginning to accelerate early is to enter the corner at one MPH, right? You could turn in and stand on the throttle! But you would never be able to make up for the extremely slow speed, no matter how early you got to full throttle, and you'd be slow heading down the next straight.

On the other side of this challenge is entering the corner at a speed that results in not being able to get to full throttle until very late in the turn. Despite carrying more speed through the corner, the lateness of full throttle means the straightaway speed will not be what it could be, and therefore the overall lap time will suffer.

Looking at these two extremes, it's easy to see that there is an ideal – a vMin that will result in the maximum speed down the ensuing straightaway. With a vMin that's too slow, no matter how early one gets to full throttle, the top speed will be slow; with a vMin

that's so fast that it delays the ability to get to full throttle, that will result in slower top speed.

So, we now agree that each corner has an ideal vMin. But there's one more factor that the driver I mentioned in the beginning was guilty of not managing in an ideal way: Change-in-Speed (CiS).

We all know that when we brake, more load is put on the front tires, and they gain traction. Accelerate, and the rear tires gain grip due to the additional load transferred onto them. Go around a right-hand corner and the left side tires gain load and traction. Also, the tires that become less loaded lose traction, relative to the ones that are more loaded.

We also know (but maybe not so clearly) that when load/weight is transferred onto a pair of tires, they do not gain as much traction as the tires that become unweighted lose. If you haven't dug into this concept yet, just stick with me here (watch my Why Smooth Is Fast video here for a simple explanation, or talk to a tire engineer for the in-depth, super-technical explanation). What you need to know is that a car that's balanced — as if neither braking, nor cornering, nor accelerating — has more overall traction than one that is doing any of those things. When you brake (because weight has transferred forward), your car has less overall grip than before you braked. Same is true when cornering. Yes, just when you need all the traction you can get — when braking, cornering or accelerating — you have less.

How much less is dependent on how much weight/load has been transferred. If you drive smoother, you cause less weight transfer, and therefore have more traction. That, in a nutshell, is why smooth is fast.

Okay, back to the criminal… er, I mean driver who I mentioned earlier. Because he over-slowed for the corner — down to, let's say 56 MPH in our imaginary corner — he then felt the car had traction. And he was right. So what did he do? He accelerated. Not in an unsmooth kind of way, but it was certainly fairly aggressive. And this was done without him really thinking about it. He just felt he had grip — he felt the car was not at the very limit at that very instant passing through the vMin area of about two-thirds of the way from the turn-in point to the apex. Driving instinct then kicked in and told him to get on the throttle and use more of the car's capabilities. It was this over-slowing and then re-accelerating that caused the CiS problem — too much Change-in-Speed. The delta from low speed to higher speed was too great (imagine the weight transfer and the car going from a nose-down attitude to a rear-down one), and that caused the car to have a lower overall limit of grip.

And this is where the real crime happened: The driver felt he was driving the car at the limit, and in a way, he was. But he had created an artificially-low limit. Had he carried 4 MPH more through the vMin area of the corner, the car would have been better balanced with more traction, and he would have been driving the car at its true limit. Oh,

and he would have begun accelerating from 60 MPH, rather that 56 (guess which would result in a higher straightaway speed?).

I suspect that you're now thinking, "Okay, if that's the case, how do I know exactly what the ideal vMin truly is?" No, you can't just look at your data system and have it spit out a number for each and every corner you drive. And even if you could, do you think you'd be able to look at a speedometer while turning into a corner to make sure you were driving at the right speed?! Ha!

The only way of determining what the ideal vMin is for each corner is to do some experimenting. You have to ease off the brakes a tiny bit earlier and roll more speed through and see if it delays where you're able to get back to full throttle. You can check at some point along the following straightaway what your speed (or RPM) is to see if it's helped or hurt it. Then you need to try slowing a bit more – hang on the brakes slightly longer – and see if you're able to get to full throttle so much earlier that your straightaway speed is higher.

With that in mind, I'll leave you with two cautions: First, just because carrying more or less speed through a corner doesn't work once or twice doesn't mean that it won't if you get the technique right. Doing the right thing imperfectly doesn't make it the wrong thing, nor does doing the wrong thing extremely well make it right. When experimenting with managing your corner vMin, give yourself enough time to do whatever you're doing very well before you make a judgement on it.

Second, and finally, what comes after a corner is also a very important factor. So important, in fact, that I'm going to gloss over it a bit here, and instead dig into it another time. But for now, keep in mind that if the straight (or relatively straight) section after a corner is relatively short, the solution to what is the ideal vMin speed will be different than if the straightway is a mile long.

As you can see, this track driving stuff is not easy! That's a good thing. And that's why I say, "There's always more. Keep learning and having fun!"

Day #124: If you surround yourself with the right people, your financial challenges will be taken care of.

I'm often asked by young drivers how best to market themselves, to sell sponsorship. My response is always the same: surround yourself with the right people.

I wish someone had told me that before I had to learn it the hard way. I spent too much time and effort on selling sponsorship. I should have spent that same time and effort meeting people — the right people — building my network. When I finally learned that lesson and focused on meeting and building authentic relationships with people who had influence, funding the drives that I needed for my career to progress became much easier (but I didn't say easy).

Part of the trick is determining who the right people are. I've heard it said that you are a result of the five people you spend the most time with. If you spend the bulk of your time with people who have the network and the wherewithal to help fund your career, the likelihood of succeeding will be increased.

This applies equally to amateur racers and track day drivers as it does to pros and pros-in-training. The only difference is what you're looking for. For the pro, it might be millions of dollars; for the amateur, it might be some product.

Think about who you know, who you could get to know, and who would have a positive influence on your racing endeavors.

But never, ever be a "user" of people. Give them what they want. Provide value to them. Be authentic. Be useful to them. Focus on giving and every now and then, you'll get something in return.

Day #125: Use a delicate touch with the steering wheel.

In the 1977 movie, *Bobby Deerfield*, Al Pacino plays the part of an American (from Newark, NJ) Formula One driver. In one scene, he's giving a ride to a young woman (played by Marthe Keller - and surprise, surprise, she becomes a love interest in the Hollywood movie) who has just checked herself out of a mental hospital. As he's driving his Alfa Romeo through the beautiful European countryside, she begins a conversation.

Keller (in a French accent): *"You have a delicate touch. When you race, can you have such a delicate touch?"*

Pacino: *"Well, you hold on tighter… but there's a kind of delicacy to it, yes."*

Keller: *"It must be difficult to be delicate when death could be so close. It's even very feminine, I think."*

Pacino: *"What?"*

Keller: *"The way you touch the wheel."*

Pacino: *"What do you mean feminine?"*

Keller: *"It's alright. The best men are always feminine. In some ways."*

Pacino: *"In what ways."*

Keller: "*The ways that are feminine. I don't mean that you're a homo. You're not homo, are you?"*

Pacino: *"Nope."*

Keller: *"I did not suggest that you're a homo, I have nothing against homos."*

Pacino: *"That's good to hear."*

Keller: *"I have always heard that the man who racing car, that the car is really nothing but an extension of the man's penis. Do you think that is true? It seems like a fair question. The motorbike, for example, I'm always hearing that the motorbike is an extension…."*

Pacino: *"You know, I know you're always hearing. But you don't know anything about cars. See, and you don't know anything about racing. Do you know anything about racing?"*

Keller: *"Only, it seems boring. Crazy. And that I'm told that the car is really nothing but an extension of the man's penis. Aren't there homos in Newark?"*

As the camera closes in on Pacino you notice his delicate touch on the steering wheel, and he responds, *“Oh yes. We have many. They’re all over the streets. We build them there. We have homo factories. Make you feel better?”*

Okay, so why the movie script (with its outdated language)? While the movie was panned by critics, and Hollywood is not known for accurately portraying what race drivers (gay or straight!) really do, that scene emphasized an important driving technique: holding the steering wheel with a delicate grip.

The same can be said of the shifter (Remember those? They were what we changed gears with before paddle shifters!). Move the shifter with an open palm or with your finger tips — do not grip the shift knob.

My challenge to you is to lighten your touch on the wheel and shifter. Practice when driving on the road, and be aware of your grip when on the track. Be delicate.

By the way, the movie does have some excellent footage of Formula One racing from the 1976 season, and it’s worth watching just for that. But not much else.

Day #126: Practice being in the moment.

A lot has been said and written about how our brains are made up of two hemispheres — and I've done a lot of that talking and writing! The main message is that the left hemisphere is responsible for language, logic, facts and details, while the right hemisphere handles intuition, creativity, art, and big picture.

But a part of this division of duties by your brain that is not so well-known is that the left hemisphere tends to deal with the past and future, while the right hemisphere is more focused on the here and now — the "in the moment" stuff.

Think about your best performances behind the wheel of your car on the track. Were you thinking about what happened in the past? Or the future? I suspect not. In fact, I bet you were very much in the moment, focused on the here and now, on the act of driving.

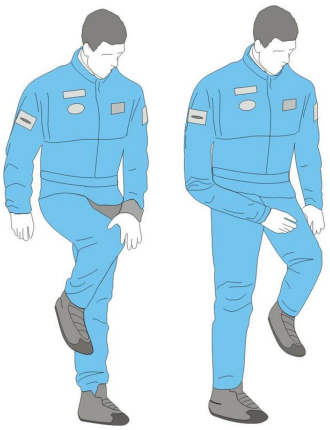
What this tells us is that driving at your best requires that your right brain is fully engaged in the activity. There are so many things we do in our lives that engage our left hemisphere, but, for many of us, less so our right brain. And that's what I want to encourage you to do — engage your right hemisphere.

You're probably aware that the left side of your brain controls the right side of your body, and vice versa. For example, if you know of someone who has had a brain injury or a stroke in the left hemisphere, the right side of that person's body is affected.

While it's your brain that controls your body, your body can awaken or engage your brain. So if you use the left side of your body more, it will awaken your right hemisphere more.

I encourage you to do more to engage your right brain, and to spend more time in your right brain. Trust and use your right hemisphere more — use your intuition and creativity, and practice seeing the big picture. And use the left side of your body more. Brush your teeth with your left hand, practice writing or drawing with your left hand, kick a soccer ball with your left foot.

Ultimately, what you're looking for is more balance between the hemispheres, and sometimes that requires over-doing it with the side that is often under-utilized. Our modern world seems to reward left brain thinking, so it's easy to let our right brain get lazy.



An exercise that I recommend to all drivers to use as a pre-drive ritual — one that will integrate your brain and make it operate more as a whole — is called a **Cross Crawl**. Here's how it works.

While standing, raise your right leg, bending it at the knee, and

bring your left hand over and touch the right knee. Return to standing. Then raise your left leg and touch the knee with your right hand. Return to standing, and then continue, alternating sides. You will find yourself marching in one place while alternately touching your knees with your opposite hand.

There is a reason this exercise is called a Cross Crawl. When babies first begin to crawl, they most often do it in a unilateral motion. That is, they move their right hand and leg forward, then their left hand and leg, and so on — one side moves, and then the other. After a few hours to a week or more for most babies, they change to a cross-lateral crawling movement, where they move the right hand with the left leg, the left hand and right leg, and so on. This cross-crawling movement is the first step in the integration process of brain development.

Children who do not do enough cross crawling (often because they go almost directly from unilateral crawling to walking) may miss out on becoming fully brain-integrated at an early age. In some cases, this leads to the child being slightly uncoordinated, or even having some type of learning challenge. By simply using the Cross Crawl exercise, many children have been able to “recover” from learning difficulties and have become far more physically coordinated. From a driver’s point of view, it triggers better performance by enabling the brain to perform at a higher level. This exercise is an extremely powerful way of “waking up” your right hemisphere, and ensuring that your brain works as a whole, rather then being either overly left- or right-brained.

Finally, spend more time in the present, in the moment. When you find yourself thinking a lot about what’s happened in the past (the stuff you can’t do anything about, anyway), or you’re worrying about something in the future, stop…. Click into the moment. Think about, and focus on, what’s happening right now. Notice the temperature where you are right now; listen to the sounds; take a big breath and smell what’s around you; feel what’s going on; look around and become more visually aware of everything around you. Practice being in the moment.

As they say, “Use it or lose it.” This certainly applies to your right hemisphere.

Day #127: Beware of your lying eyes.

The following scenario is something I've experienced many, many times over the past thirty years or so of coaching.

I'm talking with a group of drivers, and I ask them where they brake for a certain corner.

Driver #1: "At the 300 marker."

Driver #2: "At the 350 marker."

Driver #3: "At the 250 marker."

At that point, Drivers #1 and 2 look at Driver #3 as if they're either lying, or they're some kind of driving deity, achieving the impossible of braking that late.

When I look at the data, or stand at the side of the track and note where brake lights come on, I find that they're all braking at the same place. How is that possible?

Driver #1 applies the brakes (the BoB, or Begin-of-Braking) right when the 300 marker is lined up with his shoulders.

Driver #2 notices what would be the 350 marker (if there were one, but it's that point in the middle between the 400 and 300 markers), but by the time they actually react and apply the brakes, they're at the 300 marker. What registered in the driver's mind was that they were braking before the 300 marker – their perception of their BoB was before the 300 marker.

Driver #3 sees the 300 marker before arriving at it, but at the angle they're at when initially applying the brakes, they assume their BoB is at the 250 (between the 300 and 200 markers).

For each of these drivers, their perception of what they were doing did not match reality. This applies to practically all drivers, and not just for where their BoB is. It also impacts where a driver turns in to a corner, their apex, and so on — in fact, to all references.

For this reason, you need to use a bit of caution when another driver, or an instructor/coach, gives you advice on where to do something on track. I wouldn't suggest that a fellow driver would never lie to you, but sometimes the advice or information you get from others is inaccurate.

This perceptual change can happen to you alone, too. For example, in one braking zone you may notice a reference lining up with your shoulders and you pass it; in another, it may be easier to spot something beyond the actual reference, and therefore it registers more strongly in your mind. If you're consistent, lap to lap, and session to session, that's okay. It's all relative to your own consistent baseline.

However, I've seen those who are not consistent with this, and that can lead to dangerous situations. I've found that drivers who always make notes on a track map are more consistently accurate with the references they use. Of course, there are exceptions to this, but it applies more often than not.

Some of how you see references is due to, well… how you see references. Some drivers use their peripheral vision more than others, and they tend to act based on when a reference lines up with the front of their car, or their shoulders. Other drivers tend to triangulate references: they see something in the distance, a reference point to the side (it may not be directly beside them), and compare the location of those to where they are (their head/eyes) right now, in this moment. One method is not really any better than the other, they're just different. In an ideal world, you would use both.

Interestingly, drivers who use the "it's right beside me now" approach tend to report when they brake for a corner, for example, "at the 300 marker." Drivers who use the triangulation method tend to say things like, "when it looks right." I do have to say that learning to use the triangulation method could be more difficult if it's not something you would naturally do, but it can be learned (just as the "it's beside me now" method can).

Which method do you use? Would your driving improve if you combined these two ways of viewing references?

This is definitely part of your visual skills, and it can and should be practiced. Like practicing any skill or technique, the more deliberate you are, the better you'll get at it. The next time you're on track and wanting to improve your visual skills, see if you can alternate between the "it's beside me" and triangulation methods. If you do, over time you'll notice that they converge and you'll be an even better driver.

Day #128: Just breathe.

The next time you're at the track, take at least two sessions and devote them to doing only one thing: Breathing.

Yes, just focus on breathing. In through the nose and out through the mouth, aiming for about three to five seconds for each inhale and exhale. Ideally, it would be at least five seconds, but driving fast often results in faster breathing. So, three to five will do.

Now, I want you to focus on doing this in very specific places on the track...

- Just prior to heavy braking zones
- As you turn into corners, especially fast ones
- As you get to full throttle exiting corners
- In the middle of each straight

But wait, here's a super-important part of this process: Take a map of the track you'll be at (download free maps at TrackMaps), and circle with a red pen those specific places above on the track. Now, close your eyes and imagine approaching each of those areas and inhale for three to five seconds through your nose, then exhale through your mouth for three to five seconds. Then, mentally drive the next place on the track while emphasizing your breathing. Do this over and over again for at least ten minutes. Each time you inhale, say to yourself, "Breathe."

Now, go to the track. Pull your track map out just before going on track, and review exactly where you're going to focus on your breathing.

Then drive. Use this trigger word, "Breathe," to remind yourself to stay on task. You only have one job to do — drive the track while emphasizing your breathing — for at least two full sessions.

After your on-track session, take a few minutes to make note of how that session went. Only check your lap times after the sessions. Were they faster or slower? How did you feel? Make note of whatever comes to mind. This is a critical part of the process.

Did the exercise help or hurt? Were you faster or slower (only compare lap times after the fact, as your focus is on breathing while driving)? How did it feel? Did it spill over into your next sessions, where you may have still been breathing more? Did you find yourself driving slower because you were too focused on breathing? How well did your next sessions go?

Just breathe.

Day #129: Use every tool available to you to improve your chances of performing better (and winning).

In-car video, data acquisition/telemetry, track maps, notes, debrief forms, other drivers and engineers, instructors and coaches, observation from a corner, stopwatch / timer, track walks, TV coverage.... Let's see, what am I missing?

All of these are the tools I'm talking about.

There was a time in my racing career where I — like many drivers — thought that driving was something that I either had a talent for or didn't. Fortunately, though, at an early age I noticed something: the best drivers seemed to work at being the best. They actually did things to improve. Of course, at that time (in a previous century!) drivers didn't have as many tools available as they do today. Even World Champions from the 60's used tools — a notebook, in the case of Graham Hill.

"Over the years Graham has jotted all the problems he has encountered into a little black notebook. When he goes to a circuit he simply flips the pages and finds out what problems to expect." — Mike Hailwood, speaking of Graham Hill (from *Hailwood* by Mike Hailwood and Ted Macauley; Cassell, 1968).

I'd love to be able to honestly tell you that I have a perfectly organized notebook with notes and records from every track that I've ever driven, but I can't. However, I can tell you that I have notes and records from every track — just not perfectly organized. I have track maps with notes scribbled on them (both on paper and digitally), some data acquisition print-outs with areas circled and annotated, and at times I've kept track of URLs of what I think are the best in-car videos (my own personal paper version of Google!).

Finally, while I don't want to sound too old-fashioned, I want to stress the value of writing things down by hand. There's research to support my belief that when you write a note or draw a picture, it sticks in your mind better than if you simply watch a video. Do you want to know how I remembered that fact? I wrote it down.

“People live up to what they write down.” – Robert Cialdini, from the book *Influence*

Day #130: The throttle is not an on-off switch - until it is.

How many times have you been told to squeeze on the throttle, rather than flicking it on like a light switch? More than once, I'll bet. As with most advice for performance drivers, it's a great guideline, a great place to start. But you can take that too literally.

There comes a point, in most cars, where you should turn the light on. Ahhh, but where is that point?

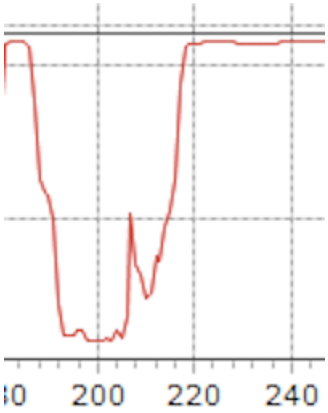
After years of studying data traces and coaching drivers, I'll share a generalization with you: Throttle movement below 50 percent have a noticeable effect on the car's weight transfer; throttle movements beyond 50 percent don't impact weight transfer as much.

Obviously, this depends on many factors, including engine torque. But once you've squeezed on 50 (or more) percent throttle, hesitating is doing little to control the balance of the car. At that point, you might as well go all the way to 100 percent — flick the switch on.

Most of the balancing of the car in a corner, where you're easing on the throttle to transfer weight to the rear, or breathing it to load the front, is done with throttle movements at or below 50 percent. So if you've applied more than 50 percent, managing the balance of the car is mostly complete, so go ahead and go to 100 percent.

Now I need to add one qualifier to this message: When you go to 100 percent throttle, the car needs to be heading in the direction you want to go. In other words, if you've turned in early or haven't rotated the car enough, going to 100 percent throttle is not a good thing! You're only going to go off track at a higher speed.

What I'm talking about here is committing to the throttle, and yes, it takes commitment.



Often, I see throttle position traces in the data that looks like the red line in the graph to the right. The driver gets on the throttle (thinking that whoever gets on the throttle soonest, wins), gets to around 50 percent, hesitates, makes a small adjustment, and then squeezes on the last half of the full throttle application. And it's this hesitation and slow squeezing on of the last half of the throttle that's costing time.

Instead, at that point when the car is on the right line, forget smooth and flip the switch. Turn on the light. Go right to full 100 percent throttle.

Yes, sometimes it'll feel a bit iffy doing that. You'll wonder whether you'll stay on track. That's why you need to be looking way ahead, tracking a line in your mind that goes

from where you are now to where you want to be down the straightaway. If you're looking far ahead, you'll feel more comfortable committing to throttle; if you're looking only at the exit/track-out point, you will hesitate.

The next time on track, focus on looking farther ahead exiting corners, and commit to switching on that last half of the throttle.

Day #131: Managing negatives are just as important as making positives happen.

Dealing with the aftermath of a crash can be challenging — mentally, physically, and financially. Through the years, I've worked with drivers who have struggled to get their mojo back after being involved in a crash, so let's talk about that.

Did you buy a ticket for the recent Powerball lottery? What were the odds of winning it? I have no idea, but being involved in a crash makes me think about that.

Why? Because if you logically look at the number of miles you've driven, the number of times you've gone through a corner, the number of times other cars and drivers have not made a mistake and hit you… well, I wonder how those odds compare to winning the lottery. I bet they're close!

My point, obviously, is that if you look at it logically, the odds of crashing are pretty slim, aren't they? And as many drivers have pointed out, the odds against crashing on the track may be better than your odds when driving on the street!

That's where I think you should begin. Look at it logically. I know that's difficult when there is an emotional piece attached to what happened. But I'm reminding you to look at it rationally.

You do what you do because you're mentally programmed to do so; you sometimes don't do what you want because you either don't have the right programming, or you access the wrong program. Up until the moment you crashed, your mental programming was "nothing will happen." Then, within seconds it changed to "it can happen." And… now your programming could be "it might happen again," and maybe even "it's going to happen again."

We develop our programming through physical and mental repetition. Fortunately, the physical repetition of being hit is a one-time thing (look at it logically). The challenge is that often we let our minds develop programming through many multiple mental repetitions — we play the bad over and over in our minds until we're convinced it will happen again. We're programming that it will happen again.

So, stop it. :) I say that somewhat jokingly because it's not easy to just "stop it." That's like me telling a depressed person to feel happy.

We cannot not think about something. If I tell you to not think about a pink elephant, you will think about a pink elephant. The only way you cannot think about a pink elephant is to think about something else — for example, a blue elephant. You can develop a mental program so that every time I said, "Don't think about a pink elephant," you would say to yourself, "Blue elephant," and that's what you'd think about. In other words, "Blue elephant" is your trigger, and the second you pull it, it fires your "blue elephant" mental program.

Same with managing a past negative incident on the track. If you spent fifteen to twenty minutes every day deliberately doing mental imagery (what is typically called visualization) — mentally seeing, feeling and hearing yourself on track with other cars, and everything going smoothly and without incident — you would be building and/or rebuilding the mental program you've had for maybe dozens of years. And most importantly, you attach a trigger phrase to it, such as, "Clean and accurate." Over a few weeks of saying "Clean and accurate" and mentally experiencing driving on track without incident, you will reinforce the "nothing will happen" program that you used to rely on.

Look at it logically, and reprogram.

Day #132: Given equal cars and equal skills, the fittest driver is going to win.

You don't need to be a pro to drive at a level where fitness plays a critical role in your performance. If you've ever felt worn out while heading home from the track, you know what I mean.

You know that your physical fitness level impacts your mental performance — if you're beginning to feel physically tired, some amount of brain power is being used up to think about that ache, pain, or just-tired muscle. And it usually impacts your mind before you even realize it.

One of the least expensive and most effective ways to improve your driving is by improving your overall physical fitness level.

What are you doing today to up your physical fitness game?

Day #133: Use your simulator to make you a better track driver.

I've been using simulators since around 2005. I'll admit that there was a time when I thought they weren't realistic enough to make it worthwhile, but that changed a long time ago. Since then, I've conducted workshops specifically designed to help track drivers use a simulator more productively, I've coached dozens of drivers on them, I've tested super high-end sim rigs (such as Dallara's in Indianapolis), interviewed pro sim and real racers, and launched the *Sim Racer Academy*. But my focus has been a little different from the many drivers who are either competing in eSports, or using it as a way to stay sharp during non-track time. My focus has been on how to use simulators as a training tool.

To really simplify things, let's consider that there are three types of drivers in the world:

1. Those who are really, really good at sim racing. Let's call them "sim natives," although some refer to them as "aliens" because of how other-worldly good they are!
2. Those who are really, really good real car/track drivers, and are learning to be good on a sim. Let's call them "sim immigrants."
3. Those who dismiss the use of simulators altogether, either because of the time and financial investment, or because "they just don't feel like a real car." Let's call them "sim naive." That's not a negative reference, only that these drivers have not embraced sims as a training tool, yet, or they would rather spend their resources on a real car and limited track time.

My experience and research has led to an observation regarding the difference in driving style and technique between sim natives and sim immigrants: Sim natives drive the car off the front tires of the car, whereas many (but not all) sim immigrants drive cars off the rear tires. What do I mean by that?

Since simulators (at least the ones you and I can afford) do not provide much sense of yaw (while the software simulates yaw, the chassis doesn't actually yaw — change direction), the only way of truly sensing the car's limit is through the steering. In a sim, you turn the steering wheel, and when the direction the car is pointing and the steering you've put in don't match, you know you're moving beyond the limits of the front tires. You're experiencing understeer. You know what you've got, and can drive the car from there. In my experience, the best sim racers tend to prefer (and maybe even induce) understeer, and then work with it. They drive the car off what the front tires are telling them — although they might not even realize they're doing this.

Many — but not all — drivers who developed their skills in real cars (and karts), and not in simulated ones, sense the limit through the car's yaw. When the rear tires have a slightly greater slip angle than the fronts — yes, a hint of oversteer — the car is now pointing further into the corner than the steering angle suggests. The car has rotated. These drivers are driving the car off what the rear tires are telling them. And yes, they may not even realize they're doing this since they don't have anything to compare their

driving style to — until they get serious about going fast in a simulator, and struggle a bit to do so.

When a driver has a finely-tuned ability to sense the car's rear tires and its overall yaw angle, that does not apply to driving a sim as easily, so they are just not quite as fast (in some cases, it's a difference of a couple of tenths or even hundredths of a second). These drivers either have to adapt to sensing the car's limits through the front tires/steering, or they'll be just a tiny tick off in speed.

By the way, this is also why some elite pro racers are very, very good in sims. Some of those pros drive their real cars off the front, so there is little to no adaptation required.

My point here is not to get buried in the comparison of sim versus real-world driving, whether one is better or worse than the other, and how accurate sims are. No, it's that to be perfect at driving a sim requires a slightly different driving technique from being perfect in a real car — even though the car setup, track conditions, and everything else is the same. Why? Because the tools the driver has are different. How a driver senses the limits of the car are different.

One could argue that the third group of drivers I mentioned above, the sim naive, are right — sims don't feel right, so don't bother with them! I disagree. I look at this situation, if you're in the middle of it, as a huge opportunity to become an even better driver. It's an opportunity to add to your driving technique toolbox, learning a new way to be even better when on a real track. Just think, if you could not only drive the car off the rear tires, but also off the front tires, just how good you could be.

Could it be that the most exceptional race drivers in history like Fangio, Clark, Stewart, Senna, Schumacher, Hamilton, and Verstappen (focusing on F1 only) had this ability to drive their car off both the front and rear tires? That they could equally sense the limit through the steering wheel and seat of the pants, and that's what made/makes them so special?

If you're relatively new to driving a sim, or struggling to find that last tenth or two or three, focus on what you feel through the steering wheel. Induce some understeer, then manage it. Feel the limit through the amount of understeer your sim car has.

Day #134: Don't set expectations - focus on the possibilities and your potential.

It's a rare thing when we exceed our expectations. In fact, I'd suggest that the only time we exceed them is a fluke. There is all sorts of brain research that backs up what I say — that our brains are incredibly powerful and accurate at making sure we match our expectations, and never exceed them.

I can't tell you the number of times that I've seen a driver head out onto the track with a specific lap time in mind — an expected result — and then match it. But here's the thing: Other drivers, because the track gripped up, went faster. Because the driver in question had put an expectation in their mind, they were unable to exceed it, even if the track and their car were capable of doing so.

Instead of setting expectations — particularly lap times or finishing positions — focus on the possibilities, on what could happen. Focus on your performance, and on its potential.

But how? The place to start is with that lap timer — turn it off, cover it up, or put it out of sight.

Let me ask this: If you turn a lap time that you want, let's say a 1:10.5, are you going to quit trying to go faster? If that's the lap time that you think is fast enough, and you do it, I guess you can quit working on going faster, right? No. No matter what lap time you do, you can very likely go faster, so why do you care whether you've matched your expectations. Why do you care what your lap time is?

Day #135: Use the dead pedal.

Many years ago, I spent a fair amount of time training police officers how to drive. They varied from new recruits to elite/special forces officers, and the training spanned from some fairly basic stuff to the really cool techniques that undercover cops needed to do their jobs.

Quite a few officers were left-foot-brakers, claiming that it helped them drive better. I went into the training process with an open mind, and had mixed feelings about the benefits of left-foot-braking for the type of driving police officers experienced. So, I spent a lot of time weighing up the pros and cons, and especially testing both options — left and right foot braking.

I was actually surprised one day when we did some back-to-back testing of stopping distances, comparing left to right-foot-braking. Consistently, the right-foot-brakers would stop a few feet shorter than the left-foot-brakers, and most importantly, with more control when having to brake while turning to avoid an object. What I had thought might happen was that the left-foot-brakers would stop shorter, having eliminated the time spent moving their right foot from the throttle to the brake. So I rode with dozens of the officers to observe, and that's when I noticed what caused the biggest benefit of right-foot-braking: keeping the body supported by bracing the left foot against the dead pedal (or, as in most cop cars, just the raised part of the floor).

When I asked right-foot-brakers to hold their left foot back by the seat, reducing the amount they supported their body in a panic stop, their braking distances increased slightly, and they could not modulate the brake pedal as well when turning to avoid an object.

It wasn't which foot they were braking with that made the difference. Instead, it was the amount of support their body had when braking heavily. I knew the importance of a well-supported body when driving on the track, but I was surprised at just how big a difference it made.

To this day, when I get in the passenger seat with a driver — on or off the track — one of the first things I notice is whether the driver supports their body with the dead pedal. And I'm surprised at how many don't. Even some experienced track day drivers and club racers don't use the dead pedal enough. I know it's basic stuff, but as I always say, the best drivers just do the basics better than the rest.

Habits are powerful things, and if you haven't made resting your left foot on the dead pedal a habit, it's going to be a challenge to build it. Fortunately, this is one of those simple things you can do when driving on the street.

I know some of you are asking, "Yeah, but what if I left-foot-brake — how do I use the dead pedal to support my body?" You can't. Well, not until you're finished braking. At that time, if at all possible, move your foot from resting just above the pedal and place it

on the dead pedal. You might think that eliminates the advantage of having your left foot poised just above the brake pedal, and you're right — to some extent. It's a trade-off. But just before you get to the next braking zone, you can move your left foot just above the brake pedal again to get ready to brake.

Now, if your seat and harnesses are supporting your body the way they should be for track driving, then the use of the dead pedal becomes less important — and possibly unimportant. In that case, if you're left-foot-braking, you're okay to keep your foot poised just above the brake pedal (but not resting on it!).

For sure, if you're a right-foot-braker, keep your left foot on the dead pedal at all times, other than to use the clutch (if you have one of those). Your left foot should come off the dead pedal to use the clutch, and the second it's done pressing and releasing it, put it back on the dead pedal. My guideline has always been that you're not allowed to turn into a corner until your left foot is braced against the dead pedal. If it's still releasing the clutch, you left your downshift too late.

Check your habit out, and make sure your left foot is placed against the dead pedal as much as possible.

Day #136: Use your mental vision.

There's nothing like facing a blind turn, is there? Turn 3b at Sonoma, 2 at Mosport, the Corkscrew at Laguna Seca, 10 at VIR, and Turn 1 at Sebring with its wall on the inside blocking your view of the exit. All super-challenging because you can't physically see where you're going.

But you can — and must — mentally see through the corner.

Physical vision is what we can physically see. "Mental vision" is what we can mentally see — what we can picture in our minds. It's also the key to being fast through those blind corners.

Drivers often ask why they struggle in certain corners more than others, and blind corners are a common concern. The difference is the ability to mentally picture oneself driving through a section of the track that can't physically be seen.

The good news is that they rarely ever move a track from lap to lap! Yes, if it was there one lap, it'll be there the following lap. You can count on it. That means that you just have to have a solid mental image of where the track goes on the other side of the hill or whatever else blocks your vision.

One thing I know 100% for sure is this: Developing the ability to mentally see a blind corner is a skill that can be learned; it's not something that some people are born with and others are not. Okay, there might be some small differences in the abilities one is born with, but I know for a fact that the ability to mentally see through a blind corner can be developed. Really, we're talking about using one's imagination. And don't tell me you don't have the ability to imagine (if you've ever worried about something, then you've imagined — you put a mental image in your mind of something going wrong in the future).

What I will admit is that mentally seeing something while sitting at home or in a chair in the paddock is different than doing it at speed while approaching a blind corner. But mentally seeing the blind corner at home or in the paddock is the start. And if you imagine it enough times, with enough detail while sitting still, it'll "magically" appear in your mind at speed.

Who doesn't struggle with blindly committing to full throttle while cresting a hill? I understand — it's not easy. But if that hill crest were not there, and the driver could see where they were going, would they have a problem? No. So having as clear a picture of where you're going as you would without whatever is blocking your view is what you need.

If you have an in-car video, sit and watch just that blind section of the track, over and over again. Then — and this is the most important part — turn the video off, close your eyes and imagine driving just that section of the track. And do this while actually moving

your hands and feet, mimicking what you want to do. If you want to be at full throttle over a crest of a hill, then hold your right foot down as if you're at full throttle.

The video is a primer, but it's not good enough. It's critical that you can do what you want in your imagination because you're going to need to do this at speed when on the track. That's why it's important that you not only mentally see it, but you also imagine the sound the car will be making, how it will feel, what physical movements you'll be making, and even how you'll feel emotionally (calm, relaxed).

Practice imagining where you want to go, even when you can't physically see it, at speed. It simply takes practice. Which takes some commitment. That's all.

Day #137: If you can't afford good safety equipment, you can't afford to go racing.

This is simple: Safety equipment is not the place to save money.

And I'm going to expand this message to include more than just racing, but also to track day and HPDE driving. In fact, it may be more important here than in pure racing — racing has sanctioning bodies which put rules in place that must be followed. Track day and HPDE events don't have the same levels of safety requirements. So it's up to you.

Recently, I was at a car club driving school and track day. Looking around the paddock, and watching the cars on track, I felt uncomfortable. Here were cars doing well over 120 MPH, turning laps faster than purpose-built race cars did just a few years ago, and the majority of them had no safety equipment beyond the driver wearing a helmet. No roll bar, no safety harnesses, no head restraints or neck devices, no window nets, no fuel cells, no steel-braided fuel lines….

The next time you're thinking about investing in some go-fast equipment for your car, take a step back and consider spending that money on additional or upgraded safety equipment.

Day #138: Accept normalizing before stretching to a new normal.

Take a look at the following lap times from a single driver over the course of a single test day.

Session #1	Session #4
1:32.5	1:32.0
1:32.7	1:31.8
1:33.0	1:31.9
1:32.4	1:31.4
1:32.6	1:31.8
Session #2	**Session #5**
1:32.5	1:31.8
1:32.6	1:31.6
1:32.4	1:31.7
1:31.9	1:31.4
1:32.4	1:31.5
Session #3	**Session #6**
1:32.2	1:31.6
1:32.1	1:31.5
1:31.9	1:31.5
1:32.0	1:31.6
1.31.9	1:31.4

In each of the real lap times from a real driver, I choose the top five laps from each session (which varied between eight and twelve laps each), and threw out any that were obviously affected by traffic. What do you notice?

Yes, the driver is very consistent. He's an experienced driver with very good foundational skills and techniques, and a calm mind behind the wheel. But what else do you notice?

In Session #1, his "normal" — what he seemed to be able to turn all day — was around a mid-32, let's say a 1:32.6. In the second session, he ran his norm, and then had one lap — the 1:31.9 — that was outside the norm. Session #3 was spent around what was becoming his new normal, a 32-flat. In Session #4, he ran mostly around this new normal of a 32-flat (or very slightly below), and then had one lap at a 1:31.4 — his new "stretch" lap time. Session #5 seemed to be spent normalizing this new mid-31 lap time, and then in the sixth session, there was another very tiny stretch, but in this case, it was normalizing just a tenth of a second or so faster.

I should mention that during the entire day, not one single change was made to the car's setup, and the fuel load was kept fairly consistent. The tires started with three heat cycles on them, and these particular tires were usually good for about ten sessions

before they started to drop off in grip again. The track temperature stayed fairly consistent, with just a small increase in the afternoon. In other words, all of the gains in lap time were made by the driver, through working on various skills and techniques.

The main message I want you to take away from this article is this: Your mind needs to normalize one thing before it's ready to move to the next level. One way of looking at this is that each of these normalization stages is a "learning plateau." It's where the brain is letting what's been learned to really sink in and become a part of the programming. It's a natural part of the human learning process, and if you try to bypass it, or even speed it up, it's likely to backfire. And by backfire, I mean a costly mistake is likely.

Your lap times may or may not be as consistent as this driver's are. That's not the point. The point is if you look at your lap times closely (well, after you're out of the car and finished with all of the other debriefing you do with yourself or a crew, including writing notes on your track map), and consider the variables (tire wear, fuel load, track conditions, ambient temperature, etc.), you will likely notice trends. And these trends may help you learn more about your own learning process. It may tell you when to relax and just let the lap times come, and when it's time to push a bit more.

Always keep in mind that your brain must have time to consolidate what it's learned — and is still learning. In fact, each of these mini-plateaus is a critical part of the learning process. Without them, your brain is not ready to learn more, and trying to force new skills, techniques, or speed is likely to result in steps backwards.

If there's one thing that I've learned from coaching thousands of drivers, it's knowing when to push for more, and when to let the learning soak in and normalize. Being self-aware about your own learning process is as important to your lap times as the line, when to trail brake, tire pressures, or anything else. Maybe more important.

Day #139: Develop your task-switching skills.

Raise your hand if you're great at multitasking. Even if you think you're not good at it, raise your hand, too. Now, while you're doing that, listen to the words of a song playing in the background while working on that Excel spreadsheet that's been a challenge to finish, while also checking your email every minute or so, and answering the question from your spouse about where you're going for dinner on Friday night.

How'd you do?

Science has proven that we, as humans, suck at multitasking, no matter what we think we're capable of. In fact, if you think you're good at multitasking, I suspect you're just better than some at "task-switching."

We can't multitask, but we can switch tasks. And in our performance driving world, the quicker and more efficient we are at task-switching, the better. The best drivers are masters of task-switching, and that's why they're so good at knowing what's going on around them, making great decisions, and being able to adapt their technique throughout their time on track.

To improve your driving, practice switching from one task to another, such as (in no particular order):

- Check your side mirrors
- Glance ahead to where you intend to be in a couple of seconds
- Focus on the next reference point
- Check your progress, comparing where you are to where you wanted to be
- Think ahead to where you want to be in a few seconds
- Mentally place an image of what's up ahead that you can't physically see yet
- Check your rear-view mirror
- Notice the car in front and the line the driver is following
- Focus on another reference point
- Think about where you're going to pass the car ahead of you
- Be aware of how you released the brakes while entering the corner
- Decide if you should attempt a pass, or wait until after the upcoming corner
- Decide where you want the faster car behind you to pass you
- Be aware of how your car is handling, making a mental note of it so you can deal with it later
- Make note of the RPM as you pass the end of the exit curbing and compare it to your best/highest RPM

These are just a few of the things that your mind is monitoring, making decisions on, and triggering some physical action. I'm sure you could add a dozen more to this list.

To develop your task-switching skills, try this:

- Play an in-car video of one of your best laps, with the audio turned down to the point where you can barely hear it.
- Using the list above, write out the main tasks you think you should be paying attention to all around the track. This will take more than one viewing of the lap to do, and you'll probably need to take a lot of editing to ensure it's in the right order for you.
- Play the video, but this time record your voice (you can do that with a smartphone app) reading the various tasks you wrote out. You'll probably have to do this more than once to get the right timing, and to match up the correct task with the correct location and timing that you're seeing in the video.
- Listen to your recording (with and without watching the video), and practice switching your attention from task to task as quickly as possible, using the recording as a guide. Make the task switch as deliberately, quick, and well-defined as possible (not overlapping one with another task).

You should also try this simple exercise:

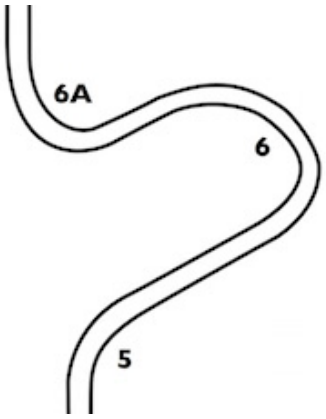
- Sit in a room and change your point of attention, for example, focus all of your attention on a screen (monitor, TV, security monitoring system, your phone), then switch to something like a tree outside your window. Spend just a second or two on each one, and then move onto something else (door, chair, person, etc.). Keep this going for at least five minutes straight. The objective is to put all of your attention on one thing for a couple of seconds, and then quickly move to something else, practicing switching from one to another. You should even change up the timing, spending approximately a second on one object, then five seconds on the next, two seconds on the next, and so on, changing how much time you spend fully focused on just that one object.

Be a great task-switcher, rather than trying to be a good multitasker.

Day #140: Ask yourself, "What can I do differently to go faster?"

While coaching at the SCCA Runoffs at Indy one year, I noticed something.

During the test days, I watched driver after driver after driver... drive Turn 6 with a very classic line. They turned in from the far-right edge of the track, arced around to a late apex, and accelerated out of the corner — only to have to make a speed adjustment for Turn 6A (the right-hander onto the back straight — Hulman Straight). The "straightaway" between turns 6 and 6A is almost non-existent.



So why, then, did so many drivers focus so much on getting a good exit off a later turn-in and apex of Turn 6, when the following straightway is almost non-existent?

Because that's the way they've been told to drive all corners. And that's the way they drive all corners. They looked like a bunch of lemmings, one after another.

The extra distance these drivers were going, at a relatively low speed, was significant. And it cost, on average (based on the dozens and dozens of segment times I took on a variety of cars), a couple of tenths of a second. Just in that one small section of the track.

Of course, there were some drivers who got this figured out. And I'm not saying that this was not the right way to drive the corner for some cars — it may have been. But I didn't see many drivers experimenting with the line.

A few years ago, I was coaching drivers competing in the Global Rally Cross series, and I remember watching Ken Block experiment with the line. Yes, Mr. Gymkhana would try more different lines than any other driver (when I asked him why he did that, he replied, "Because I read your book"), and it gave him an advantage, especially when it came to the race.

On a corner like Turn 6 at Indy (SCCA Runoffs configuration) which is practically a 180-degree turn, you could "drive it in deep, collect it in the middle, and drive it out fast." It's much like a short oval track turn, one where you "diamond" the corner, using almost a double-apex line. Yes, you might be a little wide of where the first apex is, but you'll then be braking later into the corner (carrying more speed longer on the approach and into the corner). You then rotate the car in the middle with the appropriate brake release, line the car up for a late apex, and blast out of the turn. I don't know about you, but "fast in, faster out" always seems like a better approach than "slow in, fast out!"

I think too many drivers get stuck in a rut with what they've been told to do (slow in, fast out). They don't ask themselves, "What could I do differently here that would make me faster?" Looking at Turn 6, it's easy to see that your exit speed is not super-critical, and

that the corner is long enough that you have time to carry entry speed and still have good exit speed; you have time to collect the car and rotate it in the middle of the corner.

Does this remind you of how you drive Turn 2 at Laguna Seca, 11 and 12 (Oak Tree) at VIR, 5A and B at Canadian Tire Motorsport Park, and even the Carousel at Road America?

I suggest you look at how you're going to drive a corner by starting with the end in mind, then work backwards. Ask yourself whether carrying more entry speed is going to negatively impact the exit, and if so, by how much — can you make up more on the entry than you lose on the exit?

And — this is important — does driving the extra distance to make the late turn-in actually help, or hurt? You might be surprised how often less distance traveled outweighs the classic late turn-in/apex line.

Day #141: Accept fear, then manage it.

The more you resist just about anything, the more likely it is to take control of you. This is especially true of fear, and its very close cousin, self-preservation.

It's an extremely rare person who never experiences some amount of fear when driving on a track. I think the old saying is true: There are two kinds of drivers — ones who admit being scared at times, and liars.

The sooner you accept that you're going to be scared sometime in a car, and be okay with that, the sooner you can move onto dealing with it.

There is what we could call a "fear spectrum," right? On one end, there is no fear; on the other end you're frozen with fear. Performance driving lies somewhere in the middle of these extremes. At least, it's supposed to! If you find yourself at the "frozen with fear" end too often, you're doing something wrong.



One could say that spending time somewhere just below the scariest end of the extreme is what performance driving is all about. To me, though, it's all about being comfortable with where you spend the majority of your time on this spectrum. And you can become comfortable with something just below "frozen with fear." You can become comfortable being uncomfortable.

For some drivers — and for some people — living on the edge is what life is all about. For them, flirting with the scary end is what makes them tick. For others, that's not what performance driving is about. For them, it's being comfortable, and spending too much time flirting with fear is not what they want. Neither is right or wrong. They just are.

But here's the thing: We, as humans, become accustomed to whatever we spend time doing. If we forced ourselves to stand on the very edge of a cliff overlooking a 2000-foot drop into the ocean below for a long time, and on a regular basis, we'd get comfortable with that. Some of you reading that last line probably got sweaty palms just thinking about it; for others it's no big deal. While some of that may be due to our DNA, it's a fact that we can build up a tolerance to almost anything, whether that's standing on the edge of a cliff or driving on the edge of grip through a fast sweeper.

There's a process for managing fear:

1. Accept it. Accept that every human feels fear at some point, and it doesn't make you any less of a driver if you're scared.
2. Embrace it. Not only accept that fear happens, but embrace the fact that it helps us feel more alive. It's not boring!

3. Spend time with it. The more time you spend near fear, the more comfortable you get with it. In fact, you become comfortable being uncomfortable.
4. Seek it out. Over time, being comfortable being uncomfortable becomes an addiction. Over time, you will seek out being uncomfortable because it feels good — it feels comfortable.

Notice that I'm not suggesting being afraid, of spending time in the "frozen with fear" zone. I'm suggesting spending time on that end of the spectrum, and getting comfortable feeling that way. And in the process, realizing and accepting that it's okay. Not only is it okay, it's something to seek out because you're comfortable there.

Day #142: Long races are most often decided by the team that best minimizes the effects of their problems.

In long race, you're going to have problems. That's almost a given. What matters most is how you deal with them.

It used to be, years ago, that teams in endurance races at just about every level set a pace that they felt they could turn without breaking the car. They knew that if they drove flat out, they'd break the car, so they backed off and set a pace that was fairly slow compared to what they could do if driving flat out. Nowadays, in almost every type of endurance race, it's practically a long sprint race. "Setting a pace" is "drive as fast as you can comfortably drive."

Cars, from Champ car to WEC, are more reliable than ever, and that's why we're able to push harder than ever before.

Still, it's rare for a team to have an absolutely trouble-free run in a long race. The teams that deal with these troubles the best, usually win.

Look at what Audi did at Le Mans years ago. Famously, they replaced entire gearboxes in a matter of minutes — it was part of their strategy. And the gearbox replacement is just one of many plans they had for dealing with troubles better than their rivals do.

Sure, Audi's outright speed and reliability is one reason they've won Le Mans so many times. But I'd argue that it's their preparation for handling trouble that has led to these wins more often than not.

As you prepare for an endurance race, prepare for problems. Think ahead, think about all the issues that are likely — and unlikely — to happen. Then, think through how you could best address them.

- How will you deal with your car if it's been damaged in a crash?
- How would you adapt your driving if you were running out of brakes, and had to "nurse" the car home for the final hour?
- What would you do if you couldn't use 4th gear (or some other gear)?
- How would you drive if extending a fuel stint meant one less pit stop — which would make the difference between winning or losing?
- What would you do if your radio communication failed?
- What would you do if one of your co-drivers got sick and couldn't drive?
- How would you adapt your driving if you had to make a set of tires with a bad flat spot last until the end of the race?
- What would you do if your seat belts came undone?

- What would you do if your windshield was so covered in dirt and grime that you could barely see through it?
- What would you do if your car's engine died and you pulled off to the side of the track?
- What would you do if you got seriously bad cramps in your legs?
- What would you do if you felt a suspension component was beginning to fail — one hour into the race, and/or with thirty minutes to go?
- What would you do if you felt sick while driving?
- How would you adapt your driving if your car's engine lost power and started misfiring in higher RPMs with two hours to go in the race?
- What would you do if it started to rain hard and you were on slicks, but you could see that it might clear up within fifteen minutes?
- How would you change your driving if turning a second a lap faster than you've ever driven for the final forty minutes of the race would mean you could catch and pass the leader?
- What would you do if your transmission failed just one hour into the race?
- What would you do if your dash, including the tach and any engine parameters data, quit working?
- What would you do if the fire system activated while you were driving, spraying all of its contents throughout the cockpit?

I could go on and on... but I'll leave this exercise to you and your team to do.

Day #143: Manage the fast-slow matrix.

Let's look at the difference in how you can (and should) drive slow corners versus fast ones. But first — what defines a slow corner and what defines a fast corner?

When analyzing data, the dividing line between slow and fast is around the 60-65 MPH range. Corners slower than that typically show a speed trace line that looks more like a V, and above that speed, a U. Another way of looking at this: a corner slower than 60 is more "In slow – out fast"; above 65, it's more of a momentum corner.

As with practically all things driving, there are exceptions to these guidelines, but this isn't a bad place to start.

So, with these parameters in mind, some additional guidelines…

Trail brake in slow corners; release brakes by Turn-in point in fast corners: Since a slow corner has a tighter radius, you need to change the direction of the car more than you do in a fast corner. One way to do that is to keep the front of the car loaded more heavily by trail braking longer, which helps rotate the car. In fast corners, the priority is maximizing grip — to do that, you want to balance the car as best you can (little-to-no fore-aft weight transfer). Therefore, you should release the brakes early, or most times, not trail brake at all.

Slow hands in fast corners; fast hands in slow corners: The speed at which you make your initial turn into a corner should vary based on the radius of the corner, with tighter (and by the laws of physics, slower) corners requiring a quicker turn; faster, more sweeping turns rewarding a slower initial turn of the steering wheel. This is entirely based on the amount of direction change you need.

Late apexes in slow corners; earlier apexes in fast corners: Related to the rate of turning the steering wheel is where the Apex is in the corner, and as we all know, that is dependent on where you begin turning (the Turn-in point). Typically, you'll turn in and apex later in a slow corner, and earlier in the fast ones. But never forget that it's not just where the Apex is that matters, but also the angle the car is pointing at.

Don't drive slow corners fast, and fast corners slow: If you know me, you know I hate the word "don't." I joke about "Don't use don't," when talking with instructors; it's critical to tell students what to do, and not what not to do. But I'm making an exception here, because I see far too many drivers carrying too much speed into slow corners, and not enough into fast ones. The reason may be obvious: slow corners are not intimidating, and fast ones are! Fast corners more than suggest that you should slow down for them, and slow ones do not. Next time on track, consider being a little more patient and slower entering slow corners, and carrying more speed entering fast ones.

And just to be sure, I'm going to repeat myself: There are exceptions to every rule or guideline.

Day #144: Be present.

I often ask drivers whether they think the best drivers in the world — in F1, Indy cars, NASCAR, IMSA, whatever — make as many mistakes as they do. Not surprisingly, the answer is usually, “No.” Most drivers feel they make more mistakes than the top pros, and that’s why the top pros are the top pros. If you’re making millions of dollars driving race cars, you don’t make mistakes, right?

Wrong. The best drivers make mistakes, and I believe they make as many as someone just starting out in performance/race driving. The difference is the size of the mistake, and how quickly it‘s corrected. The best recognize the mistake much sooner than inexperienced drivers, and therefore the correction is sooner and smaller — to the point where it seems they’ve never made a mistake.

This applies to both driving technique and the mental game. A while ago, I read the following quote from 7-time NASCAR champ, Jimmie Johnson, when he was talking about going into his last season or two of full-time racing.

“Chasing eight (referring to going after his eighth championship) and reading it or writing it or saying it is one piece, but I can identify with the fact that Chad [Knaus] and I in our final year, we were chasing that more than we were just being,” said Johnson. “Last year, with all the pressure and what was going on, of course we were chasing something. It just kind of all just fits together.

“My ultimate goal this year — and I think what happened through the self-awareness of the offseason — it’s just about being present. I’m going to get in that car, I’m going to give it 100 percent as I always do; who knows what the outcome is going to be. I’m going to lay it on the line and go. I’m not chasing anything; I’m just going to try and be present and just be me.”

Even multi-time champions can lose focus on what’s most important: one’s performance. And it could be argued that after winning seven championships, and being on the verge of setting a new record of eight, would make anyone focus too much on the result: on winning the championship. We all know — or at least, should know — that when you focus on the result, you most often do not perform at your best; the pressure of delivering that result negatively affects your performance. When you focus on your performance, you are more likely to perform better, and therefore get the result you want.

There are a few lessons to take away from Johnson’s quote.

- No matter who you are, you’re going to make mistakes, both physically and mentally.
- Even with the best possible support around you, you can still make mistakes and focus on the wrong things.
- When you focus on the result, the desired results don’t happen.

- When you focus on your performance, on just doing what’s necessary to perform well, you are more likely to get the results you want.
- The "woo-woo" stuff matters.

As Jimmie says, “Be present.”

Day #145: Minimize the distance you drive.

Autocrossers can teach road racers and track drivers a big lesson: Drive less distance!

Successful autocrossers have learned that the trade-off of distance against the "ideal line" is often won by traveling less distance.

The only way you're going to know for sure whether a shorter line is better than what might be considered the right line is with something other than your inner stopwatch. We can be fooled into thinking that something feels faster when it isn't really. A great example of this is when a driver uses a lower gear for a corner. The engine makes more noise, tricking the driver into thinking he's going faster when actually, the taller gear helped him carry more speed through the corner and was faster.

The problem is that the overall lap time is not sufficient. You might be faster in the corner or section of the track you're focused on, but something you do elsewhere cancels out any gain. The lap time result is the same as in the past, so you don't think you made an improvement. This is where data is so valuable, video can be really helpful, and a good old stopwatch timing specific segments works well, too.

If you're going to compare a shorter/less ideal line against the theoretical best line, then you need to look at segment times. As a general guide, start the segment just before the Begin-of-Braking point (or speed adjustment area) and end it as close to the next brake/adjustment area as you can. You'll want to include the brake zone, the corner, and ensuing straight — this will give you the best overall picture of what's working and what isn't.

As I've mentioned here in the past, you need to give each option a chance to work. If you try a different line once or twice and it doesn't result in an improved segment time, that doesn't mean that you can't make that a faster way of driving that part of the track. You might just need to give it a few more laps, so you're performing what you want as best you can.

The shortest distance between two points is a straight line; the less distance you drive, the faster you'll get to that next point.

Day #146: Use video to improve your pattern-matching skills.

I'm going to share what I think is a different way of using your in-car video, and it has all to do with one of the things that separates the best from the rest: pattern-matching.

When you approach an intersection on the street and you think "I just knew that red car was going to pull in front of me," that's pattern-matching. It's what most new teen drivers are not very good at, because they haven't yet seen as many scenarios from the perspective of being behind the steering wheel. But two drivers (who have both been driving for twenty years) are not necessarily equally good at pattern-matching, either. Some drivers are either more naturally-gifted with this all-important skill, or they've developed the ability through practice.

Approaching a corner on the track, many drivers look for singular reference points to determine where to brake, turn-in, apex, and exit the turn. But the best drivers use the pattern made up of multiple references to give them a more detailed cue as to when to begin braking, turn in, ease off the brakes, and arc in toward the apex. Think of it as watching a low-resolution (single reference) versus high-resolution (the pattern made up of multiple references) video.

This is especially true when dealing with other cars in a passing situation. How many variations are there in the positioning of just two competitors going into one corner? An infinite number of scenarios, right? So how do the best consistently make the right decisions? Pattern-matching. While no two patterns in a racecraft scenario will ever be the same, there are similarities (as you've proven when you've identified a car that was going to pull out in front of you on the street).

The good news is that your pattern-matching abilities can be developed. How? First, be aware of the pattern-matching that you're currently doing. I suspect you're doing it more than you think. So just become aware of how many patterns in the mental snapshots you have as you drive around a track. The more you're aware of it, the more it'll sink in and become part of your deep mental programming.

Again, we think that we use singular references for where we begin to brake, turn into a corner, apex, and track out of a corner. But it's not that simple — at least, it shouldn't be. Imagine stopping on a track just before the brake zone, and look ahead and into the corner. What do you see, besides your usual references? I bet when you really look at it, you'll see various angles crossing, merging, and aligning. Again, it's the relationship of one reference to another, then to another, and another, and so on. It's that image made up of many references that led to accurate, consistent, fast laps.

This is where video review can play a big role in helping you develop your pattern-matching skills.

When watching your in-car video, stop the video just prior to the Begin-of-Braking (BoB) point, and look for the pattern. Take a snapshot in your mind of what it looks like.

- If you were to extend the edges of the track past the turn-in point, where would they intersect with the horizon?
- How far to the right is that tree on the horizon from the turn-in point compared to the edge of the track on the right?
- If you extended a line perpendicular to the BoB reference, what would it line up with on the other side of the track?
- Look for the angles from the front edges of your car to the horizon — where do they cross over the apex curb?
- Do you notice subtle changes in direction of the fog lines on the edge of the track?

Forward the video to just before the turn-in point, and do the same; and then, just before the apex, at the apex, as you begin to unwind the steering wheel, and just before the track-out curb.

I know this is going to take time, but do this for multiple laps, until you notice the patterns. The pattern might be when you've gotten the car lined up just right, or you might notice that the pattern changes from lap-to-lap, telling you where you're consistent or not.

The interesting thing is the more you do this, the better you get at pattern-matching — at speed. You're improving your ability to pattern-match while driving at speed by reviewing video.

Day #147: Rely on cheap tricks!

When I travel to various events to coach or do a presentation, it's surprising and fun when a driver tells me about what helped them improve their driving. It could be something they learned on their own, a technique someone else taught them, or a tip they picked up that I shared. What's not so surprising is how often it's a simple thing that made the difference — a "cheap trick," such as:

1. *Use the dead pedal*. Whenever your left foot is not using the clutch (and/or brake pedal if you're a left-foot-braker), it should be on the dead pedal. If your car doesn't have a dead pedal, make one. What's stopping your body from shifting forward when decelerating hard? Your seatbelts/harnesses are helping, but they can't hold you still, and the steering wheel is meant to change the direction of the car, not support you. If your body shifts forward even the tiniest bit, it means that the subtle/fine movements of the throttle and/or brake pedal are going to suffer. If football is a game of inches, performance driving is a game of fractions of inches — and that includes the movement of the controls.

2. *Tilt your side mirrors out*. When was the last time your car ran into itself? Never?! Okay, then make sure you cannot see the side of your own car in your side mirrors. If you can see the side of your car, then adjust them out until you can only see it if you tilt your head in the direction of that side mirror. That way you'll see more to the side, eliminating blind spots, and being more aware of other cars around you. And this applies to when you're driving on the street, too — maybe even more so.

3. *Stay at full throttle until it's time to brake*. When driving on the street, rarely do you drive up to a traffic light or stop sign at full throttle, quickly lift your foot off the gas pedal, and apply the brakes. Instead, you see what you're stopping or slowing for, ease your foot off the gas, coast… then begin braking. And that builds a habit of doing that — a habit that often appears on the track. Be aware of whether you lift off the throttle, coast, and then brake for corners — and then stop doing that. Instead, stay at full throttle right until the point where you want to brake, quickly release the gas pedal, and immediately begin braking.

4. *Spend more time at full throttle*. Can you stay at full throttle for even a fraction of a second longer before you start braking for every corner (see previous point)? Can you get to full throttle just a fraction of a second longer when exiting corners? Are there places on the track between corners where you think you're going to full throttle, but you're not – or your mind decides not to go to full throttle because it knows you're going to lift off again in a fraction of a second from now (without you being consciously aware of this)? A few fractions of a second add up to a lot! Can you spend even one or two percent more of the lap at full throttle?

5. *Wait to get on the throttle until you're able to go to full throttle*. You've been told to never coast (I even wrote that in my *Speed Secrets* books). But there are exceptions to every piece of advice, and sometimes you're better off to hesitate for a fraction of second after coming off the brakes before beginning to apply the throttle. Why? To allow the car to finish changing direction (rotating), so when you do begin to apply the throttle, you can go all the way to full throttle (without having to ease up to avoid running off the track at the exit of the corner).

6. *Release the brakes slower*. That's it. Just be aware of how quickly you release the brakes, and slow it down a little. Be a little smoother and more deliberate with how you take your foot off the brake pedal.

7. *Think of reference points as index points*. (A hat tip to Peter Krause for this one). Rather than thinking of reference points as a place where you need to do something, think of them as indicators. A reference point could be the end of some painted curbing, but your turn-in point may be half a car's length before it. So, you're indexing your turn-in point off the end of the curbing, but you don't actually do something right at that point.

8. *Focus on the EoB (End-of-Braking) as much or more than that BoB (Begin-of-Braking).* As you approach a corner, look into where you're going to end your braking (that point where your foot is completely off the brake pedal). That's the EoB, and it's more important than the BoB. If you're focused on where you begin braking, you're not focused enough on the corner up ahead.

9. *Lighten your steering wheel grip*. On a scale of one to ten, with ten being that you're gripping the steering wheel as tightly as you can, hold the wheel with no more than a four grip. Hold the wheel with just enough effort to ensure your hands don't slip — and no more. The lighter your grip, the more sensitive you are to the feedback through the wheel. (Hint: Ensure your steering wheel is designed to assist with this — check out https://www.steeringwheelguy.com)

10. *Complete downshifts before turning into corners*. Never, ever, ever, ever… turn into a corner before you've completed your final downshift. And by complete, I mean having the clutch fully released and engaged and your left foot back on the dead pedal. That must be done before you're allowed to turn the steering wheel.

11. *Get good at the fast corners*. It's the fastest corners that usually intimidate drivers the most and that means that if you get good at them, you have an advantage over others. Often, if you need to slow for a fast corner, do it early. In our desire to take fast corners faster, we often leave our braking later and later, until we have the nose of the car loaded up as we turn in… making for an uncomfortable feeling (and a lower overall grip level). Instead, make your speed adjustment early, and then get back to power almost at the point where you're turning in. That'll make the car better balanced and give it more grip to handle the fast corner.

12. *Imagine before driving*. Prior to going on the track, take at least ten minutes to relax and imagine what it is you're going to do when you get there. Track time is expensive and limited, so make every second of it count. Taking a few laps to warm up, getting up to speed, and then starting to think about what you want to change or improve is a bad investment. Do some mental imagery of what you're going to do before you even think about starting your engine.

Based on what I've been told by drivers all around the world, relying on these "cheap tricks" will improve your driving.

Day #148: Use a Newtonian approach to developing your driving.

One of my favorite podcasts is No Stupid Questions with Steven Dubner and Angela Duckworth (guess why the title grabbed my attention?). In episode #32, Duckworth shared her research and thoughts on skill, talent, and effort, and ultimately how all of these impact achievement. If you don't know, Duckworth wrote the book, *Grit*, so you might be able to guess where she's coming from. In the podcast, her perspective seemed unique, and one that I know you'll be able to relate to driving.

The one thing that Duckworth said that really stuck with me was that we can think of talent as the rate of change of a skill. I'll come back to that shortly, but first…

Essentially, Duckworth applied Newtonian physics to human achievement. Huh?!

Galileo Galilei stated that Distance = Speed x Time. How many times in school did you use this equation to solve some problem like, "If a car travels at 60MPH from city A to city B, which are 126 miles apart, how long will it take to get to the destination?"

Duckworth relates distance to the accomplishment of a task, whether short-term or long-term; your skill is speed, and the effort you put into the accomplishment of the task is the same as time (and often is determined by the amount of time put in).

In other words, Accomplishment = Skill x Effort.

This equation didn't really include what she said about talent. So being the learning nerd that I am, I looked around for more information from Duckworth. I found a research paper, *The Mechanics of Human Achievement*, to which Duckworth contributed (Angela L. Duckworth, Johannes C. Eichstaedt, and Lyle H. Ungar). In it, it says:

"We propose organizing achievement-related traits by two possible mechanisms of action: Traits that determine the rate at which an individual learns a skill are talent variables and can be distinguished conceptually from traits that determine that effort and individual puts forth. This approach takes inspiration from Newtonian mechanics: achievement is akin to distance traveled, effort to time, skill to speed, and talent to acceleration. A novel prediction from this model is that individual differences in effort (but not talent) influence achievement (but not skill) more substantially over longer (rather than shorter) time intervals. Conceptualizing skill as the multiplicative product of talent and effort, and achievement as the multiplicative product of skill and effort, advances similar, but less formal, propositions by several important earlier thinkers."

Yes, it was Galileo Galilei who first used the Distance = Speed x Time formula, but I suppose it eventually all fit into Newtonian physics. And who am I to argue with Galileo, Newton, and Duckworth?

I love thinking of talent as the rate of acceleration. If you and I start off with the same level of skill, and we both put in the exact same amount of effort/time, where we end up

is going to be determined by the amount of natural talent we started off with. The one with the most natural talent will accelerate the improvement faster.

However, what I most love about this way of looking at accomplishing a task is that no matter how much talent we start with, we control our ability to accomplish something by the amount of skill and effort we put into it. And skill is not a static a thing either, just as the speed of a car is not static. By making changes to the car's tires, brakes, suspension, engine, aero, and so on, we can improve its "skill" - we can make it faster. And by making changes to the way we practice our driving, gaining knowledge by studying resources, developing a deeper understanding of concepts behind the skills we use, using mental imagery... we're improving our skills.

Especially with performance driving and racing, the amount of time we get on track to practice is limited. So, one part of the equation is not much of a variable! And this is where my understanding of Duckworth's concept fails just a tiny, little bit. See, time does not equal effort. We've all heard the saying, "Don't confuse movement with action," and this is the same. If I practice the wrong thing for one hundred laps, am I going to be more skilled than if I'd driven ten laps perfectly? Certainly not.

This is more than just "Practice doesn't make perfect. Only perfect practice makes perfect." Even if you and I practice doing something perfectly for the same amount of time, that doesn't mean that our practice is equally as effective. The best coaches in every activity, from sport to business, develop better, more effective practice strategies. I believe that approach is the single most important factor that separates the best coaches from the rest; the same applies to the performer (driver, athlete, businessperson, artist, etc.). Drivers who consistently perform better have learned how to practice more efficiently.

What does this mean to you? Simple (but not easy): Figure out how you can make your practice more efficient. Some examples:

- More "study" away from the track
- Using a simulator before track practice
- Working with a coach
- Using tools such as data and video during and after practice
- Using a learning management system to track and measure improvements
- Being better prepared before you begin practicing — having a practice plan
- Using a well-thought-out debrief process afterward

Whatever you decide on, you'll improve your skills faster having gone through the process of thinking about and planning a more efficient practice session. I'd even suggest that doing so will multiply your talent.

If only Galileo or Sir Isaac were around to help us put all of this into a neat little equation....

Day #149: Manage corner entry understeer with more or less trail braking.

We all know that more load on the front tires results in them having more traction. We also know that the front tires can only do so much, and too much load will overload them (leading to them sliding too much).

So, picture this: You're approaching your favorite near-180-degree, 2nd-gear corner. You're in top gear as you get to the brake zone, and you know that getting a good exit out of it onto the next straightaway is critical. Given that, you have two separate goals in mind:

1. Getting the car rotated so you can get back to full throttle early.
2. Carrying speed through the corner so you begin accelerating from as high a speed as possible.

In other words, you want it all. Welcome to the greedy world of performance driving!

As you turn into the corner, the front end starts to push. Understeer. The front tires are just not gripping the way you want, to be able to carry entry speed and get the car's direction changed enough so you can accelerate early out of the corner. What do you do?

You have two options:

1. Trail brake more to keep load on the front tires.
2. Trail brake less to give the front tires the ability to generate cornering grip.

So, the solution to understeer is to either trail brake more or less. How do you know which to do?

Let's zoom in on what you do from the turn-in point to the apex.

You begin to release the brakes as you turn in. You're fine-tuning your corner entry speed, while managing the load on the front tires. As you ease off the brakes, one of two things happen:

1. You notice the front of the car feeling more responsive, turning more.
2. You notice the front of the car feeling less responsive, turning less.

If, as you begin to release pressure off the brake pedal, you feel the front of the car gripping more, then continue to ease off. But if you feel the understeer getting worse as you ease off the brakes, just stop releasing them for a brief moment — hold the pressure there, and then slowly continue to release them. Doing this, you're managing the amount of load on the front tires, controlling the amount of grip the front tires have for changing the direction of the car. Of course, you're also controlling your corner entry speed. Therefore, if you feel you're over-slowing, ease up a little more off the brakes; if you're carrying too much entry speed, hold that pressure for a moment.

You only have two options to focus on: releasing the brake pressure more slowly, or releasing them more quickly. By doing so you're managing the load on the front tires, as well as your corner entry speed.

That's a lot to sense and process in the fractions of a second you have entering a corner, isn't it? I'm sure you can imagine Lewis Hamilton, Colin Braun, or Scott Dixon being able to do this, but you're wondering whether you can. Or maybe you're convinced you can't do that. I disagree.

You can. But it takes practice, deliberate practice.

First, go back to imagining Hamilton, Braun or Dixon doing that. Go ahead, close your eyes and imagine them doing it, again and again. Now, substitute yourself for them. Imagine yourself approaching and entering that favorite 2nd-gear corner, managing the brake release to nail the perfect corner entry speed, while getting your car to rotate — controlling any amount of understeer there is — so you can fully commit back to full throttle early in the corner.

Next, spend a few days at the track solely focused on noticing how your car responds from turn-in to apex. Play with the timing and rate of release of the brakes, and you will improve your ability to know whether to release the brakes more, or continue to trail brake more.

Day #150: Let the zone come to you.

I considered titling this article, “To zone or not to zone?” But I think “Let the zone come to you” is more representative of what I’m talking about, despite it sounding a little too Zen-like for some people.

I’m the first person to talk about the importance of being in the zone when driving (or anything else). But I do see some drivers overly-focused on getting in the zone, and that gets in the way of them driving the best they can.

Trying to get into the zone is like telling someone to relax, or trying to make yourself happy. You can’t make yourself get into the zone. It just happens — or it doesn’t. Sure, you can do things to increase the likelihood of getting into the zone, and I spend a lot of time coaching drivers on what they can do. But I see too many drivers trying to get into the zone, by being super-relaxed and flow-y with their driving. There comes a time when you need to grab your car by the scruff of the neck and make it do things. Like standing on the brakes, rather than squeezing them on gently. Like committing to full throttle a little earlier than is comfortable, rather than squeezing it on. Like turning the steering wheel hard to make the car go where you want it to, rather than being super-smooth with it.

Sometimes when you do this — when you hustle the car, when you make it do things, when you really drive it hard...you find yourself getting into the zone. Why? Because you’re not "trying" to get into the zone. Because you’re focused on the act of driving, in the moment, and in that process you stop thinking so much about the mystical state of zone and flow. And you’re fast, because of that.

Think about that. Then stop thinking about that, and just do it.

Day #151: Practice the "other stuff."

Most drivers spend most of their time on a track practicing driving as fast as possible on the ideal line. If you race, how much of the race is spent driving the ideal line? For some classes of racing, most of it. For other classes of racing, very little. You spend as much or more time driving off-line, setting up passes, discouraging being passed, being passed, getting up to speed quickly at the beginning of a race, managing your tires or other car issues, and so on. When you think of a race, very little of it is spent solely focused on turning that one fast lap.

So why not practice the “other stuff?”

This is a bit like cross training. While there has been a strong push over the past few decades for specialization in sports training (and especially in developing young athletes), the times they are a-changing. The latest research shows that athletes like Tiger Woods are the exceptions to the cross training rule. Others, like Roger Federer and Patrick Mahomes, who played all sorts of sports before focusing on their chosen sport are proving the rule: practice the other stuff.

In our world, that could be driving other types of cars. In an issue of *Speed Secrets Weekly*, my guest contributor challenged you to go drive a vintage car. Many road racers I coach go to a rally school to experience driving sideways on dirt. What made drivers like Mario Andretti, AJ Foyt, Dan Gurney, and Mark Donohue so great? They drove just about everything they could that had four wheels — Indy cars, sprint cars, NASCAR, sports cars, F1….

When Andretti drove in Formula One, how he used the brakes made him successful — something that he’d learned from driving sprint and Indy cars.

Okay, you don’t have access to a sprint, Formula One, or NASCAR car! How about a go-kart? How about a rally school? How about a skid pad? How about a vintage car? How about a rental Spec Miata? How about renting an open-wheel car?

Or how about simply practicing what you don’t usually practice? As I said, most drivers practice driving as fast as they can. What if you practiced with a car that was deliberately handling poorly, learning to adapt to it? What if you practiced driving in a taller gear than you usually use? What if you practiced driving off-line? What if you practiced braking early and light, or late and hard? What if you practiced an early turn-in? A super-late turn-in?

Think about cross training with your driving by practicing the other stuff. Get creative. What I know for sure is this: it will improve your driving faster than if you simply do what you’ve always done.

Day #152: Change is good.

Here's a situation that happens too often (and more often than one might think): A person signs up to take part in an HPDE (High Performance Driver Education) program, either with a car club or some for-profit business. He (while not always a male, it's more likely to be one) shows up at the track, and an instructor is assigned to him. Different organizations use different ways of assigning instructors to students, from alphabetical order of last names to the type of car being driven, and many more criteria in between.

The student and instructor hit the track through the first two or three sessions, and the student begins to realize that he's learning very little from the instructor. Not only that, but he doesn't particularly like or get along with him. Both are good, likable people, but they just don't mesh well. That happens in many situations in life, so why wouldn't it at the track?

The instructor, in this example, tends to be rather dominating. He's telling the student what to do, and almost intimidates the student a little (at least to the point where the student is reluctant to say anything about the instructor to the event organizer, or even question some of what the instructor is saying). In fact, the student is reluctant to ask questions at all, thereby limiting what he might learn that day.

"What do I do in a situation like this?" is a question I was asked.

I've had students come to me when I was organizing a driving school and ask for a different instructor. I've found myself in a couple of situations wanting to defend the instructor, especially when it was someone I'm friends with, and who had been very successful with hundreds of other students through the years.

But here's the thing: As I said earlier, because a student and instructor don't mesh and work well together, that doesn't mean that either one of them is a bad person. It doesn't mean the instructor is a not good at what they do. All it's saying is that these two people don't fit in a learner-teacher role. And that means a change needs to happen — as soon as possible.

The student, the instructor, and the event organizer must, and I mean must, be open to and willing to make this change.

If you're a student and you feel that you're not getting what you need from your instructor, go to the event organizer and ask for a different instructor. Be polite about it, but firm. Understand that it might take a little time for the organizer to make the change, because it could mean having to shuffle people around, but it should happen. You're the customer, right?

If you're an instructor, and your student asks for a change, don't take it personally. It just means that the fit isn't perfect from your student's perspective. It also means you get to work with another student — fun stuff. Your student is the customer, right?

If you're an event organizer, be open to a change. While you should be totally supportive (and trusting) of your instructors, you need to be ready for situations where a student and instructor don't fit — no matter who your instructor is. In my experience, the percentage of students asking for a different instructor is very low, so don't worry about it becoming a big logistical hassle. The student is your customer, right?

I'm sure that those of you reading this will nod your head and agree (well, most of you). So I'm leaving it up to you to spread the word, and make sure that others feel the same way, ultimately building a culture within the HPDE world that "change is good."

P.S. - This doesn't mean that as a student you should be switching instructors every event or session! You can go overboard with change, and not learn as much as you could be taking the time to develop a strong working relationship with an instructor.

Day #153: Adapt your braking.

Performance driving is a series of compromises.

- You trade corner entry speed for exit speed — sometimes.
- You trade corner exit speed for entry speed — sometimes.
- You trade waiting to begin applying the throttle to allow you to go all the way to full throttle sooner — sometimes.
- You trade an early, tighter cornering radius to allow you to begin unwinding the steering earlier so you can begin accelerating sooner — sometimes.
- You trade the ideal cornering line to spend more time on grippier pavement or where the track has more banking/camber — sometimes.
- You trade maximum braking for a better-balanced car to allow faster corner entry speed — sometimes.

It's this last trade-off I want to focus on today.

At the extremes of the spectrum, there are two ways to use the brakes. On one end is maximum threshold braking. This comes from a hard initial application of the pedal (yes, still squeezing on, but very quickly), and keeping the pressure at the point where the tires are right at their limit, just before beginning to lock up and skid. Of course, if you have ABS, then the system is mostly looking after keeping the tires at their threshold just before locking up. Even so, you — the driver — have to make a hard initial application of the pedal.

From a weight transfer perspective, you know what happens when you brake like this: the car's nose dives and load transfers from the rear of the car to the front, as the rear rises up; the front tires have more load on them, so their grip increases, and the rear tires' grip is lessened.

This is the fastest way to slow a car down (and why most cars have bigger brakes on the front than on the rear). In other words, you'll slow to a stop (or a set speed, such as the speed at which you want to enter a corner) in the least amount of distance. And this is why learning to brake at the maximum, at the threshold, is so critical. It's what we do to maximize the length of the straightaway approaching a corner, and minimize the amount of time and distance spent slowing the car down to the appropriate cornering speed.

At the other end of the braking spectrum, there is what I'll refer here to as the "light/long" brake application. Or, as I like to tell the drivers I'm coaching, "brush the brakes."

What's happening with the load transfer when braking lighter and longer? Obviously, there's less load transfer, right, because you're not braking as hard? The car is better balanced. By that I mean that the car is "flatter," and not nose-diving as much.

Never forget that the more balanced a car is, the more overall grip it has; the more load transfer we cause, the less overall grip the car has.

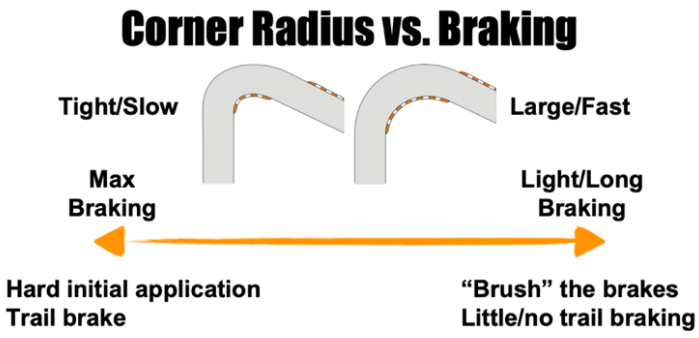
When approaching a corner that requires a big change in direction, such as a 180-degree hairpin, we trade overall cornering grip to get the car to "rotate" and point in the opposite direction. No, I'm not suggesting we drift the car around the corner, but we're okay with slightly more slip angle in the rear tires than in the front. In fact, we like that, as it helps us get the car turned and pointed in the other direction — we've rotated the car. In this case, threshold braking approaching the corner, then gradually trailing your foot off the brakes as you turn in keeps the load on the front of the car, helping it turn through the corner. We've traded some overall grip for the ability to change the direction of the car.

When approaching a fast, sweeping corner, we don't need to change the direction of the car so much. In this case, we'd rather have more overall grip, and therefore, brake lighter — even if that means braking longer. Let me be clear that by "longer" I'm not talking about really long. Just long enough. This might only be half a second longer, or twenty feet of distance. That little bit can make the difference. By braking lighter and longer, the car is better balanced, allowing you to carry more speed through the corner. And usually, when approaching this type of corner, you should also do your braking a little earlier than you might think so the car is well-balanced as you turn in.

A few years ago I was coaching two elite-level pro drivers at Sebring in a GT car. Their lap times were within a tenth of each other, then one got a slight edge — he was half a second clear of the other. In that world, that's huge. When I looked at the data, what I found was that the faster driver started braking very slightly earlier but lighter than the other driver — in most corners. This technique kept the car better balanced, which meant it had more overall grip. And grip is good (if you use it!).

This GT car was especially sensitive to balance because it was generating some aerodynamic downforce from the underside of the car, so a change in rake (nose-down, rear-up) took some aero grip away. But even in a non-aero car, this will have an impact on the amount of grip you have to work with.

I hope your take-away from what I'm saying here is this: "I need to adapt how I brake for every single corner. In some corners I want to brake late and hard, and trail brake a lot into the corner. In other corners I want to brake light and longer — just brush the brake — to keep the car balanced better. Typically, in tight and slow corners, I want to use the brakes hard and trail brake more to



change its direction; in faster corners I want to brake lighter and maybe even earlier to keep the car balanced when I turn in."

Ahhh, nice job of summing up what I wrote! Now, you just need to go put it into play. I would suggest sitting down with track maps of the circuits you'll be driving on soon, and think about each corner. Where does each corner fit on the "braking spectrum"? How might you adapt your braking for each corner?

Day #154: Dream about tire grip.

True story: I once had a dream about coaching a driver, and as I was explaining how to sense the tires' limit to him, the team engineer walked up and joined the conversation. She shared how a tire develops some grip from sliding on the surface of the track, but then kinda digs in and that's when the grip level really goes up. She drew a slip versus grip graph that had a small plateau — indicating where the sliding grip peaked — and then how it went up further as the rubber of the tire tread surface dug into the track surface.

Okay, that may seem like a bit of an odd dream to have, but I'm not normal! I woke up at this point, and lay in bed thinking about the dream, and what the engineer (I could never figure out who she was, or recall what she looked like in the dream) had said. Was there any reality to what she was saying? As I lay in bed, thoughts and images went through my head:

We, as drivers, sense grip — the peak of the grip level and going beyond it — primarily through increasing, leveling out, and decreasing amounts of g-force, and the direction change (yaw, rotation) of the car compared to where we want to be heading.

But tire grip levels are almost constantly changing. They go up with increasing temperature, and then decrease if the temps go beyond the ideal for that specific tire. Load on a tire increases its grip level, and that load comes from what we do with the brake, steering wheel, and throttle; it also changes based on track elevation and banking, and aerodynamic downforce. Track surface changes (pavement to concrete patch, sealer, cracks, painted areas, etc.) impact tire grip. Atmospheric conditions obviously affect grip levels, and this is the single biggest reason that our lap times can vary by a second or more from morning to afternoon, and day to day. It's a wonder that any human being is able to adapt to all of these different conditions!

But it was the image of the tire "digging" into the track surface that stuck with me the most from my dream (and my imaginary engineer). It reminded me of an image that is vividly stuck in my head from when I raced an Indy car at the Michigan 500 — and being passed by Nigel Mansell. As I arced into Turn 1 with my right foot nervously buried flat to the floor at something around 230 MPH, thinking that just one MPH more would result in a trip to the hospital, Mansell swooped around the outside of me. For fractions of a nano-second, my eyes went to his right rear tire. To this day, the image of the way the slick tread surface was distorted a few inches to the left of the rest of the tire is embedded in my mind. The sidewall was twisting and deforming beyond imagination, and if I'd not seen it for myself, I doubt I'd have believed it. If someone had told me that a tire could flex that much without destroying itself, let alone run lap after lap for close to a hundred miles, I'd have scoffed. The combination of 230-plus MPH, 4-plus lateral g-loads, over 5,000 pounds of aerodynamic downforce, and one incredibly skilled and brave driver led to me witness a tire on the very knife edge of the limit. A second later, I was more than a football field further around Turns 1 and 2 and onto the backstraight with that snapshot burned into my memory,

I lay in bed imagining a tire sliding on the surface of the track, with a little friction between the two, a bit like gently sliding a rubber eraser across a desk. I then saw that same tire “dug into the track surface,” as if I’d pushed down hard on that eraser while trying to move it along the desk surface. The tire sidewall and tread surface was twisting from the side loads and changing direction generated from the steering angle. I imagined the feeling from the driver’s seat. I sensed the rotation of the car, pointed just inside of where I wanted the car to go, knowing that tire slip would result in us (car and me) hitting my wished-for references. I felt the g-loads build, stabilize, go a tiny bit beyond as the tires hinted at going beyond their limit, come back to the front side of the limit curve, stabilize at the limit with the tire rubber flexing and twisting from the surface of the track.

Still dozing, but mentally in the driver’s seat, steering wheel in hands and my feet working the pedals, I could sense the most minute amount of rotation, as if I was sitting in a swivel chair. The combination of my view of the track and horizon changing just slightly from the rotation, and my internal compass, I sensed the car teetering back and forth between under and oversteer, the front and rear tires dancing on either side of the peak of the slip versus grip curve.

In my “somewhere between full consciousness and drifting back to sleep” state, my next thought was…and get this… “I need to write about all of this for Speed Secrets Weekly readers!”

But first, another lap dancing the car at the limit, tires digging into the surface of the track. Zzzzzzzzzz…. Ring, goes the alarm, as I awake, having drifted back asleep (probably with the tires at the limit through Turn 5!).

Here are a few takeaways from my dreamy night:

- Just because your car feels like it’s moving around — rotating, yawing, changing direction — that does not necessarily mean that you’re at the grip level of the tires. It may just be the “squirm” of the tread surface as it digs into the track surface and then lets go.
- The more you ask of the tires, the more they give — up to a point.
- Our brains are working while we’re asleep, and despite feeling frustrated at times when you’re not sleeping, it can be useful time spent. Sometimes what we dream about leads to breakthroughs in our thinking, understanding, and creativity (Keith Richards woke up one morning and noticed the tape recorder next to his bed had a blinking light. He hit play and then learned that he'd awoken in the middle of night and played the guitar riff that opens the song Satisfaction — he'd done this half asleep). If you take some time to decipher what was going on in a dream, you learn more.
- Yes, I’m weird to be dreaming about tire grip.

Sweet (at the limit) dreams.

Day #155: Don't anticipate, react.

You thought you should always be ahead of the car, anticipating what it's going to do, right? That makes sense… but there's a time and place where it's not the best way to drive. Let me explain.

Imagine driving into a fast corner on your favorite track, and you just know that your car is going to understeer right as you go back to power. How do you know? Because that's what it's done every lap in the past. So you apply some throttle and then ease up, transferring some weight onto the front tires, wait, and then your car goes exactly where you wanted it to go. You've adapted to the anticipated understeer, and made the most of it.

The problem is that you're losing half a second to other drivers in this corner alone. And they're in the same kind of car as you are. How is that possible?

Well, they may be reacting to what their car is doing, rather than predicting or anticipating what they think it's going to do. In doing so, they're driving their car closer to its limits than you are.

Sometimes you need to drive into a corner and not anticipate what it's going to do. Instead you need to drive it in, and then react to what it does. That'll help in two ways: First, it's the only way of really being able to read how your car is handling. And second, often it's faster because you'll be driving at the limit, and not at something you predict to be the limit.

A question I often ask drivers is, "What would happen if you took that corner at full throttle?" or "What would happen if you carried 5 (or 10) MPH more through that corner?" My goal is to get a gut feel from the driver as to what would happen. If the driver replies with something along the lines of, "Ahh, well, I think it would be okay," then we both know there's more in the car. If he replies with, "I'd crash," then I ask, "Which end of the car first?" That leads the driver into thinking about whether his car would understeer or oversteer off the track.

But if the driver answers, "I don't know," then we both know that he's anticipating that it's going to do something, so he's staying just below that limit. He's predicting what the car will do, and in doing so he's driving somewhere below the limit.

If you enter a corner and the car begins to understeer, what would happen if you allowed the push to continue? Sometimes, if you straighten the steering just a tiny bit, but keep the same speed (if you've just released the brakes, then you get back to at least maintenance throttle; if you've begun accelerating, continue with that same amount of acceleration), the car will continue to understeer, but you won't go off the track. The radius will expand, but nothing bad will happen. So don't anticipate that your car will understeer — stick with it, react slightly, but let it understeer.

If it oversteers, do the same — look and steer where you want to go (unwind the steering), and maintain or add throttle. React to it, and even let it happen — a little.

Trusting that you can react to an understeering or oversteering car, and that you don't need to anticipate it, is the most difficult part of this approach. But when you do, my guess is that you'll improve your lap times by at least half a second.

Day #156: Driving fun can be inversely proportional to the speed and cost of the car.

More than a few people have talked to me about the crazy speeds that some cars (often, without safety equipment in them) are able to achieve on the track. And how some (maybe many?) drivers are questioning at what speed the fun tapers off and starts to turn into... let's call it apprehension.

It's led me to thinking about there being (sometimes) an inversely proportional relationship between the cost/speed of the car, and fun.

The faster one goes, the higher the risk (and sometimes, fear), which often reduces the fun. That's why driving a slow car fast is so much fun. Of course, this can go too far — if the car isn't fast enough, it's no fun, either. Otherwise, we'd all be pushing little red wagons around the track.

A common meme these days goes something like, "Is the answer always Miata?" Of course, the question that goes before that is "What's the best car in which to learn to really drive on a track?" Or, "What's the most popular car at race tracks around the world?" Or, possibly, "What's the best bang for the buck when it comes to pure fun of driving on a track?"

I know, some of you don't agree that a Miata is the most fun. For some drivers, the lack of speed reduces the fun factor. But for many (as evidenced by the sheer number of Miatas at tracks), fun is not based on outright speed, but rather on being able to comfortably drive a car at or near its limits. And the beauty of a Miata is the comfort zone is larger because the cost of replacement is not so difficult to accept. And the risk of injury from a crash at its top speed is lower than one at the top speed of a much more expensive car.

Before you think I've sold my soul to Mazda and become a spokesman for the Miata, let me be clear: it's not the Miata. It's cars like them — relatively low-powered but with great cornering abilities, with limited or no driver aids and relatively lower performance tires. I'm talking about a Subaru BRZ, an E30/36 BMW, Porsche 944, or similar cars.

And this is not just me saying this. The number of drivers I've heard comments about the risk versus reward have increased dramatically over the past couple of years.

"If I was suddenly handed a Reliant Robin I would still drive my socks off and get the best out of it that was humanly possible. There's always a chance of winning no matter how badly your car is performing." - Nigel Mansell

Having said what I said, and reading Nigel Mansell's quote above, I'm reminded of Mark Donohue's famous quote: *"We're far from having too much horsepower…[m]y definition of too much horsepower is when all four wheels are spinning in every gear."* Nothing beats horsepower, right?!

Day #157: Read other cars.

My favorite part of racing endurance events is predicting what every single car/driver is going to do, and never being surprised. It's "reading other cars." After every endurance race, I review how I did, and one of the main factors I consider is whether I was ever surprised by the actions of another car/driver. It's a bit of a game to me, predicting what other drivers are going to do, and being a step ahead of them. When I've driven a 24-hour race, and gotten to the end having read every car/driver accurately, then I know I did a good job.

Some would say there is an art to noticing the subtle movements a car makes that precedes a certain line change, pass defense, or an early or late braking application. If it's an art, it's no different from learning to paint or play music. It's a skill that can be learned, just like braking, driving the line, or any other part of performance and race driving. And this doesn't just apply to racing, as reading cars at HPDE and track day events is important, too.

In Malcolm Gladwell's book, *Blink*, he shares the story of the legendary tennis coach Vic Braden. Braden surprised himself with the ability to predict — with almost 100% accuracy — when a player would double-fault service. He could tell strictly from the player's toss of the ball (before they actually hit the ball). With enough practice, we're able to notice the smallest of cues — even ones we're consciously unaware of; Braden developed this through decades of focused practice.

The key to reading other cars is picking up subtle cues.

Have you ever been in the situation where you're driving along a freeway, and you just had a feeling that the vehicle in the next lane was going to ease over into your lane? Or that a car was going to run through an intersection as its light was turning red? Or that the car behind you was almost going to rear-end you (or worse)?

What gave you those feelings? There was a subtle movement of the other vehicle that clued you in. Or it might have even been the physical actions of the driver that you noticed through a side window or mirror. Sometimes it's something as small as the tilt of the driver's head, or change of position of a hand on the steering wheel.

When you make a point of reading drivers and their car movement, it becomes an extra fun challenge.

When driving at an HPDE or track day event, where most cars are driven (in the same group) by the same driver, it's not super-difficult to start to predict their actions (and yes, every car/driver develops a reputation based on their actions — even you!). But when you're racing in a multi-class endurance race, it's a whole other game.

That Miata that is predictable and trustworthy almost turned into you as you dove down the inside entering Turn 3! What? Oh, there's a different driver behind the wheel now.

Yeah, just because a car is driven one way — cautiously, predictably — during the first half of your driving stint, that doesn't mean it will stay that way. It may now be driven erratically, and you have no idea what line the driver is going to take!

That's why you can't count on a car to be driven the same way all the time, and why it's important to continue to look for subtle cues that give you an idea as to what the driver is going to do. Don't get complacent, thinking that the green Mustang will always give you a point and let you by. Instead, pay attention to its line as it heads into the braking zone. Is it staying right next to the edge of the track, or is it angling into the turn just the tiniest little bit?

Does it look like it's under control, or is it being driven at the ragged edge — and maybe a little beyond? Could it mean that the driver is hanging on for dear life and doesn't have enough brain power left to even have a clue that you're behind them?

Has the car ahead turned in early for the next corner, indicating that it's going to slow coming out of it? Is that a hint to you that you should be a little more patient on the entry to the corner so you're able to get back to full throttle a fraction of a second earlier than usual, and get a good run on it exiting the corner to make an easy pass on the following straightaway?

Is the car ahead of you using every inch of curbing — on the entry, the apex, and track-out — bouncing the car from one to another? If that's the first time you've seen this car do that, could it be a different driver in the car? Or that the driver is trying harder than ever before, and is less aware of your passing attempts? Or that the driver is looking in the mirror at you too much? Or that the car's handling has "gone off" as the tires are used up, meaning the driver is doing everything they can to stay on line, but is unable to give you a little extra room as you get beside them?

Is the driver's head flopping from side to side, indicating that they're tired and getting sloppy with their driving?

I could go on and on....

Here's the thing: I want you to spend the next week of driving on the street being hyper-sensitive to reading other cars. Predict what every single vehicle around you is going to do, even if that means "nothing." Make a game of it, and see if you can go a full week without being surprised — even the tiniest amount — by any drivers' actions.

Remember, the reason we drive on the street is so we can practice our track driving! Good luck and have fun in your "Reading Other Cars" game for the next week.

Day #158: Visualize it.

When Vietnamese 10-meter Air Pistol athlete, Hoang Xuan Vinh, won his country's first ever Olympic medal in 2016, it made big news in his homeland. But it made news around the world for a different reason: Vinh and his teammates had not "heard the sound of gunshot for ages," according to the August 10, 2016 Saigoneer. And not only had he not used bullets to practice, he was short-sighted.

As Vinh's coach said, "As there is no ammunition, every day when they come here for training, the shooters just raise the gun, pull the trigger to hear the 'tick' sound, and that's all."

Apparently, the budget for ammunition for the Hanoi shooting team is as scarce as it is for track time for many drivers! But what Hinh and his team had an unlimited budget for was the use of mental imagery, and that's what they used to build the programming — the muscle memory — to win an Olympic medal.

Let me share a few more stories about the use of visualization, or more accurately, mental imagery.

A 2014 *New York Times* article about athletes using the mental skill: "This is, more than ever, a multisensory endeavor, which is why the term 'imagery' is now often preferred to 'visualization.' 'Visualization, for me, doesn't take in all the senses," said Emily Cook, the veteran American ski aerialist. "You have to smell it. You have to hear it. You have to feel it, everything."

The same article discussed the use of video and simulators, comparing them with mental imagery: "Video aids and video games are, for now, only out-of-competition tools. 'When you are at the actual performance, you've got to be able to pull it from here,' Nicole Detling (a sports psychologist with the United States Olympic team) said, pointing to her head.

"Detling said research had shown that athletes who were adept at imaginary play as children — 'imaginary friends, make believe, things like that' — were better at imagery."

The article addresses the fact that athletes use mental imagery for far more than the technical aspect of their sport, such as the movements they'll make when downhill skiing, or the line through the gates. Athletes also use it to deal with emotions, and help them get into the ideal performance state of mind. "Whenever fear surfaced, she (Cook) would picture herself pricking a big red balloon with a pin. 'That sound and that immediate switch would kind of snap me out of it,' she said, adding, 'The last couple years, I've definitely gotten to a point where when I'm on the hill, it's very quick for me to switch from a negative thought to a positive one."

Then, from a 2017 article in *Business Insider* about Dr. Michael Gervais (sports psychologist who has worked with NFL teams and players, Olympic athletes, UFC

fighters, and Austrian skydiver Felix Baumgarten, who completed a record-breaking dive to Earth from the stratosphere in 2012): "Gervais said that it is necessary to practice imagery on a regular basis well in advance of game day (or the day you give your presentation), because otherwise it's a 'hack' and won't be as effective."

If you drive on the road with one hand at the 12 o'clock position on the steering wheel, and only when you're driving on the track do you use two hands, what does your mind make of that? The one-handed driving is a habit, and you know how difficult it can be to change a habit. If you do mental imagery only on track/racing event weekends, it's the same thing. I suppose it's better than nothing, but it's not as good as if you did it on a regular basis.

In the *Mental Imagery Guide for Drivers* eBook, I share a program for mental imagery. Not just why you should use mental imagery, or how to do it, but how often and when. Just as you would — or have — with a physical fitness program, schedule out the various areas of your driving that you want to work on and improve using mental imagery.

For example:

- Monday evening: Improving trail braking
- Tuesday morning: Improving trail braking
- Tuesday evening: Racecraft
- Wednesday morning: Racecraft
- Wednesday evening: Track-specific for the one you're driving at next
- Thursday morning: Track-specific for the one you're driving at next
- Thursday evening: Managing confidence — feeling comfortable being a little uncomfortable driving at the limit
- Friday morning: Managing confidence — feeling comfortable being a little uncomfortable driving at the limit
- Friday evening: Regaining focus — using a Pre-Planned Thought (PPT) to trigger focus
- Saturday morning: Regaining focus — using a Pre-Planned Thought (PPT) to trigger focus
- Saturday evening: Adapting to a handling problem
- Sunday morning: Adapting to a handling problem
- Sunday evening: Having fun! Just imagining enjoying the fun of driving
- Monday morning: Having fun! Just imagining enjoying the fun of driving

This is a pretty aggressive program, so you may want to simplify yours, but this does give you an idea of how you can schedule various things you want to improve. Notice that it's not all just track, or technique-specific areas that you're mental imaging. You

can also program your mental state of mind, confidence, managing emotions, focus, and just about anything else. The key is having a plan, and then working that plan.

You may have also noticed how each area is worked on during the evening of one day, and then again the following morning. There is research that indicates that doing mental imagery just before going to sleep is the best time to have it stick in your mind (it sinks in while you're asleep), and then refreshing it the following morning will help cement what your mind learned from the evening before, and then throughout the night while you slept.

Imagine that!

Day #159: Make your mistakes learning-takes.

It's been said that there are two different types of failure:

1. You try something, fail at it, learn from it, and try again. This isn't really a failure.
2. You try something, fail at it, and don't learn anything from the experience. Or, perhaps even worse, you learn the wrong lesson from it. This really is a failure.

I like to think of the first one above simply as a "learning-take." It's a mistake that you learn from.

Of course, it's what you do with a mistake that determines whether it was a failure or not.

Far too often I see or hear of a driver who tries a new approach to driving, it doesn't work the first time, so the conclusion is "That didn't work." This could be:

- Braking lighter and earlier
- Braking later and harder
- Trail braking more or less
- Turning in earlier or later
- Carrying more corner entry speed
- Changing the line between two corners
- And so on

Any of the above new and different approaches could make you faster, if you give it a chance. But if you try something new, and it doesn't help the first or second time you try it, that doesn't necessarily mean it's not a better way. Instead of considering it a failure, think of it as an opportunity to learn how to make it work.

Imagine approaching a corner the same way you have every other time. But you've been thinking you can carry more entry speed, so you brake lighter and roll the speed in... and off the track you go. Is that a mistake? And was the extra speed the real cause of going off the track?

Most drivers, when carrying more speed into a corner than they ever have before, turn in early. It's human instinct: we all feel safer closer to the inside of the track, so if we're carrying more speed and get a little nervous and tense, we turn in early.

If you carry more speed into a corner and go off the track, does that mean that speed can't be made to work? It's possible. It's very possible you just haven't figured out how to make it work — yet. Well, I say "yet" if you don't give up on the approach, think it through -—deeply and logically — and gradually work up to making it work.

So, with our example, the increased entry speed may not be what caused the off. It may be turning in early. But without the proper thought and analysis, the conclusion might be "the speed caused the off-track excursion."

The key to making this work is the deep and logical thought process, prior to trying a different approach — and after. If you play through as many scenarios as you can before you go on track, you'll be better prepared, and more likely to make what you're wanting to do work.

If you take the time to think through what happened afterward, you may discover that what you initially thought was the cause of the problem really wasn't.

Yes, the key to going faster is often looking at mistakes as a learning opportunity.

Day #160: Would you change?

Let's say you and I are at the track and I'm coaching you. After talking with you before you go on track, observing you from a couple of different corners while you drove, then reviewing your data and video, I came to the conclusion that you needed to change a fairly major part of your driving.

Would you change if I asked you to?

Or, let's say that you're at a point in your development as a driver where you notice something that's holding you back from getting to the next level. For example, you've been driving for a while now without heel and toe downshifting; you haven't thought it was necessary to learn to do it, and it hasn't turned out too badly. You're pretty fast, and don't make many mistakes. But as you get closer and closer to consistently driving at the limit, and where you're chasing the last tenth or two of lap time, you see that your downshifting is not quite as smooth as it could be.

Would you change?

Or, you've been racing for years in cars that required you to use the clutch, but now you have a car with a sequential gearbox and your left foot seems to just be along for the ride — it's not doing anything. That means that you now have the opportunity to learn to left-foot-brake. However, after all these years your right-foot-braking is highly effective, and whenever you've tried left-foot-braking you realized you don't have the same fine muscle control and your braking is not very smooth. But you've heard from others that left-foot-braking provides an advantage.

Would you change?

What if each of these changes resulted in an improvement in your lap times? Or meant you'd have more fun from learning them? Or, what if it would be hard to prove there would be an improvement in lap times, but it could take a long time to learn to change? Or (even more important to consider), if it meant that for some period of time your lap times would get worse before they improved? What if there was a financial reward for making the change?

Of course, all of these factors come into play, right? How much time you have to learn and practice the change is another important factor to consider. If you're only going to drive a few times a year, making a major change to your driving technique may not make sense. But if your main purpose of performance driving is to challenge yourself and learn, then it makes more sense to jump at any chance to make a change for the better.

Change is not easy. We're creatures of habit. And it's been my experience that even the very best drivers are reluctant to change. In fact, often it's the better drivers who resist changing the most, since they're doing pretty darn good as it is. "Why mess with

success? Don’t fix it if ain’t broken,” is the thinking. And you can’t blame someone for thinking that way.

In business, that thinking — sometimes referred to as being risk-averse — has led to some huge problems. Kodak resisted the change to digital photography; the tech giant Digital Equipment refused to believe that the personal computer would ever catch on and stayed focused only on making mainframes. The list of companies afraid of change is a long one.

What risk are you averse to? What change in your driving are you resisting making?

If you could start fresh, what would you do differently with your driving?

What one thing done differently would make the biggest impact on your driving?

P.S. – It’s funny where inspiration comes from for improving one’s driving (if you're like me and all you ever think about is driving...). I was driving along the road listening to music, when the Tracy Chapman song, *Change*, came on. As I listened to the lyrics, I thought about the many, many drivers I see who do pretty much the same thing with their driving over and over again — often with more effort — and expect a different result. Doing the wrong thing with more effort rarely results in what you really want. "How bad, how good does it need to get?"

P.P.S. My racing career spanned the time where one had to use a clutch, and therefore right-foot-brake, to sequential gearboxes where left-foot-braking was possible. Because I had developed my right-foot-braking so well, I didn’t want to change. But I did. It meant taking a step or two backwards for a while, but I’m glad I did it. To this day, I’m still consistently better with right-foot-braking, but if I’m in a car where I can left-foot-brake and I’m looking for the ultimate in fast lap times, I will. For endurance races, where that last tenth of a second is not as important as consistently error-free driving, I go with right-foot-braking. But if I’m looking for a few super-fast laps, I use my left foot to brake. It was not an easy change to make, but I’m now fairly “ambipedal” with my braking.

Day #161: Cars don't disappear.

"I didn't see him."

There you go — an all-too-common excuse for a crash between two cars on a race track (sorry, let me make what I just said more accurate: "…between two drivers on a race track"). A driver moves to the inside of a competitor going into a corner and brakes later, the driver being passed turns in anyway, and bang… a crash.

It should not have happened – the driver being overtaken should have seen and known the passing car was there. Cars don't disappear. And drivers should never be surprised by a car attempting a pass on them. Whether you're competing in wheel-to-wheel racing, or driving in non-competitive HPDE and/or track day events, near (or at) the top of your priority list is to always know where other cars are around you.

If you look in the mirror and see another car behind you, and then you don't see it, you can assume the driver did not decide to pull off to the side of the track and quit. If you can't see the other car, you better assume it's still there, and likely in a position that is difficult or impossible to see at that moment in time. You better leave room for it, even if you don't know where the other car is. There are very few places it can go, so just leave some room, wherever that is. This is a non-negotiable option.

Now, let me flip my message around: Cars can disappear. Yes, you can disappear from view of the driver you're passing, and when that happens, there's a good chance they will turn in and hit you. Or, you'll hit them. It doesn't really matter who's at fault if the both of you are out of the race, pointing fingers at each other, both of you thinking about how you're going to pay for the damage.

Day #162: Learn the difference between using a simulator and mental imagery.

You do what you do because you're mentally programmed to do so; you don't do what you want sometimes because you don't have the right programming yet. (How many times have I said that?!) All the skills and techniques you rely on to drive fast are programmed into your mind, and as you practice, you're developing and updating your programming.

How do you program your mind? Well, physical practice, but also mental practice (mental imagery, or what's often called visualization). The physical practice can take place on the track, on the road, or on a simulator. That's why sim racing/driving is so effective — it's developing your programming.

But which is better, programming your mind by using a simulator, or through the use of mental imagery? Yes, they're both better! So, is there a difference? Let's take a look.

Let's say you're going to a track you've never been to before. It's pretty difficult to close your eyes and imagine a track you've never driven, so it's time to hit the simulator. Think about every lap you drive on the sim as programming (because that's what it is), and every time you make a mistake, you're increasing your ability to make more mistakes. Practice doesn't make perfect, only perfect practice makes perfect. And, practice makes permanent. This is why it's so important to practice the right stuff, the right way. It's why it's important to not be in a rush to turn a fast lap on your sim. You're better off to start relatively slowly, and gradually build up your speed — accurately, precisely, properly. With every corner, every technique, every lap…you're developing your mental programming to do everything right when you actually, physically, get on the real track.

You're a programmer, and in this case, you're programming yourself.

So, why would you ever want to do mental imagery when you can just practice on a simulator? Because you can improve the quality of your programming using your imagination; and because mental imagery can be even more effective for learning than using a simulator. Oh, and it's FREE!

Using a simulator is "learning from the outside in." The simulator is putting information into your mind. It's like me telling a child that 5 x 7 = 35. I put the answer in the child's brain. With a simulator, you're putting the answer in your mind, but you haven't fully internalized it. It hasn't become completely programmed.

Using mental imagery is "learning from the inside out." It's like that child figuring out that five groups of seven result in thirty-five things. The understanding is much deeper with mental imagery, and because you figured it out yourself, it'll stick with you longer.

In a perfect world, you would be merging the use of mental imagery and simulators. Again, if it's a track you've never been to before, start relatively slowly and drive lap

after lap on a simulator until you know exactly what's coming up and how you're going work with it (what techniques you'll use, and when). Then turn the sim off, close your eyes, and imagine driving laps, doing everything perfectly. Do this for fifteen minutes or so, and then fire up the sim again, and pay more attention to the details of the references around the track. Then back to mental imagery, and while doing so, think about what you're doing at a deeper level. Think about why "this" works and "that" doesn't in Turn 3, while "that" works in Turn 9.

When I talk about simulators, I'm not suggesting that you have to have a high-end unit. An XBOX with Forsa, and a simple pedal and steering wheel setup will do just fine for the process of learning. Sure, if you want to get more realism, you'll need to spend more for better equipment, but that's not necessary. In fact, I could make the argument that with a lesser system, your mental imagery becomes more important, and that in the long run will pay off. It may be that a high-end sim takes away some of the need for mental imagery, and it's that last little bit that can only come from mental imagery that will make the difference between the best driver you can be and where you're at now. See, it's possible that using a simulator makes you a bit lazy. It does do some of the heavy lifting of learning, but that last few percent comes from within.

Having said that, of course it's possible to do all of this learning without ever using a simulator! The initial learning of the track — which way it goes, references, and so on — can be learned from watching video, and then you pick it up from there using mental imagery.

If you can't close your eyes and imagine driving an entire lap while moving your hands and feet as if you're working the steering wheel and pedals — and doing it within a few seconds of a real lap time — then you haven't fully programmed what it takes to drive the track. You then need to spend more time doing mental imagery until you can imagine a lap, timed to within a second or two. Then it's truly become programmed.

Of course, there are many more uses for mental imagery and simulators than just learning a new track, and I'll write more about them in the future.

Take advantage of the time away from the track to develop your ability to use mental imagery. (If you need a push start on the subject, get my *Mental Imagery Guide for Drivers* eBook)

Day #163: Learn to be a more versatile driver.

By almost any standard, and by many opinions, the greatest race driver of the 20th century was Mario Andretti. While many put the likes of Fangio, Clark, Stewart, Senna, Schumacher, Hamilton and Verstappen (and a few others) as the best in Formula One, no one can dispute that Andretti is near or at the top. Why? Because he was so versatile.

Andretti won in dirt sprint cars, NASCAR, Indy cars, Formula One, and sports cars. There aren't many cars or series he didn't drive in and win.

"I have won on 127 different kinds of track, clockwise and counterclockwise. I have experienced the passing of the engine from the front to the back. I have raced with the greats who have since retired, and in places that are now parking lots," Andretti was once quoted as saying.

If you want to be a good driver, drive everything you can get your hands on. And it really isn't as hard as you think to get your hands on some cars.

Do track days. Go to the local indoor kart center and drive. Go to a driving school, no matter what level it's aimed at (you might be surprised at how much you learn, even after knowing it all!). Race in a Champ car or World Racing League event. Compete in autocross events. Race on an oval. Experience driving a rally car. Try drag racing! Definitely do some sim racing. Ride motorcycles.

All these things will help you improve your driving in the main arena in which you participate.

When I was racing Indy cars, I was invited to compete in a few autocross events. I learned things that helped me be a better Indy car driver. In between Indy car seasons, I went ice racing (I'm still warming up from that). Prior to racing a prototype car in the IMSA race at Lime Rock one year, I attended a Skip Barber school to learn the track, as I had not driven it before. Not only did I learn the track, I was reminded of so many important driving techniques that I became a better driver.

I've coached drivers on how to improve their corner entry speed, and specifically how to use the brakes to help rotate the car. As part of that process, I've had some drivers attend a rally school (I had them go to DirtFish Rally School, as I was involved in running the business for a few years). They learned how to enter a corner in a rally car, which resulted in doing it better on a road course.

Oval racers learn the importance of unwinding the steering coming off a corner in a way that few road racers ever fully comprehend. When you hear the engine note increase by simply straightening the steering wheel, it's motivation to never turn the steering wheel again! Anyone who has done some oval racing will tell you that it's made them a better road racer.

Indoor karts are not known for the abundance of power! And that's why it's such a great learning tool. If you over-slow by half a MPH, you've blown it. You learn to let the kart "run free." You also learn how to learn (a subject I've covered more than once in this column). When you see a 14-year-old kid turn a lap a second faster than you, you either go home and complain about the bad kart you were given, you blame it on your weight disadvantage, or you figure out how to drive faster. I recommend the third option! Figure it out. It's not impossible.

Compete in an autocross. It'll teach you how to visualize and learn a course faster, as well as how to go fast, fast. There is no time to warm up. You have to go NOW!

Drive in one of the low-cost (a relative term in our sport!) endurance series races. Due to traffic, you'll spend as much time driving off the ideal line as you do on the line (maybe more). What you learn by doing that is invaluable.

When you decide what events you're going to attend, consider doing fewer of the usual, and more of the unusual. Try something new. And be prepared to learn a lot from it that will apply to the old.

Day #164: Use a mindset shift for fast corners.

One of the most-discussed (and even argued-over) topics is, "which corner on a track is the most important?" Some options:

- Slow corners
- Fast corners
- The one leading onto the longest straight
- The one with the highest speed
- The one you spend the most time in
- The most intimidating one
- The blind corner
- Or…

Which is most important?

One simple way to answer this question is the corner you're losing the most time in. And that could be any of the ones I just listed, for various reasons (technique, confidence — or lack thereof — how well you can visualize what's coming up, your car's power versus cornering ability, etc.).

The most-often-quoted advice is that the corner leading onto the longest straightaway is the most important. And I agree that it is — sometimes. Other times, it's the fastest corner, and often that's because it's the scariest, most intimidating one. Other times, it's the corner that you have the hardest time figuring out where to place the car, how much speed to carry, what techniques it requires, and so on. And, in my experience, other times it's the corner that you find the most difficult to picture in your mind (if you're able to drive most of a track in your mind, but have a hard time visualizing one particular part of the track, that's the priority).

Most instructors and coaches observe that drivers drive the slow corners too fast, and the fast corners too slow. What does that tell you about which corner needs the most focus, and which one needs the most discipline — for you?

Let's look at the "time you spend in the corner" factor.

A good friend, Ryan Staub, once sent me an email after his first experience racing at the new Ozarks race track, and he shared a really interesting observation. Having co-driven with Ryan a few times, I know how fast a driver he is (he's fast!). Still the blind high-speed corners at a new track were a challenge to him — and to other drivers. Here's what he shared with me.

> *"On to the topic — approach to/perspective of high-speed corners. Ozarks has a ton of high-speed corners, and nearly all drivers had a tough time getting up to*

speed. There are some really intimidating ones that are off-camber, blind apex and/or exit, etc. Helping some of my co-drivers get comfortable with approaching some of the high-speed turns, and thinking about it myself while driving, I thought of a different view of the high-speed/scary corners that helped me and a couple others I was talking to.

"The perspective shift I had was about how fast and quick you go through high-speed corners, relative to low-speed corners. But, if you really think about it, and actually put a stopwatch on the amount of time you spend in a high-speed corner, it takes as much time, or even more time than many low-speed corners. You have way more real-estate to work with, so you have time to make adjustments to throttle/brake/steering. Sure, you're covering more distance for each second in the high-speed corner, but to me it seems like you have just as much time to react and adjust if you get it wrong."

My take-away from Ryan's comment is that once you shift your mindset from how fast a corner is to how much time you have to react, adjust, adapt, and fine-tune what you're doing, it makes these high-speed areas of the track much less intimidating. I thought that was a brilliant observation, and it's something that I've used when coaching drivers.

It may seem like you need to react faster in fast corners, but thinking about how much time you have — due to the distance that long, fast corners have — you may be able to improve in those turns.

Day #165: Stop worrying about your vision and just look where you're going.

I'm beginning to think that too many drivers spend too much time worrying about where they're looking, what they should be looking at, how far ahead they're looking, and who knows what else. I receive at least one email each week from a driver asking me something about where they should be looking when driving on track.

Relax.

I'm not saying that where you look, and how far ahead you look is not important. Hey, it's critical. But you'll never drive at your best when you're worrying about something.

Okay, there's a difference between focusing on something in order to improve it, and worrying about it. There's a fine balance needed.

And here's the big news: there isn't a formula for where to look. Many of the drivers asking me for advice about vision are looking for a "when approaching the brake zone, look at the markers, then half a second before you get there, move your visual focus to the turn-in point, then to the apex, then the track-out point...." But it doesn't work like that. Sure, that's the general approach, but it's not like you're clicking your vision from one point to the next, to the next, to the next, and so on.

Your vision should be more fluid than that. It glances ahead, then back, forward, then back, to the sides, forward, back, sides... all with the general flow towards where you want to go.

Right now, do this simple experiment. Look up from the device you're reading this on, and identify a point on a wall in the distance. Look at it. Now look to another point to the right of it, and focus on it. Then identify something else to your right and focus on it. With each change of focal point, notice that you're moving your vision in jumps.

Now try that again, but think about scanning from one point to another, with very subtle and quick glances back to the previous point.

Do it again, and this time make it completely fluid, with absolutely no stop-start action. It's as if you have BIG VISION, where you're "All-seeing, all-knowing." You're hyper-aware of everything, but not just one thing. You see the big picture of where you're going.

Now - and this is the most important part of this experiment — close your eyes and imagine seeing each of those points. Imagine your vision scanning and flowing from one to another in a smooth flow with subtle and quick glances to check on progress.

Were you able to imagine the visual movement with your eyes closed? If so, does that give you a clue of what you need to be able to do on track? And if you weren't able to imagine your visual picture following from one point to another in a fluid way, then all the

advice in the world about where to look and how far ahead to look will not do you any good. You need to practice imagining where you want to go, and not just seeing where you want to go — at speed. I refer to this as the difference between “physical vision” and “mental vision.” Without the ability to mentally see where you’re going on track, you’ll never be as consistently fast as you could be.

I’ve said that the real trick advanced techniques that make the best drivers the best are "just doing the basics better." The ability to mentally see where you’re going before you get there is part of the basics of driving fast on a track. If you’re relying on having to physically see something to know what to do, you’re going to be behind in everything you do. You’ll be reacting rather than driving proactively.

Day #166: Manage mental imagery errors.

Have you ever sat down to imagine driving a lap of the track, only to get to Turn 3 in your mind and begin thinking about whether you should add fuel to your car, about the mistake you made the last time you went through Turn 1, or even where you're going for dinner? Or worse, do you find yourself making a mistake in your imagination? Hey, you're human! We all do it — some more than others, but we all do it sometimes, to some extent.

Let's start with what may seem to be the "worse" situation, where you actually imagine making mistakes.

First, accept it. It's okay. It will happen in mental imagery every now and then; the most important thing is that you learn how to re-focus and turn it around. That way, in real life, when things don't go perfectly, you will actually be better at recovering from it. So, look at this as a great learning opportunity, one that will be valuable.

When you see a mistake beginning to happen in your imagination, say to yourself, "Pause." Then stop imagining what you were doing. Then say, "Rewind," and re-start just a few seconds back. This time, move through what you were doing — driving through Turn 3, for example — slowly, very deliberately, and accurately – all the while saying "Play" to yourself. Do it as slowly as it takes to do it correctly. And any time you begin to make a mistake, "Pause, rewind, play." The more you use this routine, the easier it gets. Remember that a mistake is only a "learning-take."

What about simply losing focus while imagining a lap or multiple laps? As I've written before, even the very best drivers lose their focus, whether on the track or while doing mental imagery. What matters most is how quickly you can regain your focus. So, the good news is that every time you regain your focus while doing mental imagery, you've just gotten better at regaining your focus on the track. It's like anything in which repetition builds the mental programming to do something better; the more times you lose focus, and then regain it, the better you'll be at doing that.

Using a trigger is the key to regaining your focus, so anytime that you realize your mind is beginning to wander, say, "Eyes up — look ahead," and very deliberately bring your attention back to what you were doing. If it means restarting where you were on the track, then do that. And each time you do this, use your "Eyes up — look ahead" trigger phrase.

Part of the challenge with imagining a full lap in your mind is getting the "less interesting" sections right. It's difficult to imagine how long it takes to drive down a straightaway, and that means you either lose track of time, lose your focus, or simply lose your rhythm. In most cases, when I'm coaching drivers on the use of mental imagery, I find that those who struggle the most with this are only imagining the visual picture of driving. They're visualizing, but not doing mental imagery. The difference?

Mental imagery involves more senses, and in the case of driving, you must imagine the sound and feel, as well.

If, in your mind, you're only imagining what you see, it's easy to lose track of the timing while driving down a straightaway. But if you also imagine the feel of the bumps and cracks in the track surface, and hear your engine and the shift points, it's easier to keep track of time. Move your hands and feet as if they were working the pedals and steering wheel. Replay all sounds you hear. Make it real.

To recap: If, while doing mental imagery, you either make mistakes, lose focus, or find your timing is way off:

1. Accept it — be okay with it, and don't fight it.
2. In your mind, say, "Pause, rewind, play" and begin from just before where you made a mistake or lost focus — and even slow it down to "drive" it very deliberately.
3. Incorporate what you hear and feel into your imagination. Make it mental "actualization," rather than visualization.

Mental imagery is no different from any other skill, as it takes practice to get really good at it. Is it worth the effort? It sure is. It's free, you can do it anywhere and as many times as you want, and it's almost as effective a way to learn as actually driving on track — sometimes even more effective. But it does take practice. As the comedian/actor Jim Carrey says, "I would visualize things coming to me. It would just make me feel better. Visualization works if you work hard. That's the thing. You can't just visualize and go eat a sandwich."

Day #167: Vision gets you around the track; what you feel and hear tells you whether you're at the limit or not.

We can all agree that our vision is super important to driving. Hey, if you were hearing impaired or had no sense of touch, you could still drive around a track. Yes, it would be more difficult, but you could do it. But driving around a track if you were blind would not last long!

And that's why so many suggest that your vision is the most important sense you rely on when driving. That's true, in terms of getting around that track. But the fastest drivers, the ones who drive their cars at or near the limit most consistently use their kinesthetic and auditory senses to read their cars. They rely on these senses to tell whether they're at the limit or not.

As often as you remind yourself to look further ahead (or you have someone else help you with that), remind yourself to focus on what you feel and hear.

- What do you feel through the steering wheel?
- What do you feel through your seat?
- Do you feel the g-forces build and let go?
- What do you feel through the brake pedal and throttle?
- Do you feel the weight transfer, front to rear, rear to front, side to side?
- When the car begins to slide, does it feel like a go-kart — a flat slide — or a big old Cadillac, rolling over?
- Do you hear the tires, and if so, what do they sound like before, at, and over the limit?
- Do you notice the wind noise?
- Do you notice the sound from your engine, brakes, and transmission?

If you take time on track to solely focus on these, you'll be better at sensing the limits of your car.

Day #168: Learn how to drive the ideal line.

Most high-performance and race drivers are first taught about the cornering line before pretty much everything else because it's the easiest to show and demonstrate. Illustrating load transfer, tire slip angle, controlling a slide, and vision are difficult, but drawing a line through a corner on a whiteboard is relatively simple!

Another reason is that a driver needs to know where they're going before they can work on car control and other aspects of driving fast. Knowing which way every corner goes, and the best path through them, is the foundation of getting around a track as quickly as possible.

The way I see it, there are two approaches to learning cornering lines:

1. A formulaic approach
2. A "let my car tell me" approach

The formulaic approach starts by teaching the basic concepts of cornering — references, radius, positioning of the apex, entry versus exit speed. Usually, the "school line" is then taught, with a kind of memorization-by-rote approach to driving the current track (most times by connecting the dots, or in this case, driving from cone to cone), focused on late apexes and exit speed. From there, it's all about adapting each of the variations of the basics to suit any new corners you face in the future, often by copying other drivers (videos, following, etc.) and through trial-and-error experience.

The other approach should also start with a simple understanding of the same basics, but then the driver uses more of a trial-and-error approach to learning what works and what doesn't, and how experience relates to theory. Some worry about the "error" part of this process, but approached with discipline, it's not any more dangerous than the first method. And we all know which way of learning sticks with us the best, whether it's the memorization- by-rote or learn-through-experience approach (Hint: it's learning through experience).

Ideally, there's a combination of the two, but the second relates to a Speed Secret I've shared here in the past, "Drive the car, not the track."

My point with this message is that we can get so overly-focused on driving the "perfect" line, that we don't drive the car anywhere near its limit (or the limit you want to drive). And when we drive the car at or near the limit, it tells us where the line is.

With that in mind, let's look at a few guidelines to keep in mind regarding the ideal cornering line. But first, there is no one singular ideal line. It depends! Keep that in mind while reading this.

Let's start with what I'd say are the two most important goals:

1. Spend as little time as possible in the corner.

2. Get maximum speed out of the corner by accelerating early to maximize straightaway speed. (Note: This is not always the highest priority, and sometimes corner entry and mid-corner speed is more important - but those are the exceptions to the rules).

Remember, corner exit speed is usually more important than entry speed. Until it's not.

The single most important factor in determining the line you drive through any corner is what happens after it. Does it lead to a straightaway? A long one or a short one? Does it lead almost immediately to another corner? Which way is it turning? Is it faster or slower than the one you're approaching first? What surface or shape of track (uphill, downhill, banking, negative banking, a surface with more or less grip) does the corner have after it?

Cornering is all about compromises. As I said, there is no one ideal, perfect line. Having said that, the differences in the line between cars is not as big as many will have you believe. When drivers start talking about the "front-wheel-drive line," the "Porsche line," or the "momentum car line," and how big a difference there is with each of them, I want to scream! There can be as big a difference in line between two front-wheel-drive cars as there is between a rear-wheel-drive Porsche and a rear-wheel-drive Miata. But the thing is, it's not that big a difference. It's subtle. In fact, the difference in the line between a morning session on new tires and an afternoon session on old tires can be just as different as between any two cars. In this case, the line has more to do with how you, the driver, manage the load balance, turning rate, importance of corner exit speed, and so on.

The same is true when it comes to "momentum" and "power" cars. Ultimately, all cars are momentum cars – even ones with 700 horsepower.

Every inch of track you are not using is costing you speed. You paid for it, so use it all. The key message here is to drive as large a radius as possible, which means using all the track... until using all the track increases the distance traveled to the point of it being slower.

Yes, distance matters. The shortest distance between two points is a straight line, and the shortest distance is usually the fastest, too. Again, there is a compromise here between using as much track as possible to maximize the radius, and the shortest distance. Short of copying what other drivers are doing – which may or may not give you the best answer – the only way to find the ideal compromise is to experiment and learn which is quickest. But keep in mind that it's what leads to the quickest lap time, and not just the fastest through one particular corner.

The more references you have, the more consistently fast you'll be.

For example, if you have just three reference points (turn-in, apex and exit), and you make an error and turn in too early, you may not recognize the mistake until you're

almost at the apex of the corner. At that point, your correction will need to be fairly big, causing a big loss in lap time. If, because you have more references, you recognize having turned in too early, you can make an earlier and smaller adjustment, meaning it won't hurt your lap time as much. And that leads to consistently faster average lap times.

One way of looking at cornering is that the turn-in point is the most important, as it dictates everything afterwards: where you apex and exit, and when you're able to begin accelerating. But to be consistently fast, you should be mentally- and visually-focused on the exit point, as that will make you do everything possible to get there in the least amount of time. It's starting the corner with the end in mind that really works well, and therefore, the exit is the most important reference. I do believe this is true in terms of what you have in mind: you want a mental image of the exit of the corner before you actually start turning in to it.

As a general rule, the later you turn in and apex a corner, the earlier you can begin accelerating out of it, and the higher your top speed will be down the following straightaway. So, the line you drive has a huge influence over your corner exit speed. But, one way of beginning to accelerate early is to over-slow the entry, like to 5 MPH. Hey, entering practically any corner at 5 MPH means you could start accelerating from the moment you turn into the corner! But entering that slow would defeat the purpose of what you're trying to achieve: get from point A (the start line) to point B (the finish line) as fast as possible; in other words, minimize your lap time.

Late apexes are taught and encouraged for this reason – to maximize corner exit speed – as well as being "safer" than an early turn-in. Of course, all of this is relative... what is late, and what is early? For sure, too early an apex is not as good as the ideal apex, but so is too late an apex. Some say that too late an apex is safer, but I disagree – it's neither safer nor more dangerous. Too late of an apex means you have to turn the steering more because you have to tighten up the radius of the corner, and the more steering input, the more chance of losing control. And for sure, the tighter the radius, the slower you have to be traveling. That's why a late apex is not always safer: It means that you have to turn sharper, on a tighter radius early in the corner, and a tighter radius means going slower, and possibly sliding/spinning.

Speaking of general rules, here's another Speed Secret: The faster the corner, the closer to the geometric line you should drive; the slower the corner, the more you need to alter your line with a later apex. (I define the geometric line as one with an apex on the inside at the middle of the corner, and the turn-in and exit points on opposite sides, as far to the outside as possible; the largest radius one could drive through an individual corner).

But there are exceptions to every rule. Still, this Speed Secret works more often than it doesn't: The slower the corner, the later the apex, the quicker and crisper you need to turn the steering wheel; the faster the corner, the more you need to arc or bend the car into the turn with slow hands.

While outright speed, and ultimately the shortest lap time possible, seems to be the priority, consistency also plays a big part in what you're trying to achieve. If you're racing, consistency — or the fastest average lap time — wins the most races. Many times, the driver with the single fastest lap time does not win the race, but the most consistently-fast driver does. So, what you do with your cornering line should be aimed at the fastest average lap time.

Track surface and shape will alter the consistently-fastest line as much or more than anything else. For example, a banked corner will reward a tighter line – one where you get your car into the banking as early as possible, and stay there as long as possible. In other words, it means taking advantage of the grip provided by the banking.

The same applies to pretty much any corner – take advantage of where there is more grip, and minimize the time/distance spent where there is less grip.

Give the tires a warning about what you're about to do before doing it. The idea is if you come up to a corner and very quickly turn the steering wheel, you haven't given the tires a warning about what you're about to do. If, instead, you very gently and subtly begin turning the steering wheel slightly, and then turn into the corner, the tires are ready – they will hold on longer.

This is good advice... until it's not. And it's definitely not good advice when you "crab" into a corner by easing the car away from the edge of the track before the actual turn-in point. By doing so, you tighten the radius of the corner. That's one of the most common errors I see drivers make.

It's important to ask yourself about your overall purpose? Is it to turn one fast lap, as in a qualifying session or a time trials run? Or, is your average lap time more important over the course of a race? Are you participating in HPDE/track day event where the real reward is self- improvement, and not a trophy for winning a race? Are you setting up a pass, or defending one? Are you driving in a sprint or endurance race? All of these could lead to altering your line, if even just slightly.

Finally, the cornering line should not be thought about or focused on in isolation. No, the line is affected by the amount you trail brake, for example, and vice versa; it's impacted by the way your car handles, as well as the weather you're driving in. It's a constantly-changing thing... but maybe not as much as some drivers think. That's what makes track driving and racing so much fun – there is no one single "best" line. If there was, what we do on track would get really boring!

Day #169: Understand the *Why* behind the line you drive. Then use it.

Most drivers drive the line they do around a track for one or both of these reasons:

1. It's the line they were taught (by an instructor, a sim/game, or whatever)
2. They saw others drive it (in a video, or on track) and just followed

From there, hopefully they fine-tuned it a little to better suit the conditions, their car, and even their driving style. But many don't, and continue to drive the same old line they learned from others. Why? Because they don't fully understand why they're driving the line they are.

Peeling back the layers of the onion, here are the factors a driver must consider and compromise in determining the ideal line — not just on that day, but on that very lap:

- Radius (the larger, the better)
- What it takes to get to full throttle as early as possible
- Grip (track surface, elevation changes, banking)
- Distance (the less, the better)
- What comes next

When thinking about maximizing the radius, you can also think of it as "Turn the steering wheel as little as possible; once it's turned, work at straightening it out again as soon as possible." In other words, minimize steering input. We all know that the larger the radius (and remember that a straight line is the largest possible radius!), the faster we can drive because the tires are not being asked to generate as much cornering force for the same speed; we can go faster on a larger radius with the same cornering force.

This often relates to the "What comes next" factor. After the corner you're currently dealing with, you either want to straighten the approach to the next corner or section as much as possible (minimize steering input), or place the car in position to set up for the next corner. So you might want to make as straight a line as possible through that section of track, as you typically would in a series of Esses.

Oh, but distance is also a factor here. The shorter the distance, the sooner you'll get there!

Autocrossers are often very good at getting the balance between distance and a larger radius. As I've shared with you in the past, it's important to think differently, and specifically mentioned the Turn 5-6-6A section that the SCCA Runoffs used at the Indy Road Course. Most drivers used the classic late turn-in and apex, and in doing so, were driving a significantly longer distance than others who drove what I'd call more of a "diamond" line. Shortening the distance you drive can make a huge difference in lap times, but that usually goes against the "drive the largest radius" approach. Hmmm… compromises.

By the way, when thinking about the possibility of minimizing the distance traveled, a number of corners come to mind beyond Turns 5-6-6A at Indy: Turn 2 at Thunderhill; Turn 9 at Laguna Seca; the Carousel at Road America; the Keyhole at Mid-Ohio; Turn 6 at Watkins Glen; and even Turn 17 at Sebring, to some extent (although where grip is in this corner is an even bigger factor).

A larger radius often goes against what it takes to get to full throttle earlier. As you know, the largest radius through a corner is a perfectly geometric arc from the outside edge of the track at the turn-in point, through the apex (which would be exactly in the middle of the turn in this example), and out to the very outside edge of the track again. But if you're driving your car at the limit of its grip on this radius, you won't be able to begin accelerating until you're pretty much at the very exit of the corner — at the outside edge of the track as it becomes the straightaway. And that's a LOT later than if you turned in later on a tighter radius early in the corner, and "opened up" the radius as you pass through the apex area (which will be later than the middle of the corner). Why? Because you're giving up some of the grip being used for cornering to the tires so they can begin accelerating. More compromises.

Of course, the "What's next?" question returns here, too, because if there's little to no straightaway after this corner, getting back to full throttle is likely not as important as the rate of speed you can carry all the way through the corner. But if there's a long straight afterwards, then getting to full throttle sooner is even more important. Another compromise.

Then there's grip to consider. If the corner is positively banked, you usually want to spend as much time as possible in it. If the corner begins relatively flat and then becomes more banked, you might want to dive down into the banking as soon as you can. If the banking "goes away" (the track flattens out), let's say, two-thirds of the way through the corner — just as you're exiting the turn — then you're going to want to turn the car as much as you can in that banking. And that applies even if it means turning on a tighter radius.

If there is a compression (the track goes downhill, then goes back uphill) in the track, do most of the turning there, as the car will have more grip. If there's a hill crest, do as little turning there as possible, as there will be less grip. If the track surface changes through a corner, perhaps from asphalt to concrete and then back, you're going to want to change the direction of the car the most on the grippiest surface. In most cases, grip wins, even if that means a smaller radius or turning the steering wheel more. More compromises.

All of this points out why one driver might drive a slightly different line than another. More importantly, it points out why some drivers learn tracks faster than others — they understand the "why" behind the line, consider all of these factors, and figure out where to compromise and where not to.

Radius, grip, what comes next, distance, getting to full throttle… compromises, compromises.

That's what makes this so much fun, right? Personally, I find figuring out the line myself way more fun than having someone else tell or show me where it is. But maybe that’s just me….

Day #170: Be stingy with your brain.

I think most of us would admit to experiencing brain overload at some point during our driving. And I think we'd all agree that it wasn't very helpful, and in fact, we'd like for it to never happen again.

What is brain overload? Think of the old analogy where the Zen master is pouring tea into a cup, and keeps pouring even though the cup is overflowing. The message to the young master-wannabe is that continuing to put more into your mind, beyond the full point, is a waste. So, we can think of brain overload in the driving situation as the point where your mind cannot process any more information. It's full.

Okay, I can hear some of you right now: "Hey, I thought you said that I should take in more sensory information — more references. If I take in more references, isn't it possible that I take in too much information, that I get brain overload?"

Yes, I said that. Sorta. What I really said was take in more quality information, so taking in any and all information just for the sake of taking it in is not the point. The point is to take in quality information — information that you can use. For example, driving on track and noticing a helicopter flying overhead is taking in more information, but it's not helping. That's not quality information for your driving, and it's likely to lead to brain overload.

Noticing a spectator sitting in the top row of the grandstands is not helping your performance, but the direction a flag on top of the stands is blowing might just help you adapt to the way your car will handle in an upcoming corner. It's all about quality over quantity.

Let's focus on focus for a moment (see what I did there?). One way of thinking about focus is that it's what your brain is processing. You could be sitting in your car, on the grid waiting to get on track, while thinking about how you're going to release the brakes to help rotate the car in Turn 3. What are you focused on? Or you could be driving through Turn 3 while you're thinking about the engine temperature, and that jerk in the red car who won't let you pass. What are you focused on?

We may think we can multitask, but it's been proven that we really can't. What we do is switch back and forth between mental activities — such as engine temp and the jerk in the red car — and if we switch quick enough, it feels as though we're focused on and processing two things at once. But we can't focus on two things at once; we can't truly multitask. With practice, some people are really, really good at task-switching. But even they have a limit. Go beyond that limit, and we have brain overload.

How do we manage it?

Turn this whole thing around. Rather than thinking about being overloaded, think about what you need to do to NOT have to think so much. Take heel and toe downshifting, for

example. If you have to think about this skill while driving, it's easy to get into brain overload; if you have to think about the position of your foot, how you're going to move it to blip the throttle, and when you're going to blip the throttle, well… you're thinking too much. Okay, if you're just learning to heel and toe, that's what you need to do. That's how you learn and develop any skill — consciously thinking through each piece of the puzzle, each movement. But over time, you think less and less about it, and eventually you just do it without thinking. Now you have brain power left over to think about other things.

So, one way of managing brain overload is to think about fewer things. And one way of doing that is to get as much of the basics programmed, or turned into subconscious actions, so you no longer have to consciously think about them. Then, you'll have more conscious thought available to focus on what matters.

Great race drivers and experienced performance track drivers get really good at managing what they focus on. While a less-experienced driver might make a mistake in Turn 1 and spend most of the rest of the lap thinking about it (and filling their mind up with useless information — the cup overflowing), a more experienced driver will put it aside and only think about it when appropriate. Perhaps that's on the approach to Turn 1 on the next lap, or it might not be until after the session is over and they're back in the paddock, making notes on their track map.

I do want to emphasize something here: experience is not the only thing that makes the difference. If you do the wrong thing over and over again, gaining experience doing the wrong thing, you're not getting any better. So, experience alone is not the answer.

You need to be stingy with your focus. I want you to deliberately work on this, every day, in all situations. Understand that if someone comes up to you while you're doing something at home — reading, watching a video, working on your car — and asks for your advice on something, you're not going to be able to do both at the same time. At least not at a very high level. Be stingy. Pick and choose. Focus on what's important right now.

I'm not saying that we can't do more than one thing at one time. If you're a musician, when you're playing your instrument, you're thinking about playing music. When you're playing golf or tennis, you're thinking about playing golf or tennis. Even when you're engaged in your work, you're thinking about work. But I bet when you're driving home from work you're thinking about playing music, golf or tennis, or about work. So, it can be done. But I'd also bet (a lot of money!) that when you're driving home from work thinking about everything other than driving, you're not driving at your best. And therefore, you're practicing being unfocused. You're practicing what will lead to brain overload when driving on the track.

Practice narrowing your focus, on what's important right now, in the moment. Pick and choose. Be stingy with the use of your brain.

Day #171: Build muscle memory to fine-tune your technique.

According to Wikipedia, muscle memory “is a form of procedural memory that involves consolidating a specific motor task into memory through repetition, which has been used synonymously with motor learning. When a movement is repeated over time, a long-term muscle memory is created for that task, eventually allowing it to be performed with little to no conscious effort. This process decreases the need for attention and creates maximum efficiency within the motor and memory systems.”

But, muscles don’t store memories. Brains do.

Back to Wikipedia: “Motor behavior: When first learning a motor task, movement is often slow, stiff and easily disrupted without attention. With practice, execution of motor task becomes smoother, there is a decrease in limb stiffness, and muscle activity necessary to the task is performed without conscious effort.”

Does that remind you of what you went through when you learned the various skills and techniques to get around a race track quickly? And maybe it reminds you of what you’re still working on?

A Facebook post a while ago queried whether students of driver education events should be taught a “school line” (what some believe is safer because of a later turn-in and apex), or what might be considered the ideal line (the one that leads to the fastest lap times). There is a myth among some track driving educators that inexperienced drivers are not capable of learning the right way to do things, so they choose to teach them the “school line.”

In response, I posted:

Imagine a room full of kindergarten children. The teacher shows them a picture of a dog, and says, "This is a dog." Then kids are shown a photo of a cat, and the teacher says, "This is a dog." Then a photo of a lion is shown, and the teacher says, "This is a dog." When the teacher is asked why the kids are told every animal is a dog, the response is, "They're novices. They can't be expected to learn advanced stuff, like there are different names for different animals."

Crazy, right? So why would we ever do that with drivers? Learning is programming, and programming is learning. When someone learns the wrong thing, they've become programmed to do the wrong thing. And re-programming — un-learning and then re-learning — is more difficult than learning the right things from the very beginning.

In other words, developing the wrong muscle memory is not helpful. It’s hurtful.

When I was 16, I played tennis pretty seriously, thinking about whether I could make a career hitting those silly balls over a net. Ironically, I quit my tennis career because I couldn’t afford to practice (build muscle memory) as much as my competitors could.

Many of them came from wealthy families, so they could spend all day training with a coach; I had a part-time job, so my court time was limited. So, I did the only logical thing: I committed to my number one passion, racing. Because, if you can't afford to play tennis, go racing!! Duh.

But back to tennis. During the winter, when time on the court was limited, one thing I did was place an X made of tape on the ceiling of my bedroom. With racket in hand, hundreds of times each day I would begin my serve motion, tossing a ball to the ceiling, being as accurate as possible in hitting the X. I knew that if my ball toss was just right, I was more likely to serve an ace. It's likely that I practiced my ball toss a few thousand times over the course of two or three seasons, building the muscle memory to improve my serve. And yes, my parents wanted to send me away after hearing the thump, thump, thump… of a tennis ball hitting the ceiling time after time after time.

How can you use this same kind of strategy?

Think about all the skills and techniques you use when driving on track. Now, consider which is the least consistent, or has the greatest variability. Let's say it's your brake application, and in this case, you're not quick enough to ramp up to maximum brake force. This is likely due to your daily practice: building muscle memory while driving your road car. Most drivers on the road gently squeeze on the brakes, increasing the pressure the further they travel along the "braking zone." But we know that when driving on the track, we need to make the initial application harder and quicker. No, I'm not saying you should slam on the brakes, but you should squeeze the pedal very, very quickly — and hard.

Much like my tennis serve ball toss, can you sit at home and physically imitate the action your foot needs to do when approaching a heavy brake zone on the track? You bet. And if you did, you'd be building the muscle memory so that it just happens, without thinking, when driving on the track.

So, think about your skills and techniques, and which one or ones could use some programming, some building of muscle memory?

- Initial brake application
- Brake modulation
- Brake release
- Initial rotation of the steering wheel
- Unwinding of the steering
- Transition off the brakes and onto the gas pedal (whether using your left or right foot on the brake pedal)
- Initial tip in of the throttle
- Squeezing on the throttle

- Breathing off the throttle as if the car is running wide and you need to make a correction
- Turning your head to look through a corner
- Flicking your eyes from side to side to increase your view to the sides when approaching a corner

In simple terms, there are at least three different throttle application styles (also refer to Day #31):

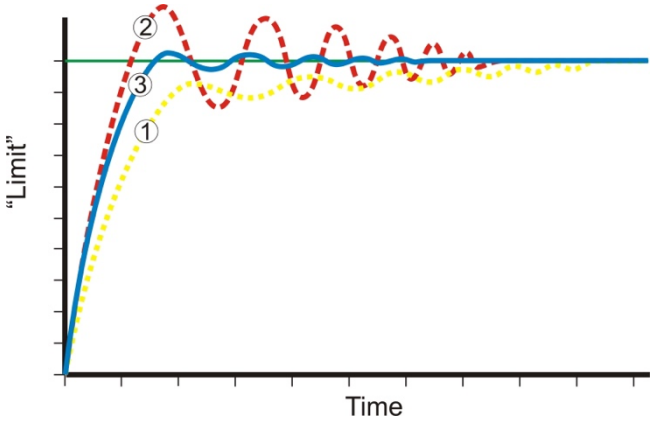
1. The Classic Squeeze
2. Stand On It
3. The Schumacher Throttle

Do you have the muscle memory to use all three styles? There's no one-size-fits-all style that works all the time, in all corners, in all cars. Rather than waiting until you get on track to practice using a different style, why not build the muscle (and brain!) memory to adapt by practicing at home.

Day #172: If you're not just slightly beyond the limit every now and then, it's likely you're not driving the limit.

Driving the limit is all about honing in on it, driving slightly below it, over it, under it, over it... and finally reducing the amount you're over and under it until you're at the limit. It's all about tuning the oscillation.

In the illustration below, the limit is represented by the green horizontal line, while the other lines represent how close three different drivers are to it.



Driver #1, represented by the yellow line consistently drives below the limit.

Driver #2, represented by the red line drives over the limit, but also under it. In fact, they're constantly over and under it. There are times when they're fast, but other times when they're either sliding too much, or not using the tires' limit.

Driver #3, represented by the blue line starts by going over the limit a little, dialing it back a little, then over, under, and eventually hones in on consistently driving the limit.

Of course, no driver, not even World Champions, drives on the limit as consistently as the blue line driver (or do they?). But my point is that you need to sense the limit to be able to home in on the limit. You need to sense when you've gone past it, and when you're not there yet.

Many drivers are so focused on driving the right line, where their braking points are, what RPM to shift at, and various other specifics, that they don't focus enough on sensing the limits of the car.

Listen to the tires and the wind noise. Note how the g-forces build up and then begin to taper off; feel the chassis pitch and roll. Look far ahead so that changes in rotation of the car are more noticeable.

The tires and your car are telling you whether you're driving the limit. Are you paying attention?

Day #173: Slowing down is a terrible thing to do to a car. Momentum is your friend.

"Oh, you drive a Miata. That's a momentum car. Keep the speed up."

Whether it's a Miata, a Formula Ford, or some other relatively low-horsepower car, the advice given above is good. But I'm here to tell you that every car is a momentum car.

A while ago, I had the privilege of taking people for rides around High Plains Raceway in a Porsche GT2RS, all for a great cause — The Race Against Kids' Cancer event in support of the Morgan Adams Foundation. While that car had somewhere around 650 horsepower, and even though I was just taking people for a few laps for fun, my focus was on carrying momentum.

I coached a driver in a small-bore sports racing car, recently; the focus was on carrying momentum. We focused on the same thing when I worked with a driver in a high-horsepower prototype sports car.

Hmmm, it seems this momentum thing might be important in all cars, no matter what the level of horsepower. Why?

The less change in speed, the less you upset the balance of the car, meaning the overall chassis stays more flat — less pitching forward and squatting down of the rear end. And any time the car stays more balanced, the more overall traction the car is going to have. And if you happen to be driving a car that generates downforce from the air passing under the car (such as a sports racing car, and some formula cars), the aerodynamic downforce is usually higher when the car stays flatter in relationship to the track surface.

Sure, you can overdo it and carry too much momentum into a corner, resulting in slower exit speeds (at best — and spinning off the track at worst). But that can be said about anything. Your goal is to balance everything - your entry speed with exit speed, and most importantly, the chassis of your car. And carrying speed and momentum through a turn does that.

Whether you have 100 horsepower or 650 (or 5 HP in a kart), the less you have to reaccelerate your car, the faster you'll be.

Day #174: Trigger your way to success.

Many drivers think that they make more errors than F1 drivers, and lose their focus more than Indy or NASCAR drivers. They're wrong. The best drivers make mistakes and lose focus like the rest of us. The difference is that they recover and regain more quickly than most drivers. They use triggers to do so.

Imagine purchasing a new piece of software for your computer, installing it, and then going to open and run it, only to find that there is no icon to click on to launch it. Well, you are similar to a computer in that you have software in your brain, and you use "clicked icons" to launch and run these programs. No, you don't actually click on an icon, but you do have something that triggers the launch of a mental program.

Think about when someone says something specific to you and how that can trigger a certain behavior from you, good or bad. For example, your child tells you that they love you. How does that make you feel? Your child's action and words trigger a reaction from you, and it's pretty consistent, because it's launching a behavioral program.

Now, think about a time when you performed at your very best when driving. Do you want to perform that way more often? Trigger it.

Let me share three triggers that I use myself, to either perform at my best, or to help drivers I coach perform at their best.

We all know that we cannot control the results of an event or race. We can't control how fast we are in comparison to other drivers, and we can't control what position we finish a race in. Sure, we can influence the results, but we can't control them. Same with our competition — we can't control them. The only thing we can control is our own performance. Interestingly, when we do — when we control our own performance — we perform at our best more often, and therefore we've increased our chances of getting the result we want. But when we try to make the result happen, by trying hard to drive fast, or forcing a pass to get into the lead, we usually don't perform as well, and don't get the result we want.

For competitive people, this is not just a hard concept to wrap their heads around, but it's especially hard to do. When I learned to fully focus on my own performance, my results improved, but it wasn't easy to do that consistently. I needed to find a way to focus on the act of driving, in the moment, and the trigger I learned to use is simply the phrase, "Car dancing." To me, dancing with the car is all about being at one with the car, in the moment, not worrying about my lap time or where I'm going to finish a race; it's me and the car, at the limit. And when I say "Car dancing," it triggers a performance that increases the chances of me getting the result I want.

The more I use this trigger, the stronger the programming in my mind becomes, and the more valuable it is.

I've noticed that some drivers are very analytical, to the point of thinking too much, and not trusting themselves to just drive at a subconscious level. Or, they may be overly focused on their car, to the point where they don't drive the way they can, and should. So, with a little time spent programming these drivers, I use the trigger phrase, "Drive stupid." That's a cue for the driver to shut off their conscious mind from thinking too much, from analyzing what they're doing, and to just do.

My final example requires a bit of insight into me and my behavioral traits. I don't like to draw attention to myself, or be a "show off." That's my natural mental programming. But there's one exception, one time when I have wanted people to notice me and what I was doing: when I was racing (having a helmet on makes it easier to "show off"). I unintentionally stumbled onto a trigger that helped me perform consistently better. I once was about to start a race from the eighth starting position, and just before the race I said to a friend, "Watch this!" I moved up to second place within a lap, and took the lead on the next lap. From that day on, prior to anything where I need to perform at my best — a race, speaking to a group of people, coaching, whatever — I say "Watch this!" to myself, and it triggers confidence, the appropriate level of effort, and a performance state of mind. And I perform better, more consistently.

These mental triggers are no different from a tennis player going through a specific routine before serving the ball, a golfer before a drive, a pitcher before throwing the ball. Triggers can be a physical routine, and/or a word or phrase.

Think about when and how you can use triggers to launch and run the right mental programs to perform at your best more often.

Day #175: Be a corner worker.

I don't want this to sound too much like a "back in my day…" kind of thing, but things were different when I started racing. One of the biggest changes between then and now is the way most drivers learn about the flags and their use.

As part of my race licensing program, I had to spend a minimum of one day standing at a couple of different corner stations at my local race track, ostensibly to assist the corner worker. The idea was that I would wave a few flags during the day under the supervision of a real corner worker. And I did that. And I learned a TON from doing so. I learned about driving as much as I did about the flags.

Not only did I gain an even deeper understanding of what each flag means, but I learned how the corner stations communicated with each other. And maybe most important is that I learned to "read" the flaggers. I noticed a pattern of how they tended to react: minor issue on the track and they moved calmly; more major issue and the movement became a little more energetic (but still — mostly — calm). I learned that by watching not just the flags, but how they were being used, I could turn a situation on the track into my advantage.

An example: We all know that a waving yellow flag means slow down, no passing, and be ready to respond appropriately to any situation on the track ahead. By paying attention to the actions of the flagger, I could get a feel for how serious the situation was, and adjust how much I slowed down. If a competitor didn't do that, and slowed down more than I did, I gained time on them.

Now, I can't stress this enough… slowing down to an appropriate speed and being under control is the priority — not gaining an advantage on others. Be safe, and even more critical, make sure the corner workers stay safe. If you think that you can just barely breathe the throttle and then go back to flat out in response to a yellow flag situation, only to find out the on-track problem was worse than you thought (and maybe that there's a worker on the track), that is not a good thing! Always err on the side of caution.

But if you're on a section of track where you can see well ahead and the flagger seems fairly nonchalant with the waving of the yellow, that may be the time to slow down a little less than if you're approaching a blind section of track with a worker very actively waving the flag.

At a track day event where I was there to coach drivers, I spent a little time observing from the control tower. The infrastructure, technology, and communication system were as good as it gets, and the workers were not only great at what they did, but they were fun to listen to. In fact, they were having fun doing a critical job well.

I'm bringing this up because it reminded me that the understanding of the flags has changed.

How is this different from when I started racing? Today many drivers manage to get into racing or upper levels of track day events without a solid knowledge of the flags and why it's important to know them. How do I know? I've coached drivers in many situations where I've witnessed them misinterpret or respond incorrectly to a flag. When asked what the flags means, they give a vague and incomplete response. It is obvious that many drivers today do not have a good enough grasp of the flags.

You should not be driving on a track if you don't have a very full and deep understanding of what every single flag means; how they're used; the variations on how they're used (different organizations have subtly different ways to use the same flag); where they're used; how the entire flagger-to-officials communication system and process works; and the care and attention the workers put in.

Many driving schools today do a poor job of preparing drivers — they don't teach them what they need to know about the flags. If you have even the slightest concern about not knowing exactly what every flag means and how they're used, ask. You will only gain respect from those you ask by doing so, especially if it's a corner worker or official. They love to help drivers.

I can't speak for every official and worker at tracks everywhere, but my experience is they welcome drivers with open arms when one asks if they could spend time with them, learning more about what they do. Most, if not all, would appreciate you even asking, and would love to share what they do. My wife, Robin, is a former corner worker and pre-grid marshal. She suggests attending the early morning workers' meeting (usually held race day). That's another way to learn and also to make yourself visible and accessible to those on the corners. You never know when you might need their help!

And here's the thing: You will become a better driver for every minute you spend on a corner station. There are not many better places to observe other cars go through a corner, and if you're smart about it, you'll ask yourself why drivers are doing what they're doing. You will then take that experience and use it to your advantage the next time you're on track. You can learn a TON about driving from the time you spend on various corner stations. And may be surprised by how much a flagger can teach you. After all, it's likely they've watched thousands of cars go through that specific corner and noticed what works and what doesn't.

I'll let you in on a little secret. Even though I've been racing and driving on tracks for forty years, and I've taught flags to thousands of drivers, even I need a refresher every now and then. And more importantly, I make sure to attend drivers' meetings (as boring as most of them can be!), since I know there will be a subtle difference in how the flags are used at the event I'm at, compared to what I thought was the case. Flags, and how they're used, is something everyone needs to stay on top of.

Finally, many corner workers are volunteers. They do it for almost the same reason you drive: the love of our sport. And you cannot do what you do without them. It’s important not to forget that, and even go out of your way to thank them (whether they’re volunteers or not). Your life depends on them.

Day #176: The time to prepare for your next event is NOW.

The next time you decide to couch-potato-it, lay back and think about what your fellow drivers and racers are doing at this moment. Are they preparing for their next time on track? If so, will they have an advantage over you because of their preparation?

Day #177: Unwind, when it's appropriate.

A funny story: At a HPDE instructor training program, an instructor told the story of telling a student to "unwind" coming out of every corner. The instructor meant for the driver to straighten the steering wheel — unwinding it — but the student thought otherwise. In fact, the student mentioned at the end of the day how he had been relaxing his shoulders all day. He thought his instructor was telling him to relax by "unwinding."

Three things came to mind when I heard this. First, an instructor should always check for comprehension, and not assume that a student knows what a word or phrase means that the instructor uses. Second, the student should ask for clarification if unsure. And third, we can laugh about this! Funny stuff!

I had been thinking about unwinding the steering since a subscriber had emailed me with an observation: A video of Ron Fellows driving Mosport apparently shows that Ron — the Master of Mosport — didn't unwind the steering exiting the corners.

I watched the videos, and while it's subtle, he does unwind the wheel. But here's the thing: He doesn't unwind the wheel as expected. So, I went to work to figure out what's going on here.

In studying the videos more closely, I noticed where Fellows unwound the wheel later than in most other corners. It almost looks as though he's only unwinding right at the end of the corners. I went back and looked at a lot of data I have of various cars and drivers at Mosport, focusing on steering angle. And I came away with two conclusions.

First, every single corner of every single lap of every single data file I have shows the steering angle being reduced from somewhere after mid-corner to the exit. There's no arguing with data. But even the data shows less unwinding of the steering wheel than I see in data from most tracks. It's very subtle. Why?

It's the long, sweeping corners at Mosport. Because the line through these long, fast corners is closer to a constant radius, it's almost a perfect geometric line. And by definition, there isn't an increase in the radius (reduction of steering angle, or unwinding of the wheel) on a line like this. It can't be; what the driver is trying to do in a long, fast corner is get from the entry to the exit in the least amount of time — in as straight a line as possible. That means that the steering will not be unwound/straightened until almost the very end of the corner.

Now, the more a driver shifts the line towards one with a later turn-in and apex, the more they will begin unwinding the steering earlier in the turn. Why would that be? To maximize acceleration out of the corner, as in a slower turn that leads onto a straightaway. The tighter the corner (if it's followed by a sizeable straightaway), the later the turn-in, the smaller the radius at the beginning of the corner, and the earlier the driver begins unwinding the steering to allow tire grip to be used for acceleration.

Conversely, the longer and faster the corner, the less the driver is concerned with acceleration off the turn; maintaining speed through the corner becomes a higher priority. And therefore, the closer the line is to a constant radius, the more it results in less unwinding of the steering as the car passes the middle of the corner and heads toward the track-out point.

Again, in a slower, tighter corner, the radius is smaller at the beginning, and then expands. Therefore, you'll need to deliberately unwind more in those types of corners. And that's where you might be told to relax… I mean, unwind the steering, exiting the corner.

In long, sweeping corners where the apex is very close to the middle of the turn, you will need to unwind the wheel - but it will be less pronounced.

Having said that, no matter how tight the radius of the corner, your goal should be to deliberately unwind the steering wheel as early as you can. Straightening the steering — unwinding the wheel — means you have more tire grip available to be used for acceleration. It also means the tires will be scrubbing less, allowing the car to "run free."

One other thought came to mind as I thought this through: You can't judge a book by its cover. Or in this case, you can't judge a steering technique by one track.

Day #178: Manage fear with your imagination.

A.J. Foyt, one of the (if not the) bravest and toughest race drivers in history, once said that if a driver says he's never been scared in a car, he's either driving slow, or lying. I take that to mean that even A.J. faced fear at times in a race car. And don't we all?

There are many kinds of fear, with the fear of injury or death being just one of them. But there's the fear of failure — not living up to expectations. There's the fear of embarrassment (again, not living up to expectations). And the fear of financial injury, and damage to our beloved car. All of these are valid and common fears.

What is fear? What do we actually fear? At the very core of the answer to that question is the fear of the unknown. We're afraid of what an injury could feel like or how it will affect us; what happens when we die; what people will think if we don't meet expectations; how we're going to deal with the financial and time challenges with a crashed car. If we knew the answers to all these questions — if they were not unknowns — the level of fear we have to manage would lessen.

Let's take the example of a crashed car. If you knew that you had a large sum of money sitting there just waiting to be spent on repairing your car, or you had an insurance package that covered absolutely everything, there would be far fewer unknowns, and you wouldn't have as much fear. I'm not saying no fear, but it would be less (and possibly only focused on other areas).

Short of actually having all that money just sitting there, or an insurance policy that covers everything, what else can you do to manage your fear? Use your imagination. No, I'm not suggesting you just imagine the answer to my question. I'm saying the answer is your imagination.

In your mind, run through all the steps, everything you'd do if you crashed your car. Imagine finding a way to deal with the financial setback, and how you'd get it back up and running in a relatively short amount of time. I'm serious. Take time to actually run this scenario through in your imagination, to the point where you have a very real plan for dealing with it. Imagine how it feels to know you're prepared to deal with it.

If you're thinking, "I can't imagine how I'd deal with the financial challenge of paying for my crashed car," then you haven't thought it through enough. I'm not suggesting it's easy to deal with the cost of a crashed car (guess how I know?). It's definitely not easy. But there's always a way, even if it's going out and getting a second job. That may not be a pleasant thought, but it's better to be prepared for it than not to be. It's better to have a plan, even it's not a fun plan.

The same thing applies to dealing with the embarrassment of not performing well, of failing. Imagine the worst-case scenario, where you completely fail at what you're trying to achieve. Now imagine how you'd deal with that. Perhaps you see yourself being embarrassed for a while, and then your friends and family comforting you. Perhaps you

see the situation the way it really is — not the end of the world! In your mind, you see yourself slowly but surely getting over it, and getting back to what you really love doing — driving/racing. Oh, and imagine you not performing as well as expected, but afterward, your friends, family, and teammates still like/love you (wow, imagine that!).

I can't advise you on imagining how you'd manage the thought of dying. Having had a pretty close brush with death — essentially being gone for about five minutes — I could suggest to you that there's not much to worry about. You likely have your own beliefs around life and death and what happens, but worrying about the unknown (because, really, does anyone know for sure what happens when we're gone?) is a waste of time. There's nothing you can do about that unknown, so why not imagine it in a way that is comforting. It doesn't matter how you imagine it, but just imagine it enough to feel prepared.

Getting injured has two issues. First, there's the pain and suffering that accompanies it, and you could imagine the most intense pain and know that it's likely never to be that bad. Second, there's how you deal with all the other "problems" that go along with such injuries, including how it impacts your family, what it does in terms of being able to work, how much your medical costs might be, and so on. Again, use your imagination and put a plan in place for all of these things.

The biggest problem with fear is that it can actually increase the chances of having something go wrong — something to really fear! The more fear we have, the more we tense up our bodies, making it more likely that we don't respond the way we should. Managing the fear that we all experience can be handled with our imagination.

And if you think fear isn't holding you back from an even better driving performance, you know what A.J. and I are thinking….

Day #179: Embrace the yin and yang of performance driving.

The yin and yang of performance driving?

We all know that smooth is fast, right? We've been told that over and over again, and we've noticed it ourselves — when we focus on driving smoother, we go faster.

Why? Because the more balanced our car is — the less weight transfer we cause with the car — the more overall traction it has. Every time we cause weight transfer, by braking, turning the steering wheel, or accelerating, we reduce the amount of traction the car has (my explanation as to why this is so is covered in the YouTube video Why Smooth Is Fast). But we can't drive without causing some weight transfer, so we focus on causing as little as possible.

However, there comes a time when we have to grab the car by the scruff of the neck and attack. When we can be too smooth. When causing more weight transfer will actually make us faster.

One area where I see some drivers be too smooth is with the initial application of the brakes. Especially with newer cars, the initial application needs to be hard, right now. Now, you can still be smooth as you apply the brakes hard and fast, but it kinda takes second priority to applying the brakes quickly and to their maximum.

There also comes a time when we want to use weight transfer to help turn the car, even though this reduces the overall level of grip that the car has. Of course, what I'm talking about is having weight on the front tires as you turn in to the corner.

But not all corners are created equal. Some will reward more of this approach, and some will reward less weight transfer at the entry.

The more you need to change direction, the more you want to use weight transfer to help with it; the less you need to change direction, the more you want to keep the car balanced. Weight transfer works best in slower, tighter radius corners, and balancing the car works best in fast, large radius corners.



Many drivers, at all levels, drive all corners the same. The best drivers adapt their driving to each and every corner, compromising traction for weight transfer to change direction of the car, and vice versa. They embrace the yin and yang of performance driving.

Day #180: Look, and think, as far ahead as possible.

It's one thing to look further ahead, but it's another to think further ahead.

Like everything, thinking further ahead takes practice. Deliberately practice it when driving on the street, then again during a few sessions on the track. Force yourself to think about what's ahead — but sooner than usual. The more you deliberately and consciously stretch how far ahead you think, the more natural and subconscious it will become.

Day #181: Carry entry speed by looking into the turn and focusing on your release of the brakes. Focus on the End-of-Braking point.

One of the most important driving techniques that has led to the biggest gains from the most drivers I've worked with is covered in the above Speed Secret.

One day I was driving along a highway in my beater Honda (every penny I had at the time was "invested" in my racing career) and up ahead the traffic light turned to red. As I started braking a lightbulb lit up above my head.

For a few years I had wondered why, when other drivers asked me what brake marker I used to start braking for a particular corner, I couldn't really tell them. I had a vague idea, but it wasn't like I could be very exact. For a while I thought there must be something wrong with me for not knowing that, but when I realized that I most often out-braked other drivers, I stopped worrying about it.

So, why the lightbulb lighting up?

When I braked for the red light I didn't look around for a brake marker to tell me when to start braking. No, I looked ahead at where I wanted to be finished braking, and then started and adjusted the pressure on the pedal to suit that.

The next time I was on track I realized I did something very similar. Okay, I used the brake markers to give me a general feeling of where to start braking, but what I focused on the most was where I was going to finish my braking — what I began calling the End-of-Braking point. It was like I was focused on the stop line at the traffic light.

Over the next few years, I asked other drivers where they started braking for certain corners, and I noticed a pattern to the answers. The best, the fastest drivers could not tell me exactly where they started braking; the slower drivers could.

Hmmm.... Where were these drivers looking? What were they focusing on mostly when heading into the turn? The fast ones were focused on the End-of-Braking point, whereas the slow drivers were focused on the Begin-of-Braking point.

I'm not saying a driver should never look at brake markers, or know where they begin braking. What I'm saying is that most of your focus should be on the End-of-Braking point, and not the Begin-of-Braking point.

When you focus on the End-of-Braking point, what do you think it does to your vision? Forces you to look ahead, into the turn, right? And guess what happens when you do that? Typically, your entire brake zone becomes more consistent, and you carry more entry speed into the turn (without it negatively affecting exit speed).

Something to think about the next time you're on the track... look for your End-of-Braking points.

Day #182: Awareness leads to improvement.

I'm sure you've noticed what happens when drivers see one of those signs at the side of the road indicating the speed they're traveling: they apply the brakes. If you ask them why, they might tell you they're concerned there is a cop nearby, ready to give them a speeding ticket. But the real reason they brake is because the sign has made them aware of the speed they're traveling. Before that moment, they had not been aware of how fast they were driving, and I'll bet that immediately after seeing the sign, they looked down at their speedometer. It's "See the sign, look at the speedometer, and slow down."

Awareness leads to behavioral changes. Without awareness, it's unlikely you'll change what you're currently doing, and likely you'll continue what you've been doing for a while.

If you want to change something about your driving, you first need to become fully aware of what it is you're doing.

I started to become aware of the power of one's awareness years ago when I was instructing drivers from the passenger seat, and specifically when getting them to use all the track when exiting corners. I would tell them over and over again to "unwind the steering," "straighten the steering," "let the car run free," "use all the track," and "drive to the edge of the track." One day, after giving this advice repeatedly — only to see the driver not get within three feet of the edge of the track — in frustration, I asked the driver, "How close is the car to the edge of the track?" Within a couple of corners, the driver was using all the track! I was onto something. I learned how a simple question can bring awareness to a behavior, opening the door for change.

I tested this instruction technique over the next few months (at the time, I was in-car instructing at least three days a week for about nine months a year), and learned another important fact: questions were more effective than telling a driver what to do. If I told the driver, "You're three feet from the edge of the track," it helped, but not as much as if I asked, "How close are you to the edge of the track?" There was something very powerful about asking a question.

A question I often get asked is, "How do I know what to work on with my driving?" Many drivers either have a hard time figuring out what they should focus on, or don't know how to prioritize when they've identified more than one thing. The solution to this situation is to ask more questions.

"If there was just one thing that I could do to become faster, more consistent, or safer, what would it be?" That's a pretty darn good starting point. Of course, you need to ask yourself this question more than once. In fact, once you've answered it the first time, ask at least five more times. Keep digging for the next level of awareness, because once you peel back half a dozen layers of the onion, it'll become quite clear as to what you should work on, and how to do it. And with these questions and answers in the back

of your mind, you will be surprised at how you'll change the way you drive during your next on-track session.

Day #183: You perform the way you practice.

If you race, what percentage of winning (or being successful with whatever resources you have) is a result of your fastest race lap? Or of your fastest average lap time? And what about the race start, how and when you pass others, whether you position yourself to discourage being passed, how you respond to yellow flag situations, re-starts, and what happens on the last lap?

I bet your outright fastest lap of the race has less to do with your results than all those other things. And yet, what do you practice the most?

How often, during a practice session do you deliberately drive off-line, as if you‘re on the inside of another car? Or brake down the inside into a corner, as if passing someone? Or drive around the outside of a corner, as if another car is on the inside?

How often do you drive up the front straight at pace lap speed and then begin accelerating as if you’d just gotten the green flag, learning how late you can brake for the first corner at this reduced speed?

How often do you practice getting up to speed on cold tires very quickly, as if you’re on the first lap of a race?

A few years ago, I was coaching a young driver who was pretty fast, but took too long to get up to speed on cold tires. So I developed a simple plan to help him improve. We went to a track for a test day (where we could come and go from the pits as often as we wanted, and do as many or few laps as we wanted; there were no scheduled sessions) with three sets of used tires, all mounted. Straight away, I sent him out for three laps with the goal of getting up to speed as quickly as possible. When he came in after the three laps, he got out of the car, took his helmet off, and walked around. Meanwhile, another set of cold tires were put on the car. He then geared up and out he went again with the goal of getting up to speed quickly.

You guessed it, he then came in and we did the same thing all over again. And while this was going on, the hot tires that had just been on the car were being doused in cold water to cool them down. And each time the driver had to get out of the car and take his gear off, because I wanted him to feel as close as possible to what jumping in the car and getting up to speed quickly really was like. We rotated through the three sets of wheels/tires at least half a dozen times. And from then on, every single time he went on track, his goal was “Fast — fast.”

No surprise, he improved his ability to get up to speed quickly. Not long after doing this exercise he was racing in a big event where the schedule was “time certain.” That meant that if someone went off early in the session and it was red-flagged, the first few laps were super-critical because the session would not restart. He nailed it. He turned a lap time on the third lap in qualifying as fast as he’d ever been, and then the session

was red flagged. That was it, and the drivers who did not get up to speed quickly, suffered.

Another time, I was coaching a driver in a rallycross event. During practice, I noticed that Ken (Hoonigan) Block was all over the place, but in a good way. I watched him try very tight lines into various corners, then drive around the outside of other corners. He slowed down and then accelerated hard into the first corner.

When I suggest to a driver that they think about where they're going to begin braking for the first corner on the first lap, there is often a "Why?" look on their face. When I explain the obvious, that starting to accelerate from pace lap speed means they will not be going as fast into that corner as usual, and therefore the Begin-of-Braking (BoB) point will likely change, the facial expression changes to "Why didn't I think of that?"

Being fast is super important, so you need to practice in a way to improve your speed. But there's much more to a race than just one or two fast laps. You need to practice every aspect of a race, not just that one fast lap.

Day #184: No matter what the level or type of motorsport you participate in, you are a team-builder.

I realize that you may not drive for a professional team. In fact, the majority of my readers don’t. And I bet there are many drivers who have to look in the mirror to talk to their team. And yet, no matter whether you have dozens of crew members supporting your racing, it's just you and your spouse or friend, or you're all on your own, you still have a team.

Think of anyone who has any influence on what you do as a driver as a team member. That could be the shop owner where you have some prep work done, an instructor who you've become friends with, a buddy who pits next to you so you can help each other, and all the way up to professional crew members. And of course, your spouse who supports your driving passion in some way is on your team.

So, don't just think of these people as team members, treat them that way, and do everything you can to bring out the best in them. In that way, you all win.

Day #185: Put energy in your car.

Imagine you're a car. Just for fun, let's say you're Lightning McQueen, the anthropomorphic race car from the animated movie, *Cars*.

Now, imagine heading down a long straightaway, approaching a slow-ish corner, one that needs a fair amount of braking and downshifting. You can choose to imagine this scenario at a track you know well, but again, picture yourself as the car.

You feel the energy of your engine powering yourself forward, your rear-end slightly squatted down, with your driver pushing fully on your throttle pedal.

Then, suddenly, as your driver lifts off your throttle and quickly applies the brakes, it's as if your eyeballs are being forced out of their sockets, and your nose is being thrust towards the ground. With each downshift, your power source is revved, and then it tapers off as you slow, slow, slow.

The energy is flowing through all parts of you: from forward thrust to nose-down deceleration; from your engine powering you forward to one revving and assisting your brakes in slowing down; and the energy from the clamping force of your brake pads on the disks turning into heat energy.

You're about the distance of the length of your body from that point where your driver turns your steering wheel and you attempt to switch directions from forward movement to a curved path through the upcoming corner, when…

Your driver releases pressure from your brake pedal.

You raise your nose. You reduce your deceleration rate.

"What's going on?! What am I to do?," you ask yourself. You feel aimless, like you no longer have a purpose in life. You feel lifeless, lethargic, like you have no energy to do anything. And that's when your driver suddenly asks you to change direction and go over there!

"Huh? You deprive me of my energy, and then ask me to do something? Sorry, but I can't react that quickly. Okay, give me a sec, I'll get around to it."

And that's when you begin to change direction — fractions of a second after your driver requests it. If only they hadn't taken that feeling, that joy of having energy in your muscles (suspension), your bones (chassis), you would have gladly and instantly changed direction for them. But it was that moment when they didn't ask anything of you —or nearly anything of you — just prior to turning your steering wheel that almost lulled you to sleep.

“See, I love being engaged with my driver, being asked to accelerate, brake and turn. But I hate the feeling of near-nothing in between.

“I have one request of you, dear driver: Put and keep energy in me, by smoothly transitioning from one control input to another — even a bit of overlap is okay sometimes — so there’s energy in me. Don’t waste my time — or your time — by having me do nearly nothing.

“The thing I hate most is when you arrive at a corner and release my brakes a fraction of a second too soon so that I feel as if you don’t know what you want me to do. Like you don’t know what you want to do! It’s very difficult for me to help you if you’re expecting me to do something after you’ve practically told me you don’t want anything from me.

“Begin braking late enough so you arrive at the turn-in point with a little more brake pressure on so I feel as if I have some energy in me — so I’m front-loaded — when you ask me to change direction. Then, I’ll be more than happy to respond to your requests.

“Energize me, please.”

Day #186: Apply the throttle at the point where you don't have to ease off it when exiting the corner.

I hear this topic discussed a lot at the track: where exactly a driver should be on the throttle to accelerate out of a corner. A common question is: "Where should I be on the throttle in relation to the apex?"

So, here's the answer: 42% of the way through the corner.

Okay, that's sarcasm! Ignore what I just said.

Driving would be easy if there was a formula for when to brake, when to turn, where to accelerate, and so on. But it would be boring. And track driving is anything but boring. Which means there is no formula to tell us when to begin accelerating.

There is a guideline you can use, though...

> *Begin accelerating at the point where you're able to apply the throttle and never have to lift (even slightly) through the rest of the corner.*

If you begin to apply the throttle so early that you then have to ease up on it to avoid driving off the track, you've probably applied the throttle too soon. You should have waited before beginning to accelerate. That's not always the case, but it happens way more often than not — more than ninety percent of the time.

I hear drivers all the time saying things like, "You should always begin accelerating before the apex." That just isn't true. Okay, it is a lot of the time — maybe even more than fifty-one percent of the time. But there are many instances when it's not. And it's impossible to put a "rule" like that into performance driving.

A generally-accepted piece of advice is the earlier you apply the throttle, the faster you'll be down the ensuing straightaway. Makes sense. But as I mentioned last week, one way of being able to apply the throttle early is to enter the corner at ten MPH. Hey, at that speed, you can probably go back to full throttle at the moment you turn in to the corner! But you'd still be slow down the next straightaway.

The main point I want to make here is that it's not when you begin applying the throttle that makes the difference. It's the speed you're at when you begin, and whether you're able to go all the way to full throttle (and stay at full throttle) all the way out onto the straightaway. That's what counts. That's what makes you fast.

I've spent a ton of time analyzing data to see what the impact is on lap times when a driver begins to feed in the throttle, and then has to either hesitate or slightly lift. The impact is bigger than most drivers would imagine. It kills exit speed, and you'd be better off waiting for a fraction of a second before beginning to apply the throttle... and then commit fully to going to full throttle.

When you head to the track this coming weekend (you are, aren't you?), become more aware of when you're at full throttle, and not just when you first begin to apply the throttle. Many times, we feel like we're doing a great job with our driving because we're tipping into the throttle well in advance of the apex, but if we really focus on what we're doing (or have a data system logging our throttle position), we find out that we're not at — and fully committed to — full throttle until much further through the corner.

Day #187: Use your imagination to drive faster.

Albert Einstein once said, “Imagination is more important than knowledge.” Coming from someone as knowledgeable as he was, that’s saying something. But what? How does this apply to driving?

When I ask drivers what they need to be faster, the answers typically fall into the following order of priority:

- Seat time
- Better car
- Instruction/coaching
- Study (some form of reading books, watching videos, attending seminars, etc.)

Rarely does imagination even make the list, despite how important Einstein thought it was. Okay, maybe there’s a big difference between performance driving and physics?! But is there? After all, when we're driving, we're working within the laws of physics.

My point here is this: How much time do you spend imagining driving, compared to actually doing it, working on your car, getting or giving instruction, reading about driving, or watching driving? All of those priorities listed above are contributing to your level of knowledge, but if you’re not using your imagination, you’re missing out.

I’ve written in the past about the benefits of writing about driving. Every single time I write about driving, I’m mentally putting myself in the car, imagining what I’m putting into words.

I’ve also written about the 10,000 hour rule, before. It’s the theory that it takes approximately 10,000 hours of practice to master something. Guess what? Even the busiest of professional race drivers who began racing karts at five years old will be in their forties before they’d ever get to a total of 10,000 hours of actual time on a race track. That’s just the nature of racing and performance driving — access to tracks, and the costs associated with it, is limited.

So how can one ever consider the idea of mastering performance or race driving if 10,000 hours is nearly unachievable for other than a few people in the world? Come on, use your imagination to come up with an answer!

I’ll bet many drivers reading this piece are nodding their heads, thinking, “Yup, I do that every single day. I imagine driving on the track when I’m driving on the road. Or I daydream about driving while riding the commuter train, or in my office.” Great! (Well, it’s great as long as you stay focused enough on what’s most important at that time, and not get yourself in a bad situation by daydreaming when you shouldn’t.)

How about setting time aside on a regular basis to simply imagine driving on track? Yes, this is essentially using mental imagery, or visualization, but I don’t want to make it that complicated. Let’s keep it simple, here. Just imagine driving on the track. That’s all. If

your mind begins to wander and you start thinking about something else, that's okay. Just say to yourself, "Eyes up — look ahead," and come back to imagining driving on track again. It doesn't matter what you imagine, just imagine it. You could imagine driving one corner of one track; a certain driving technique; racing wheel to wheel; driving an entire track; adapting your driving to a specific handling problem; whatever. Just let your mind go and imagine.

Some drivers tell me they don't have a very good imagination, and that's why they struggle to visualize. Sorry, but that's not an excuse. We all have an imagination. We were born with it. Some people don't exercise their imagination enough, and therefore it gets a bit rusty – but it's still there. It takes practice.

So, go ahead, right now, imagine yourself on a track you've never driven before. That may be more important than any amount of knowledge you have.

Day #188: Practice dealing with problems.

One of the main differences between learning to fly and learning to drive on a race track (HPDE, racing, track days) is the focus on problems. A very large portion of learning to fly is spent dealing with what might be considered emergencies. While taking helicopter pilot training, I spent a relatively large amount of time learning to deal with things like engine failure in a helicopter (an autorotation, which is essentially gliding the helicopter without the engine powering the rotor blades).

All of this training got me thinking about how performance/race drivers are trained. Well, first, many are not trained — they just start driving, get enough experience, and move up the ranks. But even those who do get training spend a relatively small portion of the time dealing with problems. Okay, an engine failure in a car isn't quite as dramatic as it is in a helicopter or airplane, but that's not what I'm talking about. I'm talking about learning to deal with a loss of control of the car.

How much time do you spend driving on the track? What percentage of that time are you "out of control"? What percentage of the time is your car sliding slightly (or more) beyond the limit? Very little, right? But I'm guessing that it's a little more often than airplanes and helicopters have engine failures. And yet, aviators practice dealing with engine failures WAY more than performance/race drivers practice dealing with a loss of control or sliding beyond the limit.

Think about it. How comfortable are you with the thought of your car suddenly sliding towards the outside of a corner? How confident are you that you can catch it? If it begins to spin, can you catch it? The next time the rear of your car steps out in the very fast corner, can you control it?

The take-away? Practice dealing with problems. Maybe "problems" is not the best word to use here, but you get the point. Learn to manage a sliding car. How? Well, you can simply get more seat time, and eventually you will acquire some level of skill in dealing with a car beyond the limit. But that's not a very efficient way of learning. Instead, a skid pad would be better. And the good news is you don't even need a purpose-built skid pad to practice controlling a sliding car. A large parking lot or the paddock at a track will often do the trick.

Or, better yet, go to a rally school where you can spend tons of time sliding sideways (and if that doesn't sound like fun, there's something wrong with you!).

The key is to deliberately spend more time with your car beyond the limit — where it's safe to do so — and get so that you know you can control it. Believe me, before I'm certified to fly a helicopter I have to prove that I'm comfortable with dealing with an engine failure. And even once I have my pilot's license, I'll practice autorotations on a regular basis so that I'm able to deal with a problem "without thinking," on auto-pilot.

Can you deal with a loss of grip or control on auto-pilot?

Day #189: The more complete a driver you are, the more likely you'll have a successful career.

Look at drivers who win championships, especially the ones who win multiple times, and you notice a common denominator: They're strong in all areas of what it takes to be a professional.

What does it take? In broad terms, a driver needs to strong in the following areas:

- Racing skills
- Testing/technical skills
- Physical skills
- Mental skills
- Marketing skills
- Career skills

Racing skills are relatively easy to define - speed, consistency, smart decisions, and racecraft.

The Complete Driver:



Testing/technical skills includes knowledge about car setup, as well as the ability to sense what a car needs and communicate it to the team.

Physical fitness and stamina is critical, as are the driver's techniques (coordination, skills, etc.).

Mental skills is a big category, and includes the ability manage one's on- and off-track personality, confidence, ability to learn, mental toughness, and consistently get into the zone and perform at one's best.

Marketing skills include business skills, understanding and working with sponsors, and media relations.

Finally, putting oneself in the right team at the right time, making smart decisions about when to move up to the next level, and surrounding oneself with the right people are all part of the career skills that lead to championships.

Drivers like Michael Schumacher, Jimmie Johnson, and Dario Franchitti were or are strong in all of these areas, and have won multiple championships. Drivers like Felipe Massa, Kyle Busch, and Paul Tracy didn't win as many championships, and I'd suggest they have or had too many weaknesses in the overall package.

Day #190: Brakes do much more than just slow you down.

After looking over some data from a very accomplished driver — one that missed making it into Formula One by a fraction of a political decision — I thought about late braking.

The data showed some of the latest braking I've ever seen - consistently. So how could one driver brake that much later, and make it work so well?

Why is braking late so difficult? For many drivers it's the mental picture — it just doesn't look right. It doesn't look like it's possible to brake later — compared to what they're used to (driving on the street, and some track driving).

If you spent a week or two driving a Formula One car every day (oh darn, wouldn't that be a shame!), your mental picture of late braking would change. What seemed normal after this two week period would have seemed absolutely unachievable before your fortnight of fun.

Driving a much faster car, one with much stronger stopping power, gives you a mental picture of slowing in a very short distance. Jump back in your usual car, and your mental picture will stretch closer to that of your F1 car's performance.

So, just go out and spend a couple of weeks driving an F1 car, and you'll be great on the brakes! What, you say that's unrealistic? Okay. What you could do, though, is gradually change your mental picture of what late braking looks like through visualization, or mental imagery. If you spent twenty minutes each evening and each morning for two weeks, seeing, feeling, and hearing yourself braking later, your mental picture of what's possible would change. Will it be the same as if you could actually drive a car with much stronger braking abilities? Not quite, but it sure beats the alternative — hoping and wishing your mental picture would change.

Something else great late brakers do involves their vision: they focus mostly on where they're going to end their braking, and less on where they start. I've often noticed that drivers who struggle to brake late focus too much on where they begin braking (the 400 marker, or a change in the pavement surface); they don't finish their braking in the right place. They start and finish early.

When I asked the came-close-to-making-it-to-F1 driver where he braked for a specific corner, he couldn't tell me exactly where he started. His response was, "When it feels right." And the only way you know what feels right is to focus on the corner, and where you finish your braking.

Developing a clear mental picture of what late braking looks, feels, and sounds like, and focusing on the end-of-braking point will result in comfort with braking late.

Of course, braking late is not always the best way of turning the fastest lap time. But that's a story for another day.

Day #191: Use your whole brain.

One thing that has changed in the scientific study of how the brain works is the emphasis on how the two separate hemispheres of our brains work. When I first started to dig into how to get my brain to help me drive faster, people like Dr. Paul Dennison had recently done a lot of research around brain integration. If you've read about, heard me talk about, or seen drivers I've coached doing Cross Crawls (the action of standing, raising one leg and touching the knee with the opposite hand, then lifting the other leg and touching it with the other hand, alternating this action as if you're marching in place and touching the knee with the opposite hand), then you know what I'm talking about here. If not, you should, as it'll make you drive better.

The main thinking around brain hemispheres is that one side tends to dominate over the other. If you were a left-brain-thinker, then you are logical, analytical, and detail oriented; if you are a right-brain thinker you are more creative, intuitive, and see the big picture more than the details. When we perform at our peak, both sides of the brain are working together, and communicating fully between each other; when we are having an "off" day (or an "off" on-track session), it is because one hemisphere is dominating over the other, and our brains are not "integrated" or working "whole-brained."

While there are some who argue that the proof of this is questionable, there are physical and mental rehab centers and therapists, along with educational systems (schools, teachers) who swear by the effectiveness of it (despite some studies that haven't produced results to prove it works). Personally, I can provide actual results that show that doing physical cross-lateral movement (such as the Cross Crawl) improves performance. This cross-lateral movement "forces" the communication between the hemispheres of the brain, causing both to work together, integrated.

Between the hemispheres of your brain is the corpus collosum, a bundle of nerves that acts a bit like a cable between a computer and a scanner that allows bio-electrical communication to flow back and forth between each side of your brain. Emotions like anxiety, fear, rage, and tension causes restriction of the flow of this communication through the corpus collusum, a little like a dimmer switch restricts the current to the light in your dining room. Also, the left hemisphere controls the right side of the body, and vice versa.

Recent research has demonstrated that our brains are not as simple as that, though. Big surprise, right?!

Through the use of fMRI (functional Magnetic Resonance Imaging), it's now believed that the hemispheres are not just looking after different types of thinking. Yes, the left hemisphere is the detail-oriented side, while the right brain sees the big picture, but the left tends to also think about what just happened or is about to happen, while the right is "in the moment"; the left is tactical while the right is strategic.

Interestingly, the left is the controlling side of the brain. It believes it's running the show, while the right is more empathetic (to others and oneself). For example, if someone's left hemisphere is damaged through a stroke, and therefore the right side of the body is affected, the brain won't even admit there's a problem. If it's a right hemisphere stroke, the brain realizes the problem, and may become depressed because of it.

Having said all this, and despite researchers not being in total agreement about the details (hmmm… are they left-brain-thinkers?) of what's going on between the hemispheres of the brain, there's no doubt that having both working at full capacity is a good thing when driving fast on a track! Whether that ensures your brain sees both the details and big picture, both analytical and intuitive, tactical and strategic, past/future thinking and in-the-moment… we want it all.

Doing Cross Crawls and other cross-lateral movements switches your brain on. It prepares your mind to perform at its best. Things like juggling, and exercises that cause you to use both hands/feet at the same time, as well as using your non-dominant hand to do things, all integrate your brain. And from my own recent experience, flying a helicopter where your left hand is controlling the collective (raising and lowering it while twisting it to modulate the throttle), your right hand is controlling the cyclic (moving it forward and aft, right and left), and each foot works the pedals (to control yaw) all integrate your brain and result in a better mental performance (I hope!).

Most people leave their performance to chance. They wait for things to align to trigger a performance state of mind. They practice physical skills, but not mental ones. They rely strictly on seat time to learn, and not mental imagery. They tend to focus on what they don't want, rather than on what they want. They don't have a mental preparation program, and that means performing at their peak is a hit-or-miss situation.

The best drivers use things like Cross Crawls and other brain integration exercises to trigger their best performances. Still.

Day #192: Check your cockpit.

It's funny how certain messages seem to pop up in clumps. For some reason, over the course of a couple of months, a number of drivers mentioned to me how they made changes to their seating position, or pedals, and it's improved their driving. Or a driver has struggled with a certain skill or technique, and by changing something in the way they sit or use the pedals, it fixed it.

So...

- Are you sitting close enough to the steering wheel that you're able to rest one of your wrists on the top of the steering wheel (at the 12 o'clock position), and still have a slight bend in your elbow without pulling your shoulder off the back of the seat at all? Check it. And make sure you do that fully belted into your seat, with your helmet and head & neck restraint device on.
- What would happen if you raised or lowered your seat a little bit? Would you have a better view through the windshield, or see around the A-pillar better? For many cars, changing the height of the seat is a major project — but it's worth it.
- When you depress the brake pedal as if you were at max braking for a corner, what's the relative height of the pedal in relation to the gas pedal? If it's lower or higher than the throttle, is that causing a problem with your heel-and-toe downshifting? And how far from the gas pedal is it? Would having a small extension on the gas pedal make it easier to blip the throttle when downshifting?
- Are you able to rest your heel on the floor while using the pedals, or do you lift your foot up to apply the brakes? Keeping your heel on the floor provides you with more subtle control over the movement of the pedal, so maybe you need to modify the floor to raise your feet up.
- How much support does your seat provide? I've seen drivers shave a second off their best lap time simply by replacing their seat with one that provides more support.
- Does your car have an adequate dead pedal, and are you using it to support your body when your left foot is not actively using the clutch (or brake pedal if you left-foot-brake)?
- Are your side mirrors adjusted far enough out to the side? If you can see the side of your own car, then you may be missing someone else's car (and your car will never run into itself, so you don't need to see the side of your own car).

Simple changes to your driving environment — the cockpit of your car — will make a difference. And it's relatively inexpensive to make improvements.

Day #193: Trust your gut.

If you feel that something is wrong with your car when you're on track, pit and check it out. Too many spins or crashes have been the result of a driver second guessing their gut feeling that there was something wrong with their car. Okay, if it's the last lap or two of a race, and you're battling for a good position (first is a good position!), then maybe, just maybe... you can take a chance and run to the end. But if it's a track or test day, a practice session, or even qualifying, you're probably better off being certain that your car doesn't have a problem that will cause you to spin or crash.

If you're an instructor, and you have a feeling that your student is getting "out of hand," pit and discuss it. I know many instructors who say, "You know, I had a feeling that crash was coming. Just the lap before my gut told me that we were trouble. I should have trusted my gut — next time, I will."

Of course, with experience, your gut feeling becomes more accurate in sensing problems, and whether you need to deal with them right now, or you can make it to the end of the session or race. But if your car is destroyed because you didn't heed the warnings, you're not going to gain much experience....

Day #194: The faster and longer the corner, the less trail braking you should use and the earlier you need to be on the power; the slower and tighter the corner, the more trail braking you should use to help rotate the car.

A common question I hear tossed around among drivers is, “how much trail braking is needed?” I wish there was a magic formula for determining exactly when and how much trail braking you should use for each and every corner — but there isn’t.

One factor that does determine how much you should trail brake into a corner is whether it’s what I call a “rotation turn” or “set turn.” Typically, but not always, “rotation turns” are shorter, tighter, slower corners; “set turns” are longer and faster.

In many fast, sweeping corners, it is best to be on power, on the throttle from the second you turn in. In other words, no trail braking. Why? Because the car is better balanced this way — the car has taken a set. If you enter the turn while trail braking and then transition to acceleration, the car’s weight distribution is changing. The weight is transferring off the front and to the rear, while the car is cornering. In most cases, a car will have more traction or grip — a higher cornering limit — when it is set, when its weight has stopped transferring from one axle to the other.

This is especially important in corners with a long mid-corner phase. Since you are spending more time in the corner, ultimate cornering grip is critical. In shorter, tighter turns whose mid-corner phase is almost non-existent, then the ability to rotate or change the direction of the car is more important than overall cornering grip. In this case, you should trail brake more. Trail braking enables you to rotate or turn and change the direction of the car quicker.

Given the choice, if you could accelerate all the way through every corner, rather than having to slow down and rotate the car, you would do that since the car would have more overall traction. You would be able to maintain a higher speed through the corners. But that's not practical or useful in every corner. The main objective for some corners is to change direction. These are Rotation Turns, since the main challenge is rotating the car. The main objective in other corners is to maintain as high a speed as possible through the turn. These are Set Turns.

Day #195: Imagine the gift of driving.

Scientific research has shown that if you receive an unexpected gift, you are more likely to perform better at whatever activity you're about to engage in. In one study, doctors were given a gift at the beginning of their day, and it was shown that they made better diagnostic decisions throughout the day than on days they didn't receive a gift.

Overall, you're more likely to be more creative, make better decisions, communicate better, and analyze information more efficiently after being given an unexpected gift. Why? Because of what that gift does to your state of mind, the emotions it triggers. Receiving an unexpected gift impacts your emotions, and your emotions impact everything else. They are hugely powerful.

Hmmm… as a performance driver looking for tiny things that will make a positive impact on your performance, how can you use this research to help you drive faster? Obviously, you can't ask someone to give you an unexpected gift!

So how can you use this to help improve your driving? Can you use this in a deliberate way?

Again, it's the little pieces, the small gains that make the difference. So if you received an unexpected gift just prior to driving, that might make the difference.

Short of setting up individuals to surprise you every now and then with a gift (wouldn't that be nice!), if you simply imagine that happening, research shows it will help. Small gains, right? And even better than just imagining someone walking up to you and giving you a gift, actually recall a time when something like that has happened to you. It doesn't even have to be a material gift, but it could be something that someone said to you that felt as good — or better — than any gift you'd received. I'm pretty darn sure that if you think hard enough, you'll be able to recall something like that from your past.

Or, how about if you turned it around and framed the fact that you get to drive on a race track as the gift. Think about it. How amazing is it that you have the gift of being able to drive fast on a track?

This isn't just a silly trick. As you think about "the gift of driving," notice how it makes you feel. Notice your emotions, your state of mind. Take a mental snapshot of that, and add a caption just below the image: "The Gift of Driving." That is now your trigger. Use it.

When you're at the track, take a few minutes prior to each session to say "I've been given the gift of driving" to yourself, then recall that snapshot, and the state of mind and emotions that go with it.

You may be thinking, "Hey, I'm at the track! I can't help but be in a great mood." Great! But, even though you're in a good mood at the track, nervousness, anxiety, frustration,

anger and fatigue can creep in. Even a tiny amount. And it’s the aggregation of many tiny bits that add up and make the difference.

Here’s the thing: Many drivers will read what I’ve written and think, “Okay, that's nice,” and then do nothing with it. They will go back to installing stickier tires on their car (that will wear out soon). They will spend hours talking to other drivers about the trick line through Turn 5 and where to brake for Turn 9. They will download and go through their data and video, looking for clues as to where to find time. And all of these things help.

But a few will do that little bit more, and actually develop the mental programming that goes along with “The gift of driving.” And they will find just a tiny bit more lap time in each and every section of the track, and that may add up to a whole second a lap faster!

Day #196: Learn what your performance diet is.

Why are some very successful drivers vegans, some eat a keto diet, some like carbs, and some are very picky about what they eat, while others just seem to stuff anything in their mouths?

They've learned what works for them.

I am far from a nutritionist, so take my advice about what to eat over the course of a race or driving event with a grain of salt (but watch your sodium intake!). What I'm about to share here is based on what has worked for me and many of the drivers I've been around and coached.

Be aware of how you feel and perform after eating. That's it. That's my advice — the advice that seems to work most often. But only if you really pay attention.

Most often, during and after eating, we focus on — are aware of, pay attention to — how the food tasted. When we're aware of how something tastes, we tend to gravitate towards food that we like the taste of, even to the point of eating food that doesn't help us perform well. But when you focus on how food makes you feel, and how you perform afterwards, you will begin to crave foods that make you feel good and perform well. We crave things that match what we're aware of.

If, while eating a bacon cheeseburger, you focus on the taste — and you really enjoy the taste — you're likely to want to eat another soon. Instead, if you focus on how you feel after eating the bacon cheeseburger — and you feel bloated and slowed down — you're more likely to want to stay away from it in the future.

If you eat something that makes you feel like you're ready to take on the world, guess what you'll crave more of? That's your performance food.

From my non-expert (but keen observer) perspective, that's why some drivers perform at their best after eating red meat, or a vegan meal, or anything else. Everyone's body is different, and what works for you may not work for me, and vice versa. So, if I or anyone else tells you that you should only eat this or that, it could be misleading.

At one time, I tried a vegan diet, but I lacked energy. I tried carb-loading before big physical challenges, and it wasn't bad, but my body tended to "crash," eventually. I've noticed that if I eat red meat the evening before racing, it can make me a little more assertive, and I have more long-term energy. I'm not saying that is the solution, and that everyone should eat red meat. Absolutely not. What I'm saying is it works for me, and that's a conclusion I've come to after many years of paying attention to how certain foods affected the way I felt and performed.

As the real experts who have written articles about nutrition, fitness, and performance here in Speed Secrets Weekly have said, hydration is perhaps the most important part

of what you need to do while at the track. Obviously, there are hundreds, if not thousands, of different drinks that make all sorts of claims about how they're going to help you recover from the stresses and strains of performance driving. Do some research — try various products before, during, and after tough physical workouts. Again, pay attention to how you feel.

One drink that does very little self-promotion (but is very effective), is good old water. Plain water. There's nothing wrong with Gatorade and drinks like that, but ordinary water is something you must be drinking. If all you drink are "energy" or "recovery" beverages, my experience is your body will get out of balance. You can get too much of what's in those products. Some is good, but too much is too much. It's extremely rare that you'll ever drink too much plain water.

If you're going to use a drink additive to help replace your electrolytes, or some of those goo packs to help your body recover, test them ahead of time. Don't try them for the first time during an event at which you want to perform at your best. When I did the Thunderhill 25-hour race in 2017, I thought I should be prepared with some new replacements, since I hadn't recently used any of the latest stuff. Over the course of the weeks leading up to the race, I tested a few different ones during my training to learn how I felt. By the time that race came around, I knew what my body preferred.

Many experts agree that it's better to eat numerous small amounts throughout the day than it is to eat a couple of big meals. Snacking is great, as long is it's not "couch snacks." Stick to trail mix, fruit, protein bars, and veggies between your meals. Oh, and hydrate (did I mention that?). Avoid the typical, greasy track food.

In my opinion, there is not a one-size-fits-all approach to diet, the amount you consume, or the type of fluids you take in over the course of a race or driving event. You have to learn what works for you, and the best way of doing that is quite simple: Pay attention to — be aware of — how you feel and perform after eating and drinking various foods and fluids.

Day #197: Go ahead, be superstitious.

Name a successful athlete who doesn't have a superstition. I bet you can't. Even if you find someone who says, "I don't believe in superstitions. In fact, I deliberately avoid doing anything that could be considered a superstition," you've found a superstitious person. See, believing that not having a superstition is, actually, a superstition!

What are superstitions? In a performance environment, such as driving, it's a ritual. Often, it's a pre-performance ritual. It's a routine that triggers the performance.

Tiger Woods always wears a red shirt during the final round of a tournament. Michael Jordan wore his University of North Carolina shorts under his uniform. Serena Williams brings her shower sandals to the court, ties her shoelaces a specific way and bounces the ball five times before her first serve and twice before her second. Wade Boggs ate chicken before each game, always took batting practice at 5:17 and ran sprints at 7:17, and drew the word "Chai" (Hebrew for "life") in the dirt before going up to bat. Hockey goalie Patrick Roy would skate backwards towards the net before turning around at the last second — an act he believed made the goal smaller; during the game, he would talk to the goal posts, thanking them when a puck was deflected.

Some of these things border on the bizarre, while others are simply a routine. But what they really do is mentally prepare the athlete to perform at their best. They're triggers, actions that begin the performance.

As long as you can control the routine, go for it. In fact, I strongly recommend using some form of pre-performance ritual as a trigger. The more you do it. the more consistent your performance will be because it will consistently trigger the performance you want.

Of course, if you can't control it, don't use it. A silly example would be a routine of saying something specific to a specific friend, and believing that's the key to you performing well. The problem with this is that you can't guarantee that your friend will always be there. Or believing that you can perform better when you have a cold, or when the sun is shining.

Notice that I used the word "believing" in that last sentence. Ultimately, a superstition is a belief. If you believe strongly enough that walking under a ladder will negatively affect your performance, then it will. That's actually a reason you want to use a pre-performance ritual; it triggers a positive belief.

If you want, think of a pre-drive routine that helps as a pre-performance ritual. And if you want, think of things that don't help as superstitions.

But in the end, managed properly, and even developed with intent, superstitions used as a pre-drive routine will consistently improve your performance.

Day #198: Brake... wait... then downshift.

Okay, I get it. More and more cars do the shifting for you. At least, partially.

But still, there are many, many cars hanging around race tracks that require manual shifting. And when that's the case, when you downshift is critical.

One of the most common mistakes I see inexperienced (and even experienced) drivers make is rushing their downshifts. I know you know this, but the reason for downshifting is not to slow the car down, but to get the car into the proper gear to begin accelerating after a corner.

When you drive down a highway in 4th or 5th gear and a traffic light turns red up ahead, you probably brake and immediately downshift a gear. And that's okay because your engine is not at high RPM. What's not okay is that you're building a habit, and that habit often shows up on the track: you hit the brakes for a corner, and immediately downshift, over-revving the engine.

You need to reinforce the habit of braking... waiting... and then downshifting. Yes, this is basic stuff, but I strongly encourage you to check the timing of your downshifts.

Day #199: The best drivers have a growth mindset.

One of my favorite books of all time is *Mindset*, by Dr. Carol Dweck. And fortunately, I was able to do a small study with Dweck, professor of psychology at Stanford, a few years ago. In it, we found - not surprisingly — that drivers with a "growth mindset" got better results than those with a "fixed mindset."

Someone with a fixed mindset believes that what we're able to accomplish is due to what we were born with, that we have a fixed amount of intelligence and talent, and working to develop that intelligence or talent will do very little.

Someone with a growth mindset believes that they're able to take what they were born with and improve, to develop it, to grow.

Obviously, this is not an on-off switch, from growth to fixed. We all fall somewhere along a continuum, from growth to fixed — and it depends on the activity we're focused on, too. For example, with driving and coaching, I'm fully up to the growth end of the continuum, but when it comes to playing music I tend more towards fixed (beyond pushing the Play button in iTunes — I know I'm great at that!). Sure, I know that with focus and effort I could learn to play an instrument, but I can't help but feel deep down inside that I "just don't have an ear for it." And while some of that might be true, a growth mindset tells me that I can improve.

To understand the growth/fixed mindset concept better, and to learn more about how you can use it in your everyday life, watch this short, fun video: Growth Mindset vs. Fixed Mindset.

If you reflect on your season so far, then on what's ahead, think about what you can do to improve. Make sure you're not putting any limitations on how you look at your abilities to learn. I hear drivers often make remarks like, "I'm just not very good at driving in the rain," or "I never could brake late like some others drivers can." Those comments are tending towards the fixed mindset end of the continuum. And they're limiting your progress.

Even worse, I've heard instructors at race and performance driving schools refer to a driver and say, "OSB" (short for Other Sport Beckons). I hate that! Not only is it insulting, it's limiting. And it says the instructor has a fixed mindset, perhaps even more than the student. So, who is really limiting the progress of the student, the driver or the instructor? If you're an instructor, stop that! While some students may get it sooner and more easily than others, there is not a driver in the history of the world who cannot improve. Everyone can improve! That's the only thing we know one hundred percent for sure about them — they can improve.

A growth mindset is NOT just positive thinking. Positive thinking is "I'm going to make a million dollars this year." Having a growth mindset is "If I put the right kind of focused effort into the appropriate things, making more money — even a million dollars — is

possible." It's knowing, "If I focus my practice the right way, I can improve my braking and corner entry speed." It's not a lack of talent that limits us, it's the limits we put on our learning.

Risk is always a factor in learning anything, especially with what we do on track. But often, it's the risk or fear of failure that really holds us back. We're afraid of looking bad and that limits us from progressing. I've seen more drivers held back by their fear of looking bad than their fear of an actual crash. That's not always the case, obviously, but it is with a surprising number of drivers I've worked with.

Success is often the root of a fixed mindset. We see this all the time...

- A child who does well in school, and is then told that she is intelligent, even brilliant. When she gets good grades, it confirms that she is smart. But over time, she tries to stay away from tests because she fears getting less than a "brilliant" grade, and possibly showing that she's not naturally intelligent. She also doesn't work hard at her schoolwork because she hasn't needed to, and to do so would indicate that she's not naturally smart. That leads to a habit of not putting the effort in. When she gets to a situation where she really has to work hard, such as going to an elite college (where she is no longer above average, but simply average at best), she struggles. She even avoids it. And don't even get started on what having to work hard indicates (that she's not naturally brilliant). I don't need to tell you some of the mental challenges she will likely face — from a lack of self-confidence to depression.
- A driver who has early success and driving fast seems to come naturally and easy for him. The comments he hears over and over again are, "He's a natural." Every driver, no matter how naturally gifted they are, needs to put the effort in to learn and improve. But if he's been told he's naturally gifted, then his mindset is one of "I don't need to put effort into this. I don't need to try anything new. What I'm doing is working, so don't mess with success." It's this "don't mess with success" that limits growth and improvement. The single biggest challenge I see successful drivers face is the willingness to try something new, to experiment.

Look at how you've been successful as a driver. Celebrate it. Enjoy the feeling of what you've accomplished. Then think about the amount of effort you've put into it to get to this level. It may be a lot, or it may seem like it's been relatively simple. But it's this effort that made the difference. It's what you've done with your natural abilities, intelligence, and talent.

Now think about the limitations you put on yourself...

- How you drive in the rain.
- How good you are on the brakes.
- Your ability to get up to speed quickly.
- Your ability to feel what the car is doing so you can tell whether it needs setup tuning.

- How well you learn new tracks.
- How well you get the corner entry just right.
- Your consistency.
- Your ability to visualize a lap (or more) in your mind.
- How often you get on the throttle before being able to commit 100% to it.
- How your budget — financial or time — impacts your ability to learn.
- Your ability to maintain or regain mental focus.
- Whatever....

You're human, so I know you have some limitations that you put on yourself. You've fixed some beliefs about yourself in your mind.

Now, think about what it would take to work through each of these challenges. Is it possible? As an example, let's take the obvious "excuse" — a lack of time and money to practice as much as you'd like, or you feel is necessary to get better. What can you do that costs little to nothing to improve each of the areas of your driving that you've identified? I know for a fact that you can do a lot with nothing.

- Mental imagery (visualization) costs nothing, and is the single most powerful tool out there. And yes, it takes practice, too. It's not something that some were born being able to do and some others not. If you can imagine anything, you can use mental imagery. If you have ever worried about something in your life, then you can use mental imagery!
- Practicing and building muscle memory by "driving" your car with the engine turned off, just sitting in the garage. Footwork is critical to driving fast, and you can practice without the car even running or moving.
- Developing your depth of knowledge through reading and watching videos.
- Reviewing data and video, but with a plan. Taking what you see and asking why it is what it is, and then thinking through what you'd change. And then, actually imagining doing those changes.
- Writing out a "driver development plan" for the new year. As Robert Cialdini (author of the book *Influence*) says, "People live up to what they write down."

There is no limit to what you can learn and improve without it costing you a penny, if you have a growth mindset.

There’s always more.

Day #200: Recovering quickly from a problem is more important than avoiding it in the first place.

It's been said many times, but it's worth repeating: There are two types of drivers — those who have crashed, and those who will. No matter how skilled you are, and even how cautious you are in your driving, the odds are you will have an "off" someday if you spend enough time driving fast around a track. It happens.

Please note that an "off" does not have to be a bad crash. It could just be a spin through the dirt, a lock-up and slide off the outside of the track, or something similar.

And when I say it happens, fortunately, it happens much, much less often than the average person on the street imagines, right? In fact, think about how many miles you're driven on the track, and compare that to the number of crashes you've had. I'm not the only one who suspects that track driving is safer than sharing the roadway with the average motorist (I wish someone would do the actual research on this and prove us right or wrong).

The same can be said about mechanical problems. Drive on a track long enough, and it's likely you'll have some type of mechanical issue. It could be a big one, or just a minor issue. Either way, it could lead to a worse problem, such as running off the track or spinning in your own oil.

My point here is not to dwell on the fact that this can happen, but on how to recover from it.

For instance, in a major endurance race, it's usually the team who deals with the problems most efficiently (and quickly) that wins. If you recall, years ago Audi designed their Le Mans-dominating cars so that the entire gearbox and rear suspension could be replaced in minutes, rather than hours. They knew that despite all the testing, development, and preparation, it was still possible to have a mechanical failure. So they prepared to recover.

As a driver, it's important to do the same: prepare to recover. The more you prepare and practice, the better you'll get at recovering.

Not all mistakes or problems result in a crash or mechanical failure. In fact, a much more common situation is simply losing a few positions in a race or having a slower lap time.

We humans are weird (there are no exceptions!). We find it easier to focus on what could go wrong than what could go right; we find it easier to tell ourselves what not to do, rather than what to do; we recall what we did poorly more easily than what we did right.

Stop it!

Right now, I want you to stop and think about the times that you're recovered. It could be from a mistake you made turning into a corner, and instead of sliding off the track, you balanced the brakes just enough, drove wide around the outside of the apex, and delayed applying the throttle; instead of spinning, you simply were slow on that lap. Or, you made a pass on the first lap, only to slide off the track at the exit and get passed by four cars; you then focused on your driving technique, looked ahead, and over the next twelve laps you picked each of those cars off, plus a couple more and finished on the podium. It could even be a car that you bought and then sold for a big loss, but in the process you met a person who became a positive influence in your life. In other words, a mistake that you recovered from.

I'm sure if you take the time to think about this you'll come up with at least one great recovery, and possibly many. Now, create and attach a trigger word or phrase to each recovery. Play that recovery over and over again in your mind while saying that trigger to yourself. You've now built yourself a "recovery program." Practice using it.

Things suck sometimes. Mistakes happens. Accept that. Then move on, using your recovery program.

Day #201: You have to BE before you can DO.

Which comes first, the ability to do something or the belief that you can do it? If you don't believe you can do something, what is the likelihood you'll be able to do it? Slim, right?

So, it makes sense that you have to believe you're capable of doing something before you're able to do it. But where does that belief come from, especially if you've never done it before? Can you believe you're capable of doing something if you've never done it before?

Which comes first, the chicken or the egg? Which comes first, the belief or the ability?

Let's take a very fast corner as an example. It's a corner that can be taken at full throttle, but just barely. In fact, you've never been able to hold your foot flat to the floor through this corner, even though you were sure you did… but the data showed that you eased up on the throttle a little. Darn that data!!!

You think the corner can be taken flat out, but you haven't done it. As you approach it each lap, you tell yourself to go full throttle, but still your sense of self-preservation kicks in and you lift just a bit. Hey, even Fernando Alonso experienced the same thing when driving at Indy last year, as he confirmed when he said, "At the beginning I have to be honest, the right foot, how do you say, had a mind of its own and was not connected with my brain, so I wanted to be flat out, but the right foot has its own life."

At this point, you think it's possible.

You have another driver take your car out for a few laps, and looking at the data afterwards you see that he flatted the corner on his third lap — and every lap after that — and that resulted in a few tenths of a second gain through that segment of the track.

Now you know it's possible.

But does that mean you can do it?

Stop to think about the physical skill it takes to take a corner at full throttle. The act of holding your foot flat to the floor is not overly difficult, is it? In fact, holding your foot to the floor on the throttle may be one of the simplest physical techniques you use when driving on a track. So why is it so difficult to do?

Because you have to Be before you can Do. Right now, you are a "fast corner throttle lifter." That's who you are, and all the attempted Doing in the world isn't going to change until you Be a "fast corner full throttler."

But how can you Be something if you've never Done it before? Fake it 'til you make it. Change your mindset. Change your belief. Mentally do it enough times that you begin to

believe that you can do it. Do it enough times in your mind and you go from thinking and knowing it's possible, to you doing it.

Imagine sitting in your car in the paddock between track sessions. You close your eyes and imagine approaching that fast corner. You're actually pushing your foot flat to the floor on the throttle while sitting there with the engine turned off. Your hands are on the steering, as they would be when driving. You imagine looking far ahead, seeing the arc you're going to drive through the fast corner. You can hear the sound of your engine pulling hard at high RPMs. You feel every bump and crack in the pavement as you rush down the straightaway. Your left foot is braced comfortably against the dead pedal, and your right foot firmly planted on the throttle.

As you get to the turn-in point you physically turn the steering wheel, as you would at speed, on track. Simultaneously, you apply just a tiny bit more pressure on the throttle, to ensure you're committed to full throttle. As you look way through the corner, you feel the g-forces build, the steering get a little heavier, your body being pushed into the side of the seat. Your head is being pushed to the outside of the corner, but you contract your neck muscles to support it. You feel the tires grip the track. The car is perfectly balanced and feels surprisingly stable. In fact, without the lifting of the throttle and the subsequent weight transfer, the car has more overall grip and arcs through the corner as if on a slot car track.

You've just flatted the fast corner. You do it again. And again. And again. And again…. Sitting in your car in the paddock, you've now become a "fast corner full throttler."

It all begins with the belief that you can do it. You need to see yourself as a driver who does it. You need to *Be* before you can *Do*.

Day #202: Just because a change doesn't work the first time, re-think and try it again. Alter your technique to make the increase in speed work.

"I was going to run a marathon, so I went out running every day for a week. But I couldn't run 26 miles at that point — obviously the training wasn't working. So I quit."

Silly, right?

Why do so many performance drivers do the same thing?

I suggested to a new coaching client that he should change the way he drove a particular corner — that he should actually brake earlier but lighter. He looked at me like I'd just recommended cutting off his right foot. After all, driving fast is all about braking hard and late, right? I could see the "That'll never work" in his eyes, but we talked it over and he said he'd give it a try.

Out he went, and after 5 laps, he came back in and said braking earlier didn't work. The stopwatch agreed — he was slower. Was I wrong? (It wouldn't be the first time!)

My now-somewhat-disheartened driver and I talked a little more, and I asked him to alter the way he used the throttle to go along with the earlier/lighter braking. But mostly I asked him to really focus on braking lighter. Out he went again, and guess what? He went a little faster. We looked over the data, talked some more, and he really got a clear image of how the earlier and lighter braking managed the weight transfer better, giving the car more traction.

When he went back on track for the 4th session, using the earlier/lighter braking technique, he went over half a second quicker through that section of the track.

The moral of the story is to not immediately write off an approach to a new driving technique or a chassis setup. Sometimes you just need to alter and fine-tune how you use it.

Day #203: If you can't do something in your mind, you'll never be able to do it on the track.

For years, tennis coaches in England have noticed a trend: For a few weeks immediately after the Wimbledon tournament is complete, players of all levels tend to play better. The interesting thing is that it's not a result of more physical practice during that time. In fact, most players have spent less time on the court, and more time watching TV. And that's the reason for the improvement in play: mental practice.

It's a proven fact that what we imagine, we get more often. I'm not suggesting that if you sit on your couch and imagine a million dollars dropping out of the sky and into your lap, that this will happen. I'm not even going so far as to suggest the "law of attraction" is real (or not). What I am saying is that if you can't imagine yourself doing something, the likelihood of you being able to physically do it is extremely low.

That is why elite athletes use mental imagery — typically referred to as visualization — to learn new skills, and practice existing ones. It's why they use it to mentally program the appropriate attitude or mindset.

But here's the thing: many drivers struggle to be able to close their eyes and imagine driving an entire lap – or even part of a lap. Yes, some can do that, but many of those cannot mentally pre-play an entire race (Although, Emerson Fittipaldi sat in his car for over three hours the day before winning the 1993 Indy 500 and ran all 200 laps in his mind).

Mental imagery is a skill, just like any other. If, for example, the first time or two that you tried heel & toe downshifting, or hitting a golf ball, you weren't able to do it right, what did you do? I suspect you stuck with it, practiced it over and over again, and eventually you got good (or great, or marginally good, or…) at it.

Can you imagine a baby getting up to walk for the first time, losing its balance, falling over, and saying, "That's it, I'll never be able to walk. I'm never going to try that again!"? Silly, right? But I talk to drivers all the time who have told me that they tried doing mental imagery a few times and still struggle to imagine track detail, lose attention, only see parts of the track, or can't mentally feel or hear it. It's a skill, like any other, and it takes practice.

One could argue that some people are better at mental imagery than others, but that would be the same as saying that some four-year-olds have no imagination. We all have an imagination, and that's really all mental imagery is. It's deliberately using your imagination to build the mental programming so that you do what you want, when you want, how you want.

Day #204: Observe and make note.

I'm going to share some random observations from an endurance race I drove in at Watkins Glen.

1. Any time spent driving at Watkins Glen is special. It's impossible to drive there without appreciating the history of the place — you can almost feel the likes of the great F1, sports car, and NASCAR drivers as you blast up the Esses. If you've never driven the Glen, you need to change that as soon as possible.
2. All of the "low budget endurance" race series have done a great service to our sport. It's made wheel-to-wheel racing more accessible than ever before. Thank you to those folks who have made these series happen.
3. If you've never raced, but have track experience through HPDE and track day events, give racing a try. Whether you choose to continue racing wheel-to-wheel or not, what you learn in the process will make you a better all-round driver. In an endurance race, you are forced (in a good way) to drive off-line, co-exist with other cars, and make more decisions than you will in HPDE/track day events. It stretches your abilities.
4. If you currently sprint race, go do some endurance racing. You'll pass and be passed more in one race than you may in a season of sprint racing. It'll make you a better sprint racer.
5. There seems to be different levels of competition throughout the field at these events. No matter what your car is, and how good you and your co-drivers are, you will find someone to race with. And that adds to the fun, as well as to the learning experience.
6. Going racing with good, fun people is... well, just a whole lot of fun.
7. Driving in the rain is still the most fun and the best learning experience.
8. I observed too many drivers driving "by the book." Yes, even if it was my book, don't just blindly follow what's been written or said by others. I get it — it's important to be consistent before trying something new, but I don't see enough "experimenting" with cornering lines, or driving technique. I see many cars driven exactly the same way in every single corner, every single lap. You only learn so much by doing that. Don't be afraid to try something new: a different line; how much or little you trail brake; where you're looking; what gear you take a corner in; and so on. Endurance races are fantastic events to try subtly different approaches.
9. Some drivers used the rain line on the wet track, and others continued to drive the usual. Which was right? Great question. Automatically switching to the rain line on a wet track makes as much sense as not considering switching. Think about why the "rain line" is a thing (see the Glossary below). If a track has relatively new pavement, the dry line may be as good or better in the rain, since the surface has not been polished and worn. The Glen was resurfaced just a few years ago, so its rain line is not as dramatically different from the dry line as it

used be before that resurface (hint: but it is different). And for you drivers who drove the dry line when it was raining, did you notice those other cars that seemed to have more grip on a different line? Did you notice the cars that were faster than you?

10. During the rain, under a full course caution and following the pace car, I drove all sorts of different lines and tested the grip levels on different parts of the track by accelerating and braking hard, as well as cranking steering in (mostly at pace car speed). I learned a lot. I didn't see anyone else doing that. Drivers don't use pace laps productively enough.
11. More than once I wondered where some drivers learned about the flags and racing rules/etiquette... or whether they learned them at all. One of the things that I really don't like about how some of the "low budget endurance" race series have made it easy to get into the sport is that drivers haven't had to go through the same level of learning of the basic rules that some other forms of racing mandate. If you're going to participate in a race, you must know what every flag means, and how they're used. If you're unsure, ask. No one will mind helping you. If you can't find someone to ask, send me an email. And if you don't know what you don't know, perhaps it's worthwhile taking a step back and go through some basic racing schools.
12. Same with racecraft. Often, I observed one of the most basic guidelines being ignored: "Be predictable." Stay on your line. If a car is coming up to pass you, guess where the driver expects you to be? Yup, on the line. Stay there. It's your line. If you think you're being a nice person by moving off line to let a faster car pass, you're not. You're confusing the overtaking driver, and actually making things more dangerous.
13. I'll say it again. Driving in the rain is the best! There is nothing more fun than searching for grip on a wet track. Nothing.

Day #205: Focus on what you love.

There was a point in my racing career where I got far too caught up in chasing the goal, and I lost track of why I was chasing it in the first place. If the old saying about the journey being more important than the destination is accurate, I'd completely forgotten it. All that mattered to me was making it to the top — whatever that was at the time.

I worked so hard at finding sponsorship to fund my racing, focused so much on supporting my sponsors, and worried about what would happen if I didn't have a sponsor for the following season that I stopped enjoying why I had started racing. I wasn't having the kind of fun that I had when I began. And a funny thing happened (actually, it wasn't that funny at all): my performance behind the wheel suffered. I didn't drive as well as I had when I simply drove for the fun of it.

I can distinctly remember the day in 1994, when I was racing Indy cars, that I realized what was happening. I knew that I couldn't keep hanging on a financial thread, and after three years of trying to put together a competitive program in Indy cars, the likelihood of Roger Penske hiring me for the following season was less than slim. Then it hit me: I was racing an Indy car! Something I'd dreamt of since I was a little kid, I was actually doing. Even though my car was uncompetitive (at that time, teams purchased new cars every year, and there were new engine developments at least every year; I was driving a two-year-old chassis and my engine was at least four generations old, almost 100 horsepower down on a competitive unit), the sheer act of driving an Indy car was incredible. I was racing with guys named Unser, Andretti, Fittipaldi, Rahal, and Mansell. How bad could this be?!

So, while the competitor in me was frustrated by not being able to afford to push my uncompetitive car to the limit, for fear that I'd break it and not be able to make it to the next race — which I had signed a contract with a sponsor to be at — I started to appreciate where I was. Out of the thousands, or even millions of people who want to race Indy cars, I had actually made it there. And even my uncompetitive Indy car was a blast to drive. Eight hundred horsepower in a 1500-pound car that could generate 4.5 gs through a turn will always be the most fun thing I've ever done, or will likely ever do.

And a funny thing happened (and this was actually kinda funny): my driving performance improved. The more I focused on the fun I was having, the better I drove. While that still didn't lead to Penske offering me a contract, a couple of IMSA sports car teams did. And that led to many years of getting paid to drive, not having to worry about sponsorship. All I had to do was enjoy driving. Of course, I never would have said this to my team owners during those years, but I would have driven for nothing, I was having so much fun. In fact, it almost seemed that the more I focused on the fun I was having, the more I got paid to drive.

If you're on a path to the top, keep this mind. And even if you're not trying to make a career as a professional race driver, keep it in mind. There are times, at all levels of motorsport, when it can get frustrating, you can get discouraged, and you might wonder

why you’re bothering. Think about the pure enjoyment you have while behind the wheel of your car. Smile. Relax. Drive well. Have fun.

Day #206: Practice footwork drills.

Way back in time, when I was a teen, I played tennis seriously enough to be on the junior tour with visions of Wimbledon in my head. I knew that if I couldn't race cars, I'd be a professional tennis player. Ironically, I stopped playing tennis and went racing because I couldn't afford to compete on the court. See, other players that I was up against were in a position to have full-time coaches and they could practice all day, whereas I had a part-time job and got coaching when I could from the club pro. So, I left my racquet behind and went after my first love, racing. But what I learned from tennis has helped me immensely in my motorsport career.

In 2022, I watched Wimbledon (sitting at Centre Court at Wimbledon!), I was reminded of the number of hours I spent as a teen running around a tennis court without ever hitting a ball, and it was a lot! Why? To improve my footwork and get efficiently faster at moving around the court.

This was an early example, for me, of deliberate practice (although I didn't know it at the time). Sure, I practiced playing the game, but my coach broke the game down into specific drills and I practiced them over and over again. And over and over again. My footwork was one of my strengths.

But it wasn't just moving around the tennis court quickly that mattered, but also how efficiently I moved. Reaching the ball in three steps instead of four was usually better. I had read *Jackie Stewart's Principles of High-Performance Driving* book and applied his "economy of movement" approach from the cockpit to the tennis court. And then, a few years later, back to the cockpit of race cars.

Footwork is as important to driving as it is to tennis (although the consequences of a mistake on the track are higher than on a court!). The smooth quickness of moving from one pedal to another (for right-foot-brakers) is important for obvious reasons (no wasted time). The transition from braking to throttle should be seamless, and unfelt. As always, there are exceptions, but the times when an unsmooth transition is not the best way to drive are extremely rare. And the rate and control of the release of brake pedal pressure is critical to being consistently fast.

Just like practicing footwork in tennis without playing an actual match, you can improve your driving footwork before getting to the track. But how?

- *Simulator*: One of the best uses of a simulator is to practice footwork, and you can/should set it up so you can record video of your feet, then review it afterward to see where you can make improvements.
- *Visualization*: Yes, actually imagining driving your car on track, but moving your feet in time with it will make you better at it on track.
- *In a stationary car*: Sit in a non-moving car (in your shop/garage, the paddock, etc.) and work the pedals quickly and smoothly, moving from one pedal to the

another, pressing down on the brake or gas pedal, easing off them in a deliberate/controlled manner; this will make you better at speed.

- *Balance board*: Part of what improves your footwork is increasing the fine muscle control and strength in your feet, ankles, and calves; using a balance board is one of the best “tools” for doing that.
- *Specific exercises*: Calf raises, toe raises, squats will improve your driving, and flexibility is at least as important as strength.
- *Racquet sports* (tennis, squash, racquetball): Any sport that requires quick footwork will help your driving.
- *Dancing*: This last one is a tough one for me to recommend because I dislike dancing, but I'm told it's good for one's footwork. For an example of me on a dance floor, watch Elaine in old episodes of Seinfeld! But dancing with a car is a different story!

Watch the footwork of the best drivers and you will be entertained (that’s what YouTube is for, right?). It’ll also be a valuable educational event.

Day #207: Act the way you want to perform.

Why do some drivers take longer to learn something than others?

If there's one trait that drivers who learned quickly had (at least the ones I've coached), it would be their ability to get a strong picture in their minds of what is needed. Sure, they needed to be told what to do (or in some cases, simply nudged along to figure out on their own what to do), but once they had that information, they could turn it into a mini-movie in their minds and follow the script.

Not surprisingly, one of the first drivers I really noticed having this ability was Craig T. Nelson, the actor. You may know him as Coach Hayden Fox from the sitcom, *Coach*, that he starred in years ago, or his role in movies like *Poltergeist*, or the voice of Mr. Incredible in animated *The Incredibles*. To me, he was a driver I co-drove with and coached in the mid-90s. The funny thing is, I had no clue who he was, nor had ever seen him as an actor when I first raced with him in the Daytona 24-Hour race in 1995. He had reached out and asked me to co-drive with him, so off I went to Daytona. I checked into my hotel room, threw my bags on the bed, laid down to stretch after a long day of air travel, and flipped on the TV. Guess what was on? *Coach*, the sitcom — which I'd never seen. I decided watching the episode would be a smart thing to do, so at least I could pretend I knew something about his acting career when we met the following day at the track!

Over the next few races, I noticed a pattern: Each time I talked to Craig about something he should do to drive faster, he'd stop, look directly at me but through me, and it was almost as if I could see him running the script in his mind. It was like he was rehearsing lines for a movie. He was pre-playing the scene in his head, imagining exactly what I'd just told him to do. I was the director, he was the actor.

Craig would not only imagine his lines, doing what he needed to do visually. No, he would move his body, as I'm sure he'd do when rehearsing for a part. Actors don't just memorize lines; they also memorize the actions that go with what's being said, from the position in the scene to subtle facial expressions. That's no different from not only rehearsing the cornering line you want to drive, but also the subtle brake release, arc of the steering wheel, and progressive application of the throttle.

About fifteen years later, I found myself at Le Mans with another actor, Patrick Dempsey. While my main focus was not coaching McDreamy, he would often ask for my help. And each time I'd suggest something, I'd see the same thing from him that I'd seen from Craig. He would rehearse his part…I mean, rehearse what he needed to do in terms of driving technique and cornering line. And each time, he'd move his body while imagining the moving picture in his mind.

Over the years, I've coached a countless number of business people who came to our sport a little later in life. I noticed a similar trait in many of these successful people.

Again, it's their ability to see what the future is going to be, and then act in a way that makes it become reality.

Use what you know and apply it to your driving. If you're like Craig or Patrick, then act. If you're a business person, see the future as you do with your company, and then act in a way to make it real. No matter what you do, act as if…. And most importantly, make that a part of your daily routine, part of a regular mental imagery practice.

With that said, I'd like to challenge you to commit to doing one thing over the next month: practice your mental imagery, and specifically making better use of your kinesthetic sense. I'm talking about moving your hands and feet on your imaginary steering wheel and pedals; imagining the g-loads against your body; feeling the vibrations and feedback through the steering wheel and pedals; moving to change gears; and feeling the car's load move around from the braking, cornering, and acceleration forces.

The next time you're doing mental imagery — and that should be daily — act like an actor. Immerse yourself in the role of the best driver you can imagine. And don't just see what you're doing in your mind, but feel it. Act it. Be it.

Day #208: Tune yourself to your tires' ideal slip angle.

Let's talk tires, grip, and your ability to sense their limits.

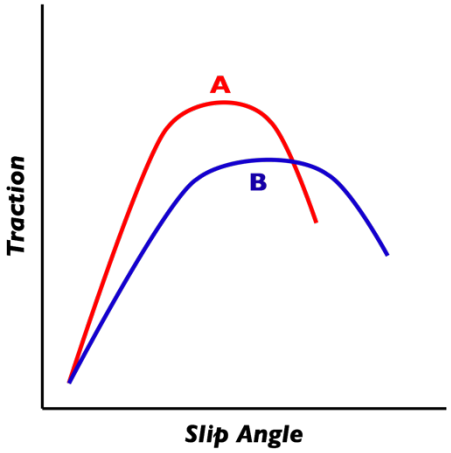
Some tires reach their peak of grip at a higher slip angle than other tires do. Some tires reach their peak grip at lower slip angles. Your "job" is to sense the difference and drive as close to that peak as you want.

I understand that some drivers don't want to drive right at that very peak, because that means you have less margin for error. But it doesn't matter… everything I talk about here applies whether you want to drive at 10/10ths or 8/10ths.

With some tires, the feeling you get when driving right at the very limit or the peak of grip, is a solid, "stuck" kind of feeling. For others, it's as if you're driving on four snow tires — the whole car is squirming around. And for different tires again, it's somewhere in between those extremes.

Some tires reach peak grip when they're barely sliding (lower slip angle), whereas others like larger slip angles and grip the most when sliding more.

Looking at the slip angle versus traction graph, you'll notice that tire A peaks at a lower slip angle, has a higher peak, but is more "peaky" (it grips, and then lets go more suddenly). Tire B likes a higher slip angle, provides less grip overall, but is more forgiving as its peak is spread over a longer range of slip. This might illustrate a range, from a race/super-high performance tire (A) to a lower performance tire (B).



If you're used to driving on tires that like large slip angles, and you change to ones that peak at lower slip angles, it's easy to over-drive them — to slide the car too much and exceed the optimum grip levels of the tires.

If you're used to driving on tires that like lower slip angles — ones that give a solid, stuck feeling — and you start driving on tires that reach their peak grip levels with higher slip angles, it's likely that you'll under-drive them. It's not unexpected for you to drive slower than your tires/car are capable of being driven, since you're not used to the amount of slipping and sliding the tires need to perform at their peak.

Okay, that's nice to know, right? But what to do with this information? Adapt.

We all have a g-load sensor in our bodies. I won't go into the physiological and psychological whereabouts of that sensor in your body, but I'm sure you would agree that you have one. You may think your sensor is pretty darn good, or that it needs a

tune-up or upgrade because it’s not as good as some other drivers'. No matter what, there’s always more. You can always improve.

As you brake, corner, and accelerate, you sense g-forces — if you pay attention. Going around a corner, g-loads (and the forces pushing against your body, sensed by your internal g-load sensor) build up as the tires generate more and more grip (moving up the slip angle vs. traction graph). They reach a peak and level off, and then begin to taper off. If you pay attention, you sense this building, peaking, and tapering off. If you don’t pay attention, you’re more likely to drive below or over the peak grip levels of your tires.

Without driving fast, you can refine your g-load sensor while driving on the street. Just pay attention to the building of g-loads. Sure, you won’t (or, at least, you shouldn’t) ever sense reaching the top and going over the peak when driving on the road, but practicing sensing this building of g-loads at low levels actually helps almost more than it does when operating in the upper levels. If you can sense this when driving at street speeds, it actually is easier when driving at speed on the track.

When you spend time on the track solely focused on sensing g-loads, you’ll improve your g-load sensor. It will become more sensitive. But you need to focus on it. You can’t be focused on tuning your sensor while thinking about changing your line through Turn 5, about your lap times, about where you’re looking, about your braking point for Turn 1, or anything else. You need to focus on tuning your g-load sensor alone, and nothing else.

And remember, every set of tires is different. From new to old, it changes, so you can imagine the differences in type and brand. Some tires have more lateral grip than longitudinal grip; some are the opposite; some are equal. Every set of tires is different. You have to adapt.

- Some tires grip better when braking than when cornering, and others corner better than when braking. Do you know how your tires perform?
- Some tires like to be slid more than others. Do you know how your tires perform?
- Some reach their peak and then fall off very quickly; some peak and stay there longer. Do you know how your tires perform?

Tune your g-load sensor through deliberate practice.

Day #209: The greater variety of driving you do, the better you'll be.

No matter what you drive, go drive something else. And then something else. And again, something else.

The best drivers can drive anything, on any surface, in any conditions.

The days of Formula One drivers crossing over to other forms of the sport are long gone, but still, the best drivers have always been able to drive anything. Think Mario Andretti, A.J. Foyt, Tony Stewart, Stirling Moss, Jimmy Clark, Jeff Gordon, Fernando Alonso.

Okay, you may not have the same opportunities for variety as some of these drivers have, but you can still change things up — and learn from doing so.

For example, the local indoor karting track, autocross events, rallies, Champ car race, riding a dirt bike, ice racing, oval track, or off-roading. Sign up for a driving or riding school. Any of these things will make you a better all-round driver, no matter what you're currently doing.

Notice that none of the suggestions in the above paragraph cost a lot of money. Compared to performance or race driving, most of them are less expensive. And I guarantee that what you learn from them will make you a better driver.

Day #210: Balance loading the car on corner entry with a disciplined turn-in.

There's a fine line between adding a small amount of load to the car before turning in and "crabbing" in to the corner with a slightly early turn-in. Yes, the smallest micro-movement of the steering wheel can shift the initial load to the outside tires and may help. But it may also hurt.

There has been talk of some F1 drivers doing a bit of the "load the steering" thing. Some have been coached to do this. What I'm talking about is just before they initiate the turn into a corner, the driver turns the steering just a tiny bit. The idea is that it puts a little load on the outside tires and it's almost as if those outside tires are not being "shocked" with a big load when turning in. And it can help the car take a set earlier so you know what you have to work with.

For drivers at the F1 level, this can work because they have the discipline to ensure they're not crabbing into the corner. I define "crabbing" as a driver angling the car away from the edge of the track just before turning into the corner - it's not a good thing. Why do drivers do this? For two reasons: First, when the driver gets near the turn-in point, he looks to the apex. As we know, our hands follow our eyes, so looking to the apex can get the hands initiating a turn before we really want it. And second, sometimes a bit of anxiety (or even fear) sets in due to the speed the driver is carrying, and it feels safer to be further away from the edge of the track.

The big problem with crabbing into the corner is it reduces the radius of the corner, meaning you can't carry as much speed. Even being a foot or so from the edge of the track at the turn-in point makes a big difference in the amount of speed you can flow through the turn — it can make the difference of three or four MPH and that can lose a few tenths of a second in just that one corner.

Back to loading the car at turn-in. When word spreads that some F1 drivers turn the steering wheel a tiny bit to load the car, many other drivers think they should do the same. And that could be like a novice skydiver trying to "fly" like an experienced pro.

I've seen some casual track day drivers who drive just a few times a year try to copy this approach of loading the car on entry, and they end up doing more harm than good by turning in early. So it can be a good thing, but only if you have the discipline to make sure you don't turn in early. I highly recommend checking whether you're doing that with data and/or video, as most drivers who crab in or turn in early don't even realize they're doing it.

That advanced stuff is usually just doing the basics better. In this case, it's making sure you're doing the basics right before ever doing something advanced like loading the car with a tiny bit of steering before turning into the corner.

Day #211: Overlap your conscious & subconscious to get into the zone.

Remember your first time driving on track? I'm going to guess you had some combination of at least two emotions. First, a sense of being overwhelmed by all there was to take in and do. This could have been a huge overload or a small one, but either way, there was some amount of it.

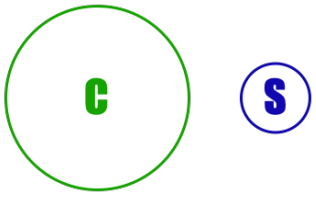
The second emotion was a sense of being where you belonged. Again, for some drivers there is a lot of this, and some little to none of it (they're just trying to survive).

Let's look at the sense of being overwhelmed — why it happens, and how you can manage it over time.

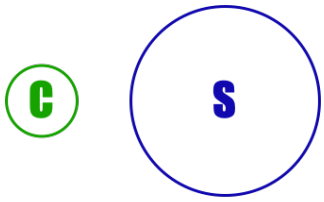
When on track, you're relying on both your conscious and subconscious minds to drive. Your subconscious is running the mental programs, the things you have developed and turned into a habit. This would be things like turning the steering wheel in a counter-clockwise direction to make the car turn to the left. You don't have to think about doing that, you just do it. How you apply and release the gas brake pedals are also done mostly at the subconscious level in the beginning because you've built habits over time from driving on the street. Again, you don't have to think about pressing down on the brake pedal to slow your car, or applying the throttle to accelerate. You do them without conscious thought because you're programmed to use them that way. Well, until an instructor tells you to be smoother with your brake release or throttle application! Then, you may have to put a little conscious thought into it, again.

After some amount of experience, both physical and mental, many of the things you had to consciously think about when you first started track driving become part of your subconscious, and you just do them.

In the beginning, the balance between conscious and subconscious looks like this:



Over time, with experience — again, both physically doing them, but also imagining doing them over and over — what was once something that required a lot of conscious thought can now be handled subconsciously, as if on autopilot. And the balance changes to look like this:



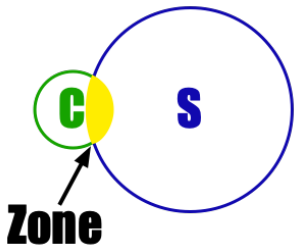
At this level, you're trusting your mental programming, or subconscious, to do most of the driving tasks; your conscious mind is being aware of what's going on around you, considering what changes in technique could do, how a change in the line you're driving could impact speed, how your car is handling, and noticing and responding (or not) to other cars around you.

You may be aware of the four learning stages from learning theory:

1. *Unconscious Incompetence* — Not only can you not do something, but you're not even aware that it's a "thing." For example, there was a time in your life when you didn't even know that heel & toe downshifting existed.
2. *Conscious Incompetence* — You're aware of something, and let's say it's the technique of heel & toe, but you're still not able to do it.
3. *Conscious Competence* — At this stage, you're doing it, but you're having to think your way through it. With heel & toe, you're consciously thinking through each step of the technique.
4. *Unconscious Competence* — For some, this is the holy grail. You no longer have to think about what you're doing — you're just doing it. Heel & toe downshifting may be at this level for you (or maybe not quite).

You can see how this relates to the conscious-subconscious balance. However, I believe there is a fifth stage that is never mentioned in any of the learning theory textbooks. As I wrote in my *Speed Secrets* books, the fifth and final stage is something I call "Unconscious competence with conscious awareness." Sure, there are a lot of big words there, but think about it… this is another way of defining "being in the zone," or "in flow." When you're in the zone, you're driving as if on autopilot, trusting your subconscious programming to drive the car, and yet your conscious mind is aware of all the important stuff around you.

Going back to our conscious-subconscious balance illustrations, this stage, being in the zone, looks like this:



It is driving with your subconscious programming doing most of the heavy lifting: the techniques, skills, actions. While at the same time, your conscious mind is aware of what's going on, and even being able to upgrade your mental programming to a newer version (v2.0, rather than 1.0; or maybe, 2.1, rather than 2.0) on the fly.

When you're in the zone, you must trust yourself to "just do" — letting your subconscious drive the car — while using your conscious mind to be aware of the important things around you (even deciding on what's important and not). Trusting yourself to "just do" and not let your conscious, analytical, thinking mind get in the way can be a challenge. In some ways, it's easier to think your way around a track. But you'll never be really fast that way, nor will you get in the zone. And it takes self-confidence not to doubt whether you have the programming to do what's necessary. It also takes self-confidence (and ego management) to know when you don't have all the programming yet, and to work on what it takes to develop it.

When Tiger Woods was at his peak for many years, he had the self-confidence to make a change to his golf swing, and the timing of these changes is interesting. At the end of a season where he'd won more tournaments than ever before, when he was apparently at his very best, he changed his swing. Understandably, his results were not as good for

a while. Then they picked up and the following season was his best ever, outdoing his previous best ever. Guess what? A year or so later, he changed his swing again, and the same pattern appeared, with him ultimately having an even better season. That took guts! He knew how to balance his conscious and subconscious minds, and trust himself to just do. But he also knew when it was time to go back and work at a conscious level before letting go and trusting his subconscious to just do. Perhaps that's why he was — for a while (and we won't go into that!) — the perfect spokesman for Nike.

Learning to balance your conscious and subconscious, while trusting yourself, takes confidence. But which comes first? The ability to do something or the confidence to do it? You have to have confidence to do this, but your confidence comes from having done it. So, it's a step-by-step process, using your awareness to realize whether you're too far in either direction.

I challenge you to work on finding the right conscious-subconscious balance as much as you do on your car balance (handling setup, managing weight transfer with your driving technique). Trust yourself, push yourself, trust yourself, push yourself.

Day #212: Treat every corner different.

A few years ago, I was coaching a young driver who showed signs of brilliance. But other times, he showed signs of the opposite — he was slow and inconsistent. Actually, that's how I got connected with him – I was asked by his family to coach him, as many people thought he was going places in racing, but he wasn't quite living up to expectations.

In the process of coaching him, I learned the importance of something that I knew deep down inside, and yet had not put enough emphasis on: Not all corners are created equal, and therefore should not be driven the same way.

I know that sounds very simplistic, and obvious. But internalizing it, and doing something with it is very different from just knowing about something.

This driver, "Mario," was blindingly fast in some corners, and not so in others. On our first day of working together, I watched from the side of the track in the slow- to medium-speed corners, and he looked great. The thing I watched closely for was the gap between the front fender and tire (he was driving a Porsche), because from that, I could get a very good idea of when and how he was releasing the brakes. I remember walking back to the paddock after a session thinking, "I don't think we're going to find much more speed in Turns 3 and 7. Mario's brake release, rotation of the car, minimum speed, and exit out of there looked great."

Back in the trailer, looking at the brake pressure trace on the data showed exactly what I had observed: a nice tapered release of the brake pressure as Mario turned into the corner.

I didn't have much time to look at the data before the next on-track session, but noticed a very similar brake trace in the two fast corners: 1 and 10. So, that's where I went to observe during the next on-track session.

Immediately, I noticed an opportunity for improvement. Mario was driving these two fast corners with the exact same technique he had in the slow corners, and that was hurting his overall lap times. He was trail braking into these corners, and late to get back on the throttle to balance the car. As I walked back to the paddock after the session, it dawned on me that many, many drivers do this exact same thing. They drive all corners with the same style or technique. And that's why they are fast in some corners, and not in others; on some tracks, and not on others.

Of course, I confirmed my observations with the data. It showed that Mario was braking late and trailing into the fast corners, the same as he had in the slower corners. This technique works well in slower, tighter corners where he needed to change the direction of the car a lot, but it didn't in the fast corners where keeping the chassis balanced is more important than changing direction.

On tracks that were mostly made up of slow- and medium-speed corners, Mario was brilliant. On tracks with fast corners, he struggled. And it was because he was driving the fast corners like slow corners should be driven. After a short conversation and explanation of the downside of late braking followed by trail braking into fast corners, I challenged Mario to start his brake zone earlier in Turns 1 and 10, and be off the pedal by the time he was turning into the corner. The goal was to begin applying a touch of throttle at the turn-in point.

Admittedly, Mario was not able to pull this change of technique off in one session, but he did feel how it might help, and made some improvement. That is, after he got used to braking so early. See, when trying to drive faster, it was in Mario's — and most all of us drivers — instinct to brake late. It feels fast. But, often, it's not in fast corners.

After doing some mental imagery/visualization that evening of what he'd begun to work on that day, the next day was one of those that makes me come back for more and more as a coach. In his first session, he knocked three-quarters of a second off his best lap time, and all of that gain came from Turns 1 and 10. The grin on Mario's face is something I'm unlikely to forget.

There was something in his smile that made me realize that it was more than just the improvement in his lap time that made Mario happy. It was a confident grin. Up until the day before, he had begun to wonder about his own abilities. He was unsure. But now he was sure. It was like he'd found that missing piece of the puzzle, and he now knew that he could be as great as everyone expected him to be. He could fulfill his potential. The small change in technique unleashed another level of confidence.

When I asked Mario to explain to me what he'd learned that weekend, he replied, "Not all corners should be driven the same. I have to adapt where and how I brake for corners. In slow corners, I want to trail brake to help rotate my car. But in fast corners, I want to get my braking done early — with no trail braking — and be on power all the way through the corner because the car feels more stable that way."

I then reminded him that there are always exceptions to guidelines like these, but he'd nailed it.

It's been fun to watch Mario go on to make a successful career as a professional race driver.

P.S. – Note that the driver's name, the turn numbers, and type of car have all been changed to protect the innocent.

Day #213: Plan to improve.

Think about your driving so far this season:

- What have you done so far?
- What could you have done better?
- If there was one thing you'd like to do better with your driving, what would it be?
- What are your strengths as a driver?
- Of the things you'd like to improve, which are physical skills/techniques, which are part of the mental game, and which fit into the preparation area?

Stop right now and think about these questions — please. Take a few minutes to write your thoughts down on paper, or type them into your phone or computer. That will make a difference — now is the time to be doing this type of preparation.

With these things in mind, what is your plan for the rest of the year (and next year)? What are your priorities for your driving performance?

There's a big difference between having a plan and wishful thinking, right?

A driver sent me his plan by email the other day, and I thought it was brilliant in its simplicity and focus. Here it is: "This year, first enjoy driving. I'm doing this for fun. Not money and not trophies. Second, it would be fun to brake a little later and commit to the throttle a little sooner. Third, don't let number 2 mess up number 1."

I love that his overall plan can be summed up into a few short sentences. It's easy to remember and keep in mind when at and on the track.

But let's take this "plan summary" to the next level, something that can be put into play more easily and effectively. If it was me, I would take some of the thoughts I'd written down from my earlier questions, and flesh out this plan with a bit more detail. Why? Because something too broad is next to impossible to act on. For example, "brake later for Turn 1" rarely works because it hasn't been defined with enough detail for the brain to use.

Taking the "it would be fun to brake a little later and commit to the throttle a little sooner" plan, here's an example of how I would clarify that further:

- I'm going to use at least one track session each day to focus on my braking technique. To begin, I'm going to experiment with making my initial application of the brakes harder and more sudden. I'll still be smooth, but more aggressive with how quickly I apply the brakes, and press a little harder — until I either experience ABS or a bit of lock-up, as I need to know the limit. I can then back off just slightly from there. In doing this, I believe that I will be able to compress my

braking zone, and make me comfortable braking later — keeping things fun. And when I brake a bit later, that will be fun to know that I'm making improvements.

- I'm going to put twenty minutes aside at least three times each week over the next couple of months to do some mental imagery of the "hard initial application of the brakes" program. I'll close my eyes, relax, and then begin imagining approaching a heavy brake zone before a corner. In my mind, I'll see, feel and hear myself quickly transitioning from the throttle to the brake pedal, and applying the brakes hard. I'll actually, physically move my foot to build the muscle memory with my imagination. I'll not just imagine it, but I'll be moving my body as I would when behind the wheel of my car.
- Back at the track, I will become more aware of the initial application of the brakes. In doing so, I'll also notice that I can now manage my corner entry speed and the load on the front tires — deliberately keeping the front loaded so the car rotates as I turn in and trail brake. This will get the car pointed in the direction I want it to go, and that gives me confidence to commit to the throttle earlier. I love this! It's fun! I'm braking later because I'm applying harder initial pressure — that's the Why I'm able to brake later — and that gives me the ability to manage my corner entry speed and rotation, resulting in getting back to throttle even earlier than before. This is cool — not only do I want to brake later and commit to throttle sooner, I know Why I can do it. I have a plan, and I understand Why this will work. This is fun!

This plan could end right there, or I could continue to dig deeper and refine and define the details of what and how I'm going to do it.

I'm sure you got the most critical part of the plan, though. It's the understanding of the Why the plan will work. Instead of wishing to brake later and commit to throttle sooner, the details give me the route to getting there, and a reason it will work. How do you think this will impact your belief system — your confidence?

What is your plan? Write it out — a summary of the overall plan, and then the details of how you're going to do it, and most importantly, Why it's going to work. If you'd like, email it to me. I'd like to see it, and maybe I'll have something to add to it that will help. If nothing else, sharing your plan with someone else might just keep you on track to follow through with it.

Have a Plan.

Day #214: Think about the rate of turn-in, rather than where you turn-in.

Wander around a paddock between sessions at a typical track day event or club race and listen in on conversations; you're bound to hear a lot of talk about the cornering line. And among that, much discussion about where to turn in for, let's say, Turn 5.

It's been said (admittedly, by me) that your turn-in dictates much of what happens throughout the rest of a corner, and especially where you apex it. What isn't focused on enough is the rate at which you turn in for a corner, rather than just on where you turn in.

Imagine approaching a right-hand corner. You apply the brakes, make a downshift or two, and then get ready to turn in. Let's say the turn-in point is at the end of the curbing on your left. If you begin to turn the steering wheel right when the front wheels are aligned with the end of the curbing, but you do so slowly and gently (trying to be smooth), where the car starts to change direction, the angle it takes towards the apex, and even how much you have to turn the steering wheel to get it to the apex, will all be very different from you turning the steering wheel relatively quickly.

Which is right? A slow, gentle turn-in, or a quick, crisp turn-in? Yup. It depends.

But my experience tells me that many drivers use the same rate of turn-in for practically all corners. Okay, perhaps the difference between a very long, sweeping, fast corner and the super-tight, slow hairpin might not be the same. But even then, for many drivers the rate of turn-in is fairly similar.

So, what is the ideal rate of turn-in?

Just fast enough to get the car to change direction the amount you want, but not so much that it causes excess weight transfer.

We all know that the less weight transfer we cause, the more overall grip that the car has. So that means we want to turn the steering wheel slowly so it doesn't upset the balance. But if you turn the steering wheel really, really, really slow… well, you won't have caused much weight transfer, but you'll be stuffed into a tire wall on the outside of the corner!

If you turn into the corner quickly, you get the car to change direction and go where you want, but it causes more weight transfer, and that means the car has less overall grip to get the car to go where you want.

You can turn the steering progressively, initiating the turn slowly and then gradually speeding up the rate at which you turn it. You can make the turning of the wheel digressive — you start quickly and then gradually slow the rate of turning down. Or it can be consistent throughout the rotation of the steering wheel.

Another compromise! Hey, that’s what performance driving is: a series of compromises.

The only way to know for sure the best rate to turn the wheel is to experiment with it. There are some general guidelines that you can start with, such as:

- In long, fast, sweeping corners, initiate the turn gently and slowly, and then speed it up (if needed). So, it's likely to be a consistent rate of turning, or possibly a progressively faster turn of the wheel for these types of corners.
- In tight, slower corners, use a crisper turn of the wheel — one that might be a consistent rate, or possibly even digressive.

Keep in mind your overall goal: cause as little weight transfer as possible, while getting the car to change directions as quickly as you can. These two factors tend to go against each other... which is why it's a compromise, and why you need to experiment with different rates of turning the wheel to see which works best. Just being aware of this, and being willing to experiment with it will likely lead to some nice improvements for you.

Day #215: WOOP it up.

You know you can brake later for Turn 1, but every time you get there your foot lifts off the throttle and applies the brake — early.

You know you can commit back to full throttle at the apex of Turn 3, but each time your foot hesitates before going to the floor.

You know you can take Turn 7 flat, at full throttle, but the data shows that each time you lift just a little. How frustrating, right?

Each of these scenarios came to mind when I was listening to the NPR Hidden Brain podcast, in which Gabriele Oettingen was interviewed. She's the author of the book, *Rethinking Positive Thinking*, and she used the acronym WOOP to explain the way she believes we should work to achieve any goal, whether braking later, committing to throttle earlier, flatting a corner (she didn't specifically mention any of these examples), or losing weight (she did mention that one).

What does WOOP stand for?

- *Wish* — This is what you want to come true, or your goal.
- *Outcome* — This is the process of clarifying what your wish is; a detailed, clear image of what it would be like, including how you would feel with the outcome.
- *Obstacle* — Getting clear on what your own personal obstacles are between where you are now and what you want to achieve.
- *Plan* — This is your action plan, what you're going to do to make your Wish come true.

Notice that nowhere does it say that wishful thinking is a good strategy! And what I took away from what Oettingen had to say was that simply thinking positively is wishful thinking.

I like this process, as I've never been a big fan of simply positive thinking. Positive thinking seems too much like a dream, rather than a goal. It's like saying, “I see myself in Paris,” and then being disappointed that you're not there. Simply thinking a positive thought doesn't make it happen. You need a plan to get to Paris, and a way of overcoming the obstacles along the way (that dreaded TSA line!). Absolutely, you need the Wish, and a clear picture in your mind of what achieving the Outcome will be like. But understanding the Obstacles within you to achieving your Wish, and then putting a Plan together to work through them… well, that's critical.

The next time you're struggling to make yourself do something that you want to do, use WOOP.

Here's an example of how I would apply this process:

- You want to brake at the 300 marker for Turn 1, but you always begin braking around 350.
- *Wish*: Your wish is to brake at the 300 marker.
- *Outcome*: You find a place where you can't be disturbed, and then you relax and clear your mind. You then imagine what the outcome of braking at the 300 would be. You imagine the sense of accomplishment, the feeling of having the car at its limit, and even seeing the difference in lap time.
- *Obstacle*: You think about what obstacles there are — in you. Fear? Apprehension? Self-preservation? Car preservation? You consider them all, thinking about them in a logical way, but not making more out of them than need be. You're realistic about the consideration of the obstacles.
- *Plan*: You now put a plan together to move your Begin-of-Braking to the 300 marker. You imagine approaching Turn 1, and having your vision aimed at the End-of-Braking point. That makes you feel more comfortable because you don't feel as though all the braking needs to be done right now. As you sit in your chair, you imagine what looking towards the End-of-Braking point would be like, and what it looks, feels and sounds like to begin braking at the 300 board. You move your body as you imagine braking at the 300, building the muscle memory and mental programming to do what you want (make your Wish come true).

Notice that this last step — the Plan — is pretty much what you may have been doing in the past, and is what most (including me) have been recommending for years. By adding the first three steps before this last one, the entire process becomes more effective.

WOOP it up!

Day #216: When learning a new track, make sure to get a "global view" of it.

I see many drivers ignore a very important part of learning a track — getting a global picture of the circuit, by looking at every part of it they can from a spectator's perspective.

It's simple enough. All you need to do is walk or drive around to every viewing area that you can to view the corners from that perspective.

It's one thing to look at and memorize a track layout from a map. It's another to view it from the perspective of an in-car video. It's another again to see it from a simulator. And another again to see it from the view you get while walking it, or even driving it. But it's completely different when you view it from the outside, watching other cars go through a corner. And that additional view helps put everything into perspective.

Prior to driving at Le Mans for the first time, I would randomly stop an in-car video and make myself quickly identify where on the track the car was. That helped a lot. But once I'd visited Le Mans and seen all the corners from the outside, it made this process easier — and helped me get a real feel for the circuit.

The next time you're learning a new track — or refreshing your memory of a track you haven't driven for a while — include taking a look at all the corners from the outside. Get a global picture of where the track goes, and what the turns look like from the spectators' perspective. It'll help you learn that track faster.

Day #217: Gut instinct is real. Trust it.

You're driving in the suburbs, along one of those four-lane roads lined with strip mall after strip mall, Chili's, Panera, Target, Starbucks, Michael's, Best Buy… LOOK OUT!

For some reason you can't quite explain, you brake. You quickly swerve to the left of your lane, checking to see if you have room to move into the next lane.

For some reason you can't quite explain, you just knew that blue SUV was going to roll right through the stop sign and right out in front of you. If you hadn't braked and swerved, there would have been a collision.

For some reason you can't quite explain, you just avoided a crash.

For some reason you can't quite explain, your gut told you that driver was going to blow right through the stop sign and out into your path. And that's a good thing.

The reason you can't quite explain this has to do with how your brain processes information, and specifically, how it recognizes patterns. Neuroscientists recognized a pattern of how we humans process information and they called it pattern recognition (hey, they're neuroscientists, not known for their creativity with naming things!).

On track, some of what we do is because of the information we consciously take in through our eyes, our sense of feel and balance, and through our ears — and then consciously process. We see a car ahead wiggle in the brake zone and we consciously think about what that could mean. We consider all the options — bumps in the surface of the track, some slippery fluid on the surface, the driver braked too hard and locked up the rear tires, the car has too much rear brake bias, and so on. Since we have time to think, we do. We think through our options — brake a little earlier than usual just in case the track is slippery; give ourselves a bit more room between us and the wiggly car; carry on as usual and take advantage of the other car's issues to get closer to it; or brake as usual but really work at getting on the throttle earlier to gain on the driver who may be recovering from the "moment" he had.

That, in fact, is not pattern recognition. That's cognitive thought and decision-making. That's the process we use when we have time.

Approaching the brake zone you glance to the rear-view mirror and see the red car lined up right behind you, staying in position. You brake at your usual Begin-of-Braking reference, apply the brakes hard, look through the turn-in point to about where the End-of-Braking is, then in towards the apex. As you move along the brake zone you ease up slightly on the brake pressure, and then as you initiate your turn and begin to ease off the brakes… Wait! Delay your turn-in, and stay on the brakes just a fraction of a second longer!! Why? Because your gut said the red car was making a last-second desperate lunge to the inside. If you'd turned in as usual you would have hit. Sure, he would have

hit you in the rear tire, which means he wasn't really in position to make a pass, but still… despite you being in the right, you would have been hit.

Here's the interesting thing: Even though you may never have been in that exact same situation before, your brain pieced together enough information that was similar to come to a conclusion that benefited you. Perhaps a flash of red in the side mirror triggered a past experience, a past pattern.

But here's the more important thing: You need to consciously and subconsciously feed your brain information over a period of time so it has enough of the pieces to put together. You need to give your brain some parts of the pattern so it can put together enough of the puzzle to recognize the big picture that it's putting together.

Let's use a puzzle to demonstrate. If you've ever tried to put together a puzzle without having any idea of what the picture will eventually turn out like, you know it's more difficult. If I dumped all the puzzle pieces on to a table and asked you to put it together, but never let you see what the picture of the completed scene was, that would be tough. But eventually, as you got some of it assembled, you'd recognize what you're aiming for, and the last few pieces would be easier.

Experience teaches us how to recognize earlier patterns. But why, then, do some drivers with little experience seem to recognize patterns better than drivers with much more experience? Some would suggest that some are born with that ability. They chalk it up to "natural talent." Pattern recognition is not something one is born with. Well, unless you were able to see, feel, and hear before you were born!

If there's something some are born with, perhaps it's a curiosity to look for and be aware of patterns. I've met some people who seem more naturally able to recognize patterns, but what stands out is their desire to recognize patterns; that seems to come from paying attention and being aware.

If you deliberately focus on collecting the pieces of the puzzle, your pattern recognition abilities will kick in earlier and you'll drive faster, and have fewer incidents. If you deliberately practice paying attention and being aware, you'll "naturally" become better at recognizing patterns. Like anything, you can improve this skill. And when you do, you'll be a better all-round driver.

Day #218: Avoid common footwork errors.

While most drivers focus a LOT of attention on what they do with the steering wheel to drive the so-called ideal line, the very best are more concerned about what they do with their feet. I'm not saying that the line and what you do with the steering wheel are unimportant, but it's what you do with your feet that makes the biggest difference to your speed, consistency, safety… and I'd even go so far as to say, your fun. After all, it's what you do with your feet that impacts the balance of your car, and ultimately, how close your tires are to their limit. If you've experienced "steering the car with your feet," you know exactly what I'm talking about; if you haven't done that yet, you don't know what you're missing!

Some very common driver-induced footwork errors I see are:

- Not keeping your left foot braced against the dead pedal. Yup, I'm starting with the most obvious and basic one. Obviously, if you're left-foot-braking, this doesn't directly apply, but even if you do brake with your left, bracing your body with the dead pedal when you're not braking will help. If you're a right-foot-braker, then your left foot should be against the dead pedal at all times, except when using the clutch. Even between shifts, when you're not actively using the clutch, get your foot back on the dead pedal. The extra bracing will support your body, but also helps connect your body to what the car is doing. Some of the feeling we think we're getting through our butts comes from having the body supported against the dead pedal.
- Sitting so far away from the pedals that your leg is fully extended when any of the pedals are fully depressed. Yes, another basic mistake (you should have a bend in your leg), but one that's not uncommon.
- Lifting the heel of your right foot off the floor to apply the brakes. You will always have more control of the pedals, and be able to finesse them, if you're pivoting your foot at the ankle, rather than pushing and releasing the pedals with your whole leg. You just don't have the same fine motor control with your thigh muscles as you do with your calf, ankle, and foot muscles. Operate the pedals with your heel on the floor. Stick a GoPro camera down in the footwell sometime to check whether your heel is staying on the floor.
- Having your whole right foot centered on the brake pedal when heel and toe downshifting. If you press the brake pedal with the ball of your foot — the part of your foot just below your big toe — you'll have better feel for the pressure being applied. And guess what? That means that no more than half your foot is on the brake pedal, leaving the right side of it to blip the throttle during downshifts. This can take a while — even a long time — to get comfortable with, as it can feel as though your foot might slip off the pedal. But if you practice it enough, you won't have to worry about that.
- Brake release either too quick, too slow, or jerky. Okay, this is the biggie. And you knew that I was going to remind you again of the importance of managing your brake release. Sometimes it needs to be relatively quick, but still controlled

and smooth. Other times it needs to be done slowly, hanging on the pedal well into a corner. It's not until you deliberately manage the brake release that you'll be a great, fast driver.

- Not enough leg, calf, ankle, foot strength. The more strength you have, the better your fine muscle control will be. Lift weights for a long time, really stressing your arm muscles, and then try to draw something that requires a lot of detail. I bet your hand will struggle to control the pencil the way it would if it wasn't tired. That same thing happens with your legs and feet — you lose control when you're tired.

Now, a couple of things that are car-specific. In other words, things that you can do to your car that will improve your footwork:

- When braking, the height of the brake pedal should be aligned with the gas pedal, allowing proper heel & toe downshifting. While less important if you're left-foot-braking, if the pedals are aligned, your body is better balanced, and will more effectively sense the limits of the car.
- Adding a heel rest. A heel rest is a small piece of metal, plastic, carbon, or whatever along the floor, just in front of the pedals. The purpose is to give your heel(s) something to brace against, allowing you to pivot your foot at the ankle.

A funny thing happens when I talk to drivers about these seemingly-basic footwork issues: most drivers claim they do all of them correctly, and yet, when I watch in-car video or ride with them, I can see that they don't. Hey, the results of poor footwork even show up in the squiggly lines of data. There's no escaping what happens if you don't use your feet properly.

Use the pedals with the finesse of a dancer, and the controlled power of a high-jumper. It will pay big rewards.

Day #219: Strengthen your leg/foot to improve your braking.

Calf and ankle strength equals foot/brake control, which equals better modulation, which equals faster corner entry, mid-corner, and exit speed.

I've spent a lot of time looking at data traces of different drivers' brake pressure, and specifically how they release the brakes. There is a big difference between the best and the rest. In other words, there are some big lessons to learn from how "the best" release the brake pedal — and this shows up in their data.

But you know where it's even easier to see this difference? Video. Specifically, video pointed directly at drivers' feet. I strongly suggest you set up a camera to record your foot movement, and then sit down and watch with a critical eye.

- How quick is the transition from the throttle to the brake pedal (assuming you're right-foot braking)?
- Is the initial application of the brake pedal quick and hard enough?
- Is the initial application of the brake pedal smooth enough (which may seem to counter the previous point, but not necessarily so)?
- How consistent is the pressure on the brake as you downshift (especially if you're heel & toeing)?
- What does your brake release look like — smooth and progressive?
- Are there any jerky movements in how you apply and release the brakes?

Much of the control you need with feet on the pedals comes from the smaller muscles in your feet, ankles, and calves. It's definitely not your bigger leg muscles that control the fine movements of the pedals. At least, it shouldn't be.

If the muscles in your feet, ankles, and calves are not adequately strong — both in outright strength, as well as endurance — you won't have the control it takes to be a really fast driver.

Fortunately, exercises to strengthen your feet, ankles, and calves are relatively easy to do and don't require a lot of fancy equipment. Calf raises, resistance training against a rubber band, balancing on a balance board (one of my favorites), and other exercises that stress those muscles should be a major focus for you. You might be surprised at how big a difference a little extra foot and calf strength/endurance will make to your lap times.

Day #220: Never lose sight of the fun.

Recently, I read another book about writing. The message the author kept pounding home throughout it was simple: focus on writing quality content, and the opportunities to be published will happen. It seems that most writers are like most drivers. They focus on the result, and too often forget why they started: for the fun of it, because it's their passion.

Focusing on the why, and not the what (the results) can be difficult — especially for competitive people who like to measure their accomplishments. But as I've written many, many times, it's when you focus on your own performance that you're more likely to perform at your best, and therefore get the result you're after; when you focus on the result, you often feel pressured and tense, and don't perform at your best — you don't get the result you want.

When I've focused on the fun of driving, and why I love it, I've driven better. Have you experienced that?

While driving has always been the best thing I've ever done, some of what's happened outside of the car — but still part of our sport — have been almost as enjoyable. And even, in some ways, more so. Or, should I say, different fun!

I hope you don't mind if I share a few personal experiences that come to mind when I think of the fun I've had in our sport, but outside of the car. Things like...

- Meeting and having lunch with Denny Hulme, 1967 World Champion and 2-time Can-Am Champion, while visiting the BMW Driver Training facility in Australia.
- Meeting and having dinner with Jim Hall, the legendary Chaparral car designer and builder.
- Meeting and having dinner with the organizers of many of the car club seminars and talks I've done over the past decade or two.
- Meeting and having lunch at the track with individual drivers of all levels, who are there for the same reason I am.
- Meeting, coaching, becoming friends with, and having lunch/coffee/dinner with Garth Stein, author of the best-seller, *The Art of Racing in the Rain*.
- Meeting and having dinner with people in Australia, South Korea, China, Sweden, Estonia, England, France, Belgium, Germany, and Italy (not to mention all over North America).

Hey, I think I see a pattern here! In fact, I could go on and on, and the message would be much the same. No, not the eating part. The meeting and getting to know people who share the same passion.

Yes, performance and race driving is a passion that everyone reading this has in common. But some of the things that happen outside of the car can be just as much as fun. And sometimes, when you're not focused on just the racing results, other cool things can happen.

Day #221: Late apexes can be dangerous!

I attend many car club driving events every year, as well as other high performance driver education events. I feel very fortunate to see for myself what the best do, and to learn from them. It's also beneficial for me to see what some can do better, and help share some best practices with them.

One topic that gets discussed in great detail at these events is the lines through the corners, and specifically where the apex of each corner is. We all know that the apex is important, as it dictates the speed a car is capable of carrying through the corner (its location in relationship to the overall length and shape of the corner determines the radius of the corner, and therefore the maximum speed a car can carry through it, given a set amount of traction), and how early the driver can begin accelerating out of the corner.

“Friends don’t let friends early apex” is a fun saying, with the intent of communicating that apexing a corner too early is a bad thing. It can be. But not always. In fact, there are times when late apexing is a bad thing — even dangerous. That’s something I never really thought enough about years ago, but have been considering more recently.

At HPDE, track day events, and club racing, it is pounded into drivers’ heads that apexing too early is dangerous. The thinking is that turning in too early results in apexing too early, which results in exiting the track too early! In other words, turn in early, apex early, and you’ll run out of track width at the exit of the corner and drive off the track. Again, that’s true — much of the time. But not all of the time.

I've witnessed many drivers being taught to turn in and apex most every corner late, and often I've seen this advice actually make things worse (and more dangerous). How could this be?

How much do you have to turn the steering wheel when you turn in late for a corner? More than you do when you turn in earlier, right? And turning the steering wheel is a bad thing to do to a car. The more you turn the steering wheel, the more likely it is that you'll lose grip and possibly control of your car.

Where this is especially "dangerous" is in fast corners. The thinking is that late apexing is safe, so the driver turns in late. To get the car to change direction enough and head towards the late apex, the driver needs to turn the wheel a lot, and at high speed this can lead to far too great a change in direction. In this case, a slightly earlier turn-in would be a more gentle arc into the corner, putting less strain on the traction capabilities of the tires.

If you're in a position of making decisions regarding the "recommended line" that is being taught to new or less experienced drivers, I urge you to revisit your current thinking. You may be perfectly correct with the line you're teaching. But maybe you're

not. Maybe, just maybe, you're actually contributing to making drivers less safe by recommending what, on the surface, seems to be safer.

As a driver, look at balancing the benefits of a late apex with one that is slightly earlier. Late apexes help you get back to full throttle sooner, and earlier apexes mean less turning of the steering wheel. Both have their advantages, as well as disadvantages.

Now, I need to be very clear with one thing: when I say earlier apex, I don't mean too early! Too early an apex, as I said earlier, leads to an early exit off the outside of the track. But too late an apex can lead to more spins, and often exiting the track back across the inside of the track. Once again, it's a balance.

Day #222: Change your techniques and beliefs in small steps.

Let's say you want to change something specific about the way you drive, such as:

- Learn to heel & toe downshift
- Take a certain corner without lifting completely off the throttle
- Learn to left-foot-brake
- Begin to brake for a corner fifty meters later
- Learn to trail brake

Which comes first, the ability to do what you want, or the belief that you can do it? If you don't believe you can do it, will you ever be able to fully develop the ability to do it?

Of course, once you've done something, it's easy to believe that you can do it. But if you've never done it, it's not easy to believe you can do it. In fact, what may be holding you back from doing something has little to do with having the skill to do it, but more likely it's not believing you can do it.

Let's take the example of lifting off the throttle less for a very fast corner (such as Turn 8 at Thunderhill, the Kink at Road America, Turn 4 at Daytona, Turn 4 at Mosport). Do you have the physical skill for simply "breathing the throttle" (just easing up on it, rather than lifting off it completely)? I bet you do. The actual technique of breathing the throttle is not that difficult. Having said that, if all you've ever done with the throttle is lifted off it completely, and never eased up part way on it, then you may have to actually develop that skill. That's not difficult, but when combined with a belief that you can't do that, it's highly unlikely that you will do it.

If you currently lift completely off the throttle for one of these fast corners, and I come along and tell you to hold the throttle flat to the floor, will you do that? I doubt it. Why? Because it's unlikely that you believe it can be done. If I then say, "Okay, then just breathe the throttle ten percent — keep the throttle at ninety percent," will you do that? Again, it's doubtful. Why? Because that's too big a gap between what you currently believe — and what you're doing — and what I'm asking you to believe is possible.

There is all sorts of research that backs up what I'm saying. Studies have shown that most people can believe a two percent change, but not more than five percent. Yeah, that's a research study, and they weren't working with performance drivers who can take on more than anyone else, right? But still....

If you moved to a country with a very different culture, let’s say Iraq, would you immediately change your beliefs? No. For most people, adjusting and adapting to Iraqi beliefs would be nearly impossible, as there’s just too big a gap between what you believe in right now, and what Iraqis believe. But, if over time your beliefs were slowly changed, step by step, you would change.

So, when you’re trying to change something, when you’re changing your beliefs, take small steps.

If you're going to decrease the amount you lift off the throttle for a fast corner, don't try to take it at full throttle! Instead, ask yourself how much you're lifting now, and then begin easing up on the throttle just a little bit less each lap — say, five percent less each time through the corner.

Of course, this approach has one other huge benefit: If you make a big change that results in going over the limit, you may be too far beyond it to bring the car back in one piece! But if you make small incremental changes, even if you go over the limit, it's likely that you'll make a correction and carry on. And that's a comforting thought to put in the back of your mind. You'll be more relaxed, and therefore, more likely to do what you want.

Day #223: Be productively paranoid - there's always more.

What do Garrison Keillor, David Dunning, and Justin Kruger have to do with your driving? They've helped us understand why we are so terrible at accurately assessing our own driving abilities.

Garrison Keillor created a fictional town, Lake Wobegon, for The Prairie Home Companion radio show, and it's here where "all the children are above-average." The Lake Wobegon Effect describes the surveys and experiments that indicate that a majority of people believe they are "above-average." It's an entertaining way of looking at the science-based Dunning-Kruger Effect.

The social psychologists, Dunning and Kruger, found that incompetent people are not only poor performers, but they were likely to inaccurately assess their abilities.

As Dunning wrote in a 2017 article for *Pacific Standard*, "In many cases, incompetence does not leave people disoriented, perplexed, or cautious. Instead, the incompetent are often blessed with an inappropriate confidence, buoyed by something that feels to them like knowledge."

"For poor performers to recognize their ineptitude would require them to possess the very expertise they lack," Dunning says. "To know how skilled or unskilled you are at using the rules of grammar, for instance, you must have a good working knowledge of those rules, an impossibility among the incompetent. Poor performers — and we are all poor performers at some things — fail to see the flaws in their thinking or the answers they lack."

Early in our development as drivers, we don't have the necessary skills to even be aware of what we don't know, and therefore, we tend to over-estimate our abilities. Just take a short drive around the block and you'll see countless drivers with under-developed abilities. But I'm sure if you asked them, they'd tell you they were above-average drivers. They don't have the ability to accurately assess their abilities!

What to do about this trait that we all have? As Dunning suggests, perhaps we should be our own devil's advocate: "to think through how your favored conclusions might be misguided; to ask yourself how you might be wrong, or how things might turn out differently from what you expect."

Okay, got it. But, rather than getting dragged down by the negativity of this very human trait, let's look at this effect from the other side. While it's common to focus on what the research shows about incompetent or inexperienced performers, the focus here is less on what this means for experienced, high-performers.

If you work on improving your driving — and you wouldn't be reading this if you didn't — there will come a time when your abilities reach a point where you can accurately assess them enough to wonder why you're not better — your skills are so well-tuned

that you can easily see what you're not so good at! It's easy at this stage to question yourself, lose confidence, and perhaps even regress. I've talked with many drivers who felt that their driving was getting worse, and yet all the evidence pointed to it being better.

The big take-away here is that we suck at assessing our own abilities. Obviously, this is why a good coach is invaluable, as they can not only use their experience to compare your abilities, but do so from an impartial perspective.

As we know, a little knowledge can lead to over-confidence. On the flip side, a lot of knowledge and skill can sometimes lead to under-confidence. Never forget that.

The challenge is knowing when you've transitioned from the inexperienced, over-confident, over-assessor to the experienced, under-confident, under-assessor. Perhaps a dose of paranoia, as Intel's Andy Grove talked about, or "productive paranoia," as management guru Jim Collins wrote, is necessary. I'm not talking about paranoia to the point of freezing up. But just enough to keep us on our toes, looking for what's next, questioning ourselves in a productive way. In fact, that's it — productive paranoia. It's just enough to handle the Lake Wobegon/Dunning-Kruger Effect.

No matter whether you're in the over-confident or under-confident stage, there's always more to learn, always ways to improve.

There's always more… especially when you think there isn't — either because you think you know it all, or because all you can see are your faults, and you've lost confidence in your ability to improve.

Day #224: Keep your heel on the floor.

“Bring your foot off the brake pedal gently and progressively. The whole sequence should be one gentle, sensitive, flowing movement. You want the nose of the car to come up gently and undramatically.”

The first time I read the quote above from Jackie Stewart, I wasn't one hundred percent sure exactly how to do what he was suggesting. Over the years, I've certainly embraced Stewart's philosophy, used it, taught it, and I think even expanded on it ("It's the timing and rate of release of the brakes that separates the best from the rest").

I can't say that brake release has been perfected by every driver I see on the track — far from it! But it's part of a driving technique that more and more drivers now understand the importance of, and a smooth brake release is almost a basic skill taught to relatively new performance and race drivers. It's an accepted and critical part of driving.

And yet, I see many drivers not doing it correctly. Often, that's due to some very basic foot placement and movement.

Do you lift the heel of your right foot off the floor of your car while using the pedals? Ever? If so, stop it. If you're unsure, or even if you're sure you don't, point a GoPro at your feet for a track session. You might be surprised at what you see!

Whether you brake with your right or left foot, the heel of your right foot should stay pretty much in the same position: on the floor, at all times.

If you left-foot brake, your right heel should never move from being lined up with the gas pedal. Your heel stays on the floor, and you pivot at the ankle to modulate the throttle. And if you're braking with your left foot, it should be in line with the brake pedal, with the heel never leaving the floor.

Same thing if you right-foot brake, with a minor adjustment: Your heel will likely be resting on the floor somewhere between the gas and brake pedals. That way, you can not only pivot your ankle to work both pedals, but you pivot or swivel it sideways, tilting your knee either out when using the throttle, or inward when using the brake pedal.

Day #225: To improve, challenge your body, challenge your mind.

Here's a challenge that will improve your driving.

Research has proven that our brains are plastic. In other words, they're moldable — no matter what your age is. The study of neuroplasticity has changed the thinking about what we can do with our brains throughout our lives. The theory that we learn up to a certain age, and then brain cells begin dying — as does our ability to learn — has been proven wrong. Yay! In fact, what we know now is that our brains can continue to learn, change, and improve until the day we die.

But here's the thing: It's a case of use it or lose it. The more you challenge your brain, the more it can take on. And one of the best ways to challenge your brain is to learn something completely new and different. Learn a new language, a musical instrument, to paint, juggle, a different sport, or whatever. The more the new thing is different from things you've done in the past, the better. Think of something that is so "not you" and take it on.

If you think you're a terrible artist, take up drawing. If you have no musical talent in your body, learn to play the guitar or piano. If you dislike using a camera, take a photography class.

Challenge your mind. In doing so, your brain will get stronger, develop new neural pathways, think better, and ultimately help you drive better.

If you've never played a racquet sport like squash or racquetball, I highly recommend them. Why? Well, first it's new and different for you. But even if you've played them before, I believe they're two of the very best sports for improving your driving. They require overall fitness, hand-eye coordination, great reflexes, and they especially challenge your visual processing. The ball bounces back off the wall at an infinite number of angles, at great speed, and your eyes need to track it perfectly. That sounds a lot like what you need your eyes to do when track driving.

One of the great things about racquetball is this: almost anyone can hit the ball and make it bounce off a wall — and that's pretty much the definition of the game! When you start playing tennis, you spend more time picking the ball up out of the net (or from over the fence!), and a long rally might be two strokes. With racquetball, you can find yourself in longer rallies the first time you play, which exercises your body and brain — and especially your vision.

I've played a lot of racket sports (tennis, racquetball, squash, badminton and ping pong) in my life, and if I had to pick just one that would help my driving, it would be racquetball. It's easier than any other racket sport to start, but just as difficult to be great at as any of those other sports.

If you're a racket sports player already, find something else that will challenge your body and brain in the off-season. A musical instrument, a language, dance, photography, computer programming, sailing, skiing, writing, etc. It doesn't matter what it is — it just needs to be something very different from what you've done in the past. Challenge your body, challenge your mind.

Day #226: Complacency is slow and dangerous.

Having spent three-and-a-half decades instructing, coaching, and training drivers of all types and levels — from soccer moms to Navy SEALs, from corporate salespeople to elite-level pro racers — I've noticed a few things. And having operated numerous driving schools and programs, I've noticed a few other things.

What are these things? Well, the number one thing is what complacency can do to instructors. It's dangerous. And it leads to drivers not learning as much as they could or should.

When you spend as much time as I have around driving schools/programs and instructors, you develop a gut feel for when things are going well, and when complacency has kicked in.

I see it when I attend some car club driving schools. I notice it when I talk to some instructors. I recognize it in some drivers.

When instructors become complacent, that's when their student makes a mistake and it turns into a "big one."

When drivers become complacent, they stop improving, get frustrated, and make more mistakes — and again, often end up in a "big one."

As you prepare for the next season, give yourself a gut check, whether as an instructor or driver — or both. If you notice yourself becoming just the tiniest bit self-satisfied, dial up your awareness a notch or two. Go back to the basics. Think about what it is that you're not thinking about.

Don't let complacency slow you down, or lead you to a "big one."

Day #227: The less you turn the steering wheel, the faster you will go.

Turning the steering wheel is a terrible thing to do to a car. It slows it down — and who wants to do that?!

The closer you drive the limit, the more you control the direction of the car with the weight transfer, and the less you change direction with the steering wheel. It's at this point when you realize just how much turning the front wheels slows the car down. It becomes a brake!

Go drive a low horsepower go-kart, and pay attention to how much you slow it down by turning the steering wheel. The same thing happens with your car — it's just harder to notice.

Day #228: Make big gains one percent at a time.

Prior to 2010, no British cyclist had ever won the Tour de France. That changed in 2012, and a rider from the UK won again in 2013 — two wins in a row. At the 2012 Olympic Games, 70 percent of the cycling medals were won by British riders.

What happened?

Dave Brailsford, and his "Aggregation of marginal gains" strategy is what happened. Taking over as General Manager of the team in 2010, he put in place his strategy of many small gains adding up to big gains.

They started with the obvious things like equipment, nutrition, and training programs, but then looked at everything from the pillows riders slept on in hotels to the ways they washed their hands to avoid infections. The strategy could be summed up by one phrase: look for a one percent improvement in tiny areas that were overlooked by others.

Most drivers, when looking to gain a second a lap look for it in one place. Have you ever done that?

If you were to look for a one percent improvement in ten areas of your driving, what would the result be? If you were to gain a tenth of a second in each corner, what would that look like?

- If you released the brake pedal a car's length earlier?
- If you trail braked half a car's length longer?
- If you looked five car lengths further ahead?
- If you committed to full throttle a car's length earlier?
- If you began unwinding the steering wheel earlier?
- If you used even more of the track, on the entry, at the apex, and at the exit — increasing the radius of the corner?
- If you were even smoother with your inputs?
- If you attacked the corners even more?
- If you began braking a car's length later?
- If you braked lighter and carried more entry speed?
- If you waited a fraction of a second before applying the throttle, allowing the car to rotate more, and then were able to go to full throttle almost immediately (rather than slowly squeezing it on)?
- If you sacrificed speed in one area of the track that allowed you to carry more speed elsewhere?

- If you trail braked less, and applied some maintenance throttle through the entry to a fast corner?
- If you were able to upshift a fraction of second quicker?
- If you changed the timing of when you downshifted, making each one a little smoother?
- If you listened to the sound of the tires to sense the limits of your car?
- If, instead of lifting for that fast corner, you left-foot-braked against the throttle?

What if? What if you could improve by one percent in ten places on the track?

Day #229: Don't hide from yourself and your weaknesses.

I'm going to try giving you a little nudge in the direction of self-awareness, if even just to crack the door open. And that reminds me of one of favorite song lyrics: Leonard Cohen wrote, "There is a crack, a crack in everything. That's how the light gets in."

The crack could be a tiny flaw in what would be near-perfection of your driving performance, or it could be the crack that allows you to see, admit to, and work on that flaw. Either way, it's a good crack.

In 2020, Lewis Hamilton won his seventh World Championship, and Scott Dixon won his sixth Indy car championship. There is at least one common trait both drivers share: Their desire to constantly improve. Neither Hamilton nor Dixon is satisfied with maintaining their current level of performance.

In an article written by Marshall Pruett for *Racer*, Dixon was quoted as saying: "One thing is the technique (of saving fuel in a race) and, maybe, the (other thing is the) will to learn about it. I love digging deep into situations like that and trying to better yourself or giving yourself more options. You need all the tools, man. There's many different strategies to win a race and you need to try to cover them all."

"…will to learn…" says it all. Oh, and it doesn't matter if you're chasing the F1 or Indy car championship, wanting to take FTD at an autocross event, shave half a second off your best lap time during a track day, or win an indoor karting league race.

Chris Medland reported that Lewis Hamilton had this to say about both racing and life outside of the sport: "This year I have had time to really spend time on learning, on trying to educate myself the best I can. Who would have thought that at 35 years old that I would learn a lot more than perhaps I have in the past 10 years?"

Hamilton went on to say, "This year has been really… it's been such a difficult year for so many people, and there's so many negatives that have happened to people — losing jobs, businesses shutting down and people becoming homeless — and there's so much struggle that's happened. But then I like to try to look at things always as glass half-full, and there's been so many positives that I hope we will be able to take into 2021. And part of that has been this learning process."

The more you learn, the more you'll improve; the more you improve, the more likely it is that you'll achieve the goals and results you want. It may also be true that the more you learn, the more you realize how much you have to learn!

Have you ever wanted to shy away from some type of test, or a situation where you would end up being compared to others? Come on, be honest with yourself, here. Even if you had just a hint of wanting to avoid those situations, it indicates some amount of having a fixed mindset. Don't worry, everyone has one to some extent, in some

situations. Especially if you have a competitive nature (like I do), and you want to prove that you're the best at everything you tackle!

In her book *Mindset*, Stanford Professor of Psychology, Dr. Carol Dweck, defined having a fixed versus growth mindset:

"In a fixed mindset, people believe their basic qualities, like their intelligence or talent, are simply fixed traits. They spend their time documenting their intelligence or talent instead of developing them. They also believe that talent alone creates success—without effort. In a growth mindset, people believe that their most basic abilities can be developed through dedication and hard work—brains and talent are just the starting point. This view creates a love of learning and a resilience that is essential for great accomplishment."

Having a growth mindset requires managing one's ego. Until you can admit that there's something more to learn, something you can do to improve, you'll plateau at your current level. And there's a fine line between self-confidence — inner belief — and ego. In fact, it takes self-confidence to admit that you have weaknesses. But you can only admit to it when you let go of your ego.

Hiding from your weaknesses is likely caused by having a fixed mindset.

With all of this in mind, here are a few questions I'd like you to ask yourself:

1. What driving techniques can I improve?
2. What part of my mental game can I improve?
3. What can I learn that will increase my depth of knowledge and understanding?
4. What areas of my physical fitness can I improve that will help my driving performance?
5. What can I do to improve my inter-personal skills that will lead to having a stronger "team" around me?
6. What tools and/or resources can I use to help me improve my driving performance?
7. What else can I do to improve my driving performance and/or simply enjoy it even more than ever?
8. What's held me back from performing at the level I want to, that I'm now willing to admit to and work on?

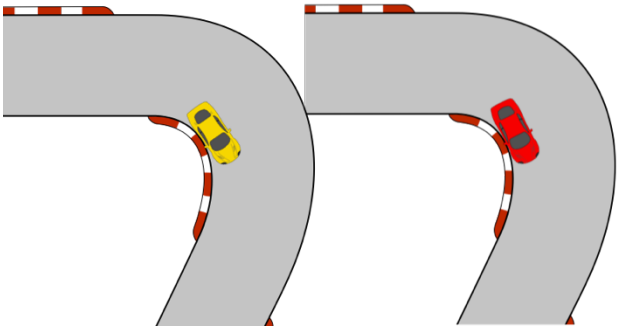
Take time to really think about these questions, and write out your answers; that will help you clarify them. While going through the answers, you may find yourself asking yourself other questions. Great! My challenge to you is to spend at least an hour or more going through this process.

Crack open your mindset, your confidence, and your ego, to let the light in.

Day #230: It's not just where you apex that matters, but also the angle your car passes by the apex.

Walking a race track with a driver is interesting, especially when the discussion turns to the line. Of course, the line one drives, and the line one walks is often different — and it will look different from the cockpit than it does when you're walking upright. But it's valuable to observe the line while walking, and to talk about it.

One thing that often comes up when I walk a track with a driver is the exact location of the apex. Drivers will point out where they're apexing, and I might suggest something just a little different. But the most important discussion I like to have with drivers on track walks is not the exact location of the apex, but the angle, or direction, that the car is aimed in at the apex. That's more important, and something many drivers don't think enough about.



For example, in the illustration the apex is at the second red panel of the inside curbing. If you look closely, you'll notice a subtle difference in the direction the yellow and red cars at heading.

It's important for you to recognize the angle that your car is on while passing the apex. To change the angle, you can turn in earlier or later, you can turn in quicker (more "crisply") or slower, you can use your brake release to change the balance of the car to help it rotate quicker or slower... or some combination of these.

Just because you're "hitting the apex," that doesn't mean you're on the right line, as the angle you pass by it is even more important.

Your apex angle is as important as where the apex is.

Get a group of drivers together at a race track and a topic that is sure to come up is where the apex is for a specific corner. What you don't hear much talk about is the angle that a car should be at when clipping or hugging an apex.

And that brings up another point: you can either clip past an apex, or you can hug or "hold" an apex. Let me explain.

First, clipping past an apex point is fairly obvious. The corner has an apex, and it's really just a point. You clip past it and immediately begin unwinding the steering.

But some apexes are more than just a point — they're an area. And you tend to make the car hug that area. In other words, you get the car in against the apex, and hold it there for some period of time and distance, and then begin to release the steering and

unwind out of the corner.

Finally (and my main focus here), there's the angle you clip past an apex. If the apex is a point, then you can clip past it on this angle… or on that angle (see the illustration above of two cars clipping the exact same apex, but on different angles). The angle your car is pointing when it clips past the apex determines how soon and quickly you can unwind the steering. And when you start unwinding the steering — and how quickly you unwind it — determines when and how hard you can begin accelerating. So, the angle you clip an apex will have a big impact on how quickly you accelerate out of a corner.

The next time you're amongst a group of drivers talking about where the apex is for, let's say, Turn 1, ask them what angle their car is at when clipping past it. My guess is that question will extend the length of that bench racing session by at least another hour....

Day #231: Exercise your vision.

When was the last time you exercised your vision?

Having your eyesight checked is a part of the medical exam that practically all race drivers go through to get their license. It generally involves reading an eye chart on the wall. But how much of race or performance driving is having clear vision of some letters twenty feet in front of you?

In fact, all this eye exam does is measure your central vision acuity. If you hold a quarter at arm's length, that small circle is all that we see with perfect clarity, with everything to the top, bottom and sides gradually blurring into our peripheral. That's why we need to move our eyes and point them directly at something to see it clearly.

I don't need to tell you that driving a car at speed has more to do with your peripheral vision, ability to quickly change focal points, depth perception, and sensitivity to contrast and movement than it does to just your central vision acuity. I refer to all of these as being your "vision."

In my Inner Speed Secrets training, I suggest a few different "exercises" that you can do to improve your vision, with the first one being the Lazy 8 exercise. Here's how it works:

Stand with one arm stretched directly out in front of you, with a slight bend in your elbow and your hand in the thumbs-up position. While keeping your head steady, trace an imaginary figure eight laying on its side (which is where the "lazy" part of the name comes from) with your thumb. Follow your thumb with your eyes, tracking this lazy 8 figure.

Do this exercise for about 20 to 30 seconds with each hand, and then with both hands. When doing it with both hands, make two fists, place the knuckles from each hand together, and make a cross with your two thumbs. Flex your arms and shoulders, and trace the lazy 8 figure, following it with your eyes (again, keeping your head still).

Have someone watch your eyes closely. This someone should notice your eyes moving smoothly, and not in a notchy kind of way, nor should they jump ahead, either. Your eyes should move smoothly, fluidly. If not, then your eye muscles are out of shape, and this exercise — done on a regular basis — will improve your overall vision.

We all know that if we exercise our bodies on a regular basis, it will stay healthy. The same thing is true of our vision. The problem is that most people don't know how to exercise their eyes. Now you know of at least one way — doing the Lazy 8 exercise.

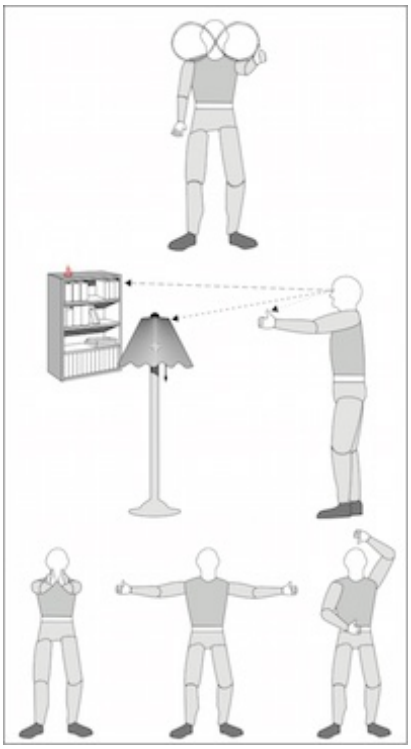
I suggest you do this once a day, and just prior to getting into your car and heading out onto the track. You may be surprised at how much of a difference this makes in your driving.

Improve your visual inputs through regular vision exercises.

Since we talk about fitness for your car, why not fitness for your vision. Is that possible, beyond having your eyes checked by a doctor or optometrist? Yes.

In my *Speed Secrets* books, I've talked about a few exercises that will help improve the quality of information that your eyes take in, and as a reminder, I thought I'd review them here to encourage you to use them to tune up your vision prior to the season.

First, let's look again at (no pun intended) the *Lazy 8* exercise.



Extend your arm directly in front of you, with your hand in the "thumbs up" position. Pretend that there is a figure-8 in front of you, but it's laying on its side (it's "lazy"), and trace that figure with your hand, moving it in a laying-down-8 path. As you do this, and keeping your head still, track your thumbnail with your eyes. To do this you'll need to have your eyes move smooth and fluidly. At least, that's the way they should move. Have someone watch your eyes, and if there is any glitchy movement at all, this exercise will help. Think of it as yoga for your eyes. Do this with your dominant hand first, then your non-dominant hand, and finally with both hands together. Do them slowly and then faster.

Next, the *Focus Stretch* exercise.

The simple way of doing this is to focus on the back of your thumb as you hold it up in front of you again. Then quickly change your focal point to something across the room. Then something in between. The goal is for your eyes to focus completely on something close to you, then far away, something in between, far away, close up... alternating quickly between them and having your eyes be completely and perfectly in focus immediately. If it takes a second or more for you to focus on any one of these objects, then you need to do this exercise more.

Finally, the *Peripheral Stretch* exercise.

Extend both arms directly in front of you with your hands in the "thumbs up" position again. Wiggle your thumbs, and while doing that gradually move your arms out to the sides of your head until you lose view of your thumbs in your peripheral vision. Bring them back, but barely into view, then move them further again, stretching the point where they disappear from view each time. Over time, you'll find that you're able to notice your thumb movement just a little further to the side of your head than you could in the beginning.

Just like any exercise, it's best to do these on a regular basis, like once a day, or every second day. The good news is that if you do all three of these exercises, it'll only take a couple of minutes at best. Yes, you only need to do them for 30-45 seconds each.

Most importantly, the more you do these, the more you'll notice that you see more. And even if you don't really notice it consciously, you will be providing your brain with better quality visual information to work with to make you consistently faster on the track.

Finally, these vision exercises cannot perform miracles and cure blindness! What they can do it keep your vision more "healthy," just like running or cycling improves your cardiovascular health.

Day #232: Go racing.

Some drivers may ignore this piece because they have no interest in ever going racing. They're happy participating in non-competitive track day events. I get it, and that's okay. But if you want to improve your driving, consider a taste of racing, if only once or twice.

I'm often asked how to find that last half second or so. "Go racing," is my usual response.

Setting up passes and being passed forces you to do things you never would if you weren't racing. And that will help you find more speed.

We identify and drive the line, but when there is another car in our way, we have to recalculate and adapt. This is one of the most challenging and enjoyable parts of racing, as it's always changing. Let's face it, if we drove the exact same line over and over and over again, it might even get a bit boring. But when we race, we have to adapt and adjust our line — it's never quite the same, so it never gets boring. Okay, I know driving on track can never get boring, but let's just say "less interesting" than driving a different line.

And the most amazing thing is that in the process of adapting and adjusting our line, we learn. We learn that what we were doing is not always the best thing to be doing. I can't tell you the number of drivers who have told me that they learned a faster way of getting through a corner by having to — by having to adapt because other cars were on the usual line.

Of course, you don't have to go racing to learn what you do when you do. But it does speed up the learning process.

Moving from track days to racing can seem daunting, confusing, and, well… "Where do I go? How do I do this? What if I don't have a car?" Those are just the beginning of the list of questions that'll be going through your head. But, I'll tell you, it's not as difficult as it seems.

First, I highly recommend that you DO NOT build your own car for racing when you first begin. Most drivers, when they do this, make big mistakes — either not following the rules, building a car for a class that doesn't really exist, or spending way more money than necessary. Find a team that you can rent a seat with. There are far more of them than you can imagine, for every driver's budget. Below, I'm listing the websites for each of the budget endurance racing series. Send an email to any of them, asking if they can recommend a team for you to do your first race with, and you'll be pleasantly surprised at the response.

For sure, I'm not listing every option here, but I am sharing the ones that I have experience with.

Budget endurance racing series options (in no particular order, and not complete):

- American Endurance Racing: http://americanenduranceracing.com
- World Racing League: https://www.racewrl.com/
- Lucky Dog Racing League: https://www.racelucky.com
- ChampCar Endurance Series: https://champcar.org
- 24 Hours of Lemons: https://24hoursoflemons.com

Besides the budget endurance racing series, there are also (again, not a complete list):

- SCCA: https://www.scca.com
- NASA: https://drivenasa.com/
- Gridlife: https://www.grid.life
- International Conference of Sports Car Clubs: https://www.icscc.com

All of these series have Facebook pages that you can use to connect with a team that will rent you a seat to get a taste of racing.

Again, you don't have to race to learn to be the best driver you can be, but it will help you get there sooner. And racing is not as "scary" as it might seem if you've not done it yet. There are many ways to get involved, and the actual wheel-to-wheel part can be less daunting than it seems, too. Even if you just do a couple of races to test it out, go for it. You may never do another race again, but at least you'll know whether you want to or not.

Day #233: When passing, always "present" yourself.

"Presenting" your car is not giving your competitor an advantage by letting him know you're making a move on him. What I mean by "presenting" yourself is putting yourself in position where you've taken the line away from him, and it's obvious.

When you don't present yourself, it's not obvious that you've taken the line, that you control the corner.

If you position your car so you're not quite beside your competitor, you're almost asking for him to cut down on you or block you. Why? Because you're either in a position where it's not easy to see you (the other driver may not even know he's chopping you off or blocking), or it's not obvious who has the line.

As this situation is usually taking place in the brake zone, the best solution is to just ease up on the brake pedal a little bit — just enough to allow your car to move a couple of feet further up alongside your competitor's car. That way you've put yourself in position to take the line away from the other driver, and it's obvious. You've "presented" yourself.

Once you've done that, the corner is yours and you dictate the line you and your competitor is going to take. Yes, by easing up on the brakes you might run into the turn a little deeper than you would prefer, but it's okay — you have the line. You've taken control of the corner.

Day #234: Your mind is a key to your speed - how open and receptive it is.

Who can honestly say they don't close their minds to new ideas or things that challenge their long-held beliefs? I wish I could, but I can't. I think I've gotten better with age at being open, but I still have a way to go to become as open as I'd like to be.

I love the Zen philosophy of having a beginner's mind. It's a mental model that I use.

I also love what the British author, Margaret Drabble once said: "When nothing is sure, everything is possible."

So, I ask you to:

- Open your mind to learning more
- Ask yourself whether what you have believed in the past should be questioned or is limiting you
- Consider whether your habits are holding you back, rather than making "everything possible."

Some questions to ask yourself:

- Is the way I'm downshifting, and the timing of it right, or should I re-visit it?
- Do I approach all corners — meaning, the amount of trail braking — the same way, rather than adapting to each corner?
- When I watch in-car video of my driving, do I notice any "bad" habits, even very small ones that I should change?
- When racing, do I have a tendency (a habit) to be too tentative, or too aggressive?
- Are there certain types of corners that I struggle with more than others, and why?
- Do I look far enough ahead (hint: every driver can practice looking farther ahead)?
- Do I over-slow for corners?
- Do I try carrying too much speed into corners, costing me exit speed?

Here's to a new and limitless driving performance.

Day #235: If you follow what everyone else does, you'll finish second at best. And second place is the first loser.

As I walked around the paddock talking to friends, old and new, I noticed a common theme among many of the conversations I heard and overheard.

The most asked question in these conversations was aimed at figuring out what others were doing — with their cars, and with their driving.

"What does ____ look like in Turn 1? Where is he braking?"

"Do you know what springs the 55 car is running?"

"It looks like most guys are running less wing than we are, so we're going go in that direction for the next session. Makes sense, right?"

"How does my line though the Esses look compared to everyone else?"

Let's assume that your car is exactly the same as a competitor's car, and he's the fastest in your class. You "overhear" that he's running 900-pound springs on the front of his car, so you decide to do the same and install 900s on your car. The first problem with this is that unless you change every other variable - rear springs, shock types and settings, anti-roll bars, ride height, toe, camber, castor, aero settings, and who-knows-what-else — going to those front springs isn't the answer. Car setup is a combination of dozens — if not hundreds — of variables, and if just one is not right, it can make the whole package not right.

But I see drivers and crews do this all the time! They glom onto one little isolated piece of information, and make all sorts of assumptions from it. And that sends them in the wrong direction.

If only they had worked on their own direction, their own program, their own performance....

And that's where the biggest problem is with trying to copy what others are doing: Even if you copy every single variable that the fastest car in your class is using, you're likely to be second to it, anyway. Why? Because if that team has been using that setup for a while, they've tuned it, learned it, and they understand it; at best, when you copy it, you're one step behind.

Of course, you can also add in the variable that no two cars are perfectly alike.

Am I saying you can never learn from what others are doing? No, I'm not saying that. But there is a difference between learning from others, and copying others.

Just because one driver brakes at the 200 marker, that doesn't mean you should, too. And it doesn't mean you shouldn't, either. But not because you're copying that brake point. It should be because you learn that someone else can brake at the 200 marker, and makes you think about your entire approach to that corner. The reason the other driver is braking at the 200, and you can't seem to get past the 250, might be because he sucks at the exit of the previous corner and is arriving at that brake zone going 5 MPH slower than you. Or, he might be braking at the 200, but he's slow to get back to power exiting the turn.

Just like the setup of a car is a combination of dozens — if not hundreds — of variables, so, too, is your driving.

Of course, you can also add in the variable that no two drivers, and driving styles, are perfectly alike.

Learning from others is okay, but blindly copying them will, at best, mean that you're just behind them, in second place.

Day #236: Braking — where you start, the pressure you apply, how and where you release them — separates the consistent winners from the rest.

I don't know of a driver today, from novice performance driver to elite-level pro racer, who didn't at one point think that brakes were simply for slowing a car down.

One thing that drivers learn over time (and the very best learn a lot about), is that brakes are used for much more than slowing the car down. Primarily...

- Changing the balance of the car.
- Changing the direction of the car.

As a driver, as you progress and develop your skills, deliberately work on:

- The overall pressure you apply to the brake system.
- How quickly you ramp up to maximum pressure.
- When you begin to release brake pressure.
- How quickly, or the rate, that you release the pressure.

What? Not how late you can brake? Well, if you focus on the above, how late you begin braking will look after itself.

Day #237: Use alternative metrics to measure success.

At the end of a day at the track, how do you measure success? Lap times, result of a qualifying session or race? How many friends you spent time with? The feeling of improvement?

I'd like to challenge the way you currently measure success. Yes, the way you're doing that now may be perfectly fine, and it may work well for you. But perhaps another way of looking at success may bring even more enjoyment to you. That can't be a bad thing!

Let's start with the obvious: lap times. The great thing about them is that they're objective, and easily measured. There's no denying a lap time. It is what it is. The downside is that conditions can affect them (more than we sometimes are willing to admit), and in either direction. I've been fooled into believing that I'd made a big improvement in either my driving, or my car's performance, only to find out that every other driver had done even better than I had. The weather and track conditions had improved so much that lap times were more than a second faster. I've also experienced the opposite, and that's a super-frustrating thing as I couldn't help but wonder if it was something I did.

The fact that lap times are easily measured is also a downside, as we can get overly focused on them.

Lap times can also fool us into believing we've performed well when we haven't, or vice versa. Have you ever turned a fast lap, while knowing deep down inside that you didn't actually drive all that well (weather/track conditions probably played a role)?

I can't tell you the number of times I've asked a driver to tell me about an instance when they performed at their very best, only to hear a story about a victory or new lap record. When I follow up with a question about whether it was truly a great performance — the best they've ever driven — the story usually changes to another time when the result (lap time or position) wasn't as good, but the driver had a huge smile on their face as they went into the details of how great it felt to drive the way they did that day. Which, then, is success?

In most cases, the thing that keeps drivers coming back to the track over and over again is improvement. If you never improved your driving, would you keep coming back? Perhaps for the social aspect of the sport, but most drivers would find something else to do.

So, how do we measure improvement? Again, we usually go back to lap times and/or results. But, depending on what type of motorsport you participate in, there are other metrics you could use to measure improvement. Let's look at a few ideas.

1. Heart rate. Let's say your all-time best lap time is 1:30.00 at your favorite track. You could say that until you got into the 29s, you haven't improved. But one simple metric

that could provide another perspective is heart rate. Let's say your average heart rate when turning 1:30s is currently 120 BPM. Then, being able to turn that exact same lap time with a heart rate of 110 BPM means you're doing it with less effort. In my experience, that also means you have more mental capacity to put towards being aware of other cars around you, of how your car is handling, or how you could change a driving technique to go even faster. So, perhaps instead of looking to turn a faster lap time, focus on doing the same lap time with less effort. And heart rate is one way of indicating that. It's very easy these days to track not just your lap times, but your heart rate at the same time. In doing so, you may even notice some trends, such as your best lap times line up with a certain heart rate, and either above or lower than that heart rate range, your lap times suffer. There's a big lesson in that.

A while after I learned how to juggle, I got to the point where I felt like I'd plateaued. I wasn't improving. Rather than doing the equivalent of trying to turn faster lap times, I focused on juggling with less effort. I would toss and catch the balls, but deliberately focus on slowing my breathing, or think about something other than juggling (I'll let you guess what I thought about, but give you the hint that it's something I do with a steering wheel in my hands!). I got to the point where I could juggle the same number of balls, at the same speed, for the same amount of time as before, but with less effort. And you know what happened after that: when I went back to focusing on juggling, I had improved — I could speed up, slow down, alter the height I was tossing the balls. I had become a better juggler by doing so with less effort.

2. Less wear on tires and brakes. If you could turn the same lap times, but with less wear on your tires and/or brakes, that could be an indication of improvement in driving. Sure, measuring tire and brake pad wear is not quite as easy as heart rate, but if you measured brake pad thickness (with calipers, to be accurate) before and after every event, you'd learn something. If you turned the same lap times, but with less wear on your car, would that be a measurement of improvement, of success?

3. Smoothness. If you have a data system, you can measure how smooth you drive. Again, if you could turn the same lap time times, but do them while driving smoother, that's another indication of improvement. I'm not going to go into the details of how to set up your data analysis software to do this, but a math channel that reports "total steering angle" (the total number of degrees of rotation of the steering wheel) of each lap is a good indication of smoothness. Looking at gSum and using the amount of g-loads generated, can indicate smoothness.

So if you keep track of your average heart rate, the wear on brakes and tires, your smoothness, lap times, and weather/track conditions, I know you're going to learn something that'll either improve your driving, and/or result in more enjoyment. And yes, you could even make note of how much socializing you did, how many things you learned (how to use data/video, a new approach to Turn 3, what to do to your car improve its understeer, etc.).

Finally, I'll leave you with this thought: There is a big difference between extrinsic and intrinsic motivation. The motivation is what brings us back to the sport; extrinsic is an external thing, while intrinsic is what's inside us. Lap times are mostly extrinsic, and intrinsic is what we take home with us — that good feeling inside. We're all different, but at the end of the day, most of what drives us is how we feel. Focusing on our own performance, and finding other ways to motivate us to continually want to improve leads to more of those Aha! moments that we love so much. Think about how you can measure your definition of success at the track.

Day #238: Make a decision. Then make that decision the right one.

In any form or level of motorsport, there is no shortage of decisions one needs to make. These range from:

- What type of car to drive or race
- Whether to race wheel-to-wheel or do non-competitive track days
- Whether to stick the car down the inside and make a pass or wait
- Should limited resources be spent on making the car or the driver faster
- Slicks or rain tires, considering the incoming weather
- Stiffen the front springs or soften the rear springs
- Prepare the car myself, or have someone else do it
- Join the social event at the end of the day or spend that time looking over data and video
- Try to catch this slide or let it spin

I could go on and on, right?

My philosophy on decision making is this: Make the decision, and then make it right.

The best racers follow this approach, as do the most successful non-racers I've seen.

Most passes on a track that end up with car-to-car contact were due to a driver hesitating, and not committing to make the decision to pass the right one. The driver attempting the pass looks to the inside of the other car, decides to out-brake it, drives it down the inside, realizes that he's not quite going to be up beside the other car, brakes harder to back out of the pass, locks up the brakes, and drives into the side of the other car. Or, because he knows he's not going to be all the way up beside the other car, he moves way over to the inside — in a position where the other driver can't see him, and so far away that when the inevitable contact happens, there's momentum behind the cars and the crash is a big one.

There's power in commitment, especially when it comes to decisions, especially when it comes to passing another car on a track.

When you make the commitment to making a pass, for example, your mindset changes. And when you have the "I'm going to pass" mindset, things happen. Things like making decisions happen. Things like making a pass stick. When you don't have that mindset, you're more likely to hesitate, second guess, and end up with a result you don't want.

Just to be clear here, I'm mostly talking about competitive driving situations. If you're in a non-competitive track driving event, the decision-making philosophy is a little different: If in doubt, don't. But actually, that's not so different, after all. You're still making a

decision — in this case, if there's any doubt, don't do it — and you still need to commit to it.

Practice your driving decisions in other areas of your life. Yes, there I go again... I look at everything as being in preparation for driving on the track. What did Steve McQueen say? "Racing is life. Everything in between is just waiting." I'd argue that everything in between is preparing for racing (or non-competitive performance driving).

So practice making a decision, and then doing whatever it takes to make that decision work, whatever you need to do to make it the right decision.

Okay, you will make the wrong decision at times. And yes, sometimes being bull-headed and refusing to change a decision is the wrong thing to do. But even a wrong decision that you've committed to will work out better than a right decision half-committed to. For sure this applies to split second decisions.

When you think about it, there are two types of decisions we make in our lives:

1. Ones we have lots of time to think about and to analyze (should I accept this job offer, should we purchase a new home, should I invest in that stock, etc.).
2. Ones that are made in a split second, like the ones we make on the track.

It's this second type of decision that I'm focused on here. I think. I mean, yes, that's the type of decision I'm talking about. Oh wait, maybe not. Maybe I'm talking about the other type. No, the split-second ones. Or the ones we have time to think about. Hmmm... I can't decide....

Day #239: How and when you release the brakes will have more effect on your lap times than where you start braking.

I'm not saying that late braking isn't a good thing. I'm not saying that by braking later, you can't improve your lap time. What I am saying is that it's an over-used method of trying to go faster.

Ask a driver how he is going to improve his lap time, and the most often given answer is "brake later for…"

If you're braking very early, then braking later is going to help, for sure. But if you're close to braking as late as anyone else, then my suggestion is to focus more on how you release the brake pedal… when you start releasing it, and how quickly you release it. When you do that, you'll find that you're able to carry more speed through the turn, while still be able to begin accelerating as early as before (and maybe even earlier because you've been able to rotate the car early in the turn, meaning you've got a better line exiting the corner).

Day #240: Understand why something does what it does.

I've spent most of the past fifty years driving, teaching driving, writing about driving, talking about driving, and thinking about driving. Yeah, I'm pretty singular in my focus. So when more than a few drivers talk about the same thing that doesn't add up for me, I'm confused.

Through the years, I've heard drivers talk about initiating oversteer or rotation entering a corner by popping their foot off the brake pedal. But that doesn't make sense from a vehicle dynamics perspective. Think about it. What happens when you pop your foot off the brake pedal? Right, the front tires become unloaded, lessening their traction in relation to the rear tires. And that results in understeer, not oversteer.

Have you experienced this?

And how does that fit with the description of getting the car to rotate by popping your foot off the brakes?

Well, here's one explanation.

If you enter a corner while braking heavily, you're asking too much of the front tires, and they'll lose grip, resulting in understeer. If, right at that moment, you pop your foot off the brakes the front tires can quickly regain grip, causing the car to change directions — to rotate.

So, popping your foot off the brake pedal can result in oversteer, despite the fact that you're actually transferring load to the rear tires. It's a sudden, short-term influence. And it's not a technique that I'd recommend. Why?

Unless there is no other way to get your car to turn the way you want (and there almost always is), popping your foot off the brakes to rotate it ultimately leads to having less overall grip. And that means that, even though you're getting the car to go where you want, you're doing it at a lower speed than if you used the best technique.

Instead, if you started to release the brakes a little sooner, but slower, you'll now enter the corner with the front tires not overworked, but loaded just enough to get the car rotated. And that will mean that you will carry more entry speed, and still get the car rotated so you can clip past the apex and get back to full throttle as soon, or sooner, than ever. In other words, it's a faster way to drive.

If you use the "pop the foot off the brakes to get the car to change direction" technique, I highly recommend changing how you manage the timing and rate of release of the brakes entering corners.

Finally, if you have something that seems to work, but it goes against what is generally thought to be the best way, ask yourself why it might seem to work. If you need help figuring it out, let me know.

Day #241: Let go of results. Relax and just drive.

A friend once said that he'd heard that a race driver loses a tenth of a second of lap time for each child he has. I said that I'd seen that happen, but also that I'd seen the opposite. Will Buxton wrote about this years ago when looking at Nico Rosberg's performance.

Buxton wrote: "Becoming a parent can do one of two things to a racing driver. For some, the knowledge that there is a small person on this earth that you have created and who needs you in his or her life, will cost them a tenth of a second. It will make them lift where they would have stayed flat, duck out of a move they'd never have flinched at previously. It dulls the sharpness that made them such a potent force."

“But for others, having a child reminds them that there is more in this world and more to this world than racing cars and winning races. Far from slowing them down, that awakening to a world far bigger and far more important than everything they've known since they were six years old actually has the opposite effect. It lifts the self-imposed burden and pressure. It frees them to do their very best, and if it doesn't work out... well, who really cares? Because there's a wonderful little human at home who is going to think they're the most amazing person in the world, whether they're a world champion or a refuse collector."

The reason I bring this up is not to debate whether children make a racer faster or slower. What comes out of this discussion, though, is the fact that when one relaxes, performance improves.

It's easy to get caught up in the pressure of "having" to turn a lap time, or get a certain result. But as Rosberg's performance has shown, it's definitely not the best way to get what you want.

You're not negotiating world peace. You're negotiating a track!

Day #242: Be crisp with your throttle release.

Picture this: You're driving your street car towards a traffic light, and the light turns red. Do you quickly and immediately go from full throttle to the brake? I don't imagine that you are coming up to the traffic light at full throttle; some of you do, maybe, but not many of you. But you will be coming up to it at part throttle.

Here's how I picture the situation: You're approaching the traffic light at part throttle, you see the light turn red, you gently ease your foot off the throttle, coast for a second or so, and then apply the brakes.

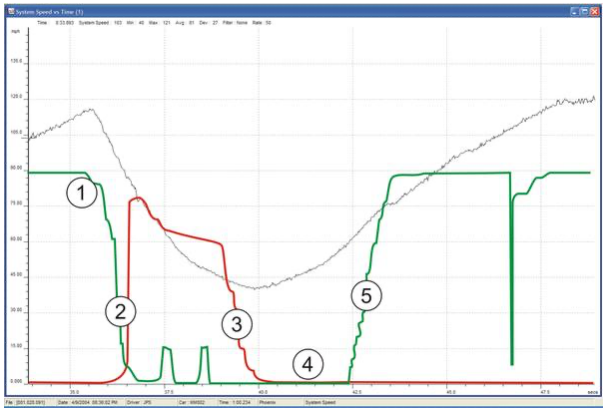
And, doing this, you're building a habit. A habit that creeps into your track driving.

Specifically, the habit is to gradually ease your foot off the throttle, and slowly move it to the brake pedal.

Now, picture this scenario: You're driving down the long straightaway at your favorite race track, approaching the brake zone. Shortly before you begin braking, your habit kicks in: you slowly ease your foot off the throttle, hesitate for a fraction of a second, and then apply the brakes (see the data trace below, noting the tapering off of the green line at the #1). That's a bad habit to use on the track. A no-no.

So, picture this scenario: You're approaching the same brake zone at the end of that same straightaway. You keep your foot flat to the floor on the throttle until the very last second, and then you quickly and crisply release the throttle and immediately apply the brakes.

That's the habit, or mental program, I want you to build.



Working with drivers of all levels, from beginners to experienced racers, I see proof of the "no-no habit" over and over again. How do I know? I hear it, I see it while riding in the passenger seat, or I see it on the data (like at #1 in the throttle trace below — the trace should be flat across, and then directly down).

I'm not suggesting that you need to approach every traffic light at full throttle! But you can build the habit of making a crisp release of the throttle without being at full throttle. And then when you get on track, pay attention to how you release the throttle and go to the brakes. You may be surprised at just how important this is to your overall lap times.

Day #243: Fine-tune your Michael Schumacher.

In the book, *Michael Schumacher — Formula for Success*, Schumacher writes, “You have to have the senses in your whole body that come up to your brain.... then transfer the information to the steering wheel. It’s how you do this that is the difference between drivers, between those who are sensitive and those who may be as sensitive but are not able to translate that into their driving. This is something you have, or you do not.”

I know it’s not nice to contradict one of the greatest race drivers of all time, but… I call B.S. While I agree with what he says about senses in the body communicating with the brain and back to the controls, the ability to sense what the car is doing is not something “you have, or you do not.” It’s possible that some people are born with an ability to sense motion better than others — maybe — but that does not mean that this sense cannot be developed. In fact, I’ve seen enough to know that it’s a skill, like any other, that can be improved.

How?

Ahhh, that’s the tough question, and why many believe this is an innate skill — because it’s not easy to describe how to develop it. And by “it,” I'll suggest that we call this skill “car sensing.”

What’s included in car sensing?

- Longitudinal g-load
- Lateral g-load
- Steering load
- Rotation
- Load transfer (balance)

With g-load sensing, we feel the loads build and build and build… peak… then taper off. Yes, sometimes it feels as though those loads taper off really fast! But they do taper off. And they build relatively slowly, too. Think about turning into a corner. Unless you snap-rotate the steering wheel super-fast into the corner, you can feel the lateral g-loads build until you’re at the peak. The same is true of longitudinal g-loads under braking.

When I talk about “steering load sensing,” I’m referring to the amount of rotational force feedback you get from the steering (remember, the steering is not only an input device, it’s also an output device — we not only turn the steering to make the car change direction, but we get feedback back from it). You know that as you turn into a corner, drive through the middle phase, and then unwind the wheel as you exit the corner, the corresponding load you feel through the steering wheel changes. It also builds, peaks, and then relaxes. But here’s the important question: When the car is at the very limit, is the load feedback heavier than ever, or is it slightly lighter than the peak? You’re right — it changes based on the corner, elevation, or camber changes in the track, and the

tires you're driving on. But there is that one level of load that indicates the car is at its limit.

Imagine what would happen if you focused in a very deliberate way to the feelings of g-loads building, peaking, and reducing while driving on the street; if you focused on the feeling back through the steering wheel; if you could sense the subtle differences in loads when driving on the road, you'd be much more sensitive to them when on the track. Sure, the loads are nowhere near as high, and that's the point. If you can sense them at a lower level, it's going to be much easier to sense the greater loads on the track. Rather than relying on whatever "g-sensing genes" you were born with, you develop them, hone them, fine-tune them. You could do what most drivers do and develop them through experience, more seat time. Of course, that's expensive, even if you do have the opportunity. Or you can practice them deliberately.

Let's look at rotation sensing, now. Imagine sitting in your car right now. You turn into a corner. Your car is now facing a different direction from where it was just before you turned in. If you're traveling slowly, it will be pointing pretty much in the direction you told it to with the steering input, but if there is some amount of slip angle (because you're driving fast), the direction the car is heading may not be the exact direction you pointed it with the steering. Think of looking directly down on your car from above — you see it's pointing at the 2 o'clock position, and yet, you'd turned the steering enough to point it at the 1 o'clock position. It has rotated. Did you feel that? Did your internal gyro sense the rotation?

Sensing car rotation is due primarily to what we see and feel, our visual and kinesthetic/vestibular senses.

Do me a favor. Read the rest of this paragraph, and then do a little test that I'll explain here. Stand next to a table, with your body perfectly perpendicular to its end. Line your body exactly with the center line of the table. Then, close your eyes. With your eyes closed, extend both arms directly out to your sides, parallel with the floor. Now, lean in one direction, and then the other. Then, rotate your body, twisting at the waist in one direction, then the other, keeping your feet in the same place. Do this, from leaning to rotating back and forth for at least 30 seconds, all the while keeping your eyes closed. Stop, and then open your eyes. How are you lined up with the table? Perfectly aligned with the center line, and perpendicular to the end of the table, still?

It's the ability to sense how much the car, and our body inside it, have rotated that is so critical to car sensing. It can be developed through simple exercises or drills like the one I just presented.

Any exercise that stresses your sense of balance, such as using a balance board, skiing, mountain biking, or forms of martial arts will improve your ability to sense the balance of your car. If you've ever wondered why one driver can get into another driver's car (or yours?) and within a few laps turn faster times, often it comes down to

this ability to sense the balance of the car. If, through training, gymnasts and dancers can improve their sense of balance, so can you.

Like building muscle, car sensing is a skill that can be developed. And just like weight lifting to become stronger, it's the process of stressing the muscle (or sense, in this case), providing recovery time (time for reflection on what has been learned), stressing it again, recovery, and so on, that leads to improvement.

If you'd like to drive like Michael Schumacher did, tune up your car sensing abilities in a deliberate way.

Day #244: Winning is often easier than losing.

Long ago, I was at home for a weekend (a rarity for me), rather than at a track somewhere. And that allowed me to watch racing on TV (I mean, what else would I do?!). I watched that year's Canadian Grand Prix, and the Texas Indy Car race. Well, the part of the Indy Car race they ran before it was postponed due to rain — for the second day in a row.

Comparing Nico Rosberg's race to Lewis Hamilton's, one would think that winning took more — more talent, more car performance, more everything. But when you consider that Rosberg had to take to the escape road in turn one on the first lap, dropped all the way back to 10th, dealt with a punctured tire, had alarms showing on his dash (with the new rules in F1, the team was not allowed to tell him what to do about them), and then almost running out of fuel in the last few laps... well, maybe his 5th place finish was just as impressive as Hamilton's win. Or maybe even more impressive.

That's not to say that Hamilton's drive was not impressive. But it was pretty straightforward.

Michael Schumacher's famous drive to second place (the first loser) at the 1994 Spanish Grand Prix is legendary. Why? Because he drove much of the race with his car stuck in 5th gear. By changing his line and driving technique, he was able to match the winner, Damon Hill's lap times. He drove his Benetton F1 car like a Spec Miata, focusing on carrying momentum through the corners!

Watch any race, and it's often the drivers at the back of the pack who are doing much more than the driver who ultimately wins the race. These "losers" are often branded as being less talented, and many claim the only reason they're there is because they bought their ride. But that's not always the case. Sergio Perez's drive to 3rd place at the Monaco GP that year was very impressive, as he managed his tires better than everyone else. And yet, he's been accused of being in F1 only because of the bank account that comes along with him. I don't imagine they're saying that, now!

What's your take-away from what I'm saying here? The best drivers, whether racing at the top levels of motorsport or participating in a track day or HPDE event, adapt and keep looking for more. More from their cars, more from the track, and more from themselves.

It's almost easy when your car is handling perfectly, the track conditions are ideal, and you have the fastest car. But it's what you do on those days when you don't have all those things going for you that matters most.

Day #245: Simulators are another tool available to you to improve your performance.

A driving simulator is a tool — a tool to help you learn and improve your driving. Here are just a few ways that I've used a simulator to prepare myself or other drivers I'm coaching:

- To learn a track. This is an obvious use. But the methods for learning a track may not be obvious — for example, driving the track slowly, just to take in all of the reference points (if you're using iRacing tracks, you know the detail is awesome). Done slowly enough, it's a virtual track walk.
- To learn how to adapt to problems. Prior to endurance races, I've simulated gearbox problems (having to drive without the use of 4th gear, for example). Then, I've used the brakes with very light pressure, as if I were conserving them at the end of a race.
- To get up to speed, quickly. For many drivers, it takes a few laps to get in a rhythm and up to speed. I get on my sim and do just three laps, seeing how close to my ultimate lap time I can get; I walk away from it for fifteen minutes, then come back and do three more laps; I walk away and come back again. I practice being 'fast-fast."
- To stay in the zone. It's one thing to turn a few laps in the zone, and another to do it for a long stint. On a sim, you can practice long stints — longer than you would ever run on the track. That way, a driving stint on the track will seem almost easy.
- To reinforce mental programming. If you use mental imagery (visualization) in a deliberate way (and you should!), you can use a simulator to reinforce what you've programmed in your mind. If you use mental imagery to pre-play how you're going to handle a qualifying session (your mindset, strategy, level of risk, etc.), for example, you can then go and practice for "real" on a simulator.
- To learn how to soak up visual and auditory sensory information.

And that’s just for starters.

Day #246: Learn to adapt your driving to your car's needs.

You're on track and you experience one of the following:

- Corner entry understeer
- Corner entry oversteer
- Mid-corner understeer
- Mid-corner oversteer
- Exit understeer
- Exit oversteer

That's a pretty good list of handling issues you can face with your car. Add in some instability under braking, and we've covered the main issues you'll experience. It would be great if we could always make adjustments to the car to get rid of the problem, but we usually can't while driving. And even after we get back into the pits, many cars (and rule books) have limitations as to what can be changed. So it's important that we learn to adapt our driving to minimize the impacts of these handling problems.

Many — maybe even most — drivers drive their cars the same way, no matter what. If the car understeers, oversteers, or whatever, they still brake, steer, and accelerate the same way they always have.

The best drivers are able to adapt their driving to minimize the negative effects of these handling problems and maximize their speed, despite them. Every one of the handling problems I listed can be improved by making adjustments to your driving.

Ultimately, there are three things you can do to adapt to handling problems:

1. Braking: How hard and the timing of the initial application of the brakes; the timing and rate of their release as you enter a corner (this is the single most important and effective adapting “tool”).
2. Steering: How crisply or quickly you turn in, the line you drive through the turn, and how you unwind the wheel.
3. Throttle: When and at what rate you apply the throttle, and whether you modulate it to change the balance of the car.

That may seem overly simplistic, or a very high-level overview, but that's all there really is that you can do. Ahhh, but it's in the details of how you use these three controls that make the difference, right?

And look at this: With these three areas, you're really just adjusting the line through the turn, and the balance of the car.

Think about what it is you want your car to do and how you can help it with the brakes, steering wheel, and throttle, to adjust the line and change the balance of the car. If you take the time to think through each scenario of each handling problem, the best response is relatively easy to identify.

Of course, figuring out what to do, and actually doing it are two different things - especially at speed! But, again, the more you think through these various handling issues, and how you can help your car do what you want it to do, the more likely you'll do them while on track.

Imagine each one of these handling problems in one single corner that you know well. What would you do with the brakes, steering, and throttle to minimize the negative effects of the handling problem?

Then imagine what you'd do in a different corner, then another, and another. If you do that now, when you get on track, you will be much better at adapting to your car's needs.

And remember, what works in one corner may not work in another.

Also, go on track and practice adapting. If you only ever drive with one technique, you won't be very good at adapting. Learn, through practice, to change:

- When and how you apply the brakes.
- When and how you release the brakes.
- When and how quickly you turn into the corners and then begin unwinding the wheel.
- When and at what rate you begin accelerating.
- Whether you can change your car's line by modulating the throttle.

I challenge you to really think through each of the handling problems I've identified above, then decide how you would adapt your driving to them. Better yet, after you've thought through each one in a logical way, close your eyes, relax, and then do a mental imagery session where you adapt to the new approach. If you do that, I'll guarantee you'll be better at adapting in the future — and that will make you a better all-round driver.

Day #247: Focus on not getting worse.

At the track, do you focus mostly on improving what you're not doing well, or fine-tuning what you're doing well? Is your attention on getting better, or not getting worse?

In striving to be an even better driver, both approaches are valid. But I read something recently that got me thinking more about this. Often, improving is not just about doing more things right. It's about doing fewer things wrong.

The New Yorker had a piece about how Japanese companies grew to dominate many industries in the '70s, stating: "Japanese firms emphasized what came to be known as 'lean production,' relentlessly looking to remove waste of all kinds from the production process, down to redesigning workspaces, so workers didn't have to waste time twisting and turning to reach their tools. The result was that Japanese factories were more efficient and Japanese products were more reliable than American ones. In 1974, service calls for American-made color televisions were five times as common as for Japanese televisions. By 1979, it took American workers three times as long to assemble their sets."

The big takeaway here was that the Japanese companies were more focused on not getting worse, rather than on getting better. If you're anything like me, you immediately think about how to apply this to track driving! In the case of the Japanese TV makers, they didn't look to make better products. Instead, they focused on making the same TV, but with fewer mistakes.

How does this apply to driving, or more specifically, improving your driving? You probably know that I'm a cup-half-full kind of guy, so looking at the negatives is not in my DNA. However, it's possible to focus on a problem without focusing on the negatives. Frequently, I've suggested experimenting with various parts of your driving as a way to find new ways of improving. But perhaps simply doing less of what's wrong would make a bigger difference. For example, instead of focusing on how to drive half a second faster for one lap, maybe focusing on consistently driving at your current pace would provide a better result. If your lap times are moving up and down by more than half a second, perhaps you'd make a bigger improvement just by being more consistent. Focus on reducing the variables, the ups and downs.

What does it take to be more consistent? First, figuring out where the inconsistencies are. This is where data acquisition can be super helpful. Overlap three, four, or five laps and look at where the biggest differences are. Those areas are prime for improvement.

Once you've identified where the biggest inconsistencies are, then you need to figure out why. Your data might help, but if not, look at — and listen to — in-car video. Do you see variances in the line? The timing of when you begin and end braking? Are these caused by not looking far enough ahead, or in the wrong direction? Are your basic skills not well-enough programmed, so you're doing different things each time? Is it because of the way your car handles?

How can you make fewer mistakes, rather than trying to improve your skills? This is like improving by doing less, rather than doing more. For many drivers, this produces the best results. Maybe even for you.

Day #248: References are static. Driving is dynamic.

Drivers are taught to use references, and as I've said, the more you have, the better you'll drive. But reference points are static (as they should be — don't use shadows as references!!), and driving is anything but static. References are suggestions, and you should adapt how you drive, based on what your car is telling you.

If you're able to turn in exactly at your Turn-in reference, lap after lap, consistently perfect each time… you may be slower than you're capable of being. If you're not adapting your line, or adapting what you need to do with the controls to get the car to go where you want, you may be slow. You may be connecting the dots, from reference to reference, but not be as fast as you could be.

Another way to think of references is how my friend Peter Krause treats them, as a place to "index" some kind of action from. As Peter says, "Index is fun because it allows for change and improvement, while still being a concrete reference." Exactly.

When driving on new tires, on a grippy track, do you turn in at the exact same spot as you would when the track grip went away from a super-hot afternoon and your tires are used up? It's unlikely. It's possible that you'll need to initiate your turn-in slightly earlier in less grippy conditions, to factor in the lack of responsiveness.

Collect and file away as many different static references as you can, but be prepared to adapt to the dynamic aspects of driving fast.

Day #249: Draw track maps.

One way to reflect on your driving to identify ways to improve is with a pencil and paper. Hey, before you accuse me of being too old-fashioned, hear me out.

I'll bet you hate to have learn the same thing twice. Or three times. The amount of time we actually spend on track is relatively short, so making the most of it is important. Many years ago, when I was racing Indy cars, I calculated that I spent about 800 hours preparing and working towards driving on track for every hour that I actually spent behind the wheel. Do the math and you may find that your prep-to-drive time ratio is no better, or maybe worse. So, it’s important to make the most of the time you get behind the wheel on track.

Here's your task: Draw all the tracks (or autocross courses) you've driven this past year, from memory. Do not look at a track map, your data, your video, whatever. Start with a blank sheet of paper and just draw the outline/layout of the track.

One of the objectives is to see how good your memory — your mental programming — is, and drawing the track from recall will tell you how well you know it. If you're unable to draw something that resembles the actual layout of the track, guess what? You need to do more mental programming. Watch some video, look at a track map (you can download most for free here Track Maps), then close your eyes and mentally drive the track again and again until you can draw the track layout.

Once you've drawn the outline of the track, start to fill in some detail: where the bumps are, elevation changes, curbs, surface changes, camber/banking.

You may be thinking, "Hey, I'm not good at drawing." It doesn't matter. You're not looking for something to frame and hang on the wall (you can if you want, though!). It doesn't have to be pretty, or even super-accurate, it just needs to represent what you see in your head. And don't say you're unable to do that, because everyone can. If you struggle with this, it's okay. That's actually good news, because it means there's an opportunity for you to improve your driving by doing it.

As Anne Quito wrote in *Quartz*, "Drawing shouldn't be about performance, but about the process. It's not just for the 'artists,' or even the weekend hobbyists." It's the process that counts, that will make you an even better driver.

Take some time over the next week with some blank pieces of paper and a pencil. Then, if you want to get fancy, scan and save them so you can refer to them prior to going to the specific track again in the future. Oh, and won't it be fun to do the same thing again a year from now and see how your track drawing has improved?

Day #250: If you seem to be on a plateau with your driving, good!

After reading *Make It Stick*, a book about how to learn more efficiently by Peter C. Brown (I highly recommend it), I wanted to share this quote with you:

"The process of strengthening these mental representations for long-term memory is called consolidation. New learning is labile: its meaning is not fully formed and therefore is easily altered. In consolidation, the brain reorganizes and stabilizes the memory traces. This may occur over several hours or longer and involves deep processing of the new material, during which scientists believe that the brain replays or rehearses the learning, giving it meaning, filling in blank spots, and making connections to past experiences and to other knowledge already stored in long-term memory."

Sound familiar? Obviously, he's writing about the learning process. It's the process that I refer to as programming of the mind. And it's what you go through with each new technique, skill, track, car, or whatever you learn.

Brown is talking about how it takes time for your mind to take new information, or a skill, and make it part of your mental programming. It explains why drivers plateau at times — like they're not improving. After a period of heavy lifting of learning, the brain needs time to consolidate everything. Learning a new track, a new technique, or changing a technique or skill all require some time for your mind to consolidate it.

Have you had the experience of going to a track that is new to you (or, at least, relatively new), you spend a day getting up to speed, sleep on it, and then come back the next day and immediately go half a second (or more) faster? If not, you will. At some tracks, especially ones with blind corners, this happens a lot. Again, it's part of the consolidation process. Sleeping on it makes a big difference.

Brown also writes, "As we've noted, sleep seems to help memory consolidation, but in any case, consolidation and transition of learning to long-term storage occurs over a period of time."

When a driver tells me they're stuck and doesn't seem to improving, I smile and say, "Great!" After every plateau, after every period of time of consolidation of a new technique, skill or acquisition of information, there is always another burst of learning. That's how humans work.

The only time we don't have another burst of learning after a period of consolidation is when we let our frustrations get in the way, so learn to relax and allow your brain time to let your new techniques, skills, or information sink in.

Having said that, I'm going to contradict myself a little. When I said there is always a burst of learning after a period of consolidation, I'm making one big assumption: that you're able to spend enough time practicing in a focused and strategic manner. If you drive three or four times each year, you can't expect to improve.

Years ago, I coached tennis, and the guideline in that sport is that an accomplished player needed to practice intensely three times per week to maintain their level of playing; it takes four times per week to see improvement once a player is at a high level. Unfortunately, very few performance or race drivers have the opportunity to drive on a track four times per week (wouldn't that be nice!). Of course, we can practice in our minds, using mental imagery (visualization), as many times per week as we want. And we can practice many of the basic skills and techniques while driving on the street (even at legal speeds!).

So, if you're frustrated because you're not improving, but you only drive on track a few times per year, you're being unrealistic. At the same time, if you feel you're not improving, understand that it may be that you're simply giving your mind time to consolidate what you've learned, and there will be another burst of improvement very soon.

Day #251: Try one gear taller.

We all love the sound of a revving engine. It's beautiful, isn't it? Exiting a corner with the engine wailing away, catching that upshift right at the track-out curb with the car on the ragged edge. It's fantastic. Except when it's slower.

Often, we confuse the sound of a revving engine with speed. And because of that, we drive through some corners in a gear lower than we should for outright lap time improvement.

Picture a corner that you take in second gear. Now drive it in third gear. Sure, it sounds slower, and maybe even feels slower. Engine torque is not so obvious, and it can even feel a bit like its bogging down as you go to full throttle. But often — sure, not always — it's faster. Why?

Mostly because the act of downshifting to second gear — or any gear that's going to cause the engine to be in the upper end of its rev range — causes the car to slow. And that often leads to driving the car into the corner at a speed lower than it's capable of. And that leads to the Change-in-Speed Problem.

What's the Change-in-Speed Problem? It's when you slow slightly below what the car is capable of, and then re-accelerate it to get it back up to the limit. For example, let's say your car can carry 50 MPH through the entry of a corner, but you slow it to 48 by downshifting one too many gears, down to second. Your "limit sensing" tells you the car is not quite at the limit, so you try to make up for it by giving it some throttle. As you over-slow and then compensate by re-accelerating, you cause weight transfer to go forward, then backward. And all that load moving around leads to less overall traction and the car feeling like it's beyond the limit — but it's only beyond the artificially low limit that you've caused. Your conclusion is that you're driving the car at the limit, so you don't make any changes to improve things — you keep doing something similar, and maybe even slow the car to 47 just to make sure the car is under control. (Note: I did a short YouTube video a while ago discussing the Change-in-Speed problem here).

If you try running the same corner in third gear instead of second, you don't cause as much weight transfer forward and enter the corner at the real limit of 50 MPH. The car is better balanced and therefore it has more overall grip, so you carry that speed through the corner and begin accelerating from 50 rather than 48. And because the car is better balanced, you feel in control, at the limit.

I encourage you to try driving some laps in one gear higher than you usually do. Don't get frustrated by the lack of acceleration or sound of a revving engine. Instead, do everything you can with your driving to make that gear work for you. Take an entire session and run it all in fourth and fifth gear, never using second or third. Force yourself to figure out how to make the most of it — how to carry momentum through corners.

My guess is that you'll be as surprised as I've been when asking my coaching clients to use a taller gear. You'll see lap times that are closer to your best times than you expected. Then, go back to using the regular gears and see what happens. I suspect that two things will happen: First, you'll turn faster lap times. And second, you'll start using a gear taller than you usually do in a corner or two, discovering that the higher gear is actually faster (and easier, since you'll be making fewer shifts).

Day #252: Drive in the moment.

A good friend shared an experience with me that I think we can all learn from. He had been competing in an endurance race, and with a couple of hours to go, the crew radioed him and said, “Keep it up. We can get a podium at this pace.” A few corners later, as he was hustling the car through the Esses at COTA, “We can get a podium” popped into his head. And he messed up the whole section of corners, almost throwing the car off the track.

The lesson is simple: Stay focused on the act of driving, and the likelihood of getting the result you want goes up. Stay in the moment, focusing on your driving.

Of course, that lesson is easier said than done.

Staying in the moment sounds all Zen-like, but it’s a simple task to do — or not do. In fact, for many people, it’s easier to not stay in the moment. We spend so much of our lives thinking about what happened in the past — what someone said, how that car cut me off on the freeway, where I was a week ago — or what will happen in the future — what am I going to say to my boss, where are we going on vacation this year, how am I going to cover that unexpected bill, and so on. In track mode, those thoughts are typically, “I keep braking early for turn one, I should have topped up the fuel before this session, that jerk keeps blocking me, that lap felt faster than a 1:43.5, it’s going to be cool to be on the podium.”

Using Steve McQueen’s great line in *Le Mans*, “Racing is life. Everything else is just waiting,” as a jumping off point, what if we use life as a tool to improve our on-track performances? If you spend more of your life in the moment, you’ll spend more of your drive time in the moment. Practice being in the moment.

I suspect you’re human, and that means you will pop out of the moment at times. It’s something we all do because we’ve practiced it many times! Not coming out of the moment is not the problem; it’s how quickly you can get back in the moment. Practice getting back in the moment.

You can actually make staying in the moment your goal for a session. Don’t worry about lap times, results, what anyone else thinks. Just spend an entire session focusing on being in the moment. And then, if you begin thinking about something in the past or future, say to yourself, “Moment,” and come back to the act of driving — your brake points; timing and rate of release of the brakes; where and how you turn the steering wheel; where you’re looking; the transition between braking and going back to throttle; how smooth or quickly you begin accelerating; where you’re back at full throttle; your shift points; any reference points; and so on.

I’m not suggesting you should be thinking about every step at a conscious level. Things happen too fast to drive by a checklist. But if you start doing some of that, over time, it reinforces the mental programming to drive in the moment at a subconscious level.

I think we all need to spend more of our lives in the moment (me, included!), and when we do that, we will drive better on the track. And the more time we spend in the moment on the track, the more time we will do that in our lives. One leads to another.

Day #253: Set performance goals, not results goals.

Setting goals is about clarifying them.

The key to setting clear goals is focusing on the "clear" part, and not the "goal". In other words, focusing on the adjective (clear) is more important than the noun (goal).

What is your performance goal for your next event? How would you need to perform to get the result you want?

"I want to finish in the top 10." That's focused on the goal, not on the clear part of setting the clear goal. It's focused on the result, and not the performance that will lead to the result.

If your goal is to finish in the top 10, or win, or get the pole... take time to clearly define what it will take to do that. Get clear on how you need to perform to get that result.

Sit down and think about it. How would you perform? How smooth would you drive? How would you release the brakes, turns the steering wheel, apply the throttle? How assertive would you be in traffic? What line would you drive through corners? Where would you be looking? What sounds would you hear? What g-forces would you feel? Would you feel bumps in the track, and if so, where? How would the tires indicate to you that they're nearing, at, or past their limits? Where would you brake at the very limit, and where would you brake lighter to carry speed? What would the transition be like between when you released the brakes, and began to apply the throttle? What would your state of mind be like? How confident would you feel? How challenged would you feel? What would you be aware of, notice, while driving?

Answer each of these questions, in detail. Write your answers down (Robert Cialdini, author of the Influence and Pre-Suasion books says, "People live up to what they write down"). If you don't write these answers down on paper or screen, you're much less likely to perform the way you want. Your clear goals will only be goals. Writing the answers to each of these answers down, you're turning your goals into clear goals.

Then, take ten to twenty minutes and do mental imagery of all of this. Close your eyes, relax, and then imagine each of your answers. Put yourself in various situations and imagine how you would perform in each.

Do this at least twice per day for at least two weeks.

Remember, wishful thinking is not a strategy. What I've described above is a strategy that will improve your performance, and therefore increase your chances of getting the results you want.

Day #254: Do two-thirds of your braking in the first third of the braking zone.

Corner entry speed is the most challenging part of driving quickly around a race track. But corner entry speed really begins with braking — and ends with the end of braking.

When we talk about corner entry speed, we're talking about:

- When and how we begin braking (Begin-of-Braking, or BoB)
- The pressure we're applying to the brakes throughout the brake zone
- Any downshifting taking place during the brake zone
- When and how we release the brakes — the timing and rate of release of them
- The End-of-Braking, or EoB
- When and how we turn the steering wheel

When "corner entry" ends is not an exact place, but generally I consider the point where you begin to apply the throttle as the next phase of the corner — the mid-corner — so let's define corner entry as everything up to the End-of-Braking point, or shortly thereafter if there is some transition time to the throttle.

When you approach a traffic light or stop sign on the street, it's unlikely that you apply maximum pressure on the brakes immediately. In fact, you're likely to squeeze on some pressure, and then add a bit more towards the end of the brake zone as you approach the stop line, finally releasing pressure at the very end to avoid a jerky stop. In this scenario, you're doing two-thirds of the braking in the LAST third of the brake zone — and you need to flip that around on the track, doing two-thirds of your braking in the FIRST third of the brake zone. That means that your initial application of the brakes needs to be hard enough so that you can release a bit of pressure as you move along the brake zone as the car slows.

And here's the most important part: When you get the majority of your braking done early in the brake zone (two-thirds done in the first third), you then have the ability to fine-tune both your speed and the car's balance when entering the corner. If you haven't slowed the car enough early in the brake zone, you're going to enter the corner with "your hair on fire," in full (or near) panic mode, just trying to survive.

Getting your corner entry speed just right is a matter of three things:

- Sensing speed - and much of that comes from what you hear (wind noise passing your car/helmet, engine note, sound of the brakes, etc.)
- Managing the load on the front of the car, whether you want/need more or less
- Fine-tuning speed to match the grip level

The great Mark Donohue once said: “Exiting a corner on the limit is like tightrope-walking; entering a corner on the limit is like jumping onto a tightrope while blindfolded.”

Once you're in a corner, you know how much grip your car has. But entering the corner, you don't. Well, not completely. But you do, to some extent, because it rarely changes too much from lap to lap (unless it's raining, which is a whole other story for another day). If you're able to sense and adjust your entry speed within a couple of MPH each lap, you can dial it up or down until you hone in on what feels "just right." And this is why it's so critical to get most of your braking done early in the brake zone, because if you're still trying to get the car "Whoa-ed down!" as you're turning into the corner, you're in survival mode, not "fine-tune corner entry speed" mode.

Here are two things I recommend you do, beginning right now:

1. As you're driving on the street, be aware of how the pressure you're applying to the brakes build, maintain, and release. Experiment with this, and even go so far as turning around the usual and making the initial application of the brakes harder and then begin releasing. In other words, do two-thirds of your braking in the first third of the brake zone. Note: Do not do this in rush hour traffic, and be aware of others around you!
2. The next time you're on track, be aware of how hard your initial application of the brakes is. Experiment with a quicker, harder application. Yes, you can still "squeeze" the pedal, but do it quicker and harder. Notice how when you do this you have more time to deal with the corner entry speed — you can fine-tune it by the timing and rate of release of the brakes. And it's even possible that you'll find yourself slowed so much earlier that you'll have to actually ease up on the brakes to get to the corner!

Day #255: Think where you want to go.

How many times have you been told to look where you want to go? Tons, I bet. Good, as it's good advice. But it's incomplete.

Is it possible to looking at something, while thinking about something completely different? Yes!

"Look where you want to go and your hands will follow." The idea is that if you look to the right, you'll turn the steering wheel to the right. But what if you're looking to the right, but thinking about what's to the left? At best, your mind will process the information going in more slowly (rarely a good thing while driving on a track at speed). At worst, you'll do the wrong thing.

Last week, I wrote about practicing "positive target fixation," and I want to expand on that by having you mentally target on what you want.

If you've ever seen two drivers who seem to continually run into each other (as I've seen more than once), it's not just bad luck. Imagine that you're racing wheel-to-wheel with a pack of cars. You and the yellow car bump into each other (let's not worry about whose fault it was, here). A few laps later you're mixed up in the same pack of cars, and that yellow car is right there again! And you bump into each other again.

Now, it's the next race event and there you are mixed up with the same pack of cars (since you're all about the same speed, it makes sense, right?). Your eyes are drawn to the yellow car, and you think, "There he is again! Stay away from him." In that very instant, what is your mind focused on? Yes, the yellow car. And if your mind is focused on something, it's as if that something is a magnet. Once again, you and the yellow car bump into each other.

It's not until you focus your mind on something else that you avoid bumping into the yellow car. Oh, and if the other driver is also thinking about you, you've doubled your chances of bumping into each other.

Practice thinking where you want to go, then look there — your hands will follow. Again, it's the positive target fixation thing I wrote about last week, but taken to the next level.

Day #256: Teach to learn.

It's been said, "Those who can't do, teach." I'm changing that to, "Those who can do, do so because they teach."

The best way to learn something is to teach it to someone else. This applies to driving. And there's research to back this up. It's real science, and not just my opinion.

In two separate experiments, participants were asked to study material in preparation for either taking a test, or for teaching the material to other students to prepare them to take a test. Guess what? The participants who were preparing to teach others out-performed the others. It was the expectation that they would have to teach others that made them learn the material better. By simply expecting to teach something, you learn more.

This is referred to by scientists as "the protégé effect."

When you have to teach something to someone, you think differently. You prepare differently. You typically study the material more and think about how to communicate it to others. In doing so, you're changing the neuropathways in your mind. You're programming yourself to be able to do it better yourself.

The point where you learn to explain something in very simple ways to a student is when you fully understand it. If you've ever experienced a teacher explaining something to you where it was confusing and complex, I suggest that the teacher did not know the material as well as they should have.

As Albert Einstein reportedly said, "Everything should be made as simple as possible, but not simpler." He also said, "If one studies too zealously, one easily loses his pants." To me, he's saying two things:

1. Your job as an instructor is to simplify things as much as possible, but don't try to be overly-clever with how you do that.
2. You must take the time to think about things. If all you do is read and study, but don't set time aside to just think about what it is you've studied — and explain it to someone else — you're wasting time (and maybe your pants?).

There are some of you reading this who are thinking, "There's no way I'm getting in the passenger seat with a driver on a track!" I understand. It's not for everyone. Some people are not able to stomach it — literally. For those of you it is for, it's one of the most rewarding things you can ever do. But even if you're not comfortable instructing from the passenger seat, there are other ways you can "teach to learn."

Most car clubs and HPDE (High Performance Driver Education) event organizers would be happy to have you volunteer as a mentor, where you take time to sit and answer

questions from the newbies coming into the sport, and help them through their progression as drivers.

There is one very strong requirement for this mentor volunteer position: You stick to the party line. In other words, do not contradict what is being taught by the instructors (no matter what your opinion is), whether in the classroom or in-car. Your job is to support, clarify, and emphasize what's being taught by others. That means that you must know exactly what's being taught by the instructors. Sit in on the classroom sessions, make sure you have an outline of what is being taught, and get a copy of the materials. Know it. Talk to the in-car instructors when they come off the track and at the end of the day. Attend their debrief sessions with their students. Know what they're focusing on.

Most important, if you're going to be a mentor, you should definitely participate in any and all instructor training sessions. Just don't plan on being an in-car instructor. But you need to know what they know, even if you're not able or willing to instruct from the passenger seat.

Although I've made it sound like the benefit to teaching is for you, and what you learn from it, ultimately the focus of teaching must be on the student. If the only reason you're instructing is to gain something from it for yourself alone, you'll never be a good instructor. Teaching anything well should be student-focused and student-motivated. But there's no reason you can't focus on helping your student, and have a great learning experience for yourself, as a side benefit.

Thank you for reading this, as it's given me an opportunity to learn more by understanding the topic well enough to explain it to you.

Day #257: Successful race starts are more about your mindset than they are any specific technique.

Your goals for race starts and restarts (in order of priority):

1. Gain positions
2. Maintain your position
3. Minimize any loss of position

Gaining positions is as much about your mindset as any technique. How many different scenarios are there for the start of a race? An infinite number. So, you can't imagine every single one of them - you can't imagine every technique you'll need. But you can imagine (mentally program) being ready, no matter what happens. It's amazing what doing some mental imagery of "I'm ready" and having the mindset of "I'm going forward" does.

This mindset — one of "owning" the positions in front of you, that you're gaining on or pulling away from your competitors — will do more for being ready and taking advantage of what happens at the start.

Much of this mindset comes into play on the pace lap. Other drivers will try to send you a message on pace laps, by taking up your space while warming their tires. They'll try to block you a little, and move you over just a little when you're packed up two-by-two. You need to push back and let them know you're not going to let them dominate you. Better yet, you need to start this game before they do.

I wish I could tell you where to line your car up in relation to other cars, but I can't make such a sweeping rule. For example, should you always be right on the rear of the car in front of you, or hang back and try to get a bit of a run? It all depends... Sometimes yes, sometimes no.

I think the "safest" place to be is closed right up behind the car in front of you. If you hang back, trying to get a run on others, there's a chance you'll miss the timing and actually lose positions. That's why I say it's safer — there's less likelihood that you'll lose anything. I think that being all over the car in front of you sends a message to other drivers, too — "I'm here to do battle, so don't mess with me."

Sure, you can hang back a little, and then just before you think the green flag will come out you can begin to accelerate even a little bit (having a one MPH advantage over others will give you a chance to pass them). But again, if you time it wrong and have to lift off the throttle just as the flag falls, you're likely to lose positions.

Day #258: Speed comes from experimenting, but you have to be consistent before you can experiment. Consistency first, then speed.

There comes a time in every driver's progression where the way to go faster is to experiment. In the beginning, when you first start performance/race driving, your goal is to "follow the book." In other words, it's to do things the way you're taught. It's like an art student studying, and copying, the great painters.

Eventually, though, the art student begins to experiment, developing their own style. It's through this experimentation, with a solid baseline built on studying and mastering the basics — what's taught "in the books" — that leads to greatness.

Experimentation — trying a different line, starting to brake earlier and lighter or later and harder, releasing the brakes earlier or later, turning the steering quicker or more gently, waiting longer but then immediately going to full throttle, or more gently squeezing on the throttle, and so on — can lead to you discovering new and faster ways to get around the track.

Having said that, without a solid foundation — the process of studying and getting good at the basics — one will never be great. One must be able to consistently perform the basics before moving onto experimenting and developing one's own style.

And I want to make this very clear: the advanced stuff, the stuff that every performance/race driver is dying to get to, is simply the basics done well. Again, advanced techniques are simply the basics done well.

That is why being consistent with the basics is critical. If you can't consistently perform the basics, and you begin to experiment, how will you ever know if what you did worked or not, or whether it was just your inconsistency showing up?

Too many drivers are in such a hurry to experiment, and to try to go fast, that they never develop the consistency to know what's working, and what's not. And by the way, this doesn't just apply to inexperienced drivers. I've seen many very experienced drivers who lack consistency because they never built a solid foundation.

To be faster, start off by ensuring your consistency is where it should be before working on going faster. In my experience, when drivers practice being consistent, speed just happens.

Day #259: The best cure for any illness is driving on a track!

I'm going to share a few memories that make me think about focus... and they're not all great ones!

When I was doing a webinar a few years ago, about ten minutes into my presentation, my stomach began cramping up and within a few minutes, I felt like I was going to "empty my stomach." But the show had to go on, despite something that I ate earlier disagreeing with me — in a big way. I had to focus, and ignore the sweat that was beginning to form on my forehead. Swallowing hard, many times, I used my "focus trigger" and focused on my presentation and the questions that participants were posting in the chat area. I'm sure that if I had not been focused, I would have spent at least an hour in the bathroom.

Another time, at Sebring, I had a really bad flu. My sinuses were completely clogged up, I could barely breathe, and I had a fever. I got into the IMSA prototype car I was paid to drive and went out for practice. At the end of that session, I felt at least 50 percent better. Later that day I went out and qualified the car on the front row, and by that evening I was feeling great.

I'm convinced that driving on a track can cure any illness! Someday, when I'm about to die, I plan to immediately get in a car and drive, putting off death for a while!

The key is having something to focus on, instead of NOT focusing on something.

I've written here in the past about having what I call a Pre-Planned Thought (PPT). Instead of finding yourself in a situation where you need to think about and focus on something else (i.e., don't think about being sick) and then spending time trying to figure out what you can focus on rather than on what you don't want to focus on, you use your PPT to trigger focus onto what you want.

In my case, in both situations I used my favorite PPT: "Car dancing." No matter what, if I need to focus on something other than what I don't want to focus on, I just say to myself, "Car dancing." This triggers a Pre-Planned Thought, or mental program for bringing me into a great performance state of mind. Just typing "Car dancing" brings a little smile to my face, and it triggers a great state of mind.

Over many years of deliberate practice, I've developed the ability to focus on what I want, and not on what I don't want. I can use it to immediately and quickly switch my focus, keeping me focused on what I want.

No matter what you're involved in, or how mentally strong you are, life — and especially performance driving — comes with many distractions away from what we want to focus on. It's more critical than ever to be able to trigger the focus that we want instead of what we don't want.

Using a PPT to trigger a performance state of mind is one of the keys to high performance. And for many of us here, simply putting a helmet on, strapping in our cars, and driving on a track triggers that same state. That state is very powerful. It can even help us recover from an illness, as I've proven to myself more than a few times!

Day #260: Synchronize your driving with the data acquisition.

Data acquisition is an extremely powerful self-coaching tool... but only if you synchronize it with your driving.

What do I mean by that?

The first time you come off the track knowing absolutely, 100%-for-sure, no doubt about it, you took that super-fast sweeper with your foot flat to the floor on the throttle, only to have the data acquisition lie... I mean, tell you that you didn't — that you lifted — well, that's what I mean about synching your driving with the data. Make sure that what you feel in the car corresponds with the data.

With time and consistent review of the data after every session on track, your driving will be synched — you'll almost be able to draw the throttle, brake pressure, steering angle, and even speed traces without looking at the data. Eventually your mind is so in tune with what you're doing that the data acquisition is simply backing up what you've already felt and reported. But — and this is critical — data acquisition will give you another dimension of detail and accuracy that you just can't develop on your own.

You know you're really switched on to what you're doing (and what the data will report) when you're able to draw a speed, throttle, braking, and steering trace for a particular corner, then go out and match it.

Day #261: Ask yourself, "Is it the car, or is it me?"

The first thing you need to do whenever you feel your car has a handling problem is to determine whether you're contributing to it, or even causing it, or whether it's the car's setup.

Ask yourself, "Is it the car, or is it me?" Then dig deep to answer that question. Yes, it's easy to blame it on the car — in fact, it's easiest to blame the car. But what if it's not really the car's fault? If you make changes to the car's setup because you put the blame there, you may actually make the problem worse. You need to be honest with yourself. And you really need to analyze your driving to see if it's something you're doing.

I've experienced the entry-understeer/exit-oversteer scenario myself as a driver, and many times with drivers I've coached. And more times than not, it was the driver that was causing the problem, not the car's setup. Okay, the initial entry understeer was the car's setup, but it was the driver (including me) who caused the exit oversteer.

When a car understeers early in a corner, the natural instinct is to turn the steering wheel a little more. With more steering lock, the front tires scrub off speed, and eventually gain grip — and the front of the car responds. And when it responds, it can cause the rear to lose grip, causing oversteer.

What do most drivers want to fix first — the entry understeer or exit oversteer? The exit oversteer, because it can make a driver feel most uncomfortable. So the driver asks for a change to the car's setup to fix the oversteer.

And guess what that does? It makes the understeer even worse, resulting in even worse exit oversteer because of the driver's reaction. So the driver asks for another change to help the rear gain grip. And this scenario continues until it's determined that it's the driver who is causing the exit oversteer.

Can you see why it's so important to determine what you're doing when the car has a handling problem? If you can figure out exactly what you're doing, then you can fix the real problem — and not just what you think the problem is.

Day #262: Watch your language.

"I suck at braking"
"I'm not a good qualifier."
"My tires are trashed."
"If only I had your horsepower."
"I'm just not feeling good today."
"I had a great lap going until I got blocked by slower traffic."

The list of excuses goes on and on... as do the actions that lead to them.

The words we use have a bigger impact on our performance than we're typically aware of. Our language can hurt our performance, make us drive slower, make more mistakes, drive inconsistently, and even crash.

When I ask drivers to tell me about their driving so I can understand how I can help them improve, I request a list of their strengths and weaknesses. Nine times out of ten, I see a common pattern. So let me ask you, "What are your strengths and weaknesses?"

My guess is that your list of things you need to improve would be much longer than your list of things you're good at. I don't know if that's human nature, modesty, a form of reverse-ego, how we've been trained, our culture, or what... but it's consistent with the hundreds (maybe even thousands of drivers) I've asked about this over the past few decades.

Every single time you talk about what you're doing wrong, you've reinforced that action. You've given yourself confirming feedback to do what you don't want to do! This is a well-studied response in the field of behavioral psychology. When you say to yourself, "Darn. I always brake early for Turn 1," you've confirmed that action and increased the chances of doing more of it.

For the next week, I challenge you to really listen to the language you use. Notice how often you use the word "don't." Every time you tell yourself or someone else to not do something, or say, "Don't do...," make note of it. Become aware of every time you make an excuse for some action or behavior. There's no need to be judgmental about it; just become aware of it.

Once you're aware of how often you use this type of language, switch it around. Whenever you're about to tell someone (or yourself), "Don't do that," turn it around and rephrase it as what to do. State what you want to do, not what you don't want to do.

Replace "I suck at braking" with "I'm going to brake one car length later." Instead of "I'm not a good qualifier," say "I'm improving my qualifying."

I like to jokingly say, "Don't use don't." So… don’t!

Day #263: Finish it off by focusing on what matters - a good clean exit.

Give me a little nod of the head if you've had this experience: you're near the end of a lap, and you either sense or notice your predictive lap timer telling you that you're on a great lap… and then you make an error in the last corner or two and ruin the lap. Have you ever done that?

It reminds me of the stories I've heard from golfers who have been playing the best game of their life, with only a couple of holes to play… and the next thing they know they've hit the ball into the woods and double boogied the last few holes, not even matching their best score.

What we have here is a loss of focus on what matters.

What were you focusing on up until the last corner or two when you were on that great lap? I bet it wasn't really on the lap time, but most likely just on driving, and perhaps specific techniques. It was the focus on the act of driving that got you so close to turning a brilliant lap time, and a focus on the brilliant lap time that cost you the brilliant lap time!

"Good clean exit." Repeat after me, "Good clean exit, good clean exit, good clean exit." The instant you begin to think about or notice your predictive timer telling you that you're on a great lap, tell yourself, "Good clean exit." Then focus on what it takes to make that happen in the final section of the track.

The more you practice doing this, the better you'll get at focusing on what matters, and not on what doesn't matter. Build a habit — a mental program — of telling yourself to focus on a "Good clean exit" out of the last corner or two to finish off the lap. Then, when you really are on a brilliant lap, you'll finish it off by focusing on the act of driving, and not the result — the lap time — you think you're going to get.

"Good clean exit."

Day #264: If the car feels like it's on rails, you are probably driving too slow.

The first race I ever drove in the rain was in a Formula Ford at Westwood Motorsport Park... and I got my butt royally kicked. That was probably the best thing that could have happened, because I decided then and there to never let it happen again. Especially in the rain.

Growing up and racing in the Pacific Northwest, I knew that wasn't going to be the last time I had to race in the rain. So every time it even hinted at rain (which was often — hey, it was Vancouver), I packed up my Formula Ford and headed to the track to practice.

My big Aha! moment was when I overcooked it into Turn 1, entering a good 5 MPH too fast for the grip level. The second I turned in, the car began to slide. And that was the best lesson I ever learned. Once the car was sliding, I knew exactly what to do to control it. On the previous laps, though, I didn't. I would enter the turn at just less the speed it took for the car to slide — it was on rails — and then I'd spend the rest of the corner waiting for it to do something. And when it did something, it took me by surprise. For sure, I was reactionary.

What I learned was that if I entered each turn just slightly faster than the grip level dictated (even just half a MPH too fast), I was ready for what the car was going to do. In fact, I made the car do what I wanted it to do. I was proactive.

And here's the cool thing: driving a car that's on rails is nowhere near as much fun as making the car slide from the very instant you enter a turn.

Of course, this approach applies to driving on a dry track just as much as it does a wet one. Woohoo!

Day #265: When you're maintaining your pace, you're actually getting slower.

Throughout a session on track, whether it's a race, a practice or qualifying session, or the first session of a track day, you should continually get a little faster and faster.

As the fuel burns off, your car is getting lighter. Lighter cars accelerate, brake, and corner better.

As more and more rubber gets put down on the track, it should get grippier. The more grip, the later you can brake, the more speed you can carry through the corners, and the earlier you can begin accelerating.

With each lap you should be learning how to go faster, and eliminating what might be slowing you down.

Yes, there are exceptions to the rules, and there are times when track and car conditions outweigh the benefits, and you'll go slower. Tire wear or overheating are examples of things that might contribute to going slower.

But in most cases, with each lap you drive, you should be getting faster. And even if you turn the same lap time, you're actually going slower, because the conditions are enabling you to go faster.

If your attitude is that staying at the same lap time is being consistent, you're wrong. If the conditions are enabling you to go faster, but you stay at the same speed, then you're actually going slower — that's not being consistent.

With each and every lap you should be able to go a little faster. If either mental or physical fatigue is slowing you down over the course of a session on track, then you know what your weakness is, and what to focus your attention on to improve.

Simply keeping in mind that you should be able to go a little quicker with each lap you do, that might be the biggest contributor to your improvement.

Day #266: Prime the flow.

I once wrote in *Road & Track* magazine (you can read it here), about what the flow or zone state is like, and how drivers like Simon Pagenaud get there so often. But that's all well and good for drivers who race practically every weekend and have the ability to test and tune what triggers that magical state of mind that leads to their peak performance. What about those who drive a few times a year? How do we figure out what triggers a flow performance?

Let's make this a practical, hands-on process to help you perform consistently better.

Take out a notepad or open a document on your computer and start writing. Not a lot. All you have to do is write out a minimum of six words or phrases that describe what it's like when YOU are in the zone. Here are some prompts for this process:

- What does it feel like?
- What's your state of mind like?
- What are your emotions?
- Are you confident?
- What happens with the timing and speed?
- What's your focus like?
- What words and phrases describe being in the zone, for you?

The reason I'm asking you to do this is to make sure you have a clear mental model of what you're aiming for. It's priming you to get there more often. Think of it like the old saying, "How can you get somewhere if you don't know where you're going." This gives you a target, and place to head towards.

You may think you can just think about these words/phrases, but you have to write them down to make this process really work.

Next, think about one of the greatest moments or experiences in your life, a time when you performed fantastically, at your best. It could be driving, but it could also be playing another sport, playing music, engaged in a hobby or something at work, or something to do with family. It should be something you were actively involved in, even to the point where you “lost yourself” in it. Think about just one of those great experiences.

- Where was it?
- What happened?
- How did you feel beforehand?
- How did you feel during?
- How did you feel after it?

In your mind, replay that experience. Notice how your face feels, how your body feels. Notice that by replaying that past success you're in a better state of mind.

Put a name to that experience. For example, if it happened in a specific place, you may want to name it after that location. Or if it was a specific event, name it after that. By adding a name to this past successful experience, you're building a trigger — a method of immediately recalling it and the state of mind that goes with it.

The next time you're at the track (and prior to driving), take a few minutes to relax, and read the six words/phrases you used to describe being in flow. Then recall and replay that past success. By doing so, you'll be triggering a “performance state of mind,” and you've increased your chances of getting into the flow. Let me know how it goes.

Day #267: Entry speed comes from focusing on your release of the brakes.

Slow in, fast out. That's the advice every driver is given when they first start performance driving — as they should be. Then they're reminded of that over and over again — as they should be. But at some point, a driver needs to also focus on carrying corner entry speed. Usually not to the point where it negatively impacts where one begins to accelerate (the "fast out" part), but better than "slow in, fast out" is "fast in, faster out."

Sure, there's definitely a limit to how far one can go with that, but it's something to be aware of. And depending on where you are in your development as a driver, it may actually become a major focus of your improvement plan.

If you carry so much speed into a turn that it delays where you can begin accelerating, that may be too much. But if you're able to carry just a little more entry speed — and are still able to get back to throttle just as soon — then you'll be starting to accelerate from a slightly higher speed — and that'll be a gain.

Nine times out of ten, the key to carrying corner entry speed is being aware of, and focusing on, how you release the brakes as you turn into the corner. Release them too soon or abruptly, and the car will not respond well because it won't be balanced the way you want. It may cause the car to understeer. Release them too slowly — hanging on the brake pedal too long — will over-slow the car. It could also cause the car to rotate too much, feeling like it's oversteering on entry.

Being aware and focusing on your brake release is best done by looking for the point at which you'll actually finish the braking, having fully released the brake (the End-of-Braking point), and paying attention to your foot movement. There's no right or wrong way to do it that applies to all corners, but if you experiment with it, you'll learn what you need.

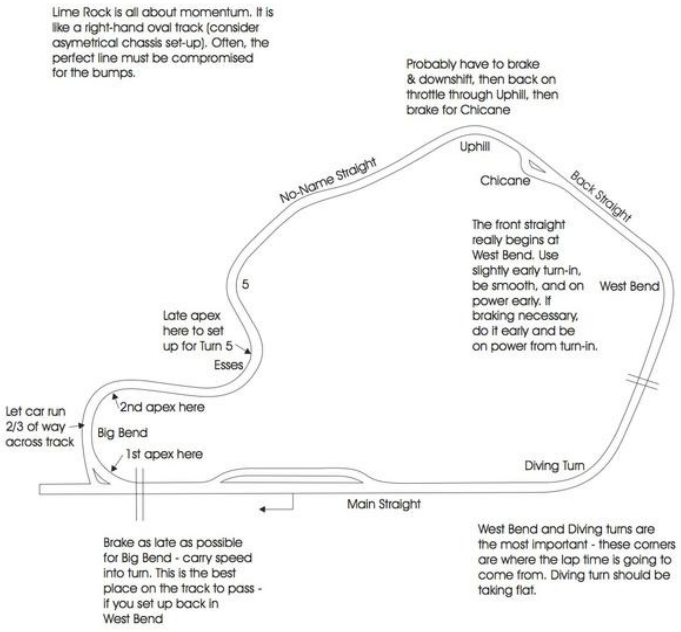
Day #268: Learning is programming.

Until a track has become "programmed" in your mind, you really haven't learned it. So, let's review a plan (but not the only plan) for learning a new track quickly:

Preparation - Review any information you can get your hands on, such as track maps, in-car video, computer simulation games, and descriptions of driving a lap written by someone with lots of experience there. The best place to start is the track's website, then an online search, and then computer games. The main objective during the preparation stage is to become as familiar as possible with the direction the track goes, and any references you may be able to use. One thing that should not be an objective at this stage is to set too much in stone — to program every little detail you pick up from an in-car video, for example. Unless you're driving the exact same car, only use the reference points you see on the video for what they are — reference points. Just because a driver (in a car that may or may not be similar to yours) is using such-and-such as a turn-in point doesn't mean you should. But -—and this is the real key to this — you can use the same reference point, but you may have to adjust exactly how you use it (turn-in just before it, just after it, or whatever). Again, the objective in preparing to drive a track is to minimize the amount of brain power you have to put into what direction the track goes, and what reference points you can use initially.

Be a sponge - During the first few sessions, focus on soaking up as much information about the track as possible, just like a sponge soaks up water. Take in as many reference points as you can, and focus on specific senses to do that. In fact, take a session or two to focus solely on soaking up more visual information, then more kinesthetic (feel, balance, g-forces), and finally more auditory information.

Download - After each and every session, make notes on a track map of every minute detail of what you're doing where on the track (shift points, where you begin braking, where you end braking, where you're back to full throttle, etc.), and all the reference points you've soaked up (every crack in the pavement, the shape and placement of every curb, turnworker stations, signs, bridges, changes in track surface, marks on the surface, etc.).



Mental imagery - After driving the track for a session or more, take all the information you know about the track from your preparation and your actual experience and replay it in your mind. The

more repetitions you do during your mental imagery sessions, the more effective it will be. In other words, you will learn the track faster — and be faster.

Drive the car, not the track - Finally, it's time to stop thinking about the track and focus simply on driving the car to its limit. Usually, if you drive the car at its limit, even off-line, you will still be faster than if you drove the perfect line, but with the car not at its limit. It's at this stage where you really forget about driving the track, and you just trust your mental programming to drive the car. Yes, you need to be consciously aware of how close the car and its tires are to their limits, but you don't want to be consciously thinking about which way the track goes and where your reference points are. By now — if you've done all the previous steps well — the track should be well programmed into your mind, and it will be time for you to simply trigger it and go.

Day #269: If you really want to improve, get out of your comfort zone.

Practicing what you normally do only makes you better at what you normally do. If you want to do something different — for example, improve — then you need to practice something different.

I don't know what kind of driver Eleanor Roosevelt was, but when I read her quote, "You gain strength, courage and confidence by every experience in which you really stop to look fear in the face. You must do the thing which you think you cannot do," it makes me think she would have been pretty damn fast.

I was listening to a podcast about small business marketing and the guest, Brooke Castillo, said this: "Discomfort is the currency of your dreams."

When you're comfortable, you're not learning much other than how to be comfortable. Of course, being uncomfortable while driving on a race track can be a bit risky. The key is knowing just how uncomfortable you can be without going over the top. How do you know that? Use some self-coaching questions:

- Am I holding my breath?
- Am I tensing up?
- Has my vision dropped and I'm not looking far ahead?
- Am I turning in early — "crabbing" into the turns?
- Do I feel out of control?

A "yes" to any of these is a fairly good sign you're too far outside of your comfort zone.

- Am I almost bored by the way I'm driving?
- Is the way I'm driving easy to replicate over and over again?
- Does my car feel totally predictable — almost like it's on rails?
- On a scale of 1 to 10, with 10 being my car's limits, am I driving at less than an 8?
- Am I very consistent — my lap times never vary much more than half a second?

Mostly "yes" answers to these questions suggests you're not pushing the limits of your comfort zone enough to be learning.

- Do I feel energized by what just happened?
- Are the inconsistencies in my driving due to deliberately experimenting with new techniques and pushing the limits?
- Am I able to turn consistent lap times — within half a second — and then turn a few laps half a second faster than that every now and then?

- Are my tires generating slip angle from the very second I turn into every corner all the way out to the track-out point?

If you answered “yes” to some or all of these questions, that’s a good sign that you’re expanding your comfort zone.

Awareness is what changes our behavior. Just look at what happens when drivers see one of those speed read-out boards on the side of the road. They see a number, become aware of their speed (I know, hard to believe that most drivers on the road are unaware of what they’re doing!), and they slow down. The awareness led to a change in behavior.

When drivers ask me how to get more comfortable with driving closer to the limit, I coach them towards being more aware of where they are currently. The self-coaching questions above may seem simple, but if you use them deliberately and honestly, you will see a change in behavior — one that’s more comfortable being uncomfortable. One that’s more comfortable being a little closer to the limit.

Obviously, the closer you drive to the limits of your car, the more risk there is. You have to decide how much risk you’re willing to take, but inching up to the limit, bit by bit, may not be as risky as you initially think. So, ask yourself these self-coaching questions, and expand your comfort zone.

Day #270: The more time you spend with the front wheels pointed straight ahead — or near straight — and the throttle to the floor, the faster you'll be.

The steering wheel. It's almost a symbol for driving. And for good reason: it determines where you go. Well, actually you do, by how you turn it. And that's what this Speed Secret is all about. Of course, a discussion of the steering wheel — and how to use it — could go on for hours, so I'm going to focus on just two key areas.

First, the steering wheel can be a brake — it can slow you down by scrubbing speed with the front wheels turned. So minimize your steering input. How? By looking further ahead. The farther ahead you look, the fewer "last second" adjustments you'll need to make — your path, or line, will be better planned out. Focus on putting as little input in to begin with, and then deliberately work at unwinding, and straightening the wheel as soon as possible.

Second, slow corners usually require quicker steering movements, while faster corners reward slower inputs. To get a car to rotate, or turn sharper in slow, tight corners, you'll need to turn the steering wheel quickly. But in long, fast, sweeping corners, you'll want to turn the steering wheel more slowly. So, turn the steering wheel slowly in fast corners, and quickly in slow corners.

Think about it: Practically everything you do with the steering wheel is designed to help you get your foot to the floor on the throttle. Yes, it's to initiate a change in direction, but then your goal is to unwind it as soon as possible to allow you to get to full throttle.

Believe me. I wouldn't steer you wrong!

Day #271: Develop and use a pre-drive ritual to trigger a peak performance.

When I think of pre-game rituals, preparation, or even superstitions, I think of Tiger Woods always wearing a red shirt on the final day of a tournament. I think of the way tennis players routinely bounce the ball before serving (for an extreme example of this, search for “Rafa Nadal routine” on YouTube). I think of all the open-wheel race drivers who have to get into their cars from the left side.

Most superstitions are simply pre-performance rituals or routines. As long as one can control the routine, then there's nothing wrong with it. In fact, it's just an action that triggers a desired performance.

I strongly recommend a pre-drive ritual, but what you include in this routine is almost unimportant. Sure, there are some things that would not be helpful — things like drinking alcohol, screaming and having a fist fight with someone on the grid, or recalling and talking about all the failures you've had in your life!

What matters most is that you do the same thing each time, and in doing so, you trigger an "I'm ready" mindset.

But there are some things that are more likely to be positive and trigger a great performance. Some that I've found help — either with myself, or with the drivers I've coached — and make for a great pre-drive ritual are:

- Putting your safety gear on in the same order/method
- Saying a certain phrase to yourself and/or others around you
- Cross Crawls and Lazy 8s (as I've written about previously and in my books)
- Visualization (mental imagery), especially pre-playing how the race/session is going to look, feel, and sound
- A visual check of your car while walking around it

I've talked to a few drivers who claim they do not have a superstition or ritual they rely on before driving. They say that they purposely do things differently each time, just to prove that they don't have a routine. They feel that if they have a certain way of doing things, and then for some reason they can't follow through with it, they'll feel unprepared. Ironically, they do have a pre-drive ritual: doing things differently each time. In fact, their "superstition" is that if they do things differently each time, they'll perform better. And that's okay. As long as they (or you), believe that is what you need to do to perform at your best, then use it.

Obviously, as I stated earlier, make sure your ritual only includes things that you can control. If, for example, your routine includes saying something to a specific person, and that person can't be at all events, you're setting yourself up for feeling unprepared. Of course, if you imagined saying your pre-drive phrase to that person, whether they were there or not, then you can control that.

Develop a pre-drive ritual, and stick to it.

Day #272: Three strikes? Back off or park it.

Physical fatigue most often leads to mental fatigue, and we all know what happens then: Loss of focus.

Like most things, recognizing the problem is the first step in fixing it, and losing one's focus is no different. The good news is that when you lose focus on the track, it's usually fairly easy to recognize. Lap times are slower, you make more mistakes, and you might even crash. Oh, and that's the bad news! But at least you recognized it!

Seriously, never forget that a mistake that leads to a spin or off-track excursion is rarely ever a singular mistake. Most times the final "off" is the result of a few small mistakes. And that really is good news if you're able to recognize them soon enough, and... have the discipline to do something about it before it turns into a big mistake.

It's this second part — having the discipline to do something about the small mistakes — that is the hardest part for most drivers. It's easier to think, "I got it. I'll correct it. I got it. I'll correct it..." until you find yourself saying, "Uh-oh, I don't got it!"

Same with fighting off fatigue. We think, "One more session. One more lap." And we all know that it's usually the last lap of the last session when the crash happens (of course, that's also why it is the last lap...).

If you make one mistake, that doesn't mean you're going to make another. If you make two mistakes in a row, the chances of you making a third go up dramatically. And if you make three mistakes in a row — no matter how small they are — your chances of NOT making a fourth big mistake are not good. In my experience, three mistakes in a row almost always lead to "The Big One."

What does that tell you? If you make one isolated mistake, carry on. If you make two mistakes in a row, be especially sensitive to a third — and even consider slowing down or taking a break. If you make three mistakes in a row, definitely back off or park it.

The biggest challenge after making a mistake is regaining your focus so you can carry on. For many drivers, a mistake throws off their rhythm, their flow, their momentum. It's hard to get that back. That is, unless you've developed and practiced using a trigger to regain your focus.

I've written before that losing your focus is not the biggest problem; it's regaining it that counts the most. Every driver loses focus, but it's how quickly you can regain it that is most important. Having some form of trigger phrase or word is the key to making that happen.

Here's a common scenario — a likely internal conversation in your head (as you're driving at high speed): "I messed up Turn 1. Darn, that cost me a lot of lap time. Gotta make sure I don't do that again next lap. I think I turned in too early, and that's why I

dropped wheels off the edge of the track at the exit. Yeah, don't want to do that again. Okay, hey, I need to get focused here. Come on, get focused. I gotta get back into the rhythm again. Come on! I need to forget about what I did back there. I need to focus on what's coming up... Oh wait, ahhhh... I screwed that up, too. Man, I'm out of it now. I can't get back into the flow. But I have to. Ahhh..." And just how far have you travelled around the track while having this non-productive conversation with yourself?

Instead, it should go something like this: "Messed up Turn 1. Eyes up — look ahead." Boom, you're back focusing on what's coming up. The internal dialogue is short, and focused on what's happening ahead. You identified that you made a mistake, and then you moved on by saying to yourself your refocus trigger phrase: "Eyes up — look ahead."

Of course, that refocusing trigger can be almost anything, but I recommend that it's a quick, short phrase or word which focuses on the act of driving in the moment. That's why "Eyes up — look ahead" works well, as do "Slow hands," or "Smooth footwork." Each of these phrases focuses you on what you need to focus on.

The more you practice using your refocusing trigger phrase, the better you'll get at regaining your focus on track. It could get to the point where you regain it so quickly that it seems as though you never lost it.

Having said that, there is a limit to what every driver can do, and that's where the "3 strikes" guideline comes back in. If you've made three mistakes, or if you've lost focus three times, back off or park it. And depending on your motivation (if you're ten laps from the end of the Indy 500 and you're in the lead, that's a different motivation than if you're at your third track day event!), two strikes might even be better threshold.

Day #273: It's not where you start to apply the throttle that matters; it's where you get to — and commit to — full throttle that matters most.

Every driver knows that the sooner you begin accelerating out of a corner, the faster you'll be down the following straightaway. It's what we do, right? We focus on getting back on the throttle as soon as possible exiting corners.

You've also been told to squeeze on the throttle. Heck, I've written that many times in my *Speed Secrets* books.

All of this is good advice, but it can lead to the feeling that giving your car some throttle is a good thing, when in fact, some throttle might be stopping you from giving it all the throttle.

If you squeeze down on the throttle too soon, you may have to hesitate just a fraction of second before you're able to get to full throttle. And what matters most, in many cars, is when you get to 100% throttle — not how much time you spend at part throttle.

Think about this, and work on it the next time you're on track. See what happens if you hesitate just for a fraction of a second before beginning to squeeze on the throttle. See if this allows you to get to full throttle sooner.

This may not work in all cars, but it's worth experimenting with to see if it results in a faster overall lap time.

Day #274: Physical preparation prepares you mentally.

While watching qualifying for the Indy 500, I was thinking about what it takes for a driver to lay it all on the line for four laps — with very little practice in a car that has had its setup changed dramatically due to a rule change just hours before qualifying started. I thought about my own brief experience at Indy (laying in the ICU at Methodist Hospital over qualifying weekend), and managing my own emotions — and fears.

In Chris Hadfield's book, *An Astronaut's Guide to Life on Earth*, the Canadian astronaut wrote, "In my experience, fear comes from not knowing what to expect and not feeling you have any control over what's about to happen. When you feel helpless, you're far more afraid than you would be if you knew the facts. If you're not sure what to be alarmed about, everything is alarming."

It's no wonder that the most experienced drivers managed the unknown of qualifying at Indy the best. They were better prepared, simply based on their years of experience.

One of the many things that made Ayrton Senna a better racer than most was his level of preparation. He was almost fanatical about it. Was that why some claimed he didn't seem to have a sense of self-preservation (a form of fear)? Was it that his physical preparation, from the direction his bed lay in a hotel (the head of the bed needed to point north) to his detailed examination of the tires, helped him feel mentally prepared? Could it be (opposite to the fear response Hadfield refers to), that Senna felt in control because he had taken control of as many things as possible? He knew what to expect because he had pre-played as many scenarios as he could in his mind. He didn't seem to have any fear because he knew what to expect.

The more you prepare, physically and mentally, the less chance of feeling alarmed and triggering a sense of self-preservation — often referred to as fear.

Day #275: Focus on being better, not faster.

Don't you just love chasing the lap time? Okay, not every driver does. Some like to chase the "feeling" that comes with driving quickly on a track. Others love the competition, and turning a fast lap isn't that important — just being in front is what matters. But for many drivers, a new personal best lap time is a big goal.

This came to mind as I was reading *Company of One: Why Staying Small Is the Next Big Thing for Business* by Paul Jarvis. In it, he wrote:

"Pat Riley, the Hall of Fame basketball coach who led five teams to the NBA championship, coined the term "the disease of more." He noticed time and time again that winning players, just like some startups, focused on *more* instead of *better*. Once they won, they'd let their own ego get in the way of all the tasks that had helped them win in the first place—like practice and focus—and instead become lured into more endorsements, more accolades, and more media attention. As a result, they ultimately lost to internal forces, not to competitors."

"More," in our case, is the lap times, and faster is the goal. But "better" is, well… better. And, usually, focusing on being a better driver results in faster lap times more often than when you're focused on chasing the lap time. Yeah, that's something that might take time to think about and internalize, though, so it's okay to go back and read that last sentence again.

My point is that a sole focus on a faster lap time, without the accompanying "how" — what you're going to do to be faster — rarely results in the result you're after. Focusing on more rarely results in better.

The "disease of more" could also apply to the old belief that all it takes is more seat time to be the best driver. Seat time for the sake of more time driving is rarely more important than quality seat time. We all know that track time is tough to come by, either for logistical or economic reasons, so it's easy to get caught up in the thinking that all one needs to get better is more seat time. But as I've mentioned more than once (maybe even more than a thousand times!), the kind of practice a driver needs is deliberate, planned, strategic practice. It's because track time is limited that this is so very important. If we could drive on track any time we wanted, quality would be less important. But we can't, so quality usually beats quantity.

If all it took was more seat time, then the best would always be the ones who drove the most. But that's not the case. I know drivers who started their career late, and had significantly less seat time than others whom they consistently out-performed. If you drive on track once a year, then it's even more important that you make that one time high-quality.

The key take-aways from what I've written here are:

1. When you focus on the process, the act of driving better, you're more likely to get the "more" you're looking for — the faster lap times.
2. Spend time planning your practice, making sure it's quality practice.

If you ever get to the point where you're the best, and there is no way you can get better, then go for the "more." Oh, and let me know, because I'd love to meet anyone who has gotten to this point! But until then, focus on being better.

Here's a very simple but effective way to help you stay focused on better, rather than more. Use triggers.

Right now, imagine yourself on track, and you're in the head-down, determined mindset, with the lap time being all that matters. You're hyper-focused on turning a lap time that is half a second faster than you've every turned. You're trying as hard as you can. As you pass the Start/Finish and notice that your lap time is not faster — in fact, it's half a second slower than usual — you realize that you're getting frustrated. So, you say to yourself, "Better," and your focus shifts to what it takes to drive better. You look just a bit further ahead; you make sure your initial application of the brake pedal is firm and quick, and the release of the pedal as you turn is smooth, deliberate, and timed just perfectly to get your car to rotate in towards the apex; your turn-in is crisp and smooth, and you dial in just the right amount of steering to arc towards the apex; you give the car a tiny fraction of a second to set, then begin to feed in the throttle, continuing until you're at full throttle, driving the car out to the edge of the track as you unwind the steering. You think, "That was better," and smile. Oh, and as a byproduct of this "Better" mindset, you just happen to turn a new personal best lap time. Your smile gets even bigger.

Back in the paddock after the session, you sit down and plan out your next track session. As you pull out your track map and begin making notes about what worked in the past session and what needs improvement, you say to yourself, "Quality." That's your trigger word to help you focus on making the most of every second you get on track. It's all about having a deliberate, strategic plan to improve your driving performance, rather than just getting more seat time. It's all about "Quality."

Now you're consistently driving with Better Quality. And you're smiling.

Day #276: Build up the tires' cornering forces slowly.

I'm sure you've been told to drive smoothly, and "smooth is fast" dozens of times. But why?

I think of the way tires grip the track surface like the action of breaking a piece of string. If you hold the ends of a piece of string in each hand and quickly snap them apart, it's easy to break the string. Try it. However, pull slowly on the ends of a piece of string, and it will take more effort to break it.

Tires are like that, too. If you quickly snap the front tires into a turn (jerking the steering wheel to the right, let's say), they will break loose more easily than if you slowly turn the steering into the corner.

Of course, there's a practical limit to how slowly you can turn into a corner. Turn too slowly and you'll never make it around the corner! But turn too quickly and your tires will begin to slip sooner.

It's a balance, obviously, which is why driving a car at speed around a track is such a interesting and fun challenge. But if you keep in mind that your goal is to turn the steering wheel as slowly as you can, while getting the car to go where you want it to go, that's not a bad approach.

Day #277: Avoid the fear of the unknown by learning more.

Some drivers love driving at night, and others don't. Many racers find it easier to get in the zone and have a great rhythm in their driving at night, since there can be fewer distractions. For other drivers, the inability to see as much as they do in the daytime creates anxiety, so they avoid driving at night.

The key to driving at night is what you do during the day. If you soak up more reference points during the day — ones that you can rely on at night — night driving is relaxing and fun. References like the distance past the end of a pit wall before turning into a corner, paint marks on the track surface, a turnworker station, bumps in the track that tell you you're on line or not. The more of these you have, the easier it is to drive at night.

Before getting into the car to drive a night stint, avoid looking directly into bright lights. These lights will dilate your pupils, making it harder to focus in the dark. Some drivers go so far as to wear sunglasses right up until the time they hop in the car to drive to keep their eyes used to the darkness.

Driving at night can be like driving in fog or in the rain: it's the fear of the unknown that is the real challenge. In either case, it's the fear there is something you can't see that makes you want to drive conservatively. You look ahead and can't see anything, so you assume there's something you should slow for, but the best night drivers turn that around and assume that if they can't see anything, it's because there's nothing to see. It's a bit like a pilot using just the instruments to fly. In this case, the only instruments you have are the reference points you can see (which are limited), the ones you feel, and the ones you hear. Notice that the things you can feel and hear are not affected by the darkness. What does that tell you? Yes, you need to pay more attention to those things when night driving.

Okay, there is one other "tool" that you rely on when driving with restricted vision, and I'm talking about your memory. If you're able to close your eyes and imagine driving a lap of the track — all the things you see, what you feel, and what you hear — then driving at night is not as scary. I highly recommend doing mental imagery, or visualization of driving the track. The more you do this, the more detail you'll fill in about the track. And that will help you greatly at night.

Day #278: Make your own decisions.

It’s all relative.

I hear drivers and coaches saying things like, “Get off the brakes sooner in that corner to reduce the understeer,” “Release the brakes later — trail brake more — to reduce the understeer.” Which one is right?

Or, “I brake at the 200 marker,” when you know you can’t seem to get past the 300 board before starting to brake. How is that possible?

It’s all relative.

There are some drivers and coaches who believe you should always trail brake less to minimize understeer — that releasing the brakes sooner will help the car turn more. That works if the front tires are overloaded – releasing the brakes gives some grip back to those tires so they can turn the car. Other drivers and coaches say you should trail brake more to get the car to turn, and that works because you’re keeping the front tires loaded to give them more grip.

Which is right? Both. Understeer can be caused by either unloading the front tires, or overloading them (of course, it may be a car setup issue, too, but you can't do much about that while driving, so you have to adapt). You need to experiment to find the right solution.

But what about the driver who swears up and down that they brake at the 200 marker, when you can’t get past the 300 board? Are they (or their car) just better? Could be. But it’s more likely that it’s a matter of perspective. I’ve been in a situation where I was coaching two drivers who said just this — one braked at the 200, and the other braked at the 300. When I looked at the data, they braked at exactly the same place.

The first driver started braking when he saw the 200 marker; the second driver braked when he noticed the 300 board passing by in his peripheral vision. It was a matter of perspective.

When someone tells you to turn in at a specific point, brake at the 200 marker, trail brake more or less, or whatever, remember that this advice may be given with the best of intentions — but it might not be right for you and your perspective. It’s all relative.

Day #279: You have to drive over the limit at times before you can consistently drive at the limit.

When I talk to drivers about driving the limit, I'm talking about the limit they want to drive. For pro racers, the limit is the absolute limit — 10/10ths. For some track day drivers and club racers — those who don't want to risk everything to drive at 10/10ths — I'm talking about consistently driving at their limit. If they've chosen (perhaps not consciously, even) to drive at 8/10ths, for example, that's their version of 10/10ths. And I'd like to see them drive at that limit all the time, and not at 6/10ths, then 9/10ths, then 7/10ths, then 10/10ths, then 8/10ths....

In other words, when I talk about driving the limit, I mean driving your limit all the time. And to do that, you do have to drive just over the limit every now and then. Of course, the trick is to not drive WAY over the limit. The only way to do that is to develop the sensitivity so you can step just over the limit — let's say, a half point over — and still bring it back to your limit. If you deliberately focus on and practice sensing that limit, you'll be able to hone in on your limit more consistently.

Day #280: There's usually more in your driving than there is in any car modification.

A driver should leave the car alone — not worry about modifying it to make it faster — until they are getting the most out of their own driving on a consistent basis. I'd rather see a driver in an older car with lower performance limits — think of an early-90s 3-series BMW on stock street tires — getting the maximum out of that car before moving up to a higher performance car. Too many drivers get cars that mask what they're doing — both the good and bad stuff — because the car's performance levels are so high.

Weekly, I receive emails from drivers asking my opinion on what they should change, upgrade, or modify on their cars. I have a difficult time answering that question because:

- I think it's more important to focus on driving first, then upgrade the car.
- I don't know what the driver's objectives are, their motivation, and most importantly, what their budget is.
- I'm not an expert on car modifications.

Having said that, I try my best to provide some advice or guidance based on my own experience, especially if the question focuses on just one aspect of upgrading, such as tires.

Upgrading from street to R-compound tires? For sure, they will have more grip and allow you to carry more speed. In my experience, some provide a little less feedback, making it a little harder to sense their limits, but not all do. It depends on what you're switching from and to. You can look at that in two ways: First, that might make driving more difficult and less enjoyable. Second, it might make it more of a challenge, and more fun. Personally, I look at something like this as a fun challenge.

Now, I say all of that under the assumption that you're driving your street tires fairly consistently at their limits already. I think it's a mistake to move to R-compound tires if you haven't learned to maximize what you already have. If you feel that you're consistently getting the most from your street tires — and even able to go beyond their limits, at times, and control your car without spinning — then that's a pretty good sign that you're ready to step up.

A factor you need to consider is the additional grip and speed you'll carry with R-compound tires. They'll put more stress and strain on your car — brakes especially — but also suspension and drivetrain. If that's okay with you, good. But be prepared for that, as it may require more maintenance costs.

Also — and this may be the biggest deciding factor — you'll be carrying more speed, and therefore be at higher risk. If you should go off the track, it'll be at a higher speed, meaning the results of hitting something could be worse. You need to factor that into your decision.

Some drivers have more fun, and like the lower strain on their car and lower risk levels that come with “lower performance” street tires. They also have more fun throwing the car around with larger slip angles. Other drivers have more fun going faster on the higher-grip tires. Your call…

The way I look at any upgrade is that you should always start with safety in mind. That’s why brake upgrades are almost always the best place to start. Then the priority should be on reliability. And that’s why brake upgrades are almost always the best place to start. (Are you seeing the pattern here?!) Then I would work on handling improvements (probably starting with tires, since they’re usually the simplest way to improve your car’s handling performance, then getting into anti-roll bars, springs, shocks, etc.), and finally on engine upgrades.

To me, it’s better to get the most out of a lower performance car than it is to get a little out of a higher performance car. As my friend Don Kitch, Jr. says, “Learn to drive a slow car fast so you don’t drive a fast car slow.”

Day #281: Use the Stoic's Approach to Performance Driving.

My advice here might surprise you just a little bit.

The Stoics had an approach to life that involved being prepared for the worst. In my opinion, they weren't thinking negatively, but they did pre-plan how they would deal with negative situations.

Most people who know me would likely describe me as being a pretty positive person, one who can usually find the positive in almost any situation. I've been accused of being too much of that type of person, if that's possible (see?). But I love the Stoic's approach. I love thinking through the "what if's..." beforehand, and planning for how to handle them. To me, that's "realistic positive thinking."

So here's a thought experiment. Think through all the things that could go wrong when driving on track. Most importantly, though, think about what you would do in each case. It's not difficult to think of what could go wrong (negative thinking, which comes naturally to some people). It's another thing entirely to think through how you could and should respond. This is much more than positive thinking. It's pre-planning. It's preparation.

- What if, in trying to go faster, you brake too late and enter the corner too fast? What would you do?
- What if you got on the throttle too soon and ran out of track at the exit of the corner? What would you do?
- What if you turned in too late for a corner and couldn't get the car down to the apex? What would you do?
- What if you turned in too early and apexed too early? What would you do?
- What if you trail braked so long into a corner that the car got loose and started to oversteer? What would you do?
- What would you do if the brakes faded entering a tight corner?
- What would you do if your transmission started to fail and you only had 4thgear?
- What would you do if your car had a serious oversteer problem?
- What would you do if your car had a serious understeer problem?

To be clear, I'm not suggesting spending time visualizing in great detail something going wrong. No, that would be programming that to happen, increasing the likelihood of the wrong thing happening. Just think about what could go wrong, and then spend the time visualizing how you would deal with it.

As Ryan Holiday writes in the book *The Obstacle is The Way*, "And in the case where nothing could be done, the Stoics would use it as an important practice to do something the rest of us too often fail to do: manage expectations. Because sometimes the only answer to 'What if . . .' is, 'It will suck but we'll be okay.' But the person who has

rehearsed in their mind what could go wrong will not be caught by surprise. The person ready to be disappointed won't be."

Mentally preparing for what you want to do involves having a clear mental picture of what you want. It's imagining how everything will go, exactly as you want it to happen, everything going right. But it's just as important to imagine how you would manage things when they go wrong. That's not negative thinking, it's being prepared.

I don't recommend spending tons of time imagining every little detail of you making mistakes as part of this preparation process. Again, that would only increase your chances of doing things wrong. But running through how you would deal with a mistake or problem, that's realistically preparing. It's "constructive positive thinking."

Captain "Sully" Sullenberger averted disaster by landing his plane in the Hudson River. He had pre-planned how to deal with a huge number and variety of problems. No, he didn't imagine himself making mistakes on take-off, as that could have led to him actually making those mistakes. He did, however, mentally practice how he would manage himself in the event of a mistake or problem. To listen to his voice when communicating to air traffic control prior to his "Miracle on the Hudson," it was obvious that he was mentally prepared. He stayed amazingly calm and carried on. All great pilots mentally prepare for what could happen. Great drivers do the same.

Spend time mentally preparing yourself for a problem or a mistake. Imagine how you would deal with problems. Be a Stoic behind the wheel.

Day #282: Blend your braking with steering, and steering with acceleration.

I was a guest instructor at a car club event the day after I had conducted a seminar for their members. Throughout the day, I rode in the right seat with half a dozen drivers of varying levels of experience and expertise. But there was one common piece of advice I gave them: overlap your braking and cornering, and then your cornering and acceleration.

Okay, to be truthful, I didn't have to tell all of them to overlap the cornering and accelerating — that seems to be fairly easy for most drivers to get. But overlapping braking with cornering — trail braking — was more challenging. As Mark Donohue once said "Exiting a corner on the limit is like tightrope-walking; entering a corner on the limit is like tightrope-walking while blindfolded."

At many car club events I find myself in the discussion (argument?) about whether to teach trail braking, and if so, when. While I won't get into the full details of my opinion here, my approach to teaching novice and intermediate drivers is to have them simply focus on releasing the brake pedal slowly and smoothly. That's it. There's no need to get into the minute technical details of why and what to call it — just release the brakes slowly and smoothly. In doing so, of course, the driver is beginning to learn to trail brake, to overlap the braking and cornering. There's nothing like keeping it simple.

Day #283: Frame Being Comfortable Being Uncomfortable.

It's one thing to tell you to be comfortable being uncomfortable, but another to tell you how to do that. It's a bit like telling a depressed person to be happy. I'm sure they'd love feel better, but how?

Let me take a stab at telling you how you can improve your ability to feel comfortable being uncomfortable (I can't promise this will work one hundred percent of the time, but it's proven to work many times).

First, what exactly do I mean by feeling "comfortable being uncomfortable"?

You know that feeling when you're beyond your comfort zone, right? It's uncomfortable. You probably tense up, hold your breath, and thoughts of what could go wrong pop into your mind. Some people hate the feeling of being outside their comfort zone, and some people love that feeling. When you love it, that's being comfortable being uncomfortable.

When you're comfortable being uncomfortable, you'll consistently drive faster — actually with less risk. But how do you do that?

Recall any experience you've had in your life where you felt very uncomfortable, such as:

- Public speaking
- Sky diving
- Driving in the rain

Now recall one of those experiences where you felt uncomfortable at some point in your life, but after a while you got where you were comfortable with it. For me, public speaking fits into that category (well, I'm mostly comfortable with it, although not entirely, all the time). What made the difference? Personally, three things had the biggest impact:

1. Experience — actually doing it over and over again
2. Mental experience — mentally doing it over and over again
3. Framing

Let me explain what I mean by "framing." When I took the time to deliberately and logically think through my fears of public speaking, and compared it to what I was doing in a race car, it put things into perspective. I saw that I could drive a race car very fast, and that was something that I could actually get hurt — or worse — doing. Public speaking, in comparison, is not so dangerous. Of course, that's what I told myself, but what I felt early on was still fear, apprehension, tension, nervousness… I didn't want to do it, even though the "reward" was worth it.

When I spent time mentally rehearsing, and especially imagining what it would be like to feel uncomfortable, and then framing that as something that I could enjoy — I was conquering it, I was handling it, I was embracing it, I was using it to make me stronger; all of these perspectives were true, rather than the usual opposite.

The key here is that I didn't deny that I felt uncomfortable. I actually welcomed it. I took it as a good sign, one that meant that I cared about what I was doing — if I didn't feel uncomfortable, that would have meant that what I was doing was not challenging, and/or I didn't really care much about doing it well.

Okay, enough about me and public speaking. Let's talk about the feeling of discomfort you get when you drive a car fast on a track — when the tension builds and the muscles tense up and the breathing is restricted and… Oh man, I don't want to be here right now!!! But then, you spend enough time there — physically and mentally — and you get used to it. You get comfortable being uncomfortable.

Experience makes the difference. Both physically and mentally experiencing it. The more time you spend experiencing what it is that makes you uncomfortable, the more comfortable you get with it.

- Driving a much faster car than you're used to driving
- Driving on a skid pad or loose surface (gravel), spending more time with the car "moving around"
- Driving in more traffic, meaning you have to be more aware of what's going on around you
- Driving in the rain
- Driving a car that's not handling well

Those are all good physical experiences that will help you be more comfortable being uncomfortable. You can also mentally experience them, using mental imagery, or visualization. The more detail you imagine, and the more senses you involve in your mental imagery, the more effective it'll be.

Do this enough and you'll get to the point where driving comfortably is a bit boring. It's a bit… well, actually uncomfortable because it's not the zone you like to play in. You prefer to feel slightly uncomfortable. You're not just comfortable with it, you crave it.

Then you frame everything you do as something you look forward to:

- You look at driving faster than ever before as something you're excited to do
- You look forward to the car sliding around
- You look forward to being surrounded by other cars
- You forward to driving in the rain
- You look forward to adapting to your car when it has a handling problem

Imagine yourself as an extreme sport athlete, someone who does wingsuit flying, free climbing, big wave surfing, or base jumping. Imagine how comfortable they feel when they're feeling uncomfortable. Imagine the way they frame that feeling. Now, put yourself in that same mindset, in your car. Frame the feeling of being uncomfortable, and feeling great about it. It's your "home."

Day #284: The better you can adapt, the faster you'll be.

Drivers have databases of knowledge and experiences. Or, you can think of it as having a toolbox, and the more tools a driver has, the safer, faster, and more consistent they will be.

For example, some drivers are safe, fast, and consistent when their car is handling perfectly. But when the car is not perfect, the driver struggles because they can't adapt their driving to suit the car. There comes a time when trying to make a car do something that it's not capable of is simply a waste of time, and the only solution is to adapt your driving to it.

Other drivers are very good at adapting to whatever the car is doing. How?

Here's a list of some of the things you can do to adapt your driving to suit a less-than-perfect-handling car:

- Trail brake more or less to help a car turn-in better.
- Trail brake less to help reduce corner entry oversteer.
- Change your line to a later turn-in and apex to help a car that understeers through the turns (allowing you to unwind the steering wheel sooner).
- Same for a car that oversteers from mid-corner on out — turn in and apex later so you can unwind the steering sooner.
- Modulate the throttle through the turn to change the car's balance, to induce some understeer or oversteer (depending on what you need to counteract your negative handling issue).
- Be more patient on the entry to a turn, focusing on making the most of your corner exit speed - this can help whether your car understeers or oversteers.
- Be smoother with your inputs, whether turning the steering wheel or applying the throttle.
- Turn the wheel more or less crisply or quickly.
- Use different track surfaces to help the car, such as running over a curb to help rotate the car more.

Ultimately, adapting your driving comes down to:

- Managing your car's weight transfer.
- Changing the line.
- Altering the timing of your inputs.

This is not a complete review of every possible thing you can do to adapt, but it's enough to make you think. I wrote an entire chapter in *Ultimate Speed Secrets* about this topic, so I didn't want to reprint it all here. But I'll leave you with this obvious

thought: doing more of the same thing rarely gets you a different result. So, for example, if your car has some corner entry understeer, driving the exact same way you always have in the past will not eliminate the problem. Sure, you could, and ultimately should, adjust your car's setup to improve its performance, but you can't always do that (especially while you're driving). And that's why it's important to learn how to adapt your driving to suit your car.

Day #285: Your right foot should either be on the brakes, squeezing the throttle down, or flat to the floor.

Yes, I wrote that in my first *Speed Secrets* book. And it's a pretty good general rule. But you know what they say about rules — they're there to be broken.

What you can read into my Speed Secret is that one should never coast — you should either be on the brakes or on the throttle, but never in between with nothing going on.

However… ***that's not always true***. So I'm going to contradict myself here, or refine what I wrote in the past.

With some cars, in some corners, there is a need for coasting. Why? Because some cars will begin to understeer — even just slightly — if you immediately go to the throttle after braking. And when that happens, you have to breathe the throttle a little before getting to full throttle. In many cases — but not all — this squeeze on the throttle, breathe it, and then back on the throttle is actually slower than if you simply waited — coasted for a tiny bit and then went fully to the floor.

The problem with a "rule" is that there is always an exception to it. In this case, there are cars and corners where coasting is not only okay, it's faster. Your challenge is to test whether that's the right way to go fast or not.

Day #286: Understand when, where, and why you choose to drive at or below the limit.

What do you do if someone says to you, “Drive at 8/10ths this session?”

We all have some idea of what this means, but let’s dig a bit deeper into it. To begin, we all agree that 10/10ths is driving at the limit. It’s consistently driving with the tires at the peak of the slip angle versus traction graph. For the purposes of this article, I’m not going to get into what we sometimes do that reduces that peak (unbalancing the car); I’m going to assume that the peak is the peak, it’s the limit, it is 10/10ths.

And let’s use a corner that has a “min speed” of 60 MPH. That means the minimum speed the car travels at, at some point in the corner, is 60 MPH. This min speed point/area is usually around the End-of-Braking (EoB) point and just before transitioning back to some amount of throttle. And for simplicity’s sake, let’s say the lap time is 1:40.0 (100 seconds).

By definition, 8/10ths is driving at 80% of the limit — speed and lap time. Therefore, your min speed would 48 MPH and your lap time would be 2:00.

I don’t think there’s anyone who would feel that driving 20 seconds a lap slower is anywhere near driving at 8/10ths. So there’s something in that logic and/or our perception that doesn’t line up.

If you drive at 9/10ths while braking for a corner, your corner entry speed is 6/10ths, your mid-corner speed is 7/10ths, and your exit speed is 10/10ths, have you driven at 8/10ths? Is driving at 8/10ths an average of various sections of the track? Or, to drive at 8/10ths, does that mean you’re driving at 8/10ths in the brake zone, corner entry, mid-corner, and the exit phase of the corner?

Oh, I know! Does driving at 8/10ths means that we only have to put 80% of the effort into driving? In fact, driving 8/10ths could mean that you only have to be 80% mentally prepared and focused. I don’t know about that — it doesn’t seem like a very good idea! Driving on track with anything less than 100% mental focus is not a smart thing to do — no matter what speed you’re at.

I suspect you’ve heard me or someone else talk about how some drivers drive the slow corners too fast, and the fast corners too slow. It’s true. And it’s easy to get caught up in the trap of over-driving slow corners (typically rushing the entry, resulting in poor exit speed), and not driving those scary fast corners fast enough (usually slowing down too much for them). In other words, they drive the slow corners at 10/10ths (or sometimes 11/10ths) and the fast corners at 6/10ths. If you average this out, are you driving the track at 8/10ths? Does driving 8/10ths mean that you drive 80% of the track at the limit, and the other 20% you drive at something less — potentially at 20% less than the limit?

As you can see, when you look at this whole concept in a logical way, it doesn't make a lot of sense. Of course, one could argue that driving around race tracks at speed is not a very logical thing to do, anyway!!!

The next time someone tells you to drive at 8/10ths (or any "/ths" below the limit), or you tell yourself that, be sure you know what it means. Backing off and driving with anything less than 100% mental focus is not an option. Always drive with 100% mental focus. Remember — what you practice becomes permanent. If you drive with an "I've backed off a little this session" mental focus, that will become your norm; it will become a habit. As you know, habits are hard to break.

If I asked you what 8/10ths means in terms of corner min speed my guess is that you would suggest that you'd be 3–6 MPH slower. By the way, I've asked many drivers this question, and this has been a fairly typical response. Do the math. That 3 to 5 MPH is between 5% and 10% slower, not 20%. Are you sensitive enough to tune your speed sensing to that level? Many drivers are, and some are even better than that. In some ways, in fact, it's almost surprising how we can tell at a gut level the difference between 60 and 55 MPH. It takes practice to be sensitive enough to notice and deliberately adjust your min speed within a couple of MPH.

What does this all mean? My goal with this article is simply to get you to think. Understanding why you would want to drive at 8/10ths, 10/10ths, 7/10ths, or whatever, and then what that means to actually do that; it's important to understand before you get on track.

Let's all agree that dialing your mental focus back even 1% is not something we're ever going to do.

Never forget that how and what you practice will become your norm. If your goal is to drive at 10/10ths at some point, practicing at 8/10ths is not helping. Okay, for short periods of time and on specific areas of the track while you're developing a new skill, driving at less than 10/10ths is a solid practice strategy. If you can't do something at full speed, slow it down to practice it until it becomes well-honed. Then you can begin to speed it up. But practice the way you want to ultimately perform. If your goal is to race at 10/10ths, then you need to practice at 10/10ths. If your goal is to always drive at 8/10ths, then drive all the track, all the time, at 8/10ths.

However, picking the areas of the track where you're willing to drive closer to the limit, and others where you're not, is okay, too. If you're not getting paid the big bucks to drive (and even if you are, to some point), then balancing risk with reward is a part of your "job." For example, you may want to drive at 10/10ths in a corner that has a huge runoff area, and 8/10ths through a corner with a wall on the edge of the track.

Think about it.

Day #287: What's behind you is not as important as what's beside you.

After all the years of coaching thousands of race drivers, including some elite-level pros, I get positive feedback about one thing more than anything else; and it's not what you might think. It’s not the importance of the brake release. Not how to maintain mental focus. Not how to use one’s vision. Not the super-trick cornering line. Not how to use data and video. Not what to do with the throttle. Not how to dial in the car’s handling.

So, what is it? What has generated more comments than anything else? How to adjust and use one’s mirrors. And this applies to driving on the street as much as on the track. In fact, it starts there, to build the mental programming.

If you can see the side of your own car in your side mirrors, they are not adjusted properly. Got it? After all, when was the last time your car ran into itself? Seriously, why do you need to see the side of your own car? You don’t. And when you do, you’re setting yourself up for missing someone else’s car. In most vehicles, seeing its sides causes blind spots, and those blind spots don’t need to be there.

Let’s start with your street car. Sitting behind the wheel, look to your left-side mirror. Now, tilt your head until your left ear is over your left shoulder, and look into the mirror. Adjust it out until you can just barely see a sliver of the side of your car. Sit with your head up straight, look into the mirror, and you should now not be able to see the side of your car. Now do the same with the right-side mirror: tilt your head to the right, until your right ear is over your right shoulder, and look into the right-side mirror. Adjust it out until you can just barely see the side of your car, and it’ll be set.

Doing this, you'll find that your rearview mirror is used just for that — looking directly to the rear — and the side mirrors are used for looking to the sides. Too many drivers use the side mirrors to look to the rear. And don’t give me the excuse that you need them when backing up. That’s what the rearview camera is for, that's what turning your body and looking over your right shoulder is for, or worst case, what tilting your head to check your side mirrors is for (which is easy at the low speed you’re traveling when reversing).

There's an advantage to adjusting your side mirrors in this new way. You don’t have to turn your head to do a big shoulder check when changing lanes — a shoulder check that takes away your view of what’s in front of you. Sure, this is going to take a little time to get used to, but it’s a habit you need to build.

With blindspot detection systems on modern cars, this seems to be less of an issue, but it's still worth having your side mirrors adjusted properly. If nothing else, it's building the right habits.

What about on the race track? Same thing. But with a helmet, head and neck restraint (you use one, right?!), and perhaps a window net and wrap-around seat, your ability to shoulder check before moving to the inside in a braking zone or turning into a corner is restricted even more. That’s why it’s critical that you adjust your side mirrors to look to

the sides. Remember, it's more important to know what's beside you than it is to know what's behind you. Most crashes on the track between two cars happen side-to-side, and not like a rear-ender on the street.

So, what about keeping track of what's behind you? Yes, that's important, but not as important as what's beside you. So, use your rearview mirror for that. If you don't have one, or the view is severely restricted, then and only then do you use your side mirrors. Still, your side vision is most important. And here's the thing: when looking to the rear, you usually have a bit more time to tilt your head to look behind you. When turning into a corner, you don't have time to tilt your head to see if someone has tried to dive down the inside to out-brake you. You need to be able to see that with a glance.

Adjusting and using your mirrors is a compromise. But in the trade-off, always give the advantage to what's beside you over what's behind you.

Day #288: Corner entry speed is more important than late braking.

Ask a driver to shave half a second off their lap times and nine times out of ten they will say, "I can brake later for turn ___."

And my response to that strategy is that simply braking later — typically combined with braking harder, compressing the brake zone, and entering the corner with the same speed — will result in very little gain.

But when you think about improving your lap times using braking, you have to look at four options:

1. Compress your brake zone, by braking later and harder. As I just said, your corner entry speed will stay the same, but you will have extended the length of the straightaway.
2. Move your entire brake zone in — start and end later, extending the length of the straightaway, and increasing corner entry speed.
3. Release the brakes sooner, resulting in higher corner entry speed (but that will change the way your car handles on the entry of the corner).
4. Brake lighter — do everything the same, but just brake lighter, resulting in higher corner entry speed.

Which approach is best? I don't know, specifically for you, in your car, at the track you're at now, and in the corner you're working on right now. But let me give you a generalization: a small gain in corner entry speed — assuming it doesn't have a negative impact on your exit speed — will improve your lap times more than a big change in how late you brake. In other words, increasing corner entry speed will help more than extending the length of the straightaway by braking later.

Of course, everything is relative. If you're braking waaaaay early for a corner, then yes, braking later will gain you a lot. But if where you begin braking is close to maximum, then moving it in even more will have less positive effect than carrying a little more corner entry speed. And when I say "a little," I mean even one to two MPH.

Day #289: One driving style does not suit all cars, corners and conditions.

One of the key areas that separate good drivers from great drivers is their ability to adapt their driving style to suit the car's handling, each type of corner, and even from one car to another.

Some drivers, no matter how the car is handling, will only drive it one way — with "their" style. A driver's style will never suit every handling characteristic. And if you cannot adapt your style to suit your car's handling, a change in track conditions, or a different type of car, you'll not be as good as you can be.

In the perfect world, if your car has a handling problem, you would make an adjustment to improve it. But you don't drive in a perfect world. You have to adapt your driving to your car's needs, at times.

I'm not saying that you should always "drive around" your car's handling problems. No, you should do what you can to tune its handling to suit your needs. But there comes a point when you can't do that — while driving on the track, for example! That's when you need to adapt your driving to your car.

There's another reason not to adjust your car's setup. There are times where making it handle better in one corner could make it worse in a few other corners.

As an example, practically every car I've ever driven at Laguna Seca understeers in Turn 3. If you tune the car to get rid of that understeer in Turn 3, it oversteers in almost all the other corners, resulting in worse lap times. So instead, I've had to adapt my driving to help minimize the understeer in Turn 3, so I can have the car work well throughout the rest of the track.

All of the various skills and techniques you use to get around a race track quickly is the science of driving. But it's how you adapt, compromising one skill or technique for another, that is the real art of driving.

Day #290: Focus on what you can control, and leave what you can't behind.

I've coached more than one driver who faced the challenge of caring too much about what others thought of them. Some worried so much about it that they ended up being too nice to other drivers on the track. You could almost see these drivers say to a following racer, "There you go. After you..." as they let their competitor pass them into a corner.

Some drivers worry about appearing to be a jerk on the track, or that others might think they're not talented enough to be racing where they are. And they're worried about what others say about them.

When working with (or even observing) a driver like this, I'm reminded of the quote, "Your opinion of me is none of my business."

In other words, you can't control what others think of you, nor should you care. I'm not saying you should go around being a jerk and treating people poorly, having the attitude that no one else in the world matters. But when you're racing wheel-to-wheel, that is not the time to worry about what another driver thinks of you.

Okay, I know that you're smart enough to realize that it is important for other drivers to think of you as being a smart driver, and not one that makes stupid decisions. But that doesn't mean that you have to let others by to show how nice a person you are.

Understand that the decisions you make — whether to be a nice guy and let someone pass, or the way you respond to what others think of you — are built into your subconscious. It's part of your mental programming. It means that it's deeply ingrained in your mind, and not something that you even think about consciously. That's why you might even ask yourself after the fact why you did what you did.

But the good news is that it's part of your mental programming, and that's something that you can deliberately change through the use of mental imagery. You can adapt your behavioral traits, and how you react to what others around you say and do.

Day #291: Move on.

When some drivers are held up by a slower car on the track, they get frustrated, angry... and worse. And even when they finally get past, they're ranting and raving about the fact the other driver didn't move out of their way.

Other drivers find a way to get past, and then immediately focus ahead. They move on. They don't get caught up in the past.

If there was a lesson that performance and race driving teaches us that can be applied in many other areas of our lives, it's this: Move on.

You can't do anything about the apex you just missed, other than be aware of it and focus on it the next lap. But you can't go back and fix it as you're passing it. No, you need to move on.

If a driver blocks and holds you up, after you pass them, there's nothing constructive that you can do. Ranting about it, and waving at them will do nothing. Move on.

How many times have you been told to look farther ahead and where you want to go on the race track? Many times, I suspect (I hope). That applies to more than just the physical act you perform with your vision. It also applies to where you place your mental focus. If you're focusing on the mistake you just made, how well will you drive the next corner? If you're focusing on the driver that just held you up before you made a daring pass, how well are you setting up your next pass? Sure, physically look ahead and where you want to go, but mentally focus ahead, too.

And when stuff happens, move on. It's a life lesson you can practice on track.

Day #292: Ask yourself, "If I could have the car do just one thing better, what would it be?" Then, keep digging with more questions until the answer is obvious.

I like to use a process to figure out what the car needs to make it faster. I ask myself this series of questions:

Quick Debrief -

- Is the handling better or worse?
- If I could make the car do just one thing better, what would it be?

Detailed Debrief -

- What is the car doing? Understeer, oversteer, or neutral?
- Where is the car doing it? Which turn(s)?
- Where in the turn(s)? Entry, middle, exit?
- What am I doing when the car does this?
- Braking?
- Trail Braking?
- Releasing the brakes?
- Coasting?
- Maintenance throttle?
- On throttle, accelerating?
- Slowly turning the steering wheel?
- Crisply turning the steering wheel?
- Steady steering?
- Unwinding the steering?
- Is it the car or me? Am I inducing the handling problem, or is it the car?

By the time I've gone through all of these questions, typically the answer — what needs to be done to the car to make it faster — becomes pretty obvious.

Of course, it's one thing to know exactly what you want the car to do, and a whole other thing to know what to change on the car to make it do that. But without the first part — understanding what the car needs — you'll never figure out the second part.

Day #293: If at first you don't succeed... adjust your techniques and try again.

WD-40 was named that because the first 39 attempts failed. Its name stands for "water displacement, 40th attempt."

James Dyson's first 5,126 prototypes for a revolutionary vacuum cleaner failed before he nailed it.

Software maker, Rovio, spent 8 years, almost went bankrupt, and failed on the first 52 attempts to create a game that would catch on with the public. Angry Birds was the result.

When I watch a Formula One, Indy car, or NASCAR race, I notice the number of mistakes the drivers are making. And there are plenty of them! You may think that drivers at that level don't make mistakes, but they do. A lot of them. Perhaps more than you do.

The difference? First of all, like the makers of WD-40, Dyson, and Rovio, they didn't make a mistake once or twice and conclude that it would never work. This may be one of the most common mistakes I see non-pros make: the mistake of making a mistake and concluding that what they were trying won't ever work.

A mistake is a guide towards what will work. Just because the amount of trail braking you just tried didn't quite work, that doesn't mean it will never work. Maybe you need to change your turn-in point, and in doing so, you'll be faster. Or the earlier apex that doesn't seem to work just needs a different amount of trail braking to rotate the car enough.

The other difference between those who make mistakes work for them and those who don't is how soon they recognize the mistake. The earlier you recognize that you've made a mistake, the sooner you can begin adjusting or correcting for it. And the sooner you do that, the smaller your correction is going to need to be.

The best drivers make just about the same number of mistakes as anyone else. The difference is how they deal with them; the very best minimizing them by identifying and correcting for them almost immediately — to the point where, from the outside, it's almost impossible to notice the mistake they made.

As I've written about in the past, the more references — things you see, feel, and hear while driving around the track — the sooner you'll recognize your mistakes. When you first start to drive on a track you learn that there are three main reference points for each corner: turn-in, apex, and exit points. And those are likely the only references you have. So when you turn in too late or early, it's likely that you'll not even notice your mistake until you're passing the apex, and therefore, your correction is going to be fairly large. But if you have more reference points — you feel a crack in the pavement just after the turn-in point, you hear the tires making a different sound when going over a

different surface, you notice the angle your car is on in comparison to the turnworker station — you will begin making your correction earlier. And that means the adjustment or correction will be smaller.

The take-aways from what I'm saying here:

1. Just because a change in line or driving technique doesn't work the first time or two, don't give up on it.
2. The more references you take in with your vision, your feel, and your hearing, the better you'll manage the mistakes that you make.

Day #294: There are two kinds of drivers: Those who have crashed, and those who will crash.

Drive around a track at high speed for long enough and eventually "it" will happen.

In fact, I believe that you could take the two best drivers in the world, put them in equal cars spaced out at opposite points on a circuit that is over twenty miles long, and eventually they will find a way to run into each other! It's the nature of the beast.

The sooner you accept that it could happen, the better. Now, I'm not saying that for sure you're going to have a massive crash. In fact, when you really look at our sport, it's surprisingly safe. While I don't think anyone has ever proven this, my gut tells me it's safer than driving on the street. So, you may drive and/or race on tracks for decades and never have much more than a minor spin.

Yes, there are many levels of "crashing." It may be a heavy impact, it may be a relatively harmless spin without any impact at all, or something in between.

Still, if you drive anywhere near the limit, there is risk.

The fastest drivers I know have the attitude that "it will happen, but not to me." Yes, there's a bit of denial there, and perhaps even an unrealistic amount of it. But if you're constantly believing that you're going to crash — and be hurt bodily or financially — it's unlikely that you'll consistently drive at the limit. Of course, that's okay if your goal is not necessarily to drive at the limit.

I think it's important for drivers to take a little time to consider what level of risk they are willing to take. And the time for doing that is not while trying to out-brake another car into a hairpin turn, or while taking that high-speed corner at full throttle. No, you should have had some thought about this topic before you ever get on track.

And what if you do have some type of crash? From talking to many drivers through the years after they've had an incident that resulted in some amount of damage to their car (and whatever else came as a result of that), it seems there are those who drive slower, and those who drive faster. It's easy to understand why someone might drive slower after having a crash, but why drive faster?

For many drivers, it's the realization that it's okay, that it wasn't as bad as they thought it would be. I've talked with many drivers who came to the realization that their car can be repaired (or replaced), and that with the use of proper safety equipment in a well-designed car, being injured is extremely unlikely. And now that it's "out of their system," they actually feel "free" — like they don't need to tense up and fight the chances of crashing.

Some food for thought....

Day #295: Use Traction Sensing Sessions to improve your ability to drive at your limit.

Let’s talk about "skill development,” specifically as it applies to sensing the limits of a car.

Some drivers seem to have the ability to sense traction — how much their car has, and where the limit is — better than other drivers. But is that something they were born with, or something they developed? I'm not going to argue the case of nature versus nurture here, but I will tell you that the best drivers have developed their ability to sense the traction limits of their car.

How? Some through massive amounts of seat time. Others through very deliberate and strategic seat time. In other words, they made the most of the amount of seat time they've had.

If you deliberately practice sensing the limits of traction, you will become more sensitive to it. In fact, if you take your very next session on track and focus entirely on sensing which tire has the most amount of traction at various parts of the track; on what the tires sound and feel like when they approach and possibly even go beyond the limit; on the weight transfer and dynamics of the chassis when at or near the limit; of what the steering feels like there… you will become more sensitive to the limits of your car. You will become one of those drivers who just seem to have a better feel for the limits of the car (or enhanced the ability you already have).

If you do what I've detailed above, you will have completed what I call a Traction Sensing Session. It's one of the best ways to develop your ability to sense the limits of your car.

Day #296: Learning a track is a process, not a talent.

A subscriber sent me an email a while back and said, "Some drivers just seem to learn tracks faster than me. That must be natural talent."

I don’t think so. Some may have a better ability to visualize a track and line, but is that natural ability? My experience tells me that it has more to do with what they've learned and practiced throughout their lives than it does talent. And there's a lot of research that confirms my experiential findings.

I read basketball hall-of-famer Bill Russell's autobiography a number of years ago, and one thing really stuck with me. He told the story of sitting on the bench in college, watching a teammate (one of the best rebounders at the time) play. He noticed his positioning, and the way he moved. He would then go home and illustrate what he saw, actually drawing on a piece of paper where this other player stood in preparation for a rebound. He mapped out all of his moves, and then mentally put himself in his place. He mentally rehearsed the positioning and movements. Russell went on to become one of the greatest rebounders the game has ever seen. Was that natural talent, or a learned process and mental program?

When I as a kid, around ten years old, a relative gave me a book for Christmas that had layouts of all the tracks in England. I traced those tracks onto paper, and then drew what I thought the ideal line would be around all those tracks. I would "dream" about driving those lines around those tracks, behind the wheel of a Lotus Formula One car (Lotus was my favorite!). To this day, something that I'm good at is learning a track quickly. Is that natural talent? No. I've developed a process for learning a track, and it works for me.

So, what is it - what is my process for learning a track quickly?

1. I start with a track map. I memorize the layout, imagine the line that I'll drive, and then actually draw it on the map.
2. I spend a lot of time (and I mean a lot) imagining driving the track, doing mental imagery of the track, and me driving it. At this point, if I've never seen the track, the mental image I'm driving is vague, and I'm focused on just the direction and basic line. I'm not worried about elevation, banking, or anything else. I'm just getting a rough layout programmed into my head, and I think this is important to have before moving to the next step.
3. I watch almost any in-car videos that I can find. Some are better than others, but I can learn something from even the worst ones.
4. If possible, I'll drive the track on a simulator. I drive the track at speed, but also at walking speed. I do a "virtual track walk" on the sim, as long as I know the quality of the track images are good (iRacing is the best, in my experience).
5. I do more mental imagery of driving the track, based on what I learned from the videos and simulator.

6. When I first get to the track, I look at it from as many perspectives as I can from the spectator areas. This helps give me the "global perspective" of the track.
7. When I first get on track, I focus on soaking up as many references (things I see, feel, and hear) as possible.
8. After the first session on track, I sit down with a track map and make notes about what I learned, and what I think I need to do next time on track.
9. After I make notes — and not before — I look at data and video if I have it. I don't look at it before making notes on a track map because I don't want it to influence my thinking, my own perceptions.
10. Before I go back on track, I do some mental imagery, preferably for a minimum of twenty minutes, imagining the details of how I'm going to drive the track in the next session, based on what I learned from my notes, and the data and video.
11. Now that I have the basic line somewhat down to the point where I don't have to put a lot of attention in it, I just drive the car. I focus on driving the car as close to the limit as I can. As I do that — driving the car, not the track — the car tells me where it wants to be. Throughout this session, I'll alter my line a little, based on what the car tells me. I experiment with the line during this session, but pay close attention to what feedback I'm getting from the car.
12. After that session on track I sit down with a fresh, clean track map and make notes again — of what I learned, what worked, what didn't, references, and what I want to work on the next session.
13. I look at data and video again to see what I can learn about my driving.
14. At this point I try to do a track walk. Yes, sometimes I do it before driving the track for the first time, but I find the track walk to be most useful after I have a session or two under my belt. I look for more references, surface changes, elevation and banking, and even where it's okay to go off track (and where not to). I like to walk the track, but I also like to ride it on a bike, as I can feel changes in the track surface better that way. And, I make a point of rubbing my shoe over the surface, and noting the difference in grip levels between different surfaces.
15. After thinking through the track walk, data and video, and my notes on the track maps, I sit down with another track map and think about what changes I'm going to make the next time on track — and make more notes (the act of writing things down cements them in my mind). Once I've thought about it, and put a specific, well-defined plan together for the next session, I get in position, close my eyes, relax, and do mental imagery of exactly what I'm going to do. I see, feel, and hear myself doing exactly what I thought about, but at a deeper level than just thinking about it. I'm pre-playing it, creating a virtual reality in my mind.
16. The process continues, looping through preparation (thinking about it, then doing mental imagery) driving the track (focusing on gathering more references, and driving the car at its limit so it tells me where we need to be), and debriefing (note-taking on a track map, reviewing data and video, and even talking to other drivers).

When you look at my list above, I hope two things become especially evident:

1. It takes effort, and saying, "I should get around to preparing to learn that new track some day" is not part of the process.
2. Never once do I say, "Rely on natural talent."

Finally, one thing that I've not mentioned yet, but is critical, is my belief system. I know I learn tracks quickly. I know that I have a process for learning tracks, that I've used it for many years. And that I've taught myself how to learn tracks quickly. I have confidence in my ability to learn tracks quickly because of all these things. With that confidence comes the ability to do it; with the ability to do it comes confidence.

Day #297: There's Always More.

Some recent “lessons” I’ve learned (or re-learned):

Strategy usually beats tactics. When I’m coaching a driver, it’s rarely the specific driving techniques or track knowledge that makes the biggest difference. No, it’s the strategy I’ve put in place for how to learn, for when to learn. There was a time this year when a driver I was coaching was getting good advice from another coach. That advice helped. But not as much as the overall strategy we had for the entire event, and I think this is something too many drivers and coaches forget about. It’s easy to just keep fixing things (brake later here, turn-in sooner for Turn 5, use 4th gear instead of 3rd, etc.), but it’s more important to prioritize these things, and have an actual plan for how to improve. If your engine leaks oil, you can keep adding more to it between sessions, but executing a plan to rebuild the engine will have a bigger impact in the long term.

Focused effort is super-powerful. Many drivers talk about wanting to be better (or the best), but fewer are willing to do what it takes to make it happen. Sure, putting more effort into the wrong thing is not going to help, but once you have the right focus, work ethic is what separates the best from the rest.

The basics still matter. No matter what level a driver is, focusing on the basics pays off. I’ve said it before, but I will repeat it here for emphasis: The advanced stuff that the best drivers do is simply the basics done better.

Patience. I hate this one! But it’s critical. Many people think I’m a patient person because of the way I coach. Wrong! I’m not sure there are any successful drivers who are super-patient people. But learning to adapt to become patient when it’s important is what leads to success. This can apply to when you’re driving (trying to get too much out of corner entry speed, setting up a pass, etc.), how you tackle car prep and setup, or your overall approach to your driving career. There’s a time for patience, and a time to not be patient. Learn to identify the differences, and adapt.

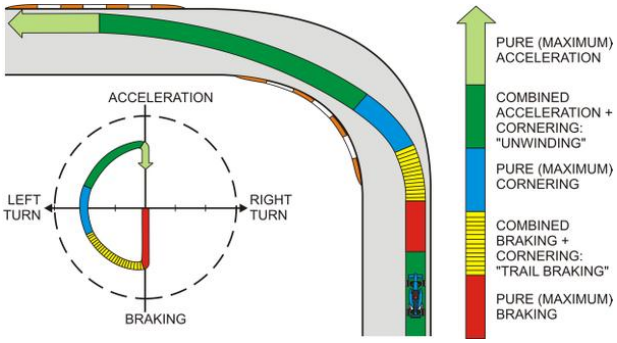
People still matter more than anything. Okay, I learn this every single year. It’s more important to have good people around you than some super-trick technical advantage or skill. Good people, all pulling together, will always win more.

Why. That was one of three key words that I focused on throughout 2019 (I pick 3 words each year to keep in the back of my mind to help me focus on what’s important), and it reminded me to drill down even deeper than usual into problems. And it resulted in better solutions. It kept reminding me of the importance of the basics, patience, strategy, effort, and people. In other words, if I hadn’t kept asking myself “Why?”, I would not have learned as much as I did — and I don’t believe the drivers I helped would have gotten as much from me as they did. I hope that applies to you.

Day #298: Overlap braking, cornering and acceleration to get the most of the traction circle.

You may or may not have heard of the "traction circle" (sometimes called the “friction circle”). Basically, it is a simple graphical measurement of the longitudinal and lateral g-forces your car experiences as you drive around a track. It's an X-Y graph, with the Y-axis representing longitudinal forces, and the X-axis the lateral forces. Imagine a heavy pen or marker hanging from a string attached to your mirror, with X-Y graph laying flat on top of your dashboard. Now imagine braking, and the marker swinging forward, tracing a line straight up the Y-axis; accelerating, and the marker tracing a line straight back and to the bottom along the Y-axis. Finally imagine go around a right-hand corner — the marker would swing to the left, tracing a line along the X-axis.

Of course, you don't have to imagine strings, markers, and graph paper on your dash, as there are a number of simple, but effective, smartphone apps available. They measure the g-forces as you drive, in real time, showing them on your phone screen (often with your location on the track, using the phone's GPS, as well as video looking out the windshield).



But what the traction circle really tells us is that we can use the tires' traction for braking, cornering, acceleration, or a combination of two. For example, we can use half the tires' traction for braking and half for cornering at the same time; we can use half for cornering and half for accelerating. And when we do that, we'll be using all of the capabilities of the tires and car — everything within the circle. If we only use a quarter of the tires' traction for braking while cornering at 50 percent, we're not using all of the traction circle.

In other words, if we want to use all of the tires' and car's capabilities, we need to overlap our braking with cornering, and then our cornering with acceleration. I often see drivers who are not using all of the traction circle, when that's what they want to do. For example, a driver who is following the "In slow, out fast" advice too carefully will not be filling the entire traction circle. This driver may brake at the limit, but by over-slowing for the corner, he then eases off the brakes and begins turning in. Eventually he is cornering at the limit — on the edge of the traction circle, but in between maximum braking and maximum cornering — and the tires were not at their full capability.

Your goal is to keep the tires on the edge of the traction circle. The cool thing is you can now use an app on your phone to give you feedback on whether you're using it all, or not. Hey, there's an app for that!

Day #299: Peel The Onion.

Far too often I hear a driver or engineer or crew member talk about the cause of a crash being a "mistake," and how important it is to learn from one's mistakes.

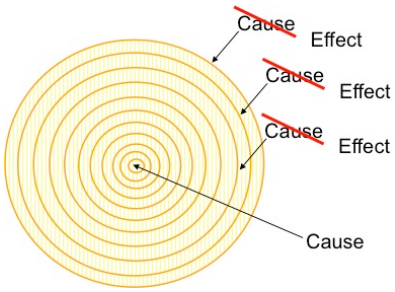
Here's the problem: How can you learn from a mistake when you haven't really identified what the cause of the mistake was?

Let me use an example to illustrate what I'm talking about.

Let's say you spin off the track and crash into a tire wall (I know, not a fun thought...). You made a mistake, right? But is that mistake the cause? No, it's an effect.

And what can you learn from thinking, "I made the mistake of spinning off the track and hitting a tire wall"? Nothing.

If you saw the movie, *Shrek*, you'll recall that the leading man... er, leading ogre said, "Ogres are like onions. They have layers." Well, so do mistakes. And to learn from a mistake, you have to peel back the layers of the onion to get to the core of the onion — the true cause.



So, you spun off the track. Peel off a layer of the onion, and you might find that you lost focus. But was that the cause — the cause that you should learn from? No, it's an effect. Peel another layer back and you may find the reason you lost focus; let's say it was because you were fatigued. So, is that the cause? Perhaps not.

Peel back another layer of the onion and you may find the reason you were fatigued: towing your car to track last night, and only getting four hours sleep. Lack of sleep is the cause of your crash. Or is it?

Putting off working on your car last weekend, so you had to finish preparing it just before leaving for the track yesterday resulted in you not getting enough sleep, leading to your fatigue in the last session of the day, causing you to lose focus, missing your turn-in point, running out of track at the exit, turning the steering wheel more, causing the car to spin.

Okay, now that you've peeled the onion, you can learn from the crash: Get your car prepared early.

Take a driving lesson from Shrek. Peel the onion, and learn from the true cause of your mistakes.

And remember, they're not mistakes. They're learning-takes... but only if you know the true cause of them.

Day #300: Make mistakes 15 percent of the time.

Daniel Pink, author of books such as *Drive: The Surprising Truth About What Motivates Us*, and *A Whole New Mind*, writes, "Researchers have long understood the general principle: We learn best when we teeter on the 'edge of our competence.' If a challenge is too difficult, we become discouraged and give up. If it's too easy, we become complacent and stagnate.

"But where exactly is the sweet spot?

"In a paper in *Nature Communications*, a team of psychologists, linguists, and neuroscientists have produced an early answer. They discovered that learning progresses most quickly when our error rate is about 15 percent. In other words, when we're practicing something new (a language, a musical instrument, a swimming stroke), we learn the most when we get it right about 85 percent of the time."

Hey, that's easy to say when one is talking about learning a language, a musical instrument, or swimming, right? None of these have the same consequences as when driving on a track (well, drowning while learning to swim can ruin your day). So how can we use this information to improve our driving?

Anyone who races or participates in any form of performance driving wants to do the best they can. It seems to be in our DNA to want to drive well, and that's why we get coached or instructed, and we practice. But I wonder if striving to be the best we can, and not wanting to make any mistakes means we're missing out on a great opportunity to be even better?

I talk about mistakes as being learning-takes, because they're one of the best ways to learn, but I know you want to avoid the costly ones. But what about the non-costly ones? This is why we all should spend time on a skid pad with a great coach (let me know if you need one, as I have one to recommend — and no, it's not me). It's why you should go to a rally school to spend time sliding a car on a loose surface. It's why you should be excited about driving in the rain, and not dread or avoid it.

If you find yourself not making mistakes, how much are you learning and improving? Again, while you strive to be the best you can (and that means not making mistakes), if you're not making errors, you're not improving as much as you can.

"Practice doesn't make perfect, only perfect practice makes perfect." But if you're making mistakes, aren't you practicing making mistakes, and therefore not "practicing perfect?" Sure. But I'm not suggesting that you repeat mistakes over and over again, practicing and solidifying them again and again. But if you never make mistakes, you're not learning. Whether you're willing to go as far as the researchers suggest and make mistakes 15 percent of the time, that's up to you.

Day #301: Check your mirrors as often as it takes to always know where others cars around you are.

While driving your everyday vehicle, sitting upright and happy behind the steering wheel, when you look into the side mirrors, do you see the side of your own car? If you answered "yes," my second question to you is, "Why? When was the last time your car ran into itself?" If you answered "no," and you cannot see the side of your own car without leaning your head toward the mirror, a high-five for you.

If you can see the side of your own car, you could be missing the side of a car next to you. In fact, by having your side mirrors adjusted to see the side of your own car, you've created the deadly blind spot.

This applies to track driving as much as it does to driving on the street.

Okay, many new cars are equipped with a small convex mirror, or warning devices to let you know if a car is in your blind spot. But the reason you need these is because you've had your side mirrors mis-adjusted.

Here's how you should adjust your side mirrors:

- Sitting behind the steering wheel, lean your head to the left, until your left ear is about in line with the outside of your shoulder. Look to the left side mirror, and tilt it out until you can then just barely see a small part of the side of your car. Straighten your head up, and look to the left side mirror — you will not see the side of your car. Don't worry, it's still there!
- Now, lean your head to the right, until your right ear is about in line with the outer portion of your shoulder. Look to the right-side mirror, and adjust it out until you can then barely see the side of your own car. Sit up straight and you'll notice you can't see the side of your own car.

That's it. You've adjusted your side mirrors to do what they were meant to do, which is to provide you with a view to the sides of your car, and not behind you (that's what the rear-view mirror is for, if you have one).

Think about this. When on the track, what do you care the most about, cars directly behind you or the ones to the sides? The ones to the sides of you, right? Because they're the ones trying to pass you; the cars directly behind you are behind you — not much need to worry about them.

For sure, when you switch over to this mirror adjustment, it will take a little time to get used to it. But not long. For most drivers, it's a matter of days to a week or so for this to become a habit, and for you to feel more confident knowing what's around you, whether you're driving on the street or the track.

So after you've set the side mirrors properly on your street car, drive along a two-lane road or highway and make note of how much better a view you have of the cars around you (without having to turn your head so far to check over your shoulder that you risk not seeing a car stopping in front of you). As you pass a car that's in the right lane, notice that you'll see the car in your peripheral vision, and then just before it disappears from view, you'll pick it up in your right-side mirror. Then, just before it disappears from your right-side mirror, you'll see it in your rearview mirror. All of this without turning your head excessively.

Do the same check when being passed by a car in the left lane. The car will go from your rearview mirror, to your left side mirror, to your peripheral vision.

If you're driving a race car that only has side mirrors, this applies just as much. Again, the cars to your sides are more important than the ones directly behind you. You want to be able to notice movement of a car to the side in a fraction of a second, without moving your head. For cars directly behind you, if you have to tilt your head a little to see them, that's okay. You have more time to notice them because they're not passing you right now, so taking the time to tilt your head to see where they are is fine.

Again, this may take a little while to get used to, but once you do, you'll wonder why you didn't do this years ago.

Day #302: Use your internal data acquisition system — your visual, kinesthetic and auditory senses.

Using my imagination, I see Mario Andretti and A.J. Foyt standing in Gasoline Alley, talking. It's the end of another day of practice during the "year of May" at Indy, and they're sharing stories about how their cars are handling, and how close to full throttle they were through Turn 1. A.J. is bragging that he flatted it once. Mario looks directly into his eyes, and says, "Yeah? Prove it. Show me your data. Let's overlay our data traces."

Of course, that never happened. Sure, they didn't have the opportunity to use data because it didn't exist back then, but I wonder just how much drivers like Andretti and Foyt would have used it, had it even been available.

Did Jackie Stewart compare data from his laps to another driver's to see where he could go faster? No. Even Ayrton Senna had little use for data overlays (yes, it was fairly primitive compared to today's systems, but he had it). They felt how to go faster. They figured it out themselves.

Today, too many drivers rely on simply copying what another driver is doing. They overlay their data trace over another driver's, see where he's going faster, and then go out and try to imitate the faster trace. That's learning from the outside-in, not from the inside-out. Mario and A.J., Jackie and Ayrton, and even Max and Lewis today, learned how to drive from the inside-out. They figured out how to go fast by sensing the limits of the car, by adjusting their line through a corner, by compromising speed in one place to gain in another. Data and video didn't tell them how to sense that. Their eyes, ears, hands, feet, and the rest of their body did.

That's all well and good for F1 and Indy car drivers, you say, but what about me? I don't have the time they do to develop those skills. I need to use data to speed up my learning process.

Nothing wrong with that. Believe me, I'm not saying you shouldn't use data and video. In fact, I'm a proponent of using it. What I'm not a proponent of is using it as a "crutch" for not learning to drive from the inside-out. I do believe there are drivers who over-rely on data and video, who essentially drive by copying what others have done.

There are club-level and some lower-level pro racing series where the process works like this: A hired "coach" gets into a client's car and goes out and sets a "data lap." Then, the "coach" tells the client/driver to go out and match that lap. The driver spends the rest of the event working on copying the coach's lap, learning from the outside-in. Don't get me wrong. It helps the driver go faster. But it's a short-term solution, because the driver hasn't learned how to figure it out, like Mario or Max have. Far too often, I've seen drivers who can go fast when they have someone set a data lap for them to copy, but they can't do it on their own.

Before you jump up and down and point out that Verstappen and Hamilton and every other F1 driver uses data these days, yes, I know that. And I'm sure they compare their laps to their teammates. But then they go figure out how to go faster than their teammate, not just copy them. And for the teammate that is already faster, he also goes out and figures out how to go faster still — using their eyes, ears, hands, feet, butt, and everything else. The data they all most rely on are their senses, not their data and in-car video.

Think about it this way: there are three ways to learn to go faster:

1. I can sit beside you and tell you what to do. You will learn what I tell you, but not much beyond how to listen to me and how to do what I say.
2. You can use data to compare yourself to other drivers and then go copy them. But you won't learn how to find the last little bits of speed that only comes from figuring it out on your own.
3. You can figure out on your own how to go faster. Maybe this is "old school," but ultimately you'll learn more.

Yes, there is a fourth, and that's some combination of these three. And I agree that you should use a combination of the three. Use someone talking in your ear, as well as data and video, but do not ignore the third option. Use data and video to confirm what you felt, and to look for trends in the data that tell you where you can improve.

Do I sound like an old fogey, pining for the good ol' days? If so, I don't mean to, because I'm a strong believer in the use of data and video. They're great tools, and I rely on them heavily when coaching, whether I'm working with a beginner track day driver or a pro racer. I'm sure if Mario and A.J. had had these tools back in the day, they would have been using them, too — anything to get an edge. But at the end of the day, they're tools, and what made Mario and A.J. as good as they were was all the time driving a car on the edge, sideways, locked up, standing on the throttle. They saw, felt and heard the edge. They figured it out.

I know that not every driver is a Mario or an A.J., and many participate in HPDE and track day programs — tens of thousands for every single top level pro racer. With the availability of low-cost data and video, it's time for a transition model of instruction. For a beginner, an instructor in the passenger seat, telling them what to do in real time, can't be beat. As this driver reaches the stage where an in-car instructor is less valuable, there should be a transition to instruction from outside the car — corner observation (exactly how most elite-level pros began when driving open-wheel cars), and data and video coaching. The transition continues with mentoring and real coaching, and then eventually, the driver is on their own.

That's a hugely valuable use of data and video. See, I'm not just an old fogey!

Use data and video, but don't do so blindly. Use it to reinforce what you've seen, felt, and heard. Use it to enhance what you see, feel, and hear.

I'll end with a funny quote I heard years ago — so long ago that I can't remember who said it, or the exact words. It went something like this: "Today, we have sensors on everything — the oil and water temperatures, the throttle, the fuel mixture. Everything. And they can tell me on the radio about all of this. It don't matter though. I'm gonna drive the son of a bitch till it blows up, anyway."

Day #303: There are two kinds of competition: external and internal. Learn how to manage them.

If we capture what we do under the umbrella of “motorsport,” then by definition, we participate in a sport with motors. Of course, that could include boating, airplanes, snowmobiles… and the list goes on. The list may be a little too broad for you (it is for me).

One thing that is a constant for most people participating in our sport is competition. But here’s where it gets tricky: There’s competition with others, and there’s competition with ourselves — external and internal competition.

I’m sure you'll agree that track day and HPDE (High Performance Driver Education) events are motorsport. But note that these are not competitive events, as in competing against others. Interestingly, if you look at pure numbers, there are more people participating in motorsport for the internal competition than there are the external kind. If you compare the number of people who participate in track day and HPDE events with those who compete in wheel-to-wheel or timed competition... well, there are many more of the former.

Look at running, and specifically marathons. Out of the hundreds of thousands (millions?) who run marathons every year, how many are competing to win? Or even to place well in the standings? Most compete with themselves, challenging themselves to beat their personal best.

Motorsport is similar. In fact, the ratio of external to internal competitors may be very similar to marathon running.

My point here is simple: Just because someone doesn’t race wheel-to-wheel with others, or against the clock, doesn’t mean they’re not participating in motorsport. On the flip side, those who do — while actually in the minority — are no less important to the overall motorsport world.

There are as many reasons for participating in motorsport as there are participants.

You’ve probably heard me say this before: Focus on your own performance, and the results will look after themselves. This is aimed at those who compete with others, and what I’m saying is that when you focus on maximizing your own performance, you’re more likely to beat your competitors; when you focus on your competitors too much, you’re likely to not perform as well as you could, and, therefore, not get the result you’re after.

As I watched the marathon from the London Olympics on TV (years ago!) I thought about how similar it is to auto racing (the external competition side of motorsport). For the top dozen or so runners, much of the race is spent focused on their own performance, pacing themselves, making sure they have the reserves for the final mile

or so. Then, as the marathon gets closer to the end, there is a gradual switch, as a lot of strategy is being played out — when to show your strength, when to reserve it; making sure to be in the right position; understanding the competition's strengths and weaknesses; strategizing when to make the final sprint to the finish, while keeping in mind one's own performance abilities. This last piece may be the most important, as you see runners make the mistake every now and then of pushing too hard, too soon, and fading at the very end — they didn't stay focused on their own performance enough, and got caught up in the competition too much.

This same thing happens in auto racing, but it's often harder to see from the outside than it is when watching runners.

To sum up, whether you're competing against others or not, ultimately you're competing against yourself first. If you're participating in non-competitive forms of motorsport, then you should stay focused on bettering your own personal best. If you're participating in motorsport where there is external competition, the longer you stay focused on your own performance, the more likely you are to get the result you're after.

Day #304: Turn the steering wheel quickly in slow corners, and slowly in fast corners.

Have you noticed that not all corners on tracks are the same? Well, duh!

Some corners reward a fairly quick, or what I'd call a "crisp" turn-in. Some corners require a slower, or more progressive turn-in. My point is that you should not be turning the steering wheel the same in every corner.

I've said it before, and I'll say it again: I don't like rules. I don't think of my Speed Secrets as rules. They're guidelines. They apply more times than they don't apply, but there are always exceptions to the rules (and the guidelines!).

With that in mind, think about how you turn the steering wheel when entering a fast corner. Most of the time it's a gentle, relatively slow (nothing is done too slow on a track), and progressively. You don't snap the steering wheel; you build up the cornering forces; you arc the car into the corner.

Now, think about how you enter a slow, tight corner. You turn the steering wheel quicker and crisper. There's not enough time to slowly arc the car into the corner — you need to change its direction, now.

Having said that, every movement you make with the steering wheel should be done as smoothly and gently as possible. But the speed at which you turn the steering wheel should be different, depending on the shape and radius of the corner.

A number of years ago, I viewed more than an hour of in-car video clips from every driver in Formula One over the course of a season. That year, Michael Schumacher fought with Mika Hakkinen; Damon Hill, Jacques Villeneuve, and others were on the grid, as well. But it was obvious to me, watching all of these great drivers, that one moved the steering wheel less than the others: Schumacher.

Being aware of how you turn the steering wheel (and making any changes) can be... well, a bit distracting. It's something you probably haven't thought about in a very long time — maybe since you drove that little toy car around in the driveway when you were a kid. So, as you head to the track for your next race or track day, be aware of the rate you turn the steering wheel, but also know that it might feel awkward for a while. Like any activity, when you think about it — and do it at a conscious level — you don't perform it as well. But with practice, it becomes ingrained, and you'll be even better than you once were.

Day #305: The best way to learn is to ask questions. Isn't it?

Here's a question I received from a reader: *"How can I get back into my groove after a horrible season of racing? Car broke too often; I was distracted by events at work; when I did drive, I drove poorly; made mistakes that I shouldn't have; lost overall speed; lost confidence in the car; all of which resulted in poor results. Much of the above is easily addressed (like work, changing how I prep the car to add reliability, performing some off-season maintenance that is due), but I'm at a loss for how to get the mojo back without wasting lots of time."*

Unfortunately, this is not the only time I've been asked this question, so here's my response.

Sorry to hear your season didn't go as you hoped. But let me ask you a question: At any time during this year, did you ever wake up in the morning, look at your bed and notice all of your talent and skills had drained out of your head overnight and dripped onto your pillow? No? Then you have all the talent and skills you've ever had — they don't just go away, you don't lose them. That's the good news.

The challenging news is that sometimes it gets harder to access all of your talent and skills, and that might be what you're feeling. But again, you still have the ability to drive as well as you ever have.

You do what you do because you're programmed to do so. You sometimes don't do what you want because you either don't have the programming (the skills) to do it yet, or because every now and then you access the wrong program — or you just don't access a program that you have. It's important that you understand that.

If you feel it's hard to get back in the groove, it might be because you're just not accessing the right mental programs. But again, they're still in your brain. What gets in the way of accessing the right programming? Often, your state of mind — your emotions, your mood. For example, if you're frustrated or anxious, there is going to be some restriction in how you access your mental programming. So be in a good state of mind! Okay, that doesn't really help much, does it?

One way — the best way I know of — to trigger a "performance state of mind" (the state of mind that will allow you to access all of your skills and talent — programming) is to replay a past success. When you vividly replay a past success in almost any activity, it puts you in a great state of mind. It's hard to be in a bad mood when you're seeing, feeling, and hearing yourself doing something in the past that just felt fantastic. It could be driving, it could playing another sport, a musical instrument, in your work life, in school, whatever. But when you recall it, and you recall the emotions that you felt at that time, you almost can't help but have a little smile on your face, and be in a performance state of mind. So, do that.

Then, do mental imagery (what's typically called visualization) of you getting back in the car, all of your car prep having paid off, your work under control, and you driving as good — or better — than ever before. In fact, by using mental imagery, it's possible that you will perform even better than you ever have by simply replaying a past success, and then pre-programming how you're going to drive. Remember that mental imagery should use as many senses as possible, so you even feel the emotions that go along with it. Do mental imagery of this every evening for about fifteen minutes, every day for three or four weeks (okay, you can take a day off every now and then — but no more than once per week), and I know for sure that you'll drive as well as you ever have.

Will things go "wrong" every now and then in the future, and you'll feel like you're going to go back into that "out of the groove" mode again? Probably. But the good news is that if you do what I suggest, you'll have a strategy for dealing with it.

When you get back in your car, focus on what you can control: your own driving, your own performance. You can't control what happens with work (when you're at the track), or problems with your car (once you're on track). As you say, manage those things up front as best you can, and then when you're on track, focus on what you can control — you.

And never forget that we're not curing cancer here, and no one is going to die if you don't drive quite as well as you have in the past for a session or two. It'll be okay. It'll work out. You have the mental programming to drive the way you have when you performed at your very best — and you'll do that again. Especially if you trigger a performance state of mind by replaying a past success just before you go on track, and you do the mental imagery of performing great again for a few weeks (and on a regular basis after that), you'll return to performing well, again.

Day #306: Use your age to your advantage.

We've all heard the stories about Paul Newman not starting to race until he was 48 years old, and Juan Manuel Fangio winning five World Championships while in his forties. But these seem to be exceptions to the rule in a sport where youth is the norm — at least in professional racing. Hey, Max Verstappen made his debut in Formula One at age 17. But we're not talking Formula One here.

Motorsport — from HPDE events to pro racing — may be more accessible to older participants than other sports. No matter what your age — or level of ability — there is almost always some type of motorsport in which you can participate. But isn't there an advantage to being young in this sport?

Comparing older drivers to younger ones, I think it comes down to desire, effort, resources, experience, and fitness (physical and mental).

Desire
To me, the single biggest difference between a young driver and an older one is simply the level of risk one is willing to take. This mostly comes down to an awareness of what can happen, and how much the driver has to "lose" if something goes wrong. Many young drivers are naïve about what can happen – they don't think anything can happen to them — whereas some older drivers have seen enough in their lives to believe that it can happen to them.

This is the risk-reward balance, and ultimately it comes down to just how bad you want it. It's how strong your desire is, and whether you're willing to accept the risks that go along with it.

Effort
Related to this awareness of risk is the desire to put in the effort to do what it takes. Often, the older driver has many other demands on their time, so even if they want to put in the effort, it's difficult to do. But there's no doubt that to be the best you can be, it's going to take effort. It's going to take time and effort, both at the track, and away from the track. In fact, I'd say the biggest difference can actually be made away from the track, as the time on the track is so limited. If you want to improve, either to get ahead of younger drivers or to keep up, what you do away from the track (mental and physical training) is likely to make the biggest difference.

Resources
When I talk about resources, I'm referring primarily to money and time, as well as access to other people.

This can work in a few different ways: The driver with money to spend but very little time due to family and/or business life commitments; the driver who has gotten their professional life to a point where they have time on their hands but must watch the budget very closely; or some other combination. One thing that most older drivers have

as an advantage over younger drivers is access to other people — additional resources to help them improve — through a larger network they've built over many years.

Of course, desire, effort, and resources are interrelated.

Experience
In motorsport — no matter what type or level, competitive or non-competitive — experience is an advantage. Unfortunately, age doesn't necessarily equal experience. Nope, simply having another birthday doesn't result in more driving experience — only driving does.

What is experience? It's mental programming. Taken to an extreme, if you were to watch videos of cars crashing on a track over and over again, all day long, for months and months, your mental programming would be that "cars crash on the track." And, because that would be your experience, that would be your mental programming. Would that impact the amount of risk you'd likely be willing to take when driving on the track? Yes.

So, needless to say, I don't recommend watching video of cars crashing over and over again! In fact, what I do recommend is playing video of cars not crashing over and over again. How do you do that? With mental imagery.

We develop experience through physically and mentally doing things, so build your experience by playing "video" in your mind, with your imagination. How far you take this is completely up to you, but you can develop physical skills with mental imagery. You can build on your experience. This goes back to what I said earlier about how much you're willing to do away from the track.

Fitness
What about reaction times? You can look at this in two ways: your physical reaction to something (research has shown that the best drivers don't necessarily have superior reaction times), and your "anticipatory reactions." It's this second one that we rely on the most when driving (and it seems this is the area that the best drivers have an edge over others). If you've ever had the experience of "just knowing" that a car was going to change lanes quickly in front of you on the highway, and you avoided it, then you know what I'm talking about. This comes from experience (your mental programming), so the more you drive — a benefit of getting older — the better your anticipatory reactions are going to be. Oh, and you can pre-play scenarios in your mind, experiencing reacting in the right way over and over again, thousands of times.

Of course, physical reactions are important in our sport. If you stay physically active, playing other sports and working on your fitness, you can maintain your reactions far longer than most people think. I strongly recommend playing sports that require good hand-eye coordination and reaction times, such as tennis, squash, basketball, and volleyball. Mountain biking may not put as much stress on your reaction times

(depending on the type and level of riding), but it helps — and it's great for maintaining and developing your sense of balance, and great for overall fitness.

What about eyesight? Don't we lose our vision when we age, and doesn't that impact our driving? Sure, there are some physiological changes to the cones and rods in our eyes, and that does impact how well we can see. Typically, it hurts our central vision acuity the most, the part of vision that is tested when we read the eye chart at the doctor's office.

But there are things we can do to maintain our overall vision skills (peripheral, contrast, color, depth perception, and the ability to sense movement — the stuff that matters most, more than just being able to read the eye chart). Borrowing from the field of sports vision therapy, the two exercises that I recommend to drivers are Lazy 8s and Focus Stretches. Done on a regular basis, you will definitely notice either that your vision is staying strong, or it actually improves (if it has deteriorated to some extent, over time). Now, these exercises cannot perform miracles and cure blindness, but they do make a difference. Combined with experience, your vision may not have any limiting effect on your driving performance. But you do need to do the work. (Again, I highly recommend these two exercises, but it's also worth doing some research on sports vision therapy, and working with a therapist).

I also recommend brain integration exercises, like Cross Crawls. They help your overall mental fitness, and we all know that performance and race driving heavily stress our mental capacity.

To recap, what specifically can an older driver do to keep up and ahead of younger drivers — even if "ahead" simply means learning more in the non-competitive track day world?

- Consider just how much risk you're willing to take, and how much effort you're willing to put in, and be realistic about how far you can go, with these in mind.
- Get as much experience as you can, both on the track and off.
- Work on your fitness level with regular exercise.
- Play sports that benefit your reaction times and balance.
- Use mental imagery to manage your risk assessment, and develop your anticipatory reactions.
- Use mental imagery to build on your experience, to develop new skills and techniques.
- Use exercises such as Lazy 8s and Focus Stretches to maintain and improve your visual abilities.
- Use Cross Crawls and Centering to enhance your brain's ability to process information, and enhance your sense of balance.

Research over the past couple of decades about neuroplasticity, or our ability for our brain to adapt and grow is emphasizing the opposite of what we've been told in the past: You can't teach an old dog new tricks. You can. You just need to use the right strategy.

Having said that, as the saying goes, all good things must come to an end. How does one know when to hang up the helmet?

Ultimately, that's a personal decision, but to me when the effort outweighs the benefit, it's time to call it quits. When desire is overwhelmed by the physical and mental strains (and possibly the strain on resources).

What about when one's skills have diminished to a point where there is a danger to yourself, and others on the track? For sure, if the risk has risen to a level that you and others find unacceptable, it's time to find something else. But that doesn't mean you have to leave motorsport completely. There are other types of motorsport that might fit your level better, and even if you do stop driving there are many things you can continue to do (anything from organizing events to mentoring and instructing).

Becoming a better driver is all about listening and learning from others. The same is true when deciding when to quit: Ask and listen. Make learning an objective — use your participation in motorsport to practice learning.

Ultimately, how long you participate in motorsport is up to you and your desire. If the desire is strong enough, you can overcome most age-related so-called limitations — and maybe even turn them into an advantage. Imagine that.

Day #307: After eating, focus on how you feel and perform, rather than on how the food tasted.

When I was in the prime of my racing career (if I ever had prime!!!), I experimented with a number of different diets with the goal of finding the one that would give me even the slightest edge over my competition. Vegetarian, vegan, high-protein, high-carbs, blood-type... you name it, I tried it. And I gave each a proper test drive, over many months (or years).

Now, I'm not a nutritionist, diet expert, fitness trainer, or anything like that. I'm a guy who used his own body and performance as a laboratory. I experimented on myself to see what would help me perform at the highest level. So I'm not saying that what I recommend is necessarily the right thing for you. Personally, I don't believe there is a one-size-fits-all diet that suits everyone (and remember what most "diets" are meant to do: help you lose weight, not perform an athletic activity better).

I found, over time, that when I focused on how I felt and performed after eating certain foods, I began to crave food that helped me perform better. When I simply focused on how the meal tasted that I had just eaten, I craved food that tasted the best — which was often not the food that was ideal for my personal performance.

Try it for yourself. Over the next month, at 30, 60, and 120 minutes after eating a meal, ask yourself how you feel. Then ask yourself how you've been performing during that period of time. Become hyper-aware of how both your mind and body feel and perform after the meal. You may find that an hour after eating a big, fat, juicy cheeseburger, that your mind doesn't feel as sharp, and that your body feels sluggish (or worse!). Pay no attention to how the meal tasted.

Again, run through this experiment for at least a month. And don't be surprised if after a month or more, you begin to crave foods that help you perform better.

Day #308: Ask. Dig. Ask More.

The best way to learn is by asking questions.

If you want to learn what your car needs to make it handle better, ask questions. Who are you going to ask?

First, ask yourself. Ask questions about what you're feeling, what you're hearing, what you're seeing.

Second, ask other people. It's not surprising that most people appreciate being asked a question, as it shows that you respect their knowledge and experience.

Of course, not all of the answers you're going to get from others are useful. Just because something works on their car, it doesn't mean it will work for your car. So, you have to filter the answers you get. But it's better to have the information, and filter it, than to not have it at all.

The best race engineers I know are very good at asking their drivers questions. They dig and poke at the driver, getting them to replay in their mind what the car was doing at various areas around the track. And once a driver answers a question, these engineers dig a little more by asking a follow up question, and then another, and then another….

You don't need an engineer to ask you these questions, though. If you ask yourself about what the car is doing while braking, during the corner entry phase, in mid-corner, and while exiting the turn — and most importantly, what you're doing with the brake pedal, the steering wheel, and the throttle in each of these areas — it becomes a little easier to understand what your car needs to go faster.

Ask the right questions, and the answers become obvious.

As noted football coach Lou Holtz said, "I never learn anything talking. I only learn things when I ask questions."

Day #309: Don't drive like a rat.

Here's a recap of a study done way back in 1907:

"Rats were trained to run in a straight, alley-like maze for a food reward which was located at the end of the alley. Watson found that once the rat was well-trained, it performed almost automatically on reflex. Upon learning the maze over time, they started to run faster through each length and turn. By the stimulus of the maze, their behavior became a series of associated movements, or kinaesthetic consequences instead of stimulus from the outside world. This routine continued until the length of the path changed, either farther or shorter.

"If the conditioned rats were released into an alleyway or path that was shortened, they would run straight into the end of the wall making a "kerplunk" sound. The first trial found that they would run full speed, passing up the food that had been moved closer. Shortening the alleyway, and moving the food closer was an early signal that was ignored by the rats.

"If the path was longer, the rats would run as usual until it reached their customary distance, the distance at which the food would normally be. They would then pause to sniff the area even though they had not reached the end of the alley, often ignoring food that was farther away."

(Click here to read the full Wikipedia page about the study)

It's kinda easy to laugh at the silly rats and think about how superior we humans (drivers!) are, but hold on a second. See, I've seen a lot of drivers do very much the same thing.

Many drivers ask me why they seem to be slower in relation to other drivers at their home track than they are at one that's less familiar to them. The rat study explains this.

I've observed more than one driver who drove their home track the same way they had over and over again in the past. They were relying on their habits, their mental programming, and that meant that they braked early and carried less speed into corners than the track and car could accept. Rather than fully using their visual, kinesthetic, and auditory senses to determine how much grip the track and their tires had (then using that), they did what they had always done in the past — began braking at the X marker, and slowing to X MPH entering the corner. They were essentially running "straight into the end of the wall, making a 'kerplunk' sound."

Ask yourself if you're driving like a rat, relying on your habits or mental programming, or whether you're using your senses to determine if the "alleyway" had been lengthened, allowing you to brake later and carry more speed into the corner?

Day #310: Plan your practice to maximize your learning opportunities.

Rather than going to the track with a practice strategy that is simply driving around and around to get seat time, develop a better plan. For example...

Start off with a basic setup on your car (one that you know will produce a fairly neutral-handling car), and run enough laps to get a baseline time. Let's say, you turn consistent 1:30 second lap times. Then:

1. Purposely make your car understeer by making a setup change. Change the handling by adjusting your tire pressures (make big changes, like 5 psi), shock absorbers (make sure you know what an adjustment will do to your car's handling), ride height, spring change, or whatever you know will cause your car to understeer. The goal is to make your car understeer, and not just a small amount.
2. Head out on the track and make note of how your car is handling, getting a feel for it as you note how much the understeer is affecting your lap times. Perhaps you're now lapping 1.5 seconds slower than you were before.
3. Continue lapping, but work at adapting your driving to the understeer. Experiment with trail braking more, and then less, to see which helps. Adjust your line, turning in slightly later or earlier. Alter how quickly you turn the steering wheel, and when and how you use the throttle. Your goal is to get back as close as possible to the baseline lap time of 1:30. With time — and adapting your driving technique — you may find you're able to get back down to within half a second of your baseline lap time.
4. Now, put your car back to its baseline setup (eliminate the understeer), and then make changes to cause your car to oversteer.
5. Head back on the track and do the same thing: find out what difference it makes, and then adapt your driving to get as close as you can to your original baseline lap time.
6. Finally, put your car back to its baseline setup, and head back on track to see how quick a lap time you can drive, now. Most drivers, having learned to adapt their technique, will turn faster lap times at this point — perhaps down in the 1:29 range. That's because you've fine-tuned your abilities to adapt to your car's handling, and you've built up your database of driving techniques. As a bonus, this practice strategy will make you better at adapting to a handling problem — something you likely can't do anything about if it happens in the middle of a race.

This is just one example of a practice strategy you can use, but it demonstrates the process of setting a plan, and then working that plan.

Day #311: Remember the Rule of 3.

The "Rule of 3" states that you should never go onto a race track trying to improve more than three things at a time. If you try to tackle four or more things, you'll be lucky to manage one. Our brains just aren't able to handle that much information all at one time without blowing up!

The second part of the Rule of 3 states that the three things you're working on need to be very clearly defined, and as specific as possible. A vague objective like, "I'm going to work on driving smoother" doesn't work. It's not specific enough.

Take the time prior to each and every session to identify one, two, or three things — but never more than three — and then define them as clearly and in as much detail as you can. Just the process of doing that will increase your chances of accomplishing these objectives.

The Rule of 3, and this process could be applied to an entire season, too. You could identify three main objectives, three specific techniques, three skills, three mindsets, or three strategies that you're going to apply to your driving this year.

I guess it comes as no surprise that I highly recommend you do this.

The author of a blog that I regularly read wrote about how he never wastes his time on New Years resolutions. But he started a routine a few years ago that made a big difference in his life: At the beginning of each year he decides on three words that he wants to influence his life for that year. An example he shares is a few years old. He chose “Ask. Do. Share.” Throughout that entire year, he reminded himself of these three words, and he found that it had a huge positive impact on his life. He asked to help others, and he asked others to help him; rather than thinking about doing something, he just went ahead and did it; he took the time to share with others what he had learned.

How can you apply this same routine to your driving? What are your three words for your driving performance?

Choose just three words that relate to the three areas of your driving that you want to improve. Write them down. Post them on your bathroom mirror. Write them in your "at track notebook" (you do have one of those, don't you?). Recite them to the tune of a favorite song, so you never forget them. Use them as a trigger to remind yourself what you want to do to improve your driving.

After all, it's that constant "drive to improve" that keeps us coming back for more, and more, and more.... And your "3 words" will help you with that.

Day #312: Copying others will rarely take your driving beyond theirs.

Many drivers have a pro or some other "better" driver take their car out and set a baseline lap; then they try to copy this lap. It's a relatively quick way of going faster.

But it's not the longest-lasting way of improving your driving.

If I tell you the answer to a question on a test or I ask you (and help you) to figure out the answer, which one will result in you remembering the answer the longest? Which results in real learning, and which just a "quick fix"?

So why do so many drivers take this approach? Because it is quick and easy — like many things that don't last.

In many race series, this is almost the norm, now. And it's a shame, because many of these drivers are not truly learning how to go fast on their own. They're simply copying what someone else is doing. And this is not limited to just racing — I often see it at track days, too.

The number of drivers I see who have hired a "coach" to drive their car, set a baseline/datum lap, then go out and try to copy what that "coach" did is surprisingly large. And guess what? There is a very large percentage of these drivers who make some quick gains, only to stall in their development as drivers, eventually getting frustrated and sometimes even quitting the sport. I see it all the time.

The driver who takes the opposite approach wins in the end. They do the work themselves, learning how to go fast through developing their skills, their ability to sense the limits of their car, how to figure out the line themselves, how to describe how the car is handling... everything that the best drivers have always done.

At the risk of repeating myself, I'll say this anyway: Did Mario Andretti, AJ Foyt, Ayrton Senna, or Jimmie Johnson have someone get in their car and set a data lap that they could then go out and try to copy? Can you imagine even suggesting that to someone like AJ Foyt?! He'd smack you with the nearest hammer!

Every now and then having someone else drive your car, so you can see what it's capable of, is not a bad idea. But if all you do is that, then go out and try to copy where that other driver went faster, you're not driving the car faster because you sense that it's capable of going faster. You're just going faster because someone else said it could go faster. And the next time you want to go faster, you'll need someone to set a fast time for you to copy again, and again, and again.

Instead, focus on developing your own ability to sense the limits of your car, how it responds to the line, and how your car is handling. Then, when you want to go faster (and when don't you?), you'll be able to do so with or without someone telling you how.

A "coach" who offers to drive your car to show you what it can do so you can copy them is doing you a disservice — and they're likely not capable of helping you learn on your own. I'd go so far as to question whether they're a real coach, or just someone who wants to drive other people's cars.

One of the main reasons people stay in this sport is because they enjoy the fun of improving. When I see drivers simply copying what others can do, I know that they'll likely hit a plateau... and often fade away from the sport.

There's a saying in racing: "If you copy your competitors, the best you'll do is finish second."

Day #312: Know and use trail braking when appropriate.

When I talk to drivers about trail braking, it's obvious that they have a few misconceptions — at least three.

The first one is what trail braking really is; second, if and when it should be used; third, whether a driver actually uses it or not.

All trail braking is just easing off — trailing off — the brake pedal while turning the steering wheel into a corner. It's blending the release of the brakes with the progressive turning of the steering wheel.

The reason to use trail braking is twofold: First, it keeps some weight transferred onto the front tires, which helps the car turn into the corner. Second, it ensures that you're using all of the tire's traction. If you lift off the brakes, and then start turning the steering, no matter how quickly you do that, there will be a small amount of time where the tires are not used to their maximum potential. By slowly releasing the brakes as you turn the steering wheel, you're trading off some braking traction for cornering traction — you're using all of the tires' capabilities.

Should trail braking be used all the time? Or none of the time? Remember the guideline: The faster and longer the corner, the less trail braking you should use and the earlier you need to be on the power. The slower and tighter the corner, the more trail braking you should use to help rotate the car. Of course, it depends on so many variables — how your car is handling; the radius and length of the corner; changes in surface, elevation or camber of the track; wet or dry track; and your driving style.

Finally, I've talked to many drivers who swear up and down that they never trail brake. But when I stand on the side of the track and watch their brake lights, it's obvious that they do. Or when I look at a brake pressure trace on their data acquisition system, we find that they do. Many drivers are unaware of how much they do or don't trail brake.

The next time you're on the track, simply become aware of how much you trail brake — be aware of how quickly and when your foot actually comes off the brake pedal, and where you are in relation to when you began turning the steering wheel. Some drivers are surprised that they actually trail brake more than they were aware of, or that they do it less than they thought. Either way, if you first become aware of what you're doing now, and then experiment with doing it more or less, you may find that you're able to drive a little faster than you have in the past.

I talk a little more about trail braking in a driving tip that I've posted on my YouTube channel at https://youtu.be/JfhNwKF4Csw.

Day #314: Know when to anticipate, and when to respond.

We're taught to anticipate what's going to happen, but sometimes it's best not to, and to just respond.

Example: You know that when you turn into that fast, sweeping corner, your car is going to understeer. So, what do you do? You ease out of the throttle before the corner. You anticipate having to slow for the understeer, so you do that — even before you need to. It makes sense, since you know you're going to need to take some speed off and correct the line due to the understeer.

But if you want to be fast (safely), often — but not always — you're better off carrying speed into the corner, and then dealing with the understeer. In this case you're responding to what you have, instead of assuming you're going to have it. Hey, who knows? Maybe the understeer will not be as severe as you assumed?

I know this is "radical" thinking, because we're taught throughout most of our lives to try to anticipate danger, problems, or whatever. But when I observe drivers, I notice that many of them are anticipating the worst, and responding in a way that doesn't help them. Okay, it might keep them safe, but at the expense of finding that next little bit of speed. And I say "might" because it's possible that it makes it more dangerous (I'll come back to this below).

I'm not suggesting that you take bigger risks and drive in an unsafe way. But anticipating, too much, what your car is going to do will result in being much slower, with very little added safety margin, if any.

Let me use a few corners as examples.

First, Turn 2 at Thunderhill. If you're not familiar with it, it's a fast, long, 180-degree left-hand turn. Many drivers anticipate what the car is going to do when entering the corner, and slow to a speed to make sure that that doesn't happen. They're definitely taking the "in slow — out fast" approach. The problem is that the "in slow" part is costing them a lot of lap time, since it's such a long corner — they're spending a lot of time (relatively speaking) driving "slow." Instead, if you enter the corner with more speed (I'd recommend a lighter brake pedal pressure so the car stays balanced better), then "learn" what the car is going to do (understeer, oversteer?), and respond to it. This corner has enough room for you to respond to most understeer or oversteer situations.

Next, the Carousel at Road America, and it's the same kind of thing. Drive it hard into the corner, collect the car in the middle, then drive it hard out of the corner.

And how about Turn 1 at Road Atlanta? If you've driven this amazing circuit, you know that when you turn into this corner, you're going slightly downhill; but shortly afterward, the track begins to climb up a hill. If you turn into Turn 1 with the speed that's appropriate if the corner had no elevation change, you'd be much slower than what is

possible. You need to turn into this corner with more speed than seems appropriate, and then respond to what the car is doing when it “hits” the uphill portion of the track. You need to trust that the track will catch the car, and then you’ll just need to respond by making small adjustments.

Turns 16-18 at COTA (also known as the Carousel) is another example, just like Road America’s Carousel: Drive it in hard, let the car tell you what it needs, and then respond.

You may have noticed the pattern here: these are all long, fast, sweeping corners. And you’ve probably heard the saying that many drivers drive the slow corners too fast, and the fast corners too slow. That applies to these faster, longer corners.

You may also be thinking, “That’s okay for someone who doesn’t need to drive their car home at the end of a track day, or for a pro with backup cars and a team to fix it when things go bad, but that’s too risky for me.” I get it. I understand. But it’s possible that over-slowing is actually increasing your chances of losing control. I’ve written and talked about what I call the “change in speed” problem in the past, but it works like this. If you slow too much when entering a corner, your “traction sensing” tells you that you have grip, so use it. At that point, you dip into the throttle, and begin to reaccelerate the car. Think about the car’s balance as you do this. First, the front is loaded, as you’ve transferred weight to the front while slowing; then, as you feed in some throttle, the load transfers to the rear. The shifting of the load from the front, then to the rear, unsettles the car, reducing its overall grip. You may feel like you’re driving the limit, and in a way you are. But you’ve created this lower limit by unbalancing the car. In doing so, you’ve actually created a situation where you’re driving less safely. If you’d braked just a little bit lighter, you would have carried a touch more speed into the corner and the car would have stuck — because it was balanced. Or, possibly it would have understeered or oversteered a little, and you’d have responded in a way that resulted in carrying more speed through the entire corner.

Anticipation is most often a good thing, but not when waiting to learn what you have to work with, and then responding appropriately.

Day #315: Get excited and carry on.

During World War II, England's Ministry of Information commissioned a public safety slogan: "Keep Calm and Carry On."

According to research, a "better" slogan might have been, "Get Excited and Carry On." That is, if they had wanted the public to have performed at their best, at their peak.

Alison Wood Brooks of the Harvard Business School in the *Journal of Experimental Psychology* (2013) wrote, "Individuals often feel anxious in anticipation of tasks such as speaking in public or meeting with a boss. I find that an overwhelming majority of people believe trying to calm down is the best way to cope with pre-performance anxiety. However, across several studies involving karaoke singing, public speaking, and math performance, I investigate an alternative strategy: reappraising anxiety as excitement. Compared with those who attempt to calm down, individuals who reappraise their anxious arousal as excitement feel more excited and perform better. Individuals can reappraise anxiety as excitement using minimal strategies such as self-talk (e.g., saying "I am excited" out loud) or simple messages (e.g., "get excited"), which lead them to feel more excited, adopt an opportunity mindset (as opposed to a threat mindset), and improve their subsequent performance. These findings suggest the importance of arousal congruency during the emotional reappraisal process."

Prior to heading out onto the track, do you feel nervous and anxious? Instead of trying to eliminate that feeling, embrace it.

If you're nervous, that means that what you're about to do matters. It means something to you. If it didn't, you wouldn't care, and you wouldn't be nervous. And when it matters to you, you're more likely to perform at your best.

Research by Mihaly Csikszentmihalyi (author of the book, *Flow*, and considered the leading expert on how humans get into flow or the zone) around the subject of flow has shown that if something doesn't matter to you, you're unlikely to get into the flow state, or into the zone.

If you feel challenged by what you're about to do, and yet you have confidence in your ability to do it, you're more likely to perform in the zone. If you don't care how well you do and don't feel challenged by it, you're less likely to perform well; same is true if you feel challenged but you don't believe you're up to the challenge. In other words, when the sense of challenge and your belief are balanced, you're more likely to perform at your best.

When you're excited by the challenge, that's a good thing. If you're too calm, that's often a sign of not feeling challenged. While being so nervous that you feel as though you're going to freeze up is not good, neither is being too relaxed.

As the research suggests, embrace your nervousness. It’s a sign that you care — it’s a sign that you feel challenged.

Day #316: Look for and drive the grippiest surface.

While everyone loves driving a dry racetrack, there are some who enjoy a wet track. In fact, there are a few drivers who revel in driving in the rain.

But there are very few who enjoy that in-between stage, when a track is going from wet to dry, or dry to wet. Even the best wet weather drivers would prefer not to deal with an in-between track, as it's a constant unknown.

After I returned from coaching at the test day at Le Mans, where most of the day was spent somewhere between wet and dry, I started thinking...
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I believe the main reason some drivers don't enjoy driving in the rain is because the grip level changes. It's not a constant. So, rather than knowing what to expect, drivers are constantly feeling out the limit, and some drivers don't like that unpredictability.

And, if a wet track is unpredictable, then an in-between track is even more so.

If you look at driving from the perspective that you're always dealing with the unpredictability of the grip level, then that should give you a clue about how to drive a drying track. The only difference between a dry track, a fully wet one, and one that is drying is the amount of unpredictability.

No matter what condition the track is in, the driver's job is to look for and use the grippiest part of the track. You need to sense where the grip is, first, by looking for it, and second, by feeling and listening for the grip level.

You look for the grip level in the wet by looking for glare or shininess of the surface. Then you test it. Drive the car in a little faster than you think is appropriate, and trust that you'll catch it if it slides too much. It's all about knowing that the car can move around and you'll catch it. It's not just that it can move around, but it should move around. If not, then you're not driving near the limit. That's okay if that's what you want. But if you want to drive the limit, you need to have the car moving around, to test the limits.

The final part of driving a drying track is not to be consistent. Yes, after all the time you've spent on practicing being consistent, now I'm telling you not to be. But if you're searching for grippy pavement, you need to change your line, and that will appear inconsistent.

You must experiment with the line, where you brake, how crisply you turn the steering wheel, and where and how hard you accelerate out of the corners. If you're doing the same thing you always do — being consistent — you will be fast sometimes, but not always. If you want to be fast all of the time (or at least, most of the time), then you'll need to do different things because the track is never the same.

Day #317: You can't completely control whether you win, or turn a faster lap time. But you can control whether you learn or not.

"I don't think you could teach me anything I don't already know." That's what I received in an email in response to my message about a resource I was creating to help drivers.

At first, I was taken aback by that comment. I thought it was a little "closed-minded," and I wondered how to respond.

The more I thought about it, though, the more I realized that he might be right, and it would be closed-minded of me to not be open to the possibility that he knows more than I do.

In fact, as a self-proclaimed learning junkie, I wondered what I could learn from this fellow.

Sure, it's possible that this person was talking with his ego, and that he still has a lot to learn — and that I could even help him. It's possible.

It did make me wonder why anyone would not take advantage of practically any opportunity to learn something that could make them an even better driver, no matter how good they currently are?

The more you learn, the better you get. The better you get, the more likely it is that you'll turn faster laps and/or win more races.

So, make learning your objective.

As a practical, simple-to-do start tactic, make more notes. Keep a small notebook with you at the track and ask yourself (at least) these three questions after every session:

1. On a scale of 1 to 10, how well did I perform in that session?
2. If I could improve just one part of my driving in the next session, what would it be?
3. On a scale of 1 to 10, how enjoyable, how much fun was that last session? (You might be surprised at how big an impact this simple question has on your state of mind, and how that will positively affect your next session).

Why not take every opportunity you can to learn even just one more thing that will make your driving better and more fun?

Day #318: Learn through osmosis.

Nico Rosberg won the 2014 World Drivers Championship. Nico's father, Keke, won the World Championship in 1982.

Damon Hill, 1996 World Champion, is the son of 1962 and 1968 World Champion, Graham Hill.

Michael Andretti won the 1991 Indy Car Championship, and of course, is the son of Mario Andretti - who won a few things in his career, too.

Al Unser, Jr. won the Indy Car championship and the Indy 500, as did his father and uncle.

While Dale Earnhardt, Jr. did not win as many races and championships his father did, he was still pretty successful.

The list of sons of fathers (and sometimes grandfathers) who have gone onto great success makes one wonder what's in the DNA of these families. Or the water they drink.

But just imagine being the son of someone like Keke Rosberg. Picture sitting around the dinner table, and listening in on the conversations that took place. Imagine...

Little Nico: *"Dad, why did Piquet not try passing you on lap 67?"*

"Dad, why didn't you run more wing in the car at Silverstone?"

"Dad, what's the most difficult corner you've driven?"

"Dad, what do you do when the brakes start to fade?"

"Dad, can you tell me again about that race at Spa in the rain?"

"Dad, what did you learn racing sports cars that helped you when you drive in Formula One?"

"Dad, who was the toughest driver you ever raced against, and why?"

"Dad, pass the sauerkraut."

"Dad..."

Now just imagine what Keke had to say, and all of the knowledge and wisdom he shared with Nico over the years.

By the time Nico sat his butt down into a kart, let alone a few years later in a car, imagine what his level of understanding of driving was like compared to others who did not have a father with the experience of Keke. Or Graham. Or Mario. Or Al. Or Dale, Sr.

How much of Nico's success is due to his DNA, and how much is due to learning through osmosis?

If you want to improve your own driving, how much can you learn through the experience of others? Read, watch, talk to... soak up the knowledge and wisdom of others.

You may be surprised at how easy this is, and how helpful it is. And how few drivers take advantage of this — and, therefore, how easy it is to gain advantage over others by doing so.

Day #319: Use reflection to deepen your understanding of driving technique.

I never set out to write a book, but I've written eleven, now. My first *Speed Secrets* book came about in two ways. First, I had a high-performance driving school, and I needed some form of information to give to our students, so I wrote a very basic handbook. Then, over the next few years, particularly when I was racing Indy cars, after each session, day and/or event I'd reflect on what I'd learned. I asked myself questions like, "Why did the car do what it did in Turn Three?", "When I changed my line through the Esses, I went faster — why?", "Why could Jimmy Vasser make his car do what it did in the hairpin?"

I had a pretty good understanding of the theory of driving, but this reflection deepened that understanding. Teaching it to others over a hundred and fifty days a years (for many years) also provided me with plenty of opportunity to reflect on not just the what and where of driving, but also the why. And it's the why that matters most.

Teaching an at-track class classroom session recently for a car club's novice students at a HPDE event I was reminded once again how important the understanding of the why is. I asked the group of newbies why they drove the line they did, and we all concluded that it was because that's where the instructor told them to drive.

One difference between a novice and a more advanced driver is the level of understanding of the why we do what we do. Interestingly, I've asked the same question (about why a driver drives the line they do) of "solo advanced" students, and got pretty much the same answer (sometimes having read between the lines of what they said) — because that's where they were told to drive.

In my experience, there are many drivers who have been given the label of advanced (experienced racers, even) who are not able to explain why they drive the line they do. They're also not able to explain why one would trail brake in one corner, and not another; they're not able to figure out the line through a corner they've never seen before without someone showing them; or why a car with its engine in front of the front wheels or behind the rear wheels handles much differently than one with its engine in the middle of the car.

There's nothing wrong with these drivers. But there is a huge opportunity for them to improve. By taking time to reflect more on why they do what they do, on why things work and don't work, and trying to explain these things to others, well, they will become much better drivers.

When I started doing webinars, that was something I spent a lot of time focused on: providing a balance between "what to do" and "why to do it."

The latest in learning science points out two very important things a learner should do. First, spend time reflecting on what you've read, heard, or been told. Simply taking information in does not necessarily result in long-term learning. And second, taking in

information in chunks, spaced out over time, results in better retention. Reading a book in one sitting, then reading it again immediately is no more effective than reading it once. If you were to read it, then re-read parts of it spread out over time, that's more effective. That's why I've written my books the way I have; that's where the value of this eBook comes from; that's how my Playbooks have been designed.

I strongly encourage you to take a little time to stop and reflect on what you've learned, why things work the way they do or don't, and do this after each and every on-track session, each day, each event, and so on. It's very powerful. Write your thoughts down. Find a willing volunteer to whom you can explain what you've learned, and especially focus on explaining the why behind what you're talking about.

Revisit information, and take it in in different ways. Find a variety of sources for it, as it's the interleaving of information that gives you the deepest understanding. And spread out the learning — don't take it all in at one time.

Day #320: Use all the track.

"You paid for it, so use it." That's a well-used phrase to encourage drivers to use all the track, from the very outside edge when approaching it, to the very inside edge at the apex (and maybe even inside the edge!), to the very outside at the exit.

Let's take a look at a 90-degree right-hander that has a theoretical maximum cornering speed of 80 MPH. Knowing the radius of the corner and the maximum g-force capability of a car, we can calculate this theoretical maximum cornering speed. Assuming your car's maximum cornering capability is not going to change while you're driving it (you can't adjust the aerodynamic downforce or suspension settings while moving, or mount grippier tires), then the radius is all you can work with.

If you don't use the entire track surface -—entering it just one foot away from the edge of the track, and exiting it one foot away — you have reduced the radius of the turn to the point where the theoretical maximum cornering speed is now slower, just over 79 MPH. That doesn't sound like much difference, but it's a little more than one percent decrease in speed. And one percent of a 1-minute lap time is over half a second. That's a lot of time to give away by simply not using all the track, by being just one foot away from the track edge on the entry and exit of a corner.

Okay, this is a pretty simplified way of looking at a simple topic. But my experience behind the wheel, as well as observing cars on track for four decades (and thousands of data reviews), I know that even a few inches can make a big enough difference to make it worthwhile to focus on using a little more track.

The next time you're on track, use all of it.

Day #321: Maintaining focus is not as important as how quickly you're able to regain it.

For a few years now I've been saying that one of the main benefits of simulators (or almost any game) is the ability to practice maintaining and regaining focus.

When I ask drivers of all types and levels what their biggest mental game challenge is when driving, most comment on their ability to maintain focus — they get distracted too easily. It's definitely the number one problem that drivers face (according to drivers themselves).

If you think about a mistake you made in Turn One while driving through Turn Three, that's a focus problem. If you're thinking about what someone said, where you're eating dinner that evening, or whether that car in front of you has more horsepower than you, then those are focus problems, too.

Think of focus as simply "whatever your mind is processing." If it's processing good stuff — your turn-in point for the upcoming corner, the sound coming from your tires, or the way g-forces build up and then taper off — then you're "focused." If your mind is processing the error you made a couple of corners back, what other people have said, or anything about another driver's car, then you've "lost focus."

But as I've written about many times before, you cannot not think about something. In other words, you can't tell yourself to stop thinking about that mistake, those words, that other driver's car. Well, I guess you can, but it doesn't help! In fact, it makes matters worse, since that means your mind is actually processing that information rather than what you want.

And that's where the ability to regain focus comes in. If you have some type of trigger — a word, phrase or physical action that cues your mind back on what you want — you'll regain your focus quickly. And what you want is to focus on the act of driving, right here and now, in the moment. When you practice regaining your focus with a trigger, you get better and better at doing so. That practice can be done, over and over again, on a simulator. At relatively low cost (almost nothing compared to driving on the track).

Practicing regaining your focus on the track takes time (meaning it costs money, and it's difficult to practice as many times since access to a track is limited compared to access to a simulator), and the consequences of not doing it right are low.

I've heard a few drivers almost brag about their ability to maintain focus. If you have the ability to focus longer than most, good for you. But if you're like most drivers, your ability to regain focus is more important. In fact, practice regaining focus by using a trigger to the point where you can brag about this ability. Make it one of your strengths.

If you regain your focus fast enough, it'll seem as though you've never lost it. Use a trigger, and practice that on a simulator.

Day #322: Use your brake release to help rotate the car - you'll improve your corner entry speed, and your exit speed.

"Turn In Early With Big Speed" (as seen on a T-shirt). That's either brilliant or crazy advice. But it's also a great target.

If you can turn into a corner early enough to make your line through the corner a perfectly geometrically radius, that's the fastest way possible to get through that individual corner. The problem is that there is always something after that corner on a road racing or oval track, and it's often a straightaway of some length. While the geometrically perfect radius line makes you fast through the corner, it doesn't necessarily make you fast around the entire track. As I'm sure you know, that's why a late apex line is most often the fastest way to get around the track — it allows you to accelerate out of the corners earlier.

But what if? Wouldn't it be great if you could turn in relatively early, taking a line close to the geometric radius in the first third or so of the corner, and yet still be able to get to a late apex… AND accelerate early? Rather than "slow in — fast out," you would be "fast in — faster out." But how?

It's all in how you manage the load transfer in the car — specifically how you manage the brakes as you enter the corner — along with how quickly you turn the steering wheel. The fastest drivers time the release of the brakes in just the right way to allow them to rotate the car in a way that sets the car along an early/geometric line, and yet, gets the car to rotate so it exits like a late apex line would allow.

I wish I could tell you exactly what it is you need to do in your car to make this happen, but I can't. There are too many variables from car to car, and corner to corner. Yes, this is dependent on the corner, as well. Some corners reward this approach more than others.

So why, then, am I even bringing this up, if I can't tell you how to use it? I'm setting a target for you and your learning. Think "fast in — faster out," rotating the car through the "timing and rate of release of the brakes" as you enter the corner. Get it right, and magic happens.

Of course, getting "fast in..." wrong, and well, bad things can and will happen! That's why this advice must be taken the right way, with respect, and by the right drivers. If you're relatively inexperienced, leave this for a while — until you're ready. Be patient, as the moment to take this on will tap you on the shoulder one day and tell you "it's time."

But if you're a little more experienced with your driving, then begin to experiment with when and how you release the brakes… AND how that can change where you turn in for the corner, the line you drive through it, and when you're able to begin accelerating. Get it right, and you'll find yourself turning the car almost as much with the brakes as you do with the steering wheel — and maybe more.

Finally, something very, very important to keep in mind: When I talk about “fast in…”, I’m not talking about tossing the car into a corner 10 MPH faster than you ever have before. But even one MPH more entry speed, added to getting to full throttle exiting the corner as early as ever, will pay off big time. And, if your entry speed is so high that it delays your ability to get to full throttle, most times (but not all) that's a bad trade-off.

So, if you’re not able to sense, set, and adjust your corner entry speed as finely as a couple of MPH, then you’re not ready to take this on, yet. If your corner entry speed varies by 5 or more MPH, work on developing your speed sensing before you tackle the “fast in — faster out” approach.

And remember, it's a target, and not necessarily a “must-do.”

Day #323: Limit the input of new information - program common tasks, filter new ones.

Brain overload. Anyone who has driven around a race track for more than two laps at speed has experienced this. For some drivers, it's a common occurrence; for others, a rarity.

As Eric Ravenscraft wrote in his *Limit How Much Information You Have to Process to Avoid "Mental Fog"* article on LifeHacker, "Put simply, the more information you have to handle, the less time you have to give it the thought it deserves. If you start to feel you're entering a mental fog—and you'll notice if you start making poorer decisions, respond more emotionally than rationally, or make frequent errors—slow down and try to limit how much information you have to process. Drop a task or two, or simply take a break from the screens."

While Ravenscraft is not talking about race or performance driving, he could be, right? It sure sounds like it.

Common tasks, skills that you use over and over again, techniques that are the tried and true... all of these must get to the point where you no longer have to think about them. You just do them, as if on auto-pilot.

- Do you have to think about how you turn the steering wheel?
- How about shifting, up or down — how much thought do you need to put into it?
- Do you need to remind yourself to look further ahead?
- Is where and how you begin braking something you need to think about?
- Do you need to think about where and how you release the brakes, or is that something you can do without thinking about it?
- Are you able to check your mirrors while staying on the line you want to drive?

As news site Big Think reports: "Our minds are designed to process one very simple train of thought at a time. In fact, most of the time we think we're 'multitasking', we're really just rapidly switching from one task to another... This overload leads to a cognitive slowdown called mental fog... In fact, multitasking was found to increase the production of cortisol, a stress hormone, as well as adrenaline, which can overstimulate the brain and cause mental fog."

This is exactly why all of your common techniques and skills must get to the point where you no longer have to think about them. They need to become programmed in your mind; that comes from physically practicing them, or mentally practicing them (visualizing).

But what about the uncommon tasks and new skills? The key is filtering them in, one at a time, without overloading your brain.

If you're learning a new skill or technique, break it down into the smallest pieces possible. Practice that piece until you no longer have to think about it, and then move onto the next piece. Then the next piece. And so on.

Day #324: Experiment with maintenance throttle versus hesitating before applying the throttle.

I've written that it's not when you begin to apply the throttle that matters most. No, it's when you get to full throttle that it makes the biggest difference.

As my race engineer friend Jeff Braun points out, guidelines (and I'll add, almost any advice) are guidelines because they work 51% of the time or more often. They're not absolutes. What I said then, and what I'll say here, are guidelines or advice.

There are times when, immediately after releasing the brake pedal, going to maintenance throttle — or applying just a little throttle — helps you maintain speed and momentum. And that works sometimes. If you hesitate for a fraction of a second before applying some throttle, your car continues to decelerate. And then when you begin to accelerate, you're starting from a slightly lower speed. That's why (sometimes), if you apply some throttle — what is often referred to as maintenance throttle, since you're maintaining speed — you're faster.

Other times, if you hesitate slightly after releasing the brakes, and before you begin applying the throttle, you'll be faster. Why? Because the hesitation allows your car to rotate slightly more, and then you're able to go from no throttle almost immediately to full throttle. And again, as a guideline (remember Jeff's definition of a guideline), whoever gets to full throttle soonest, will be quickest down the following straightaway.

My message? Experiment with it. Try applying some maintenance throttle, and see what that does to your straightaway speed and overall lap time. Then try hesitating a little, and focusing on getting to full throttle quickly, and see what that does.

Of course, what works in one corner may not work in another. Or in another car.

Day #325: Doing the wrong thing with more effort will not result in the desired performance.

Have you ever noticed how practically every great athletic performance looks almost effortless? Great performances are always achieved when the right amount of effort is used in the right places. This right amount of effort is usually less than you think necessary. The less unnecessary effort you spend, the better you'll perform. The key is to use appropriate effort, or as Jackie Stewart referred to it, "economy of movement."

Great drivers use less effort to produce great performances.

One of the most common mistakes I see drivers of all levels make is "trying." Trying is a conscious act, not an automatic, subconscious, programmed act. The second you try, your body tenses and therefore is unable to perform smoothly. Trying is one of the main causes of errors, especially if you're under pressure (from others, from yourself). Learn to relax, and let your body and the car flow.

By the very definition of the word, trying gives you a way out, an excuse. Trying means "to attempt." And when you have an excuse, your mind will often take it.

Think back to some of your best driving — or any great performance in your life. Were you tense and aggressively trying, thinking hard, and forcing yourself to perform well? Or were you relaxed, calm, focused, and simply doing what seemed to come naturally? I bet you were in the latter mode, not even aware that you were trusting your subconscious programming to perform.

Yes, driving at your best comes from driving subconsciously. It comes from the "programming" in your mind. Trying to drive fast is like trying to make a computer with no software do something. It just isn't going to happen. Trying is driving consciously. Instead, focus on giving your bio-computer (mind) more input.

Focus on what you can see, feel, and hear; become more aware; and visualize the act of driving.

Day #326: Understand what you're doing that may be causing handling problems.

Listed below are the most common driving errors that I see drivers make. I provide them simply to help you become more aware of possible mistakes you're making that can lead to handling problems. By asking yourself whether you're making these, you'll be better at identifying the real issue — is it me or the car's fault?

- "Crabbing" into a corner. By "crabbing," I mean you have eased the car away from the edge of the track before turning into a corner. Often, this results in you having to increase the amount the steering wheel is turned as the car heads towards the apex (the radius of the corner is tightened up), and that could cause understeer.
- Turning in early. This reduces the radius of the corner, and will often induce understeer, too, since you're likely to have to turn the steering wheel more in the middle of the corner.
- Turning in late. By turning in too late, you have to turn the steering more, and drive a tighter radius. In doing so, you could cause understeer by having to turn the steering wheel too much or oversteer since you've turned the steering abruptly.
- "Pinching" the exit of a corner. Pinching the exit is a way of describing when you do not begin unwinding or straighten the steering early enough. This often leads to oversteer due to the steering angle not being reduced while accelerating.
- Early and/or abrupt brake release. Popping your foot off the brake pedal typically leads to understeer due to unweighting the front tires.
- Early throttle application. This can lead to understeer due to unweighting front tires while turning (not reducing the steering angle).
- Aggressive throttle application. This is the classic power-oversteer, where you've over-worked the rear tires and they begin to slide, causing oversteer.
- Mid-corner steering increase. This usually happens when you try to force the car to a missed apex, leading to understeer or oversteer.
- Over-reaction to understeer. This is the "understeer leads to oversteer" problem. It begins with the car understeering (this could be caused by the car's setup, not one's driving), then you turning the steering wheel more (a natural reaction, but the wrong one); when the front tires scrub some speed off and then grip up, they quickly change the direction of the car, leading to the rear of the car sliding — oversteer. Often, you'll complain about the oversteer because it's scarier, but it's not really there if you wouldn't increase the steering when experiencing understeer.

These are not all the mistakes that you could ever make, but they're the most likely — the ones I see most often. Notice that each one can be fixed early enough that you don't end up stuffed into a tire wall, and each will teach you something that will make you a better driver. In fact, remember that all of these are "learning-takes," not "mistakes."

Day #327: Knowledge and practice leads to adaptability.

One of the key areas that separates good drivers from great drivers is their ability to adapt their driving to suit the car's handling, or from one type of car to another.

Some drivers, despite how the car is handling, will only drive it one way — with their style. And guess what? A driver's style will never suit every handling characteristic. If you cannot adapt your style to suit the car's handling, a change in track conditions, a mechanical problem, or a different type of car, you'll never be the best you can be.

Although it would be impossible to list every potential problem scenario you may someday face when driving, I'm going to attempt to identify the most common ones. Then, I'll give you some suggestions as to what you may be able to do to adapt your driving to help the situation.

My overall objective is to give you some knowledge about what you might do to reduce the effects of the problem. In other words, what can be done so the problem has the least effect on your lap times?

Corner Entry Understeer:
If your car is understeering in the entry portion of a corner, consider what you can do to induce some forward weight transfer and what you can do to lessen the weight transfer to the rear.

To increase the forward weight transfer, you can increase or lengthen the amount of time spent trail braking into the corner. That means not trailing, or easing off the brake pedal so quickly, keeping a little more pressure on the pedal for a little longer. And, if it is a corner that requires little-to-no trail braking, then it may be a matter of waiting a little longer — being more patient — before beginning to accelerate; or, squeezing on the throttle a bit more gently.

The key, then, is to be patient. You will probably end up entering the corner a MPH or so less. If you focus on increasing the forward weight transfer, and decreasing the weight transfer to the rear, you will be able to get the car rotated (turned) earlier in the corner and get back to throttle solidly — without having to come back off it to control the understeer. That will improve your acceleration out of the corner and down the straight.

Avoid the temptation to turn in a little earlier. If anything, you want to turn in a little later, opening up the exit line so that you can focus on the acceleration phase of the corner.

If the weight balance of the car is not the cause or the cure of the understeer, then you have to consider one other thing. It doesn't matter where in the corner the understeer is, think about what you are doing with the steering wheel. Oftentimes, corner entry understeer is caused by the driver cranking in too much steering input, or cranking it in too abruptly. Turn the steering wheel a little less, and a little more gently. I know it may not feel right — the car is not turning enough (understeering), so you turn the steering

less? Exactly. Keep the front tires at an angle that they can work at. If you steer the front tires too much, they can't help but give up their grip and begin to slide.

Again, be aware of how much steering you have input, and try taking some out. Or, turn the steering wheel a little slower, a little more gently as you initiate your turn-in. Give the tires a chance to build up their cornering grip.

Corner Entry Oversteer:
Oversteer is often a result of too much weight on the front tires, and not enough on the rears. If that is the case during the entry phase of a corner, that probably means you are braking too hard into the corner — you are trail braking too much.

The cure, then, is pretty simple. Just begin braking slightly earlier, and trail off the brakes a little sooner as you enter the corner. Perhaps, especially if it is a corner that requires little-to-no braking, it is a matter of beginning to accelerate sooner (but very gently), transferring more weight onto the rear tires.

One other thing that may help reduce corner entry oversteer is to turn the steering wheel less abruptly. Ask the car to change directions, from straight forward to a curved path, a little more progressively.

Mid-Corner Understeer:
Usually the best way to handle mid-corner understeer is simply by smoothly modulating the throttle to change the weight balance of the car. In other words, just breathe — gently lift off — the throttle to cause some forward weight transfer, giving the front tires more grip.

Often, the understeer is not related to the car's set-up, but that you have gotten back on the throttle to begin accelerating just a little too abruptly or early. Again, just breathe the throttle to transfer some weight forward.

Also, just like with the entry understeer, be aware of the amount of steering input you've dialed in. Perhaps the cure to your car's mid-corner understeer problem is just unwinding the steering a little bit to allow the front tires to get some grip.

Mid-Corner Oversteer:
Dealing with a mid-corner oversteer is almost always done by changing the car's weight balance. In this case, that means squeezing more throttle on. However, one of the reasons the car has begun to oversteer is that the speed you are carrying is slightly more than the rear tires can handle. So, the last thing in the world you need right then is a bunch more speed. That is why it is critical to squeeze on just a little more throttle.

The mid-corner oversteer could also be caused by wheelspin (in a rear-wheel-drive car, of course), which should be dealt with by being a little easier on the throttle.

If your car's set-up is the cause of the wheelspin, about all you can do is be as gentle as possible with the acceleration, and possibly alter your line slightly. If possible, drive the car a little deeper into the corner before turning in, make the initial turn radius a bit sharper, aim for a later apex, and then let the steering unwind as early as possible. This makes for a straighter acceleration line, meaning that there will be less cornering force to combine with the acceleration force that you are asking from the rear tires.

Exit Understeer:
If your car has an exit understeer problem, the best thing you can do without reducing your acceleration is to alter your line. Your prime objective is to lessen the amount of time you are turning the car while accelerating. So, if you turn in a little later and sharper (even if this means slowing the car down a little), and aim for a later apex, it will allow you to unwind the steering a bit earlier. That means you will be accelerating in a straighter line, reducing the harmful effects of the understeer.

And one more thing. The more gentle you are with the acceleration, the less understeer you will have. If you jump on the throttle, the understeer is going to be exaggerated. So, squeeze on the throttle.

Exit Oversteer:
Exit oversteer can be related to one of two things: either it is power-oversteer, caused more by the car's inability to put its acceleration traction to the ground; or it is due to the weight balance.

Usually, the way to deal with either type of exit oversteer is much the same as with exit understeer. The goal is to open up the exit of the corner, increasing the radius of the corner as soon as possible, by using a later turn-in and exit.

Another thing you have to keep in mind with exit oversteer is to be gentle with the throttle under acceleration. If you stand on the throttle, even if you have altered your line, you are going put a big load on the rear tires. In time, this will overheat them, making the oversteer problem worse, even causing it to oversteer in other parts of the corner.

There is one other approach to dealing with a car that oversteers at the exit of a corner — especially one that has an extreme oversteer problem — and that is to almost give up, or sacrifice that part of the corner. Instead of slowing the car down and using a later turn-in and apex, you pretty much do the opposite. As you approach the corner, you brake later and carry much more speed into the corner, taking an earlier apex, and then get the car straightened out and pointed down the straight well after the apex. The idea here is that since the car will not accelerate out of the corner very well, you might as well try to take advantage of where the car is working — the corner entry.

Before using this technique I would make sure that every other technique didn't work, as you will not be setting any track records using this approach. It is a bit extreme! Perhaps the biggest challenge in using this technique is that it's unlikely you have a mental

program for it. Therefore, try it out when and where it’s safe, so that you are prepared for it.

Day #328: Experiment with using the curbs.

In most situations, running over the inside curbs around the apex is a benefit, and that's because it allows you to drive a larger radius (straighter line) through the corners. Just about any increase in the radius of a turn is going to result in a worthwhile increase in speed that you're able to carry through it.

Some drivers worry that hitting the inside curb too hard/too much will result in less grip from the tires, which might lose contact with the track surface. That is the trade-off which determines how much of the curb should be used. But because the tires on the inside of the turn are so lightly loaded (from the lateral weight transfer), the loss in grip for a small amount of time is often made up for by the larger radius through the turn. You'll notice that most good drivers clip the inside curbing, and bounce past it, in a fairly short amount of time. So the loss in grip during that period is not enough to hurt the overall benefit of the larger radius.

So much of this depends on the type of car (and how compliant the suspension is), and the height and shape of the curbing. If the curbing is too high or too aggressive, or the suspension is too stiff, then the driver would be better off avoiding the curb (or at least, part of it). This really is the deciding factor.

There are times, in some cars, with some curbs, where a driver will almost try to "hook" the inside tires inside the edge of the curb — the inner sidewall of the tires on the inside of the turn hooking on the inner edge of the curbing. As you can imagine, this can be hard on the car and tires, but for a few laps in qualifying, it's sometimes worth it.

How do you know when to use the curbs, and when not to? Experiment with it. I recommend "nibbling" at them at first, then start using more and more until the car and lap times (or better yet, segment times that isolate just that corner) tell you that you're losing more from the lack of grip than you're gaining from the larger radius.

Years ago, I drove in a World Racing League race at Hallett — a series I'd never competed in before, at a track I'd never been to before. It was a blast!

We were driving a BMW E36; it was fun to experiment with using the curbs. Some corners didn't benefit from using them, and others did.

In the corners that benefitted, not only did the larger radius help, but the curbing helped turn the car. By running the inside tires along the curbing in Turn 4, for example, the rear of the car slid out a little, helping to rotate the car, allowing me to carry more speed and still get set up toward the right side of the track to turn in for Turn 5.

Having an almost three-hour driving stint on Saturday (racing the track in the counter-clockwise direction), and then nearly two hours on Sunday (in the clockwise direction), I had lots of laps to experiment with using different amounts (and even angles) of using the curbs. It paid off.

Finally, all of this advice is aimed at the apex curbing on the inside of the corner, but the same kind of reasoning applies to exit curbing at the end of a corner. Experiment in a smart way.

Day #329: Identify the problem, then quickly switch to focusing on the solution.

You've turn into the corner, the car on the edge of grip with a hint of understeer as you clip past the apex... Well, almost. See, you're two feet from the apex curb. That's a problem.

Do you spend the rest of the lap thinking about that missed apex?

Some drivers do. The best drivers identify a problem, then focus on the solution.

Once you've missed an apex, you can't go back and fix it. All you can do is manage the effects of missing it. That might mean delaying the application of the throttle, holding some steering angle in for a little longer, or using a little "extra" track at the exit (in other words, more curbing and maybe even a little of the dirt beyond it).

If your car is not handling well, it's easy to get mentally spun out of control by thinking about the problem. I've overheard drivers in the paddock telling others about the handling issues over and over and over and over... If they put that much thought and attention into figuring out what to do to fix the problem, they'd be miles ahead.

It's no different for the problems around you at the track, from personality conflicts to the rules you need to operate within.

One of the great things about driving around a track at speed is it forces you to learn to switch to the solution quickly. But not all drivers make that switch quick enough, or do the same off the track.

Use the "I can't go back and fix a missed apex" attitude that you've developed while driving on the track.

Day #330: Be fast on cold tires. Make cold tires hot.

One of the most challenging situations for a driver is getting up to speed quickly on cold tires. The car is moving around, the grip level is low, and you just don't know what you've got to work with. What a blast!

Yes, the first step in being fast on cold tires is learning to love the feeling of driving on cold tires! But how do you learn to love something you don't? Fake it until you make it.

See, you do what you do because you're programmed to do so; you don't do what you want sometimes because you don't have the right mental program to do it. If you don't love driving on cold tires, that's because you don't have a program to love it.

Where do your mental programs come from? Repetition. Yes, doing something over and over again builds programming. The good news is that repetition can be done either physically, or mentally. By playing a scenario over and over again in your mind, you're building mental programming. That's what visualization is all about. It's even better when you use your sense of feel and your hearing when imagining loving driving on cold tires. Soon, you will truly enjoy driving on cold tires (okay, you may not ever love it, but you can learn to enjoy it).

So, the first step in being fast on cold tires is learning to enjoy it. Pre-play in your mind all of the fun of a car moving around on less-than-hot-and-grippy tires.

The second step in learning to be fast on cold tires is practice. I see a lot of drivers who go out of their way to have warm tires on their cars before hitting the track, robbing themselves of the opportunity to practice driving on cold tires.

I've coached drivers who were terrible getting up to speed on cold tires, so I strategized a practice session to change that. On a test day, we had three sets of tires. I sent the driver out for a maximum of four laps on cold tires, then come back in (I was timing his out-lap, from the second he began to move out of the pit stall to when he passed the Start/Finish line, and a goal was to lower this time each time out). I had the driver get out of the car, take his helmet off, and walk around. Then, he got back in the car as we put another set of cold tires on, and out he went for another four-lap run. Then again, with the third set of cold tires. Between runs, the crew was pouring cold water over the tires and wheels to prepare for the next run. Each time, I had the driver get out of the car, walk around, and then gear up and go again. This way, it was simulating being "cold" himself.

After a couple of hours of this practice, the driver had improved his out-laps, and his ability to sense how much grip the cold tires had. At the next race, he did much better at being fast on cold tires, which began to change his belief about how fast he was on cold tires, and that led to him enjoying it more... and that kept building throughout the season.

That leads to the third step: sensing the tires' grip level. The challenge with cold tires is you just don't know how much grip you have — unless you really focus on it. When you spend time, and practice sensing the grip level of the tires, you get so you can predict how much grip they have — and how quickly it builds.

Of course, ambient and track temperature plays a big role in how quickly your tires come up to their peak grip level. The more you focus on sensing this, the better you'll get at it.

In many ways, I'm suggesting that you do what I call "Traction Sensing Sessions" - deliberating focusing on what you see, feel and hear from the tires. Typically, this is not something that you consciously focus on. Instead, you just drive, and if anything, you focus on the grip the tires don't have. In other words, you think about how little grip you have, and feel uncomfortable because of it. Flip that around: Focus on the grip you do have, and enjoy the feeling of the car moving around, sliding.

Logically, you know that the harder you drive on cold tires, the sooner they will build temperature, and the more grip they will have. This is where you have to trust your ability to control the car when it slides — as it will, on cold tires. In fact, if you deliberately make the car slide, then you know what you have — a sliding car. If you gingerly enter every corner trying to avoid the car sliding, it'll catch you off guard, and you'll have to react quickly to control it.

Make the tires slide from the second you enter a corner so you can be proactive with the car, instead of being reactive.

Let's go back to visualizing, or more accurately, using mental imagery. If you spend time every day for a month or so mentally seeing, feeling, and hearing yourself and your car driving on cold tires, you'll actually improve. As I said earlier, repetition is the key to building programming, and you can do the repetition in your mind, with your imagination. As I've said many times before, your brain can't tell the difference between a real and an imagined event. So, the more you do mental imagery, and the more senses you involve in it, the more effective it will be.

One of the best drivers on cold tires I've ever seen is Juan Pablo Montoya. Was he born with the ability to drive fast on cold tires? I don't think so. I think he developed those skills over the years — and I bet he loves it. And you can, too:

- Mentally build your "cold tire" programming using mental imagery
- Use physical practice to sense your tires gaining grip with increased temperature
- Practice driving quickly on cold tires — stage practice sessions where you can drive only on cold tires
- Proactively make your car slide to build heat in your tires

And have fun!

Day #331: Write about driving once a week.

Do you really want to improve your driving and have more fun? Then do what I do. Write about driving every single week, for years on end.

I'm not suggesting that the reason I write is to improve my driving, but it has. The process of thinking about some driving topic every single week, and to get clear enough on it that you can explain it in words on a page, will take your driving to a whole new level.

I don't care if you're not a good writer. You don't need to be. You don't need to share your writing with anyone else. Keep it private. But you must actually write it down. Just thinking about it is not enough.

Pretend you're a writer. Pretend that you write a column for a weekly magazine, and that people will read what you write. It's important to write so someone reading it will understand what you're saying. You can't just write gobbledy-gook. No, grammar and spelling and typos are not that important. But it's critical to get down on a page what you're thinking, in a way that someone will understand your main message. And it'll make you a better driver.

I hadn't raced a prototype sports car for over a decade when I was asked to drive one in the Thunderhill 25-hour race. I wasn't the fastest driver, but I wasn't the slowest. Most important, I was good in traffic, I was calm and cool throughout, and I continually worked away at driving fast with less effort (more efficient driving). Thinking about, talking about, and writing about driving made the difference.

Make a commitment to writing one article every single week. In fact, put it in your calendar to sit down every Wednesday morning, or Thursday evening, or Saturday afternoon, or whatever, and write a minimum of 500 words. Pick a topic you've been thinking about lately, or an experience you've recently had, or something you've seen another driver do, and then write about it as if you had to explain to someone who's barely ever been on a track.

How you write doesn't matter. You can scribble it out in a notebook with pen, or you can type it out. I do recommend you keep what you write, though, and go back every now and then to review what you wrote. It'll trigger new thoughts, clarify your thinking, and provide new insights and focus for your driving.

I've noticed a trend. The best coaches I know have the ability to explain things in writing. Which came first, the ability to write about driving, or the ability to coach? It doesn't matter what the answer is, but I know it's not a coincidence that great coaches have the ability to write about driving. Putting into words what a driver should do is a practiced skill, and something that makes one not just a better coach, but also a better driver.

Write on!

Day #332: Use your left foot as the third leg of a 3-legged stool.

When sitting in your car, think of your left foot as the third leg of a stool. The other two legs are your right foot and your butt.

To sense what your car is doing, you need all three legs on the “floor.”

I’ve written about the importance of keeping the heel of your right foot on the floor while working the gas and brake pedals (I’ll get to left-foot braking, shortly). If you lift your heel off the floor while modulating the gas and brakes pedals, it’s like trying to balance on two legs (or maybe just one — your butt). You soak up a lot of information about what your car is doing with that foot and leg, especially if it’s on the floor. If your heel is not on the floor, you’re missing out on feedback.

Your butt in the seat is another leg of the stool, and as you know, it’s critical to sensing your car and the track. Keeping it in contact with the seat is relatively easy, but the more it’s “in” the seat, rather than on the seat, the better.

The third leg of the “car sensing stool” is your left foot. At least, it should be. One of the most common errors I see drivers make is not placing — and bracing — their left foot on the dead pedal (the rest pad to the left of the pedals). And many of those drivers are surprised when someone makes them aware that their left foot is hovering just above either the clutch or brake pedal, or that sometimes it’s tucked back near the front of the seat.

More than one driver has found that they’re more sensitive to what their car is doing simply by bracing their foot on the dead pedal - completing the three-legged stool.

I’ve said it before and I’ll say it again: The trick stuff in driving is just doing the basics better. Using your left foot to brace your body, and sense what the car is doing, is a basic that drivers often overlook.

Why do drivers not use the dead pedal? Habit. It’s a mental program they’ve developed over (typically) a very long time. Resting your foot on the dead pedal may even feel uncomfortable for a short period of time, but like any habit, it’s just a matter of sticking with it for a while until it feels natural.

What about left-foot braking (LFB)? How can you sense what your car is doing while left-foot braking? That’s one of the downsides of LFB. You don’t have the same level of sensitivity to the feedback you get through the dead pedal. That's why it’s critical to get your left foot back on the dead pedal after every single LFB application, and not have it simply hovering over the brake pedal. And… it’s critical that the heel of the left foot is resting on the floor — again, it completes the three-legged stool.

I know some drivers will argue that it takes times to move their left foot from the dead pedal to the brake pedal and back again, and that using the dead pedal is a detriment for this reason. I disagree. Strongly.

Think about it: The only time you need to move your foot in a quick reactionary movement is if your car is spinning and you're trying to depress the clutch. The rest of the time you know when you need to move from the dead pedal to the brake pedal, so it doesn't have to be a fast movement.

And yes, I'll recognize that there are some cars with such a small, narrow foot box area that there isn't room for a dead pedal. Okay, that's the exception to the rule.

The next time you're driving — on track or street — do a quick check and make sure you're left foot is resting on the dead pedal.

Day #333: Make changes.

I was sent a question about how to make progress — get faster, rather than staying at the same level. In the email, the driver wrote this (and I think it's great stuff):

"Let me frame my question in terms of what a colleague always asked software engineers who claimed to have five years of experience. He'd ask, 'Have you really had five years of experience, or have you had one year of experience five times?'"

I see drivers doing this all the time. They do the same thing over and over again, thinking that more seat time will make them faster. But unless they make changes, they're not going to go any faster.

But "change what?" is the big question, right?

Here's a list of things to change that can help lead to you being faster:

- Your mindset. That's where it starts. First, is getting faster the right priority right now? Or is getting consistent at your current pace more important, knowing that you need that before moving onto that next level in outright speed? If the timing is right to work on getting faster, then you first need to ensure your belief system is ready for this change. If you truly, deep down inside don't believe that you can't drive faster, then you won't. How do you change your belief system? Start with mental imagery to change this mental programming. You need to be able to see, feel and hear yourself being faster in your mind, in your gut. You need to truly believe it's not only possible, but it's going to happen. But you need to give your belief system ammunition so it can go along with what you want to believe. Wishing and hoping is not believable ammunition. But doing your homework, studying what other drivers are doing, practicing your basic driving skills and techniques, analyzing data and/or video, reading advice, talking with other drivers, being coached... all of these things, if you do them, help your belief system buy in. Without them, well, that's wishing and hoping.
- Break your driving down into the basic components, analyze, and practice each piece, and then put it back together when the time is right. Pick just one piece of your driving, or one area that you know (not just think, but really know) that you're losing time to other drivers, and work on it. Work on it in a very deliberate way. For example, if you're late to getting to full throttle, find a corner, skid pad, or autocross course where you can practice committing to throttle. Many drivers will get to 75-or-so percent throttle, hesitate, and then finally go to full throttle. For many cars, the difference between 75 and 100 percent throttle is not going to cause you to spin, and if you just go to full throttle you'll be faster (without any problems).
- Be logical. Think about what is likely to happen if you brake two car lengths later, carry one MPH more into a corner, and commit to full throttle a few car lengths sooner. In most cases, not much will happen — at least, nothing bad. And the worst case, most times, and the car will slide around a little more, but you'll

control it. Far too many drivers feel that if they did go a little faster, they'd spin or crash. That happens much more rarely than you think — if you think logically. Ask yourself how many corners you've driven at high speed, and what percentage of them have you spun or crashed in? Ask yourself what usually happens when you go a little too fast (you usually manage it with a correction, and you come back the next run or lap to do over again)? Be logical.

Change your beliefs about your ability to be faster, change how you approach going faster, and change how you think about the consequences of going faster.

Day #334: Think in terms of brake zones, as opposed to brake points.

Where do you brake for Turn 1? For Turn 5? (Yes, I'm guessing that you brake for turns 1 and 5 at the track you're thinking about when I ask that question. If you don't, then think about this as any corner).

My point here is to think more about where you begin braking (Begin of Braking, or BoB), where you finish braking (where your foot comes completely off the brake pedal, the End of Braking, or EoB), and the distance in between as a whole, rather than two distinct points.

Peter Krause has talked about this before, and the most important thing he said was to be consistent and deliberate about the pressure you apply throughout the brake zone. Want to do that...consistently? Then think about...no, picture brake zones, rather than when you begin braking. And most important, look for the End of Braking point as you approach the corner.

Focusing on the EoB forces you to think about what you do with the brakes. It forces you to modulate the pedal, using all the available deceleration grip. It forces you to be smoother with the release of the pedal. It forces you to think about your brake zone, rather than just on where you start braking.

I'm not saying you shouldn't use a Begin of Braking reference point. What I am saying is that it should be a secondary reference, with your End of Braking as your primary. As you approach a corner...

- Look for and identify the BoB
- Then look for the EoB
- Be aware of your BoB with your peripheral
- While focusing in on the EoB, and noticing your BoB, start braking with a hard initial application of pressure
- As Peter says, use consistent pressure and then begin gradually releasing the pressure as you slow
- Manage the last little bit of pressure in the brake zone to fine-tune your corner entry speed and the amount of load you want on the front tires as you turn in
- End the brake zone deliberately, modulating the timing and rate of release

Many drivers are so focused on when they begin braking that they don't focus enough on how much initial pressure, the consistent gradual release throughout the zone, and the final timing and release ending with the EoB. Oh, and ensuring that there are no inconsistencies caused by a heel & toe throttle blip.

The super-trick advanced stuff is just doing the basics better.

Day #335: Plan to be prepared.

Most people would agree that when you prepare for something, you usually do better at whatever it was you prepared for. But how do you prepare for something that has so many variables that it's impossible to cover them all?

Take the start of a race, for example. How many different scenarios, how many different outcomes could there be? An infinite number of them, right?

So, should you give up on planning?

I don't think so.

What I think you should do is plan to be prepared for whatever could happen. Notice I said to be prepared for whatever could happen — and not on something that you think will happen. I don't care how experienced you are, the likelihood of you predicting what will happen when driving on a track is not good. So there is little point in planning for that. But there is huge value in planning for anything, and going into the situation with the mindset that you're ready for whatever is going to happen.

So, instead of planning for what you think will happen, plan to be prepared for whatever happens.

When I'm talking about planning, I'm talking about mentally preparing by using mental imagery (visualization). Run through as many scenarios as you can in your mind, but focus on the fact that you're mentally prepared for whatever happens. In fact, plan for things not happening the way you think they might, and instead, plan to be ready and prepared for anything.

The main reason that preparation improves a performance is the confidence that comes with it. If you know you've prepared, you feel ready and confident, increasing your chances of performing at your best. If you don't prepare at all, then you'll feel less ready, and less confident.

So, the value of mentally preparing is more in the confidence it builds, and the "I'm ready for anything" mindset.

Day #336: Focus on a harder initial application of the brakes.

If I said to you, “Brake half a brake marker later for a Turn...”, how would you feel?

I suspect you’d feel a bit tense, and maybe not even want to do what I told you to do.

But if I said, “Make your initial application of the brakes much harder,” that would be much easier. And you’d do it.

This is the exact scenario that played out while I rode in the right seat of a Porsche with a very good, experienced driver. It was an easy request and an even easier thing for my friend to do. So he did. And guess what? He immediately realized that he was braking too early. So he started braking later — comfortably, relaxed.

A couple of weeks later, I was working with a group of four drivers and one of the on-track “exercises” I asked them to do was to simply focus on making the initial application of the brakes harder.

I’ll never forget the look on one of the drivers' faces when he came off the track after just five laps. His smile filled his helmet and when I asked how that session went, he rambled on for ten minutes about all that he learned from that one simple session. He said, “Wow! I realized that when I brake harder, I can brake a LOT later. I’m braking waaaaay later than ever before. And it’s easy. Wow!”

Again, if I’d told him to brake later, he would have approached every brake zone with apprehension, and likely been all tensed up. And I suspect he would not have braked much, if any later, than before. But by focusing on making his initial application of the brakes harder, he realized that he could also brake later; he fixed that part of his driving all on his own.

If you’re thinking you want to brake later for a corner, ask yourself if you can apply more pressure to the brakes upon initial application of the pedal. You may not think you can, but eight out of ten drivers I see can. Try it. If your car has ABS, what’s the worse thing that can happen? If you’re without ABS, take a few brake zones, or laps, to gradually increase the initial application pressure. And then just see what happens.

If you don’t apply enough initial pressure on the pedal, you’ll spend the rest of the brake zone making up for it.

If you use this approach, I bet you’ll find yourself braking later without all the stress, tension, and anxiety that usually goes along with late braking.

I’m sure you’ve been told to squeeze the brakes on, to be smooth. That’s a good place to start, but like many things with driving, you may be overdoing it. Work on how hard you can quickly ramp up the pressure on the brakes.

Day #337: Understand the details and nuances of every corner.

Corners like Turn 11 at Watkins Glen, 10 at High Plains Raceway, 1 at Mid-Ohio, and 4 at Sonoma have one thing in common: The speed required to get to the apex is slower than most cars can handle at the exit.

Let's take the Glen's Turn 11. Let's say you turn into this right-hander that leads onto the front straight at 80MPH, and that's about the limit. You're traveling 90MPH on the approach, so you brake lightly and for a short amount of time, adjusting your speed to 80, and turn in. At that speed your car tracks, with a little slide, to the apex, where you're hard on the throttle.

But from the apex out to the exit/track-out curb, as you unwind the steering wheel, you feel you could have carried more speed. And that's where the challenge lies....

As you approach Turn 11 the next lap, you recall your feelings that you could carry more entry speed, so brake a little less and turn in carrying 82MPH. And guess what? You're not able to get to the apex, and ultimately delaying when you get to full throttle.

If you look at the detail of the layout of that corner, as Peter Krause and I do in the Virtual Track Walks, you notice something: the width of the track in the first half of the corner is not as wide as the last half of the corner. So, if you carry enough speed to have your car on the ragged edge/limit from apex to exit, you'd have too much speed to ever make it to the apex!

That's what makes that such a challenging corner. Every time you come out of Turn 11, you feel you left something on the table.

Okay, this is not true with every single car, but it is for most. And it is for some other corners, such as the ones I mentioned above. That's why it's so important to study the details of a circuit. If you don't, you'll find yourself looking for more speed in places where you shouldn't.

Day #338: Driving is a 2-handed sport. Use both to turn the steering wheel.

One of the strangest things taught by some schools to performance and race drivers is how to turn the steering wheel. Some tell their students to push up on the wheel with their outside hand (using the other hand to dampen the movement), and some teach pulling down with the inside hand.

The "push up" teachers suggest that it gives more precision, while the "pull down" group says you have more strength or leverage that way.

This is something I've personally experimented with, thought long and hard about, and listened carefully to the opinions of others. And still, I think both of these camps are crazy silly.

First of all, I think it's something that gets far too much attention. I think that if we let people push or pull naturally, they will do the right thing.

Second, driving a car is a two-handed sport, and that applies to which hand leads the way when turning the steering wheel. I don't think one should do the bulk of the work — I think both should. In other words, pushing up or pulling down is not something we should be emphasizing. We should be emphasizing that both hands should do an equal amount of work. Think about it: If we have a round steering wheel, and we have two hands equally spaced out on either side of that "circle" at the 9 and 3 o'clock position, why would we not use both hands equally? It makes no sense to me to use one more than the other. I hear the arguments for both the push up and pull down methods, but they really don't outweigh the benefits of using both hands equally.

Finally, from a learning factors perspective, telling someone — especially someone fairly new to the sport, and someone already feeling the stress of doing something relatively risky — to push up on the steering wheel, or pull down on the steering wheel is overcomplicating things. It's taking the act of driving and making it too granular. Sure, I believe in teaching good habits from the very beginning, but not to the point of making a student have to overthink what comes naturally to them (if they have two hands on the steering — a habit that I definitely do focus on building — then they will use both hands relatively equally).

I used to teach/coach tennis years ago, and I learned a lot about how to help people learn skills. I also learned how to slow the process of learning down (unfortunately!). It's critical to determine what skills/techniques should be taught from the very beginning, and what should be left to the person's natural tendencies.

As an example (and if you've played much tennis, you'll get this), during a serve, one should exhale as you hit the ball. That's a good habit to develop. But if you tell a relatively new tennis student to exhale while hitting the ball, you'll spend more time picking those balls out of the net — the results are terrible. Why? Because the focus is on the wrong things. The focus should be on hitting the ball over the net and into the

service court. And I noticed something interesting when teaching tennis (something that I later read that great tennis coaches know): when people hit a serve, they naturally exhale. So, to tell them to exhale would only slow the learning process down.

A funny thing related to that: When playing a “fun” tennis match, I would sometimes say to my opponent during a break, “You have a great serve. I’m curious, do you exhale or inhale when you hit your serve?” From that point on in the match, my opponent would miss more serves than ever before — because he was focused on the wrong thing — his breathing. And my question made him start to do things consciously that he would normally do subconsciously.

Do you want to slow a competitor down on the track? Ask him if he pushes the steering wheel up, or pulls down on it! Rather than focus on what’s important, he will then be focused on what’s not.

Day #339: Set goals, but focus on the performance to achieve the goal.

A driver once asked about what I have said regarding focusing on your performance rather than your competition or the result. He wondered if setting a goal of beating his competition, and winning a championship, was going against what I was saying.

Yes and no.

See, there is a big difference between setting a goal of beating another competitor/team (or winning a race/championship), and focusing on them. One is a goal that is in the future, and the other is right here and now. Focusing on beating another driver means that is your "right now," and that will take away from your performance — it's likely you won't perform at your best because your focus is on someone/something else, and not on something you can control.

You can't control your competition, or the results, right? You can influence them, but only by performing at your best. But you can't control them any more than you can control the weather (which also influences the results).

It's okay to have a goal of beating others. You probably wouldn't be racing if you didn't want to win! One could even argue that is why we race — to compete against, and beat, others. But the best way to do that is to perform at your very best, at your peak. And when you focus on others — your competition — you rarely perform at your best. Usually, you get caught up in either copying what they're doing, or being distracted by them. And that results in performing at less than your best.

When you focus on your own performance, you're more likely to perform at your best — and more likely to beat your competition.

Go ahead and set a goal. Then focus on what you have to do to accomplish that goal — how you need to perform, and what you need to do to make it happen.

Day #340: Develop your visual skills.

Whether you're in the off-season, or you're already back at the track, I want to give you a challenge.

Over the next four weeks, develop your visual skills while driving on the street:

- Practice looking at nothing. Even as you pull into a parking spot, exaggerate looking at the gap between cars. Look at the empty space, ignoring what's on either side (this is critical if a car spins on the track right in front of you). When changing lanes on the highway, look at the gap between the lane separating reflectors.
- Practice moving your head. Keep your neck flexible and look around corners.
- Practice being hyper-aware of everything around you, especially behind and beside you. Make sure that you notice every minute movement in your mirrors, but without staring into them.
- Practice looking further ahead. Sure, you've been told to look further ahead over and over again, but there's not a driver in the world who can't be reminded of its importance again — and practice it even more.
- Practice scanning. Keep your eyes moving. I almost hate to use the word "focus" when we talk about vision, because it makes you think you should focus your vision in one particular place.

Use your time driving on the street to improve your driving on the track.

Day #341: Learn from every source you can.

“He's an aggressive driver, but he makes it work. I'm not that kind of driver, so maybe I need to step up my aggressive level. That's the kind of things you learn. Also, it's good to have young blood coming in. It pushes you further, pushes your limits away, and he's a very smart young man, so obviously you're always learning, and if you think you're not learning, it's probably time to quit. I'm going to keep looking and learning from him, and other drivers.” - Simon Pagenaud, talking about Josef Newgarden

Indy 500 winner and Indy Car Champion, Simon Pagenaud, is willing to learn from his toughest competitors, admitting that he has more to improve.

How about you? And how can you learn and improve?

I'm often asked where I get the content, tips, and materials that I do. It's not just from one place! Obviously, there's my experience, both as a driver and a coach who works with drivers over a hundred times each year. Then I think about driving at least 364 days or year (more about this shortly), and I think that makes the biggest difference. And when I say I think about driving, it's during the process of writing about it, talking about it, and illustrating it. To teach something, one really needs to think deeply about the subject. And that leads to more insights.

I was recently asked whether I write everything that I appear to write. Yup. Everything. The conversation then went on to asking whether I have times when I can't come up with something, whether I get writer's block. Do I ever run out of ideas or content? Do I have to just go back and recycle past content? Writing here, every single week, on top of the books and articles I write elsewhere is a challenge at times. But rarely do I run out of things to share.

I will admit, there are times when I look at driving and think, "There is only so much to it, and that means that, yes, there's only so many ways I can say the same thing." Then I read a book or article, I watch a driver at a race or autocross event, I hear someone say something in a podcast, or someone tells me about an experience they've had... and it triggers another thought, often one that I've never had before. A fresh idea or piece of content to share with the world!

As I've said many times before, I'm a learning junkie. And I learn as much about driving from sources that aren't related to driving as I do from those specifically about driving. Reading a business book or listening to a podcast about entrepreneurship triggers just as many driving insights as what I learn when coaching a driver, or driving on track myself.

I encourage you to look outside the driving world for insights that you can apply to your driving.

For example, one of my favorite books of all time is *Thinking Body, Dancing Mind* by Chungliang Al Huang and Jerry Lynch. It has nothing to do with driving. Or does it? One of the messages the authors share shaped my thinking towards competition and performance. Until I read that book, I thought that being extremely competitive was always an advantage, but this book helped me see the downside of being overly-competitive — or, at least, focusing on the competition rather than on my own performance.

Another favorite book, *Lessons From the Art of Juggling* by Michael Gelb and Tony Buzan taught me more about how to learn — from a driving technique to understanding data — than any driving book.

I mentioned earlier about slacking off and not thinking about driving one day a year. Our brains need time to synthesize what we learn, too. I've had to learn to take in information, and give it time to sink in. Because I love to learn, I've had a habit of finishing one book and immediately picking up the next one to devour. It's like a race to read as many books as I can! But I'm trying to give myself more time to reflect on what I've read recently. It's made a difference.

One of my goals is to be the most complete resource of information for performance and race drivers to learn from. That's a bit like trying to achieve the perfect lap... I doubt I'll ever be the most complete, but as long as I keep striving to be, I'll end up being better than if I didn't bother.

And here's the thing: Like Simon Pagenaud, I learn a ton from other people. And as I said, I read a lot, and much of it is not just about driving. Of course, then I read driving into what I read!

Day #342: Filter the information you take in.

Guess what? Not all information that you find on the internet is accurate and worthwhile. I'm not talking specifically about "fake news," but close. There is fake information out there on YouTube. Yup, I know, shocking, right?!

How do I know this? Well, I've been poking around the interwebs lately to see what's out there, to learn from others, and find resources that I can share with you. And I've found more than a few resources that are saying some things that are either “wrong” or not very helpful (and therefore, not the best use of your time in viewing/reading).

You might be thinking, “What makes Ross the expert? Who says that his opinion is any better than anyone else’s? Maybe his content and resources are fake!” That may be true, and you're smart to question what I say. And that's my point!

You should question every piece of advice, content, resource, whatever. You need to filter what you take in and do your best to determine whether it's the right information for you.

So many drivers head to the track and practice what they think is the right stuff, before making sure it is the right stuff. And that's a mistake.

For example, you go to the track and focus on driving all corners with a late apex (because that was your take-away from sitting through a classroom session, or an instructor told you to do this). You spend lap after lap “perfecting” late apex corners. Great. Well, sort of great. See, not all corners should be driven with a late apex. If you'd taken some time before going to the track, watching videos, reading articles and books, and building your database of knowledge, you'd know this, and therefore, wouldn’t practice things that are not always true. Of course, you know that.

Another example: You read an article or book that says the line into a corner is a spiral. So you think about the line you drive and change the way you drive all corners. But here's the thing. Not all corners use a line that is like a spiral. Some are, some are not. You need to filter what you take in.

I've had far too many drivers read one of my books and take what I said too literally, or as a “rule.” Sure, there are some things that are certain. The laws of physics fit this category. But some things I've written are guidelines, and not rules or laws. If it applies 100% of the time, like the laws of physics, then it’s a rule. If it applies more than 51% of the time, but not necessarily 100% of the time, it’s a guideline. My Speed Secrets are guidelines.

For example, I've written “Never coast,” and “Smooth is fast.” I strongly believe these are great guidelines. But there are exceptions, which is why I say they're guidelines. There are times when you should coast, and times where you need to be less smooth. These guidelines apply at least more times than they don't, but not all of the time.

The more you learn before you go to the track, the more you'll learn in the limited amount of (expensive) time you have behind the wheel on the track. But don't take everything too literally. Think about what you take in and, especially, question it. Understand the why behind what's being said. That's part of the process of filtering it.

I don't know if you've noticed but just about anyone can record a video, a podcast (hey, even I can do that!), write an article, or write and publish a book. And for some reason, we tend to think that someone who is published is an expert. What I've noticed lately is there are fewer experts than there are people publishing content and resources. As part of your learning process, be selective about who you learn from.

Now, question what you just read and decide whether it's for you or not. And if so, why?

Day #343: Find and drive the rhythm of the track.

Every track has a rhythm. It's your mission to learn it, to make it part of your driving. Pay attention to it.

As you head to the next race or driving event, imagine how the track flows: how one corner leads into the next straight or another corner; how elevation changes relate to traction; the way surface changes relates to traction; how the banking of a turn flattens onto a straight section; where the straight brake zone blends into a corner, and the way multiple turns links with others.

How do you currently look at a track? Most drivers look at tracks corner by corner, straight by straight. And that's a good way to begin. But as you get the feel for the individual corners, straights, and sections, pay attention to the way they relate to one another. Pay attention to the flow, the rhythm of the track. Look at the track as a whole.

As drivers, we can often get too caught up in the mechanics of our driving techniques, and the technical aspects of the corners. It's like we're focused on the musical notes, and not on the music; like the brush strokes of a painting rather than taking in the entire piece of art.

Be aware of the flow of the track, play it in your mind, and listen to the rhythm of the circuit. Then let that come through in your driving.

Day #344: Learn from the near misses.

"The crashes people remember, but drivers remember the near misses." – Mario Andretti

When I look at some of the quotes that are attributed to Mario Andretti, I have to wonder… did he really say that, or was it something that was made up from things he said? It's the same with Albert Einstein quotes. There are many that are attributed to him, inaccurately.

But who cares who said it if it's a good message, right?! So I'm going to run with what Mario supposedly said. The quote above is so true! While I don't actually recall much about the few crashes I've had in my career, I can vividly recall the near misses.

Why is that?

Because I didn't learn much from the crashes, and did learn a lot from the near misses. It's the near misses that have taught me how to drive faster.

I recall popping up over the top of Turn 3 at Sears Point/Sonoma in a Riley & Scott World Sports Car, almost getting airborne from the speed, plowing off the outside of the exit curbing, and brushing the tire wall. It was during qualifying for an IMSA race, and I was looking for another tenth or two.

If I'd focused on what I learned from the crash (not a bad one, since I just brushed the wall, but it was enough to end that session), there wasn't much to learn. It was the near miss in Turn 2 coming up the hill that I learned from, and I remember most.

In that car, I'd be easily flat through Turn 1, arcing the car up the hill through 2, while lifting off the throttle, then giving the car another short burst of full throttle as I lined up with the curbing on the left setting up for Turn 3. I felt as though I needed to lift the throttle to get the car to turn through 2, but then I'd give it a small squirt of full throttle before braking for Turn 3. Until that lap….

Since it was qualifying, and it was time to really go for that one flier lap, I decided I'd flat Turn 2 without a lift — then I'd figure out how to get the car "whoa-ed down" for Turn 3. It worked… for a little while. I actually took Turn 2 at full throttle. The problem was that I arrived at Turn 3 at a higher speed than ever before and I just ran out of room to get it slowed down enough to make that corner without spearing off the track at the exit.

That was what I'd call a near miss, followed by a crash. I learned from the first part, but not so much from the second part. The next morning, in the warm-up practice session, I did Turn 2 flat again, but this time was better prepared for the speed I was carrying into Turn 3, and made it work. I did that once, then put it aside for the race (I had a brilliantly fun battle in the race with Johnny O'Connell and Jim Pace (RIP), using to my advantage what I'd learned, with a few laps to go).

As with most mistakes, it’s not what happens at the “scene of the incident” that makes the difference. It's before that — usually well before it. Most crashes are the result of more than one misjudgment that's made many seconds earlier.

The next time you have an “off” (spin, crash), rewind to the near miss that might have started it all. Spend time reviewing what led up to the “off,” and focus on learning from the near miss that likely led to it.

The next time you have a near miss — one of those close calls that gets your heart pumping just a little faster — think about what you can learn from it. It’s often during these near misses that you learn that you can actually brake later, take a corner faster, or get on the throttle sooner.

Day #345: Check progress with your peripheral vision.

Driving the ideal line consistently is an important part of performance and race driving. But how do you know whether you're on the right line or not? Or even the line you intended to drive, whether it's the right/ideal one or not?

One of the trickiest parts of track driving is that you can't go back and fix something you did wrong, or even check on how you're doing while you're doing it until it's too late to do something about it! Of course, that's part of the appeal of track driving.

We're constantly told to look ahead and where we want to go, so how do you know whether we're clipping past the apex without looking at — and not ahead or where we want to go? We check progress with our peripheral vision.

I've spent a lot of time observing drivers, researching vision, talking to vision therapists, and learning from eye tracking technology. Oh, and testing different approaches and techniques myself. On one hand, using our peripheral vision is something we do naturally, and on the other it takes deliberate practice to use it as well as we should. And sure, some people seem to naturally be better at using their peripheral vision than others, but I'd argue that it's because of how they've trained themselves, whether they're aware of that training or not.

Although it may be completely unscientific, it seems to me that drivers who are very good with their peripheral vision have spent a lot of time playing sport where that's important — sports like squash, racquetball, basketball or hockey. One cannot be a good player of any of these sports unless you have "eyes in the back of your head." Other sports, such as running, put less stress on our peripheral vision.

Used properly, the second you've looked ahead and picked up a reference point that you're approaching, that forward visual focus should be moving ahead and looking for the next one. But as you're approaching and passing by the first reference point, you should be aware of how your car is placed in relation to it. Is it where it should be, right next to or on top of that reference? And are you on the intended path or direction in relation to it?

Your reactions are actually quicker when using your peripheral vision than they are when using your central vision acuity (or, foveal vision, that small area we see in perfect detail when focusing on something). Just think about the times you've reacted to something you caught out of the corner of your eye.

In addition to exercises, deliberately use your peripheral vision while driving on the road; play racquet sports, basketball, or hockey; and as you read this article, be aware of what you notice with your peripheral vision. By focusing on not focusing, but rather, using your peripheral vision more, you become more visually aware of what's going on around you on the track, you'll be able to check your progress in relation to references at speed. After all, while driving past an apex, you can't turn your head and look down at

it, can you? Well, I guess you could, but who knows where you'd end up after doing that!

Think of your vision as a process of glancing ahead, focusing ahead on where you want to go (there is always some form of reference ahead, even if it's a wide-open track surface), and then checking your progress peripherally.

And remember, the sooner you recognize that you're not where you want your car to be, or at the right angle, the sooner you can make an adjustment to your line. And the sooner you make that adjustment, the smaller that adjustment needs to be. Ultimately, this will result in fewer offs and spins — and driving faster.

Day #346: Give yourself enough recovery time.

One of the biggest advances in athletics has been in the area of training methods, and specifically, the understanding of the importance of recovery. “No pain, no gain” is no longer the approach to improved athletic performance. “Some pain, followed by adequate recovery leads to gain” is more like it. Much of the research into improving athletic performance is focused on ways to help an athlete recover faster between training, since a body that has not recovered and then is stressed again through training only breaks down further.

What does this have to do with drivers, with you? Obviously, it applies to any physical training you currently do. It’s critical that you manage your recovery time, as simply pushing and pushing and pushing will not lead to big gains, and may actually go the opposite way. This is one of the big reasons working with a knowledgeable fitness trainers is so effective, as they can help you manage this recovery time.

But what about non-physical training?

I was thinking about how much time the average track day, club racer, and even professional race driver spends NOT driving, and how to make the most of that time. Compare the amount of time you spend driving on the track with the amount you don’t spend on the track! If you’re like most, you spend many, many, many times more time NOT track driving than you do doing what you love most.

So, how can you make the most of the time when you’re not on track? I started making a list...

- Reading
- Watching video
- Watching racing on TV
- Observing other cars/drivers on track
- Reviewing/analyzing data
- Participating in webinars
- Completing e-learning programs
- Attending training seminars
- Talking to drivers, engineers, officials, etc.
- Mental imagery
- Non-track driving (street driving)
- "Cross training" (driving karts, riding bikes, skiing, etc.)
- Note-taking (and reviewing past notes) — I would include drawing in this, as sketching out the line you drive on a track map will build mental programming, as well as a deeper understanding

Many of these may seem like no-brainers, and you didn't need me to tell you about them. And I know I've missed some (any suggestions to add to this list?). But my guess is there are some on my list that you don't use — but could quite easily.

Recovery time is not just for physical training. Research by neuroscientists have shown that our brains require recovery time, as well. When I talked with Matt Walker, the author of the great book *Why We Sleep*, he spoke of what happens to the information we take in during the day while we sleep (listen to Speed Secrets Podcast #101 for some amazing information). You know that sleep is physical recovery time, but an important part of sleep is mental recovery time, and specifically, time for our brains to "make sense" of what we took in during our waking hours.

Interestingly, reading is a form of recovery time from physical activity. When you practice something on track, and then read about it, the learning that occurs is much deeper. Often, the reading provides the Why behind what you experienced on track.

The same thing happens when you review data, video, listen to a speaker talk about driving, and even watch racing on TV. All of it contributes to the mental recovery time, and a deeper understanding.

Perhaps nothing contributes to the learning recovery process like writing notes and making illustrations that help clarify your thinking. A while ago, I suggested a simple way of improving your driving: write five hundreds words every single week about something to do with driving. It doesn't matter what it is, but just write your thoughts down. You may hate writing, or think you suck at it. That's doesn't matter. It doesn't need to be worthy of anyone else reading it. It's the process of getting an idea clear enough in your mind that you can then put it on paper (digital or not). That's what makes the difference. And it's part of a good mental recovery routine.

Finally — and related to my last point about writing notes — take time to reflect. Personally, this is something I'm working at putting more time towards. A major part of my personal recovery time in 2019 will be spent simply giving myself more time to think. I'm so addicted to learning that I often, for example, jump from reading one book to the next without giving myself enough time to fully reflect on what I really learned from the first one. I'm giving my brain more recovery time this year, and hope that I'll learn even more by doing so. I'm making the most of the time that I'm not driving on track. How about you?

Day #347: Think deeply about how you prioritize corners.

In my first *Speed Secrets* book (way back in 1997), I wrote that the most important corner on a track is NOT the one leading onto the longest straight. And yet, I hear the "most important corner is the one leading onto longest straight" advice being dished out left, right, center and over the top. Either drivers and instructors do not believe me and think my opinion is wrong (I'm okay with that), or maybe the definition of "most important" is not clear or agreed upon.

To me, the most important corner is the one that will make the biggest difference to one's overall lap time on a consistent basis.

I've coached thousands of drivers, in large groups, small groups, and individually, and the corner they struggle with the most — are challenged or intimidated by — is rarely the slow corner leading onto the longest straight. It's usually the fastest corner, or a blind corner, or one with a concrete wall on the outside of it. And that's where the most time can be made up.

Turns 7 and 11 at Sonoma are the two corners that lead to the longest straightaways on this wonderful, undulating circuit. I'm not going to suggest that they are easy corners, because they have some tricky bits about them, and getting a good exit out of them is critical to a good lap time. But most drivers, short of Scott Dixon and Alexander Rossi, leave more on the table in Turn 6 — the Carousel — than they do in 7 or 11. Why? If you've driven Sonoma, you know why; if you've never driven Sonoma, it's a fast, downhill more-than-180-degree corner. And it has surface changes in it. And it leads onto a straight that is almost as long as the ones after Turns 7 and 11.

How about Sebring? The corner that leads onto the longest straight is Turn 16, a 90-degree corner that exits onto a super-long straight. Follow the classic advice and it's obvious it's the most important corner. And yet, anyone who has driven Sebring will tell you that Turns 1 and 17 have a bigger impact on lap time, simply because they are so darn difficult to get right. And they're intimidating. And bumpy (that goes without saying when talking about Sebring)!

Lime Rock? Yay, the corner that leads onto the longest straightaway (the Downhill) is also the fastest one on the track! So it's a no-brainer that it's the most important, right? Maybe. Even then, it's arguable whether it's the corner that will make or break your lap times. If you drive a car that can driven at full throttle — or very near to full throttle — through the Downhill, maybe there are other corners that are more difficult. Or that you spend more time in. What? Yes, that's another factor to consider: how much time you spend in the corner, because the more time you spend in it, the bigger the impact is going to be on lap time if you're even one percent off the pace through it. Could Big Bend (Turn 1) be more important than the Downhill for lap time? An LMP3 spends about 3 seconds in the Downhill, and just under 6 seconds in Big Bend. Something to think about….

Turn 11 at COTA leads onto a very long straight. And yet, listen to F1 drivers talk at the U.S. Grand Prix, and they're not focusing so much on it. The medium to high-speed Esses (Turns 3 to 9, actually), and the high-speed carousel (Turns 16, 17, 18) get their attention. It also gets the attention of the engineering staff, as they spend more time working to make the cars fast through these corners than they do the slow Turn 11.

Time to get personal. In 2017, I co-drove a Ginetta LMP3 car with Colin Braun in the Thunderhill 25-Hour race. I hadn't driven a car like that for a few years, so I didn't expect to be breathing down Colin's neck in terms of lap times! But anytime a driver has a chance to compare themselves with one of the best in the world, it's a cool experience. As I write this piece, I'm looking at a data overlay of my best lap with Colin's. In Turn 14-15, the corners that lead onto the longest straight, I'm pretty close. Where is the biggest difference between us? Turn 8. The fast, intimidating, challenging corner. Sure, for Colin, he needs to focus on getting good exits out of 15 and onto that long front straight — as he does everywhere. But even when I overlay his best five laps, guess where the biggest differences are? Turn 2, a corner that you spend a lot of time in (a 180-degree corner that takes almost 8 seconds to get through, with a minimum speed of 85 MPH), and Turn 8, a relatively short corner, but one with a minimum speed of 135 MPH in an LMP3 car.

Just to be clear, I'm not saying that the advice about the corner leading onto the longest straight is wrong. I'm just saying it's not the only thing to consider when prioritizing which corners to focus on.

At the beginning of the year, I wrote that a goal of mine is to push you to develop a deeper understanding of driving, and specifically to build a deeper level of Why. The corner that leads onto the longest straight is one layer of understanding. Considering which corners are the most challenging, the most intimidating, and the ones you spend the most time in add another dimension to the process of prioritizing corners. Think about it.

Day #348: Cut your goals in half.

Cut your goals in half. That doesn't sound like the typical type-A personality approach to competitive performance driving, does it? And to be honest, I'm not even 100 percent sold on this approach, but....

I was reading a book, *Finish*, by Jon Acuff, and then listening to him be interviewed on a podcast; "Cut your goals in half" was some of the advice he gives.

Later, I was talking with a driver who had shown up at a track to drive a type of car he had never driven before. He had a solid goal in his mind to turn a set lap time. He didn't achieve that goal. Even worse, as he was quite a way off of the lap time goal, he got mad at himself. It seemed to me that the madder he got, the more he struggled with his driving.

So when I heard Acuff's advice about cutting our goals in half so we would be more likely to achieve them, it made me think. I've never been one to look for the easy goals, but I'll take them when I can get them! And I do believe that as a coach, it's important that I help my drivers get a few "wins" under their belts — whether that's braking one car length later for a corner, turning a lap time that is a couple of tenths faster, or winning a race.

I know you understand why I'm suggesting the easy wins are important. It builds momentum through increased confidence.

The difference between a driver with confidence and one without is dramatic! So, when we set a goal... and then cut it in half, it's more likely to result in a "win." That builds confidence, and that builds momentum.

Now, I'm not suggesting that you set a goal of "I'm going to drive through pre-grid and onto the track." That's silly easy. But if you set a goal that is too difficult, you sometimes set yourself up for failure. That's silly hard.

Part of the reason I'm writing this piece is to help me figure out exactly how I feel about the advice of cutting our goals in half. Like I said earlier, I'm not 100 percent convinced it's the right thing to do, but it's worth considering if it could work for you. If you're the type who is pretty hard on yourself, back off a little, get some easier wins (notice I said easier, and not easy), build confidence, and then momentum.

Let me know how it goes for you.

Day #349: Draw your data traces with your hands and feet.

While not every driver has a data system, many do (I see more and more each time I'm at the track). With lower cost options, and the systems becoming easier and easier to use, it's almost a no-brainer that you should use one. (I'm not going to get into what system and why, here, though).

The simplest data system records speed and lateral and longitudinal g-forces, and that's enough to help any driver improve the way they drive. If you want to go to the next level, add in throttle position and brake pressure (and even steering angle), and you have everything you need to coach yourself. But speed and g-loads is enough... if you know what to do with it. If you don't know what to do with your data, then no amount of channels, information, squiggly lines on a screen, sensors, and whatever else will help. I've seen drivers spend a lot of money on higher-end systems when they were getting less than ten percent of the information out what they currently have.

Learn how to download and look at the speed, and lateral and longitudinal g-forces traces on your smartphone or laptop. Do that consistently, after every single session on track. Okay, I know that on some days there isn't enough time to download and analyze data between track sessions. Then at least download at the end of each day.

There are many serious data gurus out there. Soak up every bit of information you can from them. There are Facebook pages and YouTube channels devoted to sharing of knowledge around the use of data. Follow them. There are live and recorded seminars, workshops, and webinars available almost all the time (I might have a Masterclass soon). The knowledge and information is readily available.

But here's what I think is the most important thing to do with your data: calibrate what you feel with what the squiggly lines on the screen look like.

Your speed trace line went down, flattened out, and then came back up – a bit like the shape of the letter U? Why? And that line went down and back up again, leaving something like a V shape? Why? What did those corners feel like? What was different? What did you do differently with your driving?

Ask yourself those questions, and don't move on until you can answer it with a decent amount of understanding. The more you think about why one line looks the way it does, and how that relates to what you felt when driving — and what you were doing — the more useful the data becomes.

Ultimately, your goal is to get so you can go on track and deliberately draw a line — speed, g, throttle, brake, whatever — with the way you drive. When you can do that, then you're really ready to upgrade your current data system or app.

You want the speed trace to look like a V in a corner? Draw that by braking, turning the car, and accelerating hard out of the corner. You want it to look more like a U? Brush the brakes, roll momentum into the corner, and squeeze back on the power.

You notice that the car turns into a tight corner best when the speed trace slopes down on an angle, then changes to a slightly different slope just before the bottom of the V shape (the min speed). What were you doing with your driving when this speed trace looked like that? Oh yeah, you trail braked longer into the corner, being slower with the release of the brakes, keeping load on the front tires while turning in.

Now you're trying to get your car to turn in better in a different corner… what kind of speed trace are you looking for? Yes, that slope with a short kink at the bottom (a bit like a hockey stick). Go draw that with your brake foot, then come in to check how you did, and relate that to how the car responded.

If you really want to have some fun (okay, all of this is fun, but this might be extra fun), then follow this process:

1. Drive a session or two on track
2. Analyze your data after each session
3. Decide what you want your speed, long/lat g, brake pressure, throttle, and/or steering angle to look like after the next session on track — what the data is telling you to do to go faster
4. Pre-play in your mind (visualize) doing exactly what it is you want to do on track
5. Go on track and "draw the trace" with your hands and feet (steering wheel, brake and gas pedals)
6. When you get back to the paddock, and before you look at the data (this is critical!), take a piece of paper and draw what you think the data will look like
7. Now, look at the data and see how you did
8. Go back to step 2 and go through this process again

Eventually your mind will be so in tune with what you're doing that the data acquisition is simply backing up what you've already felt and reported. But — and this is the key — data acquisition will give you another dimension of detail and accuracy that you just can't develop on your own.

I see and talk to many drivers who get totally absorbed into looking at their data, trying to find the one secret bit of information that will tell them how to drive faster. But I see very few going through the process I've just outlined. I don't think it's a coincidence that the best drivers I coach have the ability to draw their data before they actually see it on the laptop or smartphone screen.

Data acquisition is an extremely powerful self-coaching tool... especially if you synchronize it with your driving.

Day #350: Master Your Braking.

When you think about the various ways you can apply, maintain and/or modulate pressure, and release the brake pedal, you realize that there's an infinite number of ways you can use the brakes.



Typically, when you first begin performance or race driving, you have one way to brake, and that one way comes from how you've driven on the road. It's a habit, a mental program. Most drivers have the habit of gradually applying the brakes until the peak pressure is just shortly before reaching what I call the End-of-Braking (EoB) point. In this case, the brake pressure trace on a data acquisition system would look like the one to the right.



An inexperienced track driver will begin to build a new habit of applying the initial pressure more quickly, but at first, they will maintain that pressure until they release the brakes relatively quickly. The brake pressure trace to the right illustrates how an inexperienced track driver typically brakes — not quite hard enough, maintains that pressure for a while, and then releases the pedal without much thought at all as to how to use this all-important portion of the brake zone.

As a driver gains experience and begins to learn the subtleties of braking and what it takes to drive consistently fast on a track, you find that the driver has almost flipped the brake pressure trace: the initial application of the pedal is harder, and then they taper or ease off the pedal the further along the brake zone they travel. See the illustration of the "Hard initial, smooth release," below.

With even more practice, experience, and coaching, track drivers start to use the initial application and the timing and rate of release in a very deliberate manner to slow the car most efficiently, and to balance the car in just the right way to either emphasize outright cornering grip or rotating the car into the turn.

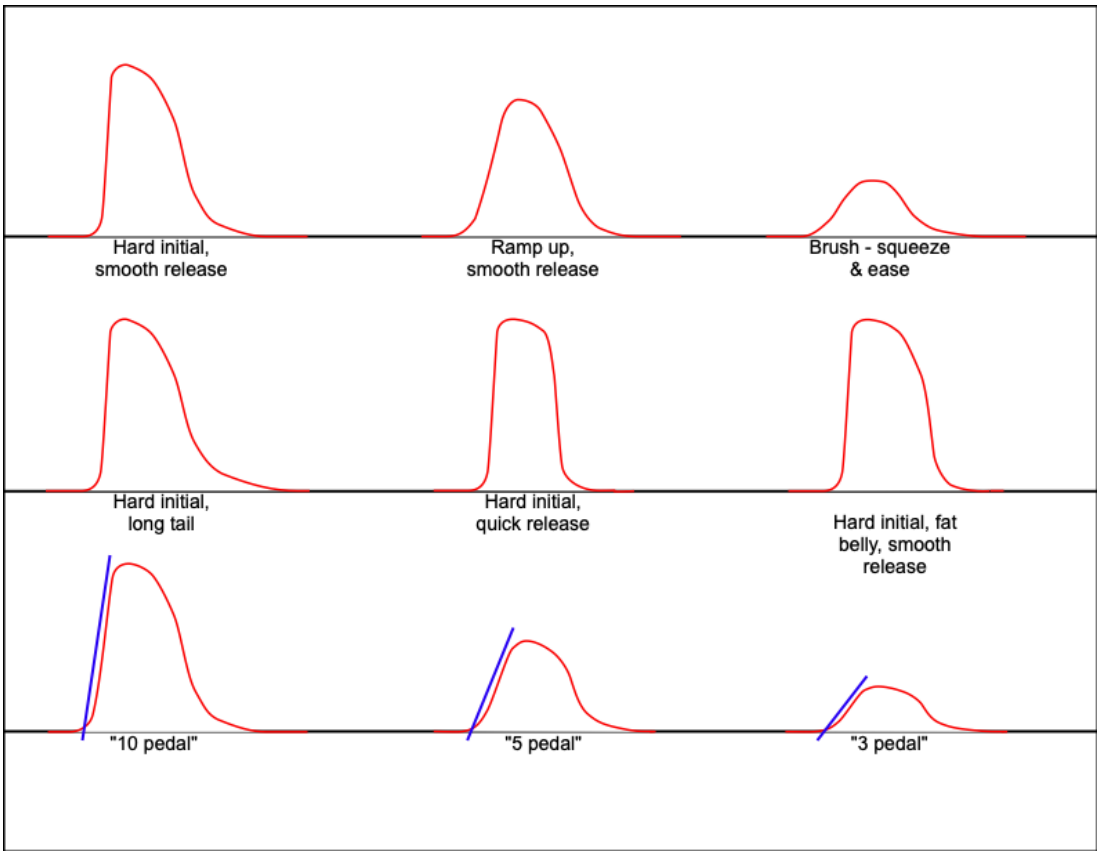
With experience, you gradually learn multiple different techniques to use the brakes, but it's a random and not very precise way to develop your braking skills. For most drivers, they still stick to just a small number of variations of how they use the brakes. Depending on a driver's experience level, their own level of comfort with experimenting, the coaching/instruction they've received, and the variety of cars and tracks they've driven, they might only have a half dozen or so different approaches to it.

Drivers who get paid big bucks to drive race cars have an almost countless number of ways to use the brakes, and they pick and choose from their library of braking techniques for each and every corner. They may even choose to change or adapt how

they use the brakes throughout a single session on track, due to the changing ambient and track temperatures, and tire conditions.

While it would be nearly impossible to catalog every single way the best drivers use the brakes, I've illustrated the most common ways below.

Before I get into the examples, if you've never used a data system that shows brake pressure, imagine a line that goes upward on a graph when pressure is applied, and back downward as you release pressure. See the illustration below, reading each brake pressure trace from left to right.



Now, look at the illustration and note the examples.

The initial application of the brakes can be quick, or a more gradual build-up of pressure. The brake pressure can then be maintained for some amount of time, and then the release can be gradual or quite rapid.

There will always be a difference between squeezing the pedal and stabbing or smacking it down; and there's a difference between squeezing the pedal too slowly versus quickly and smoothly applying it.

The amount of time you spend at or around maximum pressure is partly dependent on the length of the brake zone (which depends on how much slowing is required, and the effectiveness of your car's brakes and tires).

The timing and rate of release is the next thing to pay attention to. You can begin releasing almost the instant after reaching maximum pressure; it can be a gradual thing; or it can be some combination of these. Or the near-maximum pressure can be maintained for a short period of time — notice what we refer to as a pressure trace with a "fat belly" as an example of combining timing and rate.

Another part of the timing and rate of release is the final part — the "tail" of the release/trace. It can be almost non-existent (a quick release), or long — a slow release, meaning a lot of light trail braking.

The overall pressure can be thought of on a scale of 1 to 10, with 10 being maximum pressure — threshold braking, on the verge of locking up or fully activating the ABS — and a "1 pedal" would mean barely touching it, with just a tiny amount of pressure.

As a general guideline, the less overall pressure, the more gradual the application will be. Notice the difference in the slope of the brake traces, between a "10," "5," and "3" pedal.

As always, there are exceptions to everything I've written here! And that's one of the many reasons driving is so much fun — there are an infinite number of ways to use the brakes, so we need to constantly experiment, test, and adapt our braking technique. And that means you have to drive a lot (awww, too bad!). If there was only one way to do it, driving would be boring.

Some of you reading this will be able to look at these brake traces in the illustrations, then go "draw them with your brake foot." In other words, look at the shape of the line in the graph, and imitate it when you get on track. That's because of the way your brain works, and how visual a person you are. For others, these illustrations may not be so intuitive, and in that case, I suggest you read through what I've written a couple of times, and turn what I'm saying into the action and movement of your foot.

How to Practice

Looking at these examples, you can see that there are three phases of a brake zone:

1. Initial application
2. Maintenance
3. Timing and rate of release

As with any skill development, the more you're able to break it down into its most basic steps, the better. In other words, the more deliberate you are in how you practice, the more effective and efficient it will be.

Pick no more than two places where you have to brake for a corner, and experiment. Ideally, these are brake zones that lead to corners with some margin for error. Also, the longer the brake zone is, the more time you'll have to try different approaches.

Spend an entire on-track session solely focused on varying how quickly and how hard you apply the initial pressure. Deliberately pounce on the pedal, then slowly squeeze it, and see how quickly you can squeeze it.

Even though the maintenance phase is next in line, I recommend practicing it last.

Spend an entire session focused on experimenting with the timing and rate of release of the brakes.

Begin releasing the pressure:

- prior to the turn-in point (this is the timing part), but make the rate of release slow and long.
- right at the turn-in point, and using a slow, long release.
- after the turn-in point, using a slow, long release.
- prior to the turn-in point, releasing the pressure relatively quickly.
- right at the turn-in point, and using a relatively quick release.
- after the turn-in point, using a relatively quick release.

Finally, it's time to focus on the maintenance phase of the brake zone. Having said that, most of this phase will be taken care of by the initial application and the timing and rate of release. However, I still recommend you spend an entire track session being aware of whether you immediately begin releasing pressure after reaching the peak, or you maintain pressure near the peak for some amount of time. As I said earlier, most of the maintenance phase will be dictated by the length of the brake zone, which is influenced by the speed you're approaching the corner, how much you need to slow, and the effectiveness of your car's braking system.

A goal for you is to develop as many different ways to use the brakes to improve your consistent speed, so you'll have the ability to adapt to what your car and the track want.

Day #351: Not all corners are created equal. Treat each one differently.

As part of a seminar/workshop I was doing, I used data to coach drivers. While doing this, I found myself repeating over and over again: “Brake late, hard, and short for slow corners; brake early, light, and long for fast corners.”

As a reminder, the more balanced (with weight equally distributed over all four tires) your car is — as opposed to when it’s pitched forward from braking, squatting in the rear from acceleration, or leaning from cornering — the more overall traction it has. In other words, the less weight transfer you cause, the more grip you have to work with (if you want more background on this, I explain it in the “Why Smooth is Fast” video posted here). Flip this concept around and you get: The more weight transfer you cause, the less traction you have to work with.

But… there are times when you’re better off trading off some overall traction to get your car to change direction.

See, when you turn into a corner with weight transferred to the front — even though you have less overall traction — it’s easier to get your car to change direction, to “rotate” around the corner.

In tight, slow corners, your priority is getting your car to change direction quickly enough so you can begin to accelerate. Therefore, it’s worth it to compromise overall grip for rotating the car, by using weight transfer. That’s why it’s usually best to trail brake some amount into tight/slow corners.

Fast corners are the opposite. Because you don’t need to change the direction of the car so much in fast, larger radius corners, your number one priority is maximizing traction — and using it. So, your goal in fast corners is to minimize weight transfer. To do that, you should make any speed adjustments (braking, or even a lift of the throttle if that’s all the corner requires) early, and small. If you’re braking, an early, light brake usually works best, even if that means braking for a little longer time.

Trade off grip for rotation in slower corners; trade off rotation for grip in fast corners.

When I review drivers' data, I see they often drive all corners the same. They either brake late and hard for every corner, or light and early. But most tracks have a combination of slow and tight, and fast and large-radius corners. Which means you need to change how you approach each one. For some corners, you should brake late, hard and in as short a distance as possible (and carry a little of that braking into the turn — a little trail braking to help rotate the car). For others, you should brake earlier, lighter, and even for a longer period of time.

Think about this as you look at the next track you’re planning to drive. As with everything, there are exceptions to the rules — and even to “guidelines” like this Speed Secret.

Day #352: Check your min speed with both your gut and report cards.

We've often been told that "In slow, out fast" is the way to drive. The next piece of advice — tied into the "In slow, out fast" saying — is, "Whoever begins accelerating first is going to win the race down the straight to the next corner." And there's some truth in all of this advice. But nothing in our sport is simple — which is what makes it fun and challenging. Balance is not only critical in terms of how you manage your car's weight and traction, but there's a balance between entering a corner too fast, or too slow. I'm going to steal from four-time SCCA and SRO Champion Kevin Boehm's words, and refer to this balance as the Goldilocks zone: not too little, not too fast, but get the corner entry speed just right and life is good.

See, one way of beginning to accelerate early is to enter the corner at 10 MPH. Hey, you could tromp on the throttle the moment you turned into the corner. But you would never be able to make up for the extremely slow speed, no matter how early you got to full throttle, and you'd be slow heading down the next straight. The porridge is "too cold."

On the "too hot" side of this story, you're entering the corner at a speed that delays your ability to get to full throttle until very late in the corner. Despite carrying more speed into — and even through — the corner, the lateness of full throttle means the straightaway speed will not be what it could be. In both cases – too slow an entry speed, and too fast an entry speed - the overall lap time will suffer.

Enter Goldilocks. We want just the "right" corner entry speed that allows us to get to full throttle early, while beginning to accelerate from the highest speed possible. That's our goal. Simple, right? If it was, everyone would be Lewis Hamilton, Scott Dixon, Colin Braun, or Jimmie Johnson.

If you have a data system, it's easy to look at your speed trace on the graph and see what your minimum speed is in each corner. You can even compare it over multiple laps, noticing what the highest min speed was. If your best lap time came from laps where your min speed was consistently the fastest, then it's a good clue that you might be over-slowing at other times. That, then, becomes your focus: carrying a slightly higher min speed. If you do this for a while, and notice that your lap times are now beginning to suffer on the laps with the highest min speeds, you've gone too far. It's time to think, "In slow, out fast."

Of course, "slow" is a relative term. What is slow to you may be fast to another, or vice versa. So never forget that, especially when you're talking to another driver, or an instructor.

Sticking with the speed trace on a data graph, the other thing you need to look at is where the min speed is. You can have two drivers with the exact same min speeds, but one is consistently faster because of where the min speed is in the corner. For example, one driver's min speed could be half-way between the turn-in point and apex, while the

other driver's min speed is at the apex. Which do you think will usually be the fastest? Usually, but not always, it'll be the driver with the min speed earlier in the corner. Why? Because earlier min speed allows the driver to begin accelerating earlier.

Remember our goal: to maximize our min speed while allowing us to get to full throttle as early as possible. Get the position of the min speed just right, and that will happen.

So, you don't have a data system to work with? The same thing applies, but you're going to need to rely on your gut feel. And by gut feel, I actually mean your sense of speed and timing. Oh, and the use of other "report cards." As you exit a corner, make mental note of where you truly got to full throttle. You can usually reference this point to the apex — at full throttle just before, at, or some distance after it. But also make mental note of how long it took to get from the turn-in point to the apex — was it in more or less time than the last lap? And as you exit every single corner, use some type of report card to see how well you did: what RPM are you at as you run past the end of the exit curbing, where did you upshift compared to previous laps (an earlier upshift — as long as you're consistent with when you make the shift — means you got out of the previous corner faster), what's your speed or RPM as you pass a steady reference point nearing the end of the straightaway (again, the higher the speed, the better your exit speed was).

I've seen tons of drivers who get on the throttle early in corners because their entry speed was so slow. I've seen not quite as many who are late to throttle because they carried too much entry speed. And I've seen a very few drivers who get it right consistently: just the right entry speed so they can get back to full throttle early, and from a higher speed.

If you pay attention to what your personal sense of speed and timing tells you, what your "report cards" tell you, and what the data tells you, you will improve your ability to get the ideal min speed, at the right place in the corner.

Day #353: Make practice more difficult than what you're practicing for.

When conducting a recent "How To Use Simulators To Improve Your Driving" workshop in Boston, I was pleased to see the number of drivers who showed up with their driving shoes. If you're going to practice driving on a sim, wear shoes that are appropriate. Okay, you don't have to wear racing shoes (fortunately, I've not heard of a simulator bursting into flames, so their fire retardant benefits are not so important!), but it can't hurt.

Hmmm… but I could make the argument that there is a benefit to using a simulator with shoes with less feel than proper driving shoes. The same thing could be said for driving on the street. Here's why.

Say what you will about what Tiger Woods has done over the past few years, on and off the golf course, but there's no doubt that he's one of the greatest golfers to have ever lived. How much of that is due to the way he practiced?

Tiger's practice routine as a child was to have his father tap him in the back of the leg just as he was drawing his club back to drive the ball off the tee. His father — his coach — felt that if he made practice far more challenging than Tiger would ever face, it would make him a better golfer. Tiger got so he could stop, refocus, and start his pre-shot routine over again without it bothering him, all because his father made practice more difficult.

All pro golfers practice — a lot. Tiger didn't necessarily practice any more than anyone else, but what he did consistently was make his practice more difficult. He's famous for dropping a ball in a sand trap, stepping on the ball to sink it into the sand, and then hitting from there; for dropping a ball behind a tree, and practicing hitting around it.

There's no doubt that Michael Jordan is one of the greatest basketball players of all time. Sure, there are stories of him showing up to practice thirty minutes earlier than his teammates, but he also went out of his way to make practice more difficult. He practiced those falling-away, desperate-looking jump shots from the corner over and over again. When he made one of those in a game, it really wasn't surprising.

Michael Schumacher, one of the greatest race drivers of all time (are you seeing a pattern here?), made practice more difficult. After a day of testing where he'd driven the distance equivalent to two full Grand Prix races, he'd go to the gym and workout for an hour, and then work with his "guru" on his mental training program. Race day was easy for him.

Okay, I understand that you're not getting paid what Tiger, Michael, or Michael got paid, so your commitment to practice time is not going to be the same. I get that. But the point is not the amount of time you spend, but what you practice. When you go to the track to drive, I suspect you practice what's easy — driving the right line, with a car that's as well set-up as you can make it, with the right equipment.

What if you made practice more difficult? What if you purposely made your car handle poorly (I can hear it now… “But my car does handle poorly!”), and learned to drive around those problems? What if you wore three layers of gloves so you couldn’t feel the steering wheel as well? What if you wore extra ear protection so the auditory feedback was restricted? What if you purposely drove off line to develop more and different “correction” skills?

Back to your footwork. If you practice applying and releasing the throttle and brakes; practice quick and seamless transitions; if you practice blending when it's an option; if you drive on the street; drive with less-than-ideal shoes; if you drive a less-than-ideal vehicle... then you’re making your practice more difficult. And that will make you that much better on the track. But you have to be deliberate about your practice. You have to be present. And how much of your street driving are you actually present for, and how much are you essentially on auto-pilot?

Be present when practicing on the street. Be aware of your footwork, and practice it.

Hey, a thought just hit me: When Tiger Woods wasn't on the golf course, he wasn’t practicing. When Michael Jordan wasn’t on the basketball court, he wasn’t practicing. Well, they were in their minds, because they used mental imagery. But get this. When you’re not on the track, you’re often driving on the street, so you’re still practicing. Earlier, I said that you can’t practice as much as these superstars do, but in a way you can. You have the advantage of being able to practice your footwork when driving on the street (legally!).

Day #354: Use the brakes to turn the car more or less.

Let's pretend that as you turn into a corner, you experience understeer. So, as you arc your way from the turn-in point towards the apex, your car doesn't go exactly where you want — it tends to push or plough on a wider path than you want. Short of stopping your car right then and there and making an adjustment to tire pressures, anti-roll bars, springs or shocks, what can you do with your driving to reduce understeer and get your car to rotate more towards the apex?

- Trail brake **more**. In other words, delay the release of the brake pedal, slowing down the rate that you ease your foot off the pedal. Of course, what this does — and how it helps — is keep more load on the front tires, increasing their grip. At the same time, it unloads the rear tires to some extent, and by doing so you're aiming to have a slightly higher slip angle on them than the front tires have.
- Trail brake **less**. Yes, it's possible that the understeer is caused by the front tires being overloaded with too much load on them. So, the best thing you can do, then, is to begin releasing the brakes sooner and/or quicker. By asking the front tires to do less decelerating, you're giving some traction back to them for cornering.

Confusing, right? To reduce understeer early in a corner, you should either trail brake more or less! It would be easier if there were only one option to consider, but the nature of performance driving is not easy (but it's simple, right?).

Now, you might be thinking, "Hey, I've made my car rotate — turn more — by popping my foot off the brake, too."

I have to confess that this is something that I've rarely experienced — maybe never. When I really think back through my four-plus decades of driving on tracks, I can't recall ever making this work. And yet, I've had drivers tell me they've done this. So, either I've missed out on something, or what some drivers believe they've experienced is not accurate. Okay, I'm not saying they're lying! But is it possible that what they interpreted as popping the foot off the brake causing some initial oversteer is not really popping the foot off the brake at all?

Let's take a look at the physics of this, and start by asking, "How could quickly unloading the front tires increase their grip in relation to the rear tires — the rear tires that are now being loaded more heavily (which we know should be generating more grip)?"

As you can tell by thinking through this question, it doesn't make sense that quickly popping your foot off the brake pedal would help rotate the car. But how, then, can we explain what some drivers claim to work? I think you know that I spend time every day — and have for decades — either driving or thinking about driving, and this is a question I've wondered about for years.

Here's one possible explanation: Related to the second possible solution to dealing with understeer (which I mentioned above) is that by quickly releasing the brakes, you're giving back grip to the front tires so they can corner. We all know that tires can only generate a certain amount of grip. We'll call that certain amount "100%" — because it is. That's the limit of the tires' grip. And it makes sense that we can't get a tire to generate more than 100% grip. Of course, we also know that we can use that 100% in different ways. For example, we can use 10% of the available grip for braking while turning into the corner and use 90% for cornering. That 10% braking is trail braking, right?

If you're braking at, let's say, 50% — using 50% of the tires' grip for braking — and turning at a rate that is asking for 55% of the grip to be used for cornering, then there's going to be some understeer. If you quickly pop your foot off the brake pedal, even though the load/weight is shifting off the front tires and onto the rears, you're giving the front tires all the grip they need to change the car's direction. Bingo, no understeer because you're not asking for more than 100%.

From a vehicle dynamics perspective, this is the only explanation that I can come up with, but let me know if you have another theory.

Having said that, I strongly believe that a more controlled release of the brakes than simply popping your foot off the pedal will manage your grip level better, resulting in driving more consistently at the limit. Sure, it may be possible to rotate the car by popping your foot off the brakes, but in doing so, you may have an instant where the tires are being used at something less than 100%. And a more deliberate and controlled release of the pedal allows you to manage how much rotation more accurately. As I've said before, think of the brake release as a tool to help you manage the direction your car is going, and to therefore enable you to get back to full throttle sooner, exiting the corner.

Day #355: Advanced techniques are mostly just doing the basics really well.

It's human nature, I guess, to want to learn and master the "advanced" stuff — often before we've gotten the basics down pat.

When you really look at what are often called advanced techniques you realize that they're mostly just the basics, done well.

I'm going to talk about the basics and the "advanced stuff." For example:

- The basics tell us to go into corners slow and come out fast; that exit speed is usually more important that entry speed. Then, as we learn and develop our skills we realize that our corner entry speed can be too low, so we work at improving it. But the basics still apply: if your corner entry speed is too fast (trying to take on the advanced stuff) and is negatively impacting your exit speed, you're usually better off sticking to the basics (in slow, out fast). Of course, the real challenge is finding the perfect balance between quick corner entry speed and fast exit speed. So the advanced stuff is doing the basics right, and finding that perfect balance.
- The basics tell us to be smooth. I'm all for that. But there are times when you need to be a little more aggressive with the controls (steering wheel, brake and throttle pedals) than we were first taught and that's the advanced stuff. Your initial application of the brakes need to be very strong — as strong and aggressive as your tires/brakes can handle. Many drivers (which might include you) are too smooth and slow to ramp up to maximum brake pressure — they need to be more aggressive with the initial application of the brakes. But here's the thing: you still need to do that smoothly. And that's the basics! So, your initial application of the brakes should be hard and smooth — the basics done well.
- The basics for many of you is to do your braking in a straight line before the corner. And here's where I totally, 100% disagree with anyone ever teaching that. That's just plain wrong. The basics should say to do most of your braking in a straight line before the corner, and then gently and smoothly release the brakes as you begin turning into the corner. No, I'm not teaching beginners to trail brake — I don't mention anything about that. I only mention a gentle and smooth release of the brakes. I'm simply building good habits — a smooth release of the brakes. I'm not teaching advanced stuff. But guess what? A smooth release of the brakes builds good habits — the habits that turn into what most think are advanced techniques.

My point is that if you practice the right stuff, it doesn't matter whether you call them basics or advanced techniques, just do them the best you can. As you do this, you're developing your driving skills.

When you compare a novice to an advanced driver, the thing that separates them the most is not that the advanced driver can do super-exclusive techniques that the novice can't. No, the difference is that they do the basics really well, and consistently. In fact, if

I had to pick just one thing an advanced driver does that a novice doesn't, it's the consistency of the basic techniques.

One of the biggest mistakes I see drivers make is trying to take on too much of what they think is the advanced stuff, when what they really should be doing is getting better at the basics (which will lead to the advanced stuff).

So, before you go looking for some new trick driving technique, take a look at your basics, and refine them to the point where you could do them practically in your sleep, consistently, over and over again with the precision and accuracy of a neurosurgeon.

Day #356: Be consistent — then improve.

From the beginning, you're told to be consistent with your driving. Good advice. But what does it mean?

Typically, early in your driving career, it means to be consistently driving the car on a given line, and with your controls (to begin and end braking about the same place each lap, to apply the throttle about the same place and at the same rate each time, to turn the steering wheel the same place and way each lap). Essentially, it's being consistent with your physical skills — and the line is the most obvious, and therefore the one that gets the most attention.

Being able to consistently drive a given line around the track is important. It demonstrates (to you and any instructors in the general vicinity) that you're able to do what you want with your car. It means your basic car control skills are adequate. Perhaps not good enough to be really fast, but certainly good enough to be able work on driving faster.

Once you're consistent with your physical skills, then you can begin to experiment with other techniques that may lead to an improvement in speed. If you experiment too much before you're consistent, how are you going to know whether something you changed actually helped or hurt? That's why being consistent with your physical skills is important.

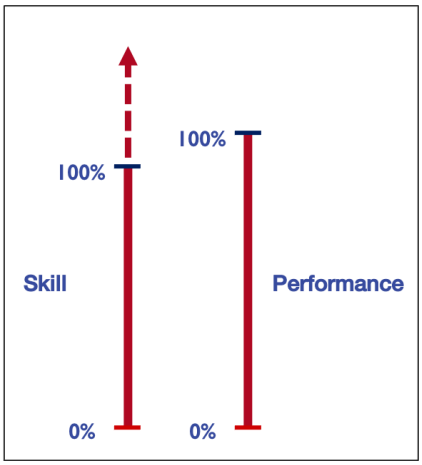
But there's another way to look at what being consistent means.

Have you ever had a day, a session, a lap, or even just a corner where you felt fantastic, and everything you did seemed almost effortless, everything flowed, everything just happened without you over-thinking it or trying super hard? Unless you're in that state every second of every lap of every session of every day at the track, your performance is not consistent. If you are in that state all the time, well, I want to meet you because not even highly-paid elite-level drivers like World Champion F1 drivers perform at that level every time they're in the car.

Notice that your performance is more than just your physical skills. Even drivers with World Champion-level physical skills can perform great one day and poorly (relative to their best) the next. It's their overall performance that isn't consistent.

When I ask drivers how often they perform at their very best, in that state I described above, the most common answer is, "Not enough!" Which means that you're not alone if you don't perform at your very best every second you're on track.

I look at performance and race driving this way: You first want to consistently perform at your very best, at your 100%. Then, you want to improve your best, your 100%.



I see too many drivers mixing up which of these two directions they should be focused on. I see some drivers who are not consistent with their performance, and yet they're focused on improving their physical skills. Instead, they should be focused on becoming more consistent with their overall performance. I see other drivers who perform consistently, and yet are not improving much. They should be experimenting, pushing the envelope with their skills to raise their 100%.

Another way to look at this is that you have your physical skills, and then you have your overall performance. You can be fantastically consistent with your physical skills, and yet be "on" one day and not another. In other words, you're consistent with your skills, but your overall performance is up and down. There's no surprise that the difference is in that area that we lump into the category called "the mental game." And yet, so many drivers don't have a clue where to start with their mental game.

Obviously, the mental game is kind of my sweet spot in terms of coaching drivers, providing resources for drivers to coach themselves in this area. After all, "Gray Matters," right?! It's your mind that's driving your car.

Take a few minutes today to consider what your priorities should be. Should you focus on performing more consistently with your current level of physical skills, or should you work at improving those skills? Come on, it's okay, put off your work for fifteen minutes or so while you focus on what's most important!

What are you waiting for? In the next ten to fifteen minutes, I want you to think about and decide exactly whether you should be working on improving your physical skills, or ensuring your overall performance is more consistent. Now, write it down — make notes of what you decided — as that'll help you stay focused on the right priorities.

Day #357: Learn how to be passed.

Put your hand up if you love to be passed by other drivers on the track. What? I don't see any hands! Could it be that none of us really like to be passed? I'm guessing that's the case. But, it happens. No matter how good a driver you are, there are others in faster cars — and maybe, just maybe, there are drivers who are just a tiny bit faster than you (I know, hard to believe!).

There are ways to be passed, and ways not to be passed. And I'm not talking about blocking someone from passing you. I'm talking about ways to be passed so that you lose the least amount of time.

If you've ever gone into a corner with another car beside you, out-braking you into the turn, then you know what I'm talking about. If you get to the turn-in point and the other car is level with you, you don't have any choice but to ease off, wait, and then tuck back in behind. Doing that is one of the best ways of losing the most (something you don't want to do). So, what could you have done?

If, as you get to the brake zone with another car beside you, you brake a little early (sometimes just a car's length or two earlier), the other car will now be positioned in front of you. Yes, you don't like being behind another car, but your goal at this point is to minimize the damage. By letting the other car get in front of you, it isn't controlling your corner speed as much as if you were side by side.

One approach to being passed is to ensure that the passing car does not impede your progress by making you over-slow for a corner. And you want to be able to begin accelerating out of the corner as early as you usually do, so do whatever it takes to make that happen. Most often that means letting the passing car go by you early enough in the brake zone and/or corner entry, so that you're back on your line and accelerating as early as possible.

It's often tempting at that point to try the over-under move where you then re-pass the other car on the exit. Since the passing car had a tighter line entering the corner, its exit speed will be hurt. But you have to be realistic and make a smart decision. I often see drivers try to re-pass a car exiting the corner, and in doing so, actually lose more time than if they'd just given that position up and focused on getting a good launch out of the corner. By trying the over-under move and moving to the inside exiting the corner (thinking that a pass back can be made), you've hurt your exit speed by not tracking all the way to the outside of the track.

There comes a time to fight for a position, and there's also a time to give it up. You can lose the war by winning the battle. This is especially true in endurance racing where there are multiple classes. Trying to re-pass a car from a faster class might hurt your overall lap time, giving competitors in your class a chance to catch you or gap you even more.

Often just a slight breathe of the throttle on a straightway will allow a faster car to get past you and out of your way. That's the key — doing what it takes to ensure a passing car is not in your way, negatively impacting your lap times. Drive smart by thinking strategically.

Day #358: Use deliberate practice. Take time to lay out a practice strategy.

Over the years I've often referred to the research conducted by Dr. Anders Ericsson about how high performers in any activity get that way, and especially his conclusions regarding how much of a superstar's abilities are natural talent versus developed skills. Much of his research has been shared in an easy-to-read book co-authored by Robert Pool. It's called *Peak*, and I highly recommend it.

The main take-aways from the book, and Ericsson's research can be summed up like this:

- The amount and type of practice someone puts into an activity is the number one determinant of one's ability.
- There is little to no evidence that one's DNA has any impact on how well someone ultimately performs an activity, other than one's basic physical make-up (if your DNA means you're 5 foot 6 inches tall, you're less likely to make it as a professional basketball player).
- There are three types of practice: the kind that most people put into an activity, and what he calls "purposeful practice," and "deliberate practice". It's this last type (what I've called "strategic practice" in my books) that is the most effective — and it's what separates the best from the rest.
- "The 10,000 hour rule," as popularized by Malcolm Gladwell in his *Outliers* book, is inaccurate, and took Ericsson's research out of context. While superstars of any activity practice more than others, it's how they practice that matters the most. And the 10,000 hours of practice that Gladwell wrote about was an average — there are superstars who practiced less than 1,000 hours, and some who practiced for 30,000 hours. Simply practicing for 10,000 is not a guarantee that you're going to become a high performer or expert.

What does this mean to you? What I've been saying for years: You need to take a more strategic or deliberate approach to how you practice, to how you spend your time improving.

Many schools, many instructors, many coaches have you do the same things over and over again. Yes, that's building mental programming, but often that's not the most efficient way to improve. In other words, you could build that mental programming — those skills — more quickly. If, for example, you're driving around and around the track, trying to practice everything all at once, that's not very effective. It's practice, but it's not purposeful or deliberate practice. I relate this to the way football teams practice. They don't show up to a practice session and play a game, do they? No, they do blocking drills, passing drills, kicking drills, catching drills, and so on. In other words, they break the game down into discrete skills and work at refining just those, over and over again — deliberate practice. And then, once those skills are well-refined, they put the whole game together in a practice game, a scrimmage.

Performance and race drivers, on the other hand, typically go on the track and work on developing all their skills, all at one time.

Of course, this assumes that the driver even focuses on their performance improvement. Beyond a few basic things, football teams don't have a lot of equipment that can affect their overall performance, which is not the case for us drivers. It's very easy — too easy — to focus all or most of your attention on making the car faster. If there's some focus aimed at improving your driving, it's things like changing the line you drive through a corner. Using Ericson's definition, this is purposeful practice. It's good, but it's not as effective as deliberate practice.

BMW CCA instructor and regional DEC director, Steve Stepanian, once told me that he will ask, "What is the one performance improvement you can buy for your car that will never wear out?" The answer, of course, is "driver development." Everything else you spend your money on will wear out someday.

When was the last time you took an entire session, or day, or even weekend to solely focus your practice on just one aspect of your driving? As an example, what if you took a weekend and only drove on old tires (I know, some of you are saying, "That's all I ever drive on!"), developing your ability to get the very most out of old tires. I know many drivers who are very fast on new or relatively new tires, but very few who are fast on old tires. What if you deliberately made your car handle poorly, and you worked at adapting to it? What if you experimented with different ways of applying and releasing the brakes, then with how you turn the steering wheel? One thing that separates the best drivers from the rest is the size of their database of driving techniques, and their ability to use different techniques in different corners (many drivers use the same technique, no matter what corner they're driving).

When I suggest this type of deliberate practice to some drivers, they respond by saying they need to just focus on shaving half a second off their lap times. Or that's what they focus on when on track. Or they just ignore me. They're unwilling to step back from the constant chase of the almighty lap time to focus on things that will ultimately lead to consistently faster lap times in the future.

If you've read what I've written or heard me talk over the years, what I'm saying here is not new, right? But when scientific research over decades, and with performers in practically every form of sport, game, music, or whatever clearly states this is the best way to improve, I get frustrated when I see drivers not using this approach — especially when I see them getting frustrated with not improving.

Day #359: Improve your corner entry speed.

To begin, we have to ask whether we are truly over-slowing our corner entries, or not. It may seem slow, but is it? What is “over-slowed”?

Let’s look at the ideal corner entry speed this way:

- If you had carried even one MPH more speed into the corner, you would have to delay applying the throttle and hurt your exit speed (I'm assuming that exit speed is more important than entry speed, due to the length of the following straightaway).
- If you had slowed even one MPH more (one MPH less speed), you wouldn't begin applying the throttle any earlier to improve corner exit speed. And you would have begun accelerating from a lower speed, negatively impacting the top speed on the following straightaway.

We all know that one way to start accelerating earlier out of corners is to enter too slowly — let’s say entering at just five MPH! Hey, at that speed you could then stand on the throttle right at the turn-in point! But if you enter and carry just one MPH more through the corner, and still apply and get to full throttle at the same place, you’re going to be faster down the following straightaway. The faster you are when you begin to accelerate, the faster you’ll be on the following straight.

Of course, if you carry too much speed into and through a corner, you'll have to wait before getting on the throttle, and that will hurt your exit and straightaway speed. This is the core reason for the “In slow, out fast” advice that is sometimes over-prescribed. Yes, for some drivers, the reason for over-slowing corner entry is the “In slow, out fast” message has been driven into their heads too much. They’re programmed to believe that entering slow is the only way to get a good, fast exit. It’s not.

The ideal corner entry speed is a compromise. Too little and you’ll be slow down the following straight. Too fast and you’ll be slow down the following straight.

Keeping the ultimate goal of the shortest lap time in mind, it’s easy to understand why finding the ideal compromise is so difficult — and I’d suggest that it’s what draws us back for more and more driving on a track. Sure, we can begin accelerating earlier if we slow more on the entry, but what do we lose in doing so? And sure, we can carry more entry speed, but how much later does that cause us to begin accelerating?

As I hinted at above, there are segments of some tracks that include corners and straightaways where it is better to sacrifice exit speed for entry speed, specifically when the straightaway after the corner is not long. While these are rare, they do exist, and should be considered.

Why Do We Over-Slow?

As I said, for some drivers, the reason for over-slowing is that they have had it too deeply programmed in their minds that they have to slow to be able to get a good exit out of a corner. It's been pounded into their heads! So, the first step in improving one's entry speed is to be open to the idea that you don't necessarily have to over-slow to get on the throttle early exiting the corner.

It's possible to be "fast in, and faster out" of corners.

Another reason for over-slowing is self-preservation — and possibly even fear. Once you're in a corner, you know what you've got, and accelerating out of it provides a certain level of risk — and therefore, less fear and self-preservation. But entering a corner, you don't know exactly what you have until you're there. You predict what's going to happen, and in most cases, you'll leave a margin for error. It's this margin that leads to over-slowing.

This reminds me of a quote from Mark Donohue which I've referred to more than once here and in my webinars: "Exiting a corner with the tires at their limit is like tight-rope walking. Entering a corner with the tires on the limit is like jumping onto a tight-rope, blindfolded." It's this sense of not knowing what you have until you get into a corner that leads to over-slowing. It's the fear of the unknown.

I also see a lot of drivers who over-slow because of how they use their vision. First, when approaching a corner that requires braking, they focus far too much on the BoB (Begin-of-Braking), rather than the EoB (End-of-Braking). Therefore, their attention is on where they begin braking, and not on the upcoming corner. When they do get to the corner, their attention is rapidly trying to switch to the corner itself. But more importantly, they have not managed their braking to suit the corner, but rather some pre-determined point where they first applied the brakes. Is it possible that your car is traveling faster or slower than usual when you get to this pre-determined BoB, in which case it's not the appropriate place to begin braking?

I've written in the past about starting a corner with the end in mind, and this one strategy has helped many drivers improve their corner entry speed. Practically all performance and race drivers were taught there are three key references for every corner: turn-in, apex, and exit or track-out. And it's presented in that order because, well, that's the order we experience them. But the best drivers work backwards from the end of a corner, and that helps them manage their entry speed. The old saying, "You can't get somewhere if you don't know where you're going" could be reworked into "You can't set your corner entry speed if you don't know where you're going."

Beginning the brake zone too early seems like an obvious reason for slow entry speed, but maybe not for the obvious reason. The obvious reason is that if you begin braking early, you'll be slow by the time you get to the turn-in point, and this is accurate. But unless you're braking way too early, there's a less obvious reason for over-slowing. Let's say you begin braking even one car length too early. When you get to the turn-in point, your speed will not be much slower than ideal. But what early braking leads to is

early ending of braking (remember your brake zone is from the BoB to the EoB, and you need to think in terms of the brake zone, rather than brake points), and when you release the brakes early, you unload the front tires and your car will likely not respond to your steering input the way you want. It won't turn in very well, and that leads to your brain coming to the conclusion that you haven't slowed enough. So, the next time you come to this corner, you won't want to move your brake zone into the corner further. In fact, you might even begin braking earlier, leading to even more over-slowing.

When you have a picture of your brake zone — from BoB to EoB — in mind, and you shift it further down the track, into the corner a little more (maybe just a car length or so), you'll turn in with an additional MPH or so, the front tires will be loaded enough that they respond to your steering input, and the car changes direction the way you want — towards the apex. This starts with having both physical and mental focus on the EoB.

How to Improve Your Corner Entry Speed
Now that I've defined what over-slowing is, and what some of the main causes are, I'll list five ways to improve your corner entry speed:

- Be open to the "fast in, faster out" idea. Embrace it.
- Breathe. Fear and self-preservation can be controlled by breathing. Combined with thinking and looking further ahead, entry speed will naturally increase.
- Have a mental image of you and your car moving around the instant you turn into corners. In fact, make it do that by carrying enough speed into the corner that it begins to understeer or oversteer immediately. That way you're expecting it, and you know what to do — you're proactive, rather than reactive. Instead of fearing the unknown, create the known by carrying enough entry speed so that you know what you're going to have to do.
- Focus your vision and mental attention on the EoB as you approach corners.
- Use the brakes as a fine-tuning tool for corner entry, and ultimately, to help you carry higher minimum corner speed, while getting to full throttle earlier.

Finally, the ideal corner entry does not necessarily mean higher speeds. Your corner entry speed may be perfectly fine right now, and raising it may lead to slower lap times (or worse). Of course, the only way to know for sure is to go slightly over the top, and carry just a touch too much speed. Experiment with this, using your brakes to manage corner entry speed, but do it in small bites. Add a couple of MPH because you know you can control your car if you enter with two MPH too much speed, but not necessarily if you carry five MPH or more. Work your way up to it.

Day #360: Understand what's causing a problem before looking for the solution.

Driving fast is not easy. Well, duh! Consider this:

- Trail braking too much can cause your car to understeer or oversteer on corner entry.
- Trail braking too little can cause your car to understeer or oversteer on corner entry.
- Getting on the throttle too hard can cause your car to understeer or oversteer exiting a corner.
- Turning the steering wheel too quickly at turn-in can cause your car to understeer or oversteer on corner entry.
- Getting on the throttle too early can delay getting to full throttle.

If an action can cause one of two reactions, how do you know how to fix it?

First of all, you need to understand the why behind each scenario — what's really causing the problem - so let's look at each individually.

Too much trail braking causing entry understeer: You're asking the front tires to do too much — to brake and steer at the same time — so the tires give up and begin to slide. Reducing the amount of brake pressure will lessen or eliminate the understeer.

Too much trail braking causing entry oversteer: You're turning into a corner with too much weight on the front end, and not enough on the rear, causing the rear tires to slide. Reduce brake pressure to transfer some weight to the rear.

Too little trail braking causing entry understeer: You've come off the brakes, unloading the front tires, resulting in them sliding. Continue to trail brake a little more or harder.

Too little trail braking causing entry oversteer: You've come off the brakes fairly abruptly, causing the balance of the car to be unstable, resulting in less overall traction (and due to the way your car handles, it oversteers). Release the brakes more gently.

Too hard on the throttle causing exit understeer: Weight transfer issue — you've un-weighted the front tires, and they slide more than the rears. Be more gentle with your throttle application.

Too hard on the throttle causing exit oversteer: You're generating power oversteer, where you're breaking the rear tires loose (exceeding their traction limit), resulting in wheelspin and oversteer. Be more gentle with your throttle application.

Turning the steering wheel too abruptly causing entry understeer: The sudden turn of the front tires means they haven't been given enough time to generate the side forces, so they "give up" and slide. Be more gentle with your turn-in.

Turning the steering wheel too abruptly causing entry oversteer: The sudden change in direction results in the rear being "whipped" to the side, and that momentum continues, resulting in the oversteer. Be more gentle with your turn-in.

Getting on the throttle too early results in being late to full throttle: You apply the throttle early (because you've been told over and over again that exit speed it critical) — so early that the car has not rotated enough around the corner that you have to breathe the throttle (to avoid driving off the track) before you can fully commit to full throttle. Getting to full throttle is what matters most, not when you start giving it a partial throttle. Hesitate very briefly, to the point where you can go all the way to full throttle in one application.

I've found that in most cases, once a driver understands the cause and effect of each of these scenarios — the why behind them — they become more aware and begin to sense what leads to what. And that's the key to addressing each of these issues.

Nobody said it would be easy.

Day #361: Save your tires.

If you've ever driven an endurance race, you know that the car suffers over the course of a stint, let alone the whole race. And the tires get the worst of that abuse. I think it was Derek Bell who joked years ago (I think it was a joke) that the best endurance drivers know when the car is going to break, so they come in and hand it off to a co-driver just before it does. I know some drivers who apply that to the tires — use them up and then hand the car off to a co-driver. Depending on how often you change tires in an endurance race — or whether you do at all — this can definitely work against you, as you may be the last one driving on those tires.

A driver asked me how some F1 drivers seem to be able to save their tires better than others, so I'll use that as my point of reference. Here are some things drivers can do to save their tires:

1. Rate of steering input. Some drivers make the initial turn of the steering more abruptly than others. If you very subtly begin turning the steering, and then build up the cornering force, that's easier on the tires. It often leads to higher cornering forces, too. The analogy I have used before is this: Take a piece of string and gently and slowly pull both ends apart; it takes a lot of strength/force to snap it. But quickly pull or snap both ends apart and the string will break more easily. If you gently build up cornering force, the tires will hang on longer, and the longer they hang on, the less wear and tear you're putting on them.
2. Turn the steering wheel less. If you feel the tires sliding, do what you can to unwind/straighten the steering wheel. Stating the obvious, tires running in a straight line don't wear out as much as ones on an angle to the direction of travel. Think about letting the tires "run free."
3. Timing and rate of release of the brakes. You knew that was coming, right? I know you've heard me talk/write about this often, and there's a reason for it. This is where the "magic" is of driving fast. It also helps save the tires. If you enter a corner and snap your foot off the brake pedal, what's going to happen? The front of the car is going to unload, and you'll be relying on the steering angle alone to turn the car. That means extra work for those front tires. If you release the brakes in a way that transitions the workload the front tires are doing, from braking to cornering, you're helping those tires with their workload.
4. Squeeze the throttle. The above three techniques are focused on saving the front tires, so what about the rears? If you're driving a rear-wheel-drive car, it's mostly about how you apply the throttle, and specifically how much longitudinal tire slip you generate. A drag car generates lots of tire wear in a short period of time due to tire slip, or wheelspin. That's at the far end of the spectrum, but even if you don't have 5,000 horsepower, being too aggressive with the throttle will use up the rear tires more than if you squeeze on, or feed in, the throttle. Unrelated to throttle usage (but still focused on rear tire usage), if you're driving a front-wheel-drive car, rotating the car with too much trail brake will overwork the rears, using them up more.

5. Ease on the brakes. If you're looking for outright speed, being easy on the application of the brakes is not the thing to do. But if you want to save the tires, then easing them on will help; it's much the same as building up cornering forces.
6. Be sensitive to tire temp. We've all experienced driving on cold tires, and then having them heat up and gain grip. The same thing happens on the other side of the traction versus tire temperature relationship; as the tires build heat beyond their ideal or peak, they lose traction. Another way of thinking about that is tires build more heat as they lose grip. So if you notice the tires beginning to lose grip, there's a very good chance they've overheated. If you deliberately practice paying attention to the tires' grip level as they heat up, stay around their ideal, and then become overheated from too much sliding, you can adjust your driving to help them. Most drivers are surprised at how little you need to back off to bring the tire temps down to their ideal range. Turning the steering wheel less, being more progressive and smooth in your throttle, and your brake application and release will make a big difference.
7. Be sensitive to the grip level. If you feel the grip level falling off, use the techniques mentioned above. It's best not to wait. The sooner you deal with the grip falling off, the better.

There are times to save the tires, and times not to (when all that matters is going as fast as possible). I've noticed something interesting: the fastest drivers are also the best at saving tires. Think of Scott Dixon and Colin Braun. While they use the driving techniques (tactics) I mention in 1 through 5 above, ultimately it's their sensitivity to tire temp and grip that tells them when to use those techniques. That's strategic driving.

Day #362: Listen up.

Research has shown that we process information with our hearing faster than we do with any of our other senses.

Looking far ahead is stressed over and over again to drivers of all levels. For sure, looking farther ahead makes a difference — a big difference. It's our vision that gets us around the track, and you can't drive smoothly if you're looking just in front of your car.

But when it comes to sensing how close you're driving to the limits of the tires, could it be that your ears play a bigger role than your eyes?

When I ask drivers which sense is most important, I've never had one tell me it's their hearing. In fact, I don't recall a single driver not tell me that vision is most important. And since it would be difficult to make a lap of a track if you were blind, it's easy to back up the argument that our eyes are the most important. Every now and then a driver will talk about how important one's kinesthetic sense is to feeling the limits of the car, but when pushed they'll go back to claiming vision is the number one sense they use when driving.

I'm not going to say that that logic isn't strong, or that it's wrong. If your only goal is to get around a track, then vision is certainly the most important sense. But if your goal is to drive your car at its limits, then your kinesthetic and auditory senses become as important, and perhaps even more so.

Think about driving through a corner, with your car on the edge of traction. It understeers a little, oversteers a little... it's at that point where another MPH will make it slide too much, but it's also moving around — it doesn't feel like it's on rails, for sure!

You feel a slight lack of responsiveness through the steering wheel, indicating the front tires are sliding. You feel the g-forces build, then begin to reduce just slightly, as if a piston has gotten to Top Dead Center and begun moving back down the cylinder. Your internal gyro senses yaw — the car has rotated, as if viewed from above. These are just some of the kinesthetic feedback you're taking in, and responding to.

Meanwhile, you hear a different sound from the tires, the pitch changing slightly; the wind noise is also changing. You don't think about these things on a conscious level. In fact, if you're asked what sound your tires make through the corners, you say you can't hear them. But you can — just not at a conscious level.

And remember, your brain is processing this auditory information faster than the kinesthetic and visual information also going into your mind.

Your ability to sense the limits, and to drive at the limit is impacted significantly by what you hear — both consciously and subconsciously.

The great thing is you can fine-tune and develop your auditory senses, just as a person who lost their sight would do — by practicing. Or, by practicing deliberately. Spend a session or two or three or more focused solely on listening to your car, your tires, the sound of the wind passing by.

Day #363: Use a "3" grip on the steering wheel.

A friend of mine is both an avid guitar player and golfer. He told me about his music teacher advising him to hold his guitar with "3" pressure. In other words, on a scale of 1 to 10, with 10 being maximum grip strength, he should use "3" pressure.

It was interesting to him that his golf instructor had told him to do the same thing when holding a golf club.

Of course, everything in my life revolves around driving, so I thought about how this applied to holding the steering wheel.

There are times when you might need to hold the steering wheel with slightly more than a "3" pressure, for instance, when there are lots of kickback from bumps in the track combined with heavy g-loads. But there are also times when slightly less than "3" would be best. So, on average, I'd agree that playing the guitar, holding a golf club, and gripping the steering wheel requires about the same pressure.

When you grip a steering wheel tighter than a "3" — let's say, a "6" or higher — do you think you receive less or more feedback through the wheel and up through your hands and arms? Less, right? In fact, the lighter your grip on the wheel, the more sensitive you'll be to vibrations, feedback and effort. All of those give you hints toward how close your tires are to their limit.

It's easy to think of a steering wheel as an input device, but it's an input-output device. Too many drivers think of it only as an input advice. However, we not only use it to make the car do something, but we get feedback from it to determine the limits of the tires. But you only get output — the feedback — if you're holding the wheel with a relaxed, light grip.

The next time you're on track, be aware of, and rate, how tightly you're holding the wheel. A "3" pressure is a pretty good target.

Day #364: Replay your successes.

Here's some homework for you: Make a list of all the things you did well this past year, in and out of the car (but only to do with driving!).

For some reason, we tend to focus too much on what we do wrong. It seems easier to see all that we do wrong than what we do right.

But think about this: If we do enough of the right things, we don't have time to do the wrong things. And if we replay the things we've done right, we tend to do more of them.

One of the best ways to get into a performance state of mind is by replaying in your mind one of your greatest past performances. By making a list of everything you've done well this past season, it'll help put you into a performance state of mind — it can't help but do that.

But this will take some effort. Take it seriously (but have some fun, too). Put some time aside, sit down with a blank sheet of paper and just begin writing everything that comes to mind when you ask yourself these questions:

- What did I do well this past season, in and out of the car (related to driving)?
- What skills and techniques did I perform well most of the time?
- What mental "skills" did I perform well?
- What did I do well with my team this year?
- What car feedback and setup work did I do well this past season?

Don't be critical as you write your answers out — just write them down. Sure, you can always do better, but I'm sure there are many more things that you did well than you give yourself credit for. So, in keeping with the theme of the season, be generous with yourself.

Now, sit back and think about everything you've written on your list. Just enjoy it. Enjoy the state of mind you're in as you re-read your list. Feels pretty good, right?

Congratulations to you for all you did well this past season!

Day #365: Ancora imparo.

I'm still learning. "Ancora imparo" is what Michelangelo apparently said. It's what I say a lot. Well, not exactly those words, as it's roughly Italian for "Yet, I am learning," and I don't speak Italian. But the sentiment is dead on.

I have a long list of things that I've either learned, re-learned, clarified, re-interpreted or gained a deeper understanding of in the past ten years, such as:

1. Why braking lighter and longer for a corner sometimes results in a faster lap time than leaving your braking to the last second and using threshold braking. The basic reasoning: a car that is better balanced (with less weight transfer) will have more overall grip, often providing better braking and cornering.

2. Why coasting — or "hesitating," as I prefer to think of it — between fully releasing the brakes and beginning to apply the throttle can sometimes lead to faster lap times. It is usually advantageous to allow the car some time to finish rotating and get it pointed, so you're able to go to full throttle sooner.

3. When to compromise corner minimum speed so you can get to full throttle sooner. This is the classic "slow in, fast out" balance. But one way to get to full throttle sooner is to enter every corner at ten MPH! If you over-slow, it doesn't matter how early you get to full throttle; you're not going to make up for what you lost at the beginning of the corner. Of course, if you carry too much entry speed, it delays getting back to full throttle, making you slower on the straightaway. Compromises!

4. The last little bit of the brake zone should be used to fine-tune one's corner entry speed and car rotation. That means you need to make your initial application of the brakes hard, so you get most of the speed taken off early. Then you can use the last bit for finely adjusting entry speed and the balance of the car (which affects how the car turns in).

5. Early turn-in is not all bad. In fact, if used correctly, along with the right amount of rotation through managing the car's weight by trail braking, you can carry more entry speed without it negatively impacting your exit speed (sometimes it even helps!).

6. The positive effect of "collecting" (and filing away in your mind) more reference points around a track. One thing that separates the best from the rest is the quantity and quality of references.

7. The End-of-Braking is more important that the Begin-of-Braking. Yup, simple as that.

8. How little deliberate effort most drivers put into the mental game, and how huge the opportunity is for those drivers who do incorporate this into their preparation (even if they're not in competitive driving, they're taking advantage of the opportunity to learn and improve their driving…). One driver I coach has made the use of mental imagery a big part of his development program, and his performance is far beyond what most drivers achieve with much more seat time. He's a world-class driver, and yet drives on track less than many HPDE and track day drivers I know. I understand that many drivers don't know what to do with their mental game, despite knowing it is critical. I also know that it feels as though it's easier to work on the things you can hold in your hands, like car parts, and even the line through a corner (holding your brain in your hands is not recommended!), but the biggest opportunities for improvement are usually in the mental game.

9. The importance of preparation prior to going to the track.

10. The importance of how you practice — being strategic and deliberate with your approach, as well as with what you practice.

11. How the rate of turning the steering wheel is often more important than when you turn it.

12. The "vision process," which is more than just looking farther ahead.

13. What you learn at the track, and especially behind the wheel when on track, applies to just about everything else in your life. In fact, people most often comment that the concepts I teach have helped them in their professional careers, family life, and other activities. That's a keeper.

There's always more.

If you're racing, and you put your car on the pole by over a second, you lead every lap of the race, lap the entire field while setting and re-setting the fastest lap of the race over and over again… there's still more.

There's always more.

Okay, there's a limit, but my point is that there's always more to learn.

If you find yourself pushing too hard for a lap time or a position, then you've gone too far. But in terms of learning, or improving, there's always more.

That is, unless you've learned everything there is to know about driving, and you've absolutely perfected it. Then I guess there's no more. I hope you never get there. (But to help, go to SpeedSecrets.com for more)

BONUS Day #366: Have fun!

If you're not having fun doing what you're doing at and around race tracks, there's something very wrong.

I can't tell you the number of drivers I've coached who had forgotten why they were doing what they were doing. They were so focused on going fast, and winning races, or chasing down their buddies on a track day event that they had completely lost focus on why they were doing what they were doing. But then, when they focused on having fun, guess what happened? They went faster, won more races, and chased down their buddies on a track day.

Oh, I had to learn this lesson the hard way. When I got to race Indy cars, even though I had achieved something I'd wanted to do since I was a kid, I forgot to enjoy it. I was racing for an underfunded team, and it was always a struggle, but still... I was driving an Indy car, for cryin' out loud! Then, it hit me. I "woke up" and realized that I was in a very special place, thought about what else I could be doing (having a regular job!), and focused more on the fun I was having driving an Indy car. And guess what? I drove better.

So, the Speed Secret above may sound simplistic and unneeded, but I don't think it ever hurts to be reminded of it.

Have fun!